

Die Ausdehnungslehre von 1844  
oder  
die lineale  
Ausdehnungslehre  
ein  
neuer Zweig der Mathematik  
dargestellt  
und  
durch Anwendungen auf die übrigen Zweige der Mathematik,  
wie auch  
auf die Statik, Mechanik, die Lehre vom Magnetismus und die  
Krystallonomie erläutert  
von  
Hermann Grassmann.

Zweite, im Text unveränderte Auflage.

Mit 1 Tafel.

Leipzig  
Verlag von Otto Wigand.  
1878.

# Contents

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |    |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| Foreword to the First Edition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 9  |
| Foreword to the Second Edition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 19 |
| Introduction                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 23 |
| A. Derivation of the Concept of Pure Mathematics                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 23 |
| B. Derivation of the Concept of Extension Theory                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 25 |
| C. Exposition of the Concept of Extension Theory                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 28 |
| D. Form of Presentation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 30 |
| Survey of the General Theory of Forms                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 33 |
| § 1. Concept of Equality.—§ 2. Concept of Conjunction.—§ 3. Combinability of Factors.—§ 4. Interchangeability of Factors. Concept of Elementary Conjunction.—§ 5. The Synthetic and Analytic Conjunctions.—§ 6. Uniqueness of Analysis; Addition and Subtraction.—§ 7. The Indifferent and Analytic Forms.—§ 8. Addition and Subtraction of Similar Forms.—§ 9. Conjunctions of Different Order. Multiplication.—§ 10. General Law of Multiplication.—§ 11. Law of Division.—§ 12. Real Concept of Multiplication. |    |

## PART ONE

### EXTENSIVE MAGNITUDES

#### CHAPTER ONE

|                                                                                                                                                                                                                                                                                     |    |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| ADDITION AND SUBTRACTION OF ELEMENTARY EXTENSIONS OF FIRST ORDER, OR DISPLACEMENTS                                                                                                                                                                                                  | 45 |
| A. Theoretical Development                                                                                                                                                                                                                                                          | 45 |
| §§ 13, 14. Extensive Structure, Displacement, and the System of First Order.—§ 15. Addition and Subtraction of Similar Displacements.—§ 16. Systems of Higher Order.—§§ 17–19. Addition and Subtraction of Dissimilar Displacements.—§ 20. Independence of the Higher-Order System. |    |
| B. Applications                                                                                                                                                                                                                                                                     | 60 |
| §§ 21–23. Inadequacy of the Present Foundation of Geometry and Attempt at a New Formulation.—§ 24. Geometric Problems and Theorems; Mean Between Several Points.—                                                                                                                   |    |

§ 25. The Newtonian Principles of Mechanics.—§ 26. Collective Motion, Motion of the Center of Mass.—§ 27. Remark on the Applicability of the New Analysis.

## CHAPTER TWO

OUTER MULTIPLICATION OF DISPLACEMENTS 73

§§ 28–30. Product of Translations in Geometry, Preliminary Treatment 73

A. Theoretical Development 76

§ 31. Generation of Extensions of Higher Order.—§ 32. Extensions of Higher Order as Products.—§§ 33, 34. Fundamental Law of Outer Multiplication.—§§ 35, 36. Principal Laws of Outer Multiplication.

B. Applications 85

§§ 37–40. The Sign Change Law in the Interchange of Spatial Factors.—§ 41. The Static Moment.—§§ 42, 43. Theorems on the Collective Moment. Equilibrium of Rigid Bodies.—§ 44. The Interchange Law Corroborated by Statics.—§§ 45, 46. Solution of Algebraic Equations of First Degree with Several Unknowns.

## CHAPTER THREE

CONJUNCTIONS OF EXTENSIVE MAGNITUDES OF HIGHER ORDER 97

A. Theoretical Development 97

§§ 47, 48. Sums of Extensions in a System of Next Higher Order.—§§ 49, 50. Validity of the Addition Laws for This New Sum.—§ 51. Formal Sum or Additive Magnitude.—§§ 52, 53. Multiplication of Extensive Magnitudes.—§§ 54, 55. Principal Laws of Outer Multiplication.

B. Applications 108

§ 56. Product of Translations in Space.—§ 57. General Concept of Collective Moment.—§§ 58, 59. Dependence of Moments.

## CHAPTER FOUR

### OUTER DIVISION, NUMERICAL MAGNITUDES 113

#### A. Theoretical Development 113

§ 60. Concept of Outer Division.—§§ 61, 62. Reality and Ambiguity of the Quotient.—§ 63, 64. Expression for the Unique Quotient.—§ 65, 66. Concept of the Quotient of Two Similar Magnitudes.—§ 67. Proportion.—§ 68. Numerical Magnitude: Its Product with an Extensive Magnitude.—§ 69, 70. Product of Several Numerical Magnitudes.—§ 71. Validity of All Laws of Arithmetic Multiplication and Division for Numerical Magnitudes.—§ 72. Addition of Numerical Magnitudes.—§ 73. Relation of This Addition to Multiplication. General Law.

#### B. Applications 133

§ 74. Numerical Magnitudes in Geometry.—§§ 75–79. Pure Geometric Representation of Proportion in Geometry.

## CHAPTER FIVE

### EQUATIONS, PROJECTIONS 139

#### A. Theoretical Development 139

§ 80. Derivation of New Equations from a Given One by Multiplication.—§ 81. Recovery of the Original Equation.—§ 82. Projection or Shadow. Shadow of a Sum.—§ 83. When the Shadow is Zero, and When it is Not Possible.—§ 84. Shadow of a Product and of a Quotient. General Law.—§ 85. Analytic Expression for the Shadow.—§ 86. Derivation of an Assembly of Equations that Replaces the Original.—§ 87. Reference Systems (Coordinate Systems), Reference Domain, Reference Measures, Principal Measure.—§ 88. Reference Terms, Indicator.—§ 89. Equations Between the Reference Terms and Between the Indicators.—§ 90. Shadows of an Equation in the Sense of a Reference System. Expression for the Indicator.

#### B. Applications 149

§ 91. Shadow in Geometry.—§ 92. Transformation of Coordinates.—§ 93. Elimination of an Unknown from Equations of Higher Degree.

## PART TWO

### ELEMENTARY MAGNITUDES

#### CHAPTER ONE

##### ADDITION AND SUBTRACTION OF ELEMENTARY MAGNITUDES OF FIRST ORDER 154

###### A. Theoretical Development 154

§ 94. Law for the Sum of Displacements Drawn from a Moving Element to a Series of Fixed Elements.—§ 95. Deviation of an Element, of an Elementary Assembly. Weight.—§ 96. Concept of Elementary Magnitudes and Their Sums.—§ 97. Composition of These Magnitudes.—§ 98. The Elementary Magnitude as Multiple Element.—§ 99. The Elementary Magnitude with Weight Zero is a Displacement.—§ 100. Sum of a Displacement and a Simple or Multiple Element.

###### B. Applications 162

§ 101. Mean of an Assembly of Points.—§ 102. The Mean as Axis.—§ 103. Center of Gravity. Axis of Equilibrium.—§ 104. Magnetism, Magnetic Axis.—§ 105. Application to Differential Calculus.

#### CHAPTER TWO

##### OUTER MULTIPLICATION, DIVISION, AND SHADOW OF ELEMENTARY MAGNITUDES 169

###### A. Theoretical Development 169

§ 106. The Extent to Which the Displacement can be Interpreted as a Product.—§ 107. Elementary Systems.—§ 108. Outer Product of Elementary Magnitudes Formally Defined.—§ 109. Realization of This Product; Divergence, Rigid Elementary Magnitude.—§ 110. The Vertex Structure.—§ 111. Comparison of the Vertex Structure with the Product. Extension of the Elementary Magnitude.—§ 112. Equal Extensive Magnitudes have Equal Divergences.—§ 113. Sums of Elementary Magnitudes.

###### B. Applications 182

§ 114. Elementary Magnitudes in Space; Linear Magnitudes, Planar Magnitudes.—§ 115. Products and Sums of These Magnitudes:—§§ 116, 117. Reference Systems for Elementary Magnitudes.—§ 118. Transformations of Coordinates.—§ 119.

Equation of a Plane.—§ 120. The Static Moment as Deviation.—§ 121. New Procedure for the Treatment of Statics.—§ 122. General Law for Equilibrium.—§ 123. General Relation Between the Static Moments.—§ 124. When an Assembly of Forces is Equivalent to a Single Force.

### CHAPTER THREE

#### THE REGRESSIVE PRODUCT 199

##### A. Theoretical Development 199

§ 125. Formal Interpretation of the Regressive Product; Degree of Dependence and Multiplication.—§ 126. Relation Between the Common and Nearest Covering Systems.—§ 127. Introduction of the Relative System.—§ 128. The Unity of Outer and Regressive Multiplication is Mediated Thereby.—§ 129. The Regressive Product in Subordinate Form.— §§ 130–132. Real Significance of the Regressive Product; Its Characteristic Value With Respect to a Principal Measure.—§ 133. Introduction of Supplementary Numbers.—§ 134. Multiplication of Products That Appear in Subordinate Form.—§ 135. Every Real Product Can Be Brought to Subordinate Form.—§ 136. Multiplication by Mutually Incident Magnitudes.—§ 137. Characteristic Value of a Regressive Product of Several Factors. Pure and Mixed Products.—§ 138. Law for the Supplementary Numbers of Pure Products.—§ 139. The Factors of a Pure Product can be Arbitrarily Combined.—§ 140. Relation to Addition and Subtraction. —§ 141. Division With Respect to a System; Degree of the Relative Magnitude.—§ 142. Complete Analogy Between Outer and Regressive Multiplication.—§ 143. Double System and Products with Respect to It.

##### B. Applications 233

§ 144. Pure and Mixed Products in Geometry.—§ 145. Center of Gravity.—§ 146. Determination of the Straight Line and Plane.—§ 147. Shadow of a Point.—§ 148. Collinear and Concurrent Point Systems.

**CHAPTER FOUR**

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |     |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| AFFINITIES                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 248 |
| A. Theoretical Development                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 248 |
| § 149. Concept of Outer Quotient of Extensive Magnitudes.— § 150. Conjunction of Extensive Magnitudes by Outer Division. — § 151. Affinity of Extensive and Elementary Magnitudes.— § 152. Outer Division with Respect to a System, and Its Relation to Outer Multiplication.— § 153. Outer Quotient and Shadow of the Product.— § 154. Affinity. Structure of Affine Assemblies.— §§ 155, 156. Correspondence Between the Products of Corresponding Magnitudes of Two Affine Assemblies.— § 157. Direct and Reciprocal Affinity. General Theorem.— § 158. Connection Between Shadow and Affinity.— § 159. Affinity in Geometry.— § 160. Linear Relations. Collineation and Reciprocity According to the Principle of Equal Indicators.— §§ 161, 162. Collineation According to the Principle of Equal Indicators and According to the Principle of Equal Constructions. Identity of the Two Concepts.— § 163. Identity of Reciprocity According to the Two Principles.— § 164. Identity of the Concept of Affinity for Point Assemblies According to the Two Principles.— § 165. The Metric Relations of Two Collinear Point Structures.— § 166. Connection Between Collineation and Projection (Perspectivity).— § 167. Harmonic Equations, Construction of the Harmonic Mean. Harmonic Sums, Harmonic Coefficients. Pole System.— § 168. Reformulation of Pure Harmonic Equations.— § 169. Transformation of the Pole System of a Harmonic Equation.— § 170. Transformation of Harmonic Systems with Unaltered Pole Systems. General Theorem of Harmonic Means.— § 171. Application to Crystal Structure.— § 172. Note on Open Products |     |
| Appendix I. On the Relation of Non-Euclidean Geometry to Extension Theory                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 279 |
| Appendix II. On the Regressive Product                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 281 |
| Appendix III. Brief Survey of the Essentials of Extension Theory                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 283 |
| Index of Defined Terms                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 296 |

# Foreword to the First Edition

If I call this work, whose first volume I herewith offer to the public, the exposition of a new mathematical discipline, the justification for such an assertion can only be provided by the work itself. Since I permit myself no other justifications, I must proceed forthwith to indicate the way I was led, step by step, to the results presented here, and at the same time to illuminate the extent of this new discipline, insofar as possible at this point.

The initial incentive was provided by the consideration of negatives in geometry; I was used to regarding the displacements  $AB$  and  $BA$  as opposite magnitudes. From this it follows that if  $A, B, C$  are points of a straight line, then  $AB + BC = AC$  is always true, whether  $AB$  and  $BC$  are directed similarly or oppositely, that is even if  $C$  lies between  $A$  and  $B$ . In the latter case  $AB$  and  $BC$  are not interpreted merely as lengths, but rather their directions are simultaneously retained as well, according to which they are precisely oppositely oriented. Thus the distinction was drawn between the sum of lengths and the sum of such displacements in which the directions were taken into account. From this there followed the demand to establish this latter concept of a sum, not only for the case that the displacements were similarly or oppositely directed, but also for all other cases. This can most easily be accomplished if the law  $AB + BC = AC$  is imposed even when  $A, B, C$  do not lie on a single straight line.

Thus the first step was taken toward an analysis that subsequently led to the new branch of mathematics presented here. However, I did not then recognize the rich and fruitful domain I had reached; rather, that result seemed scarcely worthy of note until it was combined with a related idea.

While I was pursuing the concept of the product in geometry as it had been established by my father<sup>1</sup>, I concluded that not only rectangles but also parallelograms in general may be regarded as products of an adjacent pair of their sides, provided one again interprets the product, not as the product of their lengths, but as that of the two displacements with their directions taken into account. When I combined this concept of the product with that previously established for the sum, the most striking harmony resulted; thus

---

<sup>1</sup>Cf. J. G. Grassmann, *Raumlehre*, Teil II, p. 194; and *Trigonometrie*, p. 10 (Berlin: G. Reimer, 1824 and 1835)



whether I multiplied the sum (in the sense just given) of two displacements by a third displacement lying in the same plane, or the individual terms by the same displacement and added the products with due regard for their positive or negative values, the same result obtained, and must always obtain.

This harmony did indeed enable me to perceive that a completely new domain had thus been disclosed, one that could lead to important results. Yet this idea remained dormant for some time since the demands of my job led me to other tasks; also, I was initially perplexed by the remarkable result that, although the laws of ordinary multiplication, including the relation of multiplication to addition, remained valid for this new type of product, one could only interchange factors if one simultaneously changed the sign (i.e. changed  $+$  into  $-$  and vice versa).

A work on tidal theory, which I undertook at a later time, led me to Lagrange's *Mécanique Analytique*, and thence back to the ideas of this analysis. All the developments in that work were transformed by the principles of this new analysis into such simple procedures that the calculations often came out one-tenth as long as there.

This encouraged me to apply the new analysis to the difficult theory of tides. There were manifold new concepts to develop and to express in the new analysis; thus the concept of rotation led me to geometric exponential magnitudes, the analysis of angles and of trigonometric functions, and so on.<sup>2</sup> I was delighted to see that not only were the often very complex and unsymmetric formulas fundamental to tidal theory<sup>3</sup> changed by the reformulated and extended analysis into the simplest and most symmetric formulas, but in addition the technique of the development proceeded in parallel with the concepts.

Thus not only can each formula that arises in the course of a derivation be most easily expressed verbally, but in addition each step from one formula to another appears at once as just the symbolic expression of a parallel act of abstract reasoning. The methods formerly used require the introduction of arbitrary coordinates that have nothing to do with the problem and completely obscure the basic idea, leaving the calculation as a mechanical generation of formulas, inaccessible and thus deadening to the intellect. Here, however, where the idea is no longer strangely obscured but radiates through the formulas in complete clarity, the intellect grasps the progressive development of the idea with each formal development of the mathematics.

Thus I feel entitled to hope that in this new analysis I have found the only natural way in which mathematics should be applied to nature, and likewise that in which geometry should be treated if it is to lead to general and fruitful results.<sup>4</sup> I therefore decided to

---

<sup>2</sup>For a more specific discussion, see below.

<sup>3</sup>Cf. Laplace, *Mécanique Céleste*, livre V.

<sup>4</sup>In fact it is easy to show that the difference between the analytic and synthetic treatment of geometry completely vanishes in this analysis.

make the presentation, extension, and application of this analysis one of my life tasks. When I came to devote all my free time to this subject, many of the gaps left in the more casual earlier development were filled in. Thus from the method and with the modifications that I have presented in the work itself it follows that the center of gravity of several points can be interpreted as their sum, the displacement between two points as their product, the surface area lying between three points as their product, and the region (a pyramid) between four points as their product.

The interpretation of the center of gravity as a sum led me to examine Möbius' *Barycentrische Kalkül*, a work of which until then I knew only the title; and I was pleased to find there the same concept of the summation of points to which I had been led in the course of this development. This was the first and, as I subsequently learned, the only point of contact that my new system of analysis had with one that was already known. Since, however, the concept of a product of points does not occur in that work, and since with this concept, when it is combined with that of the sum of points, the development of the new analysis begins, I found that I could not expect any advancement of my ideas from that source.

Thus when I proceeded to work out my results consistently and from the beginning, being careful to appeal to no principle proven in any other branch of mathematics, I found that the analysis I had discovered did not touch only on the subject of geometry, as it had seemed before. Rather, I soon realized that I had come upon the domain of a new science, of which geometry itself is only a special application.

For a long time it had been evident to me that geometry can in no way be regarded as a branch of mathematics like arithmetic or combination theory; instead, geometry relates to something already given in nature, namely space. I had also realized that there must be a branch of mathematics that yields in a purely abstract way laws similar to those that in geometry seem bound to space. The possibility of constructing such a purely abstract branch of mathematics was provided by the new analysis; indeed this analysis, developed without the assumption of any principles established outside its own domain, and proceeding purely by abstraction, was itself this science.

The essential advantages of this approach are, in regard to form, that all principles expressing perceptions of space are now entirely omitted, and consequently the origins of this science are as immediate as those of arithmetic; and in regard to content, that the limitation to three dimensions is absent. Only thus do the laws come to light in their full clarity and generality, and are their essential interrelationships revealed; and many regularities, which for three dimensions either do not appear at all, or at best obscurely, present themselves perfectly clearly with this generalization.

In addition, in the course of the discussion, and with the appropriate definitions as given

in the work itself, it is shown that the intersection point of two lines, the intersection line of two planes, and the intersection point of three planes can be regarded<sup>5</sup> as products of those lines or planes, from which follows an extremely simple and general theory of curves.<sup>6</sup>

I now proceed to discuss and motivate the material that I have decided to put into the second volume of this work, in which I include everything that in any way requires the concept of rotation or angle. Since this second volume, which will complete the work, will only appear in print later, it seemed to me necessary, for an overview of the whole, to note somewhat more precisely the relevant results. To this end I begin by summarizing the results that have already been obtained before the completed exposition.

I have just shown how the parallelogram can be interpreted as the product of two displacements, provided, as is always the case here, that the direction of the displacements is retained; however, this product is exceptional in that the factors can only be interchanged with a sign change, whence that of two parallel displacements is clearly zero. This concept stands side by side with another that also refers to displacements with directions retained.

Thus if I project one displacement perpendicularly onto the other and then take the arithmetic product of this projection with the displacement on which it is projected, this too can yield a product of these displacements inasmuch as the multiplicative relation to addition is valid here as well. But this product is of a completely different type from the first, since its factors can be interchanged without sign change and is zero for two mutually perpendicular displacements. I call the former the *outer*, the latter the *inner* product, since the former has a nonzero value only with directions proceeding differently, the latter only with similar, that is, with directions proceeding in part together. This concept of the inner product, which I had already obtained in essence with the elaboration of the *Mécanique Analytique*, also leads to the concept of absolute length.<sup>7</sup>

Likewise I had already obtained the geometric exponential magnitude with the work on tidal theory; thus if  $\mathbf{a}$  denotes a displacement (with direction retained) and  $\alpha$  an angle (with the plane of rotation retained), then on purely inner grounds, the specification of which had however led me thus far, one finds that  $\mathbf{a} \cdot e^\alpha$ , where  $e$  can be taken as the basis of the natural system of logarithms, denotes the displacement that results from a rotation of  $\mathbf{a}$  generated by the angle  $\alpha$ ; that is,  $\mathbf{a} \cdot e^\alpha$  denotes the displacement  $\mathbf{a}$  rotated through the angle  $\alpha$ . Furthermore, if  $\text{Cos } \alpha$ , where  $\alpha$  expresses an angle in the geometric sense, represents the same number as  $\cos \alpha'$ , where  $\alpha'$  is the arc of the angle measured by

---

<sup>5</sup>Cf. Chapter 5 of part 2.

<sup>6</sup>Cf. the same chapter.

<sup>7</sup>This concept also belongs in volume 2 since it depends on rotations.

the radius, then from that concept of the exponential magnitude the equation

$$\text{Cos } \alpha = \frac{e^{\alpha} + e^{-\alpha}}{2}$$

follows at once.<sup>8</sup> Likewise, if  $\text{Sin } \alpha$  represents the magnitude by which the displacement it multiplies changes at 90° to its direction in the rotational sense of the angle  $\alpha$ , and at the same time its length changes by  $\sin \alpha'$  in the same way, then

$$\text{Sin } \alpha = \frac{e^{\alpha} - e^{-\alpha}}{2}.$$

From these follows the equation

$$\text{Cos } \alpha + \text{Sin } \alpha = e^{\alpha},$$

all equations that reveal the most striking analogy with familiar imaginary expressions.

So far, these concepts had already been obtained earlier. As I now tried to generalize them, the concept of the inner product was first extended in a way corresponding to that which I have indicated above for the outer product in reference to the intersection of lines and planes; thus I came next to the concept of the quotient of differently directed displacements, and understood by  $\frac{b}{a}$ , where  $a$  and  $b$  represent differently directed displacements of equal length, the magnitude by which each displacement lying in the same plane evolves by the angle  $ba$  (reckoned from  $b$  to  $a$ ), so that, as it must,  $\frac{a}{b} \cdot b = a$ ; and thus immediately followed the concept for the case in which  $a$  and  $b$  are of unequal length. This simple concept was the source of a whole series of interesting relations.

First it yielded a new type of multiplication, to which this division corresponded, and which distinguished itself from the earlier types in that this new type of product could only be zero if one of the factors were zero, and the factors remained interchangeable; thus, briefly, it is a multiplication that in all its laws remains analogous to the usual arithmetic; and its concept easily followed by progressively multiplying a displacement by other such quotients and then interpreting the result itself as a quotient that could be used instead of the progressive factors. Now by definition, if  $ab$  is the angle between two displacements of equal length, then

$$\frac{e^{ab} \cdot b}{a},$$

---

<sup>8</sup>Thus if  $AB$  (in figure 1) is the original vector, and it is rotated through the angle  $-\alpha$  to the position  $AC$ , and through the angle  $+\alpha$  to the position  $AD$ , and one completes the parallelogram  $ACDE$ , then  $AE$  is the sum of the displacements  $AC + AD$ , and half  $AF$  of this sum is the cosine of the angle  $\alpha$ .

*Fig. 1.*

whence one also has

$$\log \frac{b}{a} = ab.$$

Further, if the angle  $ab$  is the  $m$ -th part of  $ac$ , then one has

$$\left(\frac{b}{a}\right)^m = \frac{c}{a},$$

since if a displacement progressively undergoes the rotation  $\frac{a}{b}$  a total of  $m$  times, it completes the rotation  $\frac{a}{c}$  in all. Thus also, if the angle  $ab$  is half as large as  $ac$ , then

$$\left(\frac{b}{a}\right)^2 = \frac{c}{a} \quad \text{and thus} \quad \frac{b}{a} = \sqrt{\frac{c}{a}}.$$

In particular, if  $a$  is a right-angle rotation then  $c$  equals that through two right angles, and thus, since  $c = -a$ ,  $\frac{c}{a} = -1$  and  $\frac{b}{a} = \sqrt{-1}$ ; that is, the expression  $\sqrt{-1}$  multiplied by a displacement changes its direction by 90° into another one, but always into the same side.

This beautiful interpretation of the imaginary magnitude is completed by requiring that

$$e^\alpha \quad \text{and} \quad e^{(\alpha)}$$

signify the same value if  $\alpha$  denotes the angle and  $(\alpha)$  its arc divided by the radius; in fact one thereby finds the corresponding formulas

$$\cos x = \frac{e^{x\sqrt{-1}} + e^{-x\sqrt{-1}}}{2}$$

and similarly

$$\sqrt{-1} \sin x = \frac{e^{x\sqrt{-1}} - e^{-x\sqrt{-1}}}{2}$$

which therefore have a purely geometric significance, since  $\sqrt{-1}$  denotes the rotation through an angle given by the radially measured arc  $x$ .

From this, all imaginary expressions now acquire a purely geometric meaning and can be described by geometric constructions. At the same time the angle is defined as the logarithm of the quotient  $\frac{a}{b}$ , and thus also the infinite number of its values correspond to the same position of the side. On the other hand it is likewise now evident how, according to the meaning of the imaginaries thus discovered, one can derive the laws of analysis in the plane; however, it is not possible to derive the laws for space as well by means of imaginaries. In addition there are general difficulties in considering the angle in space, for the solution of which I have not yet had sufficient leisure.

This then is somewhat the material (at least to the extent that I have worked it out so far) that I have withheld for the second and last volume, with which the work will be completed. I cannot yet determine the time when this second volume will appear, since with the manifold duties in which my present position involves me it is impossible to find the repose which is necessary for the work itself. The first volume was therefore organized to stand on its own, complete in itself, and I deemed it appropriate to let this first volume appear together with its relevant applications, rather than having both volumes together and separate from their applications.

Indeed, for the position and significance of a new science to be properly recognized it is absolutely necessary to show its applications and its links to related subjects simultaneously with its presentation. The introduction serves this purpose as well. It presents the nature of the subject from a more philosophical viewpoint, and if I thus separate it from the rest of the work, this is done so that mathematicians will not be frightened off by its philosophical form.

For there prevails among mathematicians (and not without some justification) a certain aversion to the philosophical discussion of mathematical and physical subjects; and in fact most analyses of this type, as for example those of Hegel and his school, permit an obscurity and arbitrariness that nullifies any fruits of such an analysis. Notwithstanding this, I felt obliged to indicate the place of the new science in the field of knowledge. I therefore inserted the introduction at the beginning to satisfy both needs. It can, however, be omitted without essentially compromising an understanding of the whole. Similarly, among the applications, those that refer to subjects on nature (physics, crystallography) can be omitted without upsetting the presentation as a whole.

I believe the importance, and even the absolute necessity, of the new science and the analysis it provides to have been particularly well illustrated by these applications to physics. At another time I hope to demonstrate that this science can furnish an excellent didactic subject in its concrete form, that is, in its application to geometry, for which a completely elementary treatment is possible. But no place could be found for such a demonstration in the present work because of its format. Particularly in a simple treatment of statics (where the results obtained are themselves intuitive and generally known—and also describable by constructions), it is absolutely necessary to take up the concepts of the sum and product of displacements, and thus to develop the fundamental laws; and I am convinced that whoever has once sought to absorb these concepts will never give them up.

If therefore I have fully discerned the claims of the new science whose treatment is at least partially presented here, and have in no way diminished its pretensions in the field of knowledge, I believe I have not thereby invited blame for presumptuousness; for the

truth demands its rights. It is not this work that brings it into existence or recognition; it has its essence and existence in itself, and it is a betrayal of the truth to minimize its just claims from false modesty. But I must accept all the more indulgence for my work on this science. For, notwithstanding all the pains taken on its form, I am myself yet aware of its great imperfections.

Indeed, I have worked through the whole many times in various forms; now in Euclidean form for the greatest rigor in demonstrations and theorems, now in the form of a narrative development for the greatest clarity of arrangement, again with both woven together, and yet again, letting the development follow after the Euclidean form. To be sure, I am well aware that, with further revision, many things could be presented in a form that is somewhat more rigorous and better arranged. But from the utter conviction that I cannot hope for complete satisfaction, and from the simple complementary fact that the presentation will always remain insufficient, I decided on the form appearing here, which now seems to me the best.

On one particular ground I hope to find indulgence, that the time for my research was extremely paltry and measured out piecemeal by virtue of my official position. Also my position offered me no opportunity, through reports on the domain of this science, or even related matter, to obtain that vivid freshness which must as an enlivening breath inspire the whole if it is to appear as a living limb of the organism of science. Although an appointment in which my primary responsibility would be such scientific communications stands before my eyes as the goal of my wish and striving, I still believe that the treatment of this science ought not be delayed until the attainment of that goal, particularly since I can hope that the treatment of this part might smooth the way to it.

Stettin, 28 June 1844

# Foreword to the Second Edition

The work whose second edition I herewith present to the public had, for the first twenty-three years after it appeared, received only insignificant and for the most part incidental recognition.

I cannot place the burden of this utter failure on the science it treats, for I know its fundamental importance and indeed its complete necessity; rather, I know that the reason thereof is found only in the rigorously scientific method of treatment, starting from the most primitive concepts.

Such a method of treatment requires not merely an incidental appreciation of this or that result, but a complete immersion in the fundamental ideas and, based on this foundation, a consistent conception of the whole structure, a full appreciation of whose individual parts is supported only by an overview of the whole. With the enormous advance of mathematics in recent times, with the appearance of ever newer areas of mathematical research whose study requires the most intense work, with the struggle for new results presented in and attracting the spirit of research, mathematicians found no rest and repose to delve into such a tightly interconnected structure.

My hope of winning an academic position, to be able to introduce youthful vigor into the science and to encourage its wider consideration, missed its mark. And as could not fail to happen, later other mathematicians by different routes reached isolated results that had already been treated in my *Ausdehnungslehre* of 1844; but it almost never happened that in the process my work was mentioned. On the contrary, it demonstrates that this work remained almost completely unknown, since otherwise the results would have been obtained much more easily and fruitfully from the inner connections found there.

In such circumstances no one can blame the publisher for shredding some of the copies of my book, or me for not continuing on the same basis in the promised second volume of my work. Instead, in 1862 the whole *Ausdehnungslehre* was reconstructed on a new basis (which, I hoped, mathematicians would find more agreeable) and carried through to the end.<sup>9</sup> But this new work also initially received as little attention as the first. Only

---

<sup>9</sup>*Die Ausdehnungslehre vollständig und in strenger Form bearbeitet.* Berlin: Enslin, 1862



since 1867 have things proceeded otherwise.

Hermann Hankel was the first who, in his *Theorie der complexen Zahlensysteme* (Leipzig, 1867), stressed the fundamental significance of my Ausdehnungslehre (see pp. 16, 112, 119–140, 140). Even more decisive was Clebsch’s recognition. Shortly before his death Clebsch, in his “zum Gedächtniss an Julius Plücker” (Göttingen, 1872) (see the notes on pp. 8 and 28), emphasized the significance of my Ausdehnungslehre of 1844 in a very favorable way. Thus in his second note Clebsch stated: “In a certain sense the coordinates of the straight line, as well as a large part of the basis of modern algebra, are already contained in Grassmann’s Ausdehnungslehre of 1844. The more precise statement of these relations would however lead too far at present.” In view of his loving and ever fruitful participation in the works of others, which distinguished this most eminent of the more recent mathematicians, it is certain that Clebsch would later have found space to present these relations and, as was his way, to fructify the Ausdehnungslehre with new and far-reaching ideas. But alas, he was suddenly snatched away in the midst of his energetic efforts on behalf of science.

However, three years earlier (in 1869) Victor Schlegel had begun to develop the ideas suggested by Clebsch. In his *System der Raumlehre nach den Prinzipien der Grassmann’schen Ausdehnungslehre und als Einleitung in diesselbe dargestellte von Victor Schlegel* (Leipzig: Teubner), the first part of which appeared in 1872 and the second in 1873, Schlegel presented with great clarity, and for the most part by an independent method completely appropriate to the subject, the significance of the Ausdehnungslehre, both for the most recent geometry and for algebra. It is particularly to be emphasized that this book by Schlegel is the first that has comprehended the essential ideas of the Ausdehnungslehre in their inner connections and has given them a presentation.

[Since this time not only has the significance of the Ausdehnungslehre repeatedly been emphasized, but very successful work using its methods has appeared in various areas: H. Noth in Freiburg, in his article “Die vier Species in der Elementen der Geometrie” (*Schulprogramm* 1874), lays down the basis of a simplified presentation of the geometry of position; R. Sturm in Darmstadt, in his essay “Sulle forze in equilibrio” (*Annali di Matem.*<sup>10</sup> 7, 217ff., 1876), turns to the methods of the Ausdehnungslehre for the solution of problems in mechanics; finally, W. Preyer in Jena, in his “*Elementen der reinen Empfindungslehre*” (Jena: Dufft, 1877), has given a presentation of the theory of sensations based on the principles of the Ausdehnungslehre, and is thus the first to succeed in an independent area by using the concepts of space.]<sup>11</sup>

It is plain that in the Ausdehnungslehre, treating a science still young, there lay concealed

---

<sup>10</sup>2nd Series

<sup>11</sup>This footnote was added in 1877.

many embryonic ideas that need and deserve further development, and thus after a ten-year interruption, in 1872 I again returned to that study. My earlier investigations in this area are explained in my *Ausdehnungslehre* of 1862, and thus I have only to note the more recent works here. Of these there are in the first place two articles in the *Gottinger Nachrichten* of 1872, “Zur Theorie der Curven dritter Ordnung” (p. 505) and “Über zusammengehörige Pole und ihre Darstellung durch Produkte” (p. 567); then in the *Mathematische Annalen* “Die neueren Algebra und die Ausdehnungslehre,” 7, 538, “Die Mechanik nach den Prinzipien der Ausdehnungslehre,” 12, 222, and “Der Ort der Hamilton’schen Quaternionen in der Ausdehnungslehre,” 12, 375. Finally, I am preparing an article for Borchardt’s *Journal* in which I seek to establish Reye’s beautiful work on surfaces upon a new and simple basis by the further development of ideas in my *Ausdehnungslehre* of 1862, no. 392, where functions are treated as extensive quantities.<sup>12</sup>

[Thus it has happened that, in the course of the last year, interest in the *Ausdehnungslehre* has extended to ever wider circles. And as, in the same measure, the demand for the now scarce first edition grew, the publisher most generously agreed to prepare a second edition.]<sup>13</sup>

In this second edition I have left unchanged the text of the first (naturally excepting a few misprints), since the idea, and also the method of treatment, which I believe is completely justifiable, is such as is certain to be more pleasing to more philosophically inclined readers than the method of presentation of the *Ausdehnungslehre* of 1862, which is more accommodated to mathematicians. On the other hand, when it seemed appropriate to me to add new notes under the text, I have marked them with the year 1877.

I have placed two extensive notes at the end as appendices, the better to maintain the pagination of the first edition. Also reprinted as an appendix is the survey of the essentials of extension theory I wrote for Grunert’s *Archiv*, vol. 6, which aids considerably the understanding of the work, as has been communicated to me from various sides. Finally, I have added an index of defined terms appearing in the work.

To facilitate comparison with the *Ausdehnungslehre* of 1862 I give here a survey of results in corresponding sections of both, noting however that not only is the basis of each completely different from the other, but also that results are derived in the one which in the other are either completely omitted or summarized with other results, whence it is not always possible to group them together. Here I indicate the *Ausdehnungslehre* of 1844 by  $A_1$ , that of 1862 by  $A_2$ :

---

<sup>12</sup>Printed in vol. 84, pp. 273–283.

<sup>13</sup>This passage was added in 1877.

|                 |   |                          |
|-----------------|---|--------------------------|
| $A_1$ §§ 13–20  | — | $A_2$ no. 1–9, 14–24     |
| $A_1$ § 24      | — | $A_2$ no. 216–223        |
| $A_1$ §§ 28–36  | — | $A_2$ no. 52–61, 66–68   |
| $A_1$ §§ 37–40  | — | $A_2$ no. 254, 262       |
| $A_1$ §§ 45, 46 | — | $A_2$ no. 134, 135       |
| $A_1$ §§ 47–55  | — | $A_2$ no. 69–85          |
| $A_1$ §§ 60–73  | — | $A_2$ no. 10–13, cf. 377 |
| $A_1$ §§ 80–90  | — | $A_2$ no. 27–36          |
| $A_1$ § 93      | — | $A_2$ no. 136            |
| $A_1$ §§ 94–119 | — | $A_2$ no. 224–286        |

In the  $A_2$  the laws of elementary magnitudes are grouped with those of extensive magnitudes and are only separated from them in the application to geometry.

|                  |   |                  |
|------------------|---|------------------|
| $A_1$ § 126      | — | $A_2$ no. 25, 26 |
| $A_1$ §§ 128–142 | — | $A_2$ no. 94–132 |

$A_1$  §§ 127 and 143 are abandoned as sterile.

|                  |   |                   |
|------------------|---|-------------------|
| $A_1$ § 144      | — | $A_2$ no. 287–305 |
| $A_1$ §§ 145–148 | — | $A_2$ no. 306–329 |
| $A_1$ §§ 149–165 | — | $A_2$ no. 401–409 |

The note on open products at the end of  $A_1$  is further discussed in  $A_2$ , no. 353–363.

The new subjects that were treated in the *Ausdehnungslehre* of 1862 are only mentioned in part in the foreword to the *Ausdehnungslehre* of 1844.<sup>14</sup> The second section (no. 348–527) is a completely new addition that treats function theory and the algebraic multiplication basic to it, along with differential calculus, infinite series, and integral calculus, and examines especially deeply the applied areas of analysis and both modern geometry and algebra; and, in various sections, as for example in the treatment of quotients (no. 377–391), the interpretation of functions as extensive magnitudes (no. 392–400), and the integration of differential equations (no. 491–527), are contained the seeds of developments which still await further treatment.

Stettin, Summer 1877.

**HERMANN GRASSMANN**

---

<sup>14</sup>Cf. foreword to the 1844 edition.

# Introduction

## A. Derivation of the Concept of Pure Mathematics

1. The principal division of the sciences is into the real and the formal. The real represent the existent in thought as existing independently of thought, and their truth consists in the correspondence of the thought with that existent. The formal on the other hand have as their object what has been produced by thought alone, and their truth consists in the correspondence between the thought processes themselves.

Thought exists only in reference to an existent that confronts it and is portrayed by the thought; but in the real sciences this existent is independent, existing for itself outside of thought, whereas in the formal it is established by thought itself, when a second thought process is confronted as an existent. Now if truth is in general based on the correspondence of the thought with the existent, then in particular in the formal sciences it is based on the correspondence of the second thought process with that existent established by the first, that is, on the correspondence of the two thought processes. Thus proof in the formal sciences does not extend beyond the sphere of thought, but resides purely in the combination of different thought processes. Consequently, the formal sciences cannot begin with postulates, as do the real; rather, definitions comprise their foundation.<sup>15</sup>

2. The formal sciences treat either the general laws of thought or the particular as established by means of thought, the former being the dialectic (logic),<sup>16</sup> the latter, pure mathematics.

The contrast between the general and the particular thus produces the division of the formal sciences into dialectics and mathematics. The first is a philosophical science, since it seeks the unity in all thought, while mathematics has the opposite orientation in that

---

<sup>15</sup>Although postulates have been introduced into the formal sciences, for example in arithmetic, this is to be regarded as an error, only to be explained by the corresponding treatment of geometry. I will return later to consider this in more detail. Here it is enough to have demonstrated that postulates are necessarily absent from the formal sciences.

<sup>16</sup>Logic has a mathematical aspect that one can call formal logic, whose content has been developed jointly by my brother Robert and myself, and is presented in an original form by him in his second book, the *Formenlehre*, Stettin: 1872. (1877) New edition in 2 vols.: *Logik u. Formenlehre*, Stettin: 1890, 1891.

it regards each individual thought as a particular.

**3.** Pure mathematics is therefore the science of the particular existent that has come to be by thought. The particular existent, viewed in this sense, we call a thought form or simply a *form*; thus pure mathematics is the theory of forms.

The name “theory of magnitude” is inappropriate for all of mathematics, since one finds no use for magnitude in a substantial branch of it, namely combination theory, and even in arithmetic only in an incidental sense.<sup>17</sup> On the other hand the expression “form” might seem rather too broad, and the name “thought form” more appropriate; but the form in its pure meaning, devoid of all real content, is precisely nothing but the thought form, and thus the expression is suitable.

Before we proceed to the division of the theory of forms we have to separate out one branch that has hitherto incorrectly been included in it. This branch is geometry. From the concepts set out above it is evident that geometry, like mechanics, refers to a real existent; for geometry, this is space. This is clear since the concept of space can in no way be produced by thought, but rather emerges as something given. Anyone who would maintain the contrary must undertake the task of deducing the necessity for the three dimensions of space from the laws of pure thought, a problem whose solution is patently impossible.

One who, despite being obliged to admit this, prefers to extend the name “mathematics” to geometry may indeed do so if in return he allows us our name “theory of forms” or its equivalent; but we must point out to him that his name, in being made to comprehend too much, must ultimately be abandoned as superfluous.

The position of geometry relative to the theory of forms depends on the relation of space perception to pure thought. Although just now we said that that perception confronts thought as something independently given, it is not thereby asserted that space perception emerges only from the consideration of solid objects; rather, it is that fundamental perception imparted to us by the openness of our senses to the sensible world, which adheres to us as closely as body to soul. It is the same with time and with the perception of motion based on time and space, wherefore one could count the pure theory of motion (phorometry) among the mathematical sciences with as much justice as geometry. The concept of motive force flows from the idea of motion through the contrast between cause and effect. Thus geometry, phorometry, and mechanics appear as *applications* of the theory of forms to the fundamental perceptions of the sensible world.

---

<sup>17</sup>The concept of magnitude is replaced by that of number in arithmetic; language thus differentiates very well “to add” and “to subtract” as pertaining to number, “to increase” and “to decrease” to magnitude.

## B. Derivation of the Concept of Extension Theory

4. Each particular existent brought to be by thought (cf. no. 3) can come about in one of two ways, either through a simple act of generation or through a twofold act of placement and conjunction. That arising in the first way is the continuous form, or magnitude in the narrow sense, while that arising in the second way is the discrete or conjunctive form.

The simple act of becoming yields the continuous form. For the discrete form, that posited for conjunction is of course also produced by thought, but for the act of conjunction it appears as given; and the structure produced from the givens as the discrete form is a mere correlative thought. The concept of continuous becoming is more easily grasped if one first treats it by analogy with the more familiar discrete mode of emergence. Thus since in continuous generation what has already become is always retained in that correlative thought together with the newly emerging at the moment of its emergence, so by analogy one discerns in the concept of the continuous form a twofold act of placement and conjunction, but in this case the two are united in a single act, and thus proceed together as an indivisible unit. Thus, of the two parts of the conjunction (temporarily retaining this expression for the sake of the analogy), the one has already become, but the other newly emerges at the moment of conjunction itself, and thus is not already complete prior to conjunction. Both acts, placement and conjunction, are thus merged together so that conjunction cannot precede placement, nor is placement possible before conjunction. Or again, speaking in the sense appropriate for the continuous, that which newly emerges does so precisely upon that which has already become, and thus, in that moment of becoming itself, appears in its further course as growing there.

The opposition between the discrete and the continuous is (as with all true oppositions) fluid, since the discrete can also be regarded as continuous, and the continuous as discrete. The discrete may be regarded as continuous if that conjoined is itself again regarded as given, and the act of conjunction as a moment of becoming. And the continuous can be regarded as discrete if every moment of becoming is regarded as a mere conjunctive act, and that so conjoined as a given for the conjunction.

5. Each particular existent becomes such through the concept of the different, whereby it is coordinated with other particular existents, and through this with the equal, whereby it is subordinated to the same universals with other existents. That arising from the equal we may call the *algebraic* form, that from the different the *combinatorial* form.

The opposition between equal and different is also fluid. Equals are already different insofar as the one and the other equal to it are at all separate (but without this separation there would be only a unity, and thus no equality); two different ideas are already equal insofar as they are conjoined by the activity of relating them, that is both are equally

conjuncts. However the two concepts by no means thereby blend together so that one could define a scale with which to specify how much equality obtains between two ideas, and how much difference; rather, although the different always somehow adheres to the equal and conversely, at the moment of consideration only the one is encompassed, while the other only appears as the requisite basis of the first.

By the algebraic form is here understood not only number but also that corresponding to number in the continuous domain, and by the combinatorial form not only combination but also its correspondent in the continuous.

6. From the interaction of these two oppositions, the former of which is related to the type of generation, the latter to the elements of generation, arise the four species of form and the corresponding branches of the theory of forms; thus the discrete form thereby separates into number and combination. Number is the algebraic discrete form; that is, it is the unification of those established as equal. Combination is the combinatorial discrete form; that is, it is the unification of those established as different. The sciences of the discrete are therefore number theory and combination theory (relation theory).

It scarcely needs further demonstration that the concept of number is hereby completely exhausted and precisely delimited, and likewise that of combination. And since the oppositions from which these definitions follow are the most elementary imparted directly in the concept of the mathematical form, the above derivation is indeed thereby sufficiently justified.<sup>18</sup> I note in addition only that this opposition between the two forms is expressed very clearly by the different notations for their elements, since that conjoined to number is designated by a single symbol (1), that conjoined to combination by different ones from among the remaining completely arbitrary symbols (the letters of the alphabet).—That each set of things (particular existents) can be interpreted as number as well as combination scarcely requires mention.

7. In precisely the same way, the continuous form or magnitude separates into the algebraic continuous form or *intensive* magnitude and the combinatorial continuous form or *extensive* magnitude. The intensive magnitude is thus that arising through generation of equals, the extensive magnitude or extension that arising through generation of the different. As variable magnitudes the former constitute the foundation of function theory, that is differential and integral calculus, the latter the foundation of extension theory.

The first of these two branches is usually regarded as subordinate to number theory, a higher branch; the second however appears to be a previously unknown branch, whence it is necessary to explain this difficult conception through the notion of continuous flow.

---

<sup>18</sup>The concepts of number and combination were developed in a completely similar way seventeen years ago in a paper written by my father on the concept of pure number theory, published in the *Programme* of the Stettin Gymnasium for 1827, without however coming to the attention of a wider public.

As with number there prevails the unification of that imagined together, and with combination the separation; so also with the intensive magnitude there appears the unification of elements, indeed separate conceptually, but which form the intensive magnitude only in their essential equality. In contrast, with the extensive magnitude there prevails the separation of elements that are indeed unified insofar as they form a single magnitude, but which constitute that magnitude precisely in their mutual separation. It is thus somewhat as if the intensive magnitude is number become fluid, the extensive magnitude combination become fluid. The latter is essentially a proceeding of elements mutually apart, retaining them as being mutually parted. With it, the generating element appears as changing, that is as passing through a variety of states, the collection of these various states forming precisely the domain of the extensive magnitude. With the intensive magnitude, its generation produces a series of states equivalent to itself, whose quantity is precisely the intensive magnitude. The best example we can offer for the extensive magnitude is the line segment (displacement), whose elements proceed essentially apart from each other and thus constitute precisely the line as extension; on the other hand, an example of the intensive magnitude is perhaps a point associated with a specific force, since in this case the elements are not removed, but rather are presented only as an intensity, thus forming a specific order of intensification.

In this case the difference so established is also expressed beautifully in the notation; thus with intensive magnitudes, which comprise the subject matter of function theory, one does not distinguish the elements by different symbols, but rather, where a particular symbol appears the complete variable magnitude is represented. On the other hand, with extensive magnitudes, or with their concrete realizations (lines) different elements are designated by different symbols (letters), just as in combination theory. It is also clear that each real magnitude can be viewed in two ways, both as intensive and extensive: thus lines may also be regarded as intensive magnitudes if one removes from their nature the way their elements lie apart, and retains simply the quantity of their elements; and in the same way a point associated with a specific force can be thought of as an extensive magnitude, since one can represent the force in the form of a line.

Historically, among the four branches of mathematics the discrete appeared earlier than the continuous (since the former is closer to the analytical sense than the latter), the algebraic earlier than the combinatorial (since the equal is easier to grasp than the different). Thus number theory is the earliest, combination theory and differential calculus appear simultaneously, and extension theory in its abstract form is necessarily last of all, although its concrete (but limited) image, the theory of space, already belonged to the earliest time.

8. Antecedent to the division of the theory of forms into its four branches is a more general subject that we may call the *general theory of forms*. In it are presented the



general conjunctive laws that apply to all branches alike. This preliminary subject is not intended simply to save repeating the same material in all four branches and thus to condense the treatment of the different parts, but also permits what naturally belongs together to appear together, and acts as the foundation of the whole.

## C. Exposition of the Concept of Extension Theory

9. Continuous becoming, analyzed into its parts, appears as a continuous production with retention of that which has already become. With the extensive form, that which is newly produced is always defined as different; if, during this process, we no longer always retain what has already become, then we arrive at the concept of *continuous evolution*. We call that which undergoes this evolution the *generating element*, and the generating element, in any of the states it assumes in its evolution, an *element* of the continuous form. Accordingly, the extensive form is the collection of all elements into which the generating element is transformed by continuous evolution.

The concept of continuous evolution can only arise with extensive magnitudes; with the intensive magnitudes, always dropping that which has already become would leave only the completely empty continuous tendency to become.

In the theory of space, the point appears as the element, the evolving locus or motion as its continuous evolution, and the various positions of the point in space as its different states.

10. The different must be produced according to a law if it is to have a specific resultant. For the *elementary* form this law must be the same for all moments of becoming. The elementary extensive form is thus the form that results from an evolution of the generating element according to this same law; the collection of all elements generated according to the same law we call a *system* or a *domain*.

Since that different from a given may be any one of an infinite manifold, it would become completely indeterminate were it not constrained by a fixed law. This law is, however, not at all defined by the content of the pure theory of forms; rather, the concept of extension is defined by the purely abstract idea of lawfulness, and the concept of the elementary extension by the same law for every moment of evolution. Accordingly the elementary extension has the property that if, from an element  $a$  of the extension there results another element  $b$  of the same extension, then by an identical process a third element  $c$  of it results from  $b$ .

In space the theory of the uniformity of direction is that single law governing the individual evolutions; thus the displacement represents the elementary extension in space theory, and the infinite straight line the whole system.

**11.** If one uses two different laws of evolution, the collection of elements generated by the two laws is a *system of second order*. The laws of evolution by which the elements of this system are produced from each other depend only on these two laws; if one adds a third independent law one arrives at a system of third order, and so on.

The theory of space may again serve as an example. Here the collection of elements of a plane are generated from a single element together with two directions when the generating element progresses by arbitrary amounts in the two directions, and the totality of points (elements) so generated are collected together as a single object. The plane is thus the system of second order; in it there is an infinite set of directions dependent on those two original directions. If a third independent direction is added, then by means of this direction, the whole of infinite space (as the system of third order) is produced. In this example one cannot proceed beyond three independent directions (evolutionary laws); but in pure extension theory their number can be infinitely increased.

**12.** The more precise determination of the difference between the laws requires a method of generation by means of which one system transforms into another. This transformation of different systems into one another thus forms a second natural order in the domain of extension theory, and the elementary presentation of this science thus concludes with it.

Rotational motion corresponds to this transformation of one system into another in the theory of space; and connected with this are angular magnitude, absolute length, orthogonality, and so forth. All of this will be worked out, but only in the second volume of the *Ausdehnungslehre*.

## D. Form of Presentation

**13.** The philosophical method characteristically proceeds by oppositions, and thus progresses from the general to the particular; the mathematical method on the other hand proceeds from the simplest concepts to the more complex, and thus in the opposite direction from the philosophical. The former method appears advantageous for the development of the subject, since its path is the natural way in which one arrives at knowledge; the latter is more appropriate for the presentation of the already completed structure, since its path is the way one surveys what is complete, but from which one could not have constructed the edifice. The present work, in that it must first erect the building, while at the same time its results have not yet become common knowledge, must combine both methods; it therefore adopts the philosophical method as far as possible in the introduction and again here and there in the work itself, and then the mathematical method in the individual treatments. But if the presentation is thus less simple than if either method alone were adopted, and if thereby the communication of the subject is

made more difficult, still this disadvantage is not decisive; for it is the same as in the construction of a city, where the path by which one arrives at the buildings, all overgrown with brush, is different from that by which one later gets around in the completed city.

# **A New Branch of Mathematics**

The Ausdehnungslehre of 1844 and Other Works

by

Hermann Grassmann

Chunk 2: Pages 51–100

Linear Extension Theory — Part One, Chapters 1–2

# Introduction (concluded)

## § 15

which, were the conclusion not stated at the beginning, one would understand nothing of where it is leading or how, and only after an extended period of imitating each step blindly and as it were randomly would one suddenly, and before one realized it, arrive at the truth sought. Such a proof may lack nothing in point of rigor, but it is not scientific; for it lacks the second requirement, that of the clarity of an overview. Whoever therefore imitates such a proof does not arrive at a clear cognition of the truth but, if he does not subsequently produce that overview himself, remains completely dependent on that particular method by which he found that truth established. And the feeling of confinement, which in such cases arises at least during the acquisition of the method, is most oppressive for one accustomed to thinking freely and automatically, and to acquiring all he absorbs spontaneously and vividly. On the other hand, if the reader is placed in a position such that at each point of the development he can see where it is going, then he remains master of the material, is no longer bound to the particular form of presentation, and that which is assimilated is a faithful reproduction.

**15.** For each given part of the presentation, the nature of its further development is essentially fixed by a dominant idea that is either nothing more than a supposed analogy with cognate and already familiar branches of knowledge, or, and this is the better case, is a direct intuition of the next succeeding truth.

As it plays into a cognate domain, the analogy is only a makeshift if it is not really important to emphasize throughout the relation to a cognate branch and thus continually to draw the analogy with it.<sup>1</sup>

---

<sup>1</sup>In the science treated here this occurs in its relation to geometry, on which I

Intuition might seem to be alien to the domain of pure science, and most of all to mathematics; but without it, it is impossible to discover any new truth. One cannot be led to a desired result by blind combinations; rather, what to combine and how must be determined by the dominant idea, and again this idea can, before it has materialized in the science itself, only appear in the form of an intuition. Intuition is therefore nearly indispensable in the domain of science. In particular, if it is of the right sort, it is what in an overall view of the entire course of development leads to the new truth, but not yet at an opportune moment for its development, and thus initially only as a dim presentiment; the detailing of that moment includes both the discovery of the truth and the critique of the presentiment.

---

have mainly drawn by way of analogy.

## § 16

**16.** The essence of a scientific presentation is thus an interlocked pair of approaches, one of which leads consistently from one truth to another and forms the actual content, while the other controls the treatment and thus determines the form. In mathematics these two separate lines of development diverge most acutely.

Following the model originated by Euclid, it has long been the practice in mathematics to present only the approach that comprises the actual content; the reader is left to puzzle out the other by reading between the lines. However perfect both its order and presentation may be, for someone wishing to learn the science it is impossible thus to maintain an overview at each point, and to be placed in a position to proceed freely and automatically. It is thus the more necessary that, as far as possible, the reader be placed in the position in which a discoverer of the truths is certain to be found in the best cases. In these circumstances, there occurs a continual reflection about the course of development as the truths are discovered; a characteristic line of thought develops in the reader about the procedure followed and about the idea lying at the foundation of the whole; and this line of thought forms the true nucleus and spirit of his activity, while the consistent detailing of the truths is only the embodiment of that idea.

As a consequence of the method of presentation, the reader may then expect to be able to progress independently, without having to be guided along such lines of thought, regarding himself as independent of the discoverer of the truth and thus reversing the relation between himself and the author, whereby the whole composition of the work appears superfluous. Accordingly, the modern mathematicians, and particularly the French, have begun to weave the two approaches together. The attraction of their works consists precisely in that the reader feels free and is not constrained to forms he must slavishly follow while not being fully conversant with them.

That these approaches diverge most acutely in mathematics originates in the characteristics of its method (no. 13); that is, since it progresses by the connection of particulars, the unity of the idea coming at the end. The second approach has a character completely opposite to the first, and their interweaving appears more difficult in mathematics than in any other science. One ought not, however, abandon the attempt on account of this difficulty, as often happens

with German mathematicians.

In the present work I have therefore adopted the method suggested, and this seems to me particularly necessary with a new science, as the idea is itself simultaneously being brought to light.



# Survey of the General Theory of Forms

## § 1. Concept of Equality

**§ 1. Concept of Equality** By the general theory of forms we mean that series of truths that relate to all branches of mathematics in the same way, and which thus assume only the general concepts of equality and difference, conjunction and separation. The general theory of forms must therefore precede all special branches of mathematics.<sup>2</sup> Since however that general branch does not yet exist as such, and we cannot omit it without entangling ourselves in lengthy digressions, we have no choice but to develop this subject to the extent required for our science.

Here we first establish the concepts of equality and difference. Since two equals must appear as different in order to stand out as two, and two differents must appear as different aspects of equals,<sup>3</sup> it seems necessary from a superficial consideration to formulate various relations of equality and difference. Thus for example in comparing two line segments one can assert equality of direction or length, of direction and length, of direction and position, and so forth; and in comparing other things, further relations of equality emerge. But already the fact that these relations change according to the character of the things being compared is proof enough that these relations do not belong to the concept of equality itself, but to the objects to which this concept of equality is applied. Thus, for example, for two equally long displacements we cannot say that they are equal as such, but only that their lengths are equal, and so these lengths themselves

---

<sup>2</sup>Cf. intro., no. 8.

<sup>3</sup>Cf. intro., no. 5.

stand in the absolute relation of equality. Thus we have preserved the simplicity of the concept of equality and can define it: *Those are equal of which one can always assert the same*, or more generally what in any judgment can be substituted one for the other.<sup>4</sup>

It plainly follows from this that, if two forms are each equal to a third, they are also equal to each other; and that those generated in the same way out of equals are again equal.

---

<sup>4</sup>This is not intended as a philosophical definition, but only as an understanding of the word, so that it will not be taken in different ways. The philosophical definition would have to take up the opposition of equal and different in its fluidity and its strict limits, for which a considerable apparatus of definitions not belonging here would be required.

## § 2. Concept of Conjunction

**§ 2. Concept of Conjunction** The second opposition whose consideration we have to take into account here is that of conjunction and separation. If two magnitudes or forms (which name we prefer, cf. intro., no. 3) are conjoined, they are called the *factors* of the conjunction; the form that is produced by the conjunction of the two, the *product* of the conjunction. If the two factors need distinguishing, we call the one the *prefactor*, the other the *postfactor*.

As the general sign for conjunction we take the symbol  $\cap$ ; now if  $a$  and  $b$  are the factors, with  $a$  the prefactor,  $b$  the postfactor, then we indicate the product of their conjunction as  $(a \cap b)$ , where the parentheses here express that the conjunction indicates that the factors are no longer separate, but that their concepts are unified.<sup>5</sup> The product of the conjunction can be further conjoined with another form, whence one proceeds to a conjunction of several factors, which however initially appears as just the sequential conjunction of two at a time. For our convenience we use the usual abbreviated parenthetical notation, whereby we omit the parentheses around two conjoined symbols if the opening parenthesis  $[$  stands at the beginning of the whole expression, or follows after another open parenthesis; for example, instead of  $((a \cap b) \cap c)$  we write  $a \cap b \cap c$ .

## § 3. Combinability of Factors

**§ 3. Combinability of Factors** The particular type of conjunction will now be specified by what maintains the same product, that is under what circumstances and to what extent the product remains equal to itself.

The only changes one can undertake without changing the individual forms conjoined are a change in the parentheses and a reordering of factors. We consider first the conjunction for which, in a product of three factors, the placement of the parentheses makes no real difference; that is to say, no difference in the product is found. In this case we have

$$a \cap (b \cap c) = a \cap b \cap c,$$

---

<sup>5</sup>How this unification is effected, and what is thereby imposed in each case on the idea of the individual conjunctions, depends on the nature of the particular type of conjunction.

from which it follows at once that the parentheses may also be omitted in every multifactor conjunction of this type without changing its product. Thus first of all, by virtue of the above assumption, each pair of parentheses enclose a two-factor expression, and this expression must itself be combined as a factor with another form; briefly, it appears as a combination of three forms for which we assume that one can omit the parentheses without changing the product of their conjunction. Thus, since one can replace each form with its equal, it follows that the combined product is not changed by omitting the parentheses. Therefore:

*If a conjunction is of the type that the parentheses may be omitted for three factors, then the same is true for any number of factors.*

Since, according to the above theorem, one is permitted to omit the parentheses in each of two expressions that differ only in the placement of their parentheses, the two are equal since they are both equal to the same expression (without parentheses); one thus has the foregoing theorem in a somewhat more general form:

*If a conjunction is of the type that the placement of parentheses is of no real consequence for three factors, then the same is also true for any number of factors.*

## **§ 4. Interchangeability of Factors. Concept of Elementary Conjunction**

**§ 4. Interchangeability of Factors. Concept of Elementary Conjunction** Now if, on the other hand, only the interchangeability of the two factors of a conjunction were established, then no further conclusion could be drawn. On adding this assumption to that made in the previous paragraph, however, it follows that the ordering of factors is also unimportant for the combined product in multifactor expressions, since one can then easily show that the interchange of any two factors in the expression is permitted.

Thus, according to the theorem previously established (§ 3), one can enclose two factors, whose interchangeability is known, in parentheses without changing the combined product, interchange the two factors without changing the product of their conjunction (by assumption), and thus without changing the product of the whole conjunction (since one can replace each form with its equal), and finally

the parentheses can be restored to their original positions. Thus we establish the interchangeability of two adjacent forms. Since however one can now continue this process to bring each factor to any arbitrary position, the order of factors is necessarily unimportant. Combining this result with that of the preceding paragraph:

*If a conjunction is of the type that one can arbitrarily place parentheses around any three of its factors and change the order of any two of its factors without changing the product, then it is also true that the placement of parentheses and the order of factors is unimportant for the product of any number of factors.*

For brevity we call a conjunction satisfying the given conditions *elementary*. Without referring to the nature of the forms conjoined, a further specification of the nature of the conjunction is not possible. We therefore turn to the inversion of the form obtained, that is to the analytic process.

## § 5. The Synthetic and Analytic Conjunctions

**§ 5. The Synthetic and Analytic Conjunctions** The analytic process consists in seeking one factor of a conjunction in terms of its product and the other factor. Thus to each conjunction there belong two types of analytic process, one for the prefactor and one for the postfactor. They both yield the same product if the two factors of the original conjunction are interchangeable. Since this analytic process can also be regarded as a conjunction, we distinguish the original or *synthetic* conjunction from the inverse or *analytic* conjunction.

In the following we suppose the synthetic conjunction is elementary in the sense of the preceding paragraph, and retain for it the symbol  $\cap$ ; in this case the two general inverses merge into one, and we adopt the inverted symbol  $\cup$  for the corresponding analytic conjunction, so that we make the prefactor of the synthetic conjunction that which is given by the analytic. Accordingly  $a \cup b$  denotes that form which, conjoined synthetically with  $b$ , gives  $a$ , that is  $a \cup b \cap b = a$ . From this there follows at once that  $a \cup b \cup c$  denotes that form which gives  $a$  when it is synthetically conjoined with  $c$  and then with  $b$ , and therefore according to § 4 also that form which gives  $a$  when conjoined synthetically with the same values in inverted order, or with  $b \cap c$ , thus:

$$a \cup b \cup c = a \cup c \cup b = a \cup (b \cap c);$$

and since the same result is true for any number of factors, it follows that the order of factors that are preceded by analytic symbols is also not important, and that one may enclose these factors in parentheses, provided one inverts the enclosed symbols. Furthermore it follows that

$$a \cup (b \cup c) = a \cup b \cap c.$$

Thus from the definition of the analytic conjunction one has

$$a \cup (b \cup c) = a \cup (b \cup c) \cap c,$$

and this expression is again, by virtue of the laws just established,  $= a \cup (b \cup c) \cap c$ . Finally, by virtue of the definition of the analytic conjunction, this  $= a \cup b \cap c$ , the first expression being equal to the last. Expressing this result in words, and combining it with the previous results, we have then:

*If the synthetic conjunction is elementary, then the order in which one synthetically or analytically conjoins is unimportant for the product. One can also insert or omit parentheses after a synthetic symbol if the conjunction contains only synthetic factors; but after an analytic symbol parentheses can be inserted or omitted only if the symbols preceding the factors in the parentheses are simultaneously interchanged, that is, the analytic symbols are changed into synthetic and vice versa.*

This is the most general result we can obtain from the assumptions given. We have yet to establish that one can omit parentheses that follow a synthetic symbol and enclose an analytic symbol. This requires a new assumption.

## § 6. Uniqueness of Analysis; Addition and Subtraction

**§ 6. Uniqueness of Analysis; Addition and Subtraction** The new assumption we add is that the product of the analytic conjunction is unique, or in other words that if one factor of the synthetic conjunction remains fixed while the other changes, then the product always changes as well. From this, results

$$a \cap b \cup b = a;$$

now  $a \cap b \cup b$  denotes the form that gives  $a \cap b$  when synthetically conjoined with  $b$ . But  $a$  is such a form, and by virtue of the uniqueness of the result, the only one; the validity of the above equation is thus established. It then follows as well that

$$a \cap (b \cup c) = a \cap b \cup c.$$

To show that the right side of this equation equals the left, we replace  $b$  with  $((b \cup c) \cap c)$ , thus:

$$a \cap b \cup c = a \cap ((b \cup c) \cap c) \cup c;$$

according to § 4 this is

$$= a \cap (b \cup c) \cap c \cup c,$$

and again according to the theorem just established, this  $= a \cap (b \cup c)$ , thus establishing that in fact the right side of the equation equals the left.

Now since one can repeat this procedure when several factors are found in parentheses, one has the theorem:

*If the synthetic conjunction is elementary and the corresponding analytic conjunction is unique, then one can insert or omit parentheses after a synthetic symbol.* In this case (provided that that uniqueness is generally valid), we call the synthetic conjunction *addition* and the corresponding analytic conjunction *subtraction*.

As for the order of factors, one finds  $a \cap b \cup c = a \cup c \cap b$ ; for  $a \cap b \cup c = b \cap a \cup c = b \cap (a \cup c) = a \cup c \cap b$ , whence we have also proved the interchangeability of two factors, one of which is preceded by a synthetic, the other by an analytic symbol, provided the uniqueness of the analytic product is assumed. And the theorems

of this paragraph are only true under this assumption, while those of the preceding paragraph remain valid if the product of the analytic conjunction is multivalued.<sup>6</sup>

---

<sup>6</sup>Examples of such multivaluedness appear not only in great abundance in extension theory, as will be shown later, but also in arithmetic, so the distinction is important there as well. In particular, addition and multiplication are taken as elementary conjunctions; and while subtraction is always unique, division is only unique so long as zero does not appear as divisor. For that reason only the theorems of the previous paragraph apply to division in general, whereas the theorems of this paragraph are only valid under the restriction that zero never appear as divisor. Failure to observe this condition necessarily results in the most awful contradictions and embarrassments, as has indeed already happened on occasion.



## § 7. The Indifferent and Analytic Forms

**§ 7. The Indifferent and Analytic Forms** One is led by the analytic process to the indifferent and analytic forms. One obtains the former by the analytic conjunction of two equal forms; thus  $a \cup a$  represents the indifferent form, which is in fact independent of the value  $a$ . Thus one has  $a \cup a = b \cup b$ , since  $b \cup b$  represents the form that gives  $b$  when synthetically conjoined with  $b$ ;  $a \cup a$  is such a form since  $b \cap (a \cup a) = b \cap a \cup a = b$ . In those circumstances where the analytic conjunction is unique,  $a \cup a$  must therefore be set equal to  $b \cup b$ . Since the indifferent form always has a unique value under the assumptions made, there follows the necessity to mark it with its own symbol. For the present let us take this symbol to be  $\text{—}$ , and designate the form  $(\text{—} \cup a)$  as  $\ominus a$ , calling  $\ominus a$  the pure analytic form; in fact, if the synthetic conjunction is addition, it is the *negative form*. It follows directly that  $(a \cap \text{—})$  and  $(a \cup \text{—})$  are both equal to  $a$ , and further that  $\cap \ominus a$  equals  $\ominus a$  and  $\cup \ominus a$  equals  $\cap a$ , since one need only substitute the complete expressions for the forms just given and the validity of these equations becomes obvious.<sup>7</sup>

The form analytic with respect to addition we call the negative form, and the indifferent form with respect to addition we call *zero*.

---

<sup>7</sup>(1877) A subsequent investigation, establishing the law for the conjunction of multivalued magnitudes, has convinced me that in general one must always transform the multivalued magnitudes into single-valued ones before one can employ any of the theorems on conjunction. I have incorporated this conjecture into my *Ausdehnungslehre* of 1862 in the notes to nos. 348 and 477, and at the same time in the first of these showed how one can transform multivalued magnitudes into single-valued ones. This conjecture is also among the fundamentals in my *Arithmetik* (Stettin: 1860, printed and published by R. Grassmann [since 1861 by Enslin in Berlin as well]).

## § 8. Addition and Subtraction of Similar Forms

**§ 8. Addition and Subtraction of Similar Forms** So far we have developed the concept of addition in a purely formal manner, since we have defined it from the validity of certain laws of conjunction. This formal concept also remains the only general one. Yet it is not the way we arrive at this concept in the individual branches of mathematics. Rather, in them a characteristic method of conjunction is obtained from the generation of the magnitudes itself, which is then interpreted as addition in precisely the general sense given, since those formal laws apply to it.

Specifically, if we consider two magnitudes (forms) that result from the continuation of the same method of generation, and which we therefore call “generated in the same sense,” then it is clear that one may arrange them so that the two of them comprise a single whole, in such a way that their total content, that is the parts comprising the two of them, become one, correlated in thought. This whole may then be regarded as generated in the same sense as those two magnitudes. Now it is easy to show that this conjunction is an addition, that is that it is elementary and that its analysis is unique. First, I can arbitrarily combine and interchange, since the parts which are correlated in thought remain the same thereby, and their result cannot change since they are all alike (being produced by like generations). Its analysis is also unique; for were this not the case, then while the product and one factor of the synthetic conjunction remained fixed, the other factor could assume a different value. But one of these two values would then necessarily be greater than the other, whence yet more parts must have been added to it. But then the same parts must have been added to the product, whence the product would then have changed, contrary to assumption. Thus since the corresponding analytic form is unique, the synthetic conjunction is interpreted as an addition, and for these conjunctions all the laws of §§ 3–7 are valid. It therefore follows that the laws of this conjunction also remain unchanged when the factors are negative. If we compare the negative magnitudes with the positive, we can say that they are generated in the opposite sense; and we can combine both like and oppositely generated magnitudes under the name *similar magnitudes*, whence in this way the real concepts of addition and subtraction are defined for similar magnitudes in general.

## § 9. Conjunctions of Different Order. Multiplication

**§ 9. Conjunctions of Different Order. Multiplication** So far we have considered only one method of synthetic conjunction and its relation to the corresponding analytic conjunction. Now we turn to establishing the relation between two different methods of synthetic conjunction. To this end the one must be specified through the other in accordance with its definition.<sup>8</sup> This definition depends on the process by which an expression containing both types of conjunction can be transformed without altering the combined product.

The simplest way that both conjunctions can appear in a single expression is that in which the product of one conjunction is subjected to the other. Thus if  $\cap$  and  $\cap'$  are the symbols for the two conjunctions, then the relation between them depends on the transformations assumed to be permitted in the expression  $(a \cap b) \cap' c$ . If the second conjunction should be symmetrically related to the two factors of the first, then the simplest transformation is that in which one subjects each factor of the first conjunction to the second, and then applies these individual products as factors in the first method of conjunction. If this transformation can be undertaken without altering the combined product, that is if

$$(a \cap b) \cap' c = (a \cap' c) \cap (b \cap' c),$$

then we call the second a conjunction of the next higher order than the first. In particular, if both factors of this second conjunction depend on the first in the same way, so that that definition of the new combination is valid for both the prefactor and the postfactor, and if in addition the first conjunction is elementary and its corresponding analysis unique, then we call the second conjunction *multiplication* since we have already adopted the name addition for the first. This is generally the way that initially, that is when no species of conjunction is yet given, such a conjunction of next higher order is defined. We therefore regard addition as the conjunction of first order, multiplication as that of second order.<sup>9</sup>

---

<sup>8</sup>This definition depends on the process by which an expression containing both types of conjunction can be transformed without altering the combined product.

<sup>9</sup>On the same principle one can take that of third order to be raising to a power, which however we omit here for brevity. But it is in the nature of the subject that the definition for these conjunctions is only formal, and cannot be embodied by real definitions except in the individual sciences.

From now on we agree to replace the general conjunctive symbols with the usual conventions for these species of conjunctions, and indeed for multiplication we choose simple juxtaposition.

## § 10. General Laws of Multiplication

**§ 10. General Laws of Multiplication** We have specified the relation of multiplication to addition as

$$\begin{aligned}(a + b)c &= ac + bc, \\ c(a + b) &= ca + cb;\end{aligned}$$

and we have thereby fixed the definition of multiplication. By repeated application of these fundamental laws, one is led at once to the more general theorem that if both factors are expanded out, then each term of the one can be multiplied by each term of the other, and these products can then be added. From this follows the corresponding law for the relation of multiplication to subtraction, that is

$$(a - b)c = ac - bc.$$

That is, to reduce the second member to the first, replace the  $a$  in it with its equal  $(a - b) + b$  to get

$$ac - bc = ((a - b) + b)c - bc;$$

but, according to the laws established previously, the second member  $= (a - b)c + bc - bc$ . Now according to §6 this expression  $= (a - b)c$ , whence the first expression equals the last. If the second factor is a difference the corresponding law follows in the same way. By repeated application of these laws one is led to the more general theorem:

*If the factors of a product are constructed by additions and subtractions, then without altering the combined product one can multiply each term of the one by each term of the other and then conjoin the resulting products by prefixing addition and subtraction symbols, according as the symbols preceding the terms multiplied are the same or not.*

## § 11. Laws of Division

**§ 11. Laws of Division** For division, the law for the expansion of the dividend is generally valid, whether the result is unique or not;<sup>10</sup> that is

$$\frac{a \pm b}{c} = \frac{a}{c} \pm \frac{b}{c},$$

where, since the interchangeability of factors for multiplication cannot be assumed in general, two types of division must be distinguished, according as the prefactor or the postfactor of the multiplicative conjunction is sought. Since in either case both factors have the same relation to addition and subtraction, the above is valid for both types of division; and if the

---

<sup>10</sup>(1877) Cf. footnotes on p. 25 and nos. 377 to 391 of the *Ausdehnungslehre* of 1862.

## § 12. Real Concept of Multiplication

**§ 12. Real Concept of Multiplication** The laws presented in the preceding paragraphs express the general relation of multiplication and division to addition and subtraction. However the laws of multiplication as such, as arithmetic formulates them, which express the interchangeability and combinability of its factors, do not result from this general concept of multiplication. On the contrary, in our science we may learn of other types of multiplication in which at least the interchangeability of the factors does not obtain, but in which all the other theorems presented have their full application.

We have therefore formally defined the general concept of this multiplication as well; if the nature of the magnitudes so conjoined is given, then this formal concept must correspond to a real concept that expresses the method of generation of the product by the factors. The relation to real addition provides us with a general definition of this method of generation.<sup>11</sup> Thus if one of the factors is taken as the sum of its terms (according to § 8), then according to the general law relating multiplication to addition, instead of the sum being subjected to the product-forming method of generation, its terms can themselves be so subjected, and the products so formed can then be added; that is, since these products again appear as generated in the same sense, they can be conjoined as parts of a single whole. That is, the multiplicative method of generation must be such that all the terms of the factors enter it in the same way, so that if a term of the one, conjoined multiplicatively with a term of the other, generates a given magnitude, then in the multiplicative conjunction of the wholes each term of the former taken with each term of the latter generates such a magnitude, and indeed the same magnitude, that these terms taken together were originally assumed to equal. It is therefore immediately clear that if the method of generation has the given characteristics, then the method of conjunction corresponding to it has the multiplicative relation to the addition of similars, and consequently all the laws of that relation hold for it.

We therefore also call such a method of conjunction a *multiplication*, provided only that its multiplicative relation to addition is demonstrated, or in other words, provided only that the equal entry of all the terms of the conjunctive factors into the conjunction is

---

<sup>11</sup>Here the dot in the divisor indicates the position of the factor sought.

established in the above sense.

The development of the general conjunctive laws to this point essentially suffices for the presentation of our science, with which we therefore now proceed.





# Part One: Extensive Magnitudes



# Chapter 1

## Addition and Subtraction of Elementary Extensions of First Order, or Displacements

### A. Theoretical Development

#### § 13. Extensive Structure, Displacement, and the System of First Order

**§ 13** The pure, scientific way to treat extension theory would be that attempted in the introduction, that is to develop everything fundamental to this science from the concepts. However, to avoid wearying the reader with continued abstractions, and at the same time, by touching upon familiar material, to put him in a position to proceed with greater freedom and independence, I will always relate the derivation of new concepts to geometry, which is the basis of our science. However, since I always present the abstract concept as the basis for the derivation of the truths that form the content of this science, without relying on any geometrical truth, I am able to keep the content of the science completely pure and independent of geometry.<sup>1</sup>

---

<sup>1</sup>In the introduction (no. 16) I have shown that in the presentation of any given science, and in mathematics particularly, two approaches interlace, one of which furnishes the material, that is, the whole series of truths forming the proper

Thus in order to obtain the extensive magnitude I relate it to the generation of the line. Here it is a generating point that assumes a continuous sequence of positions; and the collection of points into which the generating point is transformed with this evolution forms the line. The points of a line thus appear as essentially distinct, and are also labelled as such (with different letters); however, since an equality always adheres to differences (albeit in a subordinate sense), so here the various points also appear as different positions of one and the same generating point. In the same way we now proceed to the extension in our science, if we simply replace these spatial relations by their corresponding concepts.

First, in place of the point, that is, of the particular position (locus), we here substitute the *element*, by which we mean a simple particular, conceived as distinguished from other particulars; and indeed we attribute to the element in the abstract science absolutely no other content. There can therefore be no question as to what sort of particular it actually is—for it is simply a particular, devoid of any real content—or in what sense this one is distinguished from the others—for it is merely defined as distinct, without establishing any real content to the sense in which the distinction exists. Our science has this concept of an element in common with combination theory, and thus the identification of elements (by different letters) is also common to both.<sup>2</sup> The different elements can now simultaneously be regarded as different states of the same generating element, and this abstract difference of states corresponds to difference in position (locus).

We call the transition of the generating element from one state to another an *evolution*; and this abstract evolution of the generating element thus corresponds to the evolution in position or motion of a point in geometry. Just as in geometry a line is formed by the progression of a point, and spatial structures of higher order can only be formed when one again subjects the structure so obtained to motion, so also in our science the extensive structure of first or-

---

content of the science, while the other gives the reader mastery over the material. The first approach is that which I have kept completely separate from geometry, whereas to achieve my purpose with the second I have permitted myself greater freedom.

<sup>2</sup>The difference consists only in the way forms are obtained from the elements in the two sciences; that is, in combination theory by simple conjunction and thus discretely, but here by continuous generation.

der results from continuous evolution of the generating element. To summarize the results of the foregoing argument, we can establish the definition:

*By an extensive structure of first order we mean the collection of elements into which a generating element is transformed by a continuous evolution.*

In particular we call the generating element in its first state the *initial element*, and in its last the *final element*.

From this concept it follows at once that to each extensive structure belongs an opposing one that includes the same elements produced in reverse order, so that the initial element of the one is the final element of the other. Expressed more explicitly, if  $b$  results by an evolution from  $a$ , then the opposite evolution is that by which  $a$  results from  $b$ , and to an extensive structure is opposed the one in the reversed sequence that results from the opposite evolution; which taken together imply that the relation of opposition is a reciprocal one.

## § 14

§ 14 The extensive structure will therefore only appear as elementary if the evolutions which the generating element undergoes are always equal to each other; so that, if an element  $b$  is produced by an evolution from an element  $a$ , both belonging to some given elementary extensive structure, then by an equal evolution an element  $c$  of the same elementary structure is produced from  $b$ , and indeed such an equality must continue to hold if  $a$  and  $b$  are regarded as infinitesimally separated elements, since this equality should obtain throughout for continuous generation. We call such an evolution, in which there is generated from one element of a continuous form its next adjacent element, a *fundamental evolution*, and can therefore say: “The elementary extensive structure is that which results from the continuous action of the same fundamental evolution.”

Now in the same sense in which evolutions are regarded as equal, we can also regard the structures generated thereby as equal, and in this sense, that is, that those generated in the same way by equal evolutions are themselves equal, we call the elementary extensive structure of the first order an *extensive magnitude* or an *extension of first order*, or a *displacement*.<sup>3</sup> Thus the elementary extensive structure becomes an extensive magnitude if we neglect the elements that it includes and retain only the nature of its generation; and while two extensive structures can only be regarded as equal if they include the same elements, two extensive magnitudes are already equal if they are generated in the same way (that is, by the same evolutions), while not including the same elements. Finally, the collection of all elements generated by the continuation of the same and the opposite fundamental evolutions is called a *system* (or *domain*)<sup>4</sup> of first order. The displacements belonging to the same system of first order are thus all generated by continuation of the same fundamental evolution or its opposite.

Before we turn to the conjunction of displacements we illustrate the concepts established in the previous paragraph by an application to geometry. The equality of the method of evolution is represented here by equality of direction; the infinite straight line thus appears as

---

<sup>3</sup>The abstract significance of this originally concrete expression requires no justification, since the names of abstracts all originally had concrete significance.

<sup>4</sup>(1877) I now prefer the term “domain” to “system” which is frequently used in another sense.

a system of first order, the line element as an elementary extension of first order. What there will be called similar appears here as parallel, and likewise parallelism presents two aspects, as parallel in the same and opposite senses.<sup>5</sup> The name displacement we may retain in the corresponding sense for geometry, and thus by equal displacements understand here such line segments as have equal directions and lengths.

---

<sup>5</sup>This distinction is so important in geometry that it contributes greatly to the simplification of geometric proofs and demonstrations if one fixes it by simple names, whence I would like to propose the expressions “equally-” and “oppositely-directed.”

## § 15. Addition and Subtraction of Similar Displacements

### § 15. Addition and Subtraction of Similar Displacements

If the continuous generation of a displacement is imagined to be interrupted midway in its course and then continued, the whole displacement then appears as the conjunction of two adjacent displacements, one of which appears as the continuation of the other. The two displacements forming the factors of this conjunction are generated in the same sense (§ 8), and the product of the conjunction is the displacement from the initial element of the first to the final element of the second, assuming they are contiguous; that is, that they are so produced that the final element of the first is simultaneously the initial element of the second. We tentatively represent the displacement from the initial element  $\alpha$  (see fig. 2) to the final element  $\beta$  as  $[\alpha\beta]$ , and if  $[\alpha\beta]$  and  $[\beta\gamma]$  are generated in the same sense, then  $[\alpha\gamma]$  is the product of the conjunctions indicated above, where  $[\alpha\beta]$  and  $[\beta\gamma]$  are the factors.<sup>6</sup>

We have already shown above (§ 8) that this conjunction is an addition, since it describes the unification of magnitudes generated in the same sense; its corresponding analysis must therefore be subtraction, and so all laws of this type of conjunction are valid for it. We have however yet to demonstrate the characteristic significance of the negative magnitude in our domain.

First, to clarify the meaning of subtraction, observe that since  $[\alpha\beta] + [\beta\gamma] = [\alpha\gamma]$ , provided  $[\alpha\beta]$  and  $[\beta\gamma]$  are generated in the same sense, we can conclude that in general  $[\alpha\beta] = [\alpha\gamma] - [\beta\gamma]$  (see fig. 2); thus, if we use the customary language of subtraction, “if one places the minuend and the subtrahend with their final elements coincident, then the remainder is the displacement from the initial element of

---

<sup>6</sup>This notation for the displacement is only tentative, since the appropriate notation in terms of its boundary elements can only be understood once we have learned how to conjoin the elements themselves (see § 99 in part 2). (1877) In the *Ausdehnungslehre* of 1862 the notation  $[\alpha\beta]$  is chosen for the product of the two elements  $\alpha$  and  $\beta$ , which, if  $\alpha$  and  $\beta$  are points, represents the line segment between  $\alpha$  and  $\beta$ , and is distinguished from the corresponding displacement in that the latter retains only the length and direction, the former also the position, of the infinite straight line to which the line segment belongs. It is therefore all the more important to keep in mind that the notation  $[\alpha\beta]$  for the displacement is here only a temporary makeshift; the appropriate notation  $\beta - \alpha$  can, according to the principle of the presentation, only be given in § 99.



the minuend to that of the subtrahend.”

Setting  $\alpha$  and  $\beta$  equal in the last formula, one obtains

$$[\alpha\alpha] = [\alpha\gamma] - [\alpha\gamma],$$

that is, it equals zero. Furthermore, by virtue of the concept of negatives,

$$(-[\alpha\beta]) = 0 - [\alpha\beta] = [\beta\beta] - [\alpha\beta] = [\beta\alpha],$$

that is, the displacement  $[\beta\alpha]$ , which according to its concept (§ 13) is opposed to  $[\alpha\beta]$ , also appears as the opposite of the latter with respect to addition and subtraction. Finally, since  $a + (-b) = a - b$ , if  $[\alpha\gamma]$  and  $[\gamma\beta]$  are generated in opposite senses, then

$$[\alpha\gamma] + [\gamma\beta] = [\alpha\gamma] + (-[\beta\gamma]) = [\alpha\gamma] - [\beta\gamma] = [\alpha\beta],$$

that is, if two displacements are generated in opposite senses their sum is also the displacement from the initial element of the first to the final element of the second, provided they are contiguous. And thus, combining this result with that above, we can say:

*If two similar displacements are contiguous, that is, so conjoined that the final element of the first is the initial element of the second, then the displacement from the initial element of the first to the final element of the second is the sum of the two.*

In thus being labelled as a sum it is thereby understood that all the laws of addition and subtraction are valid for this method of conjunction.

I will close with a result that will be fruitful subsequently, namely that if the boundary elements of a displacement both evolve by equal displacements in the same system, then the displacement between the new boundary elements is equal to that between the old ones. Thus if  $[\alpha\beta]$  is the original displacement (see fig. 3) and  $[\alpha'\alpha] = [\beta\beta']$ , then we must show that, if all the elements named belong to the same system, then  $[\alpha'\beta'] = [\alpha\beta]$ . Thus according to the definition of the sum, we have

$$[\alpha'\beta'] = [\alpha'\alpha] + [\alpha\beta] + [\beta\beta'],$$

and since

$$[\alpha'\alpha] = -[\alpha\alpha'] = -[\beta\beta'],$$

the addition removes  $[\alpha'\alpha]$  and  $[\beta\beta']$ , whence  $[\alpha'\beta'] = [\alpha\beta]$ .

## § 16. Systems of Higher Order

**§ 16. Systems of Higher Order** In order to proceed to the conjunction of dissimilar displacements, I consider next two dissimilar fundamental evolutions, and let the first fundamental evolution (or its opposite) arbitrarily transplant an element, and then let the element thus generated likewise be arbitrarily transplanted by the second method of evolution. In this way I can therefore generate an infinite set of new elements from a single element, and the collection of elements so generated I call a *system of second order*. I then assume a third fundamental evolution, which does not lead from that same initial element again to one of the elements of this system of second order, and which I therefore designate as independent of those first two, and let this third evolution (or its opposite) arbitrarily transplant an arbitrary element of that system of second order, the collection of elements so generated forming a *system of third order*. Since this method of generation has no limitation on its concept, I can in this way proceed to systems of arbitrarily high order.

It is important to keep in mind that all the elements so generated are not to be conceived as already given otherwise,<sup>7</sup> but rather as being generated from scratch, and thus in accordance with this concept they appear as different since they are generated by different evolutions. On the other hand, it is also clear that the elements appear as given once they have been generated, and that one can say no more about their difference or equality unless one refers back to the original generation.

Before proceeding to the problem of the conjunction of different methods of evolution, I will provide some geometric examples as an aid to the intuition. Thus the plane clearly corresponds to the system of second order, and the plane is thus regarded as generated when all the points of a straight line are moved in a new direction (or its opposite) not included in it; the collection of points so generated thus forms precisely an infinite plane. A plane thus appears as a collection of parallels, all of which intersect a given line; and clearly, since parallels do not intersect, and since in addition the original line does not intersect itself, all the points so generated differ from one another, and consequently the analogy is perfect. Likewise one proceeds to the whole of infinite space as a system of third order by

---

<sup>7</sup>As perhaps in the theory of space all points are already originally given through the presupposed space.

moving the points of the plane in a new direction (or its opposite) not lying in the plane. Further geometry cannot progress, but the abstract science knows no limitation.

## §§ 17–19. Addition and Subtraction of Dissimilar Displacements

**§ 17** Returning to our problem, if I now let an element evolve, first by a displacement  $a$ , and then the element so evolved again by a displacement  $b$ , the net result of the two evolutions is equivalent to the result of a single evolution which is the conjunction of the first two, and which, if both displacements are similar, appears as their sum (§ 15). Here we can for the moment symbolize this method of conjunction by the general sign  $\cap$  for conjunction. Since the process of combining displacements does not alter the situation of the final element, the previous law that

$$(a \cap b) \cap c = a \cap (b \cap c)$$

follows at once. On the other hand, before proceeding to the interchangeability of factors there is a gap in the definition yet to be filled.

Thus we consider the method for generating a system of higher ( $m$ -th) order such as we established in the preceding paragraph, where we assumed a given sequence of  $m$  methods of evolution by which the system was generated. The elements of the system are generated when the initial element progresses by the different methods of evolution in the given sequence, so that each element resulting from a series of evolutions either only continues the latest evolution or one of the subsequent methods of evolution, but never one of the earlier ones. Therefore, if  $a$  and  $b$  are two displacements,  $a$  belonging to an earlier,  $b$  to a later of the methods of evolution, then in the generation of the system there can indeed result an element by adjoining the evolution  $b$  to the evolution  $a$ , but not conversely. That is, it is possible for the conjunction  $a \cap b$  to occur, but not  $b \cap a$ . Nevertheless, although the concept of the latter conjunction cannot be defined by the generation of the system, it must be possible as such. Here, then, is the gap mentioned above.

To see the above more clearly, let  $[\alpha\beta]$  equal  $a$ ,  $[\alpha\alpha'] = [\beta\beta'] = b$ , so that the evolution  $[\alpha\beta']$  equals  $a \cap b$ . But  $[\alpha\beta']$  also equals  $[\alpha\alpha'] \cap [\alpha'\beta']$ , that is,  $b \cap [\alpha'\beta']$ . Thus if the factors are interchangeable, that is if  $a \cap b = b \cap a$ , then necessarily  $[\alpha'\beta'] = [\alpha\beta]$ . Now assume nothing about their interchangeability; then all we can assert about the system and its elements must follow from its method of generation, since the whole system is not given except by its generation. Since however in this nothing like an evolution such as

$[\alpha'\beta']$  appears, we are warranted and required to provide a new definition for such evolutions, and the analogy with our previous results necessarily leads us to this: To the extent that a new definition is warranted,  $[\alpha\beta]$  and  $[\alpha'\beta']$  are assumed to be equal. We can, however, only carry out this equalization in the way defined once we have ascertained the extent of that warrant.

To this end we consider two equal displacements:

$$[\alpha\beta] = [\gamma\delta] = a,$$

whose boundary elements are all subjected to one of the later evolutions  $b$  and are thereby transformed into  $\alpha', \beta', \gamma', \delta'$ , so that

$$[\alpha\alpha'] = [\beta\beta'] = [\gamma\gamma'] = [\delta\delta'] = b.$$

Now since  $[\alpha'\alpha] = [\gamma'\gamma] = (-b)$ , one has for the evolutions  $[\alpha'\beta']$  and  $[\gamma'\delta']$  the equations

$$\begin{aligned} [\alpha'\beta'] &= [\alpha'\alpha] \cap [\alpha\beta] \cap [\beta\beta'] = (-b) \cap a \cap b, \\ [\gamma'\delta'] &= [\gamma'\gamma] \cap [\gamma\delta] \cap [\delta\delta'] = (-b) \cap a \cap b; \end{aligned}$$

and so the two evolutions are equal. Thus if two element pairs are generatable from each other by equal evolutions, and one subjects all four elements to the same new evolution, then the newly resulting element pairs will also be generatable from each other by equal evolutions. Now since this law also holds if  $[\alpha\beta]$  represents a fundamental evolution, it follows not only that a displacement remains a displacement if all its elements are evolved by equal amounts, but also that, provided only that it is shown that the fundamental evolutions remain equal with that progression of elements, it then also holds for the whole displacement.

The extent of the warrant mentioned is thus given, and consequently we assert that if, in a system of  $m$ -th order, a displacement belonging to one of the earlier of the  $m$  methods of evolution defining the system is subjected to one of the later methods of evolution, and indeed all elements of the same evolution, then the corresponding fundamental evolutions in the original displacement and in the one resulting from that evolution are to be regarded as equal, or unequal if the elements are subjected to different evolutions.<sup>8</sup> By virtue of

---

<sup>8</sup>The deduction through which we are led to this definition of equal evolutions belongs to the approach (intro., no. 16) that provides an overview. For the purely mathematical approach the same definition (and in general every definition) appears as purely arbitrary.

the preceding theorem it therefore follows that this equality (and inequality) also applies to the displacements themselves under the same circumstances. Thus we arrive at the theorem: *If one subjects a displacement belonging to one of the  $m$  original methods of evolution of the system to evolutions that likewise belong to those methods of evolution, and indeed all elements of the same evolutions, then the displacement produced by that evolution is equal to its original.*

That we can drop the distinction between earlier and later methods of evolution then follows easily from the reciprocity of the relation; for if it is supposed that  $[\alpha\beta]$  is equal or unequal to  $[\alpha'\beta']$  according as  $[\alpha\alpha']$  is equal or unequal to  $[\beta\beta']$ , so conversely the latter expressions are equal or unequal according as the former are, as follows at once by the method of indirect proof (*reductio ad absurdum*). Thus if the displacement generated by an earlier evolution remains constant on being subjected to a later evolution, then that generated by a later also remains constant on being subjected to an earlier; and thus follows the theorem in the form expressed above.

Now, we showed above that the theorem  $a \cap b = b \cap a$  is true by assumption; and thus for the  $m$  methods of evolution defining the system we have generally the laws

$$(a \cap b) \cap c = a \cap (b \cap c)$$

and

$$a \cap b = b \cap a;$$

whence this conjunction is elementary; but in addition the corresponding analytic conjunction is unique. For if I keep one factor, say the first, of the synthetic conjunction unchanged, and alter the other, because I either replace the method of evolution generating it with another, or instead retain the old method of evolution but break it off earlier or continue it further than before, then this changes the resulting final element, which is simultaneously the final element of the product of the conjunction, and thus alters the product. The uniqueness of the analytic conjunction follows by the familiar method of inference (see §6). Thus, according to §6, the conjunctions in question are designated addition and subtraction, and all laws of addition and subtraction apply to them. Finally, since the same laws of conjunction that apply to the  $m$  original types of evolution also hold for their conjunctions according to the laws of addition and subtraction

tion, we can combine the results of the foregoing discussion into the following very simple theorem:

*If  $[\alpha\beta]$  and  $[\beta\gamma]$  represent any evolutions, then*

$$[\alpha\gamma] = [\alpha\beta] + [\beta\gamma].$$

*In particular, since we designate this conjunction as addition, we thereby assert the validity of all laws of addition we have presented in §§ 3–7.<sup>9</sup>*

---

<sup>9</sup>I cannot sufficiently stress that the overall development, and in particular that part treated here, which belongs to the most difficult part of our science, is clarified by the corresponding geometric development. In order not to interrupt the course of the discussion I prefer not to undertake its transposition into geometry here; it is in any case always obvious (see fig. 5).

**§ 18** In the discussion of the preceding section we had considered the evolutions resulting from conjunctions only with respect to their initial and final elements, without regard to the displacement linking them; the only displacements that appeared were those belonging to the original types of evolution of the system. To fill in this gap we must now show how two elements of a higher system define all the other elements lying with them in a single system of first order.

To this end we need only return to the concept of a system of first order, to wit that it is generated by the continuation of an evolution that remains the same. Now if an element subjected to the sequence and progression of the evolutions  $a, b, c, \dots$ , belonging to the original methods of evolution ultimately produces another element  $\beta$  from the element  $\alpha$ ,<sup>10</sup> then according to the concept of a system of first order, that element produced from  $\beta$  by the same evolutions  $a, b, c, \dots$  must belong to the same system of first order, and so on. Indeed, we can also proceed backwards from  $\alpha$  by the opposite evolution and again obtain new elements belonging to the same system, but lying on the negative side if the former are interpreted as positive. The elements on the positive side are thus produced from the element  $\alpha$  as it is subjected to this repetition and progression of the same sequence of evolutions  $a, b, c, \dots$ . Now since we can arbitrarily interchange and combine the progressing evolutions, as was shown in the previous section, we can reorder and combine the equal evolutions together so as to obtain a new construction for that sequence of elements, as we will now demonstrate in detail.

Thus if one subjects the element  $\alpha$  to each of the  $m$  evolutions  $a, b, c, \dots$  individually, there result  $m$  elements that we can place in mutual correspondence. If one again subjects each of these elements to the same evolution that it experienced before, one then obtains  $m$  new mutually corresponding elements, and so on. Now if we regard the corresponding elements of each such group of  $m$  elements as the final elements of  $m$  displacements, all of which have  $\alpha$  as initial element, and which we also place in mutual correspondence, then we obtain the same elements we found previously when we evolved  $\alpha$  progressively by the corresponding displacements of each such group; and, in this way, to each such group of mutually corresponding elements there corresponds an element in the new system of first order that results from an evolution which is the sum of the evolutions

---

<sup>10</sup>In fig. 5a it is exhibited for two evolutions  $a, b$ .



represented by those displacements.

Now if, in the constructions given, the evolutions  $a$ ,  $b$ ,  $c$ , ..., are fundamental evolutions, and thus transplant directly from one element to the one adjacent, then (simultaneously applying the same process on the negative side) one obtains the whole system of first order.

Now it must be shown that with this method one can always produce a system of first order between two elements of the higher system, but only one such system.

If  $\alpha$  and  $\beta$  are the two elements of the system, then in the method of generation of the system it is already manifest that  $\beta$  is always generatable from  $\alpha$  by the  $m$  methods of evolution of the system, and indeed by the series given in only one way. If  $a$ ,  $b$ ,  $c$ , ... are these evolutions, then the problem is to show that for these displacements one can always choose such mutually corresponding fundamental evolutions that  $a$ ,  $b$ ,  $c$ , ... are corresponding displacements and thus, by the construction just given, that  $\beta$  is an element of the system of first order generated by these corresponding fundamental evolutions. If I consider, first, two displacements  $a$  and  $b$ , each of which is produced by continuation of the same fundamental evolution, then since according to the concept of continuity the fundamental evolutions can have no inherently fixed magnitude, arbitrary fundamental evolutions may be taken as corresponding. Now if, while one fundamental evolution and the displacement  $a$  generated thereby remain fixed, the other fundamental evolution increases or decreases, so also will the thereby generated displacement  $b$  corresponding to the displacement  $a$  increase or decrease; and indeed, if the fundamental evolution increases or decreases continuously, so also will the displacement  $b$  increase or decrease continuously, as is directly implied by the concept of continuity. Thus, since the fundamental evolution for  $b$  can be chosen arbitrarily, so also the  $b$  corresponding to the displacement  $a$  can take on any given magnitude, and the same applies for each of the other displacements  $c$ , and so on. Thus for the displacements  $a$ ,  $b$ ,  $c$ , ... given above such fundamental evolutions can also be chosen so that those displacements appear as corresponding, and thus the element  $\beta$  is represented as an element of the system of first order generated by these fundamental evolutions.

It immediately follows from the above argument that only a single system of first order can lie between  $\alpha$  and  $\beta$ . Thus there could be another system of first order only if those fundamental evolutions

of the displacements  $b, c, \dots$  corresponding to the fundamental evolution in  $a$  could be chosen differently; but then, as we have shown, the other displacements corresponding to the displacement  $a$  would come out differently, whence the element  $\beta$  would also no longer be generated from  $\alpha$ .

As we have now demonstrated that in fact one and only one system of first order can lie between the two elements, this fills in the gap noted at the beginning of this section, since for the displacement appearing as the sum of two displacements not only are the initial and final elements defined, but indeed the whole displacement in all its elements. The concept of the sum is thus defined not only for the evolutions but also for the displacements themselves; in particular, if  $[\alpha\beta]$ ,  $[\beta\gamma]$ ,  $[\alpha\gamma]$  are displacements generated according to the principle just developed, then in general one always has

$$[\alpha\gamma] = [\alpha\beta] + [\beta\gamma],$$

*that is: If one adjoins two or more displacements contiguously, the displacement from the initial element of the first to the final element of the last is their sum.*

Applying the concept of dependence that we introduced in § 16 to this concept of the sum, it follows that one method of evolution depends on another if the displacements belonging to the former can be expressed in terms of displacements belonging to the latter, whereas they are independent of each other if this is not possible.

**§ 19** Thus far we have made the concept of the sum of displacements dependent on the particular methods of evolution of the whole system, since if the initial and final elements of the sum were given by the contiguous adjoining of displacements, a displacement lying between them was constructed as part of a system of first order by the  $m$  original methods of evolution of the whole system. We must, in conclusion, remove this dependence.

We have already shown above (§ 18) that if several displacements are generated in a corresponding way, then not only to each element and each term of one of them corresponds an element and a term in each of the others, but the sum is also generated in the same corresponding way, so that the sum of the corresponding terms always corresponds to these terms. Now if one has any two displacements of the system, say  $p_1$  and  $p_2$ , and if both are displayed as sums of displacements belonging to the original types of evolution, that is

$$\begin{aligned} p_1 &= a_1 + b_1 + \cdots, \\ p_2 &= a_2 + b_2 + \cdots, \end{aligned}$$

then one has

$$p_1 + p_2 = (a_1 + a_2) + (b_1 + b_2) + \cdots$$

and furthermore, if  $\alpha_1, \alpha_2, \beta_1, \beta_2, \dots$  are corresponding parts of the displacements  $a_1, a_2, b_1, b_2, \dots$ , whence also  $(\alpha_1 + \alpha_2), (\beta_1 + \beta_2), \dots$  are in the same sense corresponding parts of  $(a_1 + a_2), (b_1 + b_2), \dots$ , then according to the preceding paragraph each part of the sum  $(p_1 + p_2)$  is obtained as the sum of the corresponding parts. That is to say, such a sum is always equal to

$$(\alpha_1 + \alpha_2) + (\beta_1 + \beta_2) + \cdots,$$

that is

$$= (\alpha_1 + \beta_1 + \cdots) + (\alpha_2 + \beta_2 + \cdots),$$

where the first term represents a part of  $p_1$ , the second the corresponding part of  $p_2$ . Thus each element of the sum  $(p_1 + p_2)$  is generated when one evolves the initial element the same amount by each arbitrary part of  $p_1$  and then by the corresponding part of  $p_2$ . Thus we can establish the general result: *“If two displacements are given, and one evolves an arbitrary element by a part of the first, and then (progressively) by the corresponding part of the second,*

*the collection of elements so generated comprises the sum of those two displacements."*

Although we have now established the concept of the sum in its generality and independence, we will present yet another theorem, which we had proved previously for a special case, in a more general form; that is:

*If all elements of a displacement are evolved by the same amount, the resulting displacement remains equal to its original.*

It was already shown in § 18 that a displacement results thereby; that it is equal to the original one follows from the same formulas as at the end of § 15. Thus if  $[\alpha\beta]$  is the original displacement, and if  $[\alpha\alpha'] = [\beta\beta']$ , then

$$[\alpha'\beta'] = [\alpha'\alpha] + [\alpha\beta] + [\beta\beta'] = [\alpha\beta],$$

since  $[\alpha'\alpha]$  and  $[\beta\beta']$  are removed by the addition as opposed magnitudes.

## § 20. Independence of the Higher-Order System

**§ 20. Independence of the Higher-Order System** The development presented in the previous section prepares us for an independent description of the higher-order system. So far, this has been presented as dependent on knowing the fundamental methods of evolution by which it was generated. As we will show, we can remove this dependence since the same system of  $m$ -th order is generatable by any  $m$  methods of evolution belonging to it that are mutually independent (in the sense of § 16), that is that are included in no system of lower order (than the  $m$ -th).

First I will show that if the system is generated by any  $m$  methods of evolution whatever, I can replace any given one of them by a new method of evolution ( $p$ ) belonging to the same system of  $m$ -th order and independent of the remaining ( $m - 1$ ), and generate the given system using this in combination with the other ( $m - 1$ ).

Since by assumption  $p$  belongs to the given system of  $m$ -th order, it can be expressed (§ 18) as a sum of displacements belonging to the original methods of evolution, that is we can write

$$p = a + b + c + \dots$$

if  $a, b, c, \dots$  belong to the original methods of evolution. Now if  $a$  represents the method of evolution for which  $p$  is to be substituted, then by assumption  $p$  must be independent of those remaining,  $b, c, \dots$ ; that is,  $a$  may not be zero, whereas on the other hand any of the other displacements may be zero. I must show that each element of the system generated by  $p, b, c, \dots$  also belongs to that generated by  $a, b, c, \dots$  and conversely, provided they are both generated from the same initial element. The former is immediately clear, since  $p$  belongs to the system generated by  $a, b, c, \dots$ ; the latter requires a detailed demonstration.

Each given element of the system generated by  $a, b, c, \dots$  from any chosen initial element can be generated from the initial element by an evolution

$$q = a_1 + b_2 + c_2 + \dots,$$

where  $a_1, b_2, c_2, \dots$  are similar to  $a, b, c, \dots$  respectively. In order to be able to replace  $a_1$  with the magnitude  $p$  or one similar to it, one temporarily assumes that the magnitudes  $p, a, b, c, \dots$  correspond, and in the same sense  $p_1, a_1, b_1, c_1, \dots$  correspond respectively, so

that, since  $p = a + b + c + \cdots$ , then, according to §18, the same equation also holds for the corresponding displacements. That is

$$p_1 = a_1 + b_1 + c_1 + \cdots ,$$

whence also

$$a_1 = p_1 - b_1 - c_1 - \cdots ,$$

Substituting this for  $a_1$ , one then has

$$q = p_1 + (b_2 - b_1) + (c_2 - c_1) + \cdots ,$$

that is, the element in question is generatable by evolutions similar to  $p, a, b, c, \dots$ . In other words, it belongs to the same system generated by  $p, b, c, \dots$  from the same initial element. The identity of the two systems is thus demonstrated, and it is shown that for every arbitrary member of the  $m$  original methods of evolution generating the system one can substitute a new one, provided only that it belongs to the given system and is independent of the rest. And since one can continue this process, it follows that one can always generate the same system by any  $m$  mutually independent methods of evolution, or:

*Every displacement of a system of  $m$ -th order can be represented as a sum of  $m$  displacements belonging to  $m$  given independent methods of evolution of the system, the sum being unique for each such set.*

The system is thus made independent of the choice of the  $m$  independent methods of evolution, but we have yet to make it independent of the initial element.

Let  $\alpha$  be the initial element originally chosen, and replace it as initial element by another element  $\beta$  of the system. Now if  $\gamma$  is any third element, then one has

$$[\alpha\gamma] = [\beta\alpha] + [\alpha\gamma].$$

Wait, that should be  $[\alpha\gamma] = [\alpha\beta] + [\beta\gamma]$ . Now if  $[\alpha\beta]$  and  $[\beta\gamma]$  are representable by the assumed methods of evolution, then so also is  $[\alpha\gamma]$  as their sum; that is, each element generatable from the initial element by the assumed methods of evolution is also generatable from any other element in the same way; thus

*Every system of  $m$ -th order can be regarded as generated by those same  $m$  independent methods of evolution from any arbitrary element; that is, all other elements can be generated from a single such element by those methods of evolution.*

In this way the higher-order system is shown to have its own characteristic structure.

## B. Applications

### §§ 21–23. Inadequacy of the Present Foundation of Geometry and Attempt at a New Formulation

**§ 21** I proceed next to applications, and first of all to geometry. Nevertheless, prior to this I will attempt at least by way of suggestion, and indeed independent of our science, a new foundation for geometry itself, the better to survey the agreements and divergences between the procedures of the two disciplines. In particular, I maintain that geometry still lacks a scientific starting point, and that the foundation of the entire structure of geometry still suffers from a defect that necessitates its complete restructuring. I cannot make such a claim, which threatens a structure sanctified for a millenium, without illustrating it on the most decisive grounds.

The defect whose existence I will demonstrate is most easily perceived in the concept of the plane. As it is defined in all the treatments of geometry known to me, there is at bottom the assumption that a straight line having two points in common with a plane lies completely in that plane; now either one assumes this tacitly (thus Euclid), or it is implicit in the definition of the plane, or finally it is stated as a separate postulate. The first is obviously unscientific, but the second can make no better claim to being scientific, as I will show at once. For it is clear that the plane is already defined if it is a collection of parallels that can be drawn from a straight line in a direction not contained in that line, or if it is a collection of straight lines that can be drawn from a point to a straight line.

Thus, for example, if we stay with the first definition, then clearly it must first be proved that each straight line intersecting two of these parallels must also intersect all the rest, a result that cannot be proved without a whole series of auxiliary theorems. If perhaps one defines the plane as a surface that completely contains every straight line having two points in common with it, this illustrates how the theorem stated previously is smuggled into the domain of geometry, hidden in this definition. Just as no mathematician would agree to eliminating the proof of the theorem that opposite sides of a parallelogram are equal by defining a parallelogram as a quadrangle whose opposite sides are equal and parallel, so here no one would permit the theorem cited above to be illicitly introduced into geometry with such a definition. If one wishes to remain with the usual



procedure in geometry, the only alternative is to set that theorem up as a separate postulate. However, if a postulate can be eliminated without thereby necessitating the introduction of another, then this must be done, even if it should require a complete restructuring of the science, since science necessarily achieves its character of simplicity by such eliminations.<sup>11</sup>

If we now turn from this defect, which we hope has been demonstrated, to seek after its causes, we find these in the defective comprehension of geometrical postulates.

---

<sup>11</sup>It could of course be that there is a presentation, avoiding this defect, that has not come to my attention. Since, however, such a presentation requires that at the same time the theory of parallels, that cross of mathematicians, be cleared up, I can with reasonable confidence assume that such a presentation does not exist.

**§ 22** First of all it is striking that, besides genuine postulates expressing geometric perceptions, numerous completely abstract principles are introduced under the same name, such as: “If two magnitudes equal a third, they are equal to each other,” which do not deserve the name once one understands postulates as assumed truths. Thus I believe it was demonstrated above (§ 1) that the abstract principle just mentioned only expresses the concept of equality, the same also being true for the other abstract principles, which in essence amount to the statement that those generated in the same way from equals themselves remain equal. Euclid himself is free of blame for mixing postulates with assumed concepts, the former of which are included among his “Postulates” (*aitēmata*), the latter being included among his “Common Notions” (*koinai ennoiai*), a procedure that was no longer understood even by his commentators, and which has also found little imitation among contemporary mathematicians, to the shame of their science. In fact, the abstract discipline of mathematics is a stranger to postulates; rather, in it proof only occurs by the interconnection of assertions, since use is made of no other procedural principle than that of general logic, to wit that every assertion made of a series of things one has in mind can also be truly asserted of each individual belonging to that series. And to set up this procedural principle, which, as one sees upon reflection, incorporates what one would say with the general principle, as a postulate, as mistakenly occurs in logic if it is not first proved in it, would occur to no mathematician.

**§ 23** Thus there remain as postulates in geometry only those truths that are drawn from the perception of space. These postulates will thus be correctly formulated if in their totality they provide a complete account of space, and if nothing is included that does not support this account.

The true source of the defective foundation of the usual treatment of geometry is now apparent, since in part postulates that express primitive spatial perceptions are omitted, and must then be tacitly assumed later, when their application is needed, and in part postulates are included that express no primitive spatial perception, and which therefore appear superfluous in a more rigorous treatment; and the totality of postulates generally give the impression of an aggregate of the clearest possible principles, compiled to provide the most convenient lines of argument.— However, the postulates of geometry, as we must assume them, express the fundamental properties of

space as they are originally imparted to our imagination, that is its *simplicity* and *relative limitation*.

The simplicity of space is expressed in the postulate:

*Space is constituted equally at all positions and in all directions; that is, equal constructions can be carried out at all positions and in all directions.*

In its manner of expression this postulate immediately decomposes into two, one of which expresses the possibility of translation, the other of rotation, thus:

- 1) Within the diversity of positions an equality is possible.
- 2) Within the diversity of directions, and in particular of opposed directions, an equality is possible.

We call those constructions *equal and equally-directed*<sup>12</sup> that are carried out in exactly the same way at different positions, and which are therefore distinguished only by position; those distinguished both by position and direction are absolutely equal, and in particular those carried out in the same way but in opposite directions, even when at different positions, *equal and oppositely-directed*, or briefly, *opposed*. We retain the same nomenclature for the results of the constructions. If we set these last two up as yet another pair of partial principles, we can express both postulates more precisely as follows:

- 1) What results from equal and equally-directed constructions is again equal and equally-directed.
- 2) What results from opposed constructions is again opposed.
- 3) What results from absolutely equal constructions (even though at different positions and in different initial directions) is again absolutely equal.

The first two of these three postulates comprise the positive assumptions for that part of geometry expressing the first part of our science. The relative limitation of space is expressed by the postulate:

---

<sup>12</sup>Here we include more than the usual meaning, since we replace the mere concept “parallel” with the more rigorous “equally- and oppositely-directed” (cf. third footnote § 14); otherwise it would have been more proper to introduce a simpler expression for it, such as perhaps “fully equal.”

*Space is a system of third order.*

To understand these postulates requires such antecedent assertions and definitions as we have given above in the abstract science.

The direct evidence for these postulates and their necessity are perfectly apparent at once; for without the first, no straight line is possible; without the second, no plane (see below); and without the third, no angle; and finally, the last describes space itself in its threefold extension, and although it is usually omitted in the standard presentations, it is not difficult to identify points in them where tacit use is made of it. That they suffice for geometry can only be fully demonstrated by developing geometry itself from these seeds. Here, however, we will proceed with a treatment more suggestive than detailed.

It may seem surprising that the principle that only one straight line is possible between two points, or, as expressed in Euclid, that two straight lines cannot enclose a space (*chōrion*), is here omitted as a postulate; but in fact it is implicit in the proper interpretation of the first postulate. Thus, should two straight lines having a point in common have yet another in common, then the space out to this second point would be composed differently from that out to the first if the lines did not simultaneously have all other points in common, and thus completely coincide.<sup>13</sup> If this demonstration, which moreover looks much more rigorous in a detailed exposition of the science, seems to have too philosophical a character, then for the mathematical development one may formulate the principle as a partial postulate after all, provided one remains aware of its connection to the original postulate.<sup>14</sup>

---

<sup>13</sup>In general, division into the most detailed principles possible is the most characteristic and effective of mathematical methods, cf. also intro. no. 13.

<sup>14</sup>(1877) Cf. for this section (II 15–23) appendix 1, “On the relation between non-Euclidean geometry and extension theory.”

## § 24. Geometric Problems and Theorems; Mean Between Several Points

**§ 24. Geometric Problems and Theorems; Mean Between Several Points** We conclude with a series of geometric applications that can be solved by the methods given in this chapter, and thus suppose only that one may, without using a compass, find a straight line between two points, even if they are infinitely separated, and that it is possible to place a plane on three noncollinear points. Since in the first case we say that an infinite separation between the two points is possible, we express thereby the requirement for drawing a parallel to a given line. The above requirements are in general the only ones we formulate for the part of geometry that expresses the first part of our science.<sup>15</sup>

**PROBLEM 1.** To draw a displacement  $AX$  equal and equally-directed with a given displacement  $BC$  (cf. fig. 7a and 7b).

**SOLUTION.** If one draws  $AD$  parallel to  $BC$  and  $CE$  parallel to  $BA$ , then the intersection of the two lines is the point  $X$  sought. Now if  $A$  lies on the straight line  $BC$ , one takes a point  $D$  not collinear with  $BC$  and makes  $DE \# BC$  and  $AF \# DE$  according to the construction just given; then  $F$  is the point  $X$  sought.

**PROBLEM 2.** To divide a displacement into an arbitrary number of equal parts.

The solution can be reduced to the usual one by means of the construction given in the previous problem.

**PROBLEM 3.** (cf. fig. 8) To find the point  $X$  satisfying the equation

$$[AX] = [BC] + [DE].$$

**SOLUTION.**  $G$  is the point sought if one constructs  $AF \# BC$  and  $FG \# DE$ .

---

<sup>15</sup>One is unaccustomed to including the requirement for drawing a parallel to a given line among the postulates of geometry; but we have regarded it as simply a special case of the requirement that two points can be connected by straight lines. If one does not include this requirement, then the series of theorems and problems limited to just the drawing of straight lines remains utterly barren, since one then cannot even consider the projection by which finitely distant points can be transported to infinity and conversely.

PROBLEM 4. To find the point  $X$  satisfying the equation

$$[AX] = [BC] - [DE].$$

For the following theorems and problems I introduce a pair of expressions that are indispensable for simplifying the discussion. Thus by the *deviation* of a point  $A$  from another  $B$  I mean the displacement  $BA$  including its direction and length, and by the *collective deviation* of a point  $R$  from a series of points  $A, B, C, \dots$  I mean the sum of the deviations of that point from the individual points of the series, that is the sum

$$[AR] + [BR] + [CR] + \dots,$$

wherein it is obvious that the concept of the sum as developed above is fundamental. It is therefore clear from this that the collective deviation of a series of points  $A, B, C, \dots$  from a point  $R$  is given by the sum

$$[RA] + [RB] + [RC] + \dots.$$

Now from the equation

$$[AB] + [CD] + [EF] + \dots = 0 \quad (1.1)$$

I can deduce the equation

$$[RA] + [RC] + [RE] + \dots = [RB] + [RD] + [RF] + \dots, \quad (1.2)$$

where the two sides have the same number of terms, since according to the general concept of the sum I can write  $[AR] + [RB]$  or  $[RB] - [RA]$  instead of  $[AB]$  in (1.1) and likewise substitute  $[RD] - [RC]$  for  $[CD]$  and so on, and then bring  $[RA]$ ,  $[RC]$ ,  $[RE]$ ,  $\dots$  to the other side with a change of sign.

This very simple transformation immediately leads to a series of the most beautiful and elementary theorems, provided it is recognized that one can recover the first equation from the second by the reverse procedure. Thus, first:

*If the collective deviation of a point  $R$  from a series of points equals the collective deviation of the same point from another series of points, which however must be as many as the first, then first of all the same is also true for every other point that can replace  $R$ , and furthermore the sum of the displacements from the points of the first*

series to the corresponding points of the second equals zero, however one may establish the correspondence between the two series.

Furthermore:

*If the sum of several ( $m$ ) displacements is zero, the sum remains zero if one interchanges either the initial or final points of any of them (for example replacing  $AB$  and  $CD$  by  $AD$  and  $CB$ ), and at the same time the collective deviation of the final points from any arbitrary point  $R$  remains equal to the collective deviation of the initial points from that same point  $R$ .*

A particular case of these general theorems is that in which some or all of the points of one or the other of the series coincide. If all  $m$  points of one or the other series coincide with a single point  $S$ , then since the collective deviation of these  $m$  points equals the  $m$ -fold of the deviation of the single point  $S$ , we have the theorem in the following form:

*If the collective deviation of a series of  $m$  points from a point  $R$  equals the  $m$ -fold of the deviation of a point  $S$  from the same point  $R$ , this is also true with respect to any point that can replace  $R$ , and the collective deviation of that series from  $S$  is zero.*

Conversely:

*If the collective deviation of a point  $S$  from a series of points is zero, then the collective deviation of any other point  $R$  from that series equals the  $m$ -fold of the deviation of the same point from  $S$ .*

From the last theorem it follows that there is no point other than  $S$  for which the same condition holds; we can therefore identify it by a simple name, and call it the *mean* of the series of points.<sup>16</sup> By the *mean* of a series of points is therefore meant that point whose collective deviation from that series is zero. From the first of these two theorems follows a very simple construction for the mean. Thus to find the mean between  $m$  points one draws the displacements to those points from any point  $R$ , and on equating  $RS$  to the  $m$ -th part of the sum of these displacements (by problems 2 and 3),  $S$  is then the mean. If in all the above theorems one lets some of the points coincide, one obtains multiple points or points with associated

---

<sup>16</sup>Elsewhere, I have justified the use of this name instead of the former "center of average distances" (In the article "Theorie der Centralen," Crelles Journal für die reine and angewandte Mathematik, Bd. XXIV, cf. in particular p. 271).

coefficients, and for them the same theorems apply. For example: given  $m$  points  $A_1, \dots, A_m$  with associated coefficients  $\alpha_1, \dots, \alpha_m$ , respectively, and  $n$  points  $B_1, \dots, B_n$  with associated coefficients  $\beta_1, \dots, \beta_n, \dots$ , such that

$$\alpha_1 + \dots + \alpha_m = \beta_1 + \dots + \beta_n$$

then if the collective deviation of the first set from any point  $R$  equals that of the second from the same point, that is if

$$\alpha_1[RA_1] + \dots + \alpha_m[RA_m] = \beta_1[RB_1] + \dots + \beta_n[RB_n]$$

then this is also true for every other point that can replace  $R$ .—And the remaining theorems can be transformed in the same way.

We have anticipated somewhat in order to provide an immediate overview, as we have used the concept of number, for which we have as yet no account in the abstract science.



## § 25. The Newtonian Principles of Mechanics

**§ 25. The Newtonian Principles of Mechanics** The application of our science to statics and mechanics is particularly suited for bringing its significance to light.

In order to develop the whole subject from its beginnings we consider first the Newtonian laws of motion. Thus the first law<sup>17</sup> consists of two dissimilar parts; in the first of which, that each resting body remains in a state of rest until a force sets it in motion, is implicit the concept of force as a source of motion, while the second expresses that, so long as no force acts on it, each moving body maintains the same motion, that is, in equal times it continues to describe equal displacements (in the sense of our science, equal and equally-directed). Since this continuing motion appears as an enduring force, we can express this law more simply thus:

*Every action of a force on matter is at the same time the communication to it of a force remaining ever equal to itself (that is, remaining equal in strength and parallel to its original direction).*

It is important to distinguish this force communicated to and, after communication, resident in the body, from the force acting on the matter (and thus having its seat elsewhere).

The second Newtonian law of motion<sup>18</sup> also consists of two dissimilar parts, each of which includes a fundamental assumption, which however is somewhat hidden by the Newtonian formulation of the law. Thus, their relationship aside, it seems as if the law attempts to express no more than that if different forces are imagined to act on the same particle, the communicated motions are proportional to and equally directed with the forces; this however is not a fundamental law but rather only an application of the concept of force, since, as the supposed cause of motion, force can only be defined and measured by motion. But that this is not in fact the sense of that law follows from the context, and it is evident that on the one hand this asserts how the same force acts on different masses, and on the other how the same force acts on a single body in its different

---

<sup>17</sup>“Corpus omne perseverare in statu suo quiescendi vel movendi uniformiter in directum, nisi quatenus a viribus impressis cogitur statum illum mutare.” Newton, *Philosophia Naturalis Principia Mathematica*, Lex I.

<sup>18</sup>“Mutationem motus proportionalem esse vi motrici impressae, et fieri secundum lineam rectam, qua via illa imprimitur.”

states of motion, that is, how the acting force is combined with another already resident in the body. This latter is so expressed that the change in the motion in the direction in which the force acts, and its proportionality to that force, then follow. If one rigorously interprets this concept of the change in the resident force by that adjoined, then it is nothing but what we mean by addition, once we interpret forces as displacements. Thus we grasp this part of the law better as follows:

*Two forces imparted to the same point are summed.*

On excluding what is already implicit in the concept of a force, or what can be deduced from it, the other part of this law is transformed into the fundamental law:

*Two material particles that undergo equal actions from any given motive force also experience equal actions from any other motive force.*

Two such particles, which we imagine as points or elements of infinitesimally small extension, we say are *equal in mass*.

That this law is the proper foundation for this part of the Newtonian principles could readily be demonstrated by a rigorous analysis of them, but the proof would lead me too far afield. It is however important to remark that we hereby arrive at a definite and general measure of the force, since we can set the force equal to the displacement that a material particle whose mass is taken as the unit of mass describes in unit time, if that force resides permanently in it; that is, the resident force in the unit mass equals its velocity.

Finally, the third Newtonian law of motion, on the equality of action and reaction,<sup>19</sup> we can express thus:

*If two particles of equal mass act on each other, the sum of their motions remains the same as if they had not acted on each other.*

Moreover, it is clear that the four laws just presented, of inertia, of the summation of forces, of equal mass, and of mutual action, together express a single overarching principle, that is, that the forces maintain themselves in their totality. The principle of inertia expresses the conservation of a single force on a single particle, the summation principle the conservation in their sum of two forces on a

---

<sup>19</sup>“Actioni contrariam semper et aequalem esse reactionem; sive corporum duorum actiones in se mutuo esse aequales et in partes contrarias dirigi.”

single particle, the last the conservation of the collective force in mutual action, which again the third already assumes; the third, since it is based on the concept of mass, asserts that the collective force on a collection of points is found by addition of the forces experienced by the individual mass points.

## § 26. Collective Motion, Motion of the Center of Mass

**§ 26. Collective Motion, Motion of the Center of Mass** By combining these principles we can immediately formulate the general theorem:

*The collective force (or collective motion) resident in a collection of material particles at any one time is the sum of the collective forces (or collective motions) resident in it at any previous time, plus the net force communicated to it in the intervening time interval; provided that all forces are interpreted as displacements of fixed direction and length, and referred to particles of equal mass.*

Thus according to the previous section the resident force and the resident motion are identical.—The proof of this theorem is implicit in the fully restructured laws of motion as we have reformulated them by means of the concepts of our science. Each resident force is conserved, each newly-acting force is summed, and the mutual forces of each pair of points of equal mass do not alter the collective force of the two points, and thus also do not alter the net mutual force of the whole assembly of points.

A special consequence of this theorem is that, so long as no force is adjoined from outside, the collective force, or collective motion, resident in the assembly, remains constant. If  $p$  is the collective force resident at any one time in an assembly of  $m$  equal-mass points whose mass we take to be unity,  $\alpha_1, \dots, \alpha_m$  are the positions of these points at that time, and  $\beta_1, \dots, \beta_m$  the positions into which they pass after the lapse of unit time if the collective force remains constant, then we have the equation

$$[\alpha_1\beta_1] + \dots + [\alpha_m\beta_m] = p. \quad (1.3)$$

From among all the points of the system, we now obtain one which for the moment we will leave completely unspecified, but which we will then so define that its motion is completely determined. Suppose this point has the position  $\alpha$  at some time; with constant collective force it passes over from  $\alpha$  to  $\beta$  in unit time, whence one has

$$[\alpha_i\beta_i] = [\alpha_i\alpha] + [\alpha\beta] + [\beta\beta_i]$$

according to the general definition of the sum. Now since, upon substituting in this way for all the terms in equation (1.3),  $[\alpha\beta]$

appears  $m$  times, one has

$$([\alpha_1\alpha] + \cdots + [\alpha_m\alpha]) + m[\alpha\beta] + ([\beta\beta_1] + \cdots + [\beta\beta_m]) = p. \quad (1.4)$$

Defining the point  $\alpha$  as the mean of the points  $\alpha_1, \dots, \alpha_m$ , and  $\beta$  as the mean of the points  $\beta_1, \dots, \beta_m$ , the summed terms drop out since the collective deviation of a series of points from its mean is zero according to § 24. One then has

$$m[\alpha\beta] = p \quad \text{or} \quad [\alpha\beta] = \frac{p}{m}, \quad (1.5)$$

that is, if we introduce “center of mass” instead of “mean,” as is customary in statics, and call  $m$  the mass of the whole system:

*The path that the center of mass describes in unit time, if the collective force resident in the assembly remains constant,*

or, expressed more briefly:

*The velocity of the center of mass of a system is the same as if the collective mass of the system resides on it, and the net force acting on the system acts on it alone.*

Since the same equation (1.5) would also apply if the assembly of points were united at a single point, one may say:

*The motion of the center of mass of a system is the same as if the collective mass resides on it, and the net force acting on the system acts on it alone.*

## § 27. Remark on the Applicability of the New Analysis

### § 27. Remark on the Applicability of the New Analysis

With this extremely simple argument everything is presented that in earlier textbooks on mechanics is derived by much more extensive calculational apparatus, such as we find discussed, for example, in Lagrange, *Mécanique Analytique*, on pp. 45–48 and 257–262 of the last edition.<sup>20</sup>—And our discussion would have been even simpler if we could have availed ourselves of concepts and calculational apparatus to be developed in the following chapters.

But the essential advantage of our method is not that of brevity, but rather that each step in the calculation is simultaneously the pure expression of the corresponding conceptual step, whereas in the former methods the concept is relegated completely to the background as a consequence of the introduction of three arbitrary coordinate axes. And in the discussion given here I hope already to have brought into view the advantage of the new analysis, although with each advance of our science it will come more clearly to light, and only after completion of the whole will it become completely clear.

---

<sup>20</sup>That is the first volume, and specifically in the edition of 1811, and in the *Oeuvres de Lagrange* vol. XI. The citation is I partie, section III, § 1, and II partie, section III, § 1.

## Chapter 2

# The Outer Multiplication of Displacements

### §§ 28–30. Product of Translations in Geometry, Preliminary Treatment

**§ 28** We begin with geometry in order to obtain from it an analogy for the way the abstract science must proceed, and at the same time so as to have in mind a clear idea of the unfamiliar and often difficult course along which the abstract science leads us. We proceed from the displacement to a spatial structure of higher order if we let the whole displacement, that is each of its points, describe a new displacement dissimilar to the first, so that all its points construct equal displacements. The area so generated has the form of a parallelogram. We now assign two such areas belonging to the same plane the same sign if, in passing from the direction of the moved displacement to the direction of that constructed by the motion, both inflect to the same side (for example, both to the left), but opposite signs if they are oppositely inflected. Thus we obtain the following simple and general law:

*If, in the plane, a displacement is translated by any sequence of displacements, the whole area so generated has the same magnitude as if it had been translated by the sum of these displacements (provided one associates signs to the individual elements of the area in the manner given above).*

Or:

*If in the plane, a displacement is translated between two fixed*

*parallels so that initially it lies on the one and finally on the other, then the whole area so generated always has the same magnitude, along whatever path (straight or broken) it may have moved, so long as one maintains the above sign convention.*

This theorem follows at once from the familiar theorem that parallelo-grams extending from the same base line to a parallel to it have equal areas. How this theorem proceeds is easily seen from the figure (cf. fig. 9). Thus one first regards the infinite straight lines of  $ab$  and  $cd$  as the fixed parallels, and compares the areas resulting if  $ab$  moves either by the displacement  $ac$  or by the broken line  $aec$ , whence by means of the theorem cited inspection of the figure suffices to convince one of that equality. Likewise, if one treats the parallels of  $ab$  and  $ef$  as fixed, and compares the areas resulting if  $ab$  translates on the one hand by  $ae$  and on the other by  $ac$  and then by  $ce$ , then one also easily convinces oneself that the principle is correct in this sense, provided only that one keeps in mind that the areas appearing from the motion of the displacement  $ab$  along the directions of  $ac$  and  $ce$  have opposite signs, so that their sum is the difference of the absolute areas. Thus the principle to be demonstrated flows from repeated applications.



**§ 29** On the same grounds it is immediately clear that the theorem adduced is also valid if one simultaneously interchanges the moved side and that measuring the motion in all parallelograms simultaneously. Thus one has the theorem:

*The area in the plane described by a (translated) broken line equals that of the straight line with the same initial and final points.*

Or:

*The net area described by the translation in the plane of the sides of a closed figure is always zero.*

By means of the concepts discussed in the General Theory of Forms (§ 9), it follows from the theorems of this and the previous paragraphs that that conjunction of the two displacements  $a$  and  $b$ , whose product is the area generated by the motion of the first along the second, is multiplicative, since, as can be shown at once, the relation to addition that defines a conjunction as multiplicative is valid for it. Thus, temporarily adopting the general conjunctive symbol ( $\cap$ ) to denote that method of conjunction and writing the moved displacement first, then according to the previous paragraph

$$a \cap (b + c) = a \cap b + a \cap c$$

and, according to the theorem of this paragraph,

$$(b + c) \cap a = b \cap a + c \cap a.$$

Whence, according to § 9, these are the relations defining a conjunction as multiplicative. We will state the particular characteristic of this multiplication and establish its conventions for nomenclature and notation in the rigorous scientific presentation.

**§ 30** The most elegant justification for the concept of addition as we established it in the previous chapter is provided by the relations presented here. Thus suppose one has an equation whose terms are displacements toward different directions in the same plane, and which would no longer be valid if one were to replace the displacements by their lengths, and thus make the expression algebraic. We could immediately remove this apparent disharmony between geometric and algebraic equations if we translated the whole system of those displacements in that plane and inserted the resulting areas into the equation or, expressed otherwise, if we multiplied the equation by a displacement in that plane. As we have just shown, the equation considered also holds algebraically for the areas thus resulting, provided one observes the given sign convention. It is also clear that only now, because the areas are similar to one another as parts of one and the same plane, can the concept of algebraic addition be applicable.

That apparent disharmony persists if the displacements do not all lie in the same plane, since the areas resulting from translation belong to different planes and thus must themselves be regarded as dissimilar. Obviously this dissimilarity is removed if one moves the collection of areas in yet another direction and considers the volumes so arising, these being similar to one another, as belonging to the same single infinite space.

## A. Theoretical Development

### § 31. Generation of Extensions of Higher Order

**§ 31. Generation of Extensions of Higher Order** In chapter 1 we considered extensions as they resulted by elementary generation from the element; and we thoroughly investigated the conjunction of these extensions to the extent that it resulted once more in an extension of the same species, that is, one that is again obtainable by an elementary generation from the element. We have established that this conjunction is to be interpreted as addition or subtraction. Further development thus demands the generation of new species of extensions.

The nature of this generation follows at once by analogy with the way the extension of first order was generated from the element, since in the same way one can again subject the collection of elements of a displacement to another generation. Thus the demand that the newly generated magnitude be elementary requires the equivalence of the method of generation for all elements, that is, that all the elements of that displacement  $a$  describe the same displacement  $b$ . The first displacement  $a$  thus appears here as the *generator*, the second  $b$  as the *measure of generation*; and if  $a$  and  $b$  are dissimilar, the product of the generation is part of the system of second order defined by  $a$  and  $b$ , and must therefore be conceived as an extension of second order.

Now if, as the scientific method demands, we want the extension of second order to have the same relation to the system of second order as the extension of first order to the system of first order, then to begin with the system of second order must be conceived as elementary, that is, as composed of similar parts, and in this sense the extension of second order is interpreted as one of the parts forming this system and again as one of the parts included in it. It then follows that two extensions of second order belonging to the same system appear as similar and thus, since they are generated in the same sense, they have as sum their union into a single whole. Until we have more precisely defined the nature of this conjunction we denote the resulting product of  $a$  and  $b$  by  $a \cap b$ , and mean by  $a \cap b$ , where  $a$  and  $b$  are displacements, that extension generated when each element of  $a$  generates the displacement  $b$ ; and this extension is interpreted as one among the similar parts of the system of second

order. We now extend this definition to arbitrarily many factors, and tentatively mean by  $a \cap b \cap c \cdots$ , where  $a, b, c, \dots$  are arbitrarily many, say  $n$ , displacements, that extension resulting when each element of  $a$  generates the displacement  $b$ , all elements of the resultant generate the displacement  $c$ , and so on, and require this extension to be similar to all other parts of the same system of second order. We call the extension so generated an *extension of  $n$ -th order*.

## § 32. Extensions of Higher Order as Products

**§ 32. Extensions of Higher Order as Products** That the extensions of  $n$ -th order are regarded as mutually similar, insofar as they belong to the same system of  $n$ -th order, implies for their sum the validity of the concept that we established in § 8 for the sum of similar forms, that is, that if the similar form is generated in the same (not the opposite) sense as the whole, then those similar summands comprise its terms. Consequently all laws of addition and subtraction are also valid for this conjunction of similar extensions. In order to understand the relation to addition of the new method of conjunction presented in the preceding paragraph, we next take into account the addition of similar magnitudes.

It follows directly that, if  $A$  and  $A_1$  are two similar extensive magnitudes of arbitrary order that are in fact generated in the same sense, and if  $b$  represents a displacement, then it is always true that

$$(A + A_1) \cap b = A \cap b + A_1 \cap b,$$

where  $A \cap b$  and  $A_1 \cap b$  are similar and where the conjunctive symbol represents the new type of conjunction. Since  $(A + A_1) \cap b$  signifies the collection of elements produced when each element of  $A$  and of  $A_1$  generates the displacement  $b$ , or equivalently when each element of  $A$  generates the displacement  $b$  and likewise for  $A_1$ , the net result does in fact equal  $A \cap b + A_1 \cap b$ .

Likewise, it follows that

$$A \cap (b + b_1) = A \cap b + A \cap b_1,$$

if  $b$  and  $b_1$  are generated in the same sense. For  $A \cap (b + b_1)$  denotes the collection of elements resulting when each element of  $A$  generates the displacement  $(b + b_1)$ , that is when each element of  $A$  first generates the displacement  $b$  and then each of the elements of  $A$  evolved by  $b$  generates the displacement  $b_1$ . First, if each element of  $A$  generates the displacement  $b$ , then the collection of elements so generated is  $A \cap b$ ; thereupon each of the elements of  $A$ , after having been evolved by  $b$ , is to generate the displacement  $b_1$ . Now in § 19 we showed that if all elements of a displacement are evolved by the same amount, the resulting displacement is equal to the original. We can now apply this to extensions of arbitrary order as well, since these are represented as conjunctions of displacements and therefore must be regarded as

equal if the displacements by whose conjunctions they are formed are equal. Thus the extensive magnitude  $A$  will remain equal after all its elements have been evolved by  $b$ . Consequently, if all elements of  $A$ , after having been evolved by  $b$ , generate the displacement  $b_1$ , the same extensive magnitude will result as if all elements of  $A$  had directly generated the displacement  $b_1$ ; that is, it will produce the extensive magnitude  $A \cap b_1$ . Thus, in all, the extensions  $A \cap b$  and  $A \cap b_1$  are generated, and their totality equals  $A \cap (b + b_1)$ , that is

$$A \cap (b + b_1) = A \cap b + A \cap b_1.$$

It is clear that this law of relation can be extended to arbitrarily many factors by repeated application. Since according to § 9 this law is the fundamental law of multiplication, we say that the new method of conjunction has the multiplicative relation with respect to the addition of objects that are generated in the same sense. Consequently all laws deduced from this (§ 10) are valid here, and in particular the fundamental law also holds if some of the magnitudes are generated in the negative sense, that is in the sense opposite to the positive. Now we have combined that generated in the same and in the opposite senses under the name “similar” (§ 8), and thus we can say in general that our method of conjunction has the same relation to addition of similar forms as that which multiplication has in relation to addition.<sup>1</sup> Hence, according to § 12, our conjunction is now established as a multiplication, and thus we immediately introduce the multiplicative symbol for it.

From the concept of this method of conjunction as given in the preceding paragraph it immediately follows that “a product in which two factors are similar, or generally in which the  $n$  factors are mutually dependent, that is belong to a system of lower order than the  $n$ -th, is to be regarded as zero”; here also belongs the case in which one of the factors is zero, since on the one hand zero can always be thought of as dependent, and on the other the product formed with it is zero. But conversely it also follows that “if the factors are mutually independent, the product always has a nonzero value,” since it then represents a definite part of that system of  $n$ -th order.

It remains only for us to show that that relation is also valid for the addition of dissimilar displacements. This demonstration will be the task of the following paragraphs.

---

<sup>1</sup>Cf. in particular § 12, where the equal entry of terms in the conjunction is made the principle of development.

## §§ 33, 34. Fundamental Law of Outer Multiplication

**§ 33** This general relation for two factors is in essence based on the theorem that if  $b$  and  $b_1$  are similar displacements, then

$$(a + b_1) \cdot b = a \cdot b \quad \text{and} \quad b \cdot (a + b_1) = b \cdot a.$$

In order to prove this, let  $a = [\alpha\beta]$ , where  $\alpha$  and  $\beta$  are elements (cf. fig. 10) and  $b_1 = [\beta\gamma]$ , whence  $a + b_1 = [\alpha\gamma]$  according to the definition of the sum (§ 19). Further, let

$$b = [\alpha\alpha'] = [\beta\beta'] = [\gamma\gamma'].$$

Now according to this relation the extension  $[\alpha\beta\beta'\alpha']$  (if we mean thereby the extension bounded by the displacements  $\alpha\beta$ ,  $\beta\beta'$ ,  $\beta'\alpha'$ ,  $\alpha\alpha'$ ) equals  $a \cdot b$  and the extension  $[\alpha\gamma\gamma'\alpha']$  equals  $[\alpha\gamma] \cdot b$ , which is to say  $(a + b_1) \cdot b$ , so that the equality of these two extensions is to be shown.

Because of the assumed similarity of  $b$  and  $b_1$ ,  $\beta$ ,  $\gamma$ ,  $\beta'$ ,  $\gamma'$  are elements of the same system of first order, and if we assume that  $b$  and  $b_1$  are also generated in the same sense (§ 8), then  $[\beta\gamma]$  is also generated in the same sense as  $[\gamma\gamma']$ , that is  $\gamma$  lies between  $\beta$  and  $\gamma'$ ,<sup>2</sup> and likewise  $[\beta\beta']$  is generated in the same sense as  $[\beta'\gamma']$ , since this last equals  $[\beta\gamma]$  according to § 19, whence also  $\beta'$  lies between the same two elements  $\beta$  and  $\gamma'$ . These last are thus the outermost of the four. It thereby follows that

$$[\alpha\beta\beta'\alpha'] = [\alpha\beta\gamma'\alpha'] - [\alpha'\beta'\gamma']$$

and

$$[\alpha\gamma\gamma'\alpha'] = [\alpha\beta\gamma'\alpha'] - [\alpha\beta\gamma].$$

Now, however, the extensions  $[\alpha\beta\gamma]$  and  $[\alpha'\beta'\gamma']$  are equal, since the latter results from the former by evolution of all elements by the displacement  $b$ ,<sup>3</sup> and thus according to § 19 all displacements remain equal and hence so also do the extensions of second order since each of them only describes a collection of displacements. The extensions

---

<sup>2</sup>The meaning of the figurative expression used here in the abstract science is probably clear enough.

<sup>3</sup>It is easy to see that this is just the proof for the corresponding geometric theorem translated into the abstract science.

$[\alpha\beta\beta'\alpha']$  and  $[\alpha\gamma\gamma'\alpha']$  are therefore also equal since they appear in the equations in the same way; that is

$$a \cdot b = (a + b_1) \cdot b.$$

This demonstration only encompasses the case for which  $b$  and  $b_1$  are generated in the same sense; in order to show the validity of the same law for the case of generation taking place in the opposite sense as well, let  $a + b_1 = c$ ; then  $a = c - b_1$ , and we obtain

$$c \cdot b = (c - b_1) \cdot b$$

or

$$c \cdot b = (c + (-b_1)) \cdot b,$$

that is, the law just presented is also true if the displacements labelled by  $b$  and  $b_1$  are generated in the opposite sense, and hence in general if they are similar.

In exactly the same way it now also follows that if  $b$  and  $b_1$  are similar, then

$$b \cdot (a + b_1) = b \cdot a.$$

Here, if  $a$  is zero then one has  $b \cdot b_1$  equal to zero, that is, *the product of two similar magnitudes is zero*, just as follows directly from the concept.



**§ 34** The same result can now also be demonstrated if, in a product of several factors, any two sequential factors are expanded out. Thus, since equals conjoined in the same way with equals again yield equals (§ 1), so also, if  $P$  denotes any sequence of factors,

$$(a + b_1) \cdot b \cdot P = a \cdot b \cdot P.$$

Next it will be shown that the absolute value of the product does not change upon interchange of factors. Thus  $a \cdot b \cdot c \cdots$  means the extension resulting from a single element, taken as origin-element, first generating the displacement  $a$ , then each element of this displacement generating the displacement  $b$ , then each of the elements resulting thereby generating the displacement  $c$ , and so on. All elements of the extension so formed thus

result from the same origin-element by evolutions similar to  $a$ ,  $b$ ,  $c$ , . . . , but whose magnitudes do not exceed them, and the totality of elements so generatable is just that extension. Now since the order in which the evolutions are taken (§ 17) is unimportant for the result we seek, starting from the same origin-element any sequence of the factors  $a$ ,  $b$ ,  $c$ , . . . always produces the same totality of elements comprising the extension; that is, all such products have the same absolute value. Thus the laws obtained earlier for the first two factors of such products are also true for the other two factors, since only the corresponding algebraic sign remains to be chosen. The algebraic sign can only be chosen arbitrarily to the extent that it is not yet specified by definitions. Now in the same way as previously we could only choose the sign so that it is consistent with  $(a + bi).b = a.b$  and  $b.(a + = b.a$ , we also retain this same sign convention if arbitrarily many factors precede. Thus not only the absolute value but also the sign must be in accordance with  $P.(a + bi).b = P.a.b$  and  $P.b.(a + bi) = P.b.a$ , where  $P$  represents arbitrarily many factors. Since the same result also holds if any number of factors follow, for this particular type of multiplication we have obtained the law that “if a factor includes a summand that is similar to one of the adjacent factors, then that summand can be dropped,” which already incorporates the result that if two adjacent factors are similar, the product is zero. This law, in combination with the general multiplicative relation to addition for forms, fixes all further laws regarding the particular type of multiplication we consider here, and thus can be taken as its fundamental law. We call this type of multiplication outer, and take as its particular symbol the point (period), retaining simple juxtaposition as the general sign for multiplication\*.

## §§ 35, 36. Principal Laws of Outer Multiplication

**§ 35** From this fundamental law and the law of relation we derive the remaining laws of this multiplication in a purely formal way.<sup>1</sup>

By combination of the two one has that  $P.(a + a, + bi).b.Q = P.(a + al).b.P = P.a.b.Q + P.arb.Q = P.a.b.Q + P.(a, + bi).b.Q$ , where  $P$  and  $Q$  denote any series of factors, while  $al$  and  $b$ , are displacements

---

<sup>1</sup>(1877) Here I have retained the symbol chosen in the first edition. In the *Ausdehnungslehre* of 1862 and in my later works I have chosen brackets,  $[]$ , for outer multiplication, as generally with respect to a principal domain.

similar to  $a$  and  $b$  respectively. Now since  $a, + b$ , can stand for any displacement lying in the system of second order specified by  $a$  and  $b$  (according to the concept of these systems\*), then so long as  $a, b, c$  belong to the same system of second order one has  $P.(a + c).b.Q = P.a.b.Q + P.c.b.Q$ ; that is, the general relation of multiplication to addition holds for this case as well. From this it follows at once that  $P.a.b.Q = -P.b.a.Q$  or that one may only interchange adjacent factors of an outer product with a simultaneous change of sign, if these two factors are displacements. Thus, since  $P.(a + b).(a + b).Q = 0$ , the two adjacent factors being similar, upon applying the law just proved and using the fact that  $P.a.a.Q$  and  $P.b.b.Q$  are zero, one obtains  $P.a.b.Q + P.b.a.Q = 0$ , that is  $P.a.b.Q = -P.b.a.Q$ . I will examine this remarkable result more fully only later on, in order to go over the important inferences that flow from this interchange law right away. From it follows that, if an elementary factor (as we call a factor representing an extension of first order or a displacement) jumps over two such factors, the product retains the same sign, since the double inter- change of sign leads back to the same sign again, whence in general if an elementary factor jumps over an even number of elementary factors the overall sign is unchanged, but for an odd number it must be changed to its opposite if the whole expression is to keep the same value. The law that holds for two adjacent factors must therefore also hold for two separated <sup>2</sup>

factors; thus one can bring one of the factors up to the other, whence the sign either changes or not, according as the factor jumps over an odd or an even number of elementary factors, apply the law for two adjacent factors, and then in all products return those factors to their old places, whence the sign must obviously become as it was originally.\* Thus if any two elementary factors of a product are sums of terms belonging to the same system of second order, the law of the relation of multiplication to addition holds; and since, if two elementary factors are similar, their product is zero according to § 33, it follows that terms similar to the remaining factors can be dropped from or added to one of the factors without changing the product. Further, as already follows from the concept according to § 32, one immediately has that the product of  $n$  displacements that are dependent on each other is zero; for one of them must be expressible as a sum of terms that are similar to the others; and, according to

---

<sup>2</sup>Cf. § 16 (and § 20).

the theorem just proved, one can then drop these in the product and replace the sum by zero, whence the product itself is zero.

**§ 36** From the fundamental law of the previous paragraph follows the general theorem: If in a product of  $n$  elementary factors, one is expanded out so that all factors are terms belonging to the same system of  $n$ -th order, the multiplicative relation to addition still holds. For let  $a.b. \dots (p + q)$  be this product, in which all of the  $(n + 1)$  displacements  $a, b, \dots, p, q$  belong to the same system of  $n$ -th order. We first assume that one term of the last factor plus the remaining collection of factors represent  $n$  independent displacements, that is, that they do not belong to a system of lower order (than the  $n$ -th). Assuming this term of the last factor is  $p$ , then according to § 20 it must be possible to express  $q$  as a sum of terms similar to those displacements, thus;  $q = a_i + b_i + \dots + p_i$ , where  $a_i, b_i, \dots, p_i$  are similar to the displacements  $a, b, \dots, p$  respectively. Since  $a_i, b_i, \dots$  are similar to the remaining factors of the product  $a.b. \dots (p + q)$ , they can be dropped from the product:  $a.b. \dots (p + q) = a.b. \dots (p + p_i)$ ,<sup>3</sup>

and since  $p$  and  $p_i$  are similar, according to § 32 this is  $= a.b. \dots p + a.b. \dots p_i$ . Or, since the summands  $a_i + b_i + \dots$  can again be included with the factor  $p_i$ , we can replace  $p_i$  with  $q$  and so obtain  $a.b. \dots (p + q) = a.b. \dots p + a.b. \dots q$ . So far the validity of this equation is only proved for the case that  $a, b$ , and one of the displacements  $p$  or  $q$  are independent of each other. If on the other hand  $a, b, \dots$  are mutually dependent, or if they are independent but both displacements  $p$  and  $q$  are dependent on them, then both sides of the equation are zero, since the product of dependent displacements is zero; thus the equation also holds for this case and hence in general, provided all the displacements in that product of  $n$  factors belong to the same system of  $n$ -th order. Since however only in this case are the terms on the right side similar, and for higher orders the concept of addition is only established for similar summands, we have completely verified the multiplicative relation to addition of our method of multiplication, insofar as this is defined conceptually; and thus all the laws of this relation (cf. § 10) apply here. If later

---

<sup>3</sup>If it was not changed before, then it is also not changed now, since the factor again jumps over the same number of factors; but if it was changed before it is again changed now (on the same grounds), and thus it is again as it was originally.

on a broader definition of addition should appear, then such a conjunction would not be fully established as addition until its additive relation to the previously exhibited multiplication is also verified. I have already noted above (§ 34) that we call the product to which we are led here the outer product, since by this nomenclature we mean that this type of product only represents a new extension and has a nonzero value insofar as the factors diverge from one another, and becomes zero if the factors remain together.\* We can summarize the results of this discussion in the following theorem: If by the outer product of  $n$  displacements one means the extension of  $n$ -th order that is generated when each element of the first displacement generates the second, each element so generated generates the third, and so on, in such a way that each extensive magnitude of  $n$ -th order is interpreted as one among the similar parts of the system of  $n$ -th order to which it belongs, then for it, provided only products of  $n$  factors within the same system are considered, one has all the laws expressing the relation of multiplication to addition, and also the law that elementary factors are only interchangeable with a change of sign.<sup>4</sup>

## B. Applications

### §§ 37-40. The Sign Change Law in the Interchange of Spatial Factors

**§ 37** We have now presented in full the connection between multiplication and the previously established concept of addition, and so we proceed next to applications. We have anticipated the application of the principal theorems to geometry previously, in §§ 28-30. There remains the transposition of the nomenclature and notation just introduced to that presentation. Thus the area of the parallelogram now appears as the outer product of two displacements if one simultaneously takes into account the plane to which they belong, and likewise the volume of the parallelepiped is the outer product of three displacements without the need of introducing a further condition, since space is always one and the same. The former two displacements form the sides of the parallelogram and the latter three

---

<sup>4</sup>I have indicated in the Foreword how this outer product contrasts with the inner product.

the edges of the parallelepiped, where for the former we take the displacement by whose motion the parallelogram is produced as the first factor, and that measuring the motion as the second, and regard the signs of two parallelograms as equal if their second factors lie on the same side of their first and as of opposite sign if they lie on opposite sides. This already implies the law  $a.b = -b.a$ , since if  $b$  lies to the left of  $a$  then  $a$  must lie to the right of  $b$  and vice versa. In order to get a more intuitive basis for this interchange law; which distinguishes the multiplication presented here so remarkably from the usual one, I will also deduce the more general sign law that includes it as a special case. First, it is clear from the concept of the negative that, however the two sides may be otherwise enlarged or reduced, if the base and the elevation\* of a parallelogram maintain the same directions,\*\* then its area will maintain the same sign. Furthermore, if the final point of the elevation translates along a line parallel to the base, or if the final point of the base translates along a line parallel to the elevation, while in either case the <sup>5</sup>

other side remains fixed, the parallelogram maintains the same plane area and thus the same sign. We proceed from these two requirements to produce the geometric foundation for the general sign law. First, it is clear that with the given evolutions of the elevation always remaining on the same side of the base, that is if one first progresses in the direction of the base and then in the direction of the elevation, the parallelogram so evolved must be inflected in the same way as the original. Since one can bring the elevation, as well as afterward the base, into any arbitrary positions (so long as they do not coincide) by using only those evolutions that do not change the sign, the elevation will always lie on the same side of the base, and finally, since one can arbitrarily enlarge or reduce them without changing the sign if one keeps their directions fixed, it then follows that all parallelograms whose elevations lie on the same side of their bases must have the same sign. Conversely, that those parallelograms in which the elevations lie on the opposite sides of their bases also have oppositely signed areas follows at once by the proof just given if it has been shown for any two; but for  $a.b$  and  $a.(-b)$  it follows at once from the concept of the negative. The general sign law is thus also completely proved in a purely geometric manner. For

---

<sup>5</sup>1 introduce this name, in default of a better, to label the side (that is, the second factor) adjacent to the base. Opposite directions are naturally not reckoned as equal.

parallelepipeds we can formulate the law in a completely analogous way if we distinguish their first, second, and third edges: The volumes of two parallelepipeds are equally or oppositely signed according as (expressing it figuratively) on imagining one's body placed along the first edge (feet toward the initial point, head toward the final point), in turning from the direction of the second side toward that of the third, one turns in the same or in the opposite sense.

**§ 38** In order to get an even clearer idea of this law, we consider the problem: "To transform a parallelogram into another equal in area (and sign), whose base is given (in the same plane), but is not parallel to that of the given parallelogram." Let  $a/3$  be the base and  $ay$  the elevation of the given parallelogram,  $a8$  the base of the one sought (cf. fig. 11a). From  $a$  one draws the parallel to  $pa$ , from  $y$  that to  $a/3$ , and calls the intersection of the two  $e$ ; then  $ae$  is the elevation of a parallelogram that suffices to solve the problem: for  $[aP] \bullet [all] = [aP] \bullet [ac]$ , since  $ye$  is parallel to  $a8$ , and  $[4].[ae] = [a6].[ole]$ ,

Fig. 11a. Fig. 11b. since  $f33$  is parallel to  $aE$ , whence also  $[a8].[ae] = [aP].[ay]$ . If one wishes to have the complete set of parallelograms that solve the problem, one draws the parallel to  $a8$  from  $e$ , and lets the point  $E$  on this parallel vary. If one uses this solution for the case that the base of the parallelogram sought is identical to the elevation of the one given, then by pure construction one arrives at the formula  $a.b = -b.a$ . In fact 3 then coincides with  $y$  (cf. fig. 11b), and one draws the parallel to  $a/3$  from  $e$  that intersects  $a8$  at  $e1$ , thus easily convincing oneself that  $[(red = [3a] = -[4]$ . The above solution thus shows that  $[a3] \bullet [aY] = [aY] \bullet [ae] = [aY] \bullet [aei]$ ; thus instead of  $[ae]$  substitute its equivalent value  $-[ag]$ , and associate the negative sign with the whole product:  $[ali] \bullet [aY] = [aY].(-[4]) = [aY] \bullet [aQ]$ . Since one cannot overemphasize the law of sign change accompanying the interchange of the factors in an outer product, as it seems to clash with customary notions, I will suggest an analogy, which however must here be understood as a digression. Thus one can express the area of the parallelogram  $a.b$  as  $a.b = a b \sin(ab)$  if  $(ab)$  denotes the angle enclosed by  $a$  and  $b$ , and the lengths of the displacements  $a$  and  $b$ . Similarly

$b.a = b a \sin(ba)$ . The product of lengths is as usual, that is  $a b = b a$ . Now since the angles  $(ab)$  and  $(ba)$  have opposite signs, and the sines of angles with opposite signs likewise have opposite signs,

that is  $\sin(ab) = -\sin(ba)$ , it thereby follows that  $a.b = -b.a$ .

§ 39 The discussion presented here does not contradict the description of the rectangle as the product of the lengths of its sides, provided one retains only the lengths of its sides, relative to any common unit of measurement, as factors in the product, and means only that the absolute (sign independent) area of the rectangle contains as much of the square of this measure as amounts to the product of these numbers. If however one wishes to express more thereby, and thus to maintain that the area of the rectangle as such, that is including its sign, can be set equal to the product of those sides, and retain for this product exactly the characteristics of the algebraic product (as has always been done previously), then this stands in manifest contradiction to the truths just demonstrated. Rather, the parallelogram (and thus also the rectangle) is in an essential way such a product of its sides as that for which the interchange of its factors can only occur with an accompanying sign change. Moreover, the following example will show how easily this conception helps remove significant difficulties in which even the most distinguished mathematicians have sometimes entangled themselves. In his *Mecanique Analytique*,\* Lagrange quotes a theorem of Varignon,<sup>1</sup> which he uses to connect up the various principles of statics, and which according to him states that “if from any point in the plane of a parallelogram one drops a perpendicular on the diagonal and on the two sides that enclose (comprennent) this diagonal, the product of the diagonal with its perpendicular equals the sum of the products of the two sides with their respective perpendiculars if the point falls outside the parallelogram (hors du parallelograme), or their difference if it falls inside the parallelogram.” As can be shown immediately, this theorem is not true; for the first alternative does not obtain if the point falls outside the parallelogram, but rather if it falls outside the two angular regions formed by the angle enclosed by those two sides and the angle opposite it, whereas the latter <sup>6</sup>

obtains when the point is inside this region. It is tacitly understood that the product here is taken in its usual algebraic sense. If however one now considers those products in more detail, then in fact they represent the areas of the parallelograms that have those two sides and the diagonal as bases, and whose sides opposite those

---

<sup>6</sup>P. 14 of the new edition (I. Partie, section 1, no. 12).



bases go through the given point, according to their absolute values, that is, independent of sign. Now if one retains the signs of these areas the theorem immediately holds in general, without distinction of individual cases, since the area of the one with the diagonal as base is always equal to the sum of the areas of those that have the two sides as bases; and indeed according to our analysis the proof of this theorem is given on the spot. For if  $ao$  is the diagonal of the parallelogram,  $a/3$  and  $ay$  the two sides enclosing it, and finally if  $E$  is the arbitrary point, then  $[a] = [an] + [ay]$ , since  $[fl6] = [ay]$ , and thus according to the most elementary multiplication law  $[a8].[ae] = [an].[ae] + [ay].[ac]$ , as was to be shown. If one wishes to express the theorem in terms of absolute areas, one has only to distinguish the case where the point  $e$  lies on the same side of those two sides of the parallelogram from that where it lies on opposite sides, from which the theorem in the improved form given above easily follows.

**§ 40** I will now conclude these applications to geometry with the solution of the above problem (§ 38) for a case not included there, that is to transform a parallelogram into an equal one whose sides are parallel to those of the one given, but for which the length of one side is given. I choose the method provided by our analysis. Let  $a.b$  be the given parallelogram,  $al$  the given side of the one sought, parallel to  $a$ , and  $bl$  that sought parallel to  $b$ , for which the equation  $a.b =$  holds, or, since  $al.b = -bi.at$ ,  $a.b + bi.al = 0$ . Since one can add the term  $bl$  to the factor  $a$  and the term  $a$  to the factor  $1$ ), as in both cases the added term is similar to the corresponding other factor and thus their addition does not change the product, one has  $(a + bL).b + (a + bi).al = 0$

or  $(a + b).(a + b) = 0$ , that is,  $(a + bl)$  and  $(al + b)$  must be parallel. Now implicit in this is the following construction and the proof: If  $a = [a\$]$ ,  $b = [ay]$  (cf. fig. 11c), and  $a = [a]$ , where  $a$ ,  $p$ ,  $8$  lie on a single straight line, then one makes  $Se$  parallel to and as long as  $ay$ , so that  $[ae]$  equals  $(a + b)$ ; from  $/3$  draw the parallel to  $ay$ , which intersects  $ac$  at  $'$ ; then  $[fifl]$  is the displacement  $bl$  sought. \*  
Fig. I I e.

**§ 41. The Static Moment** In statics and mechanics the concept of the outer product is represented by the concept of the moment. We can in fact define the moment of a force with respect to a point as

an outer product, the first factor of which is the displacement drawn from that point (the reference point) to the straight line along which the force acts, and whose second factor is the displacement representing the force. Thus if  $p$  is the reference point,  $a$  the point of application, that is the point on which the force acts,  $p$  the displacement representing the force, then the moment is  $[Pa] \bullet P$ , whence, as the laws of outer multiplication make clear, it is unimportant for the result what point on the line of action of the force one may introduce instead of  $a$ . Thus if  $/3$  is any other point of the line, then  $[a/3]$  is similar to  $p$ , whence  $[PR] \bullet P = ([pa] + [aP]) \cdot 1 = [Pa] \bullet li$ , since the term  $[a/3]$ , being similar to the second factor, can be dropped <sup>7</sup>

according to § 35. And likewise by the moment of a force with respect to an axis  $po$  is meant the outer product of three factors, where the first factor is the axis taken as a displacement, the second factor is the displacement from any given point of the axis to any given point along the line of action of the force, and the third factor is the force, thus: or alternatively, it is the product of the axis taken as a displacement with the moment of the force with respect to any given point of the axis; whence again, on the same grounds as before, it is unimportant which points on that line are chosen. The moment of a force with respect to a point thus appears as the area of a parallelogram, with respect to an axis as the volume of a parallelepiped, and thus two forces that are equal as displacements have equal moments only if they act along the same straight line. Further, by the collective moment of several forces acting in the same plane, with respect to a point in the plane, we mean the sum of all the moments with respect to the same point, and likewise by the collective moment of several forces with respect to an axis, the sum of all the moments with respect to this axis. Since, according to §§ 25 and 26, force and motion are represented by the same displacement precisely because the force is the supposed cause of the motion and thus is set equal to it, it is already clear without further ado what is meant by the moment of the motion and by its collective moment; but here we remark yet again that (according to § 25) only the motion of a unit mass can be equated to the velocity, and that equal motions thus have equal moments only if they proceed along the same straight line. The following discussion will sufficiently demonstrate how easily all

---

<sup>7</sup>It is understood that one can also solve this problem by a double application of the analysis given in § 38, by taking an auxiliary nonparallel base to assist the demonstration.

general laws of statics and mechanics are now derived by means of our analysis. For the present I only remark that in part 2 of this volume\* we will learn an even simpler expression for the moment, and in the next chapter (§ 57) a generalization of the concept of the collective moment.

### **§§ 42, 43. Theorems on the Collective Moment. Equilibrium of Rigid Bodies**

**§ 42.** The principal theorem for the application of the concept of the moment is that the collective moment of all internal forces is zero with respect to any <sup>8</sup>

arbitrary axis and any point; but here we only prove the latter if all lie in the same plane.\* As is known, one means by internal forces those that correspond pairwise in the sense that the forces of each pair act along the same straight line and are equal and opposed, and we can immediately show that the moment of each such pair is zero with respect to any point and any axis. Thus considering for example those two moments with respect to an axis, which according to the earlier discussion are each the outer product of three factors, the first two factors in both are exactly the same, the first being the common axis, the second the displacement connecting this axis and the common line of action of the forces, while the third factors, representing the force in each case, are equal and opposed, so that the two moments are also equal and opposed and hence their sum is zero. Now since the collective moment of the individual pairs of internal forces is zero, so also is that of all the pairs, that is, that of all the internal forces is zero. In precisely the same way as we have shown it with respect to an axis it is established with respect to a point as well, provided all lie in the same plane; whence we permit ourselves to omit this proof.

**§ 43.** Now since the motion communicated to a given point is always equal to the force communicated to it, the collective moment of the motions communicated in a given time interval to an assembly of points will also equal the collective moment of the forces communicated to it during this time, and with respect to any arbitrary axis and, if the forces lie in the same plane, also with respect to any

---

<sup>8</sup>§ 120.

point of the plane. This law, which appears here in such a simple form, is of the greatest generality and is easily applicable everywhere. For example, should equilibrium obtain, then the communicated motions must total zero, as also their collective moment; one therefore has for equilibrium the constraint that the collective moment of the communicated external forces with respect to any axis must be zero. Thus also for rigid bodies in particular, for which the forces maintaining the rigid state are internal. If however the rigid body is fixed to a point or to a line about which it can rotate freely, then the force by which that point or that line is itself maintained in its fixed position is external; however it is only interpreted as producing resistance, and thus must initially be regarded as an unknown. One therefore has to produce that unknown force in order to find the constraint equation for equilibrium. This can be done most easily by means of our analysis.<sup>9</sup>

Thus if  $a$  is the fixed point and  $x$  the force generating the resistance that keeps the point fixed, then one must choose the axis  $pa$  with respect to which one takes the moment equation so that the moment of the force  $x$  disappears, that is, for any arbitrary value of  $x$ ,  $[po \bullet] \cdot [o-a] \cdot x = 0$ , whence  $[por] \cdot [va] = 0$ . Thus the axis  $pa$  must go through the point  $a$ . We therefore have as the only condition under which such equilibrium can occur that the collective moment of the externally acting forces with respect to any axis passing through the fixed point must be zero. Should an axis of the body be fixed, one can assume two fixed points and thus two resistive points that must drop out if the axis, with respect to which the moment equation is taken, goes through both of these points simultaneously; one therefore has as the only condition under which this equilibrium can occur, that the collective moment of the externally acting forces with respect to the fixed axis must be zero.

**§ 44. The Interchange Law Corroborated by Statics** In the concept of the moment we also have a beautiful corroboration of the law that in a given plane the outer product of two displacements maintains its sign as long as the second factor lies on the same side of the first, whereas in the opposite case it changes. For considering forces in a plane that is imagined to be able to pivot about a point, the forces will be increased if they act to the same side of a line

---

<sup>9</sup>The proof for the general case follows in § 57.

from the pivot point, whereas completely or partially neutralized if oppositely; thus that concept of the outer product is ultimately corroborated by the concept of the moment according to which nature itself proceeds. I believe that the initial surprise of the sign law has now completely vanished as a consequence of the whole series of considerations we have employed in the most • varied types of relations, and on the contrary it appears not only as conceptually necessary but also as fully corroborated by nature and having everywhere stood the test.

## §§ 45, 46. Solution of Algebraic Equations of First Degree with Several Unknowns

§ 45. Now it is of course no surprise to us that outer multiplication finds no such immediate applications in number theory as in geometry and mechanics, since it assumes the concept of the different as essential, while numbers are similar in their content. It is therefore all the more interesting to observe that, so long as the nature of the conjunction of number with other

magnitudes is retained (and in this respect the one is interpreted as formally distinct from the other), the applicability of outer multiplication to algebra emerges with such a striking application that I dare claim that by this application algebra as well will assume an essentially altered form. In order to provide some idea of this, I consider  $n$  equations of first degree in  $n$  unknowns, in the form  $a_1x_1 + a_2x_2 + \dots + a_nx_n = a_0$ ,  $b_1x_1 + b_2x_2 + \dots + b_nx_n = b_0$ ,  $\dots$ ,  $s_1x_1 + s_2x_2 + \dots + s_nx_n = s_0$ , where  $x_1, \dots, x_n$  are the unknowns. Here we can regard as dissimilar the numerical coefficients belonging to different equations, inasmuch as we still maintain the concept of this difference, and indeed all of them as intrinsically dissimilar, that is as independent, in the sense of our science—and within the same equation as similar in the same respect. Adding all  $n$  equations in this sense and denoting the sum of dissimilars by the conjunctive sign +', since the same positions in the summed expression thus formed always belong to similars, we obtain  $(a_1 + 'b_1 + ' \dots + 's_1)x_1 + ' (a_2 + ' b_2 + ' \dots + ' s_2)x_2 + ' \dots + ' (a_n + ' b_n + ' \dots + ' s_n)x_n = (a_0 + ' b_0 \dots + ' s_0)$ , or, on denoting  $(a_1 + ' b_1 + ' \dots + ' s_1)$  by  $p_1$  and correspondingly with the other sums, we have  $P_1x_1 + ' P_2x_2 + ' \dots + ' P_nx_n = P_0$ . From this equation, which replaces

those  $n$  equations, it is now possible to find each of the unknowns, for example  $x_i$ , if we outer multiply both sides by the outer product of the coefficients of the other unknowns, that is in this case by  $p_2.p_3 \dots p_n$ . Thus if one multiplies the terms on the left side individually, according to the concept of the outer product (§ 32) all products containing equal factors drop out and one obtains  $P_1.P_2 \dots P_n x_i = P_0 \cdot P_1 \cdot P_2 \dots P_n$ . Now the two products are similar as belonging to the same system of  $n$ -th order, whence one has —  $POP_2 \dots P_n \cdot x_i = P_0 \cdot P_2 \dots P_n$  <sup>10</sup>

Thus each unknown equals a fraction, whose denominator is the outer product of the coefficients and whose numerator one obtains if, in this product, one inserts the right side, that is  $p_0$ , as a factor in place of the coefficient of that unknown. All unknowns thus have the same denominator, and are undefined or infinite if this denominator is zero, that is if  $P_1 \cdot P_2 \cdot P_3 \dots p_n = 0$ .

**§ 46.** That these expressions for  $x_i, \dots, x_n$  do not perhaps represent simply calculational forms but rather contain the complete solution of the given equations is even more distinctly illustrated if for a fixed number of equations we substitute in the values of  $p_i, p_2, \dots$ . For three equations one has  $P_0 \cdot P_2 \cdot P_3 (1) x_i = P_1 \cdot P_2 \cdot P_3$  where  $P_0 = (a_0 - F' b_0 - F' c_0)$ ,  $P_i = (a_i - F' b_i - F' c_i)$ ,  $\dots$  where  $a_0$  is similar to  $a$ , and so on. If we substitute these expressions into the above equation and multiply through, then since we can drop the products of similar magnitudes, as they are zero, and correspondingly order the result while observing the sign changes obtained for outer products, we have immediately (as one using minimal dexterity can read off without further ado from the above formula)  $a_0 b_2 c_3 - a_0 b_3 c_2 + a_2 b_3 c_0 - a_2 b_0 c_3 + a_3 b_0 c_2 - a_3 b_2 c_0 (2) x_i = a_1 b_2 c_3 - a_1 b_3 c_2 + a_2 b_3 c_1 - a_2 b_1 c_3 + a_3 b_1 c_2 - a_3 b_2 c_1$ , where we have reintroduced the usual multiplication symbol since everything is correspondingly ordered. This is the familiar formula by which one unknown is specified in a set of three equations in three unknowns, and thus is shown how it is completely subsumed in the much simpler formula (1). In demonstrating the applicability of our analysis (which is not limited to three dimensions) with an example, we have anticipated somewhat here, since the concepts of number and divi-

<sup>10</sup>The laws of outer multiplication and division do not however permit cancellations between numerator and denominator; cf. chapter 4.

sion applied here will only be explicated as the subject of chapter 4; however we will return to the application of this topic when we also extend the method to equations of higher degree.







## Chapter 3

# Conjunctions of Extensive Magnitudes of Higher Order

### A. Theoretical Development

#### §§ 47, 48. Sums of Extensions in a System of Next Higher Order

§ 47 Outer multiplication results in higher extensive magnitudes, but so far we have only considered their conjunctions to the extent of adding similar extensive magnitudes, since in this case addition is based on the general concept of the correlative thought that always characterizes the addition of similars (if they are of the same sign). We have developed the laws presented in the previous chapter by means of this concept. The fundamental law of multiplication, that one can replace a factor expressed as a sum of terms by the individual terms and then add the products so formed, thus has a restriction, since, if the resulting products are to be added according to the above concept, they must be similar. To remove this restriction we must therefore generalize the concept of addition for higher extensive magnitudes. The generalized concept must be such that, first of all, it reduces to the previous law for similar extensive magnitudes, and second, that for it the fundamental relation to multiplication must be valid. It is therefore clear that, if there exists a general

addition for dissimilar extensive magnitudes of higher order, the law must be  $A.b + A.c = A.(b + c)$ , where  $b$  and  $c$  represent displacements. Tentatively calling this conjunction an addition, we can formulate the definition: One adds two outer products of  $n$ -th order having a common factor of  $(n - 1)$ -th order when one adds the

dissimilar factors and combines this sum with the common factor in the same way as were its terms.

§ 48 First we must provide this formal definition with a more intuitive significance in order to analyze its extent, that is, which extensive magnitudes one may add in accordance with it. It is immediately clear that two extensive magnitudes of  $n$ -th order are only summable according to the concept formulated if they belong to the same system of  $(n + 1)$ -th order; we will however show that they are then always summable, since any two extensive magnitudes of  $n$ -th order  $A_n$  and  $B_n$ , belonging to the same system of  $(n + 1)$ -th order, can always be brought to within a common factor of  $(n - 1)$ -th order. This is clear at once if  $A_n$  and  $B_n$  are similar, since if  $(n - 1)$  elementary factors of  $A_n$  remain constant but the  $n$ -th is arbitrarily transformed by progression or regression, the product will thereby assume every arbitrary value similar to  $A_n$ , and therefore the value  $B_n$ . At the same time, this implies that one can introduce into each extension of  $n$ -th order  $(n - 1)$  arbitrary factors that belong to the same system of  $n$ -th order and are mutually independent. If  $A_n$  and  $B_n$  are dissimilar, then let  $A_n = a_1 a_2 \cdots a_n$ , where  $a_1, \dots, a_n$  represent mutually independent displacements. Then  $B_n$  must contain at least one factor independent of the collection of displacements  $a_1, \dots, a_n$ ; let  $a_{n+1}$  be such a factor, so that  $B = b_1 b_2 \cdots b_n \cdot a_{n+1}$ . Since there can be no more than  $(n + 1)$  mutually independent displacements in a system of  $(n + 1)$ -th order, it follows that each of the factors  $b_1, \dots, b_n$ , must depend on the displacements  $a_1, \dots, a_n$ , that is, they can be expressed as sums whose terms are similar to those displacements. Now, imagining that each of these factors is represented as such a sum, in each of them one can delete the term similar to  $a_{n+1}$ , without thereby

changing the product  $B_n$  (cf. § 35). If, after this deletion, the product  $b_1 b_2 \cdots b_n$  changes into  $C_{n-1}$ , then  $B_n = C_{n-1} a_{n+1}$ . The factors of  $C_{n-1}$  can only depend on the displacements  $a_1, \dots, a_n$ , that is, on the factors of the extensive magnitude  $A_n$ ; in other words, it belongs to the system  $A_n$ , and consequently, according to the result used at the beginning of this paragraph, the factor  $C_{n-1}$  can be introduced into  $A_n$  if the  $n$ -th factor can be chosen arbitrarily. It is therefore possible to introduce the common factor  $C_{n-1}$  (which is of  $(n - 1)$ -th order) into the two extensive magnitudes, or as we

can say for short, the two have a common extensive magnitude of  $(n - 1)$ -th order. The above definition can therefore be transformed into Two extensive magnitudes of  $n$ -th order belonging to the same system of  $(n + 1)$ -th order are added when one brings them to within a common factor of  $(n - 1)$ -th order and conjoins the sum of the dissimilar factors with this common factor.

## §§ 49, 50. Validity of the Addition Laws for this New Sum

**§ 49** Now in order to demonstrate the validity of the addition laws, or rather initially only that of the fundamental laws, we must first prove the interchangeability of the terms. According to the previous paragraph these terms can be represented in the forms  $A.b$  and  $A.c$ . Now  $A.b + A.c = A.(b + c) = A.(c + b) = A.c + A.b$ , whence they are in fact interchangeable. The second law whose validity must be demonstrated is that  $(A + B) + C = A + (B + C)$  if  $A, B, C$  are extensions of  $n$ -th order in the same system of  $(n + 1)$ -th order, and addition is to have the same concept as before. To this end we must answer the question, what do three such extensions have in common? Now it was already shown in the preceding paragraph that each pair of them must have an extension of  $(n - 1)$ -th order in common; thus for example  $B$  has one in common with  $A$  and with  $C$ ; and since these two extensions, that is the one  $B$  has in common with  $A$  and the one it has in common with  $C$ , belong to the same system  $B,^*$  that is to the <sup>1</sup>

same system of  $n$ -th order, then according to the same theorem of the preceding paragraph they have a common extension of  $(n - 2)$ -th order, and this is therefore common to all three extensions  $A, B, C$ . Let  $D$  be this common factor of  $(n - 2)$ -th order; then those three magnitudes, each pair of which in addition has in common an extension of  $(n - 1)$ -th order, can be brought into the forms  $A = D.b.c$ ,  $B = D.a.c$ ,  $C = D.a.b1$ . Thus each pair have besides  $D$  a common factor of first order, whose magnitude is, however, arbitrary. Between  $A$  and  $B$  we let this factor be  $c$ , between  $B$  and  $C$  we let it be  $a$ , and indeed we let the magnitude of  $a$  be such that  $B = D.a.c$ ; the common factor, besides  $D$ , to within which  $A$  and  $C$  can be brought is then the factor  $b$  or  $b1$ , a factor similar to  $b$ ; and we let

---

<sup>1</sup>We label the system the same as the extension that is a part of it where there is little likelihood of ambiguity.

$b$  and  $b1$  be chosen so that  $A = D.b.c$  and  $C = D.a.b1$ . Now, having brought  $A, B, C$  into these forms, it can be shown that  $(A + B) + C$  can be changed into  $A + (B + C)$  by the following transformations. First,  $(A + B) + C = (D.b.c + D.a.c) + D.a.b1$ . We must now carry out the sum indicated for the terms enclosed in parentheses. Now the expression  $D.b.c + D.a.c$  can be reduced to  $D.(b + a).c$ ; for one can first bring  $c$  to the next to last place in both summands, thus changing their signs, then carry out the sum according to the definition, and finally return the summed factor to the original place again with the same sign change, and so obtain  $(A + B) + C = D.(b + a).c + D.a.b1$ . Now in order to be able to sum these terms one must replace  $D.a.b1$  with  $D.(b + a).b$ , which is allowed since  $b$  is similar to  $b1$  and one is permitted to introduce terms that are similar to the other factors without thereby changing the result (§ 35). Thus, upon carrying out the summation on the right side, one obtains  $(A + B) + C = D.(b + a).(c + b1)$ , whence one has reduced the three terms to one.\* In this term one can <sup>2</sup>

again resolve the sum  $b + a$  and on the right side obtain the expression  $D.b.(c + b1) + D.a.(c + b1)$ . In the first term of this expression one can again (§ 35) drop the term  $b1$  and in the second, resolve the terms; the whole expression thereby transforms into  $D.b.c + (D.a.c + D.a.bi)$ , that is, into  $A + (B + C)$ , whence one has in fact  $(A + B) + C = A + (B + C)$ .

**§ 50** Now the third fundamental law (§ 6) must be proved, that is that the result of subtraction is unique, or that if one term is changed while the other is unchanged, the sum must also be changed. In a system of  $(n + 1)$ -th order let  $A + B = C$ , where  $A, B, C$  are of  $n$ -th order. If  $B$  is changed to  $B + D$ , then  $A + (B + D) = (A + B) + D = C + D$ , and it is to be shown that if  $B + D$  is different from  $B$ , then also  $C + D$  must be different from  $C$ . The former assumes that  $D$  is not zero; now however we can show that if  $D$  is not zero, then its combination with another magnitude ( $C$ ) must change the value of the latter. This is clear at once if  $C$  and  $D$  are similar, since what results from the correlative thought of similars is necessarily different from each of the terms. If however  $C$  and  $D$  are dissimilar,

---

<sup>2</sup>One can now show that the expression  $A + (B + C)$  can be reduced to the same term; but we continue nonrepetitively with the progression of transformations.

then it is easy to show that their sum is dissimilar to both (always assuming that neither of them is zero). Since all terms are assumed to be in the same system of  $(n + 1)$ -th order, it is possible to bring  $C$  and  $D$  to within a common factor of  $(n - 1)$ -th order. If this is  $E$  and  $C = E.c$ ,  $D = E.d$ , then  $C + D = E.(c + d)$ . Now if  $C$  and  $D$  are dissimilar, then  $d$  cannot be included in the system  $E.c$ , whence also  $(c + d)$  cannot be included in it, and thus  $E.(c + d)$  must be dissimilar to  $E.c$ , that is they cannot be equal. Consequently the magni-

tude  $C$  will also change upon being supplemented by the magnitude  $D$ ; thus if one term of that sum is changed while the other remains the same, then the sum must also be changed. Consequently, should the sum and one of the terms remain unchanged, so also must the other, that is, the result of subtraction is unique. Since all three fundamental laws of addition and subtraction are now valid here, so also are all the other laws. The fundamental relation of addition to multiplication has not yet been fully developed. For although according to the definition we have  $A.b + A.c = A.(b + c)$ , it must also be shown that  $(A + B).c = A.c + B.c$  if  $A$  and  $B$  are magnitudes of  $n$ -th order in a system of  $(n + 1)$ -th order. However (according to § 48), one can set  $A = E.a$ ,  $B = E.b$  so that  $A.c + B.c = E.a.c + E.b.c$ . Consider the expression on the right. If one brings  $a$  and  $b$  to the final position (whereby the sign is changed), then adds according to the definition, and finally returns the factor  $(a + b)$  to the next to last place (whence the sign is restored to its original value), it is changed to  $E.(a + b).c$ , which is to say, into  $(A + B).c$ , thus showing that the equation is correct. Consequently, since the fundamental laws of the relation between addition and multiplication are valid here, so also are all laws of this relation, and our method of conjunction is therefore indeed established as a true addition in all its relations. We can therefore now extend the main theorem of the previous chapter (§ 36) thus: For outer products, regarded only as products of  $n$  elementary factors in a system of  $(n + 1)$ -th order, all laws of addition and subtraction are valid, as are all laws of the relation between them and multiplication, if one maintains the concepts established for these conjunctions.

**§ 51. Formal Sum or Additive Magnitude“** This law as well still has a fundamental restriction, since so far we are only able to

add higher extensions if they belong to a single system of the next higher order. To express the law in its generality we must now also show what is meant by the sum of extensions lying in arbitrary higher systems. If one attempts to follow the same method as in the preceding paragraph, and thus conceive the sum of two magnitudes  $A.B$  and  $A.C$ ,

which do not belong to the same system of higher order, as the magnitude  $A.(B + C)$ , this would lead nowhere, since then  $B$  and  $C$  are also extensions of higher order not belonging to a single system of next higher order, so that the significance of their sum is just as mysterious as the other. Consequently there is nothing left for us but to interpret the concept of the sum purely formally in this case, without it being possible to exhibit an extension that is represented as the sum. We therefore define the sum of extensions of  $n$ -th order, which belong to a system of order higher than the  $(n + 1)$ -th, as such that the fundamental laws of addition apply to them; that is therefore as "that which remains constant, whatever changes one may produce in the form of the sum by application of the laws of addition and subtraction." This sum therefore no longer appears as a pure extension, that is as a structure that can be obtained by the progressive multiplication of displacements, but rather as a magnitude of a new type, and indeed initially as a magnitude of purely formal significance, which we can therefore most suitably denote by the name additive magnitude; we combine it with the extension under the name extensive magnitude. In order to obtain its concrete significance we must ascertain its extent, that is seek out how the form of the sum, consisting in the value of the terms, can change without thereby changing the value of the sum itself. Thus we obtain a series of concrete representations of that formal sum, and the collection of these possible representations regarded together as the varieties of a species (and not as parts of a whole) then illustrates the concrete concept. Yet, since these additive magnitudes cannot be introduced earlier than in a system of fourth order, they find no application in space as a system of third order. This representation is thus denied us until chapter 7,\* where the significance of such a sum appears in a related domain, and will take on a fruitful form through observations both in geometry and especially in statics.

## §§ 52, 53. Multiplication of Extensive Magnitudes

§ 52. On the other hand we must not forget our problem, which is to free the law established in this and the previous chapter from all the restrictions that still surround it, and thus to interpret the relation of multiplication to this addition as well. Since however the formal sum represents no <sup>3</sup>

extension, the significance of the outer product of that formal sum with a displacement is also not yet defined. Now this too must be formally defined by continuation of the multiplicative relation, and therefore, if such a multiplication is possible in general, we have to define it so that  $(A + B + C + \dots).p = A.p + B.p + C.p + \dots$ . Yet we may only assert this provided, when  $A + B + C + \dots$  remains constant, that  $A.p + B.p + C.p + \dots$  also remains constant, since the essence of the sum consists in just this constancy, and the principle of equality requires their simultaneous constancy. Thus we have to show that if  $A+B+\dots=P+Q+\dots$  then also  $A.p + B.p + \dots = P.p + Q.p + \dots$ . This however follows easily, since if  $A + B + \dots$  is set equal to the sum  $P + Q + \dots$ , and both are formal sums, then by straightforward application of the addition laws (interchange of order, collection of terms, reduction of terms to fewer terms) the second must follow from the first. Now since each such change, permitted under the condition that it not change the net value, corresponds to a like change incorporating the factor  $p$ , it follows that if one simultaneously undertakes the corresponding operations with the latter and with the former, then when  $A + B + \dots$  is transformed into  $P + Q + \dots$ , so also  $A.p + B.p + \dots$  will be changed into  $P.p + Q.p + \dots$ . Consequently it is acceptable to assert that definition, which according to this argument is no more than an abbreviated notation.

§ 53 Further, since when proceeding with several displacements, that is multiplying so that the result obtained in each case is then multiplied by the next sequential factor, the net product always maintains the same value so long as the product of those displacements maintains its value, we can abbreviate by replacing those displacements, by which it is successively multiplied, by their product. The concept of the product of two extensions is thus defined, and so also is the product of a formal sum with an extension, a product that indeed again produces a

---

<sup>3</sup>That is, chapter 2 of part 2.



formal sum in general, but in certain cases can also be transformed into an extension.\* It now follows easily that, according to this definition, we have in general  $(A + B).P = A.P + B.P$ . For let  $P = c.d \dots$ ; then  $(A + B).P = (A + B).c.d \dots$  according to the definition just established, and further  $(A + B).c = A.c + B.c$  according to § 52, whence by repeated application of the same law  $(A + B).c.d \dots = A.c.d \dots + B.c.d \dots$ , that is,  $(A + B).P = A.P + B.P$ . If the second factor is expanded out, this only permits the corresponding law to be established for real sums; in this case it follows from the above equation by interchange (whereby the sign either changes in all terms or in none) that  $P.(A + B) = P.A + P.B$ . For formal sums nothing has yet been established about the interchangeability of factors, and so this result is also not yet applicable. Since we have as yet asserted nothing in general about the concept of a product whose second factor is a formal sum, for the case that the second factor is in fact a formal sum we are allowed to make the same assumption as for the case in which the first is, and thus to require  $P.(A + B) = P.A + P.B$  and therefore to apply it to the case where  $P$  represents a formal sum as well. <sup>4</sup>

## §§ 54, 55. Principal Laws of Outer Multiplication

**§ 54** We have now removed all restrictions previously present and established the validity of the fundamental multiplicative relation for all extensive magnitudes, in part from its concept and in part by definitions. Consequently all laws of this relation are valid, as also are all laws of addition and subtraction; in this way all the concepts introduced are justified in the most general sense. Having now arrived at the conclusion of this series of discussions, we collect their results in the following theorems: If all elements of an extension (in its elementary representation\*) undergo the same generation, that is if each of its elements is replaced by an equal displacement, the initial element of which is the original element in each case, then the collection of elements so obtained is the concrete representation of an extension which, interpreted as part of the system to which it belongs, is the product of that extension with this displacement, which we

---

<sup>4</sup>Thus if the terms of the sum are of  $n$ -th order and belong to a system of  $(n + m)$ -th order, then upon multiplication by an extension of  $(m - 1)$ -th order of the same system the formal sum would obviously turn into an extension.

call an outer product. Further: If one successively multiplies one extension by the elementary factors of another in the way specified, the result is characterized as the product of the former extension with the latter. The sum of two extensions of  $n$ -th order in a system of  $(n + 1)$ -th order is defined as that extension resulting if one brings those two to within a common factor of  $(n - 1)$ -th order and adds the dissimilar factors. The sum of two extensions of  $n$ -th order in a system of order higher than the  $(n + 1)$ -th is given by the formal sum, which represents that which remains constant under application of the addition law. Finally, the product of an additive magnitude and another magnitude is defined as the sum resulting if each term of the one factor is multiplied by each term of the other, and these products are added. The validity of all these definitions thus shows that the principle of addition and the fundamental relation of multiplication to addition have been proved, since in them is simultaneously implicit the proof that all laws of addition and subtraction and the relation to multiplication of both continue to apply here. <sup>5</sup>

**§ 55** It only remains for us to express the law that characterizes outer multiplication as such in a more general form. In § 34 above we have shown as characteristic of this type of multiplication the law that, if an elementary factor of a product includes a summand that is similar to an adjacent factor, one can drop this summand without changing the value of the product; from this it follows (§§ 35, 36) that the product of  $n$  elementary factors is always zero if and only if they are mutually dependent, that is if they are included in a system of lower order than the  $n$ -th. We can immediately extend this to factors of arbitrary order, since we can then regard several extensive magnitudes as mutually dependent if the sum of their orders is greater than that of the system to which they belong; for then the number of elementary factors contained in their product will be greater than the order of the covering system, whence their product will be zero. Thus: The outer product is zero if the factors are mutually dependent, and has a nonzero value if they are not. From the characteristics of the outer product we found (§ 35) that two elementary factors can be interchanged if one simultaneously changes the sign of their product; we then extended this law,

---

<sup>5</sup>By the elementary or concrete representation of an extension we mean the structure to which this extension belongs.

so that an elementary factor may jump over an even number of elementary factors without, an odd number with sign change. Since a sequence of elementary factors appears as an extension whose order equals the number of those elementary factors, it follows, first, that an extension of even order can jump over an elementary factor, and thus any factor, without sign change, and in addition, that with the interchange of two arbitrary adjacent factors sign change occurs if and only if both are of odd order.\* The law is also valid for additive magnitudes, as is clear since if one multiplies through by the  $A.B = (-1)^{ab} B.A$ . —If the two factors are separated by a third, then the sign of the interchange also depends on that. Thus one has for example  $A.B.C = (-1)^{ab} B.C.A$ . I remark that, once the multiplicative relation to addition is established, the formal notion of outer multiplication could have been fully characterized by the law that two elementary factors are interchangeable with sign change. For if  $a.b$  is in general equal to  $-b.a$ , or  $a.b + b.a = 0$ , then this must also be true when  $b = a$ , whence  $2a.a = 0$ , and thus  $a.a = 0$ . It then follows that the product of two similar displacements is zero in general, hence the law characterizing the concept of outer multiplication as we have presented it above.<sup>6</sup>

individual terms, what is valid for the individual products must be for the sum as well. Thus: Two adjacent factors are interchangeable with or without sign change, according as the orders of the two factors are both odd or not.

## B. Applications

**§ 56. Product of Translations in Space** The laws developed in this chapter admit only a partial application to geometry and statics, since the additive magnitude, which first appears in a system of fourth order, can find no application here. The applications are therefore limited to only the first half of this chapter (§§ 47-50), and consist in freeing the laws, established in the previous chapters for these disciplines, from their previous restrictions and in examining them from a more general viewpoint. First, for geometry we have to transfer the new concept of addition to plane areas. With plane areas we must however retain their directions, that is the directions

---

<sup>6</sup>If  $a$  and  $b$  are the orders of  $A$  and  $B$  respectively, it is possible to express this as

of the planes to which they belong, and thus interpret two plane areas as dissimilar if the planes to which they belong have a difference in direction. Now since the plane areas so interpreted are extensions of second order, two such areas, as simultaneously belonging to the same system of third order (space), can according to § 48 be brought to within a common factor of first order, and thus can be expressed as parallelograms with equal bases. Their sum will therefore be a parallelogram with the same base, whose elevation (cf. § 37) is however the sum of the two elevations of those two parallelograms. One can thereby now express the theorem on translation (§§ 28 and 29) more generally thus: The geometric\* sum of plane areas that a broken line describes with its translation equals the plane area that a straight line, having the same initial and final points as the broken one, describes when it is translated in the same way. Or even more generally, if we call the displacement from the initial to the final point of the broken line its closing side: The geometric sum of the plane areas that a broken line describes with a broken path equals the plane area which the side that closes the first describes in a path that closes the second. <sup>7</sup>

For the motion of the plane area one has the theorem: The sum of the volumes that an arbitrary broken surface describes along an arbitrary broken path equals the volume that the geometric sum of those plane areas (which form the broken surface) describes in the closing of that broken path.

**§ 57. General Concept of Collective Moment** For statics and mechanics the applications of this chapter consist in a generalization which, however, is so fruitful that only now can the whole wealth of relations appear. First of all, the restriction incorporated (§ 41) in the collective moment of several forces with respect to a point now drops out, and we can therefore say that by the collective moment of several forces with respect to a point is meant the sum of all individual moments with respect to that point; and at the same time it is clear that if one draws a displacement as axis through this point, the moment with respect to that axis is found when one multiplies this axis by that first moment. If for example the forces are  $a/3$ ,  $y8$ ,

---

<sup>7</sup>I use this adjective if the magnitudes to be summed are not yet sufficiently labelled as magnitudes with constant direction, in order to distinguish the sum from the pure arithmetic sum.

. . . then their collective moment  $M$ , with respect to a point  $p$  equals  $[Pa] \cdot [a/3] [PY] \cdot [YM + \dots]$ ; and with respect to an axis  $op$  the moment of the same forces equals  $[074 [Pa].[aig] [op] \cdot [PY] \cdot [YS] \pm \dots$ . Or  $[orP] \cdot Mp$ . That the collective moment of the internal forces with respect to an arbitrary point is also zero hardly needs proof here, since it is clear at once that the proof follows in a way similar to, but even simpler than, that given above (§ 42) for the restricted concept. Thus it is clear that the theorems formulated above (§§ 43 and 44) also remain valid in this generalization. In particular, the main theorem formulated in § 43 can now be expressed thus: The collective moment of all motions communicated in a time interval to the individual points (of an assembly of points) equals the collective moment of the collection of external forces communicated to the assembly of these points during that time, and in fact with respect to any arbitrary point.\*<sup>8</sup>

In particular, if no forces act from outside then the collective moment of all motions communicated during that time interval, that is the collective moment of all motions to which the points are subject during that time, must be zero.\* This collective moment thus represents an invariant plane and within it a constant plane area; that plane is what Laplace calls the invariable plane (plan invariable), which in our science can be produced quite easily by summation. The difficulty in the derivation using the methods previously employed is easily appreciated if one simply glances at the discussion presented in Lagrange's *Mecanique Analytique*\*\* or in Laplace's *Mecanique Celeste*, and at the complicated formulae appearing in those works.

## §§ 58, 59. Dependence of Moments

**§ 58** We could of course already formulate the main theorem of the theory of moments here; since however the treatment of moments will appear even simpler in part 2, I will only give a couple of examples now in order to illustrate how easily the relevant problems can be solved and with what abundance the most interesting theorems bubble forth from it, as it were. First, consider the following problem: Given the moment with respect to a point, to find it with respect to another point distant from the first by a displacement of given

---

<sup>8</sup>Later we will exhibit the equations resulting hereby in the application of the differential calculus to our science; cf. § 105.12

length and direction, if in addition the length and direction of the collective force (the sum of all forces represented as displacements) are given. Let  $a$ - and  $T$  be the two points,  $Mc$ , be the given moment with respect to the first,  $M$ , that with respect to the second,  $[4]$ ,  $[y8]$ , . . . be the forces,  $a$ ,  $y$ , . . . their points of application, and  $s$  be the collective force, including length and direction, so that  $s = [4] + [y3] + \dots$ . Then  $Mo = Effal.EaP1 + [0-7].[Ya] + \dots Mr = [Tol] \bullet [ate] + ITYHY5l + \dots$ . Now since  $— Era = [oa] + [aT]$  <sup>9</sup>

and so on, on subtracting one of the moment equations from the other one finds  $MQ - M = [07].[4] + [y8] + \dots = [crris]$ , whence the problem is solved, and one has obtained the theorem: Transporting the reference point by a displacement increases the moment by the outer product of the collective force with this displacement.\* Simultaneously implicit in this is that the moment remains unchanged if that outer product is zero, that is if the reference point progresses in the direction of the collective force, or expressed otherwise, that: The moments with respect to all points that lie on the same parallel to the collective force are all equal. Further: If the moment with respect to a point is zero, so also is the moment with respect to any other point equal to the outer product of the collective force with the difference of the second point from the first.

**§ 59** Another problem that incorporates the dependence of the moments with respect to axes going through the same point is the following: Given the moments with respect to three axes going through the same point but not lying in the same plane, to find the moment with respect to any fourth axis going through the same point. Let  $a$ ,  $b$ ,  $c$  be the three axes,  $A$ ,  $B$ ,  $C$  be the respective moments,  $aa + pb + yc$  be the fourth axis, where  $a$ ,  $b$ ,  $c$  represent numbers,\*\* with corresponding moment  $D$  being sought. If the moment with respect to the intersection of the three axis is  $M$ , then according to § 57  $A = a.M$ ,  $B = b.M$ ,  $C = c.M$ ,  $D = (aa + pb + yc).M$ . If we remove the parentheses in the last expression, then  $D = aa.M + pb.M + yc.M = aA + 13.B + yC$ . Expressing this result in words: <sup>10</sup>

<sup>9</sup>This is the principle of equal areas, as one can easily convince oneself. PP. 262-269 (II partie, section III, § 2, no. 7-11).

<sup>10</sup>Here the word "increase" is taken in the same general sense in which one can also say that 8 has been increased by (-3) if 5 is the result. That each displacement in space can be represented as the sum of three terms that are parallel to three given displacements was shown above (I 20); therein was implicit that it can be

Given the moments of three axes,<sup>13</sup> not lying in the same plane, going through a single point, one can find that of any other axis going through the same point, and in fact between the moments there is the same multiple equation as between the axes.\* If one of the coefficients is zero, then one has the theorem: Given the moments of two axes going through a point one can find that of any other axis going through the same point, and in fact between the moments there is the same multiple equation as between the axes. Later, with the general treatment of moments, we will be able to present this theorem in a much more general form. <sup>11</sup>

---

represented as a multiple sum of them.

<sup>11</sup>Briefly we say that between magnitudes there exists a multiple equation if the terms of the equation only represent multiples of those magnitudes.

## Chapter 4

### Outer Division, Numerical Magnitudes

#### A. Theoretical Development

**§ 60. Concept of Outer Division** The analytic conjunction belonging to multiplication is division; consequently, according to the general concept of the analytic conjunction (§ 5), division consists in seeking one factor in terms of the product and the other factor. According to this definition, to each particular type of multiplication corresponds a type of division belonging to it. Outer division thus consists in seeking one factor in terms of the outer product and the other factor. Since the factors of the outer product are not interchangeable in general, it is clear that one must in this case distinguish two types of division, according as the first or the second factor is given (cf. § 11). We denote the factor sought (the quotient) thus: The given product A (the dividend) is placed above the division line in the usual way, the given factor B (the divisor) under it, but we precede or follow this given factor by a dot, according as the factor sought is to be the preceding or following factor. Thus means the factor C which, conjoined as second factor with a B, gives A, and hence satisfies the equation

$B \cdot 0 = A$ ; and means the factor C which, when conjoined as first factor with B, a gives A, that is it satisfies the equation  $C \cdot B = A$ , both definitions being expressed by the simple formulas  $B \text{ — } A = A$ ;  $\text{— } A \cdot B = A \cdot B$ .  $\cdot B$  Thus we have simply kept track of the fact that if the orders are such that the factors are directly interchangeable both equations have the same value, whereas if they are only interchangeable with a sign change, the two quotients have opposite values.\* Consequently in the first case one can drop the dot symbol if perhaps one does not yet want to indicate that the division is an outer



division.

## §§ 61, 62. Reality and Ambiguity of the Quotient

**§ 61.** The problem is now to ascertain the meaning of the quotient from the formal definition. Since the outer product of two extensions always produces an extension to which the two are subordinate and whose order is the sum of the orders of the factors, it also follows, first, that the quotient can only represent an extension if the divisor is subordinate to the dividend, that is, it is completely contained in the system of the dividend, and second, that the divisor must at the same time be of lower order than the dividend, the order of the quotient being the difference between those of the dividend and the divisor. In all other cases, then, the quotient cannot represent an extension, but rather has only a formal significance which we will temporarily defer considering. Conversely however, it follows that the quotient must always represent an extension if that condition is satisfied, that is if the divisor is subordinate to the dividend. Thus, according to

**§ 48, one can reduce each extension of  $n$ -th order to  $(n - 1)$  arbitrary** factors subordinate to it, so long as these are mutually independent, and thus one can also reduce it to any lower number of subordinate factors,<sup>1</sup>

that is, represent it as a product, one factor of which is an arbitrary extension subordinate to it. Thus: The quotient is an extension if and only if the divisor is subordinate to and of lower order than the dividend; and then the order of the quotient is the difference between the orders of the dividend and the divisor.

**§ 62.** It now remains to investigate whether the quotient is unique or multi-valued in this case; and if the latter, how the collection of its values can be found. be the quotient under consideration, and  $B$  be subordinate to the Leta magnitude  $A$ . According to the preceding section there is always an extension which, multiplying  $B$ , gives  $A$ ,

---

<sup>1</sup>Since the interchange of factors only requires a sign change if both are of odd order, and thus the product is even, so also the two quotients will only have opposite values if the dividend is of even, the divisor of odd order; in every other case they have the same value.

that is which can be interpreted as a quotient. Let  $C$  be one such, so that  $B \cdot 0 = A$ ; then the question is whether there are yet other extensions, different from  $C$ , that can replace  $C$  in this equation. It must in any case be of the same order as  $C$  (§ 61). Every extension different from  $C$ , but of the same order, can be expressed in the form  $C + X$ , where  $X$  is an arbitrary magnitude of the same order, and thus  $X$  is to be defined so that  $B \cdot (C + X) = A$ , if  $C + X$  can also appear as a value of the quotient 41. . One then has  $B \cdot 0 + B \cdot X = A = B \cdot C$ , that is  $B \cdot X = 0$ . Now however, according to § 55, only the product of two dependent magnitudes is zero, but it is also always zero; consequently, besides the partial value  $C$  of the quotient, any other magnitude differing from  $C$  by a summand dependent on the divisor also suffices, but no others. The collection of these magnitudes dependent on  $B$ , or which can replace  $X$  in the equation  $B \cdot X = 0$  we can now denote by  ${}^\circ B$ - according to the definition of the quotient, whence we have  $B \cdot 0 \text{ 713.} = C + 7B$ “

We can express this result in the following theorem: If the divisor ( $B$ ) is subordinate to and of lower order than the dividend ( $A$ ), the quotient is only partially defined, and in fact, if one knows a particular value ( $C$ ) of the quotient, one finds the general value by adding to that particular value one of the magnitudes dependent on the divisor ( $B$ ) in the undefined expression, or  $A : B \cdot {}^* = C +$  Applied to the theory of space, this theorem expresses, first, that if the base and the area (including the plane to which it belongs) of a parallelogram are given, then the other side, which we have called the elevation, is only partially defined, and that if its initial point is fixed, the position of its final point is on a straight line parallel to the base. Second, if the base plane and the volume of a parallelepiped are given, the other side (the elevation) is only partially defined, and the position of its final point with respect to a fixed initial point is on a plane parallel to the base surface. And finally, if the elevation and the volume of a parallelepiped are given, the base surface is only partially defined, since it appears as the variable plane intersection of a prism whose edges are parallel to the elevation. This last requires a demonstration. Thus if a base surface appears as a particular value of that quotient, that is if in fact its outer product with the given elevation yields the given volume, and one imagines this base surface in the form of a parallelogram, then the outer product of any parallelogram, formed by arbitrarily adding summands parallel to the elevation to the sides of the first, with the given elevation will be the same as that of the

first, which implies the theorem stated.

## §§ 63, 64. Expression for the Unique Quotient

**§ 63.** From the theorem of the preceding paragraph it follows that one cannot apply the laws of arithmetic division to our science without generalizing them, particularly since one is not permitted to cancel equal factors in the dividend and the divisor. But since in general calculation with undefined, or even only partially undefined, magnitudes is subject to manifold <sup>2</sup>

difficulties, and finds nothing exactly corresponding in other analyses of finite quantities, it is more appropriate to replace this indefinite expression with a definite one. Thus it follows that the quotient is defined once its type, that is the system of the same order with respect to the givens, is defined, thus upon supposing that this system is independent of that of the divisor but subordinate to that of the dividend. If this assumption is satisfied, then one and only one value of the quotient lying in the given system is possible. For if one thinks of any extension (C), similar to this system, multiplied by the divisor, then the product would be similar to the dividend, and thus also by enlargement or reduction of that extension (C) could be made equal to the dividend, whence this extension (C) itself appears as the quotient. But in addition only one such value of the quotient would result. For let C be one such value of the quotient  $1'$ , so that  $B.0 = A$ ; let C be changed into a similar magnitude  $C + C1$ , where C1 is not zero, whence one has  $B.(C + C1) = B.0 + = A + B.C1$ ; thus  $B.(C + C1)$  is not equal to A, since  $B.C$ , cannot be zero as the two factors are independent of each other by assumption. Thus no other value similar to C suffices to replace C in the equation  $B.0 = A$ , that is, can be interpreted as a value of the quotient  $1'$ ; there is therefore only one such value. One can express this result thus: If two equal products have an equal factor, and the other factors of the two are similar and independent of the first, then these latter two are also equal.“ It is important to find an appropriate notation for the quotient defined above. Let P be the dividend, A the divisor, B be a magnitude similar to the quotient, A and B both being

---

<sup>2</sup>This undefined term is therefore very similar to the undefined constant in integration, and the characteristic procedure involved thereby is the same there as here. (1877) Cf. footnotes to §§ 39 and 43.

subordinate to the system  $P$  and mutually independent; then  $P$  can be represented as a product of  $A_i$  with  $B$ , where  $A_i$  is similar to  $A$ , so that the quotient is  $A_1.B$ . • Since this is similar to  $B$ , we can tentatively symbolize it as  $B A$

$AB$  Thus symbolizes the magnitude  $B_1$ , similar to  $B$ , that satisfies the equation\*  $A_1.B = A.B$ .

**§64.** Now in order to ascertain the significance of this expression, we have to investigate the combination of the single expression  $14./'$  with different magnitudes.  $A, B, C$  First, it follows that, if are mutually independent, and  $A_i, A = DI$ , then it must in addition always be true that  $B_1, \_$ ,  $A B$  For from the first equation one has by definition  $A_1.B = A.B_1$ , and if one sets  $= CI$ , then  $A_pC = A.C_1$ .  $B$  Multiplying the first of these equations by  $C$ , the second by (in the same position), one then has  $A_1.B.0 = A.B_1.C$ ,  $A_1.B.0 =$  From a glance at number theory, with which our science comes into contact here, without however borrowing its theorems, it is clear that if  $A$ , is a multiple of  $A$ , then  $B$ , must be the same multiple of  $B$ , and thus that if by  $1$  we mean the number specifying what multiple  $A$ , is of  $A$ , then  $B$ , can be represented in the form  $1B$ . However elementary this application of number theory may be, we may not adopt it here without damage to our science. In addition, this betrayal of our science would be revenged soon enough by the manifold complications and difficulties into which we would be obliged to go as a consequence of the concept of irrationality. We therefore remain true to our science, without being tempted by the deceitful prospect of an easier path. <sup>3</sup>

and thus also  $A.B_1.0 = A.B$ .  $CI$ . Now since  $B_1.0$  is similar to  $B$ .  $C$ , and the other factor ( $A$ ) as well as the products on both sides are equal, so must (§ 63)  $.131.0 = B.C$ , that is  $B, A| C = C = -A.C. -B$  Thus if  $A, B = B$ , — $A$  II then the combination of the expressions  $1 \div, /'$  and with any magnitude independent of  $A.B$  yields the same result. But we can now show that this result must also be the case if both expressions are combined with a magnitude  $C$  that is independent of  $A$  and  $B$ , without simultaneously being independent of the product  $A.B$ .<sup>5</sup> First we prove this for the case that  $C$  is a

<sup>3</sup>The notation can occasion no ambiguity, since we have not as yet learned about the quotient of two similar magnitudes. Thus it remains for the moment undecided whether to interpret the notation  $1$  as a quotient in fact, and its union with  $B$  as multiplication; yet the appropriateness of the notation will only become clear if that interpretation is accepted.

displacement that we will denote by  $c$ . Thus let  $A, = c$ , or  $A_{ix} = A.cl$ , where  $c$  is in fact independent of  $A$  and  $B$  but dependent on  $A.B$ . Now in order to show that, if  $A B = B, -A$  then also  $B, A -c = -A -c = c$ ,  $B$  must follow, we try to make the factor  $c$  itself independent of  $A.B$  by adding to it a displacement  $p$ , itself independent of  $A.B$ . Instead of  $A.c$  one then obtains the expression  $A.(c + p)$ ; this can be set equal to an

expression whose first factor is  $A$ , and whose second is similar to  $(c + p)$ , and thus can be represented as a sum of two terms similar to  $c$  and  $p$ . Let this be  $c_2 + p_i$ , so that one has  $A.(c + p) = A.(c_2 + p_i)$ . Multiplying this equation by  $p$ , one obtains  $A_i.cip = A.c_2.p$  or, since  $A.c = A.cl$ ,  $A.cl.p = A.c_2.p$  and thus, since the corresponding factors are similar, according to § 63 there follows the equation  $C_1 = C_2$ . Consequently, replacing  $c_2$  with the value  $c_1$  above, one obtains  $A_i.(c + p) = A.(c, + p_i)$ . And now, since  $p$  is independent of  $A.B$ , and consequently so also is  $(c + p)$ , we can apply the law obtained above that  $B_1.(c + p) = B.(c, + p_i)$ ; thus also, upon multiplying by  $p$ ,  $B_i.c.p = B.c_1.p$ . Since the corresponding factors are similar here, one has  $B_i.c = B.c_i$  or  $B_1 = A I -c B A c$  even if  $c$  is dependent on  $A.B$ . Now we can easily extend this result to the case that the expressions  $A$  and  $71$ , corresponding to the equation  $A_i B = B$ , or  $A.B = A.B, -A$  are combined with an arbitrary magnitude  $C$  of higher order, independent of  $A$  and  $B$ . Let  $C = c.d.e . .$ , thus permitting every magnitude  $C I$  similar

to  $C$  to be represented in the form  $c_i.d.e$ , as we have already shown several times. Thus if  $A, C = C$ , or  $A_i.0 = A.C_1, -A$  then by that substitution one now has  $A.c.d.e . . = A.c_i.d.e . .$ , whence, because of the similarity of the factors, it follows that (§ 63)  $= A.cl$ . It also follows, because of the theorem just proved,  $B_i.c = B.cl$ . And finally, by repeating the same argument,  $B_i.c.d.e . . = B.c_i.d.e . .$ , that is  $B_1.0 =$  or  $B, A I C = = C. -A B$  We have thus proved the general theorem: If  $-AB = B$ , then in addition, with respect to any magnitude  $C$  independent of  $A$  and  $B$ ,  $A I B i A = B$

**§§ 65, 66. Concept of the Quotient of Two Similar Magnitudes** §\_65. Now since the concept of the expressions and is only defined to the extent that they appear in combination with magnitudes.that are indepen-

dentpfA and  $B$ , and the equality of any two such combinations

in which “e,11. and enter with the same magnitude has been demonstrated under the assumption that  $-B = B1$ , A it then follows that we are justified by the above assumption in setting the expressions 4, and 731 equal, and thus defining the concept of these expressions as such. Thus: If  $Al, = 1001 A$  or  $Al.B = A.B$ , (A and B mutually independent), then we set  $Al B, ji \bullet$  equal to  $-1-3$ . It is clear that the meaning of “41-B is in this way also specified if B is dependent on A; for one has only to take an auxiliary magnitude C, independent of A and  $/1$ , and define CI so that according to the given definition T! is equal to whence by substitution one has  $ALB \_ CIB A C$  and thereby the concept of the original expression is also defined. In particular, it then follows that  $-A = Al. A$  For take an arbitrary magnitude B, independent of A, and set  $A, B1 = A B'$  that is  $/31, =$ , whence, according to the general concept of equality, it must follow that  $Al B, , -A A = -B A$ .

However, as we have just shown, the latter expression equals  $Al$ , whence follows that which was to be shown. A, Thereby it follows at once that the expression 74 can be interpreted as a quotient as soon as its combination with other magnitudes, as we have described it so far, is proved to be a multiplication, that is, that the relation of that combination to addition is shown to be of the multiplicative type.

**§ 66.** First of all,  $Al (b + c) = A74l -b + Ai c. -A -A A$ , Thus  $7(b + c)$  is a displacement similar to  $b + c$  that can therefore also be expressed in terms that are similar to  $b$  and  $c$ ; let them be  $b_1$  and  $c_1$ , whence (1)  $A - A(b + c) = b_1 + c_1$ , Or  $Ai.(b + c) = A.(b_1 + c_1)$ . Multiplying this equation by  $c$ , one obtains  $A_1.b.c = A.bi.c$ , whence, by virtue of the similarity of the factors,  $Ai Al.b = A.b$ , or  $-Ab = bi$ . In the same way it follows upon multiplication by  $b$  that  $Al -c = c1$ ; A substituting these expressions for  $bi$  and  $c1$  into the above equation (1), one thus has in fact  $A, Al Al (b + c) = + . -A$  This must now be extended to the case that  $b$  and  $c$  are replaced by extensions of higher order,  $B$  and  $C$ . According to § 47,16 their sum then

yields an extension only if both extensions of  $n$ -th order can be brought to within a common factor of  $(n - 1)$ -th order. Thus let  $B = b.E$ ,  $C = c.E$ . Then if  $A L A1 IV ' - A A$  it follows that  $A_1.(b + c) = A.(b_1 + c_1)$  and also, if one multiplies this equation by  $E$ ,  $Al.(b$

+ c).E = A.(b, + c,).E, or  $Al.(b.E + c.E) = A.(bi.E + ci.E)$  Or A  
 (2)  $(B + C) = b, .E + ci.E$ . However, if one represents the equations  
 defining  $b_1$  and  $c_i$  in product form, and multiplies them by E,  $Al.b.E$   
 $= A.bi.E$ ,  $Al.c.E = A.ci.E$ , then  $Al B = b, .E$ , —A and in the same  
 way  $Al C = c, .E$ . —A Substituting these expressions for  $bi.E$  and  
 $ci.E$  into the above equation (2), one has  $B Al AI AI (B + C) = + C$ .  
 —A —A —A The multiplicative relation being valid for real sums,  
 it must also be valid for formal sums, as the concept of the latter is  
 only defined by the

former; thus since in that case  $B + C$  does not represent an  
 extension, then also  $Al (B + C) A$  has only a formal meaning, that  
 is to be set  $A, A, = + —C$ . A A Thus the multiplicative relation  
 for these expressions, that is  $1:1, . . .$ , is generally valid, and their  
 conjunction, as we have interpreted it, is to be understood as true  
 multiplication. Therefore 1 is a true quotient.\*

**§ 67. Proportion** In order to get a more intuitive idea of the  
 quotient we proceed first with displacements; let  $a$  and  $b$  be mutually  
 independent, and —  $a b$  then from the latter equation one has  $al.b +$   
 $bl.a = 0$ , or, since one may add to the second factor terms dependent  
 on the first,  $al.(a + b) + bi.(a + b) = 0$ ,  $(a, + b i).(a + b) = 0$ , that is,  
 $(a + b)$  and  $(al + bi)$  are similar, or can be interpreted as parts of the  
 same system of first order. According to the method of generation  
 of a system of first order,  $al$  and  $bl$  must then be corresponding  
 parts of  $a$  and  $b$ . Now, writing the original equation as a proportion,  
 $al:a = bi:b$ , leads one to the theorem: Four displacements stand in  
 proportion if the first is the corresponding part of the second that  
 the third is of the fourth. According to the concept of the quotient,  
 two similar magnitudes main- tain the same value unchanged if one  
 multiplies dividend and divisor by <sup>4</sup>

the same independent extension, thus generalizing the quotient;  
 that is, if  $al.b = a.b$ , , and thus  $a—‘$ ,  $a b‘$  then also  $al.E.b = a.E.bl$ ,  
 and thus  $al a, — t nus = —. a.E b ‘ a$  Consequently one can generalize  
 that ratio by an arbitrary extension. Now we can say that  $a, .E$  is  
 the corresponding part of  $a.E$  that  $a$ , is of  $a$ , and we have the general

---

<sup>4</sup>Since the order of a quotient is the difference between the orders of the divi-  
 dend and the divisor, ‘1 k is understood as an extensive magnitude of order zero,  
 which also agrees with the result that if an extension is multiplied by it there is  
 no change in the order.

theorem: Four magnitudes stand in proportion if the first is the corresponding part of the second that the third is of the fourth.

### § 68. Numerical Magnitude: Its Product with an Extensive

Magnitude Now we must describe the conjunctions of these newly obtained magnitudes, which we call numerical magnitudes, both among themselves and with extensive magnitudes. We have already described their multiplicative conjunction with extensive magnitudes, and established its relation to addition. Now we must investigate the pure multiplicative laws of conjunction, that is, the combinability and interchangeability of the factors. The result is that in an outer product in which numerical magnitudes appear, these can be adjoined to any arbitrary factor without changing the value of the result. Thus, denoting  $Ia(1$  by  $a$ ,  $a(B.C) = (aB).C$ . For let  $aB$  or  $111 = B1$ , or  $a1.B = a.B$ ; then upon multiplying by  $C$  one has  $a1.B.C = a.A.C$ , whence also, according to the definition,  $2'a-1(B.C) = B1.0$

or  $a(B.C) = (aB).C0$ . As for interchangeability, the meaning of the expression  $Aa$ , where  $A$  is any extension and  $a$  is a numerical magnitude, is not yet established; but we can define its meaning by analogy. Thus since the extensive magnitude of zeroth order appears as an extensive magnitude of even order, such as may be arbitrarily ordered in an outer product, we can assert that by  $Aa$  is meant the same as  $aA$ , whence it follows: The position of a numerical magnitude in an outer product is a matter of complete indifference. Finally, as to the quotient of an extensive magnitude by a numerical magnitude, its meaning is clear at once from the general concept of division; and the uniqueness of this quotient (so long as the divisor is not zero) follows easily. Thus if  $B a \text{ — } = X$ ,  $a \text{ — } ' a a1$  where  $a$  is independent of  $B$ , then one has  $aX = B$ ,  $= B$ ,  $a.X = ai$  and we have shown above that one obtains only a single value,  $X$ , similar to  $B$ , that satisfies the last equation, where that similarity is expressed in the preceding equation.

### §§ 69, 70. Product of Several Numerical Magnitudes

§ 69. We develop the concept of the product of several numerical magnitudes from the progressive product. If we define the product (1)  $Pa/3Y \bullet \bullet \bullet =$  where the extension  $P$  is multiplied progressively by the numerical magnitudes  $a$ ,  $/3$ ,  $y$ , , that is, so multiplied that



the result of each earlier multiplication is multiplied by the next sequential numerical magnitude, then there results the problem of finding a numerical magnitude which, multiplied with  $P$ , yields the same result,  $PI$ , directly. To this end we let  $a, /3, y, \dots$  be represented in the form

$A, B, C \dots$  so that  $P, A, B, C, \dots$  are all mutually independent. Multiplying both sides of the above equation (1) by  $A.B.C. \dots$  then since, according to the preceding paragraph, one can adjoin the numerical magnitudes  $a, /3, y, \dots$ , or  $A_1 B_1 C_1$   $A C \dots$  to any arbitrary one of those factors, thus “41-1 to  $A$  and so forth, one thereby obtains  $= PI.A.B.C$ . Thus, since  $PI$  is similar to  $P$ , according to the definition of the quotient  $= P A.B.0 \dots$ . As a consequence we have the law that  $A_1 B, C_1 p A_1.B_1.C_1 P ABC \dots A.B.0 \dots$  initially of course only if  $P$  is independent of  $AA..BB..C_0 \dots$ , but then also if  $P$  depends on it. To demonstrate this, we first represent the numerical magnitudes  $a, /3, y, \dots$ , or the quotients  $71, \dots$  in the new forms  $(70 \dots)$ , so that  $P$  is independent of  $A.B.F$ . then we can apply the above law and obtain a numerical magnitude  $p$  that can replace the progressive factors  $A \dots$  (or  $\dots$ ), and which equals  $A..B_1.F_1 A.B.F \dots$ . If one now takes an auxiliary extension  $Q$  independent of both  $AA..BB..C_0 \dots$  and these new magnitudes  $A.B.F \dots$ , then, by virtue of the first magnitudes,  $QaPy \dots$  equals  $A_1.131.C_1 Q A.B.0 \dots$  but by virtue of the second it equals  $QP \dots$

Consequently  $A_1.B_1.C_1 \dots P A.B.0 \dots$ . Now however  $P.aigy \dots = Pp$  as a consequence of the first set of formulas and thus also, by virtue of the value found for  $p$ ,  $A_i C_i P \dots = P AB C \dots A.B.0 \dots$ . The above law is thus proved in its complete generality.

**§ 70.** From this immediately follow two most important results for the conjunction of numerical magnitudes, to wit, first, that if for any given magnitude  $P$ , progressive multiplication by several magnitudes  $a, J3, y, \dots$  is replaced with multiplication by a specific numerical magnitude  $p$ , then this is also valid for any other magnitude that may replace  $P$ , since the expression obtained for  $p$  in the preceding paragraph is completely independent of  $P$  and depends only on the numerical magnitudes  $a, /3, y, \dots$ ; and second, that the numerical magnitudes are completely interchangeable among themselves, since

in the product  $A \cdot B$ , .  $A \cdot B$  . . equal changes in the numerator and denominator, by producing equal sign changes in both, thus produce none in the quotient. The first of these results justifies our setting the product  $a \cdot 3y$  itself equal to  $p$ . Thus By the product of several numerical magnitudes is meant that numerical magnitude which in its multiplication with any extension produces the same result as if the extension were progressively multiplied by the factors of that product. Thus if  $A, B, C, \dots$  are mutually independent,  $A \cdot B \cdot C \cdot \dots \cdot B \cdot C \cdot \dots \cdot A \cdot B \cdot 0 \cdot \dots \cdot$  The second result, which we derived before, now expresses that one can immediately interchange numerical magnitudes as factors.

**§ 71. Validity of All Laws of Arithmetic Multiplication and Division for Numerical Magnitudes** In order to prove all the laws of arithmetic multiplication and division (cf.

**§ 6) for numerical magnitudes, we have yet to demonstrate the uniqueness** of the quotient  $a$ , so long as  $a$  is not zero. According to the general definition of the analytic conjunction, means the magnitude that gives  $/3$  when multiplied by  $a$ ; now if  $ay$  equals 0, then we have to show that if at the same time  $ay'$  is equal to  $/3$ , then necessarily  $y$  must equal  $y'$ , always assuming that  $a$  is not zero. Thus, if  $A$  stands for any extension, assume that  $Ap = A(ay) = A(ay')$ ; however, since according to the preceding paragraph one can multiply by the individual factors instead of the product, one has as well  $(Aa)y = (Aa)y'$ . But from the definition of the numerical magnitude we have established that the numerical magnitudes that give equal results when multiplying the same extensions must also be regarded as equal. Now if  $a$  is not zero, then  $Aa$  is a true extension, whence according to the definition cited  $y = y'$ , that is the quotient of two numerical magnitudes is unique so long as the divisor is not zero. Now since all the laws of arithmetic multiplication are based on the interchangeability and combinability of factors and the uniqueness of quotients under the circumstances given (§ 6), and the same laws are also valid for the conjunction of numerical magnitudes with extensions, it follows that: All laws of arithmetic multiplication and division are valid for the conjunctions of numerical magnitudes among themselves and with extensive magnitudes.

\* At the same time the essential connection between arithmetic and

outer multiplication is thus demonstrated, since the former appears as a special species of the latter, in particular for the case that the factors are extensive magnitudes of zeroth order. For the multiplication of numerical magnitudes we therefore arbitrarily make use of the dot or alternatively simple juxtaposition, since the latter is often convenient in order to avoid parentheses and make formulas easier to read.<sup>5</sup>

---

<sup>5</sup>Thus we adopt nothing from arithmetic but the name, since we have obtained the laws of this conjunction independently in the General Theory of Forms, § 6.

### § 3.72. Addition of Numerical Magnitudes

In order to proceed to the addition of two numerical magnitudes ( $\alpha$  and  $\beta$ ), we must consider the expression

$$\alpha C + \beta C = C_1,$$

and seek the numerical magnitude with which  $C$  must be multiplied to produce the same value  $C_1$ .

To this end let  $\alpha, \beta$  be represented in the form  $\frac{a_1}{a}, \frac{a_2}{a}$ , where  $a$  is independent of  $C$ . The above equation is then transformed into

$$\frac{a_1}{a}C + \frac{a_2}{a}C = C_1,$$

or

$$a_1 \cdot C + a_2 \cdot C = a \cdot C_1,$$

and thus

$$\frac{a_1 + a_2}{a}C = C_1.$$

Thus we have proved the theorem that

$$\frac{a_1}{a}C + \frac{a_2}{a}C = \frac{a_1 + a_2}{a}C,$$

initially of course only if  $a$  is independent of  $C$ ; but in the same way as in §69 we can extend this to the case of dependence.

From this theorem it now follows that if

$$\alpha C + \beta C = \gamma C$$

then, since the expression for  $\gamma$  depends only on  $\alpha$  and  $\beta$  and not on  $C$ , the same equation also holds for any value of  $C$ , whence in this case one has the right to set  $\alpha + \beta$  equal to  $\gamma$ . Thus we set

$$\alpha + \beta = \gamma$$

if

$$\alpha C + \beta C = \gamma C,$$

where  $C$  is any extension; that is, according to the definition

$$\alpha C + \beta C = (\alpha + \beta)C.$$

In order to show that this conjunction is a true addition we must demonstrate the validity of the additive principle and the additive relation to multiplication.

First, the interchangeability of the terms is immediately implicit in the definition, since the terms  $\alpha C$  and  $\beta C$  are also interchangeable. In order to demonstrate the combinability of the terms we return to the equation

$$\alpha C + (\beta C + \gamma C) = \alpha C + (\beta C + \gamma C);$$

if one applies the law displayed in the definition twice to both sides, then this equation is transformed

into

$$[\alpha + (\beta + \gamma)]C = [\alpha + (\beta + \gamma)]C,$$

from which it follows that

$$(\alpha + \beta) + \gamma = \alpha + (\beta + \gamma).$$

Finally, the result of subtraction is also unique. For if the value of  $\beta$  in the equation

$$\beta = \alpha + \gamma$$

is sought, then if

$$\alpha = \frac{a_1}{a}, \quad \beta = \frac{a_2}{a}, \quad \gamma = \frac{a_3}{a},$$

it follows that from the above equation we obtain the equation

$$a_1 + a_2 = a_3$$

or

$$a_2 = a_3 - a_1.$$

Thus  $a_2$  has a definite value, whence also  $\alpha$  or that is the difference  $\gamma - \alpha$ , so that the result of subtraction is unique.

Since the principles of addition and subtraction are valid, so also are all their laws.

### § 3.73. Relation of This Addition to Multiplication. General Law

It only remains for us to demonstrate the relation of this addition to multiplication, and to show that

$$\alpha(\beta + \gamma) = \alpha\beta + \alpha\gamma.$$

According to the definition of the product (§70),

$$P \cdot \alpha(\beta + \gamma) = P\alpha \cdot (\beta + \gamma),$$

where the dot simultaneously takes the place of parentheses; however, according to the preceding paragraph the expression on the right side

$$\begin{aligned} &= P\alpha \cdot \beta + P\alpha \cdot \gamma \\ &= P \cdot \alpha\beta + P \cdot \alpha\gamma. \end{aligned}$$

Consequently, again according to the preceding paragraph, since

$$P \cdot \alpha(\beta + \gamma) = P \cdot \alpha\beta + P \cdot \alpha\gamma,$$

then also  $\alpha(\beta + \gamma) = \alpha\beta + \alpha\gamma$ . By combining these results with those obtained earlier we arrive at the general theorem:

**Theorem 3.73.1.** *All the laws of arithmetic conjunctions are valid, for both the conjunctions of*

*numerical magnitudes among themselves and with extensions; and all laws of outer multiplication and its relation to addition also remain in effect if one regards numerical magnitudes as extensive magnitudes of zeroth order, provided the result of division by it is defined.*

If we also apply the concept of dependence, as we have presented it in §55 for extensions, to numerical magnitudes as extensions of zeroth order, then it follows that these must be regarded as independent of all extensive magnitudes, provided none of these magnitudes is zero. Zero however always appears as dependent according to §32. On the other hand, numerical magnitudes always appear as similar to each other.

## B. Applications

### § 3.75. Numerical Magnitudes in Geometry

Since we have already incorporated the more elementary material concerning numerical magnitudes into the applications of the previous chapters, it only remains for us to apply the methods chosen here to geometry.

An essential defect of previous presentations of geometry is that one usually returns to discrete numerical ratios in the treatment of similarity theory. This procedure, which at first sight seems simple, soon enough becomes entangled in complicated investigations concerning incommensurable magnitudes, as we have already hinted above; and the initial impression of simplicity is revenged upon problems of a purely geometrical procedure by the appearance of a set of difficult investigations of a completely heterogeneous type, which shed no light on the essence of spatial magnitudes. To be sure, one cannot eliminate the problem of measuring spatial magnitudes and expressing the results of these measurements numerically. But this problem cannot originate in geometry itself, but only arises when one, equipped on the one hand with the concept of number and on the other with spatial perceptions, applies them to that problem, and thus in a mixed branch that one can in a general sense call by the name “theory of measurement,” of which trigonometry is a particular branch.<sup>1</sup> To relegate the theory of similarity, and even that of surface area, to this branch as has previously occurred (not to the form but to the substance) is to steal the essential content from what is called (pure) geometry. Now in modern geometry we find considerable preliminary work in the direction of the approach we demand here; but only in our science is the approach itself prescribed most completely.

## §§ 75–79. Pure Geometric Representation of Proportion in Geometry

### § 3.76.

Two approaches are presented here, the essences of which coincide, however different their expressions may appear. Thus four displacements, of which the first two and the last two are mutually parallel but the respective pairs are not, stand in proportion according to the first approach if the parallelogram of the first and fourth equals that of the second and third; according to the

---

<sup>1</sup>The numerical magnitudes, as we have developed them in our science, do not appear as discrete numbers, that is not as a set of units, but in continuous form as quotients of continuous magnitudes, and thus in no way presuppose the concept of discrete number.

second approach, if the sum (in the sense of our science) of the first and third is parallel to the sum of the second and fourth. The essential agreement of the two approaches already follows from the discussion developed in §67, since if

$$a_1 \cdot b = a \cdot b_1,$$

it then follows that

$$(a_1 + b_1) \cdot (a + b) = 0,$$
<sup>2</sup>

that is, the two sums  $(a + b)$  and  $(a_1 + b_1)$  are parallel, and likewise the latter equation follows from the former; and it is therefore of no importance on which of the two equations we make the proportion

$$a_1 : a = b_1 : b$$

dependent.

**Fig. 12a.**

We will choose the second approach as geometrically simpler, and express it thus: If two triangles have parallel sides, then we say that any two parallel sides of them are in the same ratio as two others taken in the corresponding order; for if  $a$  and  $b$  are two sides of the one, and  $a_1$  and  $b_1$  are the respective parallel sides of the other, then and only then are  $a + b$  and  $a_1 + b_1$  parallel. Here it is well to consider that at this level four displacements, as displacements, that is with lengths and directions taken into account, only appear as proportional if they are pairwise parallel, and we then place these parallel displacements in the proportion in the two first and the two last positions.

**§ 3.77.**

The proper line of development is now based on exhibiting proportion as the equality of two ratios, so that if  $a : a_1 = b : b_1$  and  $a : a_1 = c : c_1$ , then  $b : b_1 = c : c_1$ .

In order to find the geometric expression for this theorem, we set<sup>3</sup>

$$\begin{aligned} a &= AB, & a_1 &= AC, \\ b &= BD, & b_1 &= CE, \end{aligned}$$

then, if the first proportion holds, the points  $A, D, E$  must form a straight line, since  $a + b$ , that is  $AD$ , is parallel to  $a_1 + b_1$ , that is  $AE$ . Likewise, set

$$c = BF, \quad c_1 = CG;$$

then in addition, by virtue of the second proportion, the points  $A, F, G$  form a straight line. Now if the third proportion is also correct, then  $DF$  must be parallel to  $EG$ ; it is therefore to be shown that if the vertices of a triangle progress in straight lines that intersect in a single point, and two of the sides remain parallel, then the third must also remain parallel.

This theorem follows at once if the two triangles or (what amounts to the same thing) the three lines on which the vertices move, do not lie in the same plane. In this case one need only imagine

<sup>2</sup>The formula here is just a representation of a geometric theorem anyone can easily read off for himself, cf. fig. 12a.

<sup>3</sup>Cf. Fig. 12b.

a plane lying on each pair of lines issuing from  $A$ , and through the point  $C$  a plane parallel to  $BDF$  that intersects the three first planes in edges parallel to the sides of that triangle  $BDF$ , two of which coincide with  $CE$  and  $CG$ ; then the third will also coincide with  $EG$ , whence  $EG$  will be parallel to  $DF$ .

### § 3.78.

If those lines lie in a plane, one need only draw from  $B$  and  $C$  two parallel lines, lying outside the plane, that intersect a line drawn from  $A$  in the points  $H$  and  $I$ . Then according to the theorem of the previous paragraph, first,  $HD$  parallels  $IE$ ; second,  $HF$  parallels  $IG$ ; and thus by virtue of the parallelism of these line pairs,  $DF$  parallels  $EG$  according to the same theorem.

As a consequence we have shown in general that if the vertices of a triangle proceed in straight lines passing through a point, and two sides remain parallel, the third side also remains parallel; or that, if two displacement pairs are proportional to the same displacement pair, then they must also be proportional to each other, so long as these three displacement pairs represent different directions.

### § 3.79.

The concept of a proportion between four parallel lines has not previously been subject to a definition. In fact this case, although the simplest arithmetically, is geometrically the most complicated, since for three parallel displacements the geometrically proportional fourth only emerges with the help of a new direction.

According to the principle discussed in the preceding paragraph, we must regard a displacement pair as proportional to another parallel to it if both are proportional to the same displacement pair; for if it satisfies the relation with the one, then according to the previous paragraph so it must as well with the other to which the first one is itself assumed to be proportional. With the help of this definition follows the general validity of the theorem that two displacement pairs proportional to the same displacement pair must also be proportional to each other. Consequently, since we have defined its concept geometrically, we can also represent proportion as the equality of two expressions, each of which we call a ratio.

Since one applies the proportional displacements to a single point, this result expresses geometrically, first of all, only that if the vertices of a triangle, or in general of a polygon, move on straight lines that pass through a single point so that the other sides thereby remain parallel, then the final side must also remain parallel, and likewise every diagonal. Or, if one considers this evolved polygon in two of its states, then one has the theorem: "If the straight lines connecting corresponding vertices of two polygons of an equal number of sides go through a single point, and all corresponding pairs of sides except one are parallel, then this final pair must also be parallel." Those polygons can then be called "similar and similarly oriented," that single point the "similarity point." Conversely, it follows that two triangles with parallel sides are also similar and similarly oriented, or that the straight lines connecting their corresponding vertices go through a single point. Thus it follows as well that in similar and similarly oriented figures the points of intersection of two corresponding pairs of diagonals lie on a straight line with the similarity point, and in general that if one regards the connecting lines of corresponding pairs of points as corresponding, and similarly



the intersection points of corresponding pairs of lines, then in similar and similarly oriented figures each pair of corresponding points lie on a straight line with the similarity point, and every pair of corresponding lines are parallel.

Here then are the theorems for similarity, developed as far as possible at this level (without admitting the concept of length), and based in general on the concept of the similarity point. It is however easy to perceive that if one assumes in addition the concept of length as it usually appears in geometry, then these correspond perfectly to all the theorems of similarity themselves in exactly the form in which they are usually formulated, without thereby having cause to admit any concept of number. I elaborate so much the less on the further presentation of this topic, since the development of part 2 of this work parallels it.

### § 3.80.

Having thus presented the principles of development for geometry, we can perhaps forego the trouble of extending the development of proportionality to plane areas. Furthermore, since we have formally defined the conjunctions of numerical magnitudes in the abstract science, it also seems superfluous to formulate the corresponding theorems in geometry, as their formulation only has a meaning for our analysis and appears more as a mere analytic abbreviation than as displaying characteristic spatial relations.

Yet it is interesting to note that, in both the pure geometric presentation and in the abstract science, the treatment of space leads to the plane, and only then from this to the straight line; and consequently, that treatment in which all spatials are laid out, unfolded spatially, appears as characteristic of the theory of space and as its simplest expression, whereas if the structures lie within one another, then everything appears hidden as the seed in the bud, and only achieves its spatial significance if one relates those internal structures to those spatially unfolded.

# Chapter 5

## Equations, Projections

### A. Theoretical Development

#### § 5.2. Derivation of New Equations from a Given One by Multiplication

Having in the previous chapters learned the laws of conjunction to which extensive magnitudes are subject, it only remains for us to apply these laws to the solution and transformation of the equations that may exist between such magnitudes.

Since the terms on both sides of an equation, as summands or subtrahends, must all be of the same order, we may attribute this order to the equation itself, and thus by an equation of  $n$ -th order mean one whose terms are of  $n$ -th order. We must next ask ourselves what transformations we are allowed to undertake with such equations, or how we can obtain other equations from them. It is clear that one can bring their terms from one side to the other with a change of sign, and thus it remains only a question of what transformations an equation can undergo by multiplication and division. Thus we will henceforth suppose that all terms have been brought to the same (left) side, and therefore that the other (right) side is zero.

Now it is clear that if one multiplies both sides of the equation by the same extensive magnitudes then the right side remains zero, while on the left the individual terms can be multiplied instead of the whole sum. Since one always multiplies all the terms of an equation by the same extensive magnitude, one can therefore obtain a series of new equations from it, which in general are of higher order than that of the given (if the factor introduced is not of zeroth order).

If the given equation is of  $m$ -th order, and the system to which all the terms belong, and which we call the *principal system* of the equation, of  $n$ -th order, then one can multiply that equation by a particular extension of supplementary, that is  $(n - m)$ -th, order, which likewise belongs to the principal system, and obtain thereby an equation of  $n$ -th order, all the terms of which are mutually similar. Accordingly, one can in this way obtain from any equation whose terms are dissimilar a particular series of equations, each of which includes only similar terms.

#### § 5.3. Recovery of the Original Equation

Although one can now obtain arbitrarily many equations of higher order from a given equation, one cannot conversely recover the original equation from one of the former. In fact, if one multiplies

the original equation

$$A = 0,$$

where  $A$  denotes an aggregate of arbitrarily many terms, by an arbitrary extension  $L$ , and obtains a new equation

$$A \cdot L = 0,$$

it follows that if only the validity of the latter equation is given, the validity of the former by no means follows therefrom; on the contrary, from the latter it only follows that

$$A = \frac{0}{L},$$

in which, according to the previous chapter,  $\frac{0}{L}$  represents any magnitude dependent on  $L$ , zero included. The equation  $A = 0$  thus only follows if it is assumed that  $A$  has no nontrivial values dependent on  $L$ , or in other words if the terms that  $A$  is imagined to have belong to a system independent of  $L$ , that is:

**Theorem 5.3.1.** *If the terms of an equation all have a common factor  $L$  in the same position, and the collection of the remaining factors of all the terms belong to a system independent of this common factor, then one can drop the factor  $L$  in all terms.*

#### § 5.4. Projection or Shadow. Shadow of a Sum

By combining the methods of the two preceding paragraphs we now arrive at a procedure for obtaining from one equation another equation of the same order.

Thus if

$$A + B + \cdots = 0$$

is the original equation, then (according to §80) upon multiplication by  $L$  we obtain the equation

$$A \cdot L + B \cdot L + \cdots = 0.$$

Now if we want to apply the procedure of §81 to this in order to remove the factor  $L$ , we must represent the terms of this equation in such a form that the factors with which  $L$  is multiplied collectively belong to a system independent of  $L$ .

Let  $G$  be such a system, and  $A', B', \dots$  be extensions belonging to this system, and having the property that

$$A' \cdot L = A \cdot L, \quad B' \cdot L = B \cdot L, \quad \dots$$

whence one has the equation

$$A' \cdot L + B' \cdot L + \cdots = 0$$

and therefore, according to the previous paragraph,

$$A' + B' + \cdots = 0,$$

an equation of the same order as the original. Each term of the last equation thus results from the corresponding term of the first, since one has sought a magnitude in the system  $G$  which,

multiplied by a magnitude  $L$  independent of  $G$ , yields the same as the corresponding term of the original equation. It will be shown at once that if such a magnitude is possible, then only a single one is possible. Indeed if one assumes two such, say  $A'$  and  $A''$ , resulting from  $A$  in the manner given, then by assumption they must yield the same result when multiplied by  $L$  (that is,  $A \cdot L$ ); thus we obtain the equation

$$A' \cdot L = A'' \cdot L,$$

and since the system  $G$ , to which  $A'$  and  $A''$  belong, is independent of  $L$ , one can according to §81 drop  $L$  and find

$$A' = A'',$$

that is, the two magnitudes coincide; thus in fact only a single magnitude is possible.

Here we call  $A'$  the *projection* or *shadow*,<sup>1</sup>  $A$  the projected or shadowed magnitude,  $G$  the *base system*, the system  $L$  the *guide system*, and say that  $A'$  is the projection or shadow of  $A$  onto  $G$  along (according to) the guide system  $L$ . Thus by the projection or shadow of a magnitude ( $A$ ) onto a base system ( $G$ ) along a guide system ( $L$ ) we mean that magnitude belonging to the base system that yields the same product with a part of the guide system as the projected or shadowed magnitude ( $A$ ).

We can immediately express the theorem presented at the beginning of this paragraph in the form:

**Theorem 5.4.1.** *An equation remains valid if one shadows (projects) all its terms in the same sense.*

Or also, if one imagines a single term standing alone on one side:

**Theorem 5.4.2.** *The shadow (projection) of a sum equals the sum of the shadows of its terms.*<sup>2</sup>

## § 5.5. When the Shadow is Zero, and When it is Not Possible

To lend greater clarity to the method of treatment, we must investigate when the shadow is zero and when it is not possible.

Should the shadow  $A'$  be zero, then since

$$A' \cdot L = A \cdot L,$$

the product  $A \cdot L$  must also be zero, that is  $A$  must be dependent on  $L$ ; but conversely, if this dependence holds, then since the system to which any nontrivial value of  $A'$  belongs is independent of  $L$ , the product  $A' \cdot L$  cannot be made equal to zero unless  $A'$  is itself zero. Thus the shadow is zero if and only if the shadowed magnitude is dependent on the guide system.

Finally, since each magnitude belonging to the system  $G$  belongs to the system  $G \cdot L$  when multiplied by  $L$ , so also  $A' \cdot L$  will necessarily belong to the system  $G \cdot L$ , since it equals  $A \cdot L$ , assuming the shadow is possible; consequently the zero value, as always, can be regarded as belonging to any

<sup>1</sup>The names projection and shadow do not mean precisely the same thing, but their distinction will only emerge in part 2 of this volume; applied to the magnitudes considered here, the two concepts coincide.

<sup>2</sup>I place the name shadow first in the expression of the theorems since this form of the theorems is general, and in addition they will remain valid for the magnitudes to be developed later.

arbitrary system and as dependent on it. But also conversely, if  $A \cdot L$  belongs to the system  $G \cdot L$  then the shadow is always possible; for if  $A \cdot L$  is not zero and it belongs to the system  $G \cdot L$ , then the elementary factors of  $A \cdot L$  must be representable as sums of terms that are similar to those of  $G \cdot L$ . Thus in particular  $A$  must be representable in this way; but one can drop from  $A$  those terms that are similar to the factors of  $L$  without changing the value of the product. If one does this, and lets the magnitude that in consequence replaces  $A$  be called  $A'$ , then the factors of  $A'$  depend only on  $G$ , whence  $A'$  simultaneously belongs to the system  $G$  and is therefore the shadow of  $A$ . If however  $A \cdot L$  is zero, then we have already established that the shadow is also zero, and hence possible.

Consequently we have shown that the shadow is possible if and only if the product of the shadowed magnitude with the guide system belongs to the product of the base system with the guide system. Since, if  $A \cdot L$  is not zero, the above condition is identical to the condition that  $A$  belongs to the system  $G \cdot L$ , we can combine the results of this paragraph into the following theorem:

**Theorem 5.5.1.** *If the shadowed magnitude is dependent on the guide system, then its shadow is zero. If it is independent of the guide system, then the shadow always has a nontrivial value if the shadowed magnitude is dependent on the combined base and guide system; in all other cases it is not possible.*

If we apply the concept of the shadow to a magnitude of zeroth order, that is to a numerical magnitude, we need only observe that the generality of the law demands that it belongs to every arbitrary system, but, if it is not zero, that it be regarded as independent of them (cf. chapter 4).<sup>3</sup> It therefore follows that numerical magnitudes are not changed by shadowing.

## § 5.6. Shadow of a Product and of a Quotient. General Law

We now proceed to the shadow of a product in order to compare it with the shadow of its factors.

If  $A \cdot B$  is the product and  $A'$  and  $B'$  are the shadows of  $A$  and  $B$  on the base system  $G$  according to the guide system  $L$ , then one has the equations

$$A' \cdot L = A \cdot L \quad \text{and} \quad B' \cdot L = B \cdot L.$$

The shadow of the product  $A \cdot B$  is now that magnitude belonging to the system  $G$  which, multiplied by  $L$ , yields a product equal to  $A \cdot B \cdot L$ . Now since  $A \cdot L$  equals  $A' \cdot L$ , I can replace  $A$  with the value  $A'$  in the product  $A \cdot B \cdot L$ , as follows immediately by a double interchange and combination.<sup>4</sup> Consequently I obtain

$$A \cdot B \cdot L = A' \cdot B \cdot L = A' \cdot B' \cdot L,$$

---

<sup>3</sup>Cf. chapter 4.

<sup>4</sup>That is, I can set  $A \cdot B \cdot L$  equal to either  $A \cdot L \cdot B$  or  $-A \cdot L \cdot B$ , then combine the factors  $A \cdot L$  as a product, replace this product with its equal  $A' \cdot L$ , and finally restore the original order, whereby, if the minus sign was introduced, it is necessarily restored to the original sign.

this last since  $B \cdot L$  equals  $B' \cdot L$ . Now since both  $A'$  and  $B'$  belong to the system  $G$ , so also does  $A' \cdot B'$ , and since at the same time, as we have just shown,

$$A \cdot B \cdot L = A' \cdot B' \cdot L,$$

it follows that in fact  $A' \cdot B'$  is the shadow of  $A \cdot B$ ; thus one has the theorem:

**Theorem 5.6.1.** *The shadow of a product is the product of the shadows of its factors, if all shadows are taken in the same sense (that is, with the same base system and the same guide system).*

Or, combined with the earlier results:

**Theorem 5.6.2.** *A true equation remains true if one shadows all its terms, or the factors of its terms, in the same sense.*

In particular, if one has the equation

$$A_1 = \alpha A, \quad \text{or} \quad \frac{A_1}{A} = \alpha,$$

where  $\alpha$  denotes a numerical magnitude, then if  $A'_1$  and  $A'$  are the shadows of  $A_1$  and  $A$ , the equation

$$A'_1 = \alpha A', \quad \text{or} \quad \frac{A'_1}{A'} = \alpha$$

follows, that is, the value of a quotient of two similar magnitudes does not change if one replaces them with their shadows taken in the same sense.

Or, more generally, if one seeks the shadow of a quotient  $\frac{A}{B}$  then since the quotient signifies the magnitude  $C$  satisfying the equation

$$C \cdot B = A,$$

by shadowing the individual factors in the same sense the new equation

$$C' \cdot B' = A' \quad \text{or} \quad C' = \frac{A'}{B'}$$

results; that is, instead of shadowing a quotient one can shadow the numerator and denominator in the same sense. Thus, combining addition, subtraction, outer multiplication, and division under the general concept of fundamental conjunctions, we can formulate the general theorem that includes the earlier ones:

**Theorem 5.6.3.** *Instead of shadowing the result of a fundamental conjunction, one can shadow its terms in the same sense.*

## § 5.7. Analytic Expression for the Shadow

Here we present the problem of expressing the shadow analytically, given the magnitude to be shadowed and the sense of the shadow, that is the base system and guide system. We will however limit ourselves to the case that the shadowed magnitude is of the same order as the base system, although the solution in the general case could also easily be found at this point; but it would lead to an expression much less simple than the expression to be developed here.

Thus if  $A$  is the shadowed magnitude,  $L$  a term of the guide system,  $G$  the base system, and  $A$  and  $G$  are of the same order, then the shadow  $A'$  must be similar to  $G$ ; thus one can set

$$A' = xG,$$

where  $x$  is a numerical magnitude. Multiplying this equation by  $L$ , one has

$$A' \cdot L = xG \cdot L$$

or, since according to the concept of the shadow  $A' \cdot L$  equals  $A \cdot L$ , one has

$$A \cdot L = xG \cdot L, \quad \text{whence} \quad x = \frac{A \cdot L}{G \cdot L},$$

and therefore

$$A' = \frac{A \cdot L}{G \cdot L} \cdot G$$

is the analytic expression sought. We spare ourselves the verbal expression of this result until after considering the general case.

### § 5.8. Derivation of an Assembly of Equations that Replaces the Original

We must again take up the thread of the argument we let drop above (§81). There we had shown that from an equation

$$A + B + \cdots = 0$$

one does indeed obtain a new equation

$$A \cdot L + B \cdot L + \cdots = 0$$

upon multiplication by an arbitrary extension  $L$ , but that in general one cannot recover the original from this; now it is important to obtain from that equation an assembly of equations of this type that can replace it, that is, from which the first equation can be recovered.

In particular, the factor  $L$  was chosen so that, after multiplication of the individual terms by this factor, there results an equation containing only similar terms; and since such equations seem the simplest, it will be important to replace the first equation with equations of this type.<sup>5</sup> The development in the subsequent paragraph then showed how the equation

$$A \cdot L + B \cdot L + \cdots = 0$$

could be replaced by an equation between shadows, all on the same base system, according to the guide system  $L$ , that is, if we let  $A'$ ,  $B'$ , ... represent such shadows of  $A$ ,  $B$ , ..., by the equation

$$A' + B' + \cdots = 0.$$

---

<sup>5</sup>In general we say that two sets of equations are mutually replaceable if one can obtain either of the two from the other.

And the problem that we set ourselves is thus identical to that of replacing an equation by an assembly of equations resulting from shadowing the original, and in particular an equation between dissimilar terms by such shadow equations, each of which has only similar terms.

Let the original equation be of  $m$ -th order and its principal system, that is the system to which all its terms collectively belong, be of  $n$ -th order, and indeed let this latter be represented as a product of  $n$  independent elementary factors, that is  $a \cdot b \cdots$ . Then according to the concept of a system of  $n$ -th order, each elementary factor of each of the terms of the given equation is representable as a sum whose terms are similar to those factors  $a, b, \dots$ , thus in the form  $a_1 + b_1 + \cdots$ . If one imagines each elementary factor of each term of the given equation represented in this way, and carries out the multiplication so that the parentheses disappear, then one obtains a sum of terms, each of which is similar to one of the products of  $m$  factors from among  $a, b, \dots$ . If one now multiplies the equation by  $(n - m)$  of the factors  $a, b, \dots$ , then there remain only terms of nontrivial value that are similar to the product of the  $m$  remaining factors of that series  $a, b, \dots$ , since all others contain at least one elementary factor similar to the newly introduced factors, and which therefore vanish with this multiplication. Now however, according to §81, one can again drop the factors introduced, since the system to which the remaining ones belong is independent of the system of those introduced. In this way one obtains an assembly of true equations if, after the original equation is transformed in the way specified, one combines each of the sets of similar terms into an equation. And since the collection of equations so obtained recovers the original upon addition, we have found an assembly of equations that precisely replaces the original, and the problem is solved.

Consequently we have the theorem:

**Theorem 5.8.1.** *If, in an equation of  $m$ -th order, whose terms belong to a system of  $n$ -th order, one represents each elementary factor of each term as a sum whose terms are similar to  $n$  mutually independent displacements, and multiplies through, then one can group each series of similar terms resulting thereby into a single equation, and so obtain an assembly of equations that replaces the original.*

Or, since each of those equations can be replaced by an equation that results from multiplication of the original equation by  $(n - m)$  of the factors  $a, b, \dots$ :

**Theorem 5.8.2.** *If one multiplies an equation of  $m$ -th order, whose terms belong to a system of  $n$ -th order, one by one by each product of  $(n - m)$  factors that can be formed from  $n$  mutually independent displacements of that system, one obtains an assembly of equations that replaces the original.*

Since the terms that appear in each equation obtained with the preceding theorem are immediately recognized as shadows of the terms that were found in the original equation, we can also express the theorem obtained by means of the concept of the shadow, but we must formulate another series of new concepts to express it conveniently.

## § 5.9. Reference Systems (Coordinate Systems), Reference Domain, Reference Measures, Principal Measure

The method of treatment in the previous paragraph leads us to the concept of the coordinate system or reference system, which however we interpret in a much more general sense than usual. I



also permit myself to replace the usual nomenclature with a simpler, since generalizing the former to meet the demands of our science makes it appear very awkward; besides, that nomenclature is borrowed from a foreign language.

I call the  $n$  displacements  $a, b, \dots$  that define a system of  $n$ -th order (and thus are all mutually independent) the *reference measures* of first order or the *fundamental measures* of the system, inasmuch as each displacement of the system is expressed through them; the assembly of them a *reference system*, the products of  $m$  fundamental measures (retaining their original order) *reference measures* of  $m$ -th order, the reference measure of  $n$ -th order the *principal measure*. Finally, we call the system of reference measures of  $m$ -th order a *reference domain* of  $m$ -th order, and in particular the system of the fundamental measures, *reference axes* (coordinate axes). Supplementary reference measures are such as give the principal measure when multiplied together, and the reference systems to which they belong we also call supplementary.

### § 5.10. Reference Terms, Indicator

From the discussion in §86 it is clear that each extension of  $m$ -th order belonging to a system of  $n$ -th order can be interpreted as a sum of terms similar to the reference measures of  $m$ -th order belonging to that system. We now call these terms *reference terms* of that magnitude, so that each magnitude appears as the sum of its reference terms; the numerical magnitudes resulting if the reference terms of a magnitude are divided by the corresponding (similar) reference measures are the *indicators* of the magnitude, so that each magnitude therefore appears as a multiple sum<sup>6</sup> of the reference measures of the same order. The reference terms of a magnitude of first order are also what are usually called coordinates. A magnitude is said to be shadowed (projected) in the sense of the reference system if it is shadowed on one of the reference domains according to the supplementary reference domain.

### § 5.11. Equations Between the Reference Terms and Between the Indicators

If we apply these concepts to the theorems established in §86, these are transformed into the following:

One can replace all the terms of an equation with their reference terms or indicators belonging to an arbitrary reference measure, but the same reference measure for all the terms; and if one carries this out with respect to all the reference measures of the same order, one then obtains an assembly of equations that replaces the one given.

The equations obtained in §86 are thus just these equations between the reference terms, and from them one obtains the indicator equations upon division by the corresponding reference measure.<sup>7</sup> Furthermore:

From a given equation one can obtain each of its assembly of replacement equations upon multiplication of it one by one with the collection of reference measures whose order is the supplement of that of the equation with respect to that of the principal system.

<sup>6</sup>We call each product of a magnitude with a numerical magnitude a multiple of the first, and distinguish this from the multiple of a whole number which that numerical magnitude may be.

<sup>7</sup>These indicator equations, as equations between simple numerical magnitudes, mediate most completely the transition to arithmetic.

### § 5.12. Shadows of an Equation in the Sense of a Reference System. Expression for the Indicator

If we multiply a magnitude of  $m$ -th order, expressed as a sum of its reference terms, by a reference measure of supplementary, that is  $(n - m)$ -th order, then all reference terms but one drop away, and this one then appears as the shadow of that magnitude on the reference domain of  $m$ -th order<sup>8</sup> according to the supplementary reference domain, and all reference terms of that magnitude thus appear as shadows in the sense of the resulting reference system on the different reference domains of the same order. We can therefore say:

An equation of  $m$ -th order can be replaced by a set of equations resulting from shadows on the different reference domains of  $m$ -th order in the sense of the reference system.<sup>9</sup>

Simultaneously there follows a simple analytical expression for the reference terms or indicators of a magnitude. Thus for a magnitude  $P$  let  $P'$  be the reference term belonging to a reference measure  $A$ , and  $B$  be the reference measure supplementary to  $A$ ; then since  $P'$  is the shadow of  $P$  on  $A$  according to  $B$  (cf. §85), one has

$$P' = \frac{P \cdot B}{A \cdot B} A,$$

whence the corresponding indicator equals

$$\frac{P \cdot B}{A \cdot B},$$

that is:

**Theorem 5.12.1.** *For a given magnitude, its indicator belonging to the reference measure  $A$  equals a fraction whose numerator is the product of the magnitude with the supplementary reference measure and whose denominator is the product of the first reference measure with its supplement.*

## B. Applications

### § 5.14. Shadow in Geometry

If we apply the concepts developed in this chapter to geometry, then there follows, first, only one type of projection (shadow)<sup>10</sup> for the plane, since a displacement can be projected on a given straight line according to a given direction. The reference system for the plane has only two fundamental measures and two reference axes belonging to them. The area of the parallelogram formed from the two fundamental measures appears as the principal measure.

In space three types of projection result; there are either displacements or areas projected on a given plane according to a given direction, or displacements projected on a given straight line parallel to a given plane. The reference system for space has three fundamental measures and

<sup>8</sup>That an equation of  $m$ -th order in a system of  $n$ -th order can be replaced by as many elementary equations as there are combinations of  $n$  elements taken  $m$  at a time hardly needs mention.

<sup>9</sup>That an equation of  $m$ -th order in a system of  $n$ -th order can be replaced by as many elementary equations as there are combinations of  $n$  elements taken  $m$  at a time hardly needs mention.

<sup>10</sup>We put the name projection first in this application on grounds that will become clear later.

three reference axes belonging to them; and in addition three reference planes as reference domains of second order and three reference measures of second order belonging to them, which represent the areas of the parallelograms (retaining the directions of their planes) described by each pair of fundamental measures. As principal measure there appears the parallelepiped described by the three fundamental measures. It is particularly interesting that the representation of an area of definite direction appears here as the sum of its reference terms, that is as a sum of three areas belonging to the three reference planes. Since the theorems that can be established about projections and reference systems in geometry are already fully established in our science in the same form in which they are expressed in geometry, we can omit their repetition here.

### § 5.15. Transformation of Coordinates

We will, however, next solve the problem of coordinate transformations for geometry and then in general for our science.

Let  $a, b, c$  be three fundamental measures and  $e_1, e_2, e_3$  be three new mutually independent fundamental measures given as multiple sums of those original fundamental measures; the problem is then: On the one hand, to represent a magnitude  $p$ , given as a multiple sum of the original fundamental measures, as a multiple sum of the new fundamental measures, and on the other, if it is given in the latter form, to represent it in the former; and in both cases to find the indicators. In fact these problems are solved by the theorem in §90 that instructs one how to find the indicators. Accordingly, with respect to the first problem the indicator of  $p$  belonging to  $e_1$  equals

$$\frac{p \cdot e_2 \cdot e_3}{e_1 \cdot e_2 \cdot e_3}$$

and with respect to the second, the indicator of  $p$  belonging to  $a$  equals

$$\frac{p \cdot b \cdot c}{a \cdot b \cdot c}.$$

and with these extremely simple expressions the problem of coordinate transformations is solved in its greatest generality.

The second problem is of particular importance in the theory of curves and surfaces, since these are defined by an equation between the indicators of a displacement drawn from a point taken as the initial point of the coordinates to a point of the curve or surface. Thus let  $p = xa + yb + zc$  be this displacement, and

$$f(x, y, z) = 0$$

be the equation defining a surface; if one now seeks the equation of the same surface, first for the same initial point of coordinates but with respect to new reference axes and the reference measures  $e_1, e_2, e_3$  belonging to them, then if

$$p = u_1 e_1 + u_2 e_2 + u_3 e_3$$

one has the equation

$$f\left(\frac{p \cdot b \cdot c}{a \cdot b \cdot c}, \frac{a \cdot p \cdot c}{a \cdot b \cdot c}, \frac{a \cdot b \cdot p}{a \cdot b \cdot c}\right) = 0,$$

an equation between the new variables  $u_1, u_2, u_3$  if one substitutes for  $p$  its value. If one also shifts the initial point of the coordinates, say by the displacement  $e$ , then if  $q$  is the displacement from the new initial point to the same point of the surface to which the corresponding value of  $p$  refers, and

$$q = v_1 e_1 + v_2 e_2 + v_3 e_3,$$

one simply replaces  $p$  in the above equation with its value  $q + e$  in order to obtain the shifted equation; or, if

$$e = \alpha a + \beta b + \gamma c,$$

then, as follows at once, one has

$$f\left(\frac{q \cdot b \cdot c}{a \cdot b \cdot c} + \alpha, \frac{a \cdot q \cdot c}{a \cdot b \cdot c} + \beta, \frac{a \cdot b \cdot q}{a \cdot b \cdot c} + \gamma\right) = 0$$

as the shifted equation between the new variables  $v_1, v_2, v_3$ . If one wants to represent this equation as simple indicator equations, one has only to represent the new fundamental measures in the same way as multiple sums of the originals, and introduce them into the equation. Thus if

$$e_1 = \alpha_1 a + \beta_1 b + \gamma_1 c,$$

$$e_2 = \alpha_2 a + \beta_2 b + \gamma_2 c,$$

$$e_3 = \alpha_3 a + \beta_3 b + \gamma_3 c,$$

then it is immediately evident that the shifted equation is represented in the form

$$f(\alpha + \alpha_1 v_1 + \alpha_2 v_2 + \alpha_3 v_3, \beta + \beta_1 v_1 + \beta_2 v_2 + \beta_3 v_3, \gamma + \gamma_1 v_1 + \gamma_2 v_2 + \gamma_3 v_3) = 0,$$

an equation that leaves nothing to be desired in the way of simplicity.

The solution of our problem follows just as easily for the most general case of the abstract science. In fact, if a magnitude  $P$  is expressed as a multiple sum of certain reference measures, and one wishes to express it as a multiple sum of other reference measures, then according to §90, if  $B$  is the reference measure supplementary to the reference measure  $A$ , the indicator of  $P$  belonging to  $A$  equals

$$\frac{P \cdot B}{A \cdot B}.$$

## § 5.16. Elimination of an Unknown from Equations of Higher Degree

The application to the theory of equations was already anticipated above (§45) in the method of solving equations of first degree in several unknowns with the help of our analysis. We resume this subject here by explaining the method, presented using our science, for eliminating unknowns from equations of higher degree in several unknowns.

Suppose two equations of higher degree in several unknowns are given, and one of them, say  $y$ , is eliminated; then an equation between the remaining unknowns will result. Let the given equations

be ordered according to powers of  $y$ :

$$\begin{aligned} a_m y^m + \cdots + a_1 y + a_0 &= 0, \\ b_n y^n + \cdots + b_0 &= 0, \end{aligned}$$

where  $a_m, \dots, a_0$  and  $b_n, \dots, b_0$  are arbitrary functions of the other unknowns, but  $a_0$  and  $b_0$  are not equal to zero. If one multiplies the first equation one by one by  $y, y^2, \dots, y^n$ , and the second one by one by  $y, y^2, \dots, y^m$ , one obtains  $m + n$  new equations. If one regards the coefficients of each of these  $m + n$  equations as similar to each other, but those of different equations as mutually independent (even if until then they were indicated by the same letters), then, if one adds the equations so interpreted in the sense of our science, one obtains an equation of the form

$$e_{m+n} y^{m+n} + \cdots + e_0 = 0.$$

Multiplying this equation by the outer product  $e_2 \cdot e_3 \cdots e_{m+n}$ , all the terms except the last will drop out according to the laws of outer multiplication, and we will obtain the equation

$$e_1 \cdot e_2 \cdot e_3 \cdots e_{m+n} y = 0,$$

or, since  $y$  cannot be zero, as then in the given equation  $a_0$  and  $b_0$  would be zero, contrary to our assumption, we then have

$$e_1 \cdot e_2 \cdot e_3 \cdots e_{m+n} = 0$$

as the required elimination equation.

## Part I

# Elementary Magnitudes



# Chapter 1

## Addition and Subtraction of Elementary Magnitudes of First Order

### A. Theoretical Development

#### § 1.2. Law for the Sum of Displacements Drawn from a Moving Element to a Series of Fixed Elements

I associate the concept of the elementary magnitude with the solution of a simple problem by which I first arrived at this concept, and which seems to me to be most appropriate for its natural development.

**Problem.** Let three elements  $a_1$ ,  $a_2$ ,  $\beta_1$ , and a further element  $p$  be given; one is then to find the element  $\beta_2$  satisfying the equation

$$[pa_1] + [pa_2] = [p\beta_1] + [p\beta_2].$$

**Solution.** If one shifts the terms on the left side to the right, then since

$$-[pa] = [ap] \quad \text{and} \quad [ap] + [p\beta] = [a\beta],$$

one has

$$[a_1\beta_1] + [a_2\beta_2] = 0,$$

by which the element  $\beta_2$  is specified in an elementary way.

In order to bring this result more clearly into view we apply it to geometry, and thus suppose the elements are points; thus we find the point  $\beta_2$  by making  $[a_2\beta_2]$  equal and opposed to  $[a_1\beta_1]$ .—This solution is of interest since  $\beta_2$  is defined completely independently of  $p$ ; and since by the inverse process we can recover the first equation of the solution with respect to any arbitrary  $p$  from the last equation, we simultaneously have the theorem that if the equation

$$[pa_1] + [pa_2] = [p\beta_1] + [p\beta_2]$$



is true for any given point  $p$ , it is also true for every other point that may replace  $p$ .

This theorem can be proved directly, but first we will generalize it; for it is clear that the process remains applicable if instead of the two elements  $a_1, a_2$  and  $\beta_1, \beta_2$  one introduces arbitrarily many, but an equal number on both sides. Indeed, since arbitrarily many elements may coincide, it also remains true if arbitrary coefficients are adjoined to the displacements, provided the sum of these coefficients is the same on both sides. Thus, letting

$$i_1[pa_1] + \cdots + i_n[pa_n] = k_1[p\beta_1] + \cdots + k_m[p\beta_m],$$

where the magnitudes  $i_1, \dots$  and  $k_1, \dots$  represent numerical magnitudes, and simultaneously

$$i_1 + \cdots + i_n = k_1 + \cdots + k_m,$$

then we can prove that the first equation remains true for every point  $\sigma$  that can replace  $p$ . For

$$[pa] = [p\sigma] + [\sigma a], \quad [p\beta] = [p\sigma] + [\sigma\beta].$$

If one introduces such expressions, relative to the indices  $(1 \dots n, 1 \dots m)$ , respectively, into the above equation, and removes the parentheses and combines the terms that include  $[\sigma p]$  on each side, one obtains  $[p\sigma]$  multiplied by the sum of coefficients on each side; and since these are equal, this term can be removed from both sides and one is left with

$$i_1[\sigma a_1] + \cdots + i_n[\sigma a_n] = k_1[\sigma\beta_1] + \cdots + k_m[\sigma\beta_m],$$

that is, the equation remains true with respect to any element that may be substituted for  $p$ . Thus:

**Theorem 1.2.1.** *If one draws displacements from an element  $p$  to arbitrarily many fixed elements, of which two arbitrary multiple sums (whose coefficients however have equal sums) are equal, then this equality is maintained however the element  $p$  may change.*

In particular, if the sum of coefficients in the expression

$$i_1[pa_1] + \cdots + i_n[pa_n]$$

is zero, then in the same way, that is by substituting  $[pa] + [a\sigma]$  throughout, one finds that that expression equals

$$i_1[\sigma a_1] + \cdots + i_n[\sigma a_n]$$

since the term  $(i_1 + \cdots + i_n)[p\sigma]$  is zero on account of the first factor. Thus:

**Theorem 1.2.2.** *If one draws displacements from a variable element  $p$  to arbitrarily many fixed elements, then every multiple sum of these displacements whose sum of coefficients is zero is a constant magnitude.*

Since these equations can be derived from each other, it follows immediately that, if two arbitrary multiple sums of those displacements with the same two initial elements  $p$  and  $\sigma$  are equal, then their characteristic equality must continue to hold with each evolution of  $p$ , and likewise if such a multiple sum maintains the same value with respect to two initial elements  $p$  and  $\sigma$ , its sum of coefficients is zero and it therefore maintains the same value for any evolution of  $p$ .

### § 1.3. Deviation of an Element, of an Elementary Assembly. Weight

In order to express the results of the preceding paragraph more simply, we introduce some nomenclature that we also retain for geometry.

Thus we mean by the *deviation* of an element  $a$  from an element  $p$  the displacement  $[pa]$ ; by the *collective deviation* of a series of elements from an element  $p$  the sum of the deviations of the individual elements of that series from the element  $p$ . If several ( $m$ ) of those elements coincide with one ( $a$ ), then the deviation  $[pa]$  of this element will also appear as often ( $m$  times) in that sum. By this we are led to a generalization of the concept; thus, calling an assembly of elements, each of which is endowed with a specific numerical magnitude, an *elementary assembly*, we must mean by the collective deviation of an elementary assembly from an element  $p$  a multiple sum of the deviations of the elements belonging to that assembly from the element  $p$ , whose coefficients are the numerical magnitudes with which the respective elements are endowed. We call the sum of these numerical magnitudes the *weight*<sup>1</sup> of the elementary assembly, so that the numerical magnitudes with which the individual elements are endowed are their respective weights. Thus if the elementary assembly consists of the elements  $a, \beta, \dots$ , and their respective weights are  $a, b, \dots$ , then the deviation of that elementary assembly from an element  $p$  equals

$$a[pa] + b[p\beta] + \dots$$

Thus we have the theorem:

**Theorem 1.3.1.** *If two elementary assemblies deviate from the same element equally and their weights are equal, or if they deviate equally from the same two elements, then they deviate from every other element equally, and in the latter case their weights are equal.*

And:

**Theorem 1.3.2.** *An elementary assembly with zero weight deviates from any two elements equally, and an elementary assembly deviating from two elements equally has zero weight and deviates from all elements equally.*<sup>2</sup>

### § 1.4. Concept of Elementary Magnitudes and Their Sums

Each structure is fixed as a magnitude once the range of its equality and difference is specified. Thus we regard two elementary assemblies as equal magnitudes, and indeed as equal elementary magnitudes, if their deviations from the same element always have equal values. An elementary assembly is therefore an elementary magnitude to the extent that one disregards the particular nature of its composition and retains only the value of the deviation that it forms with other elements, so that an elementary magnitude can be present in another way as an elementary assembly, and each elementary assembly is interpreted as a particular embodiment of an elementary magnitude, or, as we have indicated above, as an elementary or concrete representation of an elementary magnitude.

<sup>1</sup>The name “weight” is also ordinarily used in an abstract sense in mathematics in other contexts (in probability theory) and requires no further justification here.

<sup>2</sup>It is thus understood that each individual element, by itself or endowed with a numerical magnitude, can be interpreted as an elementary assembly, with the weights of the other elements being zero.

It is therefore already understood that by the deviation and weight of an elementary magnitude is meant what we understand by the deviation and weight of the elementary assembly belonging to it, and that two elementary magnitudes can then only be equal if they have equal weights and deviations, but that the equality of the elementary magnitudes already follows just as well if only any two such values are known to be equal.

Our problem is now to ascertain the types of conjunction into which the different elements and respective weights of an elementary assembly must enter if an elementary magnitude is to appear as the result of the conjunction.

The conjunctions are of two types; on the one hand between an element and the numerical magnitude, the weight, belonging to it, and on the other between the elements with their weights, and generally between the elementary assemblies, insofar as their deviations are concerned, that is, between elementary magnitudes as such.

We first consider the latter type of conjunction; it is clear that the collective deviation of an elementary assembly remains the same, in whatever order one takes the individual terms of this assembly, and that one may group them among themselves into particular assemblies, and finally that, if one adds elementary assemblies that have equal deviations to elementary assemblies that have unequal deviations, the collective assemblies so generated must also have unequal deviations, and indeed everything will remain that is valid for the addition of displacements. The interchangeability and combinability of terms, and on the other hand the law that if one term of the conjunction remains constant, then the result only remains constant if the other term is also constant, defines that conjunction as additive according to §6, and the laws of addition and subtraction are thus generally valid for this conjunction.

Now regarding the conjunction of the element with its weight, it is clear that if in an elementary assembly the same element is found several times and indeed with different weights, one can instead include the element once, with a weight equal to the sum of the individual weights, without the deviations of the assembly being changed, as is known from the laws for the multiplication of numerical magnitudes by displacements. Thus temporarily denoting this second method of conjunction by the symbol  $\Pi$ , if  $a$  is an element and  $m$  and  $n$  the weights, one has

$$m \Pi a + n \Pi a = (m + n) \Pi a,$$

an equation that represents the multiplicative principle relative to the first conjunctive factor. But since the concept of the conjunction of a numerical magnitude with an assembly of several elements is not yet given, the other half of that principle cannot yet appear. Yet that conjunction, to the extent that it is defined, is defined as multiplicative.

Combining all this, the elementary magnitude is an assembly of elements  $a, \beta, \dots$ , with respective weights  $a, b, \dots$ , equal to

$$aa + b\beta + \dots,$$

that is, it is represented as a multiple sum of elements whose coefficients are the weights of the respective elements, and at the same time the addition of elementary magnitudes is thereby defined.

### § 1.5. Composition of These Magnitudes

Now in order to represent the multiplicative conjunction more generally, we have to define the multiplication of a numerical magnitude with an elementary magnitude so that the other half of the multiplicative principle also remains true; this is accomplished when we assert that a multiple sum of elements is multiplied by a numerical magnitude if one multiplies its coefficients by this numerical magnitude.

Consequently if  $a$  and  $b$  are arbitrary magnitudes, that is if they represent multiple sums of elements, the validity of the two multiplicative principles

$$ma + na = (m + n)a$$

and

$$ma + mb = m(a + b)$$

follows immediately. It also follows easily that the result of division by a numerical magnitude, so long as it is not zero, is defined, since different elementary magnitudes (that is, such as have unequal deviations from the same element) must have unequal deviations, and thus remain different, when multiplied by the same nonzero numerical magnitude. And likewise, if we call similar elementary magnitudes such as proceed from the same elementary magnitude upon multiplication by numerical magnitudes, it also easily follows that the quotient of two similar elementary magnitudes yields a definite numerical magnitude if the divisor is not zero. Thus all the laws for arithmetic multiplication and division are valid for the conjunction in question.

The conjunction of the element  $p$  with other elements or elementary magnitudes, as they enter with the notation introduced above for the deviation, we reserve for the next chapter.

### § 1.6. The Elementary Magnitude as Multiple Element

So far the elementary magnitude appears in general as a multiple sum of elements, and we must set ourselves the problem of representing an elementary magnitude, given in this form, in the simplest possible form.

We begin by attempting to represent it as a single term, that is as a multiple element. Thus if

$$aa + b\beta + \cdots = x\sigma,$$

where  $\sigma$  denotes an element,  $x$  its weight, then since the collective weight on both sides must be the same we obtain at once

$$x = a + b + \cdots,$$

and thus we have only to define  $\sigma$  so that the collective deviations from any given element  $p$  for the two sides are equal, and obtain

$$a[pa] + b[p\beta] + \cdots = (a + b + \cdots)[p\sigma],$$

that is

$$[p\sigma] = \frac{a[pa] + b[p\beta] + \cdots}{a + b + \cdots},$$

whence  $\sigma$  is defined so long as  $a + b + \cdots$  has a nontrivial value, that is:

**Theorem 1.6.1.** *An elementary magnitude whose weight is not zero can be represented as an element with equal corresponding weight, and indeed the deviation of this element from the element  $p$  equals the deviation of the elementary magnitude from the same element, divided by the weight.*

Furthermore, if in this equation, which is valid for every element  $p$ , one sets this element equal to  $\sigma$ , then since  $[\sigma\sigma]$  is zero one has, upon dropping the divisor, the equation

$$0 = a[\sigma a] + b[\sigma b] + \cdots,$$

that is, the collective deviation of a multiple sum of elements from the summation element ( $\sigma$ ) is zero.

### § 1.7. The Elementary Magnitude with Weight Zero is a Displacement

If the weight of an elementary magnitude is zero, then we have already shown that the deviations of that elementary magnitude from any two magnitudes are equal; thus if this deviation is zero with respect to any given element, then it is also zero with respect to any other, and that elementary magnitude can be set equal to any arbitrary element with zero weight, as was also shown from the formulas of the previous paragraph, or it can itself be set to zero. If however the deviation of such an elementary magnitude (whose weight is zero) from any given element equals a displacement of nontrivial magnitude, then its deviation is also equal to the same displacement from every other element, and this displacement, which measures that constant deviation, thus represents that elementary magnitude completely, so that to equal elementary magnitudes whose weights are zero also belong equal deviations, and conversely.

If such elementary magnitudes are added together or multiplied by numerical magnitudes, then the deviation of the resultant of those elementary magnitudes follows from the same addition or multiplication; thus between such elementary magnitudes and their deviations no distinction appears, neither as such, that is in the extent of their concepts, nor in their conjunctions. We are therefore justified in defining that elementary magnitude and its deviation as equal; indeed we are forced to it if we are not to complicate the subject with useless distinctions. We therefore set an elementary magnitude whose weight is zero equal to that constant displacement by which that magnitude deviates from an arbitrary element, or, we mean by the deviation of a displacement from an element that displacement itself and the displacement appears as a particular type of elementary magnitude.

In order to understand this more clearly we can first establish that every elementary magnitude of zero weight can be represented as the difference of two elements ( $\beta - \alpha$ ), one of which ( $\alpha$ ) is arbitrary. Thus, since the collective weight of this difference is likewise zero, it is only important that the deviations with respect to any given element ( $p$ ) be equal. The deviation of that difference from  $p$  is  $[p\beta] - [p\alpha]$ , that is it equals  $[\alpha\beta]$ , and therefore not only is the element  $\beta$  defined if  $\alpha$  is given, but the constant deviation of the given elementary magnitude itself is found, and it thus follows in addition that

$$[\alpha\beta] = \beta - \alpha.$$

These two expressions therefore only represent different notations, and since the former is arbitrary, the latter necessary, we prefer to drop the former, regarded from the beginning as only tentative, in favor of the latter, and henceforth denote by  $\beta - \alpha$  a displacement having  $\beta$  as its final element and  $\alpha$  as its initial element.<sup>3</sup>

Combining the results of both these paragraphs, it was shown that:

**Theorem 1.7.1.** *An elementary magnitude of first order (as we designate the elementary magnitudes considered so far, in contrast to those to be considered later) can be represented as a multiple element if its weight has a nontrivial value, and as a displacement if its weight is zero, and in every case one obtains this value by equating the weights and the deviations from any given element, whence the deviation of a displacement from an element is equated with that displacement itself and the weight of a displacement is set to zero.*

## § 1.8. Sum of a Displacement and a Simple or Multiple Element

Since, according to the previous paragraph, the displacement appears as a particular species of elementary magnitude, the sum of a displacement and a simple or multiple element can likewise be interpreted as an elementary magnitude, and we will now subject the concept of this sum, which is already defined by what has gone before, to a closer inspection.

If we seek the sum  $(a + p)$  of an element  $a$  and a displacement  $p$ , then since the weight of this sum is unity it must be set equal to a simple element  $\beta$ . One therefore has from the equation

$$a + p = \beta$$

the new equation

$$\beta - \alpha = p,$$

that is,  $a + p$  denotes the element  $\beta$  into which  $a$  is transformed if it is evolved by  $p$ , or whose deviation from  $a$  equals  $p$ . If we consider the sum of a multiple element  $ma$  and a displacement  $p$ , then since the weight of the sum is  $m$  we have the equation

$$ma + p = m\beta,$$

and thus

$$m(\beta - \alpha) = p,$$

or

$$\beta - \alpha = \frac{p}{m},$$

that is,  $ma + p$  denotes the  $m$ -fold of an element  $\beta$  whose deviation from  $a$  is the  $m$ -th part of the displacement  $p$ . Or, we interpret both together and express it in a more general way by simultaneously imagining that if  $\beta$  deviates from  $\alpha$  by  $\frac{p}{m}$ , then  $m\beta$  deviates from  $ma$  by  $p$ , whence it follows that

---

<sup>3</sup>It should be mentioned here that the formulas of the previous paragraph for this case represent the elementary magnitude as an infinitely distant element with weight zero, provided one admits as valid division by zero; but the specific meaning of this expression comes to light just precisely through the presentation given here.

**Theorem 1.8.1.** *The sum of an elementary magnitude of nontrivial weight and a displacement is an elementary magnitude of weight equal to that of the first and deviating from the elements of the first by the added displacement.*

## B. Applications

### § 1.10. Mean of an Assembly of Points

If we apply the results obtained in this section to geometry, we need only replace elements by points; and we then retain the nomenclature introduced in this chapter, that is “weight,” “deviation,” “elementary magnitude,” with the same meaning, thus obtaining the same theorems, of which however we will present only the most interesting in a more intuitive form.

First of all, if one is given  $n$  points  $a_1, \dots, a_n$ , then it is always possible to find a point  $\sigma$  whose deviation from any arbitrary point  $p$  is the  $n$ -th part of the collective deviation of those  $n$  points from the same point  $p$ , and this point is completely defined by an equation such as

$$[p\sigma] = \frac{[pa_1] + \dots + [pa_n]}{n}.$$

It is this point that one usually calls the point of average distance between those  $n$  points, but I have more briefly called their mean (cf. §24).

Expressing the above theorem geometrically, we can say:

**Theorem 1.10.1.** *If one draws displacements from a variable point  $p$  to  $n$  fixed points, then the parallel to the sum of these displacements issuing from  $p$  goes through a fixed point,  $\sigma$ , called the mean of these points, whose distance from  $p$  is in the  $n$ -th part of that sum.*

Or, if we also wish to avoid the concept of the sum:

If one draws displacements from a variable point  $p$  to  $n$  fixed points, and without changing their directions or lengths connects these displacements contiguously, that is adjacent to each other so that the final point of each of the displacements is always the initial point of the next, and makes  $p$  the initial point of the first, then the line that closes the figure so formed goes through a fixed point,  $\sigma$ , called the mean of the  $n$  points, which cuts this closing side at its  $n$ -th part from the point  $p$ .

Thus there results the simplest possible construction for the mean, and simultaneously the law that the displacements drawn from the mean to the  $n$  points always give a closed figure when placed end to end, or that they are equal and parallel to the sides of a closed figure.

### § 1.11. The Mean as Axis

It is clear that the laws established in the previous section are also valid if several of the fixed points are united, provided one keeps the same number of them, and thus also if one imagines these points multiplied by arbitrary positive or negative numerical magnitudes, which we can also call weights, at least so long as the weights have a nontrivial sum; we can then call the collection of points thus endowed with weights a point assembly, whence we can state the theorem: “If one draws displacements from a variable point  $p$  to the points of a fixed point assembly and multiplies

these displacements by their respective weights, and without changing their directions arranges the displacements so produced end to end from  $p$ , then the closing side of the figure goes through a fixed point  $\sigma$ , the mean of the point assembly, whose distance from  $p$  is included in the closing side in the amount of the collective weight.”

If the collective weight is zero, then as follows from the formula

$$[p\sigma] = \frac{a[pa] + b[p\beta] + \cdots}{a + b + \cdots},$$

the point  $\sigma$  goes to infinity, and the closing side then goes through the same infinitely distant point, that is, it has a constant direction. This yields even more simply and at the same time more definitively the theorems established above for the case that the collective weight is zero, and it therefore follows at once that this closing side has a constant length. An infinitely distant point thus appears as the mean of a point assembly if the collective weight is zero, or what is the same thing, as a constant direction, thus not as a (finitely distant) center but rather as a central axis. Since this case is of particular interest, we express it once more with the greatest possible avoidance of technical terms:

**Theorem 1.11.1.** *If one draws displacements from a variable point  $p$  to a series of fixed points, to which belong a series of numerical magnitudes whose sum is zero, and if after one has multiplied these displacements by their respective numerical magnitudes, one lays the results end to end without changing their directions, then the closing side has constant direction and length and can be called the axis of that point assembly.<sup>4</sup>*

## § 1.12. Center of Gravity. Axis of Equilibrium

We establish directly the fundamental principle with respect to statics, namely:

**Theorem 1.12.1.** *If the points of an assembly are acted upon by parallel forces in any fixed direction of magnitude proportional to the weights of those points, then the collective moment of those forces with respect to the mean of that assembly is zero, and with respect to any other point is equal to the collective force applied to the mean.*

The proof is very simple. Thus if  $\sigma$  is the mean of the assembly  $aa, b\beta, \dots$ , and  $a\rho, b\rho, \dots$  are the forces acting at the points  $\alpha, \beta, \dots$ , then one finds that the collective moment with respect to  $\sigma$  is equal to

$$a[\sigma\alpha] \cdot \rho + b[\sigma\beta] \cdot \rho + \cdots = (a[\sigma\alpha] + b[\sigma\beta] + \cdots) \cdot \rho = 0,$$

since the first factor is zero according to the previous paragraph. For any other point  $p$  the moment equals

$$(a[p\alpha] + b[p\beta] + \cdots) \cdot \rho,$$

and since the first factor is just  $(a + b + \cdots)[p\sigma]$ , this equals

$$[p\sigma] \cdot (a + b + \cdots) \cdot \rho,$$

---

<sup>4</sup>If one is to express the results of this paragraph in a purely geometrical form, one must then replace weights with parallel displacements whose magnitudes represent the ratios of the weights.



that is, the moment of the collective force applied to  $\sigma$ .

It is well enough known from the former characteristic that, if the weights are interpreted as physical weights, the mean is called the *center of gravity*. Since the physical weights always appear as positive, the second case has no direct application here. If however one imagines a body in a fluid, and completely immersed in the fluid, and one calculates together the force with which each of its particles is driven downward by its physical weight and that with which it is driven upward by the pressure of the fluid (which is equal to the physical weight of the displaced fluid), and regards this collective force as the mathematical weight of the particle concerned, then indeed one has both positive and negative weights. In particular, if the body floats within the fluid then the sum of those weights is zero, and instead of the center of gravity endowed with a weight there appears a specific displacement as the sum of the point assembly representing the body floating within the fluid. This displacement can indeed be zero, in which case the body floats at equilibrium in any position; but in every other case the direction of the displacement specifies the axis of the body floating in the fluid that must be vertical if the body is to be at equilibrium.

How to find the direction and length of this displacement, which (as we will show in the next section) has a definite and simple meaning in mechanics, is clear at once from the following theorem, which is an immediate consequence of the concept of the sum of several elementary magnitudes:

**Theorem 1.12.2.** *If a body is the union of several bodies, then from the centers of gravity and weights of the individual bodies one finds the center of gravity and weight of the whole, or the displacement that replaces them both, by taking the sum of their centers of gravity combined with their respective weights.*

In our case the center of gravity of the body as such and that of the displaced fluid are taken, and each is multiplied by its respective weight, the two weights being oppositely signed; and, since the weights are equal in the case that the body floats within the fluid, one obtains as sum this weight multiplied by the opposing deviations of the two centers of gravity; the axis then goes through both centers of gravity and is zero if they coincide.

### § 1.13. Magnetism, Magnetic Axis

Another important application of the latter case, in which an axis appears instead of a center of gravity, is in magnetism.

Gauss has shown<sup>5</sup> that the magnetic intensity inside a magnetic body always sums to zero. If one imagines this intensity as attributable to the mathematical weights of the body's respective points (or particles), then the sum of the point assembly so formed is of definite direction and length.

In order to learn the meaning of this displacement for the theory of magnetism we imagine a magnetic force which, perhaps like terrestrial magnetism or the force of a distant magnet, impels the individual points along parallel directions in proportion to the magnetic intensity; then the moment of this force with respect to any given point  $p$  equals

$$a[p\alpha] \cdot \rho + b[p\beta] \cdot \rho + \dots$$

---

<sup>5</sup>In his article "Intensitas vis magneticae," [Werke, vol. V, p. 79 ff.].

if  $a\rho, b\rho, \dots$  are the forces acting on the points  $\alpha, \beta, \dots$  in proportion to the magnetic intensities  $a, b, \dots$ . This expression is however altered if one places the common factor  $\rho$  outside parentheses and then supposes that the magnitude enclosed in parentheses equals that constant displacement represented by the sum of the point assembly, which we will denote by  $\alpha$ , thus

$$\alpha \cdot \rho,$$

that is, the moment of that force is equally large with respect to any two points, so that, if we call  $\alpha$  the *magnetic axis*,  $\rho$  the *active magnetic force* (when it acts on a point taken to have unit intensity), it is equal to the outer product of the magnetic axis with the active magnetic force. Equilibrium is thus attained if the product is zero, that is if the magnetic axis lies in the direction of the active force.

The concept of the magnetic axis as I have presented it here thus only differs from that otherwise current in that it is here interpreted as a displacement of specific direction and length, while one is usually accustomed to keeping only the direction. The grounds on which I have modified this concept, without changing the nomenclature, follow easily since on the one hand science requires the conjunction of direction and length in a single concept, and on the other what is asserted about the magnetic axis always follows immediately, whether the length is included in the concept or not, so that no confusion is possible. That previously one always considered the two separately in the theory of magnetism is only a consequence of the fact that the unity of direction and length, as we have interpreted them in the concept of displacement, has so far found no place in geometry. However, the extraordinary simplicity with which it is possible to represent this concept and the conjunction with the magnetic moment offered by our science already suffices to demonstrate the absolute necessity of our science for the theory of magnetism.

## § 1.14. Note

Here we are led to the one and only point at which our science approaches something already otherwise known. Thus, in the *Barycentrische Kalkül* of Möbius an addition of elementary and multiple points is likewise presented, it is true at first only as a notational abbreviation, but nevertheless with the same calculational method we have presented, but with greater generality, in the §94 of this chapter. What is however completely lacking there is the interpretation of the sum as a single magnitude for the case that the weights amount to zero when taken together.

What prevented the ingenious author of that work from interpreting this sum as a displacement of constant length and direction is without doubt the unfamiliarity of combining length and direction in a single concept. Were that sum there established as a displacement, there would result the concepts of the addition and subtraction of displacements in geometry as I have presented them in chapter 1 of part 1, and our science would then have found a second point of contact with that work; the barycentric calculus would then have achieved a much freer and more general treatment as well.<sup>6</sup>

---

<sup>6</sup>(1877) When I wrote this note I had not seen the *Mechanik* of Möbius [Leipzig 1843], in which he treats the addition of displacements. The article in Crelles Journal (vol. 28), in which Möbius bases the barycentric calculus on the method suggested here, appeared only after the *Ausdehnungslehre*, although the date of issue of the latter is later; in any case it was written earlier. This is one of those remarkable contacts between scientific works that often occur, to the surprise of those whose works coincide. [What is meant is *Die Elemente der Mechanik des Himmels*,

### § 1.15. Application to Differential Calculus

This seems to me to be the best place to at least suggest the application of our science to differential calculus.

In order to proceed to such an application, we must represent the magnitudes obtained by our science as functions. This is done most easily if the independent variable is regarded as a numerical magnitude, perhaps equal to  $t$ . Then each magnitude  $P$  can be represented in the form

$$P = A + Bt_1 + Ct_2 + \dots,$$

or more generally in the form

$$P = A_mt^m + A_nt^n + \dots,$$

where  $A, B, C, \dots$  or  $A_m, A_n, \dots$  are necessarily magnitudes of the same order as  $P$  and must be imagined as independent of  $t$ . Now if we set this expression, as a function of  $t$ , equal to  $f(t)$ , thus

$$P = f(t),$$

and further

$$dP = f(t + dt) - f(t),$$

then in general we obtain

$$\frac{dP}{dt} = mA_mt^{m-1} + nA_nt^{n-1} + \dots.$$

The simplest case appearing here is that in which  $P$ , and thus also  $A_m, A_n, \dots$ , are elementary magnitudes of first order. If in particular one assumes that  $P$  has a constant weight, then if one symbolizes the magnitudes by lower case letters, as magnitudes of first order, it can be represented in the form

$$p = a + b_mt^m + b_nt^n + \dots,$$

where  $b_m, b_n, \dots$  represent displacements, and thus  $a$  and  $p$  elementary magnitudes of equal weight. Then one obtains

$$\frac{dp}{dt} = mb_mt^{m-1} + nb_nt^{n-1} + \dots,$$

and thus represents a displacement.

One easily perceives that if  $p$  represents the position of a point at time  $t$ , then  $\frac{dp}{dt}$  represents its velocity in magnitude and direction, and  $\frac{d^2p}{dt^2}$  its acceleration, represented in the same way. By the introduction of this procedure in mechanics one achieves, upon application of this analysis, the simplest way to the solution of many a problem that otherwise appears complicated; but the further prosecution of this subject would lead me too far from my goal.<sup>7</sup>

---

cf. Möbius Ges. Werke, vol. 4, p. 1–318. The article is in Crelles J. 28, pp. 1–9, and in the Ges. Werke, vol. 1, pp. 601–612].

<sup>7</sup>(1877) Cf. my article “Die Mechanik nach dem Principien der Ausdehnungslehre” in Mathematische Annalen, vol. XII, pp. 222–246.

## Chapter 2

# Outer Multiplication, Division, and Shadow of Elementary Magnitudes

### A. Theoretical Development

#### § 2.2. The Extent to Which the Displacement can be Interpreted as a Product

The concept of deviation, which we took as a basis for the development of the preceding chapter, contains the germ of the concept of the product of two elementary magnitudes.

There, by the deviation of an element  $a$  from another element  $p$  we meant the displacement that can lead from  $p$  to  $a$ , which we symbolized by  $[pa]$ ; likewise, by the deviation of an elementary assembly from an element  $p$  we meant the multiple sum of the deviations of its elements from the same element  $p$ , provided one takes as coefficients in this multiple sum the numerical magnitudes (weights) belonging to the respective elements. Thus we defined the elementary magnitude corresponding to an elementary assembly in such a way that it could replace that assembly, provided we are only dealing with the deviation, and established precisely the equality of deviations as the sole criterion for the equality of elementary magnitudes. It therefore follows that the elementary magnitude belonging to an elementary assembly is again the multiple sum of elements endowed with their weights as coefficients, that is the corresponding multiple sum of elements, since the collective deviation of that set is a multiple sum of the deviations of the elements.

Thus if we symbolize the deviation of an elementary magnitude  $a$  from an element  $p$  by  $[pa]$  as well, then we have

$$[p(aa + b\beta + \dots)] = a[pa] + b[p\beta] + \dots ;$$

and thus also, since the collective deviation of an elementary assembly is the sum of the deviations of its parts,

$$[p(a + b + c + \dots)] = [pa] + [pb] + [pc] + \dots ,$$

if  $a, b, \dots$  represent arbitrary elementary magnitudes. Subsequently we defined the product of a numerical magnitude with an elementary magnitude, that is with a multiple sum of elements, as a multiple sum resulting from the former by multiplying its coefficients with that numerical magnitude, whence it now follows that one finds the deviation of an  $m$ -fold elementary magnitude

if one multiplies the simple one by  $m$ , so that<sup>1</sup>

$$[p(ma)] = m[pa].$$

It can easily be shown that the fundamental multiplicative relation for the conjunction in question between  $p$  and an elementary magnitude, both in itself and with respect to the adjunction of numerical factors, is valid so long as one considers only the second factor as a sum of terms. Further, since  $[pp]$  is zero and  $[pa]$  equals  $-[ap]$ , it follows that this is an outer multiplication.

### § 2.3. Elementary Systems

Before we proceed to the full concept of the outer product of elementary magnitudes we will establish the concept of an elementary system.

This concept, like that of the extensive system (§16), is based on the concept of dependence. We call an elementary magnitude of first order dependent on other elementary magnitudes if it can be represented as a multiple sum of them, whereas we call several elementary magnitudes of first order independent if no dependence, in the sense given, exists between them, that is, if it is not possible to represent any one of them as a multiple sum of the rest. Now by an *elementary system* of  $n$ -th order we mean the collection of elements dependent on  $n$  elements, these  $n$  elements being mutually independent.

Thus if  $\alpha, \beta, \gamma, \dots$  are the  $n$  mutually independent elements and I consider two elements  $p$  and  $\sigma$  dependent on them, then their difference can also be represented as a multiple sum of those  $n$  elements; this difference, which represents the mutual deviation of the two elements, has zero weight, and one thus obtains  $p - \sigma$  represented in the form

$$p - \sigma = a\alpha + b\beta + c\gamma + \dots,$$

where simultaneously

$$a + b + c + \dots = 0.$$

If one expresses any one of these coefficients, for example  $a$ , in terms of the rest by means of the latter equation, then upon introducing this value into the former equation one obtains

$$p - \sigma = b(\beta - \alpha) + c(\gamma - \alpha) + \dots,$$

that is, the mutual deviation of two elements of an elementary system of  $n$ -th order is representable as a multiple sum of  $(n - 1)$  displacements that extend from one of the elements defining the system to the rest of them; and conversely that displacement that can be represented as a multiple sum of these  $(n - 1)$  displacements necessarily leads from an element of that system to another element of the same system.

We can therefore also say that an elementary system of  $n$ -th order is the collection of elements whose mutual deviations belong to a single extensive system of  $(n - 1)$ -th order, or if one wishes to express it thus, it is the elementary representation of an extensive system of  $(n - 1)$ -th order.

---

<sup>1</sup>From this it follows in addition that it is also correct to introduce the elementary magnitudes  $a, b, \dots$  instead of  $\alpha, \beta, \dots$  in the first equation of this paragraph.

I remark however that it is immediately implicit in the concept of the elementary system that  $n$  elements are mutually independent if and only if they belong to no lower elementary system than that of  $n$ -th order.

## § 2.4. Outer Product of Elementary Magnitudes Formally Defined

Now in order to proceed at once to the concept of outer multiplication of arbitrarily many elementary magnitudes, we need only apply the general (formal) concept of outer multiplication to these magnitudes.

Thus the concept of multiplication is already defined by the fact that in a product of two factors, one of which consists of two similar terms, one replaces it by its individual terms, and the products so formed, which are again regarded as similar, may be added. The product of several magnitudes of first order (as we have called such elementary factors) will then be defined as an *outer product* if in each elementary factor such terms as are similar to one of the two elementary factors can be dropped without changing the value of the result. We therefore also define the concept of the multiplication of elementary magnitudes of first order by this principle, and simultaneously also maintain for elementary magnitudes all the definitions given in part 1 for extensive magnitudes; and since all the laws proved in part 1 depend on those principles and the accompanying definitions, they are therefore also valid for elementary magnitudes, in particular all laws of outer multiplication, formal addition and subtraction, division, and shadow. With respect to the latter I remark only that the name “projection” cannot be used here, since, as will be shown later, relative to elementary magnitudes it comprises a completely different concept from what we have previously denoted by the name “shadow.”

Our problem thus remains in particular to give our concepts the greatest clarity, and to illustrate their concrete representation.

## § 2.5. Realization of This Product; Divergence, Rigid Elementary Magnitude

The main point here is to ascertain when two products can be set equal, since the extent of the concept of the magnitude that the product represents is thereby defined. Now since the concept of the product is to be completely defined by that formal principle, we can set two products equal if and only if, by means of that principle (or what can be derived from it), one of the products can be transformed into the other.

Thus consider a product of  $n$  elementary magnitudes of first order. First of all, it is clear that if the weights of these  $n$  elementary magnitudes are each individually taken to be zero, then each of them appears as an extensive magnitude of first order, and in addition their product produces an extensive magnitude of  $n$ -th order. In every other case, and thus if even a single elementary factor has a nontrivial weight,<sup>2</sup> that product can be represented as a product of an element with an extensive magnitude of order  $(n - 1)$ . For we can first bring that factor, whose weight we assume is not zero, to the initial position; and should the sign of the product change thereby, we can then change the sign of any one of the factors. Now if  $a\alpha$  is that factor whose weight is not zero, then we can introduce an arbitrary multiple of  $\alpha$  into the remaining factors, if the weights of the latter are

---

<sup>2</sup>That is, one that is not zero.

not zero, without changing the value of the product, and thus bring the weights of the remaining factors to zero. Once this is done, the remaining factors are displacements; if their product, which is an extensive magnitude of  $(n - 1)$ -th order, is  $Q$ , then the elementary magnitude equals

$$a\alpha \cdot Q,$$

and, since  $a$  is a numerical magnitude, this again is equal to

$$a \cdot \alpha Q = a \cdot P$$

if  $\alpha Q$  is set equal to  $P$ . This therefore proves the proposition, and even more: For since the multiple of  $\alpha$  added to the individual factor is defined, as it is to make the weight of that factor zero, it yields a definite value for  $Q$ , and thus for  $P$  as well.

Now in order to show that  $P$  always maintains a single, definite value, whatever change of form one may previously have undertaken with that product, we have only to establish that all changes of form of a product that leave its value unchanged depend on the fact that one can introduce into each elementary factor terms that are similar to the remaining factors. If at first we leave the factor as in the original product unchanged, but add to any other factor a term similar to one of the remaining factors, say the factor  $b\beta$ , that is if for example we add the term  $m\beta$  (where  $m$  denotes a numerical magnitude), then in order to bring the weight of this augmented factor to zero there must be subtracted the magnitude  $ma$  in addition to what was previously subtracted, whence the complete added term equals  $m(\beta - \alpha)$ . But by the same transformation the factor  $b\beta$  is changed into  $b(\beta - \alpha)$ ; and thus according to the transformation indicated, the term added to the one factor is similar to the other, that is the product  $Q$  and hence also  $P$  maintain the same value.

Thus we have shown that the value  $P$ , appearing as second factor, is defined if  $\alpha$  is unchanged; but now  $\alpha$  can increase by any displacement belonging to the system  $P$ ; if one such is  $p_1$ , then one has

$$(\alpha + p_1) \cdot P = \alpha \cdot P,$$

that is, the element  $\alpha$  can be changed into any element of that elementary system defined by  $\alpha$  and  $P$  while  $P$  maintains the same value, and the extent of the concept is thus defined.

Now we call a product of  $n$  elementary magnitudes of first order, or a sum of such products, an *elementary magnitude* of  $n$ -th order, and such a product whose elementary factors are not all displacements a *rigid elementary magnitude*. Thus we have proved the theorem that “a rigid elementary magnitude of  $n$ -th order can be represented as a product of an element with an extension of  $(n - 1)$ -th order, and this extension, which we call the *divergence* of that elementary magnitude, is completely defined by it, but as its element any one belonging to the system defined by the elementary factors of the elementary magnitude may be adopted.” The rigid elementary magnitude thus appears in general as the union of the elementary system depending on it and its corresponding divergence; and the unitary concept of an elementary magnitude of higher order, or what is the same thing, a product of elementary magnitudes, is given by their interrelation, that is by the unification of the two ideas into one.

We will now complete the idea of the rigid elementary magnitude by seeking to exhibit it as a definite part of the elementary system to which it belongs.

## § 2.6. The Vertex Structure

According to the concept established in the preceding paragraph, the product of two elements  $\alpha, \beta$  is the displacement  $\alpha\beta$  bound to, and thus as it were rigidified by, the elementary system defined by  $\alpha$  and  $\beta$ .

We based the concept of the displacement on that of the elementary extensive structure of first order. By that we meant the collection of elements into which the generating element transforms by continuous application of the same evolution; we called the generating element in its first state the initial element of the structure, in its last the final element, both of these boundary elements, and all other elements of the structure we designated as lying between the boundary elements. Thus we can also say that the elementary structure  $\alpha\beta$  is the collection of elements lying between  $\alpha$  and  $\beta$ , whereby according to the concept of continuity it is immaterial whether or not we include the boundary elements with them, since these latter represent no extension as such.

This structure will now be interpreted as an elementary magnitude of second order if one only retains on the one hand the elementary system of second order to which it belongs, and on the other the method of generation, so that two such structures belonging to the same elementary system of second order and generated by the same evolution are equal as elementary magnitudes, but only two such. Or, one imagines the whole elementary system generated by continuous prosecution of the same evolution, and two of its elements as corresponding, and as corresponding otherwise any two elements that are generated from the corresponding ones by the same evolution; thus two structures corresponding in this way will appear as equal elementary magnitudes of second order.

Now if we apply this to elementary magnitudes of higher order, and thus consider three or more elements  $\alpha, \beta, \gamma, \dots$ , there likewise arises here the problem of finding the collection of elements lying between these elements, and of comparing this collection with the product of the elements. We have already established what we mean by an element lying between two elements; now each element finding itself between an element  $\alpha$  and an element between  $\beta$  and  $\gamma$  we define as lying between  $\alpha, \beta$ , and  $\gamma$ , and generally an element lying between  $\alpha$  and an element found between a series of elements  $\beta, \gamma, \dots$  as lying between the whole series of elements  $\alpha, \beta, \gamma, \dots$ . We will tentatively call the collection of these elements a *vertex structure*,  $\alpha, \beta, \gamma, \dots$  its *vertices*, and these elements as well as the elements lying between a part of these vertices (but not between all) the *boundary elements*; those elements lying between all the vertices are the *inner elements* of the vertex structure.

Now our first problem is to represent all inner elements as multiple sums of those elements between which they lie, and to define the relation that must exist between the coefficients.

First, for the case of two elements it is clear that an element  $p$  lies between  $\alpha$  and  $\beta$  if and only if  $\alpha p$  has the same sign as  $\alpha\beta$ , so that the latter evolution appears as the continuation of the former. Now each element  $p$  lying in the elementary system defined by  $\alpha, \beta$  can be represented by the equation<sup>3</sup>

$$p = a\alpha + b\beta,$$

where  $a$  and  $b$  represent arbitrary numerical magnitudes whose sum is unity. According to the foregoing,  $p$  now lies between  $\alpha$  and  $\beta$  if and only if  $\alpha p$  has the same sign as  $p\beta$ , that is if and only

---

<sup>3</sup>Here and below, footnote numbers 21 and 22 in the original.



if

$$\alpha \cdot (a\alpha + b\beta) \quad \text{has the same sign as} \quad (a\alpha + b\beta) \cdot \beta,$$

or, upon applying the laws for outer multiplication, if  $b\alpha \cdot \beta$  has the same sign as  $a\alpha \cdot \beta$ , and hence if  $b$  has the same sign as  $a$ . That is, their sum is unity, and thus positive, if both coefficients or weights are positive. If one of them is zero, then the element is a boundary element.

By continuing the same procedure we can now show that an element  $p$  lies between a series of mutually independent elements  $\alpha, \beta, \gamma, \dots$  if and only if it can be expressed in the form

$$p = a\alpha + b\beta + c\gamma + \dots$$

with positive coefficients only.<sup>4</sup>

We say that an element  $p$  lies between a series of elements if and only if it lies between the first element of the series and an element lying between the following elements. Thus should  $p$  lie between  $\alpha, \beta, \gamma, \dots$ , then it must be found lying between  $\alpha$  and an element lying between  $\beta, \gamma, \dots$ ; it must therefore be representable as a multiple sum of  $\alpha$  and an element lying between  $\beta, \gamma, \dots$ , whose coefficients are both positive. First of course the coefficient of  $\alpha$  must be positive, but so also must the coefficient of the element lying between  $\beta, \gamma, \dots$ . But on the same grounds this element must be representable as a multiple sum of  $\beta$  and some element lying between the following elements  $\gamma, \dots$ , with positive coefficients. In the expression for  $p$ , however, this element lying between  $\beta, \gamma, \dots$  was multiplied by a positive coefficient; thus upon introducing the expression found for this element into the expression for  $p$  and dropping the parentheses,  $p$  will have been expressed as a multiple sum of the elements  $\alpha, \beta$ , and an element lying between the following elements  $\gamma, \dots$ , with positive coefficients. And since we can continue this procedure until the final element, it follows that each element lying between  $\alpha, \beta, \gamma, \dots$  can be represented as a multiple sum of  $\alpha, \beta, \gamma, \dots$ , with positive coefficients.

We have yet to show that each element representable in this form is also an inner element.

If an element  $p$  is represented in the above form,

$$p = a\alpha + b\beta + c\gamma + \dots,$$

where  $a, b, c, \dots$  are positive coefficients, then the sum of all terms following  $a$  has the weight  $b + c + \dots$ , and is thus a positive number, so that if one divides the coefficients  $b, c, \dots$  by  $b + c + \dots$  and multiplies the sum by  $b + c + \dots$  it is representable as the product of a positive number with an element which, for its part, again appears as a multiple sum of  $\beta, \gamma, \dots$  with positive coefficients; consequently  $p$  lies between  $\alpha$  and an element representable as a multiple sum of the following elements with positive coefficients, and since we can continue this procedure to the last two elements, and since the element representable as the multiple sum of these last with positive coefficients lies between them, it follows that  $p$  itself lies between  $\alpha, \beta, \gamma, \dots$ . The theorem stated previously is thus proved; it is also clear that if one or more coefficients is zero while the others remain positive,  $p$  appears as a boundary element.

---

<sup>4</sup>The original footnote 22 here refers to this being valid for  $n$  elements.

### § 2.7. Comparison of the Vertex Structure with the Product. Extension of the Elementary Magnitude

Now, on the other hand, suppose I consider the product  $\alpha \cdot \beta \cdot \gamma \cdot \delta \cdots$ . Its divergence equals  $[\alpha\beta] \cdot [\beta\gamma] \cdot [\gamma\delta] \cdots$  according to §109 and describes the extensive structure having this value which results when, first, the element  $\alpha$  describes the displacement  $[\alpha\beta]$ , then each element so generated describes the displacement  $[\beta\gamma]$ , and so forth. It is therefore clear that each such element ( $\sigma$ ) results by an evolution from  $\alpha$  of the form

$$p[\alpha\beta] + q[\beta\gamma] + r[\gamma\delta] + \cdots ,$$

where  $p, q, r, \dots$  are all positive and less than one, and thus satisfies the equation

$$[\alpha\sigma] = p[\alpha\beta] + q[\beta\gamma] + r[\gamma\delta] + \cdots ,$$

and that that extensive structure includes no other elements, since boundary elements are designated by the values zero and one for the coefficients  $(p, q, r, \dots)$ .

The vertex structure between  $\alpha, \beta, \gamma, \delta, \dots$  includes the collection of elements satisfying the equation

$$\sigma = a\alpha + b\beta + c\gamma + d\delta + \cdots$$

with positive values for  $a, b, c, d, \dots$ , that is, which satisfy the equation

$$[\alpha\sigma] = b[\alpha\beta] + c[\alpha\gamma] + d[\alpha\delta] + \cdots ,$$

where  $b, c, d, \dots$  are positive and their sum is less than one. If we replace  $[\alpha\gamma]$  here by its value  $[\alpha\beta] + [\beta\gamma]$ ,  $[\alpha\delta]$  by  $[\alpha\gamma] + [\gamma\delta]$ , and so forth, then for an element  $\sigma$  of the vertex structure we obtain the equation

$$\begin{aligned} [\alpha\sigma] &= (b + c + d + \cdots)[\alpha\beta] + (c + d + \cdots)[\beta\gamma] + (d + \cdots)[\gamma\delta] + \cdots \\ &= p[\alpha\beta] + q[\beta\gamma] + r[\gamma\delta] + \cdots \end{aligned}$$

with the conditions that each coefficient must be greater than the one following, the first less than one, and the last greater than zero, that is with the conditions

$$1 > p > q > r > \cdots > 0.$$

The vertex structure therefore contains only a part of the elements included in the extensive structure corresponding to that product  $\alpha \cdot \beta \cdot \gamma \cdot \delta \cdots$ , namely those for which the conditions just introduced are satisfied.

We will now tentatively symbolize that vertex structure by  $[a, b, c, \dots]$ , where we represent  $[\alpha\beta]$  by  $a$ ,  $[\beta\gamma]$  by  $b$ ,  $[\gamma\delta]$  by  $c$ , and so on, and thus mean by it the collection of elements  $\sigma$  satisfying the equation

$$[\alpha\sigma] = pa + qb + rc + \cdots$$

with the condition

$$1 > p > q > r > \dots > 0.$$

Represented in this form, boundary elements appear as those in which a partial equality of those magnitudes  $(1, p, q, r, \dots)$  occurs. It is clear that any other sequence of  $a, b, c, \dots$  produces another vertex structure having no inner element in common with the first, and that, if one uses each of the boundary elements only once, the set of elements belonging to the vertex structures of all possible sequences of  $a, b, c, \dots$  represents the extensive structure itself. In fact, each element of this extensive magnitude will certainly appear in one and only one of the vertex structures if the coefficients  $p, q, r, \dots$  are all different; and if these coefficients are in part equal, then it will be a boundary element, which will therefore have been used only once. Since the vertex structures contain no element not also contained in that extensive structure, we can therefore regard the latter as the sum of all vertex structures resulting from all possible sequences of  $a, b, c, \dots$

We can then say, finally, that all these vertex structures are equal as parts of their system.

The equality of two parts of an elementary system consists most generally in that both include equal domains from the system of elements generated in the same sense, that is, that to each element of one domain corresponds one and only one element of any other.

To state this in more detail, we assume that  $a, b, c, \dots$  are corresponding evolutions, that is, such as resulted from the corresponding fundamental evolutions in the same way, and that the system was generated out of  $\alpha$  by them, specifically so that each pair of elements that are neighbors in one of the directions  $a, b, c, \dots$  are generated from each other by the fundamental evolution belonging to this direction. It is then clear that to each element of the vertex structure  $[a, b, c, \dots]$  corresponds one and only one element of a vertex structure in which the displacements appear in another order. Thus if  $\sigma$  is an element of the first and  $[\alpha\sigma]$  is represented as a multiple sum of  $a, b, c, \dots$ , then one immediately has the corresponding element of the other if in that multiple sum one brings  $a, b, c, \dots$  to the order of the second vertex structure without changing the order of the coefficients. Consequently, at least with respect to the assumed method of generation of the system, all those vertex structures are equal as elementary magnitudes. But from the way, introduced in §20, that we have made the system independent of the fundamental evolutions, it follows that this will also be true with respect to any other elementary method of generation of the system; thus the vertex structures are equal as such.

Now since all together they equal the product, we can say that each of them equals the product divided by a number that expresses the number of different sequences that the  $n$  factors  $a, b, c, \dots$  can assume; we call this number the *sequence number* of  $n$  elements, and if the number of factors is  $n$ , it is symbolized by  $n!$ . Thus the extension of the vertex structure equals

$$\frac{a \cdot b \cdot c \cdots}{n!}.$$

We call this value the *extension* of the product  $\alpha \cdot \beta \cdot \gamma \cdots$ , that is, the extension of the elementary magnitude.<sup>5</sup> Thus:

**Theorem 2.7.1.** *The extension of a rigid elementary magnitude equals its divergence divided by*

---

<sup>5</sup>Combination theory shows that  $n! = 1 \cdot 2 \cdot 3 \cdots n$ ; if we suppose this given, then for the value of the vertex structure we obtain  $\frac{a \cdot b \cdot c \cdots}{1 \cdot 2 \cdot 3 \cdots n}$ .

*the sequence number of the order of this divergence.*

In particular, since we suppose that two elements admit two sequences, and three elements six, the extension of a rigid elementary magnitude of third order is half its divergence, and the extension of a rigid elementary magnitude of fourth order is one sixth of its divergence<sup>6</sup>; and if we assume that one element admits only one ordering, namely just the placement of that one, and if no element is present again only one ordering is possible, namely the placement of no element, then it follows that for elementary magnitudes of first and second order extension and divergence are equal.

## § 2.8. Equal Extensive Magnitudes have Equal Divergences

The divergence or extension of an elementary magnitude of first order is a numerical magnitude, the same that we have above designated its weight. Thus, for elementary magnitudes of higher orders, there arises the problem of obtaining theorems corresponding to those we have established with respect to the weights for elementary magnitudes of first order.

First of all, one has that “if all the terms of an equation include the same element  $\alpha$  as a common factor, while the other factor in each term is an extension, then one can drop the element  $\alpha$  from all terms without affecting the validity of the equation.” That this theorem is true becomes clear if, in the given equation, one shifts a single term to the left side, and combines the rest into one term with the factor  $\alpha$ , and thus produces the equation in the form

$$\alpha \cdot A = \alpha \cdot (B + C + \dots).$$

Now since the left side represents a rigid elementary magnitude, the right side does likewise, and the divergences of both sides must therefore be equal, that is

$$A = B + C + \dots.$$

Thus upon restoring the terms of this equation to their original order once more, one has the equation whose validity was to be demonstrated.

We can therefore also call the sum of the divergences of several terms, all having the same element  $p$  as factor, the divergence of the sum, if this sum represents a formal extensive magnitude, and thus express the theorem just obtained as follows: “In an equation whose terms have the same element  $p$  as a common factor, one can replace all terms simultaneously with their divergences without affecting the validity of the equation.” By virtue of this theorem it now follows that if one multiplies all the terms of any given equation by the same element  $p$ , and replaces the terms so obtained by their divergences, the equation remains true.

As a consequence of the previous chapter we now understand that the deviation of a magnitude  $B$  from another  $A$  is the divergence of the product  $A \cdot B$ , and we have therefore obtained the theorem that one may simultaneously replace all the terms in an equation by their deviations from the same element  $p$ , or expressed more simply, that equal elementary magnitudes also deviate equally from

---

<sup>6</sup>These results correspond to the theorem in geometry that the triangle is half of the parallelogram of equal base and elevation, and the three-sided pyramid is one sixth of the parallelepiped whose edges equal three adjoining sides of the pyramid.

the same element. From this one notes that by definition it follows immediately that the deviation of an extension from an element equals the same extension, and thus is completely independent of the element.

We now introduce an equation whose terms are partly elementary magnitudes, partly extensions, and in which each of the former is represented as a product of an element with an extension, that is in the form  $aA$ . Then, upon multiplying all terms by  $p$ , each such term is converted into  $p \cdot a \cdot A$  or equivalently  $p \cdot (\alpha - p) \cdot A$ , since one can introduce into each factor of an outer product terms similar to the other factors. Now since  $(\alpha - p)$  is a displacement, it follows that  $(\alpha - p) \cdot A$  is an extension, so that one can now drop the common factor  $p$  and in this way obtain the deviation equation that results from the given equation by subtracting  $p$  from all the elements of the rigid elementary magnitudes while the terms representing extensions remain unchanged. Now if one subtracts this equation from the one given, the extensions drop out and the term  $aA$  is transformed into  $aA - (\alpha - p) \cdot A$ , that is into  $p \cdot A$ . Thus, instead of the different elements by which the deviations were multiplied, the element  $p$  appears throughout; according to the previous paragraph this common element can now be dropped and thus one obtains an equation which results from the original one by dropping the extension terms and replacing the rest with their divergences. Now since the divergence of a sum of elementary magnitudes is defined as the sum of their divergences, according to which it is already implicit that the divergence of an extensive magnitude is zero,<sup>7</sup> we can say more simply:

**Theorem 2.8.1.** *Equal elementary magnitudes have equal divergences.*

Or:

**Theorem 2.8.2.** *An equation remains correct if one simultaneously replaces all its terms by their divergences.*

From this theorem, by reversing the method of derivation by which it was obtained, there results the converse theorem:

---

<sup>7</sup>The original footnote 24 here.

# **A New Branch of Mathematics**

The Ausdehnungslehre of 1844, and Other Works

By

Hermann Grassmann

*Part One: Linear Extension Theory*

Pages 201–250



# Linear Extension Theory (continued)

Two elementary magnitudes having equal divergences, and deviating equally from any given element  $p$ , are equal (and thus deviate equally from every other element by an equal magnitude).

In particular, if

$$\alpha_1.A_1 + \alpha_2.A_2 + \dots + P$$

and

$$\beta_1.B_1 + \beta_2.B_2 + \dots + Q,$$

where Greek letters represent elements and Latin letters extensive magnitudes, are two elementary magnitudes whose divergences are assumed equal, that is

$$A_1 + A_2 + \dots = B_1 + B_2 + \dots,$$

and their divergences from any given element are also equal, that is

$$(\alpha_1 - p).A_1 + (\alpha_2 - p).A_2 + \dots + P$$

equals

$$(\beta_1 - p).B_1 + (\beta_2 - p).B_2 + \dots + Q,$$

then, observing that upon dropping parentheses in the last equation one can delete the terms involving  $p$  by virtue of the previous equation, one obtains that

$$\alpha_1.A_1 + \alpha_2.A_2 + \dots + P$$

equals

$$\beta_1.B_1 + \beta_2.B_2 + \dots + Q.$$



A particular consequence of this theorem is that an elementary magnitude whose divergence is zero equals an extensive magnitude, and deviates from all elements by an equal amount, namely by precisely this extensive magnitude. For if the deviation of that elementary magnitude from any given element  $p$  equals  $P$ , then since that deviation must by definition always represent an extensive magnitude, the elementary magnitude must itself equal  $P$ , as it has the same divergence as  $P$ , namely zero, and both deviate from the same element  $p$  by an equal amount; for the deviation of any extensive magnitude from any given element is just this extensive magnitude itself; thus that equality follows according to the theorem just proved, and from this the other part of the theorem to be proved then follows at once.

## § 2.1. Sums of Elementary Magnitudes

We can also apply the theorem of the previous paragraph to the addition of a rigid elementary magnitude  $(\alpha.A)$  and an extension  $(P)$ .<sup>1</sup>

If  $A$  is the divergence of the former, then it must also be that of the sum, since the divergence of an extensive magnitude is zero; thus if the sum is again a rigid elementary magnitude, then it must be representable in the form  $\beta.A$ , and thus  $p.A$  must in fact equal that sum, since both have equal deviations from any given element, for example  $\alpha$ ; but the deviation of the magnitude  $\alpha.A$  from  $\alpha$  is zero, whence one has as the sole constraining equation

$$P = (\beta - \alpha).A,$$

that is:

The sum of a rigid elementary magnitude and an extensive magnitude is again a rigid elementary magnitude only if the divergence of the former is subordinate to the latter, and indeed the sum is then that elementary magnitude having a divergence equal to that of the former, and deviating from an element of the former by the latter.

---

<sup>1</sup>Cf. § 112.

## B. Applications

### § 2.2. Elementary Magnitudes in Space; Linear Magnitudes, Planar Magnitudes

Having developed the generation of elementary magnitudes of higher orders from those of the first by multiplication and addition, and having clarified our perception of the concept by comparison with elementary magnitudes of first order and with extensive magnitudes, we proceed to applications in geometry and mechanics, which illustrate these concepts clearly.

First, with regard to geometry, it is clear that the straight line and the plane appear as elementary systems of second and third order. Space itself appears as an elementary system of fourth order, and the true significance of space is only represented thus.

The rigid elementary magnitude can be represented most simply as a product of an element with an extensive magnitude, which we call its divergence; and it appears itself as the divergence bound to its elementary system.

Thus if we consider the product  $(\alpha.p)$  of a point  $(\alpha)$  with a displacement  $(p)$ , then  $p$  is the divergence of this product; the straight line, drawn from  $\alpha$  in the direction of the displacement  $p$ , is the elementary system itself, and the product therefore appears as a displacement comprising part of a fixed straight line and remaining bound to that line. We call this product a *linear magnitude*, since it forms a part of a straight line, and continue to call its divergence the displacement appearing in it. Likewise, the product  $(\alpha.P)$  of a point  $(\alpha)$  with a plane area  $(P)$  of fixed direction is represented as a plane area lying in a fixed plane, namely in the plane through that point in the direction of the plane area; we call that magnitude a *planar magnitude* since it forms a part of a fixed plane, and that plane area of fixed direction its divergence. Finally, the product of a point with a volume has no other meaning than this volume itself, since space is an elementary system of fourth order, and thus every volume is already bound to it as such.

### § 2.3. Products and Sums of These Magnitudes

From this the concept of the product of several points easily follows.

First, if one considers the product of two points  $\alpha.\beta$  or  $\alpha\beta$ , then

the system to which it is bound is the straight line drawn through both points; and since

$$\alpha.\beta = \alpha.(\beta - \alpha),$$

the divergence of this product is the deviation of the second point from the first; that is, the product of two points is a linear magnitude whose line goes through the two points, and whose divergence is the displacement leading from the first to the second.

The product of three points  $\alpha.\beta.\gamma$  appears as a planar magnitude whose plane goes through those three points; and since

$$\alpha.\beta.\gamma = \alpha.(\beta - \alpha).(\gamma - \alpha) = \alpha.[\alpha\beta].[\alpha\gamma],$$

the divergence of this planar magnitude is the plane area of a parallelogram that has the deviations of the latter two points from the first as sides. Since

$$[\alpha\gamma] = [\alpha\beta] + [\beta\gamma],$$

we can also set

$$[\alpha\beta].[\alpha\gamma] = [\alpha\beta].[\beta\gamma];$$

thus the divergence is the product of the adjacent sequential displacements that connect the points, taken in the sequence in which they appear in the product.

The product of four points,  $\alpha.\beta.\gamma.\delta$ , appears as a volume; and in fact, since

$$\alpha.\beta.\gamma.\delta = \alpha.(\beta - \alpha).(\gamma - \alpha).(\delta - \alpha) = \alpha.[\alpha\beta].[\alpha\gamma].[\alpha\delta],$$

its divergence equals the volume of a parallelepiped that has the deviations of the last three points from the first (taken in their proper sequence) as sides. Or, since

$$\begin{aligned} [\alpha\gamma] &= [\alpha\beta] + [\beta\gamma], \\ [\alpha\delta] &= [\alpha\beta] + [\beta\gamma] + [\gamma\delta], \end{aligned}$$

then also, upon dropping terms similar to the remaining factors,

$$[\alpha\beta].[\alpha\gamma].[\alpha\delta] = [\alpha\beta].[\beta\gamma].[\gamma\delta],$$

that is, the divergence of the product of four points equals the product of the adjacent sequential displacements that connect the points

in the sequence in which they occur in that product. In this case one does not need to include as well that this magnitude is regarded as bound to space, since all spatial magnitudes are bound to it.

Since space is an elementary system of only fourth order, the product of more than four points must always be zero. If the points being multiplied are endowed with weights as well, then one has only to multiply the product of the elementary points by the product of the weights, whereby only the divergence is changed.

Everything proceeds much more straightforwardly if we consider the extension. According to the definition of the inner or interior elements, the totality of which represent the extension, the extension of the product  $\alpha.\beta.\gamma$  equals the plane area of the triangle having  $\alpha, \beta, \gamma$  as vertices, and that of the product  $\alpha.\beta.\gamma.\delta$  equals the volume of the pyramid having  $\alpha, \beta, \gamma, \delta$  as vertices; and at the same time the theorem that the extension of a rigid elementary magnitude equals its divergence divided by the sequence number belonging to the order of this divergence implies that the triangle is half of the parallelogram and that the three-sided pyramid is one sixth of the parallelepiped whose sides are parallel to three of those of the pyramid.

The concept of a product of several elementary magnitudes of first order in space is thus defined; and we are thereby led to only two new magnitudes, namely the linear magnitude and the planar magnitude. It is also apparent that the product of a linear magnitude with a point (or an elementary magnitude of first order) always yields a planar magnitude; the product of two linear magnitudes, and that of a point with a planar magnitude, always produces a volume. But this product is zero if the orders of the combined factors is greater than that of the elementary system in which they lie, so that for example the product of two linear magnitudes is zero if they lie in the same plane. Consequently we are led to no other magnitudes than those enumerated.

By way of contrast, the addition of linear magnitudes leads us to a characteristic additive magnitude that is of particular importance, especially in statics. We showed above (chapter 3 of part 1) that the sum of two products of  $n$ -th order only<sup>2</sup> appears again as a product of  $n$ -th order if those two products belong to the same system of  $(n + 1)$ -th order, but yields a formal sum, which we called an additive magnitude, if they can only be included in a yet higher

---

<sup>2</sup>Cf. § 96.

system. The latter case can only arise in space, as an elementary system of fourth order, if elementary systems of second order, that is linear magnitudes, are added, and these do not lie in a single plane. I defer a detailed discussion of this case until the applications to statics, in which this additive magnitude acquires an independent meaning.

## **§ 2.4. §§ 116, 117. Reference Systems for Elementary Magnitudes**

### **§ 116.**

Of the many applications that the method of our analysis permits in geometry, I emphasize here only those that seem to me to be the most suitable for illustrating the essence of that method.

I first apply the concept of the reference system to the interpretation of space in order to let its relation to the commonly used coordinate definition emerge. In chapter 5 of part 1 we had formulated a reference system for extensive magnitudes, and shortly thereafter established that all the definitions we had formulated for extensive magnitudes could also be transferred to elementary magnitudes. Thus, whereas extensive magnitudes of first order appeared there as fundamental measures, here elementary magnitudes of first order appear as fundamental measures, and the meanings of all the concepts introduced in §§ 87 and 88 are thus defined for elementary magnitudes; in particular, the definitions of reference measures, reference domains, reference terms, and indicators are precisely the same here as there; only the reference domains of first order, which we called reference axes there, we must call reference elements here. In this connection I will only remark that, since displacements can also be interpreted as elementary magnitudes of first order, arbitrarily many fundamental measures may appear as displacements, and only if all fundamental measures are displacements do we obtain the reference system for extensive magnitudes. That reference system closest to this one, which suffices for the representation and definition of elementary magnitudes, is one in which a single fundamental measure is an element and all the rest are displacements, a reference system that merits particular attention on account of its simplicity.

## § 117.

Now if we apply this to geometry, then since space is an elementary system of fourth order, four mutually independent elementary magnitudes of first order appearing as fundamental measures suffice for its definition. The condition that they be mutually independent only expresses that they may not lie in a single plane, and at least one of them must be a rigid elementary magnitude (while arbitrarily many of the rest may be displacements).

If we take four rigid elementary magnitudes (that is, multiple elements) as fundamental measures, then we have the type of coordinate definition on which Möbius bases his *Barycentrische Kalkül*, which coincides in essence with that of Plücker in his *System der analytischen Geometrie*. As reference systems of second order there appear six straight lines, each of which connects two of the reference elements, and which form the edges of a pyramid (tetrahedron) that has those reference elements as vertices; as reference domains of third order are four planes, each of which lie on three of the reference elements and form the faces of that pyramid; and the reference measure of fourth order, which is the principal measure here, represents a volume. Every elementary magnitude of first order, whether it is a rigid elementary magnitude or a displacement, can be represented in space as a multiple sum of the four fundamental measures; every elementary magnitude of second order, whether it is a linear magnitude, a plane area of fixed direction, or an additive magnitude, can be represented as a sum of six linear magnitudes belonging to the six lines mentioned above; in brief, every magnitude can be represented as a multiple sum of reference measures of the same order, or as a sum of terms belonging to the reference domain of the same order.

Following Möbius, we call this reference system (whose fundamental measures are rigid elementary magnitudes, that is, multiple elements) *barycentric*. The simplest type of barycentric reference system is one in which the fundamental measures simply represent points. But the barycentric reference system itself appears only as a particular case, even though it is the widest-ranging type of general reference system consisting of four arbitrary elementary magnitudes of first order. For we showed that arbitrarily many (save one) of them can be transformed into displacements, and thus, besides the reference system mentioned, others are obtained in which the reference domains of first order are partly reference elements, partly reference

axes (fixed directions).

Under this heading we place particular emphasis on that type of reference system that has one element and three displacements as fundamental measures. As reference measures of second order there appear here, on the one hand, three linear magnitudes whose lines go through the reference element and whose divergences are the three other fundamental measures, and, on the other, three plane areas of fixed directions, which represent the three parallelograms it is possible to construct from those displacements. As reference measures of third order there appear, on the one hand, three planar magnitudes whose planes go through the reference element and whose divergences are the plane areas of those three parallelograms, and on the other a volume regarded as an extensive magnitude, which is represented by the parallelepiped constructed from those three displacements. Finally, as principal measure there appears the same volume, regarded as an elementary magnitude of fourth order. The system to which these reference measures belong then forms the proper reference domain.

The reference terms of a point with respect to such a reference system are now, on the one hand, the reference element, and on the other three displacements parallel to the reference axes; and every point in space can be represented as the sum of four such reference terms. The deviation of a point in space from the reference elements is therefore determined, according to this reference system, by reference terms of fixed directions (by parallel coordinates), and thus in precisely the same way as an extension is generally determined by reference systems which serve to determine extensions.

## § 2.5. Transformations of Coordinates

Since we present all reference systems as particular types of a general reference system whose fundamental measures are elementary magnitudes, we have, on the one hand, in this way found the most general coordinate definition with respect to which the plane still appears as a point structure of first order, and, on the other, we are put into a position to apply that procedure by which we can pass from one coordinate definition to another of the same type, and which we presented in § 92 for parallel coordinates, and we are also enabled thereby to pass from one type of coordinate definition to another, so long as both belong to the more general type we have presented. In

particular we can immediately transform the barycentric equations into equations between parallel coordinates without the need of any other special prescription.—Furthermore, since we now interpret the concept of reference terms (coordinates) in a more general sense, as we now accept reference terms of higher order, the same general type of reference system suffices to define elementary magnitudes of higher order, namely linear magnitudes and planar magnitudes.

Before we examine the significance of these definitions, we must take note of a distinction between the method of definition we give and the usual one and indicate how this distinction can be reconciled. Thus we always arrive at the definition of elementary magnitudes, that is of points with associated weights, of linear magnitudes, and of planar magnitudes. With the definition by coordinates, however, the definition of points, lines, and planes depends on their locations, and thus in our method of treatment we always obtain one more reference term or indicator than is necessary in that definition of the location. This distinction can be reconciled on the spot, since one recognizes that if all the reference terms or indicators of a magnitude are multiplied or divided by the same numerical magnitude, its location (the elementary system) is not changed thereby. One therefore immediately reduces the number of indicators by one if one always divides the reference terms (or indicators) by one of the indicators, and thereby always sets one of the indicators to unity. The indicators thus obtained then always suffice to define the location.

Since we now define the location of a plane in such a way, for example by its indicator, and establish an equation of  $m$ -th degree between the indicators taken as variables, there is in this way given an infinite set of planes whose indicators satisfy that equation; and all these planes will envelop a surface, which I will later show is the same as that which has been called a surface of  $m$ -th class. Likewise the definition of the straight line by its indicators leads to characteristic structures not previously recognized, which I first happened to investigate in an article in Crelle's *Journal*.<sup>3</sup>

Since further discussion of this subject would lie outside the limitations of this work, I will satisfy myself here with formulating the equations obtained by our science for the straight line and the plane,

---

<sup>3</sup>Crelle's *Journal für die reine und angewandte Mathematik*, XXIV. [pp. 262–282 and 372–380]. (1877) Particularly since Plücker's latest work, *Neue Geometrie des Raumes* (1868), these structures are treated by many of the most distinguished mathematicians, and form the principal subject matter of modern linear geometry.



and relating them to the equations already known for these structures.

## § 2.6. Equation of a Plane

The most general problem we can pose here is to formulate the equation of a plane going through three arbitrary given points, or the equation of a line going through two arbitrary given points.

Let the given points in the first case be  $\alpha$ ,  $\beta$ ,  $\gamma$ , and in the second be  $\alpha$ ,  $\beta$ ; let the variable point, which will be determined by an equation between it and the given points, be  $\sigma$ , then from the concept of an elementary system of first or second order one has at once in the first case

$$\alpha.\beta.\gamma.\sigma = 0,$$

and in the second

$$\alpha.\beta.\sigma = 0,$$

and the problem is solved in the most general sense by these formulas, which enjoy the greatest degree of simplicity. Now if one wishes to formulate the corresponding coordinate equations from a preference for the usual coordinate treatment, or on other grounds, then one can immediately obtain them from those simple equations if one is not averse to the trouble of writing down those lengthy formulas.

If for example one wants to exhibit the equation in parallel coordinates, then one has only to make use of the reference system mentioned at the end of § 117. In this reference system every point is represented as a sum of the reference element  $p$  and a displacement. Thus if

$$\alpha = p + p_1, \quad \beta = p + p_2, \quad \gamma = p + p_3, \quad \sigma = p + p,$$

then upon substituting these expressions into the equation of the plane one has

$$(p + p_1).(p + p_2).(p + p_3).(p + p) = 0,$$

or, after removing parentheses and dropping products that are zero,<sup>4</sup>

$$p.p_2.p_3.p + p.p_3.p_1.p + p.p_1.p_2.p - p_1.p_2.p_3.p = 0.$$

---

<sup>4</sup>In particular these are all those including  $p$  as more than one factor, and the product  $p_1.p_2.p_3.p$ .

Now one brings  $p$  to the initial position with the corresponding observation of the algebraic sign, and then drops it according to § 112, whence

$$(p_2.p_3 + p_3.p_1 + p_1.p_2).p = p_1.p_2.p_3.$$

In order to transform this equation into a coordinate equation, one then need only replace each displacement with the sum of its reference terms. Thus let

$$\begin{aligned} p &= x + y + z, \\ p_1 &= x_1 + y_1 + z_1, \end{aligned}$$

..., where  $x, y, z, \dots$  represent the reference terms. Then one has

$$\begin{aligned} &(x_2 + y_2 + z_2).(x_3 + y_3 + z_3).(x + y + z) \\ &\quad + (x_3 + y_3 + z_3).(x_1 + y_1 + z_1).(x + y + z) \\ &\quad + (x_1 + y_1 + z_1).(x_2 + y_2 + z_2).(x + y + z) \\ &= (x_1 + y_1 + z_1).(x_2 + y_2 + z_2).(x_3 + y_3 + z_3). \end{aligned}$$

Now one need only remove the parentheses, observing that the reference terms with the same letters are parallel, and at once only six nontrivial products result from those three factors; then one must order the factors of the twenty-four resulting products, observing the signs, so that the letters in each product follow each other in the same way, thus obtaining an equation in which the reference terms are replaced by their indicators, and thus produce an arithmetic equation in which again the order of the factors is unimportant. In this way one obtains the following equation:

$$\begin{aligned} &(y_2z_3 - y_3z_2 + y_3z_1 - y_1z_3 + y_1z_2 - y_2z_1)x \\ &\quad + (z_2x_3 - z_3x_2 + z_3x_1 - z_1x_3 + z_1x_2 - z_2x_1)y \\ &\quad + (x_2y_3 - x_3y_2 + x_3y_1 - x_1y_3 + x_1y_2 - x_2y_1)z \\ &= x_1y_2z_3 - x_1y_3z_2 + x_3y_1z_2 - x_3y_2z_1 + x_2y_3z_1 - x_2y_1z_3, \end{aligned}$$

where one now recognizes  $x, y, z, \dots$  as the indicators.

This equation, which cannot be reduced to a simpler form by the standard analysis, and appears so intricate, is however no more than that original equation

$$\alpha.\beta.\gamma.\sigma = 0,$$

and comprises the briefest solution of the above problem that is possible using coordinates. One sees in this particularly striking example the advantages of our method, and the complexity of the formulas one gets into as soon as this method is abandoned.

## § 2.7. The Static Moment as Deviation

Since I have deferred the development of the geometric shadow and of the projection, as well as the various systems of relationships, to a later chapter<sup>5</sup> in which these concepts will be brought to light more extensively, I now proceed to applications in statics.

The concept of the moment appears in its full simplicity only here, and likewise only here does the concept of force find its representation, since we conceive of force as a linear magnitude, that is as an elementary magnitude of second order. By the moment of a force  $\alpha\beta$  with respect to a point  $p$  we mean the product

$$[p\alpha].[ \alpha\beta] \quad \text{or} \quad (\alpha - p).(\beta - \alpha);$$

if we multiply this value again by the element  $p$ , then the moment appears as the divergence of the resulting elementary magnitude  $p.(\alpha - p).(\beta - \alpha)$ ; however, according to the known law of outer multiplication this is equal to

$$p.\alpha.\beta,$$

so that we can define the moment with respect to a point as the divergence of a product whose first factor is the reference point and whose second is the force, or as the deviation of the force from the reference point. Since each equation between the elementary magnitudes also holds between their divergences, each equation between those products will also obtain between their moments; however the converse is not true.

One might therefore wonder whether one ought rather to define that product of the reference point with the force as the moment, and represent what we had previously defined as the moment as only the divergence of that magnitude.—But we will continue to use the established concept.

---

<sup>5</sup>Chapter 4 of this part.

By the moment of a force  $\alpha\beta$  with respect to an axis  $p\sigma$  we previously (§ 41) meant the product

$$[p\sigma].[ \sigma\alpha].[ \alpha\beta] \quad \text{or} \quad (\sigma - p).(\alpha - \sigma).(\beta - \alpha).$$

Multiplying this by  $p$ , we obtain the product

$$p.\sigma.\alpha.\beta,$$

whose divergence is precisely that moment. The moment of a force with respect to an axis thus appears as the divergence of a product whose first factor is the axis and whose second is the force, or, expressed more simply, as the deviation of the force from the axis. Moreover, since an equation between elementary magnitudes of fourth order in space, as an elementary system of fourth order, has no other meaning than as the equation between their divergences, one can interpret the moment with respect to an axis directly as the product of this axis with the force.<sup>6</sup>

## § 2.8. New Procedure for the Treatment of Statics

At this point of the development a method is presented by which all the laws for the equilibrium of rigid bodies can be obtained in the simplest way without assuming any of the previously derived theorems of statics. For this we only need on the one hand the postulate that three forces acting on a single point are in equilibrium if and only if their sum is zero, or, since we call two forces or systems of forces equivalent if they can be balanced by the same force, that two forces acting on a single point are equivalent to their sum acting on the same point, and on the other hand that two forces acting on a rigid body are in equilibrium if and only if they act along the same straight line equally and in opposition. It thereby follows at once, upon retaining the concept of equilibrium just established, that two forces acting on a rigid body are equivalent if and only if they act

---

<sup>6</sup>Since the name (static) moment now appears superfluous, its being completely replaced by the name “deviation,” and this is even more easily dealt with, it is certainly appropriate to introduce the name “moment” only in the sense in which it was introduced by Lagrange, for example in his *Mechanique Analytique*, where he spoke of the moment without further qualification, and to designate the so-called static moment precisely as the deviation. Even so I have not introduced this nomenclature arbitrarily.

along the same line and are equal, or, expressed more simply, if they are equal as elementary magnitudes.

Thus if we regard the forces acting on a rigid body as linear magnitudes, it follows immediately that two forces whose lines of action intersect are equivalent to their sum; for if  $\alpha$  is this point of intersection, then both forces are represented as linear magnitudes whose first factor is  $\alpha$ ; for if  $\alpha.p$  and  $\alpha.q$  are the forces, where  $p$  and  $q$  signify displacements, then according to the first postulate they are equivalent to  $\alpha.(p + q)$  or otherwise  $\alpha.p + \alpha.q$ , that is they are also equivalent to the sum of the forces if the forces are regarded as linear magnitudes.

If the forces are parallel, for example if one equals  $\alpha.p$ , the other  $m\alpha.p$ , where again  $p$  is a displacement, then we cannot immediately find the force equivalent to the two by the same principle; we therefore take as auxiliaries two mutually balanced forces, namely  $\alpha.m\beta$  and  $m\beta.\alpha$ ,<sup>7</sup> so that those two forces are equivalent to the four forces

$$\alpha.p, \quad \alpha.m\beta, \quad m\beta.\alpha, \quad m\beta.p,$$

of which the first pair are equivalent to their sum because they act on the same point, and likewise for the second pair, whence we obtain the two forces

$$\alpha.(p + m\beta), \quad m\beta.(\alpha + p),$$

as equivalent to the given forces. Since, according to the laws of outer multiplication, we can add to the second factor in each of these two products a term equal to the first without thereby changing the value of the product, we can bring the two products to within a common factor; the forces are then equal to

$$\alpha.(\alpha + m\beta + p), \quad m\beta.(\alpha + m\beta + p).$$

Now if  $m$  is not equal to  $-1$ , the second factor represents a multiple point (with weight  $1 + m$ ); thus both forces act on the same point and are therefore equivalent to their sum. This sum is

$$(\alpha + m\beta).(\alpha + m\beta + p),$$

that is, it equals

$$(\alpha + m\beta).p,$$

---

<sup>7</sup>They cancel each other out, since  $\alpha.\beta = -\beta.\alpha$ .

and therefore the two forces  $\alpha.p$  and  $m\beta.p$  are equivalent to the single force  $(\alpha + m\beta).p$ , that is their sum, if  $m$  is not equal to  $-1$ , that is if the sum of their divergences is not zero.

Now we consider the case we have so far excluded, namely that in which the divergence of the two forces together is zero, that is in which the two forces, regarded as displacements, are equal in magnitude and oppositely directed. It is then clear that they will be in equilibrium if and only if they lie on the same straight line, that is if the sum of the forces is itself zero. In this special case we can therefore also say that the two forces are equivalent to their sum. It thus remains only to investigate the case in which the two forces have zero sum as displacements but not as linear magnitudes.

Now in this case equilibrium does not exist according to the second postulate; but we can also easily show that there is then no nonzero force that holds those two forces in equilibrium. For from the two postulates we stated at the beginning of this paragraph it follows that the divergence of the collective force is always the sum of the divergences of the individual forces. Thus the divergence of the force in question here must be zero, that is this force must itself be zero, and the given forces must already be in equilibrium, contrary to assumption. Thus we have shown that two forces acting on parallel but separate lines, and as displacements equal in magnitude and opposite in direction, cannot be reduced to a single equivalent force. This case is however the same as that in which the sum of the forces represent no linear magnitude, but rather an extension of second order; thus  $\alpha.p - \beta.p$  equals  $(\alpha - \beta).p$ , which represents an extension of second order.

In order more clearly to illustrate the significance of this case for statics, we observe that the collective moment of two such forces with respect to all points in space is a fixed magnitude (that is, their collective deviation is the same from all points). Thus, since the collective deviation equals the deviation of the sum, but the sum is now an extension of second order, and the deviation of an extension is always equal to the extension itself, it follows that the collective deviation of those two forces from that arbitrary point equals the sum of those two forces and thus remains constant so long as the sum remains constant. Consequently we say that these two forces are equivalent to this moment, which is represented by their sum.<sup>8</sup>

---

<sup>8</sup>This is therefore to be regarded as a generalization of the concept of equiva-

Thus we can now express the theorem:

Two or more forces acting in a single plane are equivalent to their sum.

In particular, this can immediately be generalized from two forces to arbitrarily many.

## § 2.9. General Law for Equilibrium

Turning now to the treatment of forces in space, we must mention in this connection that the addition of forces as elementary magnitudes of second order only has real significance if they lie in a single plane as a system of third order, whereas a purely formal significance obtains if this is not the case. By virtue of this formal significance two such sums are equal if they can be reduced to the same expression by the application of real addition and the general additive law of conjunction.

If we consider two such sums of forces in space, which can be reduced to the same expression in this way, and recall that for real addition (in which forces always lie in a single plane) the sum of the forces is always equivalent to the collection of the individual forces forming their terms: then it follows that under that transformation of the formal sum into one equal to it, the forces that form this sum always remain equivalent to one another, and thus “two assemblies of forces having equal sums are always equivalent,” and also “a series of forces whose sum is zero is in equilibrium.”

Furthermore, we can now reduce that sum of forces to a single force whose point of application is arbitrary, plus a single moment, or alternatively to two forces. Thus if we set the sum of several forces equal to

$$\alpha.p + M,$$

where  $\alpha$  represents an element,  $p$  a displacement,  $\alpha.p$  a force, and  $M$  an extension of second order, and thus a moment; then, according to the theorems presented above, the two expressions are equal if and only if they have equal divergences and equal deviations from any given element, for example  $\alpha$ . Thus  $p$  must equal the sum of all the

---

lence, since the moment itself is interpreted as a particular force magnitude that can act together with other forces; and thus the true significance of the theory of force couples, so important in statics, is understood.

deviations from the element  $\alpha$ ; since however both sums are always real, the former as a sum of displacements and the latter as a sum of extensive magnitudes of second order in a system of third order, that series of forces can always be brought to the form given, where  $\alpha$  is arbitrary while  $p$  and  $M$  are specified. Now if one can bring that sum of forces to the expression  $\alpha.p + M$ , then one can bring it to the sum of two forces; if for example  $M$  equals  $r.s$ , then one can subtract the term  $\alpha.s$  from the term  $\alpha.p$  and add the same term to  $M$  without altering the value of the sum, thus obtaining

$$\alpha.p + M = \alpha.(p - s) + (\alpha - r).s,$$

where the right side represents two forces. Finally, since two assemblies of forces having equal sums are equivalent, as we have showed above, one has the theorem that “every series of forces in space can be reduced to two forces or one force and a moment, which are equivalent to it and have the same sum as those forces.”

This result includes the result that “several forces are only in equilibrium if their sum is zero”; for they can be reduced to two forces equivalent to them, which also have the same sums, but, according to the second postulate, two forces are then only in equilibrium if their sum is zero; then however the sum of the given forces, as it is the same, will also be zero, whence the theorem is proved.

Now if two assemblies of forces are equivalent, then the forces of the first assembly taken together with the opposites of the forces of the second must (according to the definition of equivalence) produce equilibrium, that is the sum must be zero according to the foregoing theorem. Thus the forces of the one assembly have the same sum as those of the other; we have therefore shown that “two equivalent assemblies of forces necessarily have the same sum.” Combining this theorem with the converse that we proved previously, we obtain the theorem:

Two assemblies of forces are equivalent if and only if they have equal sums.

This theorem entitles us to regard the collective action of several forces as the action of their sum, even when this sum can no longer be represented as a single force; thus we have the general theorem:

Two or more forces are equivalent to their sum, and are thus only in equilibrium if their sum is zero.



This theorem encompasses all the earlier ones and appears as their end result.

## § 2.10. General Relation Between the Static Moments

Because we have represented assemblies of forces as equivalent to their sums, that two equivalent assemblies of forces have equal collective moments with respect to every point and every axis, and that two assemblies of forces having equal collective divergences and equal moments with respect to any given point are equivalent and have equal moments with respect to every point and every axis, are now only alternative ways of expressing the theorems we established in the abstract science.—Consequently we will not linger over the derivation of these laws of statics, but instead establish a more general theorem in the theory of moments, which in generality far surpasses all theorems one had previously established in this theory, and yet is obtained most easily by means of our analysis.

In order to state this theorem directly in its most easily comprehensible form, I will introduce a new concept, which is of the greatest importance in the treatment of relationships. Namely I say that an assembly of magnitudes stand in the same numerical relation as another assembly of corresponding magnitudes if each equality existing between the multiple sums of the magnitudes of the latter assembly are also valid if one replaces these magnitudes with the corresponding ones of the former assembly. The theorem that we will prove here can now be presented in the form:

The collective moments of an assembly of forces with respect to different points or axes stand in the same numerical relation as these points or axes.<sup>9</sup>

For if  $S$  is the sum of the assembly of forces, then their collective moment with respect to any given magnitude  $A$  (be it a point or an axis) equals the divergence of the product  $A.S$ . Now if different relative magnitudes  $A, B, \dots$  are given, and there exists a numerical relation between them that can be represented in the form

$$aA + bB + \dots = 0,$$

---

<sup>9</sup>Grassmann supplies no proof here; in his later work *Die Mechanik nach den Prinzipien der Ausdehnungslehre* (Teubner, Leipzig, 1867) he returns to this theorem.

where  $a, b, \dots$  are numerical magnitudes, then if one multiplies by  $S$  it is also true that

$$aA.S + bB.S + \dots = 0.$$

Now according to § 112 this equation remains valid if one replaces the products  $A.S, \dots$  with their divergences, that is the moments with respect to those magnitudes; thus those moments stand in the same relation as the relative magnitudes do.

Thus by this theorem we can use the moments with respect to two points to find the moment with respect to any other point on the same straight line; the moment with respect to three points not lying on a single straight line to find that belonging to any other point in the same plane; and the moments with respect to four points not lying in the same plane to find that belonging to any other point in space. Further, we can use the moments with respect to two intersecting axes to find the moment with respect to any other axis going through the same point and lying in the same plane; the moments with respect to three axes lying in the same plane, which do not go through the same point, to find that of any other axis in the same plane; and generally the moments with respect to a series of axes which do not stand in any relation to each other to find the moment with respect to any axis standing in a specific numerical relation to them.

### § 2.11. § 124. When an Assembly of Forces is Equivalent to a Single Force

I close these applications with the solution to the problem of finding the constraint equation that must hold if a system of forces is to be equivalent to a single force or to a moment.

In both cases the sum of the forces  $S$  can be represented as the product of two elementary magnitudes of first order, and thereby for this case the equation

$$S.S = 0$$

follows at once, an equation that is never satisfied if  $S$  represents a formal sum; for then  $S$  can be represented as the sum of two forces which do not lie in the same plane; if these are  $A$  and  $B$ , thus

$$S = A + B,$$

then

$$S.S = (A + B).(A + B) = 2A.B,$$

since  $A.A$  and  $B.B$  are zero, but  $A.B$  equals  $B.A$ <sup>10</sup>; now since  $A$  and  $B$  do not belong to the same plane,  $A.B$  cannot be zero, so that the equation  $S.S = 0$  is the necessary and sufficient constraint equation for the case that  $S$  represents a single force or a single moment; and indeed it will represent a moment if the divergence of  $S$  is zero, but otherwise a force of nonzero magnitude.

If

$$S = A + B + C + \dots,$$

then

$$S.S = 2A.B + 2A.C + 2B.C + \dots$$

equals the sum of the products of two factors that can be formed from the terms.<sup>11</sup> Thus there immediately follows the theorem:

An assembly of forces is equivalent to a single force or to a single moment if and only if the sums of the products of two factors, which can be formed from the forces, is zero.

Further:

Two assemblies of forces can only be equivalent if the products of two factors that can be formed from the forces of the one assembly have a sum equal to that formed with the forces of the other.

These theorems also remain valid if one replaces the products of two forces with their sixth part, that is the pyramids that have the forces as opposing edges.

---

<sup>10</sup>Thus  $A$  and  $B$  are magnitudes of second, and thus of even order, which according to § 55 can be interchanged without sign change.

<sup>11</sup>Thus it equals the elementary sum if one treats the products  $A.B$  and  $B.A$  as formed differently.

## Chapter 3

# The Regressive Product

### A. Theoretical Development

#### § 3.1. Formal Interpretation of the Regressive Product; Degree of Dependence and Multiplication

The concept of a product as “outer” consists in that each term of one factor that is dependent on the other factor can be dropped without changing the value of the product, which therefore implies that the product of two dependent magnitudes is zero. Real magnitudes, that is, such as can be represented as products of genuinely elementary factors, are then called “mutually independent” if each of their individual factors lies completely outside the system defined by the other factors, or expressed more abstractly, if no magnitude belonging to the system of one of the magnitudes simultaneously belongs to the system defined by the rest. Now since this definition characterizing the product as “outer” is not inherent in the concept of a product as such, it must be possible to retain the general concept of a product and yet give up that definition or replace it with another. In order to discover this new definition we must investigate the different degrees of dependence, since according to this new definition the product of two dependent magnitudes can also have a nontrivial value.<sup>1</sup>

In general, if two systems of higher order are dependent on each other, then there will be magnitudes that belong to both simultaneously. Now since each system that includes certain magnitudes must also contain magnitudes dependent on them collectively, that

---

<sup>1</sup>(1877) For this chapter, cf. appendix II.

is, the whole system defined by them, including the outer product of those magnitudes as well, it follows that systems having certain magnitudes in common will also have in common the whole system defined by these magnitudes, and thus also their outer product.

The degree of dependence can be defined by the order of this common system, and we can say that the systems are mutually dependent in the  $m$ -th degree if they have a system of  $m$ -th order in common, and likewise two real magnitudes are mutually dependent in the  $m$ -th degree if the systems defined by them are, or if they can be brought to within a common factor of  $m$ -th order (and none higher). This last follows from the foregoing, since according to § 61 that magnitude belonging to the system defined by another magnitude can also be regarded as a factor of the latter.<sup>2</sup>

To each degree of dependence corresponds a type of multiplication; we include all these types of multiplication under the name *regressive* multiplication, and in particular mean by the regressive product of  $m$ -th order one in which, without changing the value of the product, only such a term in each factor can be dropped as is dependent on the other factor in a degree higher than the  $m$ -th; and we call the regressive product of  $m$ -th order *real* if the factors are mutually dependent in at least the  $m$ -th degree; if in a lower degree, *formal*.<sup>3</sup> The value of the regressive product thus consists in precisely that which remains constant under those allowed alterations. Here however we investigate only the real product, since the formal product requires another method of treatment and notation; besides, it is of much less importance.

Now the real regressive product either has a nontrivial value or is zero, and indeed not only will that product be zero if one factor is zero, but also if the two factors are mutually dependent in a degree higher than that of the order of the regressive multiplication. This last follows since one can then consider one factor as a sum, one term of which is zero, the other the factor itself, and one can drop this term according to the above definition; the product thus appears as zero.

---

<sup>2</sup>We can also say of independent magnitudes that they are mutually dependent in the zeroth degree, that is, exactly not at all.

<sup>3</sup>(1877) In the 1862 *Ausdehnungslehre* the concept of the formal regressive product is abandoned as sterile, thus simplifying the whole subject.

### § 3.2. Relation Between the Common and Nearest Covering Systems

In order to be able to present the significance of the real regressive product we have to make the condition for it to be zero dependent on the system to which both factors belong, whereas previously we could stipulate it from the common system of the two factors or from the degree of their mutual dependence.

To this end we set ourselves the problem: “If the common system of the two magnitudes is given, to find their nearest covering system, that is the lowest<sup>4</sup> system to which both belong simultaneously.”

In this regard we mention that a magnitude belongs to a system if and only if it is subordinate to another magnitude that represents this system, that is, if it can itself be represented as an outer factor of this latter magnitude. Thus if  $A$  and  $B$  are the two magnitudes and  $C$  represents their common system, then  $C$  can be represented as an outer factor of  $A$  as well as  $B$ , whence for example  $B$  can be brought to the form  $CD$ .<sup>5</sup>

Since we define  $C$  as the common system of  $A$  and  $B$ , according to the previous paragraph we mean by this that  $C$  includes all magnitudes belonging to both  $A$  and  $B$ , but no others. It then follows that  $D$  can have no magnitudes in common with  $A$ , since otherwise  $CD$ , that is  $B$ , would also have further magnitudes in common with  $A$ , but not belonging to  $C$ , contrary to assumption. Now since this implies that  $A$  and  $D$  are independent, the product  $AD$  is nonzero as an outer product, whence both magnitudes  $A$  and  $B$  will be subordinate to the product  $AD$ . For, first of all,  $A$  appears directly as one of its factors; but also, of the two factors of the magnitude  $B$  or  $CD$ , one is included in  $A$  and the other appears directly in the outer product  $AD$ , whence  $B$  itself is also shown to be an outer factor of this product. That there is no lower magnitude to which both magnitudes  $A$  and  $B$  are subordinate follows at once, since such a magnitude would have to have  $A$  as well as  $D$  as outer factors, and thus, since these are independent, it would also have to have their outer product  $AD$  (§ 125) as an outer factor. Thus  $AD$  represents the nearest covering system of the magnitudes  $A$  and  $B$ , and the problem is solved. Thence follows the theorem:

<sup>4</sup>Naturally this is to be understood as the system with the smallest order.

<sup>5</sup>Cf. § 61.

If two magnitudes  $A$  and  $B$  have a magnitude  $C$  as their greatest common factor, and one sets one of them,  $B$  for example, equal to the outer product  $CD$ , then the product of the other with the magnitude  $D$ , that is the product  $AD$ , represents the nearest covering system.

If we denote the orders of the four magnitudes  $A, B, C, D$  by the corresponding lower case letters, and that of the nearest covering system by  $u$ , then we have  $u$  equal to  $a + d$ , or, since  $B = CD$ , and thus  $b = c + d$ ,

$$u = a + b - c$$

or

$$u + c = a + b$$

or

$$c = a + b - u,$$

that is:

The order of two magnitudes together is as large as the order of their common system and their nearest covering system together.

Or:

One finds the order of the nearest covering system of two magnitudes by subtracting the order of their common system from the sum of the orders belonging to the individual magnitudes.

Or:

One finds the order of the common system of two magnitudes by subtracting that of their nearest covering system from the sum of the orders of the two magnitudes.

The last form of this general theorem is particularly convenient in applications, as is easily seen if one applies it to geometry.<sup>6</sup>

---

<sup>6</sup>For example, if I regard the plane as the nearest covering system of two lines, then since the former is an elementary system of the third, the latter of the second order, the common system is of order  $(2 + 2 - 3) = 1$ , and consequently

### § 3.3. Introduction of the Relative System

According to § 125 the regressive product of two nonzero magnitudes has a real nonzero value if and only if the order of their common system equals the order of the regressive multiplication, or, applying the law established in the preceding paragraph, if the order of the nearest covering system and that of the regressive product together equal the sum of the orders of the two factors.

Now if in general we call that number, which together with the order of the regressive multiplication amounts to the sum of the orders of the two factors, the *relative number* of the regressive product or the regressive multiplication, then it follows that the regressive product of two nonzero values produces a nonzero real value if and only if the order of the nearest covering system is equal to the relative number of the product. If the order of the common system is greater than the order of the regressive multiplication, then according to § 125 the product is zero, while if it is less the product yields a purely formal value. Now if the orders of the two factors remain the same, then as the order of the common system increases that of the nearest covering system decreases and conversely, since they have a fixed sum, namely the sum of the orders of the two factors. It therefore follows that a regressive product of two nonzero values will be zero if the order of their nearest covering system is smaller than the relative number; and a formal value obtains if it is greater.

Thus, if a system of  $n$ -th order is given and we know that all magnitudes under consideration belong to this system as the principal system (cf. § 80),<sup>7</sup> then we can be sure that the regressive product whose relative number is  $n$  will have a real value. We then call this multiplication that relative to that system, and call this system the *relative system* of the product,<sup>8</sup> and if both factors belong to this relative system, then (consistently with the earlier nomenclature) we

---

is represented either by a point or a direction. Thus we have the theorem that “two noncoincident lines lying in the same plane either intersect in a point or are parallel.” If space is thought of as the nearest covering system, then one has the theorems “two noncoincident planes either intersect in a straight line or are parallel”; “a line not lying completely in a given plane either intersects it in a point or lies parallel to it”; “two nonparallel planes have one and only one direction in common.”

<sup>7</sup>Cf. § 80.

<sup>8</sup>The order of this system is just the number that we called the relative number above.



also call it the principal system of the product. Thus we can say that the regressive product is always real if the factors belong to the relative system, and at the same time it has a nonzero value if the nearest covering system of the factors is simultaneously the relative system of the product, and it is zero if the nearest covering system of the two factors is subordinate to the relative system of the product and (thus) is of lower order.

### **§ 3.4. The Unity of Outer and Regressive Multiplication is Mediated Thereby**

According to § 55 the outer product of two nonzero magnitudes is zero if they are mutually dependent, that is, if the order of their nearest covering system is smaller than the sum of the orders of the two factors; or, since we can regard that system to which the factors are subordinate, and whose order is as great as or greater than that sum, as the relative system for the outer product, we can extend the law of the previous paragraph to say:

A product of two nonzero values is zero if and only if the factors are mutually dependent and at the same time their nearest covering system is lower than the relative system.

In this it is then implicit that “such a product only has a nonzero value if either the two factors are mutually independent, or their nearest covering system is the relative system.” And in fact in the former case the product is an outer product, in the latter a regressive one. If both conditions obtain simultaneously, that is if the two factors are mutually independent and at the same time their nearest covering system is the relative system, then the multiplication can be interpreted not only as outer but also as regressive of zeroth degree. The second theorem of the previous section is thus extended to the following theorem:

If in a product of two nonzero values, the sum of the orders of the factors is less than the relative number, then the product is an outer product; if that sum is greater the product is regressive and indeed of order equalling the excess of that sum over the relative number; finally, if that sum equals this number, then that product can be treated both as outer and as regressive of order zero.

By introducing the relative system or the principal system we have therefore obtained the important advantage that, once the relative system is fixed as the principal system, it is no longer necessary to specify the method of multiplication for the product of two magnitudes, since it is superfluous to distinguish outer multiplication from regressive, or among the different degrees of the latter.<sup>9</sup>

### § 3.5. The Regressive Product in Subordinate Form

In order to bring the actual value of a real regressive product under a simple concept we must seek, for the given product whose value is sought, all forms into which it can be cast, without changing its value, as a consequence of the formal multiplication principles determined by the definition. Whatever all of these forms have in common will then represent the product under a simple concept.

The alterations in form permitted as a consequence of the definition are, first, the general multiplicative one that one may change the factors in reciprocal proportion, and second, that special one that one may drop from one factor a term that depends on the other factor in a higher degree than the order of the regressive product; or, reverting to the relative system, that one may drop from one factor a term which, together with the other factor, is included in a system whose order is smaller than the relative number.

The simplest case is that in which one factor represents the reference system and the other is therefore subordinate to it, or expressed more briefly, where the product appears in *subordinate form*. Since in this case the nearest covering system is always the relative system as well, no nonzero term can be added to any of the factors

---

<sup>9</sup>At the same time we have thus achieved the advantage of an easier applicability to the theory of space. If for example we consider the plane, that is an elementary system of third order, as the principal system, as is always the case in planimetry, then the product of two elementary magnitudes with respect to this system will be zero if and only if they are mutually dependent and simultaneously belong to a system of second order, that is when they have points or directions in common and at the same time lie on a single straight line. Further, if we consider space, that is an elementary system of fourth order, as the principal system, which is the case in stereometry as such, then the product of two elementary magnitudes with respect to it will be zero if and only if they lie in the same plane and simultaneously are dependent, that is if they have points or directions in common; for example, the product of two linear magnitudes that intersect or are parallel, that of two planes when they coincide, and so on.

without changing the value of the product. The only change in form that leaves the value of the product unchanged is therefore the general multiplicative one, that is that the factors may be changed in reciprocal proportion, whence it is permissible to set

$$A.B = mA.\frac{1}{m}B,$$

where  $m$  represents any given numerical magnitude. Thus the system of the two factors remains constant under all allowed alterations in form, and their magnitudes therefore only change in reciprocal proportion. The combined observation of the two systems together with the quantity to be distributed multiplicatively on both factors thus forms the value of the product.

### § 3.6. §§ 130–132. Real Significance of the Regressive Product; Its Characteristic Value With Respect to a Principal Measure

#### § 130.

In the more general case, if  $A$  and  $B$  are the two factors of the regressive product and the magnitude  $C$ , whose order is  $c$ , represents the common system of the two factors, then according to § 126, if  $B$  is set equal to  $CD$ ,  $AD$  represents the nearest covering system and thus, also according to § 126, the relative system as well, if the product is not zero.<sup>10</sup>

Now we showed in § 129 that, outside of the general multiplicative alteration, the only change in form permitted is that the one factor  $CD$  may increase by a term that is dependent on the other factor  $A$  in a degree higher than the  $c$ -th. It is clear that this term is not allowed to be similar to  $CD$ , since such a term would stand in the same degree of dependence with  $A$  as  $CD$  itself; thus it must be supposed to be dissimilar to  $CD$ . We have established a real and a formal concept for the addition of dissimilar magnitudes, the former of which applies if the two magnitudes to be so added can be decomposed into elementary factors, all but one of which they have in common. Now since formal addition only appears as an abbreviated notation, we already discover the significance of our product if we

---

<sup>10</sup>Naturally we make the assumption that the product is not zero here, since in case it is zero no further determination is necessary.

consider real addition, and thus assume that the term to be added has all elementary factors save one in common with  $CD$ . Since the term to be added depends on  $A$  in a degree higher than the  $c$ -th, this single elementary factor necessarily belongs to the system of  $A$ , while the collection of elementary factors of  $C$  must necessarily be found among the remaining elementary factors. Thus it must be possible to represent this term in the form  $CE$ , where  $E$  is dependent on  $A$ . Accordingly the product now appears in the form

$$A.(CD + CE) \quad \text{or} \quad A.C(D + E),$$

where  $E$  is dependent on  $A$ . Now if we compare the two products

$$A.CD = A.C(D + E),$$

then  $AD$  represents the nearest covering system for the factors of the first,  $A(D + E)$  that for the factors of the second; and since  $E$  is dependent on  $A$ , one has

$$AD = A(D + E)$$

and thus  $AD$  is the nearest covering system for both products.

Besides this change in form, the only general multiplicative alteration permitted is that the factors may be changed in reciprocal proportion. Since the system of the factors is not changed by this, the common and nearest covering systems also remain the same under this alteration of form; thus that system generally remains unchanged with every alteration of form of its product, and therefore belongs to what makes up the constant value of this product. If one places the common outer factor  $C$  in the middle, so that as we have demonstrated above the product appears in the form

$$A.CD,$$

then the product of the outer factors  $AD$  gives the nearest covering system; and thus the middle factor as well as the product  $AD$  of the outer two represent constant systems.

If in addition we compare the values of the two magnitudes  $AD$  and  $C$ , then not only must we consider those transformations by which the values of the regressive factors  $A$  and  $CD$  (but not their product  $A.CD$ ) are changed, but also those that leave the value of the outer product  $CD$  and the system of its first factor unchanged.

By means of the first type of transformation  $CD$  can increase by a term dependent on  $C$ , which must then likewise be dependent on  $A$ , since  $C$  is subordinate to  $A$ . Thus, if we symbolize this term as  $E$ , then in both cases the product  $A.CD$  is changed into its equal  $A.C(D + E)$ . Now since  $E$  is dependent on  $A$ , whence

$$A(D + E) = AD,$$

in both products the middle factor as well as the outer factors remain unchanged. Moreover, the general multiplicative alteration in form, according to which the factors can be changed in reciprocal proportion, is applicable to both types of transformation. If one applies this alteration to both types of transformation, then every time a number is introduced as multiplier to one factor, the same number must be introduced as divisor to one of the others; thus in addition, if one of the three factors of the product, for example  $C$ , becomes  $m$  times greater, then the product of the two others must become  $m$  times smaller, that is  $C$  and  $AD$  must be changed in reciprocal proportion.<sup>11</sup> Now since it is already implicit in this that their systems remain constant, we can express the result of the above development in the theorem that “if a regressive product is brought into the form  $A.CD$ , in which the middle factor represents the common system of the two factors  $A$  and  $CD$  of the regressive product, then  $C$  and  $AD$ , that is the middle factor and the product of the two outer factors, can only be changed in reciprocal proportion if the whole product is to remain constant.”

### § 131.

To extract the full significance of the regressive product it remains to answer the question of whether the two systems represented by the middle factor and the product of the two outer factors fully represent that which remains constant (aside from the quantity to be distributed multiplicatively between them) with the unchanged value of the regressive product, or in other words, whether, if those magnitudes  $C$  and  $AD$  change in reciprocal proportion, the product  $A.CD$

---

<sup>11</sup>For example, if  $A$  transforms into  $mA$ , then  $CD$  will transform into  $\frac{1}{m}CD$  or  $C\frac{1}{m}D$ ; now if  $C$  simultaneously transforms into  $nC$ , then  $\frac{1}{m}D$  transforms into  $\frac{n}{m}D$ , while the product of the two outer factors then transforms into  $\frac{m}{n}AD$  while  $C$  transforms into  $nC$ .

maintains a constant value, supposing the middle factor continues to represent the common system of the two factors  $A$  and  $CD$ .

We can easily show that this is in fact the case if we suppose that the regressive factors maintain their same orders. To this end, let  $A.CD$  and  $A'.C'D'$  be two such products, in which the middle factor  $C$  or  $C'$  represents the common system of the two regressive factors  $A$  and  $CD$  or  $A'$  and  $C'D'$ . We assume that, in the conversion of the one expression into the other,  $AD$  has been changed in reciprocal proportion with  $C$  (wherein it is already implicit that their system remains constant), and that the order of  $A$  and that of  $CD$  remain unchanged. We will show that the two products  $A.CD$  and  $A'.C'D'$  are equal.

To begin, we can bring the latter to a form in which the middle factor is the same as in the former product, whence the product of the two outer factors also maintains the same value in both. Thus if the latter product transforms into  $A_1.CD_1$ , then we have the simpler assumption that

$$AD = A_1D_1,$$

and  $A$  and  $A_1$ , and likewise  $D$  and  $D_1$ , are of equal order; thus it remains to be shown that

$$A.CD = A_1.CD_1.$$

Two equal outer products whose corresponding factors have equal orders (as here  $AD$  and  $A_1D_1$ ) must be generatable one from the other by a series of alterations in form, a part of which consists in the factors changing in reciprocal proportion, a part in one factor increasing by a term dependent on the other. With the former type of alteration it is immediately apparent that the value of the regressive product is not changed. With the other, either  $D$  can increase by a term dependent on  $A$ , or  $A$  by a term dependent on  $D$ . Thus, first transforming  $D$  into  $D + E$ , where  $E$  is dependent on  $A$ ,  $A.CD$  transforms into  $A.C(D + E)$  or equivalently  $A.(CD + CE)$ . Since  $E$  is dependent on  $A$ , while  $C$  is subordinate to  $A$ , and is therefore dependent on  $A$  in the  $c$ -th degree, it follows that  $CE$  is dependent on  $A$  in a degree higher than the  $c$ -th and can therefore be dropped as a term in the other factor; thus the value of the product remains the same. Second, the factor  $A$  can increase by a term dependent on  $D$ . Letting  $A$  equal  $CF$ , then if as always  $C$  represents the common system, the increase of the factor  $A$  by a term dependent on  $D$  will

be effected if  $F$  increases by a term dependent on  $D$ ; then, on the same grounds as before for the increment of  $D$ , this will leave the value of the whole product unchanged.

Thus we see that all alterations that leave the value of the middle factor and that of the two outer factors unchanged also leave the values of the whole product unchanged; or, going back yet another step, that if those two magnitudes  $C$  and  $AD$  are changed in reciprocal proportion, the value of the product  $A.CD$  is not changed under the assumption that the orders of  $A$  and  $CD$  remain the same. Combining this with the results of the previous paragraph, we can then say “the value of a regressive product consists in the common and nearest covering systems of the two factors if the orders of the factors is given, apart from the quantity to be distributed multiplicatively on the two systems.”

### § 132.

In this discussion the concept of the regressive product is still dependent on the values of the orders, since according to the foregoing definitions two products cannot yet be considered equal so long as their factors have unequal orders. This dependence of the concept on the values of the orders introduces a limitation that spoils the simplicity of the concept and is repugnant to the analytic treatment. In order to remove this limitation we require that two regressive products of nonzero values  $A.CD$  and  $A'.C'D'$ , in which the last two factors are conjoined by outer multiplication while the middle factor represents the common system of the two regressive factors (that is  $A$  and  $CD$  or  $A'$  and  $C'D'$ ), are equal so long as generally the product of the outermost factors with the middle one is equal in both expressions, or if they stand in reciprocal proportion, whether the values of the orders of the corresponding factors agree or not.<sup>12</sup> In particular, with this definition we can bring that regressive product into subordinate form (§ 129).

Thus, according to this

$$A.CD = AD.C$$

if in the first product  $C$  and  $D$  are conjoined by outer, and  $A$  and  $CD$

---

<sup>12</sup>We are warranted in such an extended definition since the comparison of regressive products with unequal orders for their factors was not previously established.

by regressive multiplication, and  $C$  represents the common system of the two regressive factors. For in the last expression  $AD$  can be imagined as the first factor,  $C$  as the middle, and the unit as the last, conjoined by outer multiplication with  $C$  (according to chapter 4 of part 1), while  $C$  still represents the common system. Interpreted in this way, the second expression has the same product of outer factors and the same middle factor as the first, and the two are therefore equal.

I have yet to mention here that if the product of the outer factors is of lower order than the relative system, then both products are simultaneously zero (according to § 127), whence their equality remains true for this case as well. Finally, if we adopt a specific part  $H$  of the principal system as principal measure (§ 87), then we can bring every regressive product with respect to that principal system to a form in which the first factor is the principal measure. Thus, according to what was said previously, we can bring each such product having a nonzero value to a form in which the first factor represents the relative system or here the principal system, and thus also, since we can change the factors in reciprocal proportion, the first factor is any given part of the principal system. If the regressive product is zero, then we can choose the first factor arbitrarily provided the second is zero, and thus in this case as well the product can be brought to the desired form.

If the product has been brought to this form, then we call its second factor the *characteristic* (specific) value or factor of that product with respect to the principal measure  $H$ , and its system, which is simultaneously the common system of the two factors, the “characteristic system of that magnitude;” its order, that is the order of the common system of the two factors,<sup>13</sup> we can interpret as the *order* of the magnitude itself. Only in this way does the value of the regressive product appear in its full simplicity.

### § 3.7. Introduction of Supplementary Numbers

From the concept of the regressive product established in the previous paragraphs we can now obtain its interchange law.

---

<sup>13</sup>Thus if the product magnitude has a nonzero value (and only in this case is it possible to speak of its order), then the order of the product magnitude equals the order of the regressive multiplication.



Thus we consider two products of nonzero value,

$$AB \cdot AC \quad \text{and} \quad AC \cdot AB,$$

where the dot indicates regressive multiplication, simple juxtaposition outer multiplication, and in which the factor  $A$  represents the common system, and thus  $ABC$  or  $ACB$  the nearest covering or relative system. Then according to the preceding paragraph one has

$$AB \cdot AC = ABC \cdot A, \quad AC \cdot AB = ACB \cdot A.$$

The two products are therefore equal or opposite according as  $ABC$  and  $ACB$  are, that is, according as the outer factors  $B$  and  $C$  can be interchanged with or without a sign change. Now (according to § 55) one must change the sign upon the interchange of two sequential outer factors if (and only if) the orders of both factors are odd. One will therefore also be able to interchange the factors of that regressive product with or without a sign change, according as the orders of  $B$  and  $C$  are simultaneously odd or not. However the orders of  $B$  and  $C$  supplement those of the regressive factors  $AC$  and  $AB$ , bringing them up to the order of the relative system  $ABC$ . Thus we call that number which supplements the order of a magnitude, bringing it up to that of the relative system, the *supplementary number* of that magnitude (with respect to that system), whence we have the law

The two factors of a regressive product can be interchanged with or without a sign change according as the supplementary numbers of the factors are both odd or not.

In this it is implicit that a factor representing the relative system can be interchanged without a sign change, since its supplementary number is zero, and thus even. This law corresponds to that established in § 55 for outer multiplication, with which the theorem in § 68 on the arbitrary position of numerical magnitudes is to be compared.

Since the supplementary number replaces the order here, it seems appropriate to seek the general correspondents to the above theorems on outer multiplication that are based on orders, which of course can only occur here with respect to products of two factors.

The order of a nonzero outer product is the sum of the orders of its factors. With regressive multiplication the order of the common system of the two factors is (according to § 132) interpreted as the

order of the product magnitude if this has a nonzero value. If  $a$  and  $b$  are the orders of the factors and  $h$  is that of the relative system, which here is simultaneously the nearest covering system, then according to § 126 that of the common system ( $g$ ) is equal to  $a + b - h$ . In order to introduce the supplementary numbers here, one can present the equation in the form

$$h - g = h - a + h - b,$$

or, if one symbolizes the supplementary numbers by  $a'$ ,  $b'$ ,  $g'$ , as

$$g' = a' + b',$$

that is, the supplementary number of a nonzero regressive product is the sum of the supplementary numbers of its factors.

It still remains for us to consider the case in which the product is zero. In regressive multiplication this case would occur (according to § 125) if the common system of the two factors were of higher order than the order of the regressive multiplication, that is  $a + b - h$ , thus if

$$g > a + b - h,$$

that is, if

$$h - a + h - b > h - g,$$

that is,

$$a' + b' > g',$$

and otherwise only if one of the factors is zero; that is, a regressive product of two nonzero values is zero if the supplementary numbers of the two factors taken together is greater than the supplementary number of the common system of the two factors. By way of contrast, an outer product of two nonzero values appears as zero if the orders of the factors taken together is greater than that of the nearest covering system of the two factors.

These laws for the two methods of multiplication thus correspond if one interchanges the concept of order with that of supplementary number, and that of the nearest covering system with that of the common system, a relation which, as we shall see, maintains its validity in further developments.

### § 3.8. Multiplication of Products That Appear in Subordinate Form

The product of three or more factors, to which we now turn, can always be reduced to that of two factors provided the multiplication of two factors can be established for the case that these factors are again products. Now since, if the factors are again regressive products, the sense of their multiplication is not yet established, we need a new definition here; and indeed we must establish what meaning an arbitrary product magnitude has as the first factor, and what as the second.

If a magnitude enters as the second factor, we will say that it is *multiplied by* it, and if as the first the factor itself is *multiplied*. I now define “multiplication by a product magnitude brought to subordinate form, that is, represented so that each subsequent factor is subordinate to its predecessor, as *progressive*<sup>14</sup> multiplication by its factors,” and “a product magnitude brought to subordinate form multiplied by any given magnitude means the last factor of the former multiplied by the latter (without changing the earlier factors).” Naturally, in order that the sense of the total multiplication be clear in this, the order of each of the individual multiplications to which that total reduces must be specified.

I will show at once that these definitions suffice for every real product. Thus the product will appear as real with nonzero value if the degree of the regressive multiplication agrees with the degree of dependence in the individual multiplications; on the other hand it will be zero if the degree of dependence with any one of these multiplications exceeds the order of the multiplication, since one of the factors will then be zero. It will have a purely formal significance if the degree of dependence is anywhere smaller than the order of the multiplication belonging thereto, unless the opposite result enters elsewhere.

---

<sup>14</sup>According to the usage introduced earlier, conjoined progressively with a series of magnitudes means so conjoined that each time the result of the conjunction is conjoined with the next sequential magnitude in the series.

### § 3.9. Every Real Product Can Be Brought to Subordinate Form

The demonstration that the definitions established suffice for the real product therefore coincides with the proof of the theorem that every real product can be brought to subordinate form.

Thus, first of all, the product of two pure factors (as we may call such factors as do not appear as further regressive products) can be brought to subordinate form, according to § 132. Now to such a product  $A.B$ , where  $B$  is subordinate to  $A$ , let a third factor be adjoined, dependent on  $B$  in the  $c$ -th degree and on  $A$  in the  $(c+d)$ -th, where its own order is  $c+d+e$ ; then it can be represented in the form  $CDE$ , where  $C$  is subordinate to  $B$  (and thus also to  $A$ ), and  $CD$  to  $A$ , but no dependence obtains otherwise, assuming that  $c$ ,  $d$ ,  $e$  are the orders of  $C$ ,  $D$ ,  $E$ . Then if this is a real product of nonzero value, that is if the order of multiplication agrees with the degree of dependence, it can be shown that

$$A.B.CDE = AE.BD.C.$$

Thus, since the product magnitude  $A.B$  appears here in subordinate form, it is then multiplied by another magnitude  $CDE$  if one multiplies its last factor by the latter; thus

$$A.B.CDE = A.(B.CDE).$$

But since  $C$  is subordinate to  $B$  and  $c$  is the degree of the multiplication,  $B.CDE$  equals  $BDE.C$  (according to § 132), whence that product

$$= A.(BDE.C).$$

Since  $C$  is subordinate to  $B$ , and hence also to  $BDE$ , it follows according to the first part of the definition (§ 134) that one multiplies by  $BDE.C$  by multiplying first by  $BDE$  and then multiplying the product of this multiplication by  $C$ . But now since  $B$  and  $D$ , and thus also  $BD$ , are subordinate to  $A$ , and  $(b+d)$  represents the degree of dependence,  $A.BDE$  equals  $AE.BD$ , and thus the above expression

$$= AE.BD.C.$$

This expression has subordinate form since  $C$  is subordinate to  $B$  and thus also to  $BD$ , while  $BD$  is subordinate to  $A$  and thus also to  $AE$ .

Consequently the progressive product of three factors can always be brought to subordinate form.

Upon introducing a fourth factor one can then begin by bringing the first three to subordinate form. Let  $A.B.C$  be this form. If one introduces a fourth factor, then in order that the sense of the multiplication be defined it must be established in what degree of dependence this factor stands with respect to each of the three magnitudes  $A$ ,  $B$ ,  $C$ , if the product is to have a real nonzero value. Suppose then that the fourth factor is dependent on  $C$  in  $d$ -th degree, on  $B$  in  $(d + e)$ -th degree, on  $A$  in  $(d + e + f)$ -th, while it has order  $d + e + f + g$  itself; then it can be represented in the form  $DEFG$ , where  $D$  is subordinate to  $C$ ,  $E$  to  $B$ ,  $F$  to  $A$ , and  $d$ ,  $e$ ,  $f$ ,  $g$  represent the orders of  $D$ ,  $E$ ,  $F$ ,  $G$ . One can then show that

$$A.B.C.DEFG = AG.BF.CE.D.$$

For

$$\begin{aligned} A.B.C.DEFG &= A.B.(C.DEFG) \\ &= A.B.(CEFG.D), \end{aligned}$$

since  $D$  is subordinate to  $C$ . Now since  $CEFG.D$  appears in subordinate form, one can progressively multiply by the individual factors  $CEFG$  and  $D$ ; but since  $C$  and  $E$ , and thus also  $CE$ , are subordinate to  $B$ ,  $B$  multiplied by  $CEFG$  gives the expression  $BFG.CE$ . One therefore finds that the above expression

$$\begin{aligned} &= A.(BFG.CE).D \\ &= A.BFG.CE.D \\ &= AG.BF.CE.D, \end{aligned}$$

since  $B$  and  $F$ , and thus also  $BF$ , are subordinate to  $A$ .

Thus the progressive product of four pure factors also appears in subordinate form, and it can already be seen that in exactly the same way it follows in general that a progressive product of arbitrarily many factors can be brought to subordinate form. But if this is the case, then since according to the definitions the multiplications can in general be reduced to the progressive multiplication of pure magnitudes, the same is also valid for arbitrary real products, that is

**Every real product can be represented in subordinate form.**

Thus the above definitions in fact suffice for the real product, and the subordinate form, as the simplest to which the real product can be brought, defines its meaning.

**§ 3.10. Multiplication by Mutually Incident Magnitudes**

We now address the problem of combining the different transformations permitted by the regressive product according to the presentation so far, and expressing them in a simple general theorem to which we can always refer in the sequel, whenever such transformations need to be considered.

To achieve this we simply continue the use of the transformations introduced in the previous section, and express them verbally. There it was found that

$$A.B.CDE = AE.BD.C$$

if  $B$  is subordinate to  $A$ ,  $C$  represents the system that  $CDE$  has in common with  $B$  and thus also with  $A$ , and furthermore the type of multiplication is taken to be such that it yields a real nonzero value under these assumptions. Under the same assumptions it also follows that

$$EDC.B.A = EA.DB.C.$$

For

$$\begin{aligned} EDC.B.A &= (EDC.B).A \\ &= (EDB.C).A, \end{aligned}$$

and since  $EDB.C$  appears in subordinate form, one multiplies it (according to § 134) by  $A$  by multiplying  $C$  by  $A$ . Since  $C$  is subordinate to  $A$ , according to § 133 the order is unimportant here; one therefore finds that the last expression

$$= EDB.(A.C).$$

Since in addition  $A.C$  has been brought to subordinate form, one can progressively multiply by  $A$  and  $C$ , to find that the last expression

$$= EA.DB.C.$$

The expression

$$EDC.A.B \quad \text{or} \quad EDC.(A.B)$$

now reduces to the same form; for, since  $EDC.A$  equals  $EA.DC$ , one has

$$\begin{aligned} EDC.A.B &= EA.DC.B \\ &= EA.DB.C. \end{aligned}$$

It therefore follows that in a product of real nonzero value one is permitted to multiply progressively by two mutually incident<sup>15</sup> factors in arbitrary order, or alternatively to multiply all at once by their product.

If  $c, d, e$  are the orders of  $C, D, E$ , then (cf. the previous section) it is assumed here that  $EDC$  depends on  $A$  in  $(c + d)$ -th degree, and on  $B$  in  $c$ -th degree; and since in the two products

$$EDC.A.B \quad \text{and} \quad EDC.B.A$$

the method of multiplication is taken to be real and of nonzero value if the degree of dependence just indicated obtains, each of the two products will be zero if and only if the degree of dependence increases, and hence if one of these products is zero then the other must also be zero. Consequently the above law also remains valid if the product is only known to be real, and since it can immediately be generalized from two mutually incident factors to several, we have the theorem:

Instead of multiplying by a product of mutually incident factors one can progressively multiply by the individual factors, and indeed in any order.

Here we have assumed the method of multiplication is such that the product appears real with the same dependence relation in all these forms simultaneously. Consequently this law expresses a generalization of the first part of the definition (§ 134), since instead of multiplying by the product appearing in subordinate form one can progressively multiply by its factors. The law that generalizes the second part of the definition (§ 134), that is, that one multiplies a product of mutually incident factors by a magnitude by multiplying the final factor by it, easily follows in the same way as the above

---

<sup>15</sup>We call magnitudes *mutually incident* when one is subordinate to the other.

law but is of lesser importance. Moreover it is clear that the above law simultaneously includes the law about the middle factor in § 132, that is

$$BA \cdot AC = BAC \cdot A,$$

since, instead of progressively multiplying  $B$  by  $A$  and the factor  $AC$  subordinate to it, one can multiply in the inverse order.

### § 3.11. § 137. Characteristic Value of a Regressive Product of Several Factors. Pure and Mixed Products

We now leave the general concept of the regressive product and restrict our treatment to the case that the multiplication can always be referred to the same principal system. Now since, according to § 132, once each such product is brought to subordinate form it necessarily has as first factor either a magnitude representing the principal system or one which can be represented in this form, it follows that if one applies the procedure given in § 135 to a product of several factors referred to the same principal system, the product can be brought to the form in which all the factors with the exception of the last represent the principal system.<sup>16</sup>

If we bring all those earlier factors representing the principal system to the same magnitude value by applying the general multiplicative alteration of form, and regard this value as the principal measure, then, as was already established in § 132 with respect to two factors, we can call the last factor “the characteristic (specific) value or factor of that product magnitude with respect to this principal measure,” and its system “the characteristic system” of the product magnitude, and we can interpret the order of this system as the order of that product magnitude itself. Further, we can call the magnitudes resulting by multiplication of pure magnitudes *relative magnitudes*, since they only have a simple meaning relative to a system or a measure. The characteristic value of a pure magnitude naturally appears as this magnitude itself.

---

<sup>16</sup>Thus for example let  $H.A.B$  be this product, in which  $H$  represents the principal system while the product of the first two factors is already brought to the required form; now let  $B = CD$ , where in the case that the whole product has a nonzero value  $AD$  represents the principal system. Then that whole product equals  $H.AD.C$ , which has the required form. If the whole product is zero, one can choose the first factors arbitrarily, provided the last is zero; thus in this case the product can also be brought to the required form.



What we said in § 128 about the notation for multiplication by two factors is also valid here, that is, once the principal system is fixed as the relative system it appears superfluous to notationally distinguish outer multiplication from regressive, or the different degrees of the latter.<sup>17</sup> On the other hand there appears here a new distinction, namely that between *pure* and *mixed* products.

Thus I call those products *pure* whose factors are always conjoined progressively by the same type of multiplication, that is either by outer multiplication (outer products) or only by regressive multiplication with respect to one and the same system (regressive product); on the other hand I call those products *mixed* whose factors are progressively conjoined either by both types of multiplication (outer and regressive) or indeed only by regressive multiplications, but with respect to different systems.

Since the pure and mixed products are subject to different laws, their distinction is very important; and although a distinction by notation is unnecessary, since whether the product is pure or mixed is also always defined by the orders of the factors if the principal system is fixed as the relative system, yet in many cases such a distinction seems very convenient. I will therefore make use of the dot in such cases to separate the factors of a pure product, and thus establish that, where dots are used to mark multiplication, then always, when they are arranged either without any parentheses or within the same pair, they separate the factors of a pure product, whence a product of magnitudes written immediately adjacent to each other always appears as a single factor with respect to this dot: For example,  $AB \cdot CD \cdot EF$  denotes a pure product whose factors are  $AB$ ,  $CD$ ,  $EF$ .

---

<sup>17</sup>This would be completely otherwise for general real multiplication, since with it one must establish the different degrees of dependence between the individual factors for which the product still has a nonzero value. The product of several factors would then be defined by a series of numbers representing those degrees of dependence; this definition would then be a composite and would no longer represent an elementary concept. And this is the reason why we have omitted this general case here.

### § 3.12. Law for the Supplementary Numbers of Pure Products

We can now apply the theorem, established in § 133 for two factors, to several factors.

First, with respect to interchange, it is evident that for several factors the position of a factor representing the relative system is completely unimportant; and it therefore follows in general that in order to multiply two product magnitudes one need only multiply their characteristic values with respect to any given principal measure, and this product has the principal measure included as many times as it appears in the two magnitudes taken together; for example,  $H^m A \cdot H^n B$  equals  $H^m H^n A.B$  or  $H^{m+n} A.B$ , where  $H$  represents the principal measure. Implicit in this is that two product magnitudes appearing together as factors are likewise interchangeable with or without a sign change according as their supplementary numbers are both simultaneously odd or not.

We can only apply the subsequent theorems of that section to pure regressive products. Thus, since the supplementary numbers of a nonzero regressive product with two factors is the sum of the supplementary numbers of the factors, this law remains true if another regressive factor is adjoined to this regressive product, and this latter product is again of nonzero value; and as is clear upon a double application of the theorem for two factors, the supplementary number of the whole product is then the sum of the supplementary numbers of its factors, and so forth for arbitrarily many factors. Furthermore, since the product of two factors appears as regressive if and only if the sum of the two orders is greater, that is the sum of the supplementary numbers is less, than the order of the principal system, the nonzero product of three or more factors also appears as a pure regressive product if and only if the sum of the supplementary numbers always remains smaller than the order of the principal system, that is, the sum of all supplementary numbers of the factors always remains smaller than the order of the principal system.

Finally, in order to apply the theorem in § 133 regarding when the product is zero, we mention in this connection that the sum of the supplementary numbers of two magnitudes having the relative system as nearest covering system, and thus having a nonzero value, is equal to the supplementary number of their common system; however, if the nearest covering system is lower than the relative system,

and the product is therefore zero, the order of the common system is greater,<sup>18</sup> and its supplementary number therefore smaller, than the sum of the supplementary numbers belonging to the factors. Now if we adjoin a factor, then the common system of all the factors is again the one that the adjoined factor has in common with the common system of all the previous factors. Thus so long as the whole product maintains a nonzero value, the sum of all the supplementary numbers will be equal to the supplementary number of the common system of the collection of factors; but if the product becomes zero upon the introduction of any factor without that introduced factor itself being zero, the supplementary number of the common system will be smaller, and thus if more factors are adjoined will remain smaller, than the running sum of the supplementary numbers of the factors. Thus a pure regressive product whose factors are of nonzero value will be zero if and only if the supplementary number of the common system of all the factors is smaller than the sum of the supplementary numbers of the factors. The same method of proof shows that the characteristic value of such a product (if it is not zero) represents that of the common system of all the factors.

If we combine the laws formulated for the supplementary numbers and include as well the corresponding laws on the orders of outer products, we obtain the theorem:

**A product of arbitrarily many factors of nonzero value is pure if either the orders or supplementary numbers of all the factors combined is smaller than the order of the principal system, and in the former case it is an outer product, in the latter a regressive, and a mixed product if the case is neither of these two. The pure product is zero in the former case if the combination of the orders of the factors is greater than the order of the nearest covering system of the factors, and in the latter, if the combination of the supplementary numbers of the factors is greater than the supplementary number of the common system of the factors. If the pure product is of nonzero value, then its characteristic value represents the nearest covering system in the former case, and the**

---

<sup>18</sup>Cf. note to § 126.

common system in the latter; and in the former case its order is the sum of the orders of the factors, in the latter its supplementary number is the sum of the supplementary numbers of the factors.

### § 3.13. The Factors of a Pure Product can be Arbitrarily Combined

We now proceed to the multiplicative law of combination, that is, we investigate whether and to what extent it is possible to set

$$PQR = P(QR).$$

From the theorem in § 136 it already follows that this law is not generally valid for the mixed product of three factors<sup>19</sup>; on the other hand we will show that it is valid for the pure product in the general sense that, using the notation introduced in § 137, one always has

$$P \cdot Q \cdot R = P \cdot (Q \cdot R).$$

First of all, it is clear that, if the validity of this law is established for the case that  $P$ ,  $Q$ ,  $R$ , are pure magnitudes, it is at once also proved for the case that they are all or partly relative magnitudes. For according to the preceding paragraph, one has to multiply relative magnitudes together by multiplying their characteristic values together with respect to one and the same principal measure and introducing the principal measure into the product as a factor in as many places as the principal measure was included in the two magnitudes together. Since, according to this, one can in general place the factor representing the principal measure in an arbitrary position and can arbitrarily move it into or out of parentheses, it follows that that law is also valid in general for relative magnitudes if it is valid for pure magnitudes. Now according to the laws of outer multiplication it is valid for outer products of pure magnitudes and thus for outer products generally. It therefore only remains to be shown that it is also valid for the pure regressive product of pure magnitudes.

---

<sup>19</sup>To be sure, cases can be found in which our law still finds its application by means of the theorem in § 136; but these cases are so isolated, and the constraints under which they occur so contrived, that no scientific advantage results from their enumeration.

In this case the problem is to show that, if the regressive product has a nonzero value,  $P$ ,  $Q$ ,  $R$  can be represented in the forms  $ABC$ ,  $ABD$ ,  $ADC$ , so that  $ABCD$  represents the principal system of them all.

Let the supplementary numbers of the magnitudes  $P$ ,  $Q$ ,  $R$  be  $d$ ,  $c$ ,  $b$ , respectively; then according to § 138 (at the end) the supplementary number of the product of the common system  $A$  of these factors is equal to the sum of these numbers, that is, it is equal to  $b + c + d$ ; and therefore if  $a$  is equal to the order of that common system, then that of the principal system is equal to  $a + b + c + d$ . According to the same theorem, two of the factors, for example  $P$  and  $Q$ , have a common system whose supplementary number is the sum of the supplementary numbers of those factors, and therefore equals  $c + d$  here; the order of this common system thus equals  $a + b$ , and therefore this system can be represented by a product  $AB$ , in which  $B$  is of  $b$ -th order and is independent of  $A$ . Likewise, the common system of  $P$  and  $R$  is of  $(a + c)$ -th order, and thus includes in itself a magnitude  $C$ , independent of  $A$  and of  $c$ -th order. And thus  $C$  must be independent of  $AB$ ; for if it were dependent, that is, if  $C$  had any magnitude in common with  $AB$ , then the three factors  $P$ ,  $Q$ ,  $R$  would have this magnitude in common, and hence a magnitude independent of  $A$ , in conflict with the assumption.

Thus there are now three mutually independent magnitudes  $A$ ,  $B$ ,  $C$  subordinate to the magnitude  $P$ , as also is their product  $ABC$ . It must therefore be possible to represent  $P$  as a product, one factor of which is  $ABC$ ; but since  $P$  itself is of  $(a + b + c)$ -th order, the other factor that  $P$  includes besides  $ABC$  is of zeroth order; that is, it is a pure numerical magnitude, and  $P$  can therefore be represented as a multiple of  $ABC$ .

Finally, on the same grounds  $Q$  and  $R$  have a common factor  $D$  independent of  $A$ , whence the magnitudes are representable as multiples of  $ABC$ ,  $ABD$ , and  $ADC$  respectively; indeed, since for the magnitudes  $A$ ,  $B$ ,  $C$ ,  $D$  only the systems represented by them are defined, they themselves may be taken to be arbitrarily great, whence also, as is easily seen, one can also assume that the magnitudes  $P$ ,  $Q$ ,  $R$  are equal to those values themselves, and thus

$$P \cdot Q \cdot R = ABC \cdot ABD \cdot ADC.$$

Since we assumed that the whole product is to have a nonzero value, so also must the product  $ABC \cdot ABD$ , for example, whence the

nearest covering system here, that is  $ABCD$ , is simultaneously the relative system. This product is therefore equal to  $ABCD \cdot AB$ ; thus the whole expression

$$= ABCD \cdot AB \cdot ADC$$

and since  $AB \cdot ADC = ABDC \cdot A$ ,

$$= ABCD \cdot ABDC \cdot A.$$

Now the other product  $P \cdot (Q \cdot R)$  can be brought to the same form; for  $Q \cdot R$  or  $ABD \cdot ADC$  equals  $ABDC \cdot AD$ , whence

$$P \cdot (Q \cdot R) = ABC \cdot (ABDC \cdot AD).$$

Since  $ABDC$  represents the principal system, according to § 138 we can multiply the characteristic values together and introduce  $ABDC$  as a factor. But we find that  $ABC \cdot AD$  is equal to  $ABCD \cdot A$ , whence the above expression

$$= ABCD \cdot ABDC \cdot A.$$

Thus since the two products  $P \cdot Q \cdot R$  and  $P \cdot (Q \cdot R)$  are equal to the same expression, they are equal to each other.

In this proof we assumed that the products had a nonzero value. Now however they can only become zero simultaneously, for according to § 138 the zero value is possible if and only if the common system of the factors is of higher order than the amount of the (order of the relative system diminished by the) sum of the supplementary numbers, and this can only occur simultaneously in both products. Thus for this case as well the equality of the two products is maintained. The law is therefore generally valid for pure magnitudes and hence must also be valid for relative magnitudes, as we have seen above; thus in general for pure magnitudes one has

$$P \cdot Q \cdot R = P \cdot (Q \cdot R).$$

Ultimately, therefore, since the law of combination must be valid for arbitrarily many factors if it is valid for three (§ 3), there follows the general theorem

**The factors of a pure product may be arbitrarily combined.**

### § 3.14. Relation to Addition and Subtraction

For the addition of relative magnitudes, the general law relating it to multiplication provides a definition. One has only to bring the two magnitudes to subordinate form. Brought to this form, they are summable if, on the one hand, the principal measure appears the same number of times in both of them, and on the other, if the magnitudes themselves have the same order; and thus they are added when one adds their characteristic values, and the principal measure is included as a factor in the sum as many times as it appears as a factor in each of the products.<sup>20</sup>

The general law relating multiplication to addition is that

$$P.(Q + R) = P.Q + P.R,$$

and

$$(Q + R).P = Q.P + R.P.$$

We have to prove the validity of this law first of all for the case that the magnitudes  $P, Q, R$  are pure, since the introduction of arbitrary factors representing the principal measure to which the magnitudes are referred can change nothing. We therefore assume first of all that  $P, Q, R$  are pure magnitudes.

In order to bring the terms of the sum

$$P.Q + P.R$$

into subordinate form, let  $Q = AB$ , where  $A$  is subordinate to  $P$ , while  $PB$  represents the principal system to which the multiplication is referred, and may be set equal to  $H$ , and likewise let  $R = CD$ , where  $C$  is subordinate to  $P$  and  $PD$  represents the principal system. Since  $D$  can assume any arbitrary magnitude here (whence  $C$  must then only be changed in reciprocal proportion with  $D$ ), one can then take it such that

$$PD = PB = H.$$

Then

$$P.Q + P.R = HA + HC = H(A + C),$$

this last according to the definition.

---

<sup>20</sup>This procedure serves precisely as a definition since we mean by the sum of two relative magnitudes the sum formed in the way given.

Now we can also bring  $P.(Q + R)$  to the same form. Thus since  $PD$  equals  $PB$ , it follows that  $D$  can also be set equal to  $B$  plus a magnitude, which we will call  $K$ , dependent on  $P$ ; consequently  $R$ , which was set equal to  $CD$ , equals  $C(B + K)$  or  $CB + CK$ . Thus

$$P.(Q + R) = P.(AB + CB + CK).$$

Since here  $K$  is dependent on  $P$ ,  $CK$  depends on  $P$  in a higher degree than  $CB$ , whence it can produce no nonzero product with  $P$ , and thus according to § 125 it can be dropped. The above expression is thus

$$\begin{aligned} &= P.(AB + CB) \\ &= P.(A + C)B. \end{aligned}$$

Since  $A$  and  $C$ , and therefore  $A + C$ , are here subordinate to  $P$ , while  $PB$  or  $H$  represents the principal system, the last expression is again

$$= H(A + C).$$

Thus the two expressions  $P.(Q + R)$  and  $P.Q + P.R$ , which are to be equated, are equal to the same third expression, whence they are equal to each other.

Now if the principal measure is adjoined several times to  $P$ , say  $m$  times, and likewise also to  $Q$  and  $R$ , but to the latter equally many times, say  $n$ , so that they remain summable, then this is as good as if  $H$  were adjoined as a factor to each of the two expressions  $(m + n)$  times, whence they remain equal if they were so before. Finally, since the same can now be said of the two expressions  $(Q + R).P$  and  $Q.P + R.P$ , it follows that the law of the relation of multiplication to addition is also valid in full generality for these new types of addition and multiplication. Consequently all laws based on it are also valid, that is:

**All laws expressing the relation of multiplication to addition and subtraction are still generally valid for each type of addition and multiplication that has been established so far.**



### § 3.15. § 141. Division With Respect to a System; Degree of the Relative Magnitude

For division<sup>21</sup> one immediately finds that it is only real if the divisor and dividend are mutually incident, that is if the divisor is subordinate to the dividend or vice versa.

In the former case it is outer division, in the latter regressive; thus if both cases appear together, that is, if the divisor and the dividend are similar to each other, then the division can be regarded as outer as well as regressive. And indeed, these definitions are not only valid if the two magnitudes to be conjoined are pure magnitudes, but even if they are relative magnitudes.

In the latter case it is important that the characteristic values stand in the relation given, while the principal system to which both magnitudes are referred is the same. Under these circumstances it can then happen that the principal measure appears more often in the divisor than in the dividend; the quotient then appears as a real magnitude divided several times by the principal measure, or multiplied by a power of the principal measure whose exponent is negative. We therefore also interpret these new magnitudes as relative magnitudes, and call the exponent of that power of the principal measure with which the characteristic value of a relative magnitude is bound by multiplication the *degree* of the relative magnitude. Consequently the new magnitude is a relative magnitude whose degree is negative, while the degree of those previously considered is positive, and in addition the pure magnitudes can now be regarded as relative magnitudes of degree zero.

In this connection I must remark that the magnitudes of zeroth order and the magnitudes representing the principal system, that is the magnitudes of zeroth and  $h$ -th order (if  $h$  is the order of the principal system) can be interpreted in a twofold way. Thus “a magnitude of zeroth order and  $n$ -th degree can be regarded as a magnitude of  $h$ -th order and  $(n - 1)$ -th degree,” since one can imagine the characteristic value of that magnitude (which is simply a numerical magnitude) multiplied by one of the factors representing the principal measure, and interpret this product as the characteristic value of that magnitude, whereby of course its degree diminishes by one. Likewise conversely, “every magnitude of  $h$ -th order and  $n$ -th degree can be

---

<sup>21</sup>(1877) See the footnotes to § 6 and § 11.

regarded as a magnitude of zeroth order and  $(n + 1)$ -th degree." In general we prefer to consider such a magnitude as a magnitude of zeroth order.

It is important now to investigate the uniqueness of the quotient. To this end let  $A$  be the dividend,  $B$  the divisor as first factor,  $C$  a value of this quotient, so that

$$B.C = A,$$

and the quotient appears in the form  $\frac{A}{B}$ . Now every value that satisfies that equation when it replaces  $C$  can also be interpreted as a particular value of this quotient. Each such value can be generated from the value  $C$  by addition, but  $B$  multiplied by the terms added to  $C$  must give zero if the product is to remain equal to  $A$ , and each such added term will then likewise permit the product to equal  $A$ ; now we can in general symbolize such a term, which gives zero when  $B$  multiplies it, by  $\mathfrak{A}$ , and therefore say that, if  $C$  is a particular value of the quotient and  $B$  the divisor, then the complete value of the quotient equals

$$C + \mathfrak{A},$$

as we have already proved for outer division in § 63. Yet here we must always keep in mind as well that by  $\mathfrak{A}$  is meant a magnitude that can be added to  $C$ , that is a magnitude equal to  $C$  in order and degree. The quotient will therefore be unique if under this assumption  $\mathfrak{A}$  is always zero, that is if there is no other magnitude  $X$  of this type, except for zero itself, that yields zero when  $B$  multiplies it.

Since the product of a magnitude of zeroth order, which itself is neither zero nor a magnitude representing the principal system, always produces a nonzero value if the other factor has a nonzero value, it follows that if  $B$  has a nonzero value and simultaneously either  $B$  itself or  $X$  is a magnitude of zeroth or  $h$ -th order,  $X$  must invariably be zero if  $B.X$  is to be zero. Thus in this case, but no other, the quotient will also be unique. For if both magnitudes  $B$  and  $X$  are of intermediate order, that is if their orders lie between 0 and  $h$ , then without its being zero  $X$  can always be taken so that  $B$  and  $X$  are mutually dependent, and their nearest covering system is not yet the principal system itself; then, according to § 128, there will be a nonzero value for  $X$  whose product with  $B$  gives zero, that is, the quotient will not be unique.

If the divisor is zero, then, as we have already learned, since zero conjoined in a product with any magnitude gives zero, the dividend must also be zero if the quotient is to be one of the magnitudes developed so far, and indeed each of these magnitudes can be regarded as a particular value of the quotient. If however the dividend is a magnitude of nonzero value while the divisor is zero, then the quotient appears as a magnitude of a completely new type, which we can call an *infinite magnitude*, while those previously considered appear as *finite*.

We can now combine the results just obtained by recognizing that if  $C$  is of zeroth or  $h$ -th order the dividend and divisor are similar, whence we are led to the theorem:

The quotient represents a finite magnitude if and only if the divisor is of nonzero value and simultaneously either it can itself be represented as a magnitude of zeroth order, or it is similar to the dividend. If the dividend and divisor are zero, the quotient is any finite magnitude. In every other case, that is if the divisor is not zero and simultaneously the divisor and quotient are both of intermediate order, the quotient is only partially defined, and indeed one then obtains the general value by adding to the general expression a magnitude that gives zero when multiplied by the divisor.

Particular interest attaches to such expressions whose dividend is unity while the divisor represents a magnitude of nonzero order, for example the quotient  $\frac{1}{ab}$ . If  $abcd$  or  $H$  is the principal measure, then

$$\frac{1}{ab} = \frac{cd + \mathfrak{A}}{ab \cdot cd},$$

where  $\mathfrak{A}$  represents any magnitude of second order dependent on  $ab$ .

### § 3.16. Complete Analogy Between Outer and Regressive Multiplication

In order for us to complete the analogy between outer multiplication and real regressive multiplication there remains one further consideration. Thus with outer multiplication all magnitudes of higher order can be represented as products of magnitudes of first order, and their

law of conjunction can be obtained formally from the laws of conjunction for magnitudes of first order. According to § 138,<sup>22</sup> in regressive multiplication there correspond to the magnitudes of first order magnitudes whose supplementary number is one, that is magnitudes of  $(h - 1)$ -th order, if the relative system for all magnitudes and products is the same, and is in fact a system of  $h$ -th order. According to § 138 their multiplication results in magnitudes whose supplementary numbers exceed unity, and whose orders are therefore smaller than  $(h - 1)$ . In order to demonstrate the full analogy it is therefore important to prove the analogy of the laws for these magnitudes of first and  $(h - 1)$ -th order.

We have established the identity of these laws insofar as they represent only the general laws of conjunction of the four fundamental operations (addition, subtraction, multiplication, division). We have also showed that the laws of outer multiplication as such, so long as they refer only to the concepts of order and the common system, are also valid for regressive multiplication with respect to a fixed principal system if one introduces the concept of supplementary number instead of that of order, and the concept of the nearest covering system instead of the common system, and conversely. Thus since the concept of dependence, on which all the special laws of outer multiplication are based, is defined by that of the common or nearest covering system, the laws of outer multiplication can also be applied to pure regressive multiplication according to that principle.

But the concept of dependence, which first appeared with magnitudes of first order, was originally defined completely differently, and many laws developed subsequently were based on this original definition. Thus originally a magnitude of first order was represented as dependent on a series of such magnitudes if it could be expressed as a sum of displacements to which these are similar, or, as we expressed it later, if it is expressible as a multiple sum of them. And thus in general we called several magnitudes of first order mutually dependent if one of them was representable as a multiple sum of the rest, and it only followed then by means of the original concept of a system that  $n$  magnitudes of first order are mutually dependent if and only if they are included in a system of lower order than the  $n$ -th; and by means of the concept of outer multiplication, that the product of dependent magnitudes, and only such, is zero. We must

---

<sup>22</sup>Cf. also § 137.

therefore seek the analogue of that original definition in our domain.

First, if, in a system of  $n$ -th order,  $n$  magnitudes of first order are given whose outer product is not zero, then it has been shown that every other magnitude of first order belonging to this system can be represented as a multiple sum of these original ones. The analogous theorem would read: If in a system of  $n$ -th order  $n$  magnitudes of  $(n - 1)$ -th order are given, whose regressive product with respect to that system is not zero, then every other magnitude of  $(n - 1)$ -th order belonging to the system can be represented as a multiple sum of those original ones.

The proof of this theorem follows from § 138. Thus, according to that section, every  $(n - 1)$  of the  $n$  factors having the property stated in the theorem have as a common system a system of first order, while all  $n$  factors have no system of nonzero order in common if their product is to have a nonzero value. Thus in all, there are given  $n$  such systems of first order, each of which is subordinate to  $(n - 1)$  of the  $n$  factors. These  $n$  systems of first order must however be mutually independent, for were one of them dependent on the remaining  $(n - 1)$ , it would then have to lie in the system defined by them (according to the original definition of the system); but these remaining systems are subordinate to one of the  $n$  factors, whence that first system must also be subordinate to this factor. That first system is however the common system of the remaining  $(n - 1)$  factors, whence this system would be common to all  $n$  factors and so their product would be zero according to § 138, contrary to assumption. Thus those  $n$  systems of first order are mutually independent. Now if we take  $n$  arbitrary magnitudes of first order belonging to these systems and thus likewise mutually independent, then first of all each of the  $n$  given factors can be represented as a multiple of an outer product of those magnitudes, since  $(n - 1)$  of those magnitudes are subordinate to it, and it is itself of  $(n - 1)$ -th order; and furthermore every magnitude of first order belonging to the principal system (of  $n$ -th order) can be represented as a multiple sum of those  $n$  magnitudes of first order, whence in addition every magnitude of  $(n - 1)$ -th order belonging to that principal system can be represented as the outer product of  $(n - 1)$  such multiple sums. The product of these  $(n - 1)$  multiple sums is however changed, upon multiplying out, into a multiple sum of outer products of  $(n - 1)$  factors of those  $n$  magnitudes of first order, whence, since these products are similar to the  $n$  given factors, into a multiple sum of

these factors as well.

Thus we have proved the theorem stated above. Our problem is however not yet solved. On the contrary, the essence of outer multiplication as outer depends on the theorem that a product of magnitudes of first order is zero if and only if one of them can be represented as a multiple sum of the rest, and since we have not applied this theorem to our domain, the analogy is not yet complete.

That a product of magnitudes of  $(n - 1)$ -th order will always be zero if one of them can be represented as a multiple sum of the others becomes clear at once from the theorem on multiplying out, if one simultaneously recognizes that the product of two similar magnitudes of  $(n - 1)$ -th order is zero. In order to prove as well that the product is only zero if one of the factors can be represented as a multiple sum of the others, we must show that if, to a nonzero product of  $m$  factors of  $(n - 1)$ -th order in a principal system of  $n$ -th order, is adjoined a factor of the same  $(n - 1)$ -th order that makes the product zero, then this adjoined factor can be represented as a multiple sum of the previous ones.

That a product of more than  $n$  factors of this type is zero is already a consequence of the general theorem of § 138; but it also follows at once from the theorem proved above. Further, if to  $n$  such factors whose product has a nonzero value a factor of the same order is adjoined, then on the one hand the product is always zero, and on the other it can be represented as a multiple sum of those  $n$  factors, as we showed above. In order to proceed with the proof of the theorem it therefore only remains for us to consider the case that the number of the factors ( $m$ ) is smaller than the order of the principal system ( $n$ ).

To proceed with the proof in this case we can take  $(n - m)$  auxiliary factors of  $(n - 1)$ -th order that yield a product of nonzero value with the  $m$  given factors. Then the factor of  $(n - 1)$ -th order, which is to be introduced into the product of the  $m$  given factors ( $P$ ) and thus make it zero, will, according to the theorem proved previously, be representable as a multiple sum of all  $n$  of the magnitudes whose product has a nonzero value, that is, a sum, one of whose terms ( $A$ ) is a multiple sum of the  $m$  given factors, the other term ( $B$ ) being a multiple sum of the auxiliary factors; and it remains to be shown that this second term is zero. Now if we multiply the product of the  $m$  given factors ( $P$ ) by this sum ( $A + B$ ), then we can drop the first term ( $A$ ), since it appears as a multiple sum of the first  $m$  factors,

and therefore gives zero when they multiply it. Now since the product of that sum and the  $m$  given factors should equal zero, that is  $P.(A + B) = 0$ , it follows that the product of its second term and the  $m$  factors must be zero; thus

$$P.B = 0.$$

The second term  $B$  is however a multiple sum of the  $(n - m)$  auxiliary factors; and we can show that, since the coefficients of this multiple sum must each equal zero, it must itself be zero. To this end, if one multiplies by the individual terms instead of the multiple sum  $B$ , one then obtains a multiple sum with the same coefficients, and indeed besides the  $m$  given factors each term contains one of those taken as auxiliary. Now in order to prove that the coefficient of any such term is zero, one has only to multiply both sides of the above equation, or rather its terms, by those  $(n - m - 1)$  of the auxiliary factors that this term lacks; then it is clear that all the terms except that one drop out, and the equation expresses that this term, and thus also its coefficient, is zero. Since all the coefficients of the multiple sum are zero, it is itself zero; the factor introduced, which was set equal to  $(A + B)$ , is therefore equal to  $A$ , that is to a multiple sum of the  $m$  given factors, as we wished to show. Thus, combining the results obtained, we are led to the theorem:

**A product of magnitudes of  $(n - 1)$ -th order with respect to a principal system of  $n$ -th order is zero if and only if one of them can be represented as a multiple sum of the others.**

This law completes the analogy between regressive and outer multiplication so long as the reference system is one and the same and at the same time represents the principal system. And the analogy demonstrated applies so extensively that all laws of outer multiplication, that is even those that refer back to the general concept of conjunction or to the concepts of super- or subordination of magnitudes and order, are valid in analogous form for regressive multiplication with respect to the principal system, when one interchanges the concepts of super- and subordination and replaces the concept of order with that of supplementary number. And also, since the adjoining of factors that represent the principal system does not change the equations if it occurs equally many times in all the terms of the equation,

those laws also still apply if one replaces the pure magnitudes with relative magnitudes whose relative system is likewise the principal system.

### § 3.17. § 143. Double System and Products with Respect to It

Having proved the complete analogy between the outer and regressive products, I indicate yet another generalization of the foregoing considerations.<sup>23</sup>

Thus if one has several magnitudes superordinate to the same system of  $a$ -th order and subordinate to the same system of  $(a + b)$ -th order, one can then represent them as products, one factor ( $A$ ) of which is of  $a$ -th order and is the same in all of them, while the other factors belong to the same system of  $b$ -th order,  $B$ , which is independent of  $A$ . It is then clear at once that every numerical relation that exists between these factors belonging to the system  $B$  must also prevail between the original magnitudes (since according to § 81 one can drop the factor  $A$  in the equation that represents that numerical relation). In particular, if we take magnitudes of  $(a + 1)$ -th order, for example  $Ac, Ad, \dots$ , where  $c, d, \dots$  belong to the system  $B$ , then the same numerical relations prevail between  $Ac, Ad, \dots$  as between  $c, d, \dots$ , and conversely.

Defining the concept of such magnitudes  $Ac, Ad, \dots$  so that it is zero if the product of the corresponding magnitudes  $c, d, \dots$  is, one can then apply all concepts and laws of magnitudes of first order in a system of  $b$ -th order, and therefore in addition all concepts and laws for magnitudes of higher order in such a system to those magnitudes of  $(a + 1)$ -th order and the magnitudes generated from them in the same way. In this way a series of new concepts is generated, the most important of which I will summarize here.

We can call the union of two such systems, one of which is subordinate to the other, a *double system*, and say that a magnitude is *incident* on this double system if it is superordinate to one of the two systems comprising the double system, and subordinate to the other. We can call the higher of the two systems comprising the double system the *superior system*, and the lower the *inferior system*. It

<sup>23</sup>(1877) I have also abandoned the conceptual generalization indicated here in the edition of 1862.



can then be shown that a product with respect to a double system of two nonzero values incident on the double system is certainly zero if and only if the common system of the two factors is of higher order and the inferior system and simultaneously their nearest covering system is of lower than the superior system; furthermore, a product of nonzero value with respect to that double system appears as an outer product if the common system of the factors is the inferior system and simultaneously their nearest covering system is the superior system.

Hermann Grassmann

# A New Branch of Mathematics

The Ausdehnungslehre of 1844

and Other Works

Pages 251–300

Chunk 6 of 12

Translated by Lloyd C. Kannenberg



# Linear Extension Theory<sup>1</sup>

## § 0.156. Direct and Reciprocal Affinity. General Theorem

By applying the law developed in § 155 to the two methods of multiplication discussed in § 156 we arrive at two principal types of affinity, that is

*Two assemblies of magnitudes, each constructed from the corresponding magnitudes of a system of  $n$ th order by the same fundamental conjunctions, are directly affine if the magnitudes occurring in the product are all of lower order than  $n$ , and reciprocally affine if they are all of higher order than  $n$ ; and it is only in these two cases that the condition set out in § 155 is satisfied.*

Since the correspondence appears only in the multiplication, we can interpret it as constant; and indeed if we denote the factors of the first type as progressive, those of the second as regressive, then a progressive product can only appear in affinity as outer, a regressive product only as inner (in the sense of § 141).

---

<sup>1</sup>The 1844 Ausdehnungslehre, Part One.

### § 0.157. Connection Between Shadow and Affinity

We must now represent the shadow as a particular type of constant numerical relation and of affinity, and specify in which case the general relationship passes over into that particular type.

If we adopt the law of shadow according to which the corresponding magnitudes are such that their difference belongs to the guide system, then we must seek a method of multiplication by which, if all corresponding magnitudes have this relationship, a definite fundamental equation can be established. Now the law on multiplying through is the same for both products, and furthermore every product between  $A_1, \dots, A_n$  that includes equal factors and is therefore zero, has a partner in the corresponding product that is also zero; and implicit in this is that the same interchange law also prevails, since in both cases  $(A+B)(A+B)$  or  $AB+BA$  is zero, so the factors are only interchangeable with a change of sign. It therefore follows that if  $PQR$  is a multiple sum of products of corresponding magnitudes, and if in it each magnitude and its corresponding partner are exchanged, then in both cases the corresponding magnitudes on both sides, as likewise the nearest covering systems and the common systems of corresponding products, coincide, whence the resulting equations coincide, and since in the expression of affinity the two sides are set equal, the one equation is a consequence of the other.

Among the methods of multiplication considered so far there are only two that satisfy the condition stated in the previous section, that is that the product appears as zero if and only if a numerical relation exists between the factors. These are progressive and regressive multiplication, and indeed for corresponding magnitudes of lower and higher order. But the condition is also satisfied for products whose factors are partly of lower and partly of higher order if one always adjoins progressive factors as progressive and regressive ones as regressive, so that the product is at the same time both

### 0.157. Connection Between Shadow and Affinity

outer and regressive, since then it is zero if and only if a numerical relation exists between those of progressive type and likewise one between those of regressive type, and thus altogether a numerical relation exists between all the factors.

In this case the two products are complementary, in the sense that they yield together a product of highest order; and it is therefore required that the factors combined progressively, and likewise those combined regressively, are so many fewer than the order of the principal system. Thus, if the principal system is of  $n$ th order and  $a$  factors are combined progressively,  $b$  regressively, then one must have  $a + b = n$ .

It is evident that this case, of all cases of affinity, is the one of most direct application.

## 252 PART ONE

If corresponding to  $n$  mutually independent magnitudes of first order are taken  $n$  likewise mutually independent magnitudes of first order, or  $n$  magnitudes of  $(n - 1)$ -th order belonging to a system of  $n$ -th order whose regressive product has a nonzero value, then the magnitudes constructed from the corresponding magnitudes by the same fundamental conjunctions form two affine assemblies of magnitudes, and every fundamental equation that exists between the magnitudes of one assembly remains true if one replaces those magnitudes with the corresponding magnitudes of the other assembly.

For it is implicit in the correspondence of the magnitudes of first order or of  $(n - 1)$ -th order that the same fundamental equations must also exist between those constructed from them by the same fundamental conjunctions. Now the products of the corresponding magnitudes are all zero simultaneously, since a product can only be zero if a numerical relation exists between its factors, and this condition is the same for both; thus the products of the corresponding magnitudes are equal, and from this equality there again follows the same relation between the constructed magnitudes.

### § 0.158. Connection Between Shadow and Affinity

Two assemblies of magnitudes, one of which is the shadow of the other, appear as affine if the nearest covering system of their magnitudes is the inferior system, that is if the magnitudes of the assemblies belong to the same inferior system; and they are reciprocally affine if their nearest covering system is the superior system; and finally, such a product can simultaneously be interpreted as outer and regressive if both conditions are satisfied, and if the magnitudes of the assemblies partly belong to one and partly to the other of these systems.

*The condition for two affine assemblies of magnitudes is that every product that is not zero in one case is also not zero*

# 0.158. Connection Between Shadow and Affinity

*in the other, that is if  $a_1, \dots, a_m$  are the magnitudes of the first assembly  $A$  corresponding to the magnitudes  $b_1, \dots, b_m$  of the second assembly  $B$ , then the condition is that*

$$a_1 \cdot a_2 \cdots a_m = 0 \quad \text{if and only if} \quad b_1 \cdot b_2 \cdots b_m = 0.$$

*In particular, for the case of direct affinity, if  $a_1, \dots, a_m$  are the magnitudes of the first assembly  $A$  corresponding to the magnitudes  $b_1, \dots, b_m$  of the second, then if the  $a_1, \dots, a_m$  are simultaneously the shadows of the  $b_1, \dots, b_m$  along the guide system, one obtains the equations*

$$L.a_1 = L.b_1, \dots, L.a_m = L.b_m$$

*or*

$$L.(a_1 - b_1) = 0, \dots, L.(a_m - b_m) = 0,$$

*that is, the magnitudes  $(a_1 - b_1), \dots, (a_m - b_m)$  are subordinate to the guide system. But these magnitudes are both mutually independent and independent of the system of the assembly  $A$ , since otherwise a numerical relation would exist between the magnitudes  $a_1, \dots, a_m$ , which is impossible by the assumption of their independence.*

*The two assemblies are therefore shadows of each other if they are affine and their common system coincides with the guide system.*

Conversely, the condition for the shadow is a special case of direct affinity; for if the magnitudes of one assembly are shadows of those of the other, then they are affine, and if their common system coincides with the guide system, then every magnitude of the one assembly differs from the corresponding magnitude of the other only by a magnitude belonging to the common system, and is therefore its shadow with respect to that system.



**254** PART ONE

*Two affine assemblies one appears as the shadow of the other if and only if in the common system of the two assemblies every pair of corresponding magnitudes coincide, and then each of the two assemblies can be regarded as the shadow of the other.*

**§ 0.159. Affinity in Geometry**

In order to elucidate the results obtained with a geometric interpretation it suffices to consider affine assemblies of both types in the plane.

It is clear that to three noncollinear point magnitudes (which however must have equal weights) correspond three likewise noncollinear point magnitudes with equal weights, and that then to each point of the one plane corresponds a definite point of the other, such that the straight lines joining corresponding points pass through one and the same point, or are parallel if this point is at infinity; this is the ordinary case of affinity. Furthermore, to three straight lines in the one plane, bounding a triangle, correspond three straight lines in the other bounding a likewise formed triangle, such that the points of intersection of corresponding straight lines all lie on one and the same straight line; this is the case of reciprocal affinity. In addition, one can conceive of a mixed case in which two corresponding point pairs lie on two intersecting lines, but the third pair lies on the line joining the two points of intersection; and then the one triangle lies in two straight lines and the point of intersection, the other in two points and the line joining them.

**§ 0.160. Linear Relations, Collineation and Reciprocity According to the Principle of Equal Indicators**

Our method of treatment thus far is distinguished from the usual geometric point of view in that we interpret the spa-

tial elements partly as extensive magnitudes of first, second, third order, and so on, partly as elementary magnitudes. The result of this was that the equal construction on both sides led directly to the concept of affinity, while in the usual treatment one arrives at the related concept of collineation only by a circuitous route through metric considerations, which we were able to dispense with entirely.

It therefore remains for us to assimilate that particular difference into the laws presented in this chapter.

We connect this treatment to the general relation of affinity, and first call the corresponding systems of two affine assemblies linearly related, and indeed, if those assemblies are directly affine, we call the assembly of their systems collinear, but if reciprocally affine, we call it reciprocal.

## § 0.161.

In order to prove this for the plane, one takes any four points in the first plane, no three of which are collinear, and likewise four such points in the other, and places them in mutual correspondence. Then according to the principle of equal indicators there correspond to the points of the first plane definite points of the second. Now since the straight lines of the first plane are likewise transformed into those of the second, the six points of intersection of the four points of the first plane correspond to the six points of intersection of the four points of the second. The four points of the first plane are however so situated that their connecting lines mutually intersect; thus this must also hold for the connecting lines of the corresponding points; and this can only occur if the four corresponding points likewise lie in one plane, that is if no three of the latter are collinear. Thus the theorem is proved for collineation.

## 256 PART ONE

*The theorem can also be proved without any assumption concerning the collineation of the four points.* For if  $A, B, C, D$  and  $A', B', C', D'$  are the two systems of four points, then according to the principle of equal constructions, since the fourth point does not depend on the first three by any linear construction: Now in each plane one can adjoin to three of the points weights such that the fourth point appears as the sum of the three elementary magnitudes so formed; for provided one takes those three points as reference elements, the three reference terms of the fourth point are the elementary magnitudes sought.

If one now takes these three pairs of elementary magnitudes, since three pairs of corresponding magnitudes with equal weights correspond to them in both assemblies, so that the three pairs of corresponding magnitudes appear in both with the same indicators. The fourth pair likewise corresponds, since the principle of equal indicators assigns to each magnitude in the one assembly a definite corresponding one in the other.

### § 0.162.

The corresponding theorem can be proved for space by simply replacing those four point pairs with five, no four of which lie in a single plane. Then it can be shown that, according to the principle of equal indicators, the lines corresponding to the lines of the first space must also mutually intersect in the second, and thus that the point of intersection of two lines in the first corresponds to the point of intersection of the corresponding lines in the second; from which it follows that the planes corresponding to planes, and the points corresponding to the points of intersection of corresponding planes, likewise correspond reciprocally; which reciprocity is then proved in the same way as was collineation.

0.163. Identity of Reciprocity According to the Two Principles

**§ 0.163. Identity of Reciprocity According to the Two Principles**

According to the principle of equal constructions, we call two assemblies mutually reciprocal if to every point of the first assembly corresponds a plane of the second, to every line of the first a line of the second, to every plane of the first a point of the second. The theorem to be proved is then:

*If to each of  $n + 1$  points of the first assembly not lying in the same  $(n - 1)$ -th order system corresponds a plane of the second assembly, such that to the points lying in a plane of the first correspond the planes passing through the corresponding point of the second, then the assemblies are reciprocal.*

For then to every point of the first there corresponds a definite plane of the second, since by the principle of equal indicators this is the case for the points of the first, and the planes corresponding to the lines connecting them are the lines of intersection of the corresponding planes. Thus the correspondence extends over the whole assembly.

Now let  $A, B, C, D$  be four points of the first assembly and  $A', B', C', D'$  the four corresponding planes of the second. Let  $A, B, C$  lie in a straight line; then  $A', B', C'$  pass through a point. Further, let the line  $AB$  intersect the plane  $D'$  in a point  $P$ ; then the plane corresponding to  $P$  in the first assembly is the one passing through  $A, B$  and the point corresponding to the line in which  $D'$  intersects  $A'$ , that is through  $A, B$  and the point of intersection of  $B'$  and  $C'$ , or what amounts to the same, the plane through  $A, B, C$ . Thus

**258** PART ONE

*the planes of the first assembly that correspond to the collection of points of the second that lie on the same straight line, pass through a straight line; and conversely the points of the first assembly that correspond to the collection of planes of the second that go through the same straight line, and can be regarded as belonging to the second assembly, are connected by a straight line.*

It scarcely needs further illustration that the structures reciprocal in this way are also such according to the principle of equal indicators, since this follows in precisely the same way as was given above for collineation.

**§ 0.164. Identity of the Concept of Affinity for Point Assemblies According to the Two Principles**

If we place three noncollinear points in correspondence with three other noncollinear points, and thus by equal indicators form two assemblies of corresponding points, then the weights of the corresponding points are equal if one assigns to the three points of the first assembly weights such that their sum is the point of intersection of the lines joining the corresponding points; and one then assigns to the three points of the second assembly the same weights. Then the same will also hold for any fourth point of the first plane and the corresponding point of the second.

For if the three noncollinear points are  $A, B, C$  and the corresponding points  $A', B', C'$ , and if  $a, b, c$  are the weights of  $A, B, C$  such that  $a + b + c = 1$ , then the point  $P = aA + bB + cC$  corresponds to  $P' = aA' + bB' + cC'$ , since the indicators of  $P$  with respect to  $A, B, C$  are  $a, b, c$ , and by the principle of equal indicators the indicators of  $P'$  with respect to  $A', B', C'$  are the same.

Now if we assign the same weights to the corresponding points of both assemblies, then the indicators of any point in the first assembly with respect to any three noncollinear

### 0.165. The Metric Relations of Two Collinear Point Structures

points are equal to the indicators of the corresponding point in the second assembly with respect to the three corresponding points. And since the indicators of a point with respect to three others are determined by the ratios in which the lines joining the point to the vertices of the triangle are divided, it follows that the correspondence between the points of the two assemblies is such that the lines joining corresponding points are divided in the same ratio.

But this is precisely the condition for affinity; and since it follows from the principle of equal indicators, the two concepts are identical.

### § 0.165. The Metric Relations of Two Collinear Point Structures

Regarding the metric relations of two collinear point structures, these are most simply expressed as follows:

*Every fundamental equation independent of the measures of the magnitudes appearing therein remains true if one replaces those magnitudes with the corresponding magnitudes of a collinear assembly.*

## 260 PART ONE

Thus since one can so arrange those measures that the two assemblies are affine, and the validity of this theorem for affine assemblies of magnitudes has been proved, it is therefore also proved for collinear assemblies.

Moreover, if  $A$  and  $C$  are straight lines,  $b$  and  $d$  points, and  $A, C, b, d$  lie in the same plane, then

$$\frac{Ab}{bC} \cdot \frac{Cd}{dA} = \frac{A'b'}{b'C'} \cdot \frac{C'd'}{d'A'}.$$

Further, if  $A$  and  $C$  are planes,  $b$  and  $d$  points, then

$$\frac{Ab}{bC} \cdot \frac{Cd}{dA} = \frac{A'b'}{b'C'} \cdot \frac{C'd'}{d'A'}.$$

Finally, if  $A, B, C, D$  are lines in space, then

$$\frac{AB}{BC} \cdot \frac{CD}{DA} = \frac{A'B'}{B'C'} \cdot \frac{C'D'}{D'A'}.$$

The added specifications correspond to the condition appended to the more general theorem that the cross ratio is to represent a number.<sup>2</sup>

### § 0.166. Connection Between Collineation and Projection (Perspectivity)

As collineation is related to affinity, so projection is related to the shadow, since, as we showed above, for elementary magnitudes the system of the shadow represents the projection. Thus all the fundamental equations independent of the measures of their magnitudes also remain valid if one replaces the magnitudes with their projections; in particular those double quotients will also maintain the same values under projection.

Further, since the assemblies generatable by the shadow on one another represent a special type of affinity, so also will the assemblies generatable by projection on one another

---

<sup>2</sup>42

0.166. Connection Between Collineation and Projection  
(Perspectivity)

represent a special type of collineation, and indeed, if we call the assemblies generatable by projection on one another *perspective*, then we can state the theorem:

*Two collinear assemblies are perspective if and only if in the intersection of the two systems belonging to those assemblies, every pair of corresponding points coincide, and the sense of the projection is thereby defined.*

This theorem is just a translation of the theorem stated in § 158 for the shadow. Thus it follows from that theorem that two collinear lines that do not lie in the same plane are always perspective, since they never intersect.

Finally, in those cases in which the collinear assemblies are simultaneously affine, projection is identical to the shadow; in particular, if the shadow and the shadowed magnitude represent points or in general elementary magnitudes of first order with equal weights. This will be the case if the guide system is an extensive system (or expressed otherwise, is an elementary system in the infinite limit). This case occurred in part I (§ 82), where the projection and the shadow coincided.



## 261 LINEAR EXTENSION THEORY

### § 0.167. Harmonic Equations, Construction of the Harmonic Mean. Harmonic Sums, Harmonic Coefficients. Pole System

If we ask in general which equations are independent of the measures of the magnitudes of nonzero order that appear in them, and therefore remain valid under projection and generally under collineation, then they are those in which every magnitude of nonzero order appearing in only a single term occurs as often as a factor in the denominator as it occurs as a factor in the numerator, and those factors that all numerators or denominators have in common can appear arbitrarily often only if they occur equally often in each of them.

The simplest form of such an equation is therefore

$$\frac{\alpha QA}{PA} + \frac{\beta QB}{PB} + \dots = 0,$$

where  $\alpha, \beta, \dots$  represent numerical magnitudes, and in order that the equation can achieve a definite sense we must also assume that the denominators are all similar without being zero. On this assumption, and taking  $Q$  to be unity, the equation goes over into

$$\frac{\alpha A}{PA} + \frac{\beta B}{PB} + \dots = 0.$$

We call this a *harmonic equation*,  $\alpha, \beta, \dots$  the *harmonic coefficients* (harmonic weights), the systems of  $A, B, \dots$  the *harmonic systems*, and  $P$  the *pole system*. If by  $A, B, \dots$  we mean only systems, then we can also write the equation thus:

$$\alpha A + \beta B + \dots =_P 0,$$

and say that the expression  $\alpha A + \beta B + \dots$  is zero with respect to  $P$ .

0.167. Harmonic Equations, Construction of the Harmonic Mean. Harmonic Sums, Harmonic Coefficients. Pole System

The condition that the magnitudes  $PA$ ,  $PB$ , ... must all be similar without being zero can also be expressed by saying that for all these products the nearest covering system and the common system of the factors must be the same. If the nearest covering system is the same for all of them, this means that it must coincide with the nearest system that covers the collection of magnitudes  $P$ ,  $A$ ,  $B$ , ..., that is, with the principal system of the equation. If the common system in one of those products (and thus of all of them) is of zeroth order, then the product is outer; and then and only then are the values of the quotients definite magnitudes (§ 141). In this case we call the harmonic equation *harmonic of pure form*.

But although in the other case the quotients only represent partially defined values, the harmonic equation nevertheless maintains its specific significance, which we shall now seek.

Since  $PA$ ,  $PB$ , ... are mutually similar without being zero, it must be possible to assume such measures of  $A$ ,  $B$ , ... that

**262** PART ONE

$$PA = PB = \dots$$

then the equation will appear in the form

$$\frac{\alpha A + \beta B + \dots}{PA} = 0,$$

whence one obtains the absolute equation

$$\alpha A + \beta B + \dots = 0$$

upon multiplication by  $PA$ . Multiplying this equation by  $P$ , one obtains

$$(\alpha + \beta + \dots)AP = 0,$$

that is

$$\alpha + \beta + \dots = 0,$$

or:

*In a harmonic equation the sum of the harmonic coefficients of the two sides are equal.*

One therefore simultaneously obtains a mean by constructing the value  $\sigma S$ , which satisfies the equation

$$\alpha A + \beta B + \dots = \sigma S,$$

that is, by finding the harmonic coefficients and the harmonic system of this term. First of all,

$$\sigma = \alpha + \beta + \dots,$$

and second, if  $A, B, \dots$  are adjusted so that their products with  $P$  are equal, and  $S$  is also so adjusted, then according to the above

$$\alpha A + \beta B + \dots = \sigma S,$$

or

$$S = \frac{\alpha A + \beta B + \dots}{\sigma}$$

which defines  $S$  itself provided  $\sigma$  is not zero,<sup>3</sup> and thus also defines the system of  $S$ . The significance of the harmonic equation is thereby demonstrated.

---

<sup>3</sup>If  $\sigma$  is zero, and also  $\alpha A + \beta B + \dots = 0$ , then  $S$  is completely

0.167. Harmonic Equations, Construction of the Harmonic Mean. Harmonic Sums, Harmonic Coefficients. Pole System

---

undetermined, as is implicit in the idea of the harmonic equation. If  $\sigma$  is zero and  $\alpha A + \beta B + \dots$  represents a nonzero value, then there is no (finite) value for  $S$  that satisfies the equation; since  $(\alpha A + \beta B + \dots)P$  is then also equal to zero, it is clear that the system corresponding to that sum also does not satisfy the condition of yielding a product with  $P$  of nonzero value.

## 263 LINEAR EXTENSION THEORY

We call the system of  $S$  the *harmonic mean* between the systems  $A, B, \dots$  with respect to the coefficients  $\alpha, \beta, \dots$  belonging to them and the pole system, and this system, combined with the harmonic coefficient  $(\alpha + \beta + \dots)$ , we call the *harmonic sum* of  $\alpha A, \dots$  with respect to  $P$ .

### § 0.168. Reformulation of Pure Harmonic Equations

In the previous section we showed that a harmonic equation is also absolute if one appends to the systems measures such that their products with the pole system become equal. Now if the equation is harmonic in pure form, then one can also set the products of the pole system with the different systems equal to unity, and then the equation takes the form

$$\alpha A + \beta B + \dots = 0,$$

in which the coefficients are no longer connected by the equation  $\alpha + \beta + \dots = 0$ .

More important than this transformation is another, which rests on the fact that one can replace the pole system by another without disturbing the form of the equation. This is implicit in the theorem:

*If  $L$  is a system subordinate to all the systems  $A, B, \dots$  appearing in a pure harmonic equation*

$$\alpha A + \beta B + \dots = 0,$$

*and one combines this system with the pole system, so that the equation remains harmonic with respect to the combined pole system.*

For it is clear that, if  $L$  is independent of  $PA$ , and  $PA$  is, as we assume, an outer product, then in addition

$$LPA = LPB = \dots,$$

## 0.168. Reformulation of Pure Harmonic Equations

whence  $LP$  can also be taken as the pole system, and furthermore

$$\alpha AL + \beta BL + \dots = 0$$

and

$$PAL = PBL = \dots,$$

so that this equation combined with  $L$  is still harmonic with respect to the same pole system.

Without question, more important than these transformation theorems is the following:

*If the harmonic systems of an equation have a system  $E$  of nonzero order in common, and  $R$  is the supplementary system of  $E$ , that is  $RE$  is of the same order as the principal system, then the equation is also harmonic with respect to the system  $QR$ , where  $Q$  represents the pole system.*

**264** PART ONE

For if  $E$  is common to all the harmonic systems, then  $ER$  has the same order as the principal system. Now  $Q \cdot ER$  is the common value of  $PA$ ,  $PB$ , ... if one replaces  $A$ ,  $B$ , ... by  $AE$ ,  $BE$ , ...; but then the equation becomes

$$\alpha AE + \beta BE + \dots = 0,$$

which is harmonic with respect to  $QE$ ; and now if one sets  $P = QR$ , then  $PA = Q \cdot RA = Q \cdot ERA = QEA$ , when  $A$  is of higher order than  $E$ . Thus the equation is also harmonic with respect to  $QR$ .

*Thus it follows that in a pure harmonic equation one can combine the pole system with a system subordinate to all the harmonic systems and can define this combination as the pole system, or generally:*

*If the harmonic systems of an equation have a system of nonzero order in common, then one can change the pole system arbitrarily without disturbing the pure harmonic form of the equation if one merely adjoins the new pole system to that common system.*

This theorem can also be expressed as follows:

*If one combines a pure harmonic equation with a system subordinate to all its harmonic systems, then one obtains an equation that is likewise pure harmonic, and whose pole system is the combination of the original pole system with that system.*

The simplest way to express this theorem is the following:

*In a pure harmonic system one can, without changing the pole system, combine the harmonic terms with any system incident on the pole system.*

Implicit in this is the theorem:

*If  $n$  variable systems  $A$ ,  $B$ , ..., whose  $n$  fixed systems are all incident on the same fixed system  $R$ , form a harmonic equation with constant harmonic coefficients  $\alpha$ ,  $\beta$ , ... (whose sum is not zero) and with respect to that pole system  $P$ , is incident on a fixed system  $Q$ , and in fact this fixed system  $Q$*

0.168. Reformulation of Pure Harmonic Equations

*is the harmonic mean of the  $n$  fixed systems with respect to the same coefficients  $\alpha, \beta, \dots$  and the same pole system  $P$ .*



**268** PART ONE

This form of expression follows from the former (established in the previous section) with perfect precision if one makes the same assumptions. For the condition that the variable system is incident on  $R$  can be expressed by saying that it is combined with  $R$ , and since one can, by the theorem of the previous section, combine the pole system with any system incident on all harmonic systems, so one can here combine the variable system with  $R$ , and thus the resulting system is incident on  $R$ .

**§ 0.169. Transformation of the Pole System of a Harmonic Equation**

If the equation

$$\alpha A + \beta B + \dots = 0,$$

together with the constraint equations

$$PA = PB = \dots$$

are given, and if the products  $PA, \dots$  are regressive, then the harmonic equation that results thereby can be represented in pure form. Thus if  $E$  represents the system common to the factors of each of these products, and  $R$  is the supplementary system of  $E$ , so that  $ER$  is of the same order as the principal system, and one defines  $Q = PE$  (so that  $QR = PER = PR \cdot E$  has the same order as the principal system), then the equation is also harmonic with respect to  $QR$ . Now one sets  $P_1 = QR$ ; then  $P_1 A = QR \cdot A = Q \cdot RA$ , if  $A$  is of higher order than  $E$ ; thus  $P_1 A = Q \cdot RA = QE \cdot RA = Q \cdot ERA = P \cdot ERA$ . Since  $ERA$  is of the same order as the principal system, the product is outer, and the equation is therefore pure harmonic.

**§ 0.170.**

*Transformation of Harmonic Systems with Unaltered Pole Systems. General Theorem on Harmonic Means*

By means of these theorems we can now present the theorem of § 168 in a form somewhat simpler and more useful, namely:

*If the equation*

$$\alpha A + \beta B + \dots = \sigma S$$

*is harmonic with respect to the pole system  $P$ , and  $L$  is a system subordinate to all the systems  $A, B, \dots, S$ , then the equation*

$$\alpha AL + \beta BL + \dots = \sigma SL$$

*is likewise harmonic with respect to the same pole system  $P$ .*

## 265 LINEAR EXTENSION THEORY

For since  $AL, BL, \dots, SL$  are subordinate to  $A, B, \dots, S$ , the equation is also pure harmonic with respect to the combined pole system  $PL$ . Since however  $L$  is subordinate to all the harmonic systems, one can, by the theorem of § 168, drop the combination with  $L$ , and the equation remains harmonic with respect to  $P$ .

*This theorem can be stated more simply if one introduces the concept of the harmonic mean:*

*If  $S$  is the harmonic mean of  $A, B, \dots$  with respect to the coefficients  $\alpha, \beta, \dots$  and the pole system  $P$ , and  $L$  is a system subordinate to  $A, B, \dots, S$ , then  $SL$  is the harmonic mean of  $AL, BL, \dots$  with respect to the same coefficients and the same pole system.*

A simple application of this theorem is the following:

*If  $A, B, \dots$  are points and  $S$  is their harmonic mean, then the line joining  $S$  to any point  $L$  is the harmonic mean of the lines  $LA, LB, \dots$  with respect to the same coefficients and pole system.*

## § 0.171. Application to Crystal Structure

I will conclude this presentation with the most beautiful application that can be made of the science treated here, that is with its application to crystal structure.<sup>4</sup>

It is known that the surfaces of crystals can be grouped, according to the law of rational indices, into systems of planes, to each of which the following law applies:

*If the planes of a system are referred to any three planes of the system that do not belong to the same zone, then the ratios of the intercepts made by any plane of the system on the edges of the fundamental parallelopiped formed by the three reference planes are rational numbers.*

In order to express this law in the simplest way, we take the three reference planes as fundamental planes and express

---

<sup>4</sup>45

each plane of the system in terms of them. If the three reference planes are  $A$ ,  $B$ ,  $C$  and any fourth plane  $D$  of the system is the intersecting plane, then we can express any plane  $P$  of the system as

$$P = xA + yB + zC,$$

where  $x$ ,  $y$ ,  $z$  are the indicators of  $P$  with respect to  $A$ ,  $B$ ,  $C$ .

Now the theorem states that for any plane of the system,  $x$ ,  $y$ ,  $z$  are rational numbers. In particular, if we take the three planes adjacent to  $D$  as reference planes, then their indicators with respect to  $D$  are 1, 1, 1, and thus  $D$  can be expressed as their sum.

*Problem.* If the indicators with respect to four planes  $A$ ,  $B$ ,  $C$ ,  $D$ , the last of which is the intersecting plane, of four other planes  $Q_1$ ,  $Q_2$ ,  $Q_3$ ,  $Q$  and another plane  $P$  are given, to find the indicators  $x$ ,  $y$ ,  $z$  of  $P$ , if  $Q_1$ ,  $Q_2$ ,  $Q_3$ , and  $Q$  are regarded as the original planes, and  $Q$  as the intersecting plane.

**266** PART ONE

*Solution.* Let the indicators of  $P$  with respect to  $A, B, C, D$  be  $p_1, p_2, p_3, p_4$ , and those of  $Q_1, Q_2, Q_3, Q$  with respect to the same planes be  $a_1, a_2, a_3, a_4; b_1, b_2, b_3, b_4; c_1, c_2, c_3, c_4; q_1, q_2, q_3, q_4$ . Then we have

$$\begin{aligned} P &= p_1A + p_2B + p_3C + p_4D, \\ Q_1 &= a_1A + a_2B + a_3C + a_4D, \\ Q_2 &= b_1A + b_2B + b_3C + b_4D, \\ Q_3 &= c_1A + c_2B + c_3C + c_4D, \\ Q &= q_1A + q_2B + q_3C + q_4D. \end{aligned}$$

From the last equation we obtain

$$D = \frac{1}{q_4}(Q - q_1A - q_2B - q_3C),$$

and substituting this value into the other equations, we have

$$\begin{aligned} P &= \left(p_1 - \frac{p_4q_1}{q_4}\right)A + \left(p_2 - \frac{p_4q_2}{q_4}\right)B + \left(p_3 - \frac{p_4q_3}{q_4}\right)C + \frac{p_4}{q_4}Q, \\ Q_1 &= \left(a_1 - \frac{a_4q_1}{q_4}\right)A + \left(a_2 - \frac{a_4q_2}{q_4}\right)B + \left(a_3 - \frac{a_4q_3}{q_4}\right)C + \frac{a_4}{q_4}Q, \\ Q_2 &= \left(b_1 - \frac{b_4q_1}{q_4}\right)A + \left(b_2 - \frac{b_4q_2}{q_4}\right)B + \left(b_3 - \frac{b_4q_3}{q_4}\right)C + \frac{b_4}{q_4}Q, \\ Q_3 &= \left(c_1 - \frac{c_4q_1}{q_4}\right)A + \left(c_2 - \frac{c_4q_2}{q_4}\right)B + \left(c_3 - \frac{c_4q_3}{q_4}\right)C + \frac{c_4}{q_4}Q. \end{aligned}$$

Thus the indicators of  $P$  with respect to  $Q_1, Q_2, Q_3, Q$  are obtained if in the expressions for  $P, Q_1, Q_2, Q_3$  we replace  $A, B, C, Q$  by the indicators of  $Q_1, Q_2, Q_3, Q$  with respect to these planes, that is by 1, 0, 0, 0; 0, 1, 0, 0; 0, 0, 1, 0; 0, 0, 0, 1.

Now if we take the three edges of the crystal that meet at the vertex formed by the intersection of  $A, B, C$ , and which represent the plane areas delimited by  $D$ , one obtains as the indicators of these edges the expressions

$$\frac{P \cdot B \cdot C}{A \cdot B \cdot C}, \quad \frac{A \cdot P \cdot C}{A \cdot B \cdot C}, \quad \frac{A \cdot B \cdot P}{A \cdot B \cdot C},$$

which are related to the edges represented by the products  $AB$ ,  $BC$ ,  $CA$ .

Since one can take any four spatially limited crystal surfaces as fundamental surfaces, there consequently flows from this problem the theorem:

*If without changing their directions, one replaces the four fundamental planes of a crystal by any other four planes of the crystal, and for both sets of fundamental planes determines the indicators of the crystal edges and all crystal planes in the same way, then the ratios of these indicators are the same in both cases.*

## 267 LINEAR EXTENSION THEORY

*The ratios of the indicators of the crystal edges and crystal planes are independent of the choice of the fundamental planes.*

This theorem, which as it were stands at the summit of the theory of crystal structure, and from which all other laws of this theory can be derived, can be proved much more directly by means of the concept of the shadow. For if one takes the fundamental planes as reference elements and expresses all other planes in terms of them, then the indicators so obtained are exactly the shadow magnitudes of the planes when the fundamental planes are taken as guide magnitudes. But between the shadow magnitudes and the shadowed magnitudes the same equations obtain as between the former, and therefore between them precisely the same equations obtain as between the shadowed magnitudes, and the assembly of these shadows thus faithfully represents all relations prevailing in the crystal structure; the crystal faces are represented by linear magnitudes, the crystal edges by point magnitudes, or, insofar as both are given by just their directions, by systems of first and second order.

### § 0.172. Note on Open Products

In the presentation above I have mostly restricted myself to those products in which the factors can be arbitrarily combined into definite products without changing the value of the product, and which I therefore called *closed* products. But it is also necessary to consider products in which one factor can be inserted in place of an empty place in the product, and which I call *open* products.

If one has a product of the form  $[A.B]$ , where a space is left open for a factor to be inserted, and one inserts the factor  $P$ , then the product becomes  $[A.B]P = AP.B$ , where the dot indicates that  $P$  is to be combined with  $A$  before multiplying by  $B$ . We let this place be indicated by empty parentheses,

so that

$$[A().B]P = AP.B$$

and

$$[A_1().B_1 + A_2().B_2 + \dots]P = A_1P.B_1 + A_2P.B_2 + \dots,$$

and call such a product with empty parentheses *open*. If several factors appear, only one of which is to be inserted into the space, it can be distinguished with the same parentheses by which the space is indicated.

Consider the sum

$$S = e_1().e_1 + e_2().e_2 + \dots = \sum [e().e],$$

in which  $e, e_1, e_2, \dots$  represent displacements in space, and the summation symbol in the last expression refers to the different indices. It is clear that if the displacements  $e$  do not by chance belong to a single plane, the magnitude  $P$  which will be multiplied with that sum must be of second order, that is



**274** PART ONE

*standing in no relation to that sum, is introduced, namely the fundamental reference system.* Those six magnitudes therefore serve only the transformation to given reference systems, whereas the elementary concept of the sum is yet to be realized.

In order to approach the solution of this problem we can first try to reduce the sum to as small a number of terms as possible.

Since every open product  $a().b$  is a magnitude of the same order as the factors  $a$  and  $b$ , it follows that each term of the sum  $S$  is of second order, and therefore the sum itself is of second order. Now a sum of magnitudes of second order can always be reduced to three terms; for three magnitudes of second order, if they are mutually independent, form a system of third order, and every magnitude of second order can be expressed in terms of three such magnitudes. We therefore assume that the sum can be reduced to three terms, and set

$$S = a().a + b().b + c().c,$$

where  $a, b, c$  are displacements. If we multiply this equation by  $bc$ , we obtain

$$S(bc) = a(bc).a + b(bc).b + c(bc).c = abc \cdot a,$$

since  $b(bc) = c(bc) = 0$ . Similarly,

$$S(ca) = abc \cdot b, \quad S(ab) = abc \cdot c.$$

Thus if we set

$$S(bc) = Aa, \quad S(ca) = Bb, \quad S(ab) = Cc,$$

we have

$$A = abc, \quad B = abc, \quad C = abc, \tag{1}$$

$$S(bc) = Aa, \quad S(ca) = Bb, \quad S(ab) = Cc. \tag{2}$$

If we now multiply equation (4) by  $a$ , we obtain

$$S(bc)a = Aa^2 = abc \cdot a^2,$$

and similarly

$$S(ca)b = Bb^2 = abc \cdot b^2, \quad S(ab)c = Cc^2 = abc \cdot c^2.$$

From these equations it follows that

$$a^2 : b^2 : c^2 = A : B : C.$$

## 275 LINEAR EXTENSION THEORY

and  $S(ab)$  will equal  $abc \cdot Cc$ . Since the measure  $c$  is also arbitrary, and  $C$  is positive, one can take  $c$  so that  $C$  is unity; and then

$$S(ab) = abc \cdot c.$$

Furthermore, if one takes

$$P = ca, \quad z = x = 0, \quad y = 1,$$

then

$$S(ca) = abc(C'a + Bb),$$

which must necessarily lie in the plane  $ab$ , but since  $B$  cannot be zero, it is independent of  $a$ . Now since  $b$  can be chosen arbitrarily within the plane  $P$ , the result of  $S(P)$  must always lie in the plane  $ab$ ; but this can only be the case if the plane  $P$  itself is the plane  $ab$ . Thus we have shown that  $c$  is perpendicular to the plane  $ab$ ; and similarly  $a$  is perpendicular to  $bc$ , and  $b$  is perpendicular to  $ca$ .

One is led directly to equations (5) once one assumes that the sum can be reduced to three terms. For if  $a, b, c$  are the displacements belonging to these terms, then one immediately has equations (5) from (4) upon multiplication by  $ab, ca, bc$ . If one considers one of these equations, for example the first, then it is independent of the measure of the factor  $(ab)$  by which  $S$  is multiplied, since if one multiplies by  $\alpha(ab)$ , the equation is only multiplied by  $\alpha$ . Thus if one takes the measure of  $ab$  to be unity, then  $S(ab) = abc \cdot c$ , where  $abc$  is a numerical magnitude. If one now sets

$$\alpha = \frac{1}{\sqrt{A}}, \quad \beta = \frac{1}{\sqrt{B}}, \quad \gamma = \frac{1}{\sqrt{C}},$$

and introduces the new displacements  $a' = \alpha a, b' = \beta b, c' = \gamma c$ , then

$$a'^2 = b'^2 = c'^2 = abc \cdot \alpha\beta\gamma,$$

and the sum takes the form

$$S = a'().a' + b'().b' + c'().c',$$

0.172. Note on Open Products

in which  $a', b', c'$  are three conjugate radii of the same ellipsoid.

**276** PART ONE

Thus we have proved that a sum of open squares in space can always be reduced to the sum of the open squares of three conjugate radii of an ellipsoid. It remains to show that this reduction can be accomplished in infinitely many ways, and that the ellipsoid is the same for all these reductions.

For this purpose we observe that if  $a, b, c$  are three conjugate radii of an ellipsoid, and  $a', b', c'$  are three other conjugate radii of the same ellipsoid, then

$$a().a + b().b + c().c = a'().a' + b'().b' + c'().c',$$

since both sums represent the same magnitude  $S$ . Conversely, if the above equation holds, then  $a, b, c$  and  $a', b', c'$  are conjugate radii of the same ellipsoid; for from the equation it follows that  $S(bc) = abc \cdot a$ , and also  $S(b'c') = a'b'c' \cdot a'$ , whence  $abc \cdot a = a'b'c' \cdot a'$ , that is  $a$  and  $a'$  are parallel, and similarly for the other pairs.

Thus the ellipsoid is uniquely determined by the sum  $S$ , and is independent of the particular reduction chosen.

To determine the ellipsoid directly from the sum  $S$ , we proceed as follows. If  $e$  is any displacement, then  $S(e)$  is a displacement that depends linearly on  $e$ . We can therefore write

$$S(e) = \phi(e),$$

where  $\phi$  is a linear function. Now if  $e$  is a displacement in the direction of one of the principal axes of the ellipsoid, then  $S(e)$  is in the same direction, and we have

$$S(e) = \lambda e,$$

where  $\lambda$  is a numerical magnitude. Thus the principal axes are given by the equation

$$\phi(e) = \lambda e,$$

which is of third degree in  $\lambda$ , and has three roots corresponding to the three principal axes.

0.172. Note on Open Products

The ellipsoid itself is then determined by the equation

$$e \cdot S(e) = 1,$$

or, if we express  $e$  in terms of its components  $x, y, z$  with respect to the principal axes,

$$Ax^2 + By^2 + Cz^2 = 1,$$

where  $A, B, C$  are the roots of the above equation.

## 277 LINEAR EXTENSION THEORY

measures  $a, b, c$  are conjugate radii. If we call an expression like  $a().a$  an *open square* of  $a$ , we can present the results obtained in the following theorem:

*A sum of open squares in space equals the sum of the open squares of three arbitrary conjugate radii belonging to a constant ellipsoid.*

As this ellipsoid is therefore the complete, true expression of that sum, we call it the *ellipsoid of the sum*.

The directions of the principal axes are determined by the condition that  $S(e)$  has the same direction as  $e$ , that is  $S(e) = \lambda e$ , whence  $e \cdot S(e) = \lambda e^2$ . Since  $e \cdot S(e)$  is a numerical magnitude, and  $e^2$  is also numerical,  $\lambda$  is determined.

If we set

$$Q = xbc + yca + zab$$

and

$$e = \alpha a + \beta b + \gamma c,$$

whence

$$eQ = (\alpha x + \beta y + \gamma z)abc,$$

if  $abc$  (as the principal measure) is set to unity, and one therefore has

$$(\alpha x + \beta y + \gamma z)^2 = 1$$

or, retaining the above notation,

$$Ax^2 + By^2 + Cz^2 + 2A'yz + 2B'zx + 2C'xy = 1.$$

Thus far we have only considered the sums of open squares. If we also take into account their differences, then the ellipsoid transforms into a hyperboloid or an elliptic paraboloid, according as the signs are mixed or not.

**278** PART ONE

*The general equation of second degree represents an ellipsoid, a hyperboloid, or a paraboloid, according as the signs of the coefficients are all positive, mixed, or the discriminant vanishes.*

---





# Appendices (1877)

## Appendix I. (1877)<sup>5</sup>

### ON THE RELATION OF NON-EUCLIDEAN GEOMETRY TO EXTENSION THEORY

(Cf. §§ 15–23)

To the detriment of science, the entire presentation in §§ 15–23 still remains almost totally unnoticed. Neither Riemann in his *Habilitationsschrift*\* of 1854, first published in 1867, nor Helmholtz,\*\* in his paper “Über die Tatsachen, welche der Geometrie zur Grunde liegen” (1868), nor even in his excellent lecture “Über den Ursprung und die Bedeutung der geometrischen Axiome” (1876) mention it, even though the foundations of geometry come into view much more simply than in those later publications.

In extension theory the straight line is quite special and, in contrast to Euclid, is the foundation for geometric definitions. In § 16 the plane is defined as a collection of parallels that intersect a straight line, and space as a collection of parallels that intersect a plane; geometry can proceed no further, but the abstract science is not so limited. Since all points of a straight line may be numerically derived from two of its

---

<sup>5</sup>49

## Appendix I

points, the straight line appears as a simple elementary domain of second order, and correspondingly the plane as a simple elementary domain of third, and ultimately space as one of fourth order.\*\*\*

Thus for example the points of a plane are numerically derivable from three noncollinear points, e.g. by numbers  $x_1$ ,  $x_2$ ,  $x_3$ . Upon establishing a homogeneous equation between these three numbers, the collection of points satisfying this equation is reduced to a domain of second order. If this homogeneous equation is of first degree, then the domain so defined is elementary, that is a straight line; if however that equation is of higher degree it forms a curved line for which only some of the longimetric laws for the straight line are valid.

\* [Here is meant his Habilitationsrede “Über die Hypothesen, welche der Geometrie zu Grunde liegen,” Ges. Werke 1st ed. p. 54ff., 2nd ed. p. 272ff.]

\*\*

[The article appears in Gott. Nachr., 1878, pp. 193–221, cf. also Ges. wiss. Abh., vol. II., pp. 618–639. The lecture is found in his Vorträgen und Reden, vol. II., p. 1ff; Braunschweig: 1884.]

\*\*\*

To forestall confusion, I observe that the displacements in a plane form an elementary extensive domain of second order, those in space an elementary extensive domain of third order, and in general the displacements in a simple elementary domain of  $(n + 1)$ -th order an elementary extensive domain of  $n$ -th order.

## 280 PART ONE

Turning to space, each of its points is numerically derivable from four points forming a tetrahedron, by four numbers  $x_1, \dots, x_4$ . If, among these magnitudes, there exist two mutually independent homogeneous equations, neither of them of first degree, we then obtain doubly curved lines for which again only a part of the longimetric laws are true.

Now if we proceed another step beyond space, as a domain of fourth order, to a domain of fifth order (which does not exist geometrically), then one has five basis numbers  $x_1, \dots, x_5$ , and if a homogeneous equation of first degree holds between them, then one returns to the simple elementary domain of fourth order, that is to Euclidean space. On the other hand, upon imposing on them a homogeneous equation of higher degree one also produces elementary domains of fourth order, but ones to which the Euclidean axioms no longer apply, and thus as it were to non-Euclidean spaces\*; furthermore, one can proceed to an elementary domain of sixth order, and between the six determining numbers assume two higher homogeneous equations to obtain once more new elementary domains of fourth order. Each such domain of fourth order appears as a space in which the geometry is different; and if we ask under what conditions all lines which lie in a plane and can be drawn through a point have an absolutely determined length, or equivalently, under what conditions the geometry in such a space is congruence geometry, it is easily shown that this occurs if and only if the determining equation is of second degree, and therefore either the space is Euclidean or it is congruent in the sense of the usual spherical or pseudo-spherical geometry.

\* See my note "Sur les différents genres de multiplication" (Crelle's Journal, vol. 49).

Appendix II. (1877)<sup>6</sup>

ON THE REGRESSIVE PRODUCT

(Cf. chapter 3 of part 2)

Since in the entire chapter only the concept of the product with respect to a principal domain is developed in full clarity, and all other formal concepts are gradually dropped, it would have been appropriate to limit the full development to that concept. Indeed, with this restriction the presentation is much simpler, as I will show below.

The concept of the regressive product can be defined as follows:

*If  $A$  and  $B$  are extensive magnitudes of orders  $a$  and  $b$ , and  $n$  is the order of the principal domain, then the regressive product of  $A$  and  $B$  is an extensive magnitude of order  $(a + b - n)$  defined by the equation*

$$A \cdot B = (-1)^{(n-a)(n-b)} B \cdot A,$$

*where the product on the right is to be understood as outer.*

However, this definition is not sufficient; for it assumes that the outer product is already defined, and that the principal domain is given. But the regressive product can also be defined independently of the outer product, namely as follows:

*If  $A$  and  $B$  are extensive magnitudes, then the regressive product  $A.B$  is defined as that extensive magnitude  $C$  for which*

$$A.B.C = A(B.C) = (A.B)C,$$

*where the products on the right are to be understood as progressive.*

This definition has the advantage that it does not assume the principal domain, but only the progressive product. However, it is not completely satisfactory either, since it assumes

---

<sup>6</sup>50

that the progressive product is associative, which is not always the case.

The most satisfactory definition is the following:

*The regressive product of two extensive magnitudes  $A$  and  $B$  is that extensive magnitude  $C$  which satisfies the equation*

$$A.B = C,$$

*where  $C$  is so determined that for every extensive magnitude  $X$  of appropriate order*

$$C.X = A(B.X).$$

This definition is independent of both the principal domain and the associativity of the progressive product. It defines the regressive product by means of the progressive product alone, and is therefore the most general and most satisfactory definition.

From this definition there follow immediately the fundamental properties of the regressive product, namely:

1. *The regressive product is distributive:*

$$A.(B + C) = A.B + A.C.$$

2. *The regressive product is associative:*

$$(A.B).C = A.(B.C).$$

3. *The regressive product is commutative up to sign:*

$$A.B = (-1)^{(n-a)(n-b)} B.A.$$

4. *The regressive product of two magnitudes is zero if and only if the two magnitudes have a common system of higher order than the sum of their orders minus the order of the principal domain.*

**282** PART ONE

These properties follow immediately from the definition, and they suffice to develop the entire calculus of the regressive product.

In particular, if  $A$  and  $B$  are of orders  $a$  and  $b$ , and the principal domain is of order  $n$ , then the regressive product  $A.B$  is of order  $a + b - n$ . If  $a + b < n$ , then the regressive product is zero; if  $a + b = n$ , then the regressive product is a numerical magnitude; if  $a + b > n$ , then the regressive product is an extensive magnitude of positive order.

The regressive product is thus a generalization of the outer product, and includes it as a special case. For if  $a + b \leq n$ , then the regressive product coincides with the outer product, up to a numerical factor; and if  $a + b > n$ , then the regressive product is a new operation, not reducible to the outer product.

The regressive product is also closely related to the concept of the common system. For if  $A$  and  $B$  have a common system of order  $c$ , then the regressive product  $A.B$  can be expressed as the product of the common system with a magnitude of order  $a + b - n - c$ . In particular, if  $A$  and  $B$  have no common system (that is, if  $c = 0$ ), then the regressive product is of order  $a + b - n$ , and is equal to the outer product of  $A$  and  $B$  divided by the principal measure.

The regressive product can also be interpreted geometrically. If  $A$  and  $B$  are interpreted as domains of orders  $a$  and  $b$ , then  $A.B$  is the domain of order  $a + b - n$  that is common to both domains, provided that they intersect in a domain of order  $a + b - n$ . If they do not intersect in such a domain, then the regressive product is zero.

Thus, if  $A$  is a plane and  $B$  is a straight line, and the principal domain is space (of order 4), then  $A.B$  is a point, namely the point of intersection of the plane and the line. If  $A$  and  $B$  are both planes, then  $A.B$  is a straight line, namely the line of intersection of the two planes. If  $A$  is a point and  $B$  is a plane, then  $A.B$  is a volume, namely the volume of the parallelepiped formed by the point and the plane.

The regressive product thus provides a unified framework for all geometric operations, and it is this unity that constitutes its greatest advantage.



Appendix III. (1877)<sup>7</sup>

BRIEF SURVEY OF THE  
ESSENTIALS  
OF EXTENSION THEORY

*(Written in response to a request from Grunert;  
published in Grunerts  
Archiv 6, pp. 297–350, 1845)*

**I. Purpose of Extension Theory as Such**

1. Extension theory forms the abstract basis of the theory of space (geometry), that is, it is the pure mathematical science stripped of all geometric content, whose special application is geometry. Just as arithmetic is the science of pure numerical magnitudes, so extension theory is the science of pure extensive magnitudes.
2. An extensive magnitude is one that is generated from elements by evolution, that is by a continuous process of generation. The simplest extensive magnitude is the displacement, which is generated from a point by translation.
3. The order of an extensive magnitude is the number of independent elements from which it is generated. Thus a displacement is of first order, a plane area of second order, a volume of third order.
4. The sum of two extensive magnitudes of the same order is defined by adding their corresponding elements. The product of two extensive magnitudes is defined by the combinatorial product of their generating elements.
5. The combinatorial product of two extensive magnitudes  $A$  and  $B$  is denoted by  $A.B$  or  $AB$ , and is defined as

---

<sup>7</sup><sub>54</sub>

the extensive magnitude generated by the combined elements of  $A$  and  $B$ . The product is zero if  $A$  and  $B$  have a common element.

6. If one subjects all the elements of a domain of  $n$ -th order to one and the same method of evolution that leads to new elements (not contained in that domain), then the collection of elements generatable by this method of evolution and its opposite is called a domain of  $(n + 1)$ -th order; the domain of third order corresponds to the plane, that of the fourth to all of space.

7. If a domain of  $n$ -th order is generated from  $n$  independent elements, then the product of these  $n$  elements is called a volume segment of that domain. The numerical magnitude that expresses how many times a volume segment contains another is called the measure of the former with respect to the latter.
8. If the straight line, the plane, and the volume are interpreted respectively as domains of first, second, third, and fourth order, then in this is implicit the suggestion of the general theorem that a domain of  $a$ -th order and one of  $b$ -th order have a domain of  $(a + b - c)$ -th order in common when they intersect in a domain of  $c$ -th order but in no domain of lower order; but the theory of space cannot proceed further. Since however the abstract science is not so limited, there follows from this the possibility of a domain of  $n$ -th order for every  $n$ , and thus the concept of an extensive magnitude of  $n$ -th order for every  $n$ .
9. The concept of the domain, which originally is a spatial concept, thus receives a purely abstract meaning in extension theory; it is simply a system of extensive magnitudes of successive orders, generated from one another by evolution.
10. The domain of  $n$ -th order generated from a single element by successive evolutions is called a simple elementary domain of  $n$ -th order. The extensive magnitudes belonging to such a domain are called elementary magnitudes.
11. The theory of curved lines and surfaces can also be developed in the context of extension theory. A curved line is defined as the collection of points satisfying a homogeneous equation of higher degree; a curved sur-

face as the collection of points satisfying a homogeneous equation of higher degree in the coordinates.

## II. Basic Forms of Extension Theory

12. The basic forms of extension theory are the extensive magnitudes of different orders, and the operations that can be performed on them. The most important operations are addition, subtraction, multiplication, and division.
13. Addition and subtraction are defined for extensive magnitudes of the same order; multiplication and division are defined for extensive magnitudes of any order, provided that the conditions for their existence are satisfied.
14. The product of two extensive magnitudes is called outer if the order of the product is greater than the orders of the factors; it is called regressive if the order of the product is less than the sum of the orders of the factors.
15. The regressive product is defined with respect to a principal domain of  $n$ -th order; it is an extensive magnitude of order  $(a + b - n)$ , where  $a$  and  $b$  are the orders of the factors.
16. The combinatorial product of two extensive magnitudes is commutative up to sign; that is,  $A.B = (-1)^{ab}B.A$ , where  $a$  and  $b$  are the orders of  $A$  and  $B$ .

**III. Basic Rules of Calculation for the New Analysis**

The conjunctions appearing in this part of extension theory are addition, subtraction, combinatorial multiplication, and combinatorial division.

**Problem 1.** To solve two first-degree equations with two unknowns by the rules of combinatorial multiplication.

Let the two equations be

$$a_1x + b_1y = c_1,$$

$$a_2x + b_2y = c_2.$$

If one multiplies the first equation by  $a_2$ , the second by  $a_1$ , and subtracts, one obtains

$$(a_2b_1 - a_1b_2)y = a_2c_1 - a_1c_2,$$

and similarly for  $x$ . Thus

$$x = \frac{c_1b_2 - c_2b_1}{a_1b_2 - a_2b_1}, \quad y = \frac{a_1c_2 - a_2c_1}{a_1b_2 - a_2b_1}.$$

This solution can be expressed in a more symmetric form by introducing the combinatorial product. If we denote by  $[ab]$  the combinatorial product of  $a$  and  $b$ , then

$$x = \frac{[cb]}{[ab]}, \quad y = \frac{[ac]}{[ab]}.$$

## 287 LINEAR EXTENSION THEORY

**Problem 2.** To solve three first-degree equations with three unknowns by the rules of combinatory multiplication.

Let the three equations be

$$\begin{cases} a_1x + \beta_1y + \gamma_1z = \delta_1, \\ a_2x + \beta_2y + \gamma_2z = \delta_2, \\ a_3x + \beta_3y + \gamma_3z = \delta_3. \end{cases} \quad (1)$$

If one multiplies the three equations respectively by three combinatory factors of first degree  $e_1, e_2, e_3$  whose product is not zero, adds them, and sets

$$\begin{cases} a_1e_1 + a_2e_2 + a_3e_3 = a, \\ \beta_1e_1 + \beta_2e_2 + \beta_3e_3 = b, \\ \gamma_1e_1 + \gamma_2e_2 + \gamma_3e_3 = c, \\ \delta_1e_1 + \delta_2e_2 + \delta_3e_3 = d, \end{cases}$$

then one has

$$ax + by + cz = d.$$

If one multiplies this equation by  $bc$ , then  $a(bc)x = d(bc)$ , whence

$$x = \frac{d(bc)}{a(bc)},$$

and similarly

$$y = \frac{d(ca)}{b(ca)}, \quad z = \frac{d(ab)}{c(ab)}.$$

If one substitutes these values into the original equations, one finds that they satisfy them; for if one substitutes the value of  $x$  into the first equation, one obtains

$$a_1 \cdot \frac{d(bc)}{a(bc)} + b_1y + c_1z = d_1,$$

and this equation is satisfied if one takes for  $y$  and  $z$  the values found above.

#### IV. Intuitive Interpretation of the Different Magnitudes and Methods of Conjunction in Geometry

13. The spatial magnitudes of first order are simple or multiple points, and straight lines of definite length and direction (§§ 13–19).

If  $A$  and  $B$  are points, then I denote the straight line between  $A$  and  $B$ , insofar as both its length and direction, but nothing else, are retained, by  $B - A$ ; that is, I regard the straight line as the difference of two points. The sum of two such lines  $B - A$  and  $D - C$  is then defined as  $E - A$ , where  $E - B = D - C$ ; that is, one adds two lines by placing the final point of the first on the initial point of the second.

14. A multiple point is a sum of the form  $\alpha A + \beta B + \dots$ , where  $\alpha + \beta + \dots = 1$ . The weight of this multiple point is  $\alpha + \beta + \dots$ , and its position is determined by the ratios of the coefficients.
15. The spatial magnitudes of second order are plane segments, that is parallelograms of definite area and direction, and line segments, that is parts of straight lines of definite length. A plane segment  $AB$  is represented by the combinatory product of two displacements  $A$  and  $B$ ; a line segment by the product of a point and a displacement.
16. Two plane segments  $AB$  and  $A'B'$  are equal if the parallelograms represented by these products lie in parallel planes and have equal and equally signed areas; and

$$a \cdot b \cdot c = a' \cdot b' \cdot c'$$

if the parallelepipeds represented by these products have equal and equally signed volume.

17. The side (right or left) toward which the construction of a spatial magnitude takes place defines its positive or negative value, that is the sign of the magnitude.



## Appendix III

### 289 LINEAR EXTENSION THEORY

18. Two displacements  $a$  and  $b$  are added by placing them contiguously; that is, if  $a = B - A$  and  $b = C - B$ , then  $a + b = C - A$ .
19. Two equal and oppositely directed displacements have sum zero; two displacements that are not equal and oppositely directed have as sum a displacement that is the diagonal of the parallelogram formed by them.
20. The sum of two displacements is commutative; that is  $a + b = b + a$ .
21. The product of two displacements  $a$  and  $b$  is the parallelogram formed by them; it is positive if  $b$  lies to the left of  $a$ , negative if to the right. The product is anticommutative; that is  $a \cdot b = -b \cdot a$ .
22. Any two magnitudes of the same order, but only such, can be added; the concept of addition for such magnitudes can always be defined if one retains the given sign of these magnitudes and applies the calculational laws valid for algebraic sums.

### V. Problems of the New Analysis

**Problem 5.** To add two displacements  $B - A$  and  $D - C$ .

If one makes  $E - B = D - C$ , then

$$(B - A) + (D - C) = (B - A) + (E - B) = E - A.$$

Thus one adds two displacements by placing the final point of the first on the initial point of the second, and drawing the line from the initial point of the first to the final point of the second.

**Problem 6.** To add two multiple points  $\alpha A + \beta B$  and  $\gamma C + \delta D$ .

If  $\alpha + \beta = \gamma + \delta = 1$ , then

$$(\alpha A + \beta B) + (\gamma C + \delta D) = \alpha A + \beta B + \gamma C + \delta D,$$

and the sum is a multiple point whose weight is  $\alpha + \beta + \gamma + \delta$ .

## Appendix III

### 290 PART ONE

**Problem 9.** To add a point  $B$  to a displacement  $D - C$ .  
If one takes  $A$  arbitrary, and sets

$$\alpha A + (C - B) = \alpha A + \alpha(D - A) = \alpha D,$$

it follows that

$$A + (D - C) = D + (A - C),$$

that is, one adds a displacement to a point by displacing the point.

For example, if  $A.B$  and  $A'.B'$  are equally directed parts of the same straight line, their sum is a similarly directed part of the same line as large as those two parts together.

**Problem 10.** To add two plane segments  $A.B$  and  $C.D$ .

If  $A, B, C, D$  lie in the same plane, and  $E$  is any point in that plane, then

$$A.B + C.D = (A - E).B + A.(B - E) + (C - E).D + C.(D - E),$$

and this sum is a plane segment lying in the same plane. If  $A, B, C, D$  do not lie in the same plane, then the sum is a volume segment.

**Problem 11.** To add a point to a plane segment.

If  $A.B.C$  is a plane segment and  $D$  a point, then

$$A.B.C + D = A.B.C + D.E.F,$$

where  $E, F$  are arbitrary points, and the sum is a volume segment if  $D$  does not lie in the plane  $ABC$ .

**291** LINEAR EXTENSION THEORY

**Problem 12.** To add a line segment and a plane segment.

If  $A.B$  is a line segment and  $C.D.E$  a plane segment, then

$$A.B + C.D.E = (F - G).(B - A).(C - A),$$

where  $F - G$  is a displacement in the direction of the line  $AB$ .

**Problem 13.** To add a plane segment and a volume segment.

If  $A.B.C$  is a plane segment and  $D.E.F.G$  a volume segment, then

$$A.B.C + D.E.F.G = A.B.C + (H - I).(F - E).(G - E).(D - E),$$

where  $H - I$  is a displacement perpendicular to the plane  $ABC$ .

**Problem 14.** To add two equally long and oppositely directed line segments  $A.B$  and  $C.D$ .

If both lie on the same straight line, the sum is zero. If this is not the case, then since  $D - C$  equals  $-(B - A)$  one has

$$\begin{aligned} A.B + C.D &= A.(B - A) + C.(D - C) \\ &= A.(B - A) - C.(B - A) \\ &= (A - C).(B - A), \end{aligned}$$

which is a plane segment.

**Problem 15.** To add a point to a line segment.

If  $A$  is a point and  $B.C$  a line segment, then

$$A + B.C = A.B.C + D,$$

where  $D$  is an arbitrary point, and the sum is a volume segment.

**Problem 16.** To add a line segment and a volume segment.

If  $A.B$  is a line segment and  $C.D.E.F$  a volume segment, then

$$A.B + C.D.E.F = A.B.C.D + E.F.G.H,$$

where  $E, F, G, H$  are arbitrary points.

## Appendix III

### 292 PART ONE

Thus, since here as well the previous equation can again be recovered from this one, it follows that *to find the sum of arbitrarily many multiple points  $\alpha A, \beta B, \dots$ , whose coefficient-sum is not zero, one lays lines from the arbitrary point  $R$  to  $A, B, \dots$ , and multiplies these by  $\alpha, \beta$ , respectively.* The lines so obtained are then laid contiguously together, without changing their lengths or directions, and the line joining the initial point of the first to the final point of the last is the required sum.

**Problem 17.** To add two plane segments  $A.B.C$  and  $D.E.F$ .

If  $A + C$  is a straight line of definite direction and length (problem 11); if this equals  $H - G$ , then

$$A.B.C + D.E.F = (H - G).(B - A).(C - A),$$

whence this sum is a volume.

**Problem 18.** To add a part  $A.B.C$  of a definite plane and a volume  $(D - A).(B - A).(C - A)$ .

$$\begin{aligned} A.B.C + (D - A).(B - A).(C - A) \\ &= A.(B - A).(C - A) + (D - A).(B - A).(C - A) \\ &= D.(B - A).(C - A), \end{aligned}$$

from which the concept of this addition is evident.

**Problem 19.** To add two volume segments.

The sum of two volume segments is a volume segment whose measure is the sum of the measures of the two volume segments.

**Problem 20.** To find the outer product of two displacements  $a$  and  $b$ .

The outer product  $a.b$  is the parallelogram formed by  $a$  and  $b$ ; it is positive if  $b$  lies to the left of  $a$ , negative if to the right.

**Problem 21.** To find the outer product of three displacements  $a, b, c$ .

The outer product  $a.b.c$  is the parallelepiped formed by  $a, b, c$ ; it is positive if  $a, b, c$  form a right-handed system, negative if left-handed.

## Appendix III

### 293 LINEAR EXTENSION THEORY

**Problem 22.** To find the outer product of a point  $A$  and a displacement  $a$ .

The product  $A.a$  is a line segment; it is the line through  $A$  in the direction of  $a$ , of length equal to the measure of  $a$ .

$$A.B+C.D = (A-C).(B-A) = S.(1+a)(B-A) = S.(B-A+D-C),$$

in which is implicit a simple construction for that sum.

**Problem 23.** To find the regressive product of four plane segments  $ABC$ ,  $ABD$ ,  $ACD$ ,  $BCD$ .

*Solution.*

$$ABC \cdot ABD \cdot ACD \cdot BCD = (ABCD)^3.$$

Note. The product of two plane segments therefore yields a line segment, that of three a point, but that line segment and that point will have a volume segment or a product of volume segments as coefficients, and if one takes a volume segment as the unit then these coefficients turn into actual numerical magnitudes.

These are perhaps the most remarkable results of the whole theory, and they show the extraordinary simplicity and elegance that the regressive product introduces into geometry.

**294** PART ONE

**Problem 24.** To find the regressive product of a plane segment and a line segment.

If  $ABC$  is a plane segment and  $DE$  a line segment, then

$$ABC \cdot DE = (ABC.D)E - (ABC.E)D,$$

which is a point.

**Problem 25.** To find the regressive product of two line segments.

If  $AB$  and  $CD$  are two line segments, then

$$AB \cdot CD = (A.CD)B - (B.CD)A,$$

which is a plane segment.

**Problem 26.** To find the regressive product of a line segment and a point.

If  $AB$  is a line segment and  $C$  a point, then

$$AB \cdot C = (A.C)B - (B.C)A,$$

which is a volume segment.

**Problem 27.** To find the regressive product of a plane segment and a point.

If  $ABC$  is a plane segment and  $D$  a point, then

$$ABC \cdot D = (ABCD),$$

which is a numerical magnitude.

**Problem 28.** To find the regressive product of two plane segments.

If  $ABC$  and  $ABD$  are two plane segments, then

$$ABC \cdot ABD = (ABCD) \cdot AB,$$

which is a line segment.



## Appendix III

### 295 LINEAR EXTENSION THEORY

**Problem 29.** To find the regressive product of three plane segments.

If  $ABC$ ,  $ABD$ ,  $ACD$  are three plane segments, then

$$ABC \cdot ABD \cdot ACD = (ABCD) \cdot A,$$

which is a point.

**Problem 30.** To find the regressive product of four plane segments.

*Solution.*

$$ABC \cdot ABD \cdot ACD \cdot BCD = (ABCD)^3.$$

*Note.* The product of two plane segments therefore yields a line segment, that of three a point, but that line segment and that point will have a volume segment or a product of volume segments as coefficients, and if one takes a volume segment as the unit then these coefficients turn into actual numerical magnitudes.

These are perhaps the most remarkable results of the whole theory, and they show the extraordinary simplicity and elegance that the regressive product introduces into geometry.

# Index of Defined Terms (1877)

The defined terms that I later abandoned are enclosed in parentheses. Those that I replaced with other expressions in the *Ausdehnungslehre* of 1862 are linked to the later terms by an equal sign.

|                                 |       |                                   |
|---------------------------------|-------|-----------------------------------|
| Abschattung=Zurückleitung       | § 82  | Shadow                            |
| (Abweichung)                    | § 95  | Deviation                         |
| Addition einfacher Ausdehnungen |       |                                   |
| erster Stufe                    | § 15  | Addition of elementary extensions |
| of first order                  |       |                                   |
| höherer Ausdehnungen            | § 48  | of higher extensions              |
| Affinität                       | § 154 | Affinity                          |
| direkt, reciproke               | § 157 | direct, reciprocal                |
| Allgemeine Formenlehre          | § 1   | General theory of forms           |
| Analytische Form                | § 7   | Analytic form                     |
| Verknüpfung                     | § 5   | conjunction                       |
| Änderung, stetige               | § 13  | Evolution, continuous             |
| Ausdehnung, erster Stufe        | § 14  | Extension of first order          |
| höherer Stufe                   | § 31  | of higher order                   |

## Index of Defined Terms

|                                    |       |                                        |
|------------------------------------|-------|----------------------------------------|
| von ergänzender Stufe              | § 80  | of supplementary order                 |
| der Elementargrösse                | § 111 | of elementary magnitudes               |
| Ausdehnungsgrösse=extensive        |       |                                        |
| Grösse                             | § 13  | Extensive magnitude                    |
| (Aussere Division)                 | § 60  | Outer division                         |
| Multiplikation der Strecken        | § 28  | multiplication of displacements        |
| der höheren Ausdehnungen           | § 54  | of higher extensions                   |
| (Ausweichung der Elementargrösse)  | § 109 | Divergence of the elementary magnitude |
| (Beziehungsgrösse)                 | § 137 | Relative magnitude                     |
| (Beziehungssystem)                 | § 127 | Relative system                        |
| (Beziehungszahl)                   | § 127 | Relative number                        |
| (Doppelsystem)                     | § 143 | Double system                          |
| (Eckgebilde)                       | § 110 | Vertex structure                       |
| Eindeutige analytische Verknüpfung | § 6   | Unique analytic conjunction            |
| Eingeordnet=incident               | § 136 | Incident                               |
| Eingewandtes=regressives Produkt   | § 125 | Regressive product                     |
| Element                            | § 13  | Element                                |
| Elementargrösse                    | § 94  | Elementary magnitude                   |
| Elementarsystem                    | § 107 | Elementary system                      |
| Erganzzahl einer Grösse            | § 133 | Supplementary number of a magnitude    |
| Ersetzender Verein von Gleichungen | § 86  | Replacement assembly of equations      |
| (Formales Produkt)                 | § 125 | Formal product                         |
| Formelle Summe                     | § 51  | Formal sum                             |
| Gemeinschaftliches System          | § 126 | Common system                          |
| Gemischtes Produkt                 | § 137 | Mixed product                          |
| (Gesamtabweichung)                 | § 95  | Collective deviation                   |

**297 LINEAR EXTENSION THEORY**

|                               |            |                                   |
|-------------------------------|------------|-----------------------------------|
| Gewicht des Elementarvereins  | § 95       | Weight of the elementary assembly |
| (Grad der Abhängigkeit)       | § 125      | Degree of dependence              |
| Grundmasse=Einheiten          | § 87       | Fundamental measures              |
| (Grundsystem)                 | §§ 82, 149 | Base system                       |
| (Harmonische Gleichungen)     | § 167      | Harmonic equations                |
| (Harmonische Koeffizienten)   | § 167      | Harmonic coefficients             |
| Hauptsystem=Hauptgebiet       | §§ 80, 127 | Principal system                  |
| Indifferente Form             | § 7        | Indifferent Form                  |
| Kollineation                  | § 160      | Collineation                      |
| (Kombination der Grössen)     | § 151      | Combination of magnitudes         |
| (Leitsystem)                  | §§ 82, 149 | Guide system                      |
| Multiplikation, siehe Produkt |            | Multiplication, see Product       |
| Nachstumfassendes System=     |            |                                   |
| verbindendes Gebiet           | § 126      | Nearest covering system           |
| Offenes Produkt               | § 172      | Open product                      |
| (Polssystem)                  | § 167      | Pole system                       |
| Produkt                       |            | Product                           |
| ausseres                      | §§ 28, 106 | outer                             |
|                               | § 125      | regressive                        |
| eingewandtes=regressives      |            |                                   |
| (formales)                    | § 125      | formal                            |
| gemischtes                    | § 137      | mixed                             |

## Index of Defined Terms

|                              |            |                            |
|------------------------------|------------|----------------------------|
| reales                       | § 125      | real                       |
| reines                       | § 137      | pure                       |
| Projektion                   | §§ 82, 151 | Projection                 |
| Proportion in der Geometrie  | § 75       | Proportion in geometry     |
| Reciprocität                 | § 160      | Reciprocity                |
| Reines Produkt               | § 137      | Pure product               |
| (Richtaxen)                  | § 87       | Reference axes             |
| (Richtmasse)                 | § 87       | Reference measure          |
| (Richtstücke)                | § 88       | Reference terms            |
| (Richtsystem)                | § 87       | Reference system           |
| Spath=Spat                   | §§ 30, 37  | Parallelepiped             |
| Spatheck=Spateck             | §§ 28, 37  | Parallelogram              |
| Starre Elementargrösse       | § 109      | Rigid elementary magnitude |
| Strecke                      | § 14       | Displacement               |
| Synthetische Verknüpfung     | § 5        | Synthetic conjunction      |
| System=Gebiet $n$ -ter Stufe | §§ 16, 20  | System of $n$ -th order    |
| Unterordnung, Form derselben | § 129      | Subordinate form           |
| Verwandschaft                | § 154      | Relationship               |
| Zahlengrössen                | § 68       | Numerical magnitudes       |
| Zeiger=Ableitungszahlen      | § 88       | Indicators                 |

# Editorial Notes

1. Part Two of this Raumlehre has the title: “Ebene räumliche Grössenlehre,” and it says in a footnote on p. 194f:

“Properly speaking, the rectangle is itself the true geometric product, and its construction as given in § 53 actual geometric multiplication. If one takes the concept of a product in its purest and most general sense, it denotes the result of a construction produced from that already generated (constructed), in the same way as that generated was itself generated from the original generator; and multiplication is therefore a construction to a higher power. In geometry the point is the original generator; from it the line results by that construction. If we make the bounded line (as that generated by the first construction) the basis of a new construction, in that we treat it by the same method as previously for the point, the rectangle results. The rectangle is therefore produced from the line just as the line is produced from the point.

“Now the same also holds in number theory. Here the original generator is the unit, which must be regarded as simply given with respect to number. From this, number results by enumeration (the arithmetic construction). If one makes this now generated number the basis of a new enumeration by putting it in the place of the unit, one obtains the arithmetic relationship to multiplication, which is therefore nothing but a number of higher order, a number whose unit is also

a number. Thus one can perhaps say that the rectangle is a bounded line in which a bounded line takes the place of the generating point. One would then also be able to grasp the two preceding theorems thus: Rectangles are the geometric product of base and height, and [otherwise] the case is as with arithmetic.”

In the *Lehrbuch der ebenen und sphärischen Trigonometrie* by J. G. Grassmann, Professor of the Gymnasium at Stettin (Berlin: G. Reimer, 1839), one reads the following in a footnote on p. 10:

“If one takes the concept of the product in its purest and most general form, then in mathematics it denotes the result of a synthesis in which that generated by an earlier synthesis replaces the original element and is treated as that was. The product must likewise result directly from that generated by the first synthesis as this was generated from the original. — In arithmetic the element is the unit; the synthesis, enumeration; and the result, number. Now if this number, as the result of the first synthesis, replaces the unit and is treated in the same way, that is enumerated, there results the arithmetic product, which can be regarded as a number of higher order than a number, whose unit is also a number. — In geometry the element is the point; the synthesis, translation of the point in any given direction; the result, the path of the point, the line. Now if the line, as the result of the first synthesis, replaces the point and is treated in the same way, i.e. it is translated in another direction, then the plane results as the path of the line; it is therefore the true geometric product of two linear factors, and appears first of all as a rectangle, insofar as none of the first direction is included in the second. If the plane replaces the point, the geometric solid results as the product of three factors, upon which geometry has its terminus since space has only three dimensions; but arithmetic is not limited in the

number of factors appearing. See my *Raumlehre*, Part Two, Berlin 1824.”



**299** LINEAR EXTENSION THEORY

2. The bracketed material was inserted by V. Schlegel. Before his death, Grassmann had only completed an outline of the foreword to the second edition. At the request of his son, Schlegel incorporated the new material as smoothly as possible.
3. The sense of the words is: The new subjects, which were to have been treated in the unpublished second volume of the *Ausdehnungslehre* of 1844, are only partially enumerated in this foreword. These subjects are treated in the 1862 *Ausdehnungslehre*; newly added are, etc.
4. It seems appropriate to make a few general remarks here.

It is well known that the semiphilosophical style of the *Ausdehnungslehre* of 1844 is the reason this noteworthy book did not receive proper recognition for so long. Grassmann's express intention was wherever possible to present all its concepts directly in the most general form possible; he always sought to abstract the common connections that would illuminate the simple basic ideas, and to keep the incidental and contingent in the background. Finally, his whole structure was to be independent of geometry and analysis, geometry being permitted to appear only as an illustrative example for the clarification of his general concepts (see the note to § 13). However much his work owes to the very essential merit of this effort, which figures in all its individual parts, and which led Grassmann to very important insights into the fundamental principles of mathematics—we mention only his theory of the synthetic and analytic conjunctions and his general conception of multiplication—, it cannot be denied that in striving for generality Grassmann occasionally cuts the ground from under his feet.

This is particularly true of the basic concepts from which extension theory is constructed in the present work. The basic concepts are too imprecise, too contentless, to support the conclusions Grassmann draws from them. To convince oneself of this, one need only consider the definition of elements in § 13 and compare with it all that is subsequently done with these elements.

The element is supposed to be that elementary particular existent without any real content; out of one such element,  $a$ , another element  $b$  is

**300** PART ONE

produced by an evolution. Now what is meant when he says that by an equal evolution a new element,  $c$ , is produced out of  $b$ ? How is it at all possible to compare an evolution of  $a$  with an evolution of  $b$  and thus to speak of equal evolutions? Finally, what is meant by the concept of continuous evolution? All these concepts, and naturally also all of their consequences are left hanging in the air.

Equally unsatisfactory is what Grassmann says about the parallel axiom and in general about the foundations of Euclidean geometry. He is of the opinion that the parallel axiom is not a basic truth but a theorem that can be proved. But his proof is based on concepts that are just as vague as those mentioned above, and therefore carries no conviction.

In spite of these deficiencies, which are partly due to the striving for generality and partly to the philosophical tendency of the author, the *Ausdehnungslehre* of 1844 remains a work of the highest significance. It is the first attempt to construct a purely abstract science of extensive magnitudes, independent of all geometric intuition, and to develop from it the whole of geometry as a special application. This attempt, even if not completely successful, opened up new paths for mathematics and exercised a lasting influence on its development.

# **A New Branch of Mathematics**

The Ausdehnungslehre of 1844, and Other Works

By Hermann Grassmann

Pages 301–350

Chunk 7 of 12



## Appendix II (1877)<sup>1</sup>

### On the Regressive Product

(Cf. chapter 3 of part 2)

Since in the entire chapter only the concept of the product with respect to a principal domain is developed in full clarity, and all other formal concepts are gradually dropped, it would have been appropriate to limit the full development to that concept. Indeed, with this restriction the presentation is much simpler, as I will attempt to demonstrate here. Thus I return to the theorem at the end of § 126, according to which, if  $a$  and  $b$  are the orders of two magnitudes,  $u$  is that of the nearest covering domain, and  $c$  that of the common domain, then  $c = a + b - u$ .

The concept of the regressive product with respect to a principal domain is then determined, since it is zero if and only if the nearest covering domain of the factors is of lesser order than that of the principal domain. The formal concept of the product under consideration is then based on this and the general concept of multiplication, that is its known relation to addition. In order to arrive at its real concept, it is then a question of presenting in the clearest possible form that which remains constant with respect to a change in the form of the product that does not change its value.

Let  $A \cdot B$  be a regressive product whose value is not zero, and let  $a$  and  $b$  be the orders of  $A$  and  $B$ , and  $u$  be that of the principal domain; then first I show that the common domain

---

<sup>1</sup>This is marked with 5° in the original.

of the two factors remains unchanged under any change in the form of the product  $A \cdot B$  that leaves its value unchanged.

This is immediately clear if the two factors maintain the same value or are changed in reciprocal numerical proportion, since then their domains remain the same. Now if one of the two factors, for example the second,  $B$ , changes its value to  $B + B_1$ , without thereby changing the value of the product, then it follows that  $A \cdot B_1 = 0$ , that is, the nearest covering domain of  $A$  and  $B_1$  must be of lesser order than the  $u$ -th, or, what is the same thing, their common domain must be of higher than  $c$ -th order. Now since, according to § 51,<sup>2</sup> the sum of two magnitudes of higher order always reduces to the case where the two magnitudes lie in a domain of next higher order, and thus for our case in a domain of  $(b+1)$ -th order, we need only consider this case here. But then  $B$  and  $B_1$  have a domain of  $(b-1)$ -th order in common, and one can therefore decompose  $B_1$  into  $b$  elementary factors,  $b-1$  of which lie in  $B$  and one outside of  $B$ . Now  $B_1$  can have only  $c+1$  factors in common with  $A$ , which is only possible if  $c$  of those  $b-1$  factors simultaneously lie in  $A$ , that is, if they belong to the common domain  $C$ , and the one factor of  $B_1$  lying outside of  $B$  likewise lies in  $A$ . The whole domain  $C$  therefore lies in  $B_1$ , that is,  $C$  is the common domain<sup>3</sup> of  $A$  and  $B_1$  and thus also of  $A$  and  $B + B_1$ ; consequently the common domain of the two factors remains unchanged if the product maintains the same value. The same argument is also valid for changes

---

<sup>2</sup>According to § 51; Here as well is meant the unproved law according to which the sum of two extensions of the same order is again an elementary extension of the same order only if both are included in the same domain of next higher order. (Study)

<sup>3</sup> $C$  is not the common domain of  $A$  and  $B_1$ , but rather is subordinate to this common domain. By comparison  $C$  is of course the common domain of  $A$  and  $B + B_1$ . In particular one can set  $C = a_1 a_2 \dots a_c$ ,  $B = a_1 a_2 \dots a_c \dots a_{b-1} u$ ,  $B_1 = a_1 a_2 \dots a_{b-1} v$ ,  $A = C v D$ , where  $a_1, a_2, \dots, a_{b-1}, u$  and  $v$  are independent magnitudes of first order and  $D$  has no factor of first order in common with  $B$ . Then  $B + B_1 = a_1 a_2 \dots a_{b-1} (u + v)$ , and since  $a_1, a_2, \dots, a_{b-1}, (u + v)$  are also mutually independent, then, other than the  $c$  factors of  $C$ ,  $A$  has no factor of first order in common with  $B + B_1$ .

in the other factor.

Now in order to find the metric value of the product we set  $B = CD$ , so that  $A \cdot B = A \cdot CD$ ; then  $AD$  represents the covering, and here the principal, domain. Now we can set a part of the principal domain (for example the outer product of the original units taken in their sequence) equal to unity. Then  $A \cdot D$  represents a number. If it is equal to 1, then instead of  $C$  and  $D$  we can introduce the magnitudes  $C_1 = AC$  and  $D_1 = \frac{1}{k}$ , without changing the value of the product; but then  $A \cdot D_1 = 1$ , and  $A \cdot CD = A \cdot C_1 D_1$ .

I now assert that  $C_1$  remains unchanged under the change of form discussed above. According to the above it is possible to set  $B_1 = C_1 \cdot E_1$ , where  $E_1$  has an elementary factor in common with  $A$ , that is<sup>4</sup>  $A \cdot E_1 = 0$ ; then in the structure  $A \cdot C_1(D_1 + E_1)$  the product  $A \cdot (D_1 + E_1) = A \cdot D_1 = 1$ , and the  $C_1$  so defined remains unchanged. We can therefore regard  $C_1$  as the true value of the product, and also, even if  $A \cdot D$  is not equal to one, always set  $A \cdot CD = AD \cdot C$ , where  $AD$  is a part of the principal domain and is therefore a number; we can therefore interpret this as the real definition of the regressive product.

---

<sup>4</sup>Here  $A \cdot E_1$  is regarded as the outer product.





## Appendix III (1877)<sup>5</sup>

### Brief Survey of the Essentials of Extension Theory

(Written in response to a request from Grunert; published  
in Grunerts Archiv 6, pp. 297–350, 1845)

#### I. Purpose of Extension Theory as Such

1. Extension theory forms the abstract basis of the theory of space (geometry), that is, it is the pure mathematical science, stripped of all geometric content, whose special application is that theory.

Because it is based on something given in nature, namely space, the theory of space is not a branch of pure mathematics but rather an application of it to nature. It is not however a mere application of algebra, even when the algebraic magnitude is regarded as continuously variable as in function theory; for algebra lacks the characteristic spacial concept of different dimensions. Thus a branch of mathematics is needed in which, along with the concept of a continuously varying magnitude, there is included the concept of differences corresponding to the dimensions of space; and this branch is extension theory.

---

<sup>5</sup>This article appeared in 1845 in Grunerts Archiv, vol. VI, pp. 337–350. It should be noted that the expressions “combinatorial product” and “combinatorial factors of first degree” do not appear in the 1844 *Ausdehnungslehre*.

2. The theorems of extension theory are however not mere translations of geometric theorems into abstract language; rather, they have a very general significance. For, while geometry is bound to the three dimensions of space, the abstract science is free from this restriction.

In space theory lines can be generated by the motion of points, surfaces by that of lines, and volumes by that of surfaces; but the theory of space can proceed no further. On the other hand, if one supposes that abstract concepts, independent of space, can be substituted for points and motion (see below, nos. 4–6), then this restriction disappears.

3. It therefore follows that the theorems of the theory of space have a tendency toward generality which is not there satisfied as a consequence of the restriction to three dimensions, but is only fulfilled in extension theory.

Two examples may clarify this:

- 1) Two straight lines in the same plane intersect in a single point, likewise a straight line and a plane, and two planes in a straight line, provided that the lines, or the plane and the line, or the two planes, do not coincide, and that intersections at infinity are included. If the point, the straight line, the plane, and the volume are interpreted respectively as domains of first, second, third, and fourth order, then in this is implicit the suggestion of the general theorem that a domain of  $a$ -th order and one of  $b$ -th order have a domain of  $(a + b - c)$ -th order in common when they intersect in a domain of  $c$ -th order but in no domain of lower order; but the theory of space can only illustrate this theorem for  $c$  less than or equal to four.
- 2) The plane area of a triangle is half of that of a parallelogram whose sides are equally long and parallel to two sides of the triangle, the volume of a tetrahedron one sixth of that of the parallelepiped whose edges are

equally long and parallel to three edges of the tetrahedron that meet at the same point. In this is suggested the theorem: The space lying between  $n$  points that are united in a domain of  $n$ -th order (and in no domain of lower order) is  $\frac{1}{1.2.3 \dots n}$  of the space in a structure (a figure, a body) whose boundary lines (sides, edges) are equally long and parallel to the straight lines drawn from one of the points to all the rest. But even here this theorem is not expressed in its complete generality.

On the contrary, in these two and all other cases the completely general theorems appear fully only in extension theory. Thus everywhere the theory of space pushes toward generality, but cannot attain it as it encounters spatially-imposed barriers through which only the abstract science of extension theory can break.

4. In extension theory the structure corresponding to the line is the collection of elements into which a continuously varying element transforms. The line may be regarded as the collection of points assumed by a point which continuously changes position. If we substitute for the point generally any given thing capable of a continuous evolution of its state, and abstract all other content of the thing and all particulars of its state, and call the thing abstracted of all other properties the element, then we arrive at the concept formulated.

5. Now if the state of the element always evolves in the same way, so that if from an element  $a$  of the structure another element  $b$  of the same structure results from such an evolution, then by an equal evolution a new element  $c$  results from  $b$ , it follows that the structure corresponding to the straight line is the domain of second order.

The straight line is constructed from the point if the point always changes its position in the same direction; thus if we substitute for direction the type and method of evolution, then the concept formulated results.<sup>6</sup>

---

<sup>6</sup>If the straight line and its corresponding structure are infinite in

6. If one subjects all the elements of a domain of  $n$ -th order to one and the same method of evolution that leads to new elements (not contained in that domain), then the collection of elements generatable by this method of evolution and its opposite is called a domain of  $(n + 1)$ -th order; the domain of third order corresponds to the plane, that of the fourth to all of space.

If the points of a straight line all move in one and the same direction that leads to new points (not contained in that line), then the collection of points generatable by this motion and its opposite is the plane; and if one proceeds likewise with the points of the plane one obtains the whole of space. If in this case one substitutes for the spatial concepts those abstract concepts previously given, and retains the advance from one order to the next higher order generally, then the above concept follows.

## II. Purpose of the Method of Calculation Used in Extension Theory, Illustrated by Geometry

7. In extension theory there appears a characteristic method of calculation which, transposed into geometry, is inexhaustibly fruitful, and here (in the theory of space) consists in subjecting spatial structures (points, lines, and so forth) directly to calculation.

For example, the magnitude and direction of the straight line drawn through two points is interpreted as the conjunction of those points and indeed as a characteristic type of multiplication (see below, no. 15); likewise, the area and position of the surface area of the triangle lying between three points is interpreted as the product of three points, so that this product is zero if the surface area of the triangle is, that is if the three points lie on a straight line; further, the point

---

both directions, then the point (the element) must also proceed in the opposite direction (method of evolution), which we have omitted here for simplicity.

of intersection of two straight lines is the product of these lines in a sense to be developed more fully below (no. 22 and problem 19).

8. The purpose of this method of calculation in geometry is to unify the synthetic and analytic methods, that is, to transplant the advantages of each into the soil of the other, so that every construction is accompanied by an elementary analytic operation and conversely.

The following example may serve to illustrate. As is known, a vertex  $\gamma$  of a variable triangle whose two other vertices  $\alpha, \beta$  are moved in fixed straight lines  $A$  and  $B$  and whose sides go through three fixed points  $a, b, c$ , describes a conic section. If  $a, b, c$  are the fixed points through which the sides opposite the vertices  $\alpha, \beta, \gamma$  go, respectively, then one sees (no. 7) that  $\gamma a B$  represents the vertex  $\beta$ ,  $\gamma a B c A$  the vertex  $\alpha$ , and since the points  $a, b, \gamma$  lie on a single straight line, their product is zero (no. 7), whence one has the equation

$$\gamma a B c A b \gamma = 0$$

as the equation of a conic section described by  $\gamma$ . One sees that this equation is of second degree in  $\gamma$ , and herein one already anticipates an important law applying to all algebraic curves.

### III. Basic Rules of Calculation for the New Analysis

The conjunctions appearing in this part of extension theory are addition, subtraction, combinatorial multiplication, and combinatorial division.

9. The usual calculational procedures are valid for all types of addition and subtraction.

10. For all methods of multiplication and division the following law holds: Instead of multiplying or dividing an aggregate

of terms by an unsigned expression in any given way, one can, without changing the final result, multiply or divide the individual terms by this expression in the same way,<sup>7</sup> and conjoin the individual products or quotients into an aggregate by preceding each one by the sign of that term by whose multiplication or division it was produced. Furthermore, if a numerical factor is adjoined to any given factor of the product it can be adjoined to any other factor or to the product; and finally  $\frac{A}{A}$  is always unity, provided  $A$  is not zero.

11. I call a product  $a \cdot b \cdot c \dots$  *combinatory* if besides law no. 10, it also satisfies the law that if a consecutive pair of the individual factors  $a, b, c, \dots$  are interchanged, the product assumes the opposite value; and in particular I then call  $a, b, c, \dots$  and their sums or differences factors of first degree.

Thus for example

$$a \cdot b \cdot c \cdot d = -a \cdot c \cdot b \cdot d.$$

12. If in a combinatory product, two factors of first degree are equal, the product is zero.

For example  $a \cdot b \cdot b \cdot c = 0$  (as also follows immediately if one sets  $b$  and  $c$  equal in the example for no. 11).

The following problems may serve to illustrate this method of multiplication:

**Problem 1.** To simplify the combinatory product

$$(\alpha_1 e_1 + \alpha_2 e_2 + \alpha_3 e_3) \cdot (\beta_1 e_1 + \beta_2 e_2 + \beta_3 e_3) \cdot (\gamma_1 e_1 + \gamma_2 e_2 + \gamma_3 e_3)$$

where  $\alpha_1, \alpha_2, \alpha_3; \beta_1, \beta_2, \beta_3; \gamma_1, \gamma_2, \gamma_3$  denote numerical magnitudes, and  $e_1, e_2, e_3$  combinatory factors of first degree.

By applying the calculational rules (9–12) one ultimately obtains the expression

$$(\alpha_1 \beta_2 \gamma_3 - \alpha_1 \beta_3 \gamma_2 - \alpha_3 \beta_1 \gamma_2 + \alpha_3 \beta_2 \gamma_1 + \alpha_2 \beta_3 \gamma_1 - \alpha_2 \beta_1 \gamma_3) e_1 \cdot e_2 \cdot e_3.$$

<sup>7</sup>This expression refers not only to the method of conjunction in general, but also to the position of factors in the product.

**Problem 2.** To solve three first-degree equations with three unknowns by the rules of combinatory multiplication.

Let the three equations be

$$\begin{cases} \alpha_1 x + \beta_1 y + \gamma_1 z = \delta_1 \\ \alpha_2 x + \beta_2 y + \gamma_2 z = \delta_2 \\ \alpha_3 x + \beta_3 y + \gamma_3 z = \delta_3. \end{cases} \quad (1)$$

If one multiplies the three equations respectively by three combinatory factors of first degree  $e_1, e_2, e_3$  whose product is not zero, adds them, and sets

$$\begin{cases} \alpha_1 e_1 + \alpha_2 e_2 + \alpha_3 e_3 = a \\ \beta_1 e_1 + \beta_2 e_2 + \beta_3 e_3 = b \\ \gamma_1 e_1 + \gamma_2 e_2 + \gamma_3 e_3 = c \\ \delta_1 e_1 + \delta_2 e_2 + \delta_3 e_3 = d \end{cases} \quad (2)$$

then one obtains the equation

$$xa + yb + zc = d. \quad (3)$$

If one multiplies this equation combinatorially by  $b \cdot c$ , then since  $b \cdot b \cdot c$  and  $c \cdot b \cdot c$  are zero according to no. 12, one obtains the equation

$$x \cdot a \cdot b \cdot c = d \cdot b \cdot c, \quad \text{whence} \quad x = \frac{d \cdot b \cdot c}{a \cdot b \cdot c},$$

and one finds  $y$  and  $z$  in a similar way, obtaining

$$x = \frac{d \cdot b \cdot c}{a \cdot b \cdot c}, \quad y = \frac{a \cdot d \cdot c}{a \cdot b \cdot c}, \quad z = \frac{a \cdot b \cdot d}{a \cdot b \cdot c}. \quad (4)$$

These expressions (in which the laws of combinatory multiplication permit no cancellation of individual combinatory factors) are most convenient for analysis. If one wants the unknowns expressed in the usual form, one need only substitute from equation (2), simplify according to problem 1, and



cancel  $e_1 e_2 e_3$  in the numerator and denominator according to no. 10; one finds, for example,

$$x = \frac{\delta_1 \beta_2 \gamma_3 - \delta_1 \beta_3 \gamma_2 + \delta_3 \beta_1 \gamma_2 - \delta_3 \beta_2 \gamma_1 + \delta_2 \beta_3 \gamma_1 - \delta_2 \beta_1 \gamma_3}{\alpha_1 \beta_2 \gamma_3 - \alpha_1 \beta_3 \gamma_2 + \alpha_3 \beta_1 \gamma_2 - \alpha_3 \beta_2 \gamma_1 + \alpha_2 \beta_3 \gamma_1 - \alpha_2 \beta_1 \gamma_3} \quad (5)$$

One sees that this procedure is not only applicable in general for  $n$  equations of first degree in  $n$  unknowns, but also how one can immediately write down the final result with some facility as soon as the  $n$  equations are given.

#### IV. Intuitive Interpretation of the Different Magnitudes and Methods of Conjunction in Geometry

13. The spatial magnitudes of first order are simple or multiple points, and straight lines of definite length and direction (§§ 13–19).

If  $A$  and  $B$  are points, then I denote the straight line between  $A$  and  $B$ , insofar as both its length and direction, but nothing else, are retained, by  $B - A$ ; thus I say that  $B - A$  equals  $B_1 - A_1$  if and only if the straight lines from  $A$  to  $B$  and from  $A_1$  to  $B_1$  have equal lengths and directions.

14. The spatial magnitudes of  $n$ -th order are formed by combinatory multiplication of  $n$  magnitudes of first order, which occur as factors of first degree.

In this case, that is when the factors of first degree are at the same time magnitudes of first order, I call the multiplication *outer*.

15. If  $A, B, C, D$  are points, then (§§ 106–115)

- 1)  $A \cdot B$  denotes the line having  $A$  and  $B$  as boundary points, understood as a definite part of the infinite straight line defined by  $A$  and  $B$ .
- 2)  $A \cdot B \cdot C$  denotes the triangle with vertices  $A, B, C$ , understood as a definite part of the infinite plane defined by  $A, B, C$ .

- 3)  $A \cdot B \cdot C \cdot D$  denotes the tetrahedron with vertices  $A, B, C, D$ , understood as a definite part of infinite space.

That is, we put  $A \cdot B = A_1 \cdot B_1$  if both products represent equal and equally signed<sup>8</sup> parts of the same straight line; further,

$$A \cdot B \cdot C = A_1 \cdot B_1 \cdot C_1,$$

if both triangles are equal and equally-signed parts of the same plane; and finally

$$A \cdot B \cdot C \cdot D = A_1 \cdot B_1 \cdot C_1 \cdot D_1,$$

if both tetrahedrons have equal and equally-signed volume.

16. If  $a, b, c$  are lines of definite length and direction, then (§§ 28–36)

- 1)  $a \cdot b$  denotes the parallelogram whose sides are equal and parallel to  $a$  and  $b$ , and is regarded as a plane area of definite magnitude and direction.<sup>9</sup>
- 2)  $a \cdot b \cdot c$  denotes the parallelepiped whose edges are equal and parallel to  $a, b, c$ , and is regarded as a volume of fixed magnitude.

That is, we put

$$a \cdot b = a_1 \cdot b_1,$$

if the parallelograms represented by these products lie in parallel planes and have equal and equally signed areas; and

$$a \cdot b \cdot c = a_1 \cdot b_1 \cdot c_1,$$

if the parallelepipeds represented by these products have equal and equally signed volume.

<sup>8</sup>I call two magnitudes equally signed if either both have a positive or both have a negative value.

<sup>9</sup>I say two parallel planes have equal planar directions.

17. The side (right or left) toward which the construction of a spatial magnitude takes place defines its positive or negative value, that is

- 1) We define two parts of the same line,  $A \cdot B$  and  $A_1 \cdot B_1$ , as equally signed if  $B$  lies on the same side of  $A$  as does  $B_1$  of  $A_1$ .
- 2) We define two parts of the same plane,  $A \cdot B \cdot C$  and  $A_1 \cdot B_1 \cdot C_1$ , as equally signed if  $C$  lies on the same side of  $A \cdot B$  as does  $C_1$  of  $A_1 \cdot B_1$ , or more plainly, if  $C$  is on the same side according to one standing on  $A$  and looking toward  $B$  as  $C_1$  is, according to one standing on  $A_1$  and looking toward  $B_1$ .
- 3) We define two finite volumes,  $A \cdot B \cdot C \cdot D$  and  $A_1 \cdot B_1 \cdot C_1 \cdot D_1$ , as equally signed if  $D$  lies on the same side of  $A \cdot B \cdot C$  as does  $D_1$  of  $A_1 \cdot B_1 \cdot C_1$ ; or more plainly, if the point  $D$  lies on the same side of a human figure with head toward  $A$ , feet toward  $B$ , and eyes directed toward  $C$ , as does  $D_1$  of a human figure with head toward  $A_1$ , feet toward  $B_1$ , and eyes directed toward  $C_1$ .
- 4) We define two parallel areas,  $a \cdot b$  and  $a_1 \cdot b_1$ , as equally signed if the direction  $b$  lies on the same side of the direction  $a$  as does  $b_1$  of  $a_1$ .
- 5) We define two volumes,  $a \cdot b \cdot c$  and  $a_1 \cdot b_1 \cdot c_1$ , as equally signed if the direction of  $c$  lies on the same side of  $a \cdot b$  as does  $c_1$  of  $a_1 \cdot b_1$ , that is, if the direction  $c$  lies on the same side of a human figure for which the direction  $a$  goes from feet to head and with eyes looking in the direction  $b$  as does  $c_1$  of a human figure, etc.

18. There are seven types of spatial magnitudes, divided into four orders:

|           |                                                            |
|-----------|------------------------------------------------------------|
|           | {1. Simple or multiple points                              |
| 1st Order | {2. Straight lines of definite length and direction        |
|           | {3. Definite parts of definite infinite straight lines     |
| 2nd Order | {4. Plane areas of definite magnitude and planar direction |
|           | {5. Definite parts of definite infinite planes             |
| 3rd Order | {6. Definite volumes                                       |
| 4th Order | 7. Definite volumes.                                       |

Volumes appear twice here, once as magnitudes of third order, once as magnitudes of fourth order, according as they are regarded as products of three straight lines of definite direction and length or as products of four points.

19. The sum of equally signed parts of the same whole is a part of the same whole with the same sign, and as large as the two parts taken together.

For example, if  $A \cdot B$  and  $A_1 \cdot B_1$  are equally directed parts of the same straight line, their sum is a similarly directed part of the same line as large as those two parts together.

20. Any two magnitudes of the same order, but only such, can be added; the concept of addition for such magnitudes can always be defined if one retains the given sign of these magnitudes and applies the calculational rules of III.

**Problem 3.** To add two points  $A$  and  $B$ .

If one puts  $A + B = 2S$ , one obtains  $B - A = 2(S - A)$ , that is,  $S$  is the mean between  $A$  and  $B$ . Thus the sum of two points is twice the mean between them.

**Problem 4.** To add two multiple points  $\alpha A$  and  $\beta B$ , if the coefficients  $\alpha$  and  $\beta$  are positive; that is, to find the point  $S$  satisfying the equation (§§ 94–98)

$$\alpha A + \beta B = (\alpha + \beta)S.$$

If this equation is satisfied, then necessarily

$$\beta(B - A) = (\alpha + \beta)(S - A),$$

and conversely one obtains the former from the latter. But from the latter follows the construction: If, on the line  $AB$  from  $A$  one takes the part  $\frac{\beta}{\alpha+\beta}$  (or from  $B$  the part  $\frac{\alpha}{\alpha+\beta}$ ), then the final point of this part is the point  $S$ .—Thus “the sum of two multiple points with positive coefficients is a point, multiplied by the sum of the coefficients, lying on the straight line between the two points such that its distance from these points is inversely proportional to the coefficients belonging to these points.”<sup>10</sup>

**Problem 5.** To add a point  $A$  and a straight line  $C - B$  of definite length and direction.

One constructs a straight line from  $A$ , equal to  $C - B$  in length and direction; if this is  $D - A$ , then  $D$  is the sum sought. For since  $C - B = D - A$ , then

$$A + (C - B) = A + (D - A) = D.$$

Thus “the sum of a point  $A$  and a straight line of definite length and direction is the final point of this line, if  $A$  is its initial point.”

**Problem 6.** To add a multiple point  $\alpha A$  and a straight line  $C - B$  of definite length and direction.

One constructs a straight line from  $A$  having the same direction as  $C - B$ , but only  $\frac{1}{\alpha}$  as long; if this is  $D - A$ , then  $\alpha D$  is the sum sought. For since

$$C - B = \alpha(D - A)$$

it follows that

$$\alpha A + (C - B) = \alpha A + \alpha(D - A) = \alpha D.$$

---

<sup>10</sup>One sees that this point is the center of gravity if the coefficients represent weights.

**Problem 7.** To add two straight lines  $B - A$  and  $D - C$  of definite length and direction.

If one makes  $E - B = D - C$ , then

$$(B - A) + (D - C) = (B - A) + (E - B) = E - A.$$

Thus “one adds two lines of definite length and direction by placing the final point of one on the initial point of the other, without changing the length and direction; then the straight line from the initial point of the first to the final point of the second is the sum sought.”

**Problem 8.** To add  $n$  straight lines of definite length and direction.

Repeated application of the solution to problem 7 leads immediately to the solution of this problem; thus “one adds  $n$  straight lines of definite length and direction by placing the individual lines in a contiguous series without changing their directions or lengths, that is laying them so that where one terminates the next begins; then the straight line from the initial point of the first to the final point of the last is the sum sought.”

**Problem 9.** To find the sum of  $n$  points  $A_1, A_2, \dots, A_n$ , that is, to find the point  $S$  satisfying the equation

$$A_1 + A_2 + \dots + A_n = nS.$$

If one subtracts  $nR$  from both sides of this equation, where  $R$  is an arbitrary point, one obtains

$$(A_1 - R) + (A_2 - R) + \dots + (A_n - R) = n(S - R),$$

and, since the previous equation is again obtainable from this one, it follows that: “In order to add  $n$  points, one draws straight lines from an arbitrary point  $R$  to the  $n$  points, and without changing their directions and lengths lays them contiguously together so that the initial point of the first falls

on  $R$ ; then, connecting  $R$  to the final point of the last one by a straight line and dividing this line into  $n$  equal parts, the first of these points from  $R$  is the point  $S$ , whose  $n$ -fold is the sum sought.”

**Problem 10.** To add arbitrarily many multiple points  $\alpha A$ ,  $\beta B$ , ..., if the sum of coefficients  $\alpha + \beta + \dots$  is not zero.

If one puts

$$\alpha A + \beta B + \dots = (\alpha + \beta + \dots)S,$$

and subtracts  $(\alpha + \beta + \dots)R$  from both sides, where  $R$  is an arbitrary point, one obtains

$$\alpha(A - R) + \beta(B - R) + \dots = (\alpha + \beta + \dots)(S - R).$$

Thus, since here as well the previous equation can again be recovered from this one, it follows that “to find the sum of arbitrarily many multiple points  $\alpha A$ ,  $\beta B$ , ..., whose coefficient-sum is not zero, one lays lines from the arbitrary point  $R$  to  $A$ ,  $B$ , ..., and multiplies these by  $\alpha$ ,  $\beta$ , respectively.<sup>11</sup> The lines so obtained are then laid contiguously together, without changing their directions or lengths, so that the initial point of the first falls on  $R$ , and  $R$  is connected to the final point of the last; then upon taking the  $\frac{1}{\alpha + \beta + \dots}$ -th part of the connecting line from  $R$ , the final point of this part multiplied by  $(\alpha + \beta + \dots)$  is the sum sought.”

**Problem 11.** To find the sum of the multiple points  $\alpha A$ ,  $\beta B$ , ..., if  $\alpha + \beta + \dots = 0$ .

If one subtracts the expression  $(\alpha + \beta + \dots)R$  from the sum  $\alpha A + \beta B + \dots$ , then since the subtracted magnitude is zero the value of the sum does not change, and thus

$$\alpha A + \beta B + \dots = \alpha(A - R) + \beta(B - R) + \dots$$

---

<sup>11</sup>As is easily seen, in such multiplication by a numerical magnitude  $\alpha$  the direction is not changed if  $\alpha$  is positive, while the length is changed in the ratio  $1 : \alpha$ ; and if  $\alpha$  is negative, then the direction is reversed.

Thus “the sum of multiple points whose coefficient-sum is zero is a straight line of definite length and direction that one finds by drawing straight lines from an arbitrary point  $R$  to the given points, multiplying them by the coefficients belonging to these points, and adding the products.”

**Problem 12.** To add two segments,  $A \cdot B$  and  $C \cdot D$ , of lines that intersect in  $E$ .

If one makes  $E \cdot F$  equal to  $A \cdot B$  and  $E \cdot G$  equal to  $C \cdot D$ , then

$$A \cdot B + C \cdot D = E \cdot F + E \cdot G = E \cdot (F + G) = 2E \cdot S,$$

if  $S$  is the mean of  $F$  and  $G$  (cf. problem 3).

**Problem 13.** To add two parallel line segments  $A \cdot B$  and  $C \cdot D$ , if they are not of equal length and oppositely directed.

If  $A \cdot B$  and  $C \cdot D$  are parallel, then  $D - C$  must be equal to  $\alpha(B - A)$ , where  $\alpha$  is some positive or negative number. Now as  $A \cdot B$  equals  $A \cdot (B - A)$ , since according to no. 12  $A \cdot A$  is zero, one has

$$\begin{aligned} A \cdot B + C \cdot D &= A \cdot (B - A) + C \cdot (D - C) \\ &= A \cdot (B - A) + \alpha C \cdot (B - A) \\ &= (A + \alpha C) \cdot (B - A). \end{aligned}$$

Now if the sum  $A + \alpha C$  equals  $(1 + \alpha)S$  (cf. problem 4), then the last expression

$$= S \cdot (1 + \alpha)(B - A) = S \cdot (B - A + D - C),$$

in which is implicit a simple construction for that sum.

**Problem 14.** To add two equally long and oppositely directed line segments  $A \cdot B$  and  $C \cdot D$ .



If both lie on the same straight line, the sum is zero. If this is not the case, then since  $D - C$  equals  $-(B - A)$  one has

$$\begin{aligned} A \cdot B + C \cdot D &= A \cdot (B - A) + C \cdot (D - C) \\ &= A \cdot (B - A) - C \cdot (B - A) \\ &= (A - C) \cdot (B - A). \end{aligned}$$

The sum is therefore an area of definite magnitude and planar direction.

**Problem 15.** To add two areas of definite magnitude and planar direction,  $a \cdot b$  and  $c \cdot d$ .

If the areas are parallel they can already be added according to no. 19; if they are not, then the two areas have a direction in common. If  $e$  is a straight line having this direction, and  $a \cdot b$  equals  $e \cdot f$ ;  $c \cdot d$  equals  $e \cdot g$ , then

$$a \cdot b + c \cdot d = e \cdot f + e \cdot g = e \cdot (f + g).$$

**Problem 16.** To add two parts,  $A \cdot B \cdot C$  and  $D \cdot E \cdot F$ , of definite, nonparallel planes.

If the planes are not parallel they will intersect. Let  $G \cdot H$  be a segment of the line of intersection, and let  $A \cdot B \cdot C$  equal  $G \cdot H \cdot J$ ,  $D \cdot E \cdot F$  equal  $G \cdot H \cdot K$ ; then

$$A \cdot B \cdot C + D \cdot E \cdot F = G \cdot H \cdot J + G \cdot H \cdot K = G \cdot H \cdot (J + K) = 2G \cdot H \cdot S,$$

if  $S$  is the mean of  $J$  and  $K$ . Thus “one adds two parts of nonparallel planes by representing them as triangles whose common side lies in the intersection of the two planes; then the sum sought is twice the triangle that has the same base and whose apex is the mean of the apexes of those triangles.”

**Problem 17.** To add two parts  $A \cdot B \cdot C$  and  $D \cdot E \cdot F$  of parallel planes.

If the planes are parallel, then  $(E - D) \cdot (F - D)$  must equal  $\alpha(B - A) \cdot (C - A)$ , where  $\alpha$  is a numerical magnitude. Thus

$$\begin{aligned} A \cdot B \cdot C + D \cdot E \cdot F &= A \cdot (B - A) \cdot (C - A) + D \cdot (E - D) \cdot (F - D) \\ &\quad \text{[Cf. no. 12]} \\ &= (A + \alpha D) \cdot (B - A) \cdot (C - A) \\ &= S \cdot [(B - A) \cdot (C - A) + (E - D) \cdot (F - D)], \end{aligned}$$

in which there is again implicit a simple construction. Now if  $\alpha = -1$ , that is if the two areas are of equal magnitude but are oppositely signed, then  $A + D$  is a straight line of definite direction and length (problem 11); if this equals  $H - G$ , then

$$A \cdot B \cdot C + D \cdot E \cdot F = (H - G) \cdot (B - A) \cdot (C - A),$$

whence this sum is a volume.

**Problem 18.** To add a part  $A \cdot B \cdot C$  of a definite plane and a volume  $(D - A) \cdot (B - A) \cdot (C - A)$ .

$$\begin{aligned} A \cdot B \cdot C + (D - A) \cdot (B - A) \cdot (C - A) \\ &= A \cdot (B - A) \cdot (C - A) + (D - A) \cdot (B - A) \cdot (C - A) \\ &= D \cdot (B - A) \cdot (C - A), \end{aligned}$$

from which the concept of this sum easily follows.

21. A combinatory product whose factors of first degree are magnitudes of  $(n - 1)$ -th order, but which lie in one and the same domain of  $n$ -th order, I call a *regressive product* with respect to that domain.

For example, a combinatory product of line segments in the plane, or parts of planes in space.

22. If from now on outer multiplication is denoted by simple juxtaposition and regressive multiplication by a dot placed between the factors, then by the regressive product  $AB \cdot AC$ ,

where  $A, B, C$  are arbitrary magnitudes, we mean the product  $ABC \cdot A$ , in which  $ABC$  is treated as a coefficient belonging to  $A$ , provided that the product is with respect to the domain of lowest order in which  $A, B, C$  lie simultaneously.

**Problem 19.** To find the product of two line segments  $AB \cdot AC$  with respect to the plane  $ABC$ .

According to no. 22 it is equal to  $ABC \cdot A$ ; that is, “the product of two line segments whose lines intersect is the point of intersection, with a part of the plane as coefficient.” If one imagines that a part of the plane is taken as the unit, then the parts of the plane with which the points are associated are real numerical magnitudes and the products appear as multiple points; but then all the magnitudes to be compared must lie in the same plane as that designated by the product (as is always the case in planimetry).

**Problem 20.** To find the product of three line segments  $AB, AC, BC$  with respect to the plane  $ABC$ .

SOLUTION.  $AB \cdot AC \cdot BC = ABC \cdot ABC = (ABC)^2$ .

**Problem 21.** To find the regressive product of two plane segments  $ABC$  and  $ABD$  (with respect to the volume).

SOLUTION.  $ABC \cdot ABD = ABCD \cdot AB$ .

**Problem 22.** To find the regressive product of three plane segments  $ABC, ABD, ACD$ .

SOLUTION.  $ABC \cdot ABD \cdot ACD = ABCD \cdot ABCD \cdot A = (ABCD)^2 \cdot A$ .

**Problem 23.** To find the regressive product of four plane segments  $ABC, ABD, ACD, BCD$ .

SOLUTION.  $ABC \cdot ABD \cdot ACD \cdot BCD = (ABCD)^3$ .

*Note.* The product of two plane segments therefore yields a line segment, that of three a point, but that line segment and

that point will have a volume segment or a product of volume segments as coefficients, and if one takes a volume segment as the unit then these coefficients turn into actual numerical magnitudes.

These are perhaps the most essential concepts appearing in volume 1 of my *Ausdehnungslehre*. But it is impossible to provide here more than a superficial idea of the unlimited fruitfulness of these new methods for the treatment, not only of the theory of space, but in general of all sciences that are based on spatial relations. Nor, by the same token, can I provide here the proofs that the calculational rules given in III for the individual methods of conjunction presented here are in fact valid; rather, I must again refer to my detailed treatise in which these proofs are given with complete rigor, and where at the same time the presentation generally proceeds in such a way that whatever arbitrariness may still seem to lie in the various concepts vanishes.



## Index of Defined Terms (1877)

The defined terms that I later abandoned are enclosed in parentheses. Those that I replaced with other expressions in the Ausdehnungslehre of 1862 are linked to the later terms by an equal sign.

| Term                                                | Translation                                      |
|-----------------------------------------------------|--------------------------------------------------|
| Abschattung= <del>Zurueckfuehrung</del><br>§82      | Shadow                                           |
| (Abweichung) §95                                    | Deviation                                        |
| Addition einfacher Ausdehnungen erster Stufe<br>§15 | Addition of elementary extensions of first order |
| hoeherer Ausdehnungen<br>§48                        | of higher extensions                             |
| Affinitaet §154                                     | Affinity                                         |
| direkt, reciproke §157                              | direct, reciprocal                               |
| Allgemeine Formenlehre<br>§1                        | General theory of forms                          |
| Analytische Form §7                                 | Analytic form                                    |
| Analytische Verknuepfung §5                         | Analytic conjunction                             |
| Aenderung, stetige §13                              | Evolution, continuous                            |
| Ausdehnung, erster Stufe<br>§14                     | Extension of first order                         |
| hoeherer Stufe §31                                  | of higher order                                  |
| von ergaenzender Stufe<br>§80                       | of supplementary order                           |

| Term                                     | Translation                              |
|------------------------------------------|------------------------------------------|
| der Elementargroesse §111                | of elementary magnitudes                 |
| Ausdehnungsgroesse=extensive Groesse §13 | Extensive magnitude                      |
| (Aussere Division) §60                   | (Outer division)                         |
| Multiplikation der Strecken §28          | multiplication of displacements          |
| der hoeheren Ausdehnungen §54            | of higher extensions                     |
| (Ausweichung der Elementargroesse) §109  | (Divergence of the elementary magnitude) |
| (Beziehungsgroesse) §137                 | (Relative magnitude)                     |
| (Beziehungssystem) §127                  | (Relative system)                        |
| (Beziehungszahl) §127                    | (Relative number)                        |
| (Doppelsystem) §143                      | (Double system)                          |
| (Eckgebilde) §110                        | (Vertex structure)                       |
| Eindeutige analytische Verknuepfung §6   | Unique analytic conjunction              |
| Eingeordnet=incident §136                | Incident                                 |
| Eingewandtes=regressives Produkt §125    | Regressive product                       |
| Element §13                              | Element                                  |
| Elementargroesse §94                     | Elementary magnitude                     |
| Elementarsystem §107                     | Elementary system                        |
| Erganzzahl einer Groesse §133            | Supplementary number of a magnitude      |
| Ersetzender Verein von Gleichungen §86   | Replacement assembly of equations        |
| (Formales Produkt) §125                  | (Formal product)                         |
| Formelle Summe §51                       | Formal sum                               |
| Gemeinschaftliches System §126           | Common system                            |
| Gemischtes Produkt §137                  | Mixed product                            |

| Term                                               | Translation                       |
|----------------------------------------------------|-----------------------------------|
| (Gesamtabweichung) §95                             | (Collective deviation)            |
| Gewicht des Elementarvereins §95                   | Weight of the elementary assembly |
| (Grad der Abhaengigkeit) §125                      | (Degree of dependence)            |
| Grundmasse=Einheiten §87                           | Fundamental measures              |
| (Grundsystem) §§82, 149                            | (Base system)                     |
| (Harmonische Gleichungen) §167                     | (Harmonic equations)              |
| (Harmonische Koeffizienten) §167                   | (Harmonic coefficients)           |
| Hauptsystem=Hauptgebiet §§80, 127                  | Principal system                  |
| Indifferente Form §7                               | Indifferent Form                  |
| Kollineation §160                                  | Collineation                      |
| (Kombination der Groessen) §151                    | (Combination of magnitudes)       |
| (Leitsystem) §§82, 149                             | (Guide system)                    |
| Multiplikation, siehe Produkt                      | Multiplication, see Product       |
| Nachstumfassendes System= verbindendes Gebiet §126 | Nearest covering system           |
| Offenes Produkt §172                               | Open product                      |
| (Polssystem) §167                                  | (Pole system)                     |
| Produkt, aeusseres §§28, 106                       | Product, outer                    |
| eingewandtes=regressives §125                      | regressive                        |
| (formales) §125                                    | (formal)                          |
| gemischtes §137                                    | mixed                             |
| reales §125                                        | real                              |



| Term                                 | Translation                |
|--------------------------------------|----------------------------|
| reines §137                          | pure                       |
| Projektion §§82, 151                 | Projection                 |
| Proportion in der Geometrie §75      | Proportion in geometry     |
| Reciprocitaet §160                   | Reciprocity                |
| Reines Produkt §137                  | Pure product               |
| (Richtaxen) §87                      | (Reference axes)           |
| (Richtmasse) §87                     | (Reference measure)        |
| (Richtstuecke) §88                   | (Reference terms)          |
| (Richtsystem) §87                    | (Reference system)         |
| Spath=Spat §§30,37                   | Parallelepiped             |
| Spatheck=Spateck §§28,37             | Parallelogram              |
| Starre Elementargroesse §109         | Rigid elementary magnitude |
| Strecke §14                          | Displacement               |
| Synthetische Verknuepfung §5         | Synthetic conjunction      |
| System-Gebiet $n$ -ter Stufe §§16,20 | System of $n$ -th order    |
| Unterordnung, Form derselben §129    | Subordinate form           |
| Verwandschaft §154                   | Relationship               |
| Zahlangroessen §68                   | Numerical magnitudes       |
| Zeiger-Ableitungszahlen §88          | Indicators                 |

## Editorial Notes

1. Part Two of this *Raumlehre* has the title: “Ebene räumliche Groessenlehre,” and it says in a footnote on p. 194f:

“Properly speaking, the rectangle is itself the true geometric product, and its construction as given in § 53 actual geometric multiplication. If one takes the concept of a product in its purest and most general sense, it denotes the result of a construction produced from that already generated (constructed), in the same way as that generated was itself generated from the original generator; and multiplication is therefore a construction to a higher power. In geometry the point is the original generator; from it the line results by that construction. If we make the bounded line (as that generated by the first construction) the basis of a new construction, in that we treat it by the same method as previously for the point, the rectangle results. The rectangle is therefore produced from the line just as the line is produced from the point.

“Now the same also holds in number theory. Here the original generator is the unit, which must be regarded as simply given with respect to number. From this, number results by enumeration (the arithmetic construction). If one makes this now generated number the basis of a new enumeration by putting it in the place of the unit, one obtains the arithmetic relationship to multiplication, which is therefore nothing but a number of higher order, a number whose unit is also a number. Thus one can perhaps say that the rectangle is a bounded line in which a bounded line takes the place of the

generating point. One would then also be able to grasp the two preceding theorems thus: Rectangles are the geometric product of base and height, and [otherwise] the case is as with arithmetic.”

In the *Lehrbuch der ebenen und sphaerischen Trigonometrie* by J. G. Grassmann, Professor of the Gymnasium at Stettin (Berlin: G. Reimer, 1839), one reads the following in a footnote on p. 10:

“If one takes the concept of the product in its purest and most general form, then in mathematics it denotes the result of a synthesis in which that generated by an earlier synthesis replaces the original element and is treated as that was. The product must likewise result directly from that generated by the first synthesis as this was generated from the original.—In arithmetic the element is the unit; the synthesis, enumeration; and the result, number. Now if this number, as the result of the first synthesis, replaces the unit and is treated in the same way, that is enumerated, there results the arithmetic product, which can be regarded as a number of higher order than a number, whose unit is also a number.—In geometry the element is the point; the synthesis, translation of the point in any given direction; the result, the path of the point, the line. Now if the line, as the product of the first synthesis, replaces the point, and is treated in the same way, i.e. it is translated in another direction, then the plane results as the path of the line; it is therefore the true geometric product of two linear factors, and appears first of all as a rectangle, insofar as none of the first direction is included in the second. If the plane replaces the point, the geometric solid results as the product of three factors, upon which geometry has its *terminus* since space has only three dimensions; but arithmetic is not limited in the number of factors appearing. See my *Raumlehre*, Part Two, Berlin 1824.”

2. The bracketed material was inserted by V. Schlegel. Before his death, Grassmann had only completed an outline of the foreword to the second edition. At the request of his

son, Schlegel incorporated the new material as smoothly as possible.

3. The sense of the words is: The new subjects, which were to have been treated in the unpublished second volume of the *Ausdehnungslehre* of 1844, are only partially enumerated in this foreword. These subjects are treated in the 1862 *Ausdehnungslehre*; newly added are, etc.

4. It seems appropriate to make a few general remarks here.

It is well known that the semiphilosophical style of the *Ausdehnungslehre* of 1844 is the reason this noteworthy book did not receive proper recognition for so long. Grassmann's express intention was wherever possible to present all its concepts directly in the most general form possible; he always sought to abstract the common connections that would illuminate the simple basic ideas, and to keep the incidental and contingent in the background. Finally, his whole structure was to be independent of geometry and analysis, geometry being permitted to appear only as an illustrative example for the clarification of his general concepts (see the note to § 13). However much his work owes to the very essential merit of this effort, which figures in all its individual parts, and which led Grassmann to very important insights into the fundamental principles of mathematics—we mention only his theory of the synthetic and analytic conjunctions and his general conception of multiplication—, it cannot be denied that in striving for generality Grassmann occasionally cuts the ground from under his feet.

This is particularly true of the basic concepts from which extension theory is constructed in the present work. The basic concepts are too imprecise, too contentless, to support the conclusions Grassmann draws from them. To convince oneself of this, one need only consider the definition of elements in § 13 and compare with it all that is subsequently done with these elements.

The element is supposed to be that elementary particular

existent without any real content; out of one such element,  $a$ , another element  $b$  is produced by an evolution. Now what is meant when he says that by an equal evolution a new element,  $c$ , is produced out of  $b$ ? How is it at all possible to compare an evolution of  $a$  with an evolution of  $b$  and thus to speak of equal evolutions? Finally, what is meant by the concept of continuous evolution? All these concepts, and naturally also all of their consequences are left hanging in the air.

Equally unsatisfactory is what Grassmann says about the parallel axiom and in general about the fundamental concepts of geometry (§ 22). The concepts “direction” and “equal construction” are just as unclear as the concepts “continuous evolution” and “equal evolution.”

Aside from Grassmann’s geometric axioms, which seem to have incurable defects, things are fortunately not as bad as they appear to be. In fact, Grassmann has with his “evolution,” on which he bases his system of extension theory, meant nothing but parallel displacement,

$$x'_i = x_i + a_i \quad (i = 1, 2, \dots)$$

in a multidimensional space. Replacing the contentless “evolution” with this concrete concept, which Grassmann refused to take over from analysis, would indeed sacrifice extension theory’s independence from the rest of mathematics, which however was only apparent, while achieving for the basic ideas a tangible content.<sup>12</sup> The subsequent development however leaves nothing to be desired in rigor, as far as we can see.

---

<sup>12</sup>One can also supply the concept “evolution” with a more general meaning. Where Grassmann says that by the same evolution that produced  $b$  from  $a$  a new element  $c$  is produced from  $b$ , he has in mind (although to be sure in a very obscure form) a “transformation of a multidimensional space”, for such a transformation indeed coordinates an evolution to each point of the space, and one can set these evolutions equal to each other since collectively they arise from the same transformation. The assumptions that Grassmann makes about his evolutions (see, e.g. § 17) are, according to Lie’s mode of expression, that in an  $n$ -dimensional space there are  $n$  one-parameter groups, whose infinitesimal transformations carry a point from an arbitrary position directly along

We call the reader's attention to another point.

Grassmann himself believed that in his extension theory he had created a universal instrument for geometric research, whose supreme merit was the avoidance of arbitrary coordinate systems. This basic conception of the character of geometry we can today express more precisely thus,<sup>13</sup> that it is basically the properties of certain transformation groups that exclusively or almost exclusively form the content of geometric research. The content of geometry is however by no means exhausted by these transformation groups. At present, thanks to the work of Lie, we have a more precise idea of the remarkable diversity of groups and the "geometry" belonging to each of them, that is "invariant theory," and we must say that there is not and cannot be such a universal instrument as Grassmann believed he had in his extension theory.

Indeed the *Ausdehnungslehre* of 1844 is mainly concerned with certain algorithms belonging to specific transformation groups. These are essentially the general projective group and the group of affine transformations.<sup>14</sup> In the *Geometrische Analyse* and in the *Ausdehnungslehre* of 1862 are added to these two groups the group of rotations about a fixed point and the more comprehensive group of Euclidean motions.

Invariant with respect to the general projective group are

---

one of  $n$  independent directions and whose finite transformations are pairwise interchangeable. These  $n$  one-parameter groups then specify an  $n$ -parameter simply transitive group with pairwise interchangeable transformations, and this group can, according to a theorem of Lie, be converted by a point transformation into the group of all translations of the space in question. Thus we return to parallel transport from a more general standpoint.

<sup>13</sup>After Lie in 1871 established the concept of a continuous transformation group, F. Klein in 1872 showed in his program "Vergleichende Betrachtungen ueber neuere geometrische Forschungen" (reprinted in Math. Ann., vol. 43) that the various geometric theories can be interpreted in the way given in the text.

<sup>14</sup>The word "affine" taken in the usual sense, and thus according to Lie the general linear group. Grassmann uses the word "affine" in a more general sense.

those equations of  $A_1$ , from which the measure (*masswerth*) drops out, that is those relations treated in §§ 165–170. To the group of affine transformations belong the remainder of the equations treated in  $A_1$ , in which the measure plays an essential role, for example in equations of the form  $a + b + c = 0$ , if  $a$ ,  $b$ , and  $c$  are extensive magnitudes. Throughout the work only the operations of addition and outer and regressive multiplication come into play. Not treated in  $A_1$  but discussed in the *Geometrische Analyse* and in  $A_2$ , inner multiplication belongs to the last two of the groups mentioned above.

The idea expressed here, that the most important discussions of extension theory relate to specific groups, and that outside this domain the calculus of extension theory loses its importance and usefulness, would be even more evident if the relation of extension theory to the invariant theory of those groups could be presented. That however demands a more extensive analysis, which must await a more detailed development. (Study and Engel)

5. The principles discussed here: Combinability of factors (§ 3), interchangeability of factors (§ 4), and the principle of the relation of multiplication to addition (§§ 9 and 10) one now denotes by the names: the associative, the commutative, and the distributive principles. According to H. Hankel, (*Theorie der complexen Zahlensysteme*, Leipzig: 1867, p. 3), the last two names were already introduced by Servois, the first probably by Hamilton, where it is already found in 1843. The completely general conception of the laws of conjunction is however Grassmann's property in any case.

6. It is not clear why one method of conjunction must be specified through the other; in any case the sentence contributes nothing to the clarity of the whole and can be omitted without harm. One then only replaces "definition" (*Begriffsbestimmung*) by "relation" (*Beziehung*) in the following material.

7. Namely those relating to opposing directions.

8. One need only think of the circle to see that this proof fails.

9. Grassmann uses the word “force” (*Kraft*) in a double sense, namely once in the usual sense and then where we would say “quantity of motion” (*Bewegungsgroesse*). Later Grassmann uses “motion” in the same sense. (Study)

10. Varignon first presented this theorem in the *Memoires de l'Academie des Sciences de Paris*, Annee 1719, p. 66 (Paris: 1721). In Varignon's *Nouvelle Mecanique ou Statique*, Paris 1725, it appears in vol. 1, p. 84, Lemma XVI.

The generally valid form of the theorem given by Grassmann one finds even earlier in the *Statik* of Moebius, § 34 (1837), see his *Werke*, vol. III, p. 51.

11. The discussion in this paragraph is not entirely complete, since there remains unsettled the question of whether the case considered in §§ 47–50 is the only one in which the sum of two extensions of higher order again gives an extension. To be sure, the question can be answered in the affirmative, but one does not find a proof of it in either  $A_1$  or  $A_2$ , although Grassmann repeatedly appeals to it later.

12. This promise is not kept in § 105, but see there the footnote.

13. “Moments of three axes” more properly reads “with respect to three axes”; similarly below, should read “going through the same point and lying in the same plane.” Furthermore, the theorems of § 59 are already found in the *Statik* of Moebius (*Werke*, vol. III), and are probably much earlier.

14. One recalls that the two products are different from zero according to the assumptions of the theorem.

15. It was assumed before that  $A$ ,  $B$ ,  $C$  are mutually independent, and this is equivalent to the assumption that  $C$  is independent of  $A$ ,  $B$  and the product  $A \cdot B$  (cf. § 55).



16. "According to § 47," cf. note 11 above.
17. That is, every numerical magnitude belongs to every arbitrary system (cf. § 83), but according to the definitions of § 54 it is independent of every extension of  $m$ -th order since its order is zero and  $m + 0$  is not  $> m$ .
18. That is, of all extensions of the system.
19. One adds: "belonging to this one reference term."
20. This elimination method was already given in 1840 by Sylvester in the *Philosophical Magazine*, but that method is not essentially different from one that Jacobi had previously developed in 1836 (*Crelles Journal* 15, pp. 101–124; *Jacobis Werke* vol. III., pp. 295–320).
21. Naturally  $p$  has unit weight.
22. Here it is to be understood that the sum  $a + b + \dots = 1$ . The same goes for § 111, third equation.
23. The extension of a rigid elementary magnitude is the true form of this elementary magnitude. Cf. appendix III, no. 15.
24. Every extensive magnitude is indeed represented in the form  $(\alpha - \beta) \cdot A$ , where  $\alpha$  and  $\beta$  are elements of unit weight and  $A$  is an extensive magnitude. Now since  $(\alpha - \beta) \cdot A = \alpha \cdot A - \beta \cdot A$ , the divergence of  $(\alpha - \beta) \cdot A$  has the value  $A - A = 0$ .
25. It is understood that  $\alpha A$  and  $\beta$  must be of the same order.
26. "Two products then only ...," the "only" is simply not proved; cf. note 11.
27. In volume 1 of his *Mecanique Analytique* (Coll. Works vol. XI.), Lagrange says: "en prenant le mot de moment dans le sens que Galilee lui a donne; c'est a dire pour le produit de la force par sa vitesse virtuelle" (I. part., sect. II., no. 2)

The moment in the now usual sense Lagrange indicates as the “moment relatif a un axe de rotation” (I. part., sect. III., no. 6).

28. “The collective moments ...”; special cases of this theorem were already given in §§ 58, 59.

29. The awkwardness of this presentation of the regressive product is due for the most part to the attempt to bring both products, outer and regressive, under a single head, and thus to have one theorem where there should have been two. In the *Ausdehnungslehre* of 1862 the theory of the regressive product is essentially simplified by the introduction of the supplement.

30. Here and subsequently Grassmann drops the dot for all outer products, since he uses it to denote the regressive product. Cf. § 133 and appendix III, no. 22.

31. Properly speaking, it was indeed just assumed that all magnitudes under consideration, and thus in particular both factors of the product, belong to the relative system!

32. Under the assumptions of § 116 there exist equations of the form

$$B = CB', \quad A = DBA' = DCB'A',$$

where  $E, D, C, B', A'$  are mutually independent. Therefore according to § 132

$$EDC \cdot B = EDC \cdot CB' = EDCB' \cdot 0 = EDB \cdot C,$$

and likewise

$$\begin{aligned} EDB \cdot A &= EDCB' \cdot DCB'A' \\ &= EDCB'A' \cdot DCB' = EA \cdot DB, \\ EDC \cdot A &= EDCDCB'A' \\ &= EDCB'A' \cdot DC = EA \cdot DC. \end{aligned}$$

33. After "the order of the common system is greater" one adds "than as in the first case."

34. Similar relative systems are of course those having the same principal system and whose "characteristic values with respect to the principal measure" (§ 132) are similar.

35. Cf. also § 133.

36. But in § 142 the shadow is not mentioned.

37. Since it was explicitly assumed that  $A$  belongs to the system  $LG$ .

38. Since  $AB = SM \cdot B = SB \cdot M$ .

39. "In the former case"; refer to the equation  $L \cdot (A' \cdot B') = L \cdot (A \cdot B)$  above, and the following two paragraphs.

40. The proof can be handled more clearly as follows: If  $M$ ,  $N$ , and  $B$  are ranked by orders  $\mu$ ,  $\nu$ , and  $\beta$ , then the common system of  $M$  and  $N$  is of order  $\mu + \nu - \beta$ . Now if  $C$  has only  $N$  in common with  $B$ , and the two of them have the principal system as nearest covering system, then  $C$  has order  $\beta + \nu - \beta$ ; on the other hand  $C$  has no other system in common with  $M$  but that common to  $M$  and  $N$ , so the order of the common system of  $C$  and  $M$  is  $\mu + \nu - \beta$ , and therefore the order of the nearest covering system of  $C$  and  $M$  equals  $\beta + \mu + \nu - (\mu + \nu - \beta)$ .

41. "According to § 153," that is, to paragraph two, thus:

$$M' = A' \cdot B', \quad N' = B' \cdot C'.$$

42. The first three of the last four equations represent a familiar double ratio of four points on a straight line, and one easily perceives to which four points the double ratio belongs in the individual cases. In comparison, the double ratio of four straight lines in space is new. Since this double ratio does not seem to have been previously noted by geometers,

it is perhaps not superfluous to show that one can give these magnitudes a simple—but nonrational—geometric interpretation. Thus they can be represented as the product of two ordinary double ratios and thus can be constructed by van Staudt's rule.

By  $a, b, c, d$  we mean four points lying on a straight line  $G$ , and use round parentheses to denote outer multiplication relative to the domain  $G$ , to distinguish it from the spatial magnitudes to be considered later. Then we have

$$(ab)(cd) + (ac)(db) + (ad)(bc) = 0.$$

Thus between the magnitudes

$$d_1 = \frac{(ab)(cd)}{(ad)(cb)}, \quad d_2 = \frac{(ac)(db)}{(ab)(dc)}, \quad d_3 = \frac{(ad)(bc)}{(ac)(bd)}$$

there are two linear equations, from which follow the relation  $d_1 d_2 d_3 = -1$ .

The magnitudes  $d_i$  and their reciprocal values are the six different double ratios that can be formed from the four points  $a, b, c, d$ .

Now just as one can form six double ratios from four points on a straight line, one can do likewise with four straight lines  $A, B, C, D$  in space, among which however two are mutually independent; thus the three magnitudes

$$D_1 = \frac{AB \cdot CD}{AD \cdot CB}, \quad D_2 = \frac{AB \cdot DB}{AB \cdot DC}, \quad D_3 = \frac{AD \cdot BC}{AC \cdot BC}$$

are interrelated by the condition  $D_1 D_2 D_3 = +1$ , and also by the three reciprocal values belonging to them. That two independent double quotients can be found among these six corresponds to the fact that four straight lines situated arbitrarily in space have two independent invariants with respect to projective transformations of the space. The  $D_i$  are evidently such invariants, and indeed are rational.

In order to correlate these two types of double ratios we consider two straight lines,  $G$  and  $\Gamma$ , that intersect the

straight lines  $A, B, C, D$  and are in general distinct. If  $a, b, c, d$  and  $\alpha, \beta, \gamma, \delta$  are the corresponding points of intersection, then we may assume that

$$A = a\alpha, \quad B = b\beta, \quad C = c\gamma, \quad D = d\delta;$$

then  $AB$ , for example, will have the value

$$AB = a\alpha \cdot b\beta = -ab \cdot \alpha\beta.$$

Now in every expression that is homogeneous with respect to both  $a, b, c, d$  and  $\alpha, \beta, \gamma, \delta$ , we can replace the products  $ab, \dots$  with respect to the space with the products  $(ab), (\alpha\beta), \dots$  with respect to the straight lines  $G$  and  $\Gamma$ , provided we adjoin an appropriate proportionality factor.

Thence follow three equations of the form

$$\begin{aligned} AB \cdot CD &= p \cdot (ab)(cd) \cdot (\alpha\beta)(\gamma\delta) \\ AC \cdot DB &= p \cdot (ac)(db) \cdot (\alpha\gamma)(\delta\beta) \\ AD \cdot BC &= p \cdot (ad)(bc) \cdot (\alpha\delta)(\beta\gamma), \end{aligned}$$

where  $p$  denotes a number on whose value we will expend no further effort.

Thus if we denote the double ratio of the points  $\alpha, \beta, \gamma, \delta$  corresponding to the  $d_i$  by  $\delta_i$ , then

$$D_1 = d_1\delta_1, \quad D_2 = d_2\delta_2, \quad D_3 = d_3\delta_3$$

from which

$$1 + \frac{1}{D_1} + \frac{1}{D_2} + \frac{1}{D_3} = 1 + \frac{1}{\delta_1} + \frac{1}{\delta_2} + \frac{1}{\delta_3}, \quad \dots$$

follows.

In modern terms the magnitudes  $d_i$  and  $\delta_i$  are therefore defined as nonrational invariants of the four straight lines  $A, B, C, D$  with respect to projective transformations of space, and are expressed by the rational invariants  $D_i$ .

If instead of the points of intersection of  $A, B, C, D$  with  $G$  and  $\Gamma$  one considers the connecting planes, then one is led to the same magnitudes, except that the  $d_i$  and the  $\delta_i$  interchange their roles.

If four given straight lines are transformable into four other given straight lines by projective (or dual) transformations, then it is necessary and in general also sufficient that the symmetric functions  $\sum D_i + \frac{1}{D_i}$  and  $\sum D_i \cdot \frac{1}{1-D_i}$  for the first four straight lines be equal to the corresponding functions for the second four. On the other hand, the mutual positions of four straight lines is no longer completely characterized by the two magnitudes chosen if two of the lines intersect. There is also yet another case. This is characterized by the existence of an equation of the form

$$\frac{1}{A} = 0 \quad \text{or} \quad \sqrt{AB \cdot CD} + \sqrt{AC \cdot DB} + \sqrt{AD \cdot BC} = 0.$$

The four straight lines are then either intersected by only a single straight line or by infinitely many; that is, they belong to a single special linear congruence of first order and first class, or they lie in a surface of second degree and are included in infinitely many linear congruences. In both cases the double ratio of Grassmann equals the square of an ordinary double ratio of four points or planes. (Study)

43. Properly speaking this equation must be written in the following way:

$$\alpha A + \beta B + \dots + (-\sigma)S = 0.$$

44. That is, the last two italicised theorems of § 169. If  $QR$  is an outer product, the first of them applies, since then  $RA, RB, \dots$  are also outer products and have the system  $R$  in common. If however  $QR$  is a regressive product, then  $RA, RB, \dots$  are also regressive products, and the system of lowest order that covers all these harmonic systems,  $RA, RB, \dots$ —it may be called  $S$ —is therefore subordinate to  $R$  and

consequently  $Q$  and  $S$  have the same system in common as  $QR$  and  $S$ ; thus the second of the two theorems then applies.

45. The general theorem on crystal structure, with which Grassmann crowns his monograph, is of course just a particular formulation of the law of rational indices, and indeed this formulation is obtained very easily if one conjoins Grassmann's rules on the multiplication of displacements and surfaces with the "law of zones."

In his dissertation "De lege zonarum principio evolutionis systematum crystallinorum," Berlin: 1826, F. Neumann had already indicated the connection between the law of zones and the law of rational indices.<sup>15</sup> On the other hand, in 1827 Moebius, in chapter 6 of part 2 of his *Barycentrische Kalkuel*, had considered the "geometric net" in the plane, a system of points and straight lines that is obtainable from four straight lines, just as the surfaces and planes of a crystal structure from four surfaces underlying it. But Moebius had not noticed that he had thereby provided the basis for a derivation of the law of rational indices from the law of zones; to have observed that is the merit of Grassmann. Moebius himself said in his work "Ueber das Gesetz der Symmetrie der Krystalle" (Leipzig: 1849, *Werke* vol. II, pp. 349–360) that in the composition of his *Barycentrische Kalkuel* he had taken the investigation of geometric nets as a purely geometric speculation, without in the least surmising the close connection in which it stood to a fundamental attribute of crystals. Only Grassmann had called attention to it (and precisely in his *Ausdehnungslehre* of 1844).

It is perhaps not superfluous if the theorems that Grassmann states without proof in § 171 are proved here using Grassmann's methods.

---

<sup>15</sup>These ingenious communications are essentially taken from *Geometrische Krystallographie* by Liebisch (Leipzig: 1881); cf. in particular p. 30. Grassmann was however completely unfamiliar with Liebisch, and in fact Grassmann's contributions are rather less than those of F. Neumann and Moebius.

In crystal structures only the directions of the edges and surfaces are important, not their positions in space; extensive structures of first and second order, that is displacements and plane areas of fixed direction, are therefore especially suitable for treating crystal structure analytically.

Now the law of zones states that every line of intersection of two crystal surfaces yields the direction of a possible crystal edge, and that every plane parallel to two crystal edges yields the direction of a possible crystal surface; besides which, the law states that on the basis of this principle all possible edges and surfaces can be obtained from four crystal surfaces, no three of which are parallel to the same straight line, or from four crystal edges, no three of which are parallel to the same plane.

Thus if we think of a tetrahedron  $ABCD$  formed of four crystal surfaces, and denote the displacements equal in length and equally directed with the edges  $DA$ ,  $DB$ , and  $DC$  by  $a$ ,  $b$ ,  $c$ , then  $ab$ ,  $bc$ ,  $ca$  are plane areas proportional to the plane areas  $DAB$ ,  $DBC$ ,  $DCA$  and similar to them. Further, the lengths and directions of the edges  $AB$ ,  $BC$ ,  $CA$  are represented by  $b - a$ ,  $c - b$ ,  $a - c$ , and likewise the direction of the surface  $ABC$  by  $ab + bc + ca$ .

If the direction of a plane parallel to the two edges  $DA$  and  $BC$  is now represented by the outer product  $a(c - b)$ , then  $ac - ab$  is a possible crystal surface, and likewise of course  $ba - bc$  and  $cb - ca$ . The combination of one of these three surfaces with the four original surfaces does not yield any new crystal edges, but one obtains three new crystal edges by pairwise combination of these three surfaces. In order to find them we must consequently, for example, seek the intersection of the two surfaces  $ac - ab$  and  $ba - bc$ . This intersection will be represented by the regressive product

$$(ac - ab) \cdot (ba - bc),$$

which, on account of

$$ac - ab = a(c - b - a), \quad ba - bc = (c - b - a)b,$$



according to § 133, fourth equation, has the value

$$a(c - b - a)b \cdot (c - b - a) = -abc \cdot (c - b - a).$$

By continuing this process one recognizes that every possible crystal surface is representable in the form  $\lambda ab + \mu bc + \nu ca$ , and every possible crystal edge in the form  $\lambda'a + \mu'b + \nu'c$ , where  $\lambda, \mu, \nu$  and  $\lambda', \mu', \nu'$  are positive or negative whole numbers. But this is precisely the content of the first and third theorems of this section.

If on the other hand one thinks of four crystal edges, no three of which lie in a plane, then one can always choose four displacements parallel to these edges such that, if  $a, b, c$  are three of these displacements, the fourth will be representable as  $a + b + c$ . The law of zones then shows again that  $ab, bc, ca$  are crystal surfaces and that every possible crystal surface is representable as  $\lambda ab + \mu bc + \nu ca$ , and every possible crystal edge in the form  $\lambda'a + \mu'b + \nu'c$ , where  $\lambda, \mu, \nu$  and  $\lambda', \mu', \nu'$  are whole numbers. From this standpoint, when one proceeds from four crystal edges, Grassmann places the fourth theorem, whereas previously he always proceeded from crystal surfaces.

Still it must be observed that the fourth and fifth theorems have to be somewhat modified. In the first in particular, the representation as a harmonic multiple sum is only possible for such edges as are not parallel to the plane in question, and in the second only for such surfaces as are not parallel to the edges in question.

46. In the *Ausdehnungslehre* of 1862 the theory of open ("place-holding") products is presented much more completely and clearly; one also finds applications of this theory there.

47. That is, without the previous expression reducing to a sum of squares.

48. One does not read this off from § 144, so for clarification let the following be noted:

Let  $u$  be a point—the origin of the representative—and  $a, b, c$  be three mutually independent displacements; then every plane not going through  $u$  can be represented in the form

$$P = xubc + yuca + zuab + abc,$$

and every finite point in the form

$$v = u + x'a + y'b + z'c,$$

where  $x, y, z$  and  $x', y', z'$  are numbers. If this point  $v$  is to lie on the plane  $P$ , then necessarily

$$(u + x'a + y'b + z'c)P = 0,$$

that is

$$xx' + yy' + zz' = 1.$$

The deviation of the plane  $P$  from  $u$  is then, according to § 112, equal to the divergence of the product

$$uP = uabc,$$

whence this deviation equals  $abc$ , that is it will be  $= 1$  if one sets  $abc = 1$ . If  $Q = xbc + yca + zab$  is fixed as a surface parallel to the plane  $P$ , and in addition  $p = x'a + y'b + z'c$ , which is a displacement parallel to the linear magnitude  $uv$ , so that the deviation in question is also representable in the form  $PQ$ , then  $PQ = abc$ .

49. Appendix I. It is not clear what Riemann and Helmholtz should have inferred for their purposes from §§ 15–23. With more justice Grassmann could have complained that Riemann had not mentioned him in regard to the introduction of the concept of an  $n$ -fold extended manifold; but Riemann probably knew nothing of the 1844 *Ausdehnungslehre*. Moreover, Grassmann only learned of the relevant works of Riemann and Helmholtz shortly before his death, and was unable to study them in detail.

50. Appendix II. The expression “original units” appearing herein is not seen in the 1844 *Ausdehnungslehre*; Grassmann introduced it only later, in the 1862 *Ausdehnungslehre*.

51. Appendix III. This article appeared in 1845 in *Grunerts Archiv*, vol. VI, pp. 337–350.

52. This is not the proper place to detail the relationship between Grassmann’s calculus and Hamilton’s quaternions. However it is not unfitting to relate here what Hamilton has said about extension theory.

In Hamilton’s *Lectures on Quaternions*, Dublin: 1853, on page (62) of the foreword one reads in a footnote the following: “It is proper to state here, that a species of non-commutative multiplication for inclined lines occurs in a very original and remarkable work by Prof. H. Grassmann (*Ausdehnungslehre*, Leipzig: 1844), which I did not meet with till after years had elapsed from the invention and communication of the quaternions; in which work I have also noticed (when too late to acknowledge it elsewhere) an employment of the symbol  $\beta - \alpha$ , to denote the directed line, drawn from the point  $\alpha$  to the point  $\beta$ . Notwithstanding these, and perhaps some other coincidences of view, Prof. Grassmann’s system and mine appear to be perfectly distinct and independent of each other, in their conceptions, methods, and results. At least, that the profound and philosophical author of the *Ausdehnungslehre* was not, at the time of its publication, in possession of the theory of quaternions, which had in the preceding year (1843) been applied by me as a sort of organ or calculus for spherical trigonometry, seems clear from a passage of his preface, in which he states (under date of June 28th, 1844), that he had not then succeeded in extending the use of imaginaries from the plane to space; and generally that unsurmounted difficulties had opposed themselves to his attempts to construct, on his principles, a theory of angles in space.”

This comment of Hamilton is supplemented in a remarkable way by a communication for which I thank the kindness

of Herr Prof. O. Henrici in London. On 18 January 1893 he wrote me the following: “In the Graves Library of University College is a copy of the original edition of the 1844 *Ausdehnungslehre*. This copy is bound together with the *Barycentrische Kalkuel* of Moebius and was evidently very thoroughly studied by an Englishman. There are throughout pencilled comments in the margins and underlinings in the text. Although Hamilton’s name does not appear, it is nevertheless certain that the comments are Hamilton’s.” Herr Prof. Henrici informed me of the most important of these comments. Here they are:

Foreword, p. XII: “Grassmann thus seems to have reinvented Double Algebra.”

P. 71: “Not seen until 1853.”<sup>16</sup>

P. 102: “His *Zahlengroessen* are my scalars.”<sup>17</sup> He does not seem to introduce the conception of equal angles into his proportion of lines. His angles are identical (see fig. 12). See also p. 111.”

P. 130: “Consider this!”

P. 145: “He thinks that he first introduced the conception of directed lines . . . the ‘Strecke.’”

P. 172: “His line-magnitudes are on lines fixed in space.”

I was informed that I could get certain communications from Henrici that would be of value for this edition of Grassmann’s works by Gordon.

---

<sup>16</sup>The year in which the Lectures appeared.

<sup>17</sup>The words “my scalars,” as Herr Henrici observes, clearly point to Hamilton.



# Geometric Analysis



Geometric Analysis

# GEOMETRIC ANALYSIS



Connected with the  
Geometric Characteristic Invented by Leibniz

PRIZE ESSAY  
BY  
H. GRASSMANN

WITH AN EXPLANATORY TREATISE

BY



A. F. MOEBIUS

LEIPZIG

WEIDMANN'SCHE BUCHHANDLUNG

1847

## Contents

|       |                                                                    |     |
|-------|--------------------------------------------------------------------|-----|
|       | Introduction                                                       | 316 |
| § 1.  | The Leibnizian Characteristic                                      | 318 |
| § 2.  | Congruence and Collineation                                        | 320 |
| § 3.  | Point Magnitudes and Linear Magnitudes                             | 324 |
| § 4.  | Addition of Point Magnitudes and of Linear Magnitudes              | 326 |
| § 5.  | The Collinear Function                                             | 329 |
| § 6.  | Return to Congruence. Equations Between Spatial Magnitudes         | 330 |
| § 7.  | Inner Products of Displacements                                    | 334 |
| § 8.  | Inner and Outer Multiplication                                     | 339 |
| § 9.  | Inner Products of Surface Areas and Displacements                  | 340 |
| § 10. | Geometric Applications                                             | 344 |
| § 11. | Applications to the Pure Theory of Motion                          | 347 |
| § 12. | The Differential Equations of Mechanics                            | 350 |
| § 13. | Application to a Problem in Statics                                | 357 |
| § 14. | Transition from Displacements to Points                            | 360 |
| § 15. | Sums of Point Magnitudes                                           | 362 |
| § 16. | Inner Multiplication of Point Magnitudes. Inner Magnitudes         | 364 |
| § 17. | First Type of Sum of Inner Magnitudes: Inner Displacement Products | 366 |
| § 18. | Second Type of Sum of Inner Magnitudes: Inner Planar Magnitudes    | 367 |
| § 19. | Addition of Inner Planar Magnitudes                                | 370 |
| § 20. | Third Type of Sum of Inner Magnitudes: Spherical Magnitudes        | 371 |
| § 21. | Geometric Representation of Sums of Inner Magnitudes               | 374 |
| § 22. | Difference Plane of Two Spherical Surfaces                         | 376 |
| § 23. | Multiple Sums of Spherical Surfaces                                | 381 |
|       | Conclusion                                                         | 382 |



## Introduction

If the distinctive intellectual power of a thinker who rises above his times is already displayed in his ability to comprehend and develop those ideas toward which the age strives, so that he becomes its representative: then all the more does such an intellect stand out in those ideas which are ahead of the times and even anticipate the course of development of centuries.

While ideas of the former type are often developed simultaneously by the leading intellects of the time (as for example the differential calculus, developed simultaneously by Newton and Leibniz) precisely because the time was ripe for them: so those of the latter type appear as the special property of such an individual, as the innermost workshop of his intellect, into which only a privileged few contemporaries are permitted to enter and discern the wealth of developments that will be manifest at a future time. Whereas those former ideas find powerful approval and produce a great response and elaboration, since they represent precisely the high point of that time: at the same time these others usually die away nearly ineffective in their day, since they were only understood by a few, and perhaps by no one in their entirety. Often only after centuries, when time has pushed development to the point anticipated by that intellectual force, will those thoughts sow a rich harvest.

That Leibniz's sublime idea, whose comprehension and completion comprises the subject of this essay, namely the idea of a genuine geometric analysis, belongs among these leading and, as it were, prophetic ideas cannot be doubted

for a moment, as they have also shared the fate of such ideas. Indeed, by a particular misfortune of circumstances it remained hidden until well past the point in time at which it could have strongly influenced the course of development. For even before that Leibnizian idea was extracted from hiding by Uylenbroek, the way was already prepared for a closely related analysis from various directions.

Even so, it seems an important task to invigorate the new analysis by that idea, and the special form which Leibniz gave it, and thus revive that seed so long dormant. For although that idea was excluded from the process of historical development by a concealment of more than a century: yet, since it has now appeared, it is by no means outdated, but rather contains a still fresh and potent seed whose right to merge with the historical development can no longer be denied. For on the one hand the ideal of this geometrical analysis, as conceived of by Leibniz for future development, has by no means yet been completely achieved, nor is it sufficiently recognized as such by mathematicians, and on the other hand the distinctive form which he gave this idea by his notational method, admittedly mostly by way of example, is by no means similarly developed by any more recent mathematician; on the contrary, all newer analyses proceed from other, if often more fruitful, viewpoints—afforded by the progress of time.

Leibniz himself distinguished most clearly on the one hand his idea of a pure geometric analysis, whose formulation and completion floated before his eyes as a distant goal, whose importance however he completely understood, and on the other his attempt at such a new characteristic, in order to make the possibility of the realization of that idea more credible, and to bequeath a memorial to posterity, should his execution have been hindered, which might subsequently motivate another to the realization of that idea. The two must therefore be sharply distinguished if one is properly to appreciate Leibniz's contribution in formulating the geometric analysis.

In fact, one easily convinces oneself that the characteris-

tic attempted by Leibniz does not in the least realize what he generally promised of geometric analysis; on the contrary, it lags so infinitely far behind its goal that it can only be viewed as a crude if worthy first approximation to that goal. Moreover, one cannot claim that Leibniz had other notational methods at hand which stood nearer the attainment of that goal, for then they would have made its attainability more believable, and thus either would have been immediately advanced by him, or at least been touched upon by way of suggestion, no trace of which is found. One would thus obtain a completely distorted and incorrect assessment of Leibniz's accomplishment in this domain if one did not keep firmly in mind that he by no means ascribed to his attempt that merit which he would ascribe in rich measure to geometric analysis in general.

Given, then, that matters stand thus, from whence comes his energetic expression of esteem for a subject whose configuration he did not know? Whence this emphatic praise of its importance, this enumeration of all the fruits, touching on unbelievability, that it must also educe for all domains of applied knowledge?

To this we can only answer that there exists a special talent of a higher intellect, whereby he grasps the importance of an idea, seeing into its most distant consequences, while smaller intellects, mistaking the insignificant for the important, pass by true greatness unmindful. Precisely this outstanding talent of Leibniz to survey at a glance a complete course of development without digging through its details and, without analyzing and dissecting it in advance, yet with prophetic spirit to imagine it and thus to know its consequent importance, this talent is precisely what led him to such imposing discoveries in almost all domains of knowledge. I hope in the course of this essay to show that, apart from isolated exaggerations, which however generally reside more in their expression than in the subject itself, Leibniz perceived correctly. This I will do by formulating, at least in outline, an analysis which in general actually accomplishes what he re-

garded as the goal of geometric analysis; indeed it will be shown that even the essential merits of this analysis were enumerated by him with a degree of completeness. In this way, then, the scientific significance of his idea of a geometric analysis can best be shown.

In order to also illuminate the scientific significance of his particular characteristic, and thereby bring into consideration this aspect of his scientific contribution, I will in the derivation and development of the new analysis proceed from the Leibnizian characteristic, and show how, from this seed, by consistent development and proper deletion of extraneous materials and fertilization with the ideas of geometric relationships, there results the analysis which I am inclined to regard, if only relatively, as the realization of the Leibnizian idea of a geometric analysis. That this is not the way in which I was led to this analysis hardly needs mention.

## 0.1 § 1. The Leibnizian Characteristic

The essential feature of the Leibnizian method of notation is that he designates points whose positions are unknown or variable by the last letters of the alphabet just as is the custom in algebra; he then indicates points with known or fixed positions with the remaining letters. This notation he then applies in particular to congruence in the examples he communicates, where with complete generality he sets two arbitrary sets of corresponding points congruent if, without changing the relative positions of the points in either one of the two sets, they can be brought into superposition, that is specifically so that each point of the one set is superposed on its correspondent in the other, whence he then always places the corresponding points on the corresponding positions of the set regarded as congruent. This simple approach and method of notation now becomes for him the seed for a series of the most astonishing results; indeed, he is thereby already in a position to show, by these examples in fact, how a purely geometric analysis is possible, and indeed one to which all

spatial relations can be subject.

If with Leibniz we choose  $\mathfrak{E}$  as the symbol for congruence, by  $x$  understand a point with variable position, and by  $a$ ,  $b$ , and  $c$  fixed points, one sees at once that

$$ax \mathfrak{E} bc \quad (1)$$

yields a sphere (whose center is  $a$  and whose radius is  $bc$ ), and

$$ax \mathfrak{E} bx \quad (2)$$

a plane (to which  $ab$  is perpendicular) as the geometric locus of the point  $x$ .

Since however all constructions in space can be carried out by means of the sphere and the plane (indeed, by the sphere alone), and thus all types of spatial dependence can be mediated by them, it follows that the principle mentioned must suffice for every spatial consideration. Indeed, these two expressions (1) and (2) can be taken as definitions of the sphere and plane, and thus the entire structure of geometry can be independently erected upon them. Formula (2) can be taken as the expression for the geometric locus of the center of all spheres intersecting the fixed points  $a$  and  $b$ .

Likewise, the formula

$$ax \mathfrak{E} bx \mathfrak{E} cx \quad (3)$$

yields the straight line as the geometric locus of the centers of all spheres intersecting the three fixed points  $a$ ,  $b$ ,  $c$ . Finally the circle appears as the intersection of two spheres, whence its expression is

$$ax \mathfrak{E} ac$$

$$bx \mathfrak{E} bc$$

or, combined,

$$abx \mathfrak{E} abc. \quad (4)$$

From this one sees that everything can be reduced to the expression for the sphere,<sup>18</sup> and thus how such a construction

---

<sup>18</sup>This footnote appears in the original text.



of geometry from the fundamental expression (1) according to this unity and simplicity of principle would be of great scientific significance. But it would lead me too far from the previously stated goal were I merely to sketch the characteristic features of such a construction of the science, which of course would necessarily entail the derivation of the requisite fundamental principles and the demonstration that they suffice for the construction of the science; and this investigation would always remain most difficult. Instead I will now investigate the applicability of the Leibnizian method of calculation to the solution of the two fundamental problems of geometry: the expression for a straight line going through two given points and for a plane containing three given points.

First, should the expression for a straight line going through two points  $a$  and  $b$  be sought, then it must have the form of expression (3), in which  $a$ ,  $b$ ,  $c$  must be replaced by three auxiliary points  $a'$ ,  $b'$ ,  $c'$ , which however may not be collinear. One then has the formulas

$$\begin{aligned} a'x \mathfrak{E} b'x \mathfrak{E} c'x \\ a'a \mathfrak{E} b'a \mathfrak{E} c'a \\ a'b \mathfrak{E} b'b \mathfrak{E} c'b \end{aligned} \tag{5}$$

which, taken together, specify the straight line as the geometric locus of the point  $x$ . In this case one can however combine the last two sets of congruences into the one

$$aba' \mathfrak{E} abb' \mathfrak{E} abc'.$$

This then expresses that the auxiliary points  $a'$ ,  $b'$ ,  $c'$  lie on a circle whose plane intersects  $ab$  perpendicularly at the center of the circle.

Equally complicated is the expression for a plane containing three noncollinear points. Since the expression must have the form (2), one must introduce two auxiliary points  $a'$  and  $b'$ , and since  $a$ ,  $b$ ,  $c$ ,  $x$  must lie in the plane defined by these

auxiliary points, one has the four formulas

$$\begin{aligned} a'x &\mathfrak{E} b'x \\ a'a &\mathfrak{E} b'a \\ a'b &\mathfrak{E} b'b \\ a'c &\mathfrak{E} b'c. \end{aligned}$$

The last three sets of congruences can be written as one, thus:

$$abca' \mathfrak{E} abcb'.$$

One therefore has

$$a'x \mathfrak{E} b'x, \quad abca' \mathfrak{E} abcb' \tag{6}$$

(of which the last is a combination of three, and expresses that  $a'$  and  $b'$  lie symmetrically opposite  $abc$ ) as the plane containing the three points  $a, b, c$ .

If one is to give the systems of formulas (5) and (6) the requisite simplicity, one must be in a position to eliminate the arbitrary auxiliary points having nothing to do with the nature of the problem, and to combine the system of formulas into a single one. One immediately sees that this is not possible in the Leibnizian notation, to the extent that he has developed it. It is therefore quite easy to see that the number of formulas then required in the case of a somewhat complicated problem must soon increase enormously, and that one must introduce a number of elements into the treatment that have nothing to do with the original problem.

## 0.2 § 2. Congruence and Collineation

Thus we recognize the inadequacy of this characteristic. Wherein does it lie?

Obviously, first of all,<sup>19</sup> in that the relation of congruence is introduced to replace the simple relation of equality, which

---

<sup>19</sup>The original has a footnote marker here.

alone should occur in mathematical formulas in order that one can substitute throughout. Even though, for this relation of congruence, the principle that what are congruent to the same third thing are congruent to each other is valid, in a congruence relation one can by no means always replace each expression with its congruent. For example, let  $ac \mathfrak{E} bc$ , and let a point  $d$  lie off the perpendicular to  $ab$  dropped from  $c$ ; then it is not true that

$$acd \mathfrak{E} bcd,$$

even though

$$ac \mathfrak{E} bc$$

so that in  $acd \mathfrak{E} bcd$  one is not permitted to replace the  $ac$  on one side with its congruent  $bc$ . Or even more simply, since all points are congruent, then if one could substitute congruents for each other it would be possible to set  $abc$  congruent with every set of three arbitrary other points, which is nonsense.

The possibility of substitution, which is cut off in the above notation, we must however necessarily retain in any mathematical notation if it is to lead to fruitful results. First of all we must therefore change the notation and introduce the essential relation of equality, in which we just set as simply equal that which we can substitute one for the other in each proposition.

If  $abc$  is congruent to  $def$ , then we ask: What is equal between the two sets of points? Obviously some geometric function of the three points  $a, b, c$  must be set equal to the corresponding function of the three points  $d, e, f$ . We will tentatively designate this function with the symbol *figure* or *fig*, and instead of

$$abc \mathfrak{E} def$$

now write

$$fig(a, b, c) = fig(d, e, f)$$

and likewise

$$fig(a, b) = fig(d, e)$$

if  $ab$  is as long as  $de$ .

It is then a question of finding the form of this function, specifically for two points the form of the function  $fig(a, b)$ . For the equation

$$fig(a, b, c) = fig(d, e, f)$$

is just a unification of the three equations

$$fig(a, b) = fig(d, e)$$

$$fig(b, c) = fig(e, f)$$

$$fig(c, a) = fig(f, d).$$

Only if we succeed in ascertaining the form of this function and the laws that underlie it will we be in a position to substitute completely freely.

In order to reach this point we will first extend the Leibnizian idea. Thus for example for two similar figures we can say that their shapes are equal, and thus instead of

$$abc \sim def$$

write

$$form(a, b, c) = form(d, e, f).$$

Thus we can express the equality of the area between any three points or the volume between any four points, or even the affinity, by special symbols.

We immediately proceed to the most general linear relationship, collineation, and for two collinear collections

$$a, b, c, d, e, f \quad \text{and} \quad a', b', c', d', e', f'$$

write

$$form(a, b, c, d, e, f) = form(a', b', c', d', e', f').$$

Now considered from a certain standpoint it is true that congruence is the simplest of all these relationships, since for it everything can be reduced to functions of two points; and

in the same respect collineation appears as the most complicated, since for it five points, no four of which are coplanar, can be set collinear to any other five such, and only for the sixth point does there appear a condition such that, if in one system a sixth point is given, then the sixth of the other is specified. For collineation in space the simplest function to which everything can be reduced is a function of six points, and one easily sees that two systems of arbitrarily many points can be related collinearly if and only if that simplest collinear relation obtains between every six points of the two systems.

Thus while congruence reduces to a function of two points, collineation reduces to a function of six points. However, if one considers the way that in congruence the second and in collineation the sixth point is specified, then on the contrary collineation appears the simpler.

For if two points of one system and only one of the corresponding points of the congruent system are given, then the remaining point can lie anywhere on a sphere that has the first point as center and the distance between the points of the first system as radius; the first condition that enters here is thus a partial one, tied to the concept of the sphere. By way of contrast, given five points, no four of which are coplanar, and five corresponding points of a collinearly related system, no four of which may thus be coplanar, then the point corresponding to the sixth point of the first system is completely specified; thus the first condition that enters in collineation is a complete one. It is also known that the way the sixth point of the collinearly related system depends on the given points is mediated only by planes (or, if one prefers, by straight lines), while congruence is mediated by spheres. But the concept of the sphere includes that of the plane, inasmuch as the latter may be regarded as a sphere with an infinitely distant center, but the converse does not obtain; for an independent part of geometry can be developed in which not even the rudiment of the concept of the sphere is presupposed, as is demonstrated by Grassmann in his *Ausdehnungslehre*. Thus

with respect to the type of dependence—and only this is of importance in the present investigation—collineation appears as the more elementary relationship.

Accordingly I will proceed with it, first however considering only collinear systems in the same plane. Thus to four points, no three of which are collinear, any four other such points can be set as corresponding points of a collinearly related system, but then the point of the collinearly related system corresponding to any fifth point of the first system is completely specified; and the question is, what is the nature of this specification?

To express myself succinctly in the following I will say that the straight line going through two points, and the intersection point of two straight lines, are derived by *linear constructions* from those points and these lines, and indeed that the straight line between two points  $a$  and  $b$ , or the intersection point of two straight lines  $A$  and  $B$ , is derived by linear constructions corresponding to those by which the straight line between two points  $a'$  and  $b'$  is derived from these points, or the intersection point of the two straight lines  $A'$  and  $B'$  from these straight lines.

Now if  $a, b, c, d, e$  and  $a', b', c', d', e'$  are corresponding points of two collinear plane sets, that is

$$\text{collin}(a, b, c, d, e) = \text{collin}(a', b', c', d', e'),$$

then it follows from the principle of collineation (cf. Moebius, *Barycentrische Kalkuel*, § 217), that, if  $e$  can be derived from  $a, b, c, d$  by linear constructions,  $e'$  can be derived from  $a', b', c', d'$  by the corresponding constructions. Further, it is known (cf. Moebius, *Barycentrische Kalkuel* § 205) that, if no three of the four points  $a, b, c, d$  that lie in a single plane are collinear, then every other point of the same plane can be derived from them by linear constructions, either exactly or as closely as is desired; thus the principle of linear constructions cited suffices for the apprehension of the above collineation equation.

One therefore sees that the entire way the fifth point depends on the remaining points is based only on those elementary constructions, and thus that, if one wishes to express the fifth point analytically by the four others, one need only express the line connecting two points as their conjunction, and the point of intersection of two straight lines as the conjunction of these lines, and formulate the laws of this conjunction.

### 0.3 § 3. Point Magnitudes and Linear Magnitudes

It is however to be remarked that neither the point as such, nor the infinite line can be regarded as magnitudes, since to magnitudes belongs the intrinsic possibility of the enlargement and reduction of measure (metric value). Thus should it be spatial magnitudes that are subject to conjunction, then one must distinguish two aspects in them, that is on the one hand their spatial orientations and on the other their measures, which may also possibly be simple numerical magnitudes; and one will then only be able to set two spatial magnitudes equal if both the spatial orientations with which they are endowed as well as their measures are equal.

Thus for example the point as such only represents a position in space, but the point endowed with a specific numerical coefficient appears as a spatial magnitude. This numerical coefficient may be unity, in which case the point itself also appears as a magnitude, but only insofar as the numerical coefficient "one" in it is regarded as a particular value of the possibly variable numerical coefficient; thereby the point also appears as capable of enlargement and reduction, and thus as a magnitude. To give another example, a limited part of a straight line appears as a spatial magnitude, since the line (as infinite) represents the spatial orientation, its length the measure. If the measure of a magnitude is zero the magnitude vanishes, and it must then be set to zero itself.

I will now call such points and lines provided with definite measures *point magnitudes* and *linear magnitudes* for short,

and understand by the combination  $ab$  of the two point magnitudes  $a$  and  $b$ , that is, two points with associated coefficients, a linear magnitude whose line goes through the two points belonging to  $a$  and  $b$ , and whose measure will be fixed later on; and likewise by the combination  $AB$  of two linear magnitudes  $A$  and  $B$  (that is, two finite straight lines) a point magnitude, whose point is the point of intersection belonging to  $A$  and  $B$ , and whose coefficient will be fixed later on. Next I will only introduce the assumption that the result of this conjunction shall always have a definite value in order to account for the case in which the measure of the magnitude obtained by the conjunction is zero.

First I consider the combination of two point magnitudes. Should they coincide, the line of their combination is undefined, and the result of the conjunction would then, contrary to assumption, be unspecified if one does not assume that the measure is zero in this case. If the above assumption is to remain in force we must therefore set the combination of two point magnitudes to zero if their points coincide, and on the same grounds set that of two linear magnitudes to zero if their associated lines coincide. Furthermore, if one of the two point magnitudes is zero, then its point can have any orientation; the line of its combination with the other point magnitude is therefore unspecified. Thus if the assumption is to be maintained the combination must be set to zero in this case as well. And on the same grounds the combination of two linear magnitudes must also be set to zero if one of them is zero. Since however in no other cases than those mentioned are the spatial orientations of the results unspecified, we can also require that in no other case is the result zero. Thus we are led to the tentative definition:

Two point magnitudes or two linear magnitudes combined give zero if and only if one of them is zero or they both have the same orientation.

With this method of notation one is now in a position to represent any linear dependence in the form of an equation,



one side of which is zero, while the other includes no other conjunctions than those designated above. Thus for example

$$ab = 0$$

signifies that the points  $a$  and  $b$  coincide; and further

$$(ab)(cd)e = 0$$

that  $e$  is the point of intersection of the two lines  $ab$  and  $cd$ . For  $(ab)(cd)$  expresses the point of intersection of the lines  $ab$  and  $cd$ , and the equation  $(ab)(cd)e = 0$  expresses that  $e$  coincides with this point. For any nonzero point magnitude or linear magnitude in such equations one can also substitute another of the same orientation, that is, that yields zero with its combination. It is also clear that one can now express the principle of collineation by means of the concept of these combination equations, so that each combination equation that obtains between the points of one system also obtains in the same way between the corresponding points of the collinear system.

The combination of a point magnitude with a linear magnitude remains to be considered; for this we have only to require, by analogy with the earlier result, that it is zero if and only if either one of the terms of the conjunction is zero or the point lies on the line.

How fruitful this simple idea is, and to what unexpected disclosures about the nature of curves it leads, Grassmann has shown in his article in *Crelle's Journal fuer Math.*, vol. 31. In addition one is thereby already in a position to formulate an equation between ten points in the plane, five of which correspond collinearly to the remaining five; for one need only convert the known construction of the fifth corresponding point by means of a straightedge into the method of notation introduced in order to arrive at this equation. It would however appear in a very complicated form and would by no means be appropriate to represent the essence of collineation directly.

## 0.4 § 4. Addition of Point Magnitudes and of Linear Magnitudes

In order to obtain such an equation as will in fact achieve this we must first extend the concept of combination to take the measure into consideration. In this the analogy with multiplication, which springs to the eye, can guide us. This analogy between multiplication and the combination of point magnitudes and linear magnitudes is indicated by the previously noted feature of it, that if one of the conjunctive terms of the combination is zero, its result is also, and that, if the result is zero it remains zero if one lets the measure of that conjunctive term which is not zero enlarge or reduce in any proportion. Thus we extend this analogy yet further by requiring that, if  $\alpha$  and  $\beta$  are numerical magnitudes and  $A$  and  $B$  are point magnitudes or linear magnitudes, then

$$(\alpha A)(\beta B) = \alpha\beta(AB).$$

Thus if, for example,  $A$  and  $B$  are points, then the combination of the point magnitudes is equal to the combination of the points multiplied by the product of the coefficients.

The essential significance of multiplication lies in its relation to addition (cf. Grassmann's *Ausdehnungslehre*, § 10), in that instead of multiplying a sum one can, without changing the product, multiply its terms individually in the same way and add these products. Thus we have to investigate whether we can realize the concept of the addition of point magnitudes and of linear magnitudes, and thus also the definition of the measure of the combination now interpreted as a product, so that that relation of multiplication to addition and subtraction retains its validity.

We assume that the sum of two point magnitudes is again a point magnitude, and consider first the sum of two points  $a + b$ . Since according to the concept of the sum one has  $a + b = b + a$  and the sum must have a definite magnitude, it follows, since the summation point must have the same orientation relative to  $a$  and  $b$  and its orientation must be

definite, that its position can only be the mean between  $a$  and  $b$ .

If  $s$  is the mean between  $a$  and  $b$ , and  $x$  the coefficient of the sum, then

$$xs = a + b. \quad (7)$$

Now if the multiplicative relation is to hold, then necessarily

$$xss = sa + sb;$$

thus, since  $ss$  is zero (as the combination of coincident points), one has

$$0 = sa + sb \quad \text{or} \quad sa = -sb,$$

that is,  $sa$  and  $sb$  (which are two equally long but oppositely directed displacements) must be interpreted as equal but opposite magnitudes in the analysis under discussion. Furthermore, upon multiplication by  $a$  one has

$$xas = aa + ab,$$

since  $as$  is zero. Now  $ab$  is twice as long and directed the same as  $as$ . If I designate twice  $as$  by  $2as$ , provided it lies on the same line in the same direction, then  $ab = 2as$ , and thus

$$xas = 2as;$$

therefore  $x = 2$  and consequently

$$a + b = 2s, \quad (8)$$

that is, the sum of two points is their mean with the coefficient 2.

Now if I consider any given point  $r$ , then if the relation of multiplication to addition is to be valid, it must be that

$$ra + rb = r(a + b) = 2rs. \quad (9)$$

Thereby it follows immediately that the sum of  $ra$  and  $rb$  is the diagonal of the parallelogram  $arbd$ , and that, if  $ra$  and  $rb$  have the same direction, the sum is as great as the two terms

taken together, and thus the sum is the same as if one had added the two displacements as numerical magnitudes.

That is, if  $a, b, c$  are points on the same straight line, then invariably

$$ab + bc = ac.$$

It therefore also follows that  $\alpha \cdot ab$ , where  $\alpha$  is a number, is interpreted as a linear magnitude which has the same orientation but is  $\alpha$  times as great as  $ab$ ; and thus that the displacement  $ab$  represents precisely the measure of the combination  $ab$ , while the line  $ab$ , as infinite, represents its orientation.

Now the sum  $\alpha a + \beta b$ , where  $\alpha$  and  $\beta$  are numbers and  $a$  and  $b$  points, can easily be ascertained, provided  $\alpha + \beta$  is not zero. Thus let

$$\alpha a + \beta b = xs;$$

then for any point  $r$  it is necessary that

$$r(\alpha a + \beta b) = xrs,$$

that is

$$\alpha ra + \beta rb = xrs, \quad (10)$$

and also in the case that  $r$  coincides with  $s$ , so that  $rs$  is zero. Then one has

$$\alpha sa + \beta sb = 0, \quad (11)$$

that is,  $s$  is the center of gravity between the points  $a$  and  $b$ , if to these are attributed weights proportional to the numbers  $\alpha$  and  $\beta$ . Furthermore, if  $r$  coincides with  $a$  in equation (10), that is if  $ra$  is zero, then one has

$$\beta ab = xas,$$

and if  $r$  equals  $b$  in equation (10) so that  $rb$  is zero,

$$\alpha ba = xbs.$$

Thus upon subtracting the second equation from the first,

$$\beta ab - \alpha ba = x(as - bs)$$

or, since  $-ba = ab$ ,  $-bs = sb$  and  $as + sb = ab$ ,

$$(\beta + \alpha) \cdot ab = xab,$$

thus  $x = \alpha + \beta$ , and therefore

$$\alpha a + \beta b = (\alpha + \beta)s. \quad (12)$$

Consequently the sum of two point magnitudes, the sum of whose coefficients is not zero, is the center of gravity with a coefficient that is the sum of those coefficients.

Now just how one can establish that this conjunction between points or point magnitudes, symbolized here as addition, is such in fact, how there emerges from it a corresponding subtraction which, as the difference of two points, appears as a displacement of fixed direction and length,<sup>20</sup> and how an addition and subtraction for such displacements follows from this, all this cannot be treated here and may be supposed already known from the works and articles of Moebius and Grassmann, and that it is clear that one can proceed in the manner adopted here to obtain these methods and laws. Even so, I will develop these laws at least for points and their multiples below.

On these same grounds I cannot deal here with showing that the conjunction here designated as combination can be unified with algebraic multiplication in a single concept, whence I retain the name outer or combinatorial multiplication for the former, nor that a multiplication of displacements (as lines of fixed direction and length) and of point magnitudes with displacements can be developed, nor finally that linear magnitudes can be combinatorially multiplied with each other and with point magnitudes, and thence as well follow

---

<sup>20</sup>In order to forestall misunderstandings I remark here that the linear magnitude  $ab$  and the displacement  $b - a$  have related but essentially different meanings, since the equation  $b - a = d - c$  expresses only that the line drawn from  $a$  to  $b$  is as long as and directed the same as that drawn from  $c$  to  $d$ , whereas the equation  $ab = cd$  not only expresses this, but at the same time that  $ab$  and  $cd$  are parts of the same infinite straight line.

corresponding laws for the multiplication of surfaces with each other and with displacements. With respect to all of this I must refer to Grassmann's *Ausdehnungslehre*.

Thus I proceed, but only after I have given the application to collineation, to the development of the really new methods of calculation, in which however I must assume the results obtained in the papers cited.

## 0.5 § 5. The Collinear Functions

We were seeking the form of the collinear function above. Now it follows, if

$$\text{collin}(a, b, c, d, e) = \text{collin}(a', b', c', d', e')$$

and by  $abc$  is understood the area of the triangle whose vertices are  $a$ ,  $b$ , and  $c$ , and is assumed to be positive if  $c$  lies to the left of  $ab$ , that (*Ausdehnungslehre* § 165) then

$$\frac{eab}{dab} \cdot \frac{ebc}{dbc} = \frac{e'a'b'}{d'a'b'} \cdot \frac{e'b'c'}{d'b'c'}.$$

Yet the point  $e'$  is still not specified by a single equation such as the above, and thus the collinear function is not completely expressed; however two such equations, that is

$$\left( \frac{eab}{dab} \cdot \frac{ebc}{dbc} \cdot \frac{eca}{dca} \right) = \left( \frac{e'a'b'}{d'a'b'} \cdot \frac{e'b'c'}{d'b'c'} \cdot \frac{e'c'a'}{d'c'a'} \right)$$

do suffice, where the parentheses on both sides expresses the equality of the two corresponding ratios joined by the symbol “:”, and we can therefore regard

$$\left( \frac{eab}{dab} \cdot \frac{ebc}{dbc} \cdot \frac{eca}{dca} \right) \quad (13)$$

as the required collinear function for the plane.

Moreover, in space, provided one understands by  $abcd$  the volume of the tetrahedron whose vertices are  $a$ ,  $b$ ,  $c$ ,  $d$ , one would find from

$$\text{collin}(a, b, c, d, e, f) = \text{collin}(a', b', c', d', e', f')$$

that

$$\left( \frac{fabc}{eabc} \cdot \frac{fbcd}{ebcd} \cdot \frac{fcda}{ecda} \cdot \frac{fdab}{edab} \right) \quad (14)$$

equals the corresponding expression in  $a', b', c', d', e', f'$ , in which the equality of the three ratios is thus expressed, and expression (14) would itself represent the collinear function for space.

## 0.6 § 6. Return to Congruence. Equations Between Spatial Magnitudes

I now once again connect the derivation of the new method of calculation to Leibniz's idea by proceeding directly from collineation, the simplest relation on the one hand, to the simplest relation on the other, congruence, from which Leibniz originally proceeded; and with conjunctions previously derived from, or rather shown to be derivable from, collineation, and which are now assumed to be known, it will be possible to treat congruence as well in a corresponding way.

The principle of congruence is not in fact derivable from the methods of conjunction considered previously, since they relate only to drawing straight lines through, or laying planes upon, given points, but with them nothing is assumed in the nature of circles or spheres, as is the case with

# **A New Branch of Mathematics**

The Ausdehnungslehre of 1844, and Other  
Works

Hermann Grassmann

**Geometric Analysis**

Part Two — Continued

Pages 351–400

(Chunk 8 of 12)





## § 6. Continuation: Congruence and Equal Lengths

congruence. Furthermore, since the published volume of Grassmann's *Ausdehnungslehre* extends only to such methods of conjunction as depend solely on the concepts of the straight line and the plane, we find in that work no further developments relevant to the methods of conjunction elucidated in the following, except for a few remarks in the forward. For that reason, to allow nothing to be omitted in respect of precision, I will from now on choose that way whereby, after the derivation of a concept, I always restate this concept in the form of a definition, and develop the laws that arise thereby in rigorous mathematical form, without assuming any theorem provisionally employing the derivation of the concept.

The simplest case of congruence, to which every other can be reduced, was that in which two equally long lines were said to be congruent, thus

$$\text{fig.}(a, b) = \text{fig.}(c, d) \quad (1.1)$$

if the line segment  $ab$  was equally as long as the line segment  $cd$ . Now we have learned about equal linear magnitudes and equal displacements. The linear magnitudes  $ab$  and  $cd$  were said to be equal if and only if both lie on the same straight line and in the same (not opposite) direction, reckoned from the initial to the final point, and have equal lengths. On the other hand we interpreted a displacement as the difference between two points, and  $a - b$  and  $c - d$  must then be regarded as equal if and only if both displacements (that from  $b$  to  $a$  and that from  $d$  to  $c$ ) have the same direction and length. From this it follows that if two bounded lines, for example that from  $a$  to  $b$  and that from  $c$  to  $d$ , are equal as linear magnitudes, that is  $ab = cd$ , they must also be equal as displacements, thus  $b - a$  must be equal to  $d - c$ , but not conversely, since if  $b - a$  is equal to  $d - c$  then  $ab$  will also equal  $cd$  only if  $a, b, c, d$  lie on the same line. Further it follows that, if those bounded lines are equal as displacements, that is if  $b - a$  equals  $d - c$ , they would also be congruent or equally long, but not conversely, since the equality of the displacements

follows from their equal lengths only if their directions are taken to be equal.

Congruence is therefore included in the equality of displacements, and since if

$$a - b = c - d$$

then always

$$\text{fig.}(a, b) = \text{fig.}(c, d)$$

as well, we regard  $\text{fig.}(a, b)$ , which represents the length of the straight line drawn from  $a$  to  $b$ , as a function of  $a$  and  $b$ , and thus if  $f$  is the symbol for this function we can instead of  $\text{fig.}(a, b)$  write  $f(a - b)$ , and instead of equation (15) obtain the equation

$$f(a - b) = f(c - d). \quad (1.2)$$

Thus we have the advantage of having interpreted length as a function of a specific magnitude.

Now we have to give this function such a form that equation (16) is always true if and only if  $a - b$  and  $c - d$  are equally long.

To find such a function we first assume that all the magnitudes considered lie on the same line. Consequently two equally long line segments can only have the same or opposite directions. In the former case they are equal as displacements; in the latter, one displacement is the negative of the other. If  $p$  denotes a displacement, then  $p$  is as long as  $(-p)$ ; but besides  $p$  and  $(-p)$  there is no displacement on the same straight line with length equal to that of  $p$ . It would thus follow, but only here, that  $f(p)$  must have such a form that  $f(p) = f(-p)$ . If one could treat displacements on a line as numbers, then  $p^2$  would be such a function, and in fact the simplest that would satisfy this condition. Thus we must next investigate whether and to what extent the laws of numerical conjunction can be applied to displacements on the same line, or generally whether they can be applied to magnitudes which, like displacements on the same straight line, can be regarded as proportional to a series of numerical magnitudes.

**Definition** (1). I regard a series of spatial magnitudes  $A, B, \dots$  as proportional to a series of numerical magnitudes  $\alpha, \beta, \dots$  if there is some nonzero spatial magnitude  $M$  of such a nature that  $A = \alpha M, B = \beta M, \dots$  and so forth,<sup>1</sup> and in this case call spatial magnitudes *similar* with respect to it; if in addition two series of spatial magnitudes are proportional to the same series of numerical magnitudes, then I call them these *mutually proportional*.

Thereby follows immediately that

**Theorem** (1). *If a series of spatial magnitudes  $A, B, \dots$  is proportional to a series of numbers  $\alpha, \beta, \dots$ , and these are again proportional to another series of numbers, then the series of spatial magnitudes is also proportional to the latter series of numbers.*

For each series of numbers proportional to a series of numbers  $\alpha, \beta, \dots$  can be represented in the form  $\rho\alpha, \rho\beta, \dots$ , where  $\rho$  can be any nonzero numerical magnitude; now if one sets  $M' = \rho M$ , then

$$A = \alpha\rho M', \quad B = \beta\rho M', \quad \dots,$$

that is,  $A, B, \dots$  are also in the same relation as  $\rho\alpha : \rho\beta : \dots$ , i.e. as any series of numbers proportional to  $\alpha, \beta, \dots$ .

Now if one had an algebraic equation in which the numerical magnitudes  $\alpha, \beta, \dots$  appear, and whose nature is such that it also applies if one replaces the series of numbers  $\alpha, \beta, \dots$  with a series of numbers proportional to it, then one could even regard the equation as correct if one substituted for the series of numbers  $\alpha, \beta, \dots$  a series of spatial magnitudes  $A, B, \dots$  proportional to it, since one has then only established that this latter equation has no other significance than the former.

---

<sup>1</sup>If I symbolize a magnitude as the product of a numerical magnitude  $\alpha$  with a spatial magnitude  $A$ , it is thereby expressed that for this multiplication the laws of multiplication and division by numbers are valid, and in particular that  $\beta(\alpha A) = (\beta\alpha)A$  and  $\frac{\beta A}{\alpha} = \frac{\beta}{\alpha} \cdot A$ , which will be used in the following argument.

Such an equation as we have imagined has the form

$$F(\alpha, \beta, \dots) = F'(\alpha, \beta, \dots), \quad (1.3)$$

where the symbols  $F$  and  $F'$  represent algebraic and homogeneous functions of the same degree, that is two functions whose terms are all of the form  $A\alpha^a\beta^b\dots$  with the same sum of exponents  $a + b + \dots$ , where  $A$  is an arbitrary numerical magnitude. If one defines the sum  $a + b + \dots = x$ , and replaces  $\alpha, \beta, \dots$  with the proportional series of numbers  $\rho\alpha, \rho\beta, \dots$ , where  $\rho$  represents a nonzero number, then  $F(\rho\alpha, \rho\beta, \dots)$  will be equal to  $\rho^x F(\alpha, \beta, \dots)$  and  $F'(\rho\alpha, \rho\beta, \dots) = \rho^x F'(\alpha, \beta, \dots)$ , and thus also

$$F(\rho\alpha, \rho\beta, \dots) = F'(\rho\alpha, \rho\beta, \dots). \quad (1.4)$$

If I now replace the series of numbers  $\alpha, \beta, \dots$  in equation (17) with the proportional series of spatial magnitudes  $A, B, \dots$ , then I call the resulting equation

$$F(A, B, \dots) = F'(A, B, \dots) \quad (1.5)$$

as well a homogeneous algebraic equation in the spatial magnitudes, and am thus led to the following definition:

**Definition** (2). I regard a homogeneous algebraic equation of spatial magnitudes that are similar, that is proportional to a series of numerical magnitudes, as true if and only if the equation that results if one replaces the spatial magnitudes with their respectively proportional numerical magnitudes is true; and I say of the resulting conjunctions of the spatial magnitudes that they correspond to the algebraic conjunctions of the numerical magnitudes.

From this follows the theorem:

**Theorem** (1B). *For similar spatial magnitudes all algebraic laws of conjunction are valid that can be represented by homogeneous equations of those magnitudes, and every homogeneous equation between similar spatial magnitudes is also valid for spatial magnitudes proportional to them.*

I have formulated this theorem (which I need here to apply only to displacements within the same straight line) so generally because it is of primary importance for the transferral of algebraic conjunctions to spatial magnitudes.

## § 1.1. Inner Products of Displacements

Returning now to the equation  $f(p) = f(-p)$ , it is clear that this would be satisfied most easily if one sets  $f(p) = p^2$ , since  $p^2 = (-p)^2$ , even when  $p$  represents a displacement. For just as  $p$  is in the same relation to  $-p$  as the numbers  $1 : (-1)$  (Definition 1), so, according to Definition 2,  $p^2$  is to be regarded as equal to  $(-p)^2$  if, as is the case,  $1^2 = (-1)^2$ .

Thus if in general  $p$  and  $q$  are equally long displacements we can also try regarding the square of  $p$  as equal to that of  $q$ . But it still remains doubtful whether we can view this square as always corresponding to that of arithmetic, or indeed generally whether, for the multiplication by which the two equal factors of this square are conjoined, any of the laws of multiplication hold once one compares it with squares of differently directed displacements; that is, whether we have formulated a natural hypothesis in regarding the function  $f(p)$  generally as the square of  $p$ . In any case we must at least temporarily designate the multiplication for which we formulated that hypothesis by a special name. We call it *inner multiplication*, and the inner product of two equal displacements the *inner square* of this displacement. Thus we are led to the following definition:

**Definition (3).** By inner products of any two parallel displacements I mean such magnitudes as are regarded as proportional to the numbers that result if one measures the two parallel displacements of one of those inner products by the same unit and the quotients of these two measures are multiplied together, all units however assumed to be equally long. The inner product of two displacements is symbolized by  $a \times b$ , the inner square  $a \times a$  by  $a^2$ .

Thus for example (cf. fig. 4)  $a \times b$  and  $c \times d$  are proportional

to the numbers resulting if one measures  $a$  and  $b$  by  $r_1$ ,  $c$  and  $d$  by  $r_2$ , where  $r_1$  and  $r_2$  are equally long and  $r_1$  is parallel to  $a$  and  $b$ ,  $r_2$  to  $c$  and  $d$ . From the definition it follows at once that  $a \times b = b \times a$  and

$$a \times (b + c) = a \times b + a \times c$$

if  $a$ ,  $b$ ,  $c$  are parallel displacements, in which case of course the equations also have the same meaning as those defined in Definition 2.

It is thus now a question of finding the inner product of two nonparallel displacements.

One arrives at the concept of such a product if one tries to establish the general relation of multiplication to addition and to regard the factors as interchangeable as well, so that the multiplication of nonparallel and parallel displacements fall within a common concept. If I tentatively impose these two assumptions in order to proceed to the concept of this product, it follows that

$$(a + b) \times (a + b) = a \times a + b \times b + b \times a + a \times b,$$

whence

$$(a + b)^2 = a^2 + 2a \times b + b^2. \quad (1.6)$$

If the displacements appear as the difference between two points, that is if (cf. fig. 5)

$$a = B - A, \quad b = C - B,$$

then

$$a + b = B - A + C - B = C - A.$$

Thus if  $a$  and  $b$  are at right angles to each other, then  $a + b$  is the hypotenuse of a right triangle whose sides are  $a$  and  $b$ , whence

$$(a + b)^2 = a^2 + b^2. \quad (1.7)$$

If we compare this equation with (20) above, it follows that  $2a \times b = 0$ , and thus also  $a \times b = 0$ , that is the inner product of two perpendicular displacements must be regarded as zero.

The concept of the inner product of two arbitrary displacements  $a$  and  $b$  can be derived from this.

Thus let (cf. fig. 6)

$$a = A - E, \quad b = B - E.$$

If one drops a perpendicular from  $B$  onto  $EA$ , with its foot at  $C$ , then

$$B - E = B - C + C - E.$$

Thus also

$$\begin{aligned} a \times b &= (A - E) \times (B - E) = (A - E) \times (B - C + C - E) \\ &= (A - E) \times (B - C) + (A - E) \times (C - E). \end{aligned}$$

The former of the last two products is zero since  $A - E$  is perpendicular to  $B - C$ , whence

$$a \times b = a \times (C - E);$$

but  $C - E$  is the orthogonal projection of  $b$  on  $a$ . Consequently there follows the definition:

**Definition (4).** By the inner product  $a \times b$  of two nonparallel displacements  $a$  and  $b$  will be meant the inner product of the first,  $a$ , with the orthogonal projection of the second on the first.

I now have to develop the laws of inner multiplication out of this assertion without assuming auxiliary theorems taken from the derivation of the concept,<sup>2</sup> and in fact first:

---

<sup>2</sup>It is extremely important that the following mathematical demonstration be strictly distinguished from the previous (more philosophical) derivation of the concept. That derivation served only to dispel the appearance of free choice that would have resulted from the immediate introduction of the new definition, while the following demonstration is completely independent of that derivation and follows at once from the definition stated. Thus for example the concept of the Pythagorean theorem was used in the derivation; however the mathematical demonstration does not assume this theorem, on the contrary it appears as a consequence of the mathematical development presented here, which therefore also includes a complete proof of that theorem.



**Theorem (2).** *One can interchange the factors of inner products of two displacements without altering the value of the product, that is  $a \times b = b \times a$ .*

For, if  $a = A - E$  (cf. fig. 7),  $b = B - E$ ,  $C$  is the foot of the perpendicular dropped from  $B$  onto  $AE$  and  $D$  the foot of that dropped from  $A$  onto  $BE$ , then

$$a \times b = (A - E) \times (C - E),$$

$$b \times a = (B - E) \times (D - E).$$

Because of the similarity of the triangles  $ADE$  and  $BCE$  it now follows that the length of  $AE$  is to that of  $DE$  as the length of  $BE$  is to that of  $CE$ , whence also the product of the lengths of  $AE$  and  $CE$  is equal to the product of the lengths of  $DE$  and  $BE$ , and in addition, if one measures  $A - E$  and  $C - E$  by a unit on the line  $EA$  and  $D - E$  and  $B - E$  by an equally long unit on the line  $EB$ , the product of the quotients of the two former measures equals the product of the quotients of the two latter measures, that is, according to Definition 3,

$$(A - E) \times (C - E) = (B - E) \times (D - E),$$

that is

$$a \times b = b \times a.$$

In addition it follows that

**Theorem (3).** *An inner product of two mutually perpendicular displacements is zero, but not that of any others whose factors are not themselves zero.*

For if two displacements are mutually perpendicular, the orthogonal projection of one on the other is zero, whence, according to Definition 4, if one replaces the second factor with its orthogonal projection on the first, this second factor is zero, and thus the product is also zero. On the other hand, in any other case the projection will not be zero (unless perhaps the displacement itself is zero), and thus the product is also not zero.

Furthermore it can be shown that

$$\begin{aligned} a \times (b + c) &= a \times b + a \times c \\ (b + c) \times a &= b \times a + c \times a. \end{aligned} \tag{1.8}$$

For, if (cf. fig. 8)

$$a = A - E, \quad b = B - E, \quad c = C - B,$$

then  $b + c = C - E$ . If in addition  $B', C'$  are the feet of the perpendiculars dropped from  $B$  and  $C$  on  $EA$ , then  $B' - E$  is the orthogonal projection of  $b$  on  $a$ , while  $C' - E$  is that of  $b + c$  on  $a$ , and  $C' - B'$  that of  $c$  on  $a$ ; thus

$$a \times b + a \times c = a \times (B' - E) + a \times (C' - B'),$$

and since all the displacements on the right lie on a straight line, the right side

$$= a \times (B' - E + C' - B') = a \times (C' - E) = a \times (b + c).$$

Finally, since the factors are interchangeable it also follows that

$$(b + c) \times a = b \times a + c \times a.$$

Equations (22) are thus proved. But then there follows the general theorem

**Theorem (4).** *All algebraic theorems that express the relation of multiplication to addition and subtraction are also valid for the inner multiplication of two displacements, since this relation depends exactly upon the fundamental theorem just proved (cf. Grassmann's Ausdehnungslehre § 10).*

The terminology and notation for this method of conjunction as a multiplication is thus justified, but at the same time, since the law that the inner product of two mutually perpendicular displacements is zero finds nothing comparable in algebra, this method of multiplication proves to be essentially different from one of the algebraic type, wherefore we retain the name *inner*

*multiplication* to distinguish it both from algebraic and from *outer multiplication*, and use the symbol  $\times$  as the symbol for this multiplication. Thus it follows that if  $a, b, c, d$  are points we not only replace the equation

$$\text{fig.}(a, b) = \text{fig.}(c, d)$$

or

$$ab = cd$$

by the equation

$$(a - b) \times (a - b) = (c - d) \times (c - d)$$

or

$$(a - b)^2 = (c - d)^2,$$

but now all previous theorems apply for this multiplication as well, and thus we can operate completely freely with this multiplication just as with algebraic multiplication.

## § 1.2. Inner and Outer Multiplication

The problem we originally set ourselves, that is to reduce the Leibnizian characteristic to its natural expression, is thus solved. At the same time, however, there is implicit in the nature of the solution the seeds of further development in two different directions.

Thus in the first instance there arises the problem of deriving from the product of two point differences that of the points themselves, so that by dropping the parentheses in which the point differences are still enclosed to arrive at a freer and more general treatment of this conjunction that represents congruence.

The other seed of further development lies in the relation of the inner product to the outer. This relation is mediated by the idea of “orthogonal proportionality.”<sup>3</sup>

Thus,

---

<sup>3</sup>Grassmann uses the term “orthogonal proportionality” (in German: “orthogonale Proportionalität”).

**Definition (5).** I call two surface areas  $A$  and  $B$  orthogonally proportional to two displacements  $a$  and  $b$  if the surface areas are orthogonal to the displacements and have such values that, if one repositions one displacement,  $b$ , and its corresponding surface area,  $B$ , without changing their relative orientations, so that  $b$  is directed as is  $a$ , and the displacement  $b$  repositioned thus is then measured by  $a$ , the surface area  $B$  repositioned thus measured by  $A$ , the quotients of the two measures are equal.

From this follows the theorem:

**Theorem (5).** *Two inner products of any two displacements  $a \times b$  and  $c \times d$  stand in the same relation as the outer products that result if in the two products, one replaces two chosen factors, for example  $a$  and  $c$ , with their orthogonally proportional surface areas  $A$  and  $C$  and changes the symbol for inner multiplication to that for outer multiplication, thus*

$$(a \times b) : (c \times d) = (A \cdot b) : (C \cdot d) \quad (1.9)$$

For, if I regard  $c$ ,  $d$ ,  $C$  as a system and arbitrarily reposition this system while keeping it congruent to its original form, then the products  $c \times d$  and  $C \cdot d$  remain unchanged under this repositioning; consequently their values are also unchanged if thereby  $c$  is repositioned into the same direction as  $a$  has. It therefore follows that equation (23) will be proved true in general if only I have proved it for the case that  $a$  and  $c$  are similarly directed.

Under this assumption, if  $c = \gamma a$ , then (according to Definition 5)  $C = \gamma A$  as well. Thus in this case, if I substitute  $\gamma a$  for  $c$  and  $\gamma A$  for  $C$  in equation (23), then it remains only to be shown that

$$(a \times b) : (\gamma a \times d) = (A \cdot b) : (\gamma A \cdot d).$$

This will be proved once I have shown that

$$(a \times b) : (a \times d) = (A \cdot b) : (A \cdot d), \quad (1.10)$$

or that, if  $a \times b = \alpha(a \times d)$ , where again  $\alpha$  is a numerical magnitude,  $A \cdot b = \alpha(A \cdot d)$  as well. But if  $(a \times b) = \alpha(a \times d)$ , then

it is also  $= a \times (\alpha d)$ , and thus  $a \times (b - \alpha d) = 0$ , that is  $b - \alpha d$  is either zero or perpendicular to  $a$  (according to Theorem 3), or, since  $A$  is a surface area orthogonal to  $a$ ,  $b - \alpha d$  is either zero or parallel to  $A$ . Now the outer product of a surface area with a displacement parallel to it gives zero (Ausdehnungslehre §§ 54, 55); thus  $A \cdot (b - \alpha d)$  is zero and so  $A \cdot b$  equals  $A \cdot \alpha d$ , which equals  $\alpha(A \cdot d)$ , proving equation (24) and therefore also (23) for the case that  $a$  and  $c$  are similarly directed, and thus generally.

We can express the above theorem even more generally by saying:

**Theorem (5a).** *In an equation whose terms are inner products of any two displacements or multiples of such products, one can, without giving up the validity of the equation, replace these inner products with those outer products proportional to them that one obtains upon replacing any one of the factors of those inner products with the orthogonally proportional surface area and replacing the symbol for inner multiplication with that for outer, and conversely one can derive the former equation from the latter.*

For, since the inner products are proportional to a series of numerical magnitudes and the outer products proportional to them are proportional to the same series, the theorem stated follows upon application of Definition 2.

Herein lies the previously indicated seed of further development, since one can transpose the idea of orthogonal proportionality to other outer products as well. I will now present this development, and then, before I pursue the other series of developments indicated above, show by applications to geometry and mechanics to what extent the analysis developed thus far possesses the advantages Leibniz attributed to such a one.

### § 1.3. Inner Products of Surface Areas and Displacements

It is clear that by the idea of orthogonal proportionality one can find an inner product corresponding in a new way to both the

outer product of two displacements and also to that of a surface area with a displacement, since one can replace the displacement with a surface area remaining proportional to it. For the derivation of the laws of these new conjunctions, and also in support of the laws of the previously considered inner product, if one converts Theorem 5 into a definition of this inner product, the following theorem suffices:

**Theorem (6).** *If  $a, b, c, \dots$  are displacements and  $A, B, C, \dots$  their orthogonally proportional surface areas, then their sums also stand in the same proportional sequence, that is if  $s = a + b + c + \dots$  and  $S = A + B + C + \dots$ , then also  $s, a, b, c, \dots$  are orthogonally proportional to  $S, A, B, C, \dots$ .*

Once this has been proved for the sum of two displacements, it is immediately clear by continuing the same argument that it is also valid for arbitrarily many; consequently we prove it at first for only two displacements, and thus set  $s = a + b$ ,  $S = A + B$ , and show that  $s, a, b$  are orthogonally proportional to  $S, A, B$ .

If  $f$  is simultaneously perpendicular to  $a$  and  $b$ , then  $f$  is also perpendicular to  $s$ , since  $s$  lies in the same plane as  $a$  and  $b$ . Intuitively one thinks of  $a, b, s$ , and  $f$  as proceeding from the same point. Now if, without altering their lengths and relative orientations, I rotate the system of the three displacements  $a, b, s$  about the perpendicular axis  $f$  until the system (and thus each displacement in it) has described a right angle, and thus transform  $a, b, s$  into  $a', b', s'$ , then obviously it is still true that  $s' = a' + b'$ . But now it is clear that  $fa', fb', fs'$  are orthogonally proportional to  $a, b$ , and  $s$ ; therefore, if  $A = \gamma fa'$ , then also  $B = \gamma fb'$ , whence  $A + B$  or  $S = \gamma(fa' + fb') = \gamma fs'$ , that is,  $S, A, B$  are orthogonally proportional to  $s, a, b$ , and thus generally for arbitrarily many terms.

Moreover, since three terms of such an orthogonal proportion determine the fourth, it follows conversely that

**Theorem (7).** *If  $s, a, b, c, \dots$  are orthogonally proportional to  $S, A, B, C, \dots$ , and*

$$s = a + b + c + \dots$$

then also

$$S = A + B + C + \dots,$$

and conversely, under the same assumption the former equation is valid if the latter is.

We will however only use the former theorem in the following development, and that even only for two terms.

The following concepts now result from the idea of orthogonal proportionality:

**Definition (6).** By inner products of two surface areas I understand such magnitudes as are proportional to the outer products one obtains if in all such inner products one replaces the first factors by their orthogonally proportional displacements while leaving the second factors unchanged, but exchanges the symbol for inner multiplication ( $\times$ ) for that of outer ( $\cdot$ ), and

**Definition (7).** By the inner products of a surface area with a displacement I understand such magnitudes as are orthogonally proportional to the outer products that result if one replaces the surface areas appearing as factors of these inner products by their orthogonally proportional displacements, and, without changing the other factor, exchanges the symbol of inner multiplication with that of outer.

Thus for example the products  $A \times b$  and  $C \times d$ , in which  $A$  and  $C$  are surface areas,  $b$  and  $d$  displacements, are orthogonally proportional to the surface areas  $a \cdot b$  and  $c \cdot d$  if  $a$  and  $c$  are orthogonally proportional to the surface areas  $A$  and  $C$ .

The laws to which these conjunctions are subject can very easily be derived by means of Theorem 6.

Thus one finds

$$A \times (B + C) = A \times B + A \times C,$$

since  $a \cdot (B + C) = a \cdot B + a \cdot C$ , and if  $a$  is orthogonal to  $A$  then the terms in the equation to be proved are proportional to those in this one, which is thus subject to the same equation (according to Theorem 1B). Likewise one finds

$$(B + C) \times A = B \times A + C \times A,$$

since, if  $b$  and  $c$  are orthogonally proportional to  $B$  and  $C$ , then also (by Theorem 6)  $b$ ,  $c$  and  $(b + c)$  are to  $B$ ,  $C$  and  $(B + C)$ , so

$$(b + c) \cdot A = b \cdot A + c \cdot A,$$

whence the same equation must also obtain between the magnitudes proportional to the terms of this equation, thus

$$(B + C) \times A = B \times A + C \times A.$$

Similarly it follows that

$$A \times (b + c) = A \times b + A \times c.$$

For

$$a \cdot (b + c) = a \cdot b + a \cdot c;$$

thus by Theorem 7 the same equation also holds for the magnitudes orthogonally proportional to its terms, that is

$$A \times (b + c) = A \times b + A \times c.$$

Finally, it also follows that

$$(A + B) \times c = A \times c + B \times c.$$

For let  $a$  and  $b$  be displacements orthogonally proportional to the surface areas  $A$  and  $B$ ; then (Theorem 6)  $a$ ,  $b$ ,  $(a + b)$  are orthogonally proportional to  $A$ ,  $B$ ,  $(A + B)$ . Now

$$(a + b) \cdot c = a \cdot c + b \cdot c.$$

Thus (Theorem 7) the same equation is also valid for the magnitudes orthogonally proportional to its terms. But according to Definition 7

$$(A + B) \times c, \quad A \times c, \quad B \times c$$

are orthogonally proportional to

$$(a + b) \cdot c, \quad a \cdot c, \quad b \cdot c,$$



since  $A + B$ ,  $A$ ,  $B$  are orthogonally proportional to  $a + b$ ,  $a$ ,  $b$ . Thus (according to Theorem 7)

$$(A + B) \times c = A \times c + B \times c.$$

In order to explain the idea of multiplication in the last case of multiplication of a surface area by a displacement  $b$  even more intuitively, I will assume  $b_1$  is the orthogonal projection of  $b$  onto the surface area  $A$ , so that  $b$  equals  $b_1 + b_2$ , where  $b_2$  is orthogonal with respect to the surface area  $A$ . Then if  $a$  is a displacement orthogonal with respect to  $A$ ,

$$a \cdot (b_1 + b_2) = a \cdot b_1,$$

since  $a \cdot b_2$  is zero as the outer product of two similar displacements. Thereby also

$$A \times (b_1 + b_2) = A \times b_1,$$

or

$$A \times b = A \times b_1,$$

and  $A \times b_1$  is to be set orthogonally proportional to the product  $a \cdot b_1$ . It is however clear that the displacements orthogonal to  $a \cdot b_1$  in the plane  $A$  lie perpendicular to the projection  $b_1$ , and thus also to  $b$  itself; thus it follows that the product  $A \times b$  is represented by a displacement lying perpendicular to  $b$  in  $A$ , whose relative magnitude is specified uniquely by its being orthogonally proportional to the surface area  $a \cdot b_1$ .

From the relation to addition shown above for these two new types of multiplication follows the general theorem:

**Theorem (8).** *For each type of inner multiplication of two extensions in space all algebraic theorems hold that express the general relation of multiplication to addition and subtraction.*

This theorem includes the earlier Theorem 4 as a special case. In addition it follows at once that:

**Theorem (9).** *For all species of inner multiplication of two extensions in space the product is zero if and only if the factors are mutually orthogonal, since then and only then the corresponding outer product contains parallel factors and thus is zero; and again this theorem appears as a generalization of Theorem 3.*

Once more it also follows at once that:

**Theorem (10).** *Each inner product of nonparallel factors is equal to the inner product of one factor with the orthogonal projection of the other factor on the first, whereby naturally, if the product considered is a surface area with a displacement, only the projection of the latter on the plane of the former can be spoken of.*

Furthermore, it is easy to show that  $A \times B = B \times A$ ; for, if the orthogonal projection of  $B$  on  $A$  is  $\alpha A$ , where  $\alpha$  is a number, then

$$A \times B = A \times \alpha A = \alpha(A \times A)$$

and

$$B \times A = (\alpha A) \times A = \alpha(A \times A),$$

whence

$$A \times B = B \times A.$$

Since the product  $\alpha \times B$  is not yet defined, we can by analogy set it equal to  $B \times \alpha$ , whence we have the theorem:

**Theorem (11).** *The two factors of any given inner product can be interchanged without changing the value of the product,*

and it is clear that this theorem is a generalization of Theorem 2.

Thus it now appears that all laws for inner multiplication are in agreement. In particular the inner multiplications of displacements and of surface areas, that is of magnitudes of equal order, are so completely parallel that there is no theorem proved for one of these conjunctions that is not also valid for the other, if only one interchanges the concepts of displacement and surface area. And thus in fact the two types of product can be derived from each other, since one can replace both factors with their orthogonally proportional magnitudes.

## § 1.4. Geometric Applications

I now proceed to applications, and first of all to those in geometry, focussing only on the inner multiplication of displacements and of surface areas.

Since  $a + b$  represents the third side of a triangle whose other two sides are  $a$  and  $b$  if one lays it from the initial point of  $b$  to the final point of  $a$ , there results from the formula

$$(a + b)^2 = a^2 + 2a \times b + b^2$$

the following theorem, which I present simultaneously with its correspondent for surface areas:

The square of one side of a triangle is as great as the sum of the squares of the two others plus twice their inner product if these two sides are taken progressively.<sup>a</sup>

In particular, since the inner product of perpendicular displacements is zero, the square of the hypotenuse of a right triangle is the sum of the squares of the other two sides.

If  $a, b, c$  are all three mutually perpendicular, then

$$(a + b + c)^2 = a^2 + b^2 + c^2, \quad (A + B + C)^2 = A^2 + B^2 + C^2,$$

that is, the square of a displacement equals the sum of the squares of its orthogonal projections on three mutually perpendicular lines.

Furthermore, in general

$$(a+b+c)^2 = a^2+b^2+c^2+2a \times b+2b \times c+2c \times a \quad (A+B+C)^2 = A^2+B^2+C^2+2A \times B+2B \times C+2C \times A$$

that is, if one projects a displacement onto three arbitrary axes parallel to the planes laid on the axes, then the square of the displacement is the sum of the squares of the projections plus twice the inner products of each pair of projections.

---

<sup>a</sup>That is, on each axis parallel to the plane of the other two; for then the displacement is the sum of the three projections (Ausdehnungslehre § 90).

The square of one face of a three-sided prism equals the sum of the squares of the other two faces plus twice their inner product if these faces are taken progressively.<sup>1</sup>

In particular, since the inner product of orthogonal surface areas is zero, the square of the hypotenuse face of a right triangular prism is the sum of the squares of its other two faces.

If  $A, B, C$  are all three mutually orthogonal, then

that is, the square of a surface area is the sum of the squares of its orthogonal projections on three mutually orthogonal planes.

Furthermore, in general

that is, if one projects a surface area onto three arbitrary planes parallel to the lines of intersection of the planes, then the square of the surface area is the sum of the squares of the projections plus twice the inner products of each pair of projections.

This theorem, which can be understood as a generalization

of the Pythagorean theorem, serves as well to express the length of a given displacement by its reference terms (coordinates), or the content of a given surface area by its reference terms.

The formula

$$(a + b) \times c = a \times c + b \times c,$$

where  $a, b, c$  lie in the same plane, furnishes the theorem:

If one lays a circular line through a vertex of a parallelogram, and draws the diagonal  $(a + b)$  from the same vertex, then the product of the diagonal with the chord lying on it is the sum of the products of the two sides  $a$  and  $b$  joined at that vertex with the chords lying on them, since these chords are the orthogonal projections of the diameter on the three lines.

If one lays a cylindrical surface through an edge of a parallelepiped, and lays the diagonal surface  $(A+B)$  through the same edge, then the product of the diagonal surface with that part of it cut off (by the cylinder) is the sum of the products of the two surface areas intersecting at that edge with the parts of them cut off by the cylinder, since these parts are proportional to the orthogonal projections of the diameter plane.

Since the parallel theorems for surface areas can always easily be derived from those for displacements, and the former offer less interest than the latter, from now on I will state only the latter ones.

From the formula

$$(a + b + c + \dots) \times p = a \times p + b \times p + c \times p + \dots,$$

where one thinks of  $a, b, c, \dots$  and  $p$  with their initial points coinciding,  $p$  projected orthogonally on them, and recognizes that the sum  $a + b + c + \dots$ , with  $A, B, C, \dots$  the final points of the displacements and  $R$  their common initial point, equals

$$A - R + B - R + \dots = n(S - R),$$

where  $S$  represents the final point of the center of gravity and  $n$  is the number of displacements  $a, b, c, \dots$ , results the theorem:

If one draws straight lines from any given point ( $R$ ) to  $n$  points ( $A, B, C, \dots$ ) and to their center of gravity ( $S$ ), and lays an arbitrary spherical surface through that first point ( $R$ ) (which intersects the lines  $RS, RA, RB, \dots$  in the points  $S', A', B', \dots$  the second time): then the product of the straight line ( $RS$ ) drawn to the center of mass ( $S$ ) with the chord ( $RS'$ ) lying on it is the arithmetic mean of the products of the lines ( $RA, RB, \dots$ ) drawn to the  $n$  points with the chords ( $RA', RB', \dots$ ) lying on them.

If one has an equation between inner products of any two displacements, then one can very easily transform this equation into numerical equations by the adoption of orthogonal coordinates. For if  $p_1 \times p_2$  is such an inner product, and if  $a_1, b_1, c_1$  are the reference terms (coordinates with directions retained of the axes on which they lie) of  $p_1$ , and  $a_2, b_2, c_2$  those of  $p_2$  with respect to the same system of axes, then

$$p_1 = a_1 + b_1 + c_1, \quad p_2 = a_2 + b_2 + c_2,$$

whence

$$p_1 \times p_2 = (a_1 + b_1 + c_1) \times (a_2 + b_2 + c_2) = a_1 \times a_2 + b_1 \times b_2 + c_1 \times c_2,$$

since according to Theorem 9 all other products are zero as inner products of perpendicular displacements. Now if one measures all reference terms on the same axis by the same standard, and assumes that the three standards for the three axes are equally long, say as long as the displacement  $e$ , and one designates the quotients of these measures by the corresponding Greek letters, then one obtains

$$\frac{p_1 \times p_2}{e^2} = \alpha_1 \alpha_2 + \beta_1 \beta_2 + \gamma_1 \gamma_2,$$

and thus upon dividing the whole equation by the square of the standard  $e$  one obtains a simple numerical equation.

## § 1.5. Applications to the Pure Theory of Motion

I now turn to applications in mechanics.

First I will consider a point in motion, without being concerned about the cause of its motion. With Möbius (*Mechanik des Himmels* § 9) and Grassmann (*Ausdehnungslehre* § 105) I assume that the concept of velocity includes direction, so that I only regard two velocities as equal if they not only have equal absolute values but also equal directions, that is, I regard the velocity of a point in motion as equal to that displacement it would describe if, in the time interval taken as the unit, it maintained that motion which it has.

Thus, if a point proceeds in the same direction so that in equal time intervals it always travels equal and equally directed paths, the paths thus being in the same ratio as the time intervals in which it travels them, then I say the point preserves its motion unchanged. Thus is established that if at any given instant  $T$  it has such a motion that, if it maintained unchanged the same motion during the time interval defined as the unit, it would proceed from the position  $p$  that it had at time  $T$  to the position  $p_1$ , then its velocity is  $p_1 - p$ .

I could now develop purely geometrically the procedure by which this velocity can be derived from the presumed motion of a point; but I would then have to return to the principles of the differential calculus in order to apply them to the analysis presented here. Since however this would not correspond to my present purpose, I will fall back on the usual concept of the differential, and to that end project the moving point  $p$  orthogonally onto an axis. Let the projection be  $p'$ . Then it is clear that if the paths that the projected point travels in equal intervals are equal and equally directed, so also the projections of these paths will be equal; thus the projection  $p'$  proceeds uniformly if  $p$  proceeds uniformly and rectilinearly, and since, if the projected point proceeds from the position  $p$  to  $p_1$ , the projection simultaneously proceeds from the position  $p'$  to the projection  $p'_1$  of  $p_1$ . Thus if  $p_1 - p$  is the velocity of the projected point, then  $p'_1 - p'$  is that of the projection, that is the velocity with which the orthogonal projection of a point onto a straight line proceeds is always equal to the projection onto this line of the velocity with which the point itself proceeds.

Now if we project a displacement orthogonally onto three mutually orthogonal axes, then the projected displacement is the sum of its three projections. Thus, if we project a moving point orthogonally onto three mutually orthogonal axes, the velocity of that point is the sum of the velocities of its projections.

We will now designate the displacement from the point of intersection of the three axes to the projection of the point  $p$  on one of the three axes by  $xa$ , where  $x$  is a numerical magnitude and  $a$  is a part of the axis (this part being treated as a displacement). Now it is known that the velocity of this projection, if  $x$  is a given function of the time  $t$ , is equal to  $\frac{dx}{dt}a$ . If we call the displacements from the point of intersection  $g$  of the three axes to the projections of the point  $p$  on the two other axes  $yb$  and  $zc$  (where  $a, b, c$  can be assumed to be equally long), then  $\frac{dy}{dt}b$  and  $\frac{dz}{dt}c$  are the velocities of the other two projections, whence the velocity of the point  $p$  equals

$$\frac{dx}{dt}a + \frac{dy}{dt}b + \frac{dz}{dt}c,$$

while

$$p - g = ax + by + cz$$

equals the sum of its projections, or

$$p = g + ax + by + cz.$$

We designate the above differential expression  $\frac{dx}{dt}a + \frac{dy}{dt}b + \frac{dz}{dt}c$  by  $\frac{\partial p}{\partial t}$ , since one also obtains it if one regards the magnitudes  $g, a, b, c$  in the function

$$p = g + ax + by + cz$$

as constants and then differentiates with respect to time. That is:

If  $p = g + ax + by + cz$  and  $g$  is a point while  $a, b, c$  are three mutually orthogonal displacements, then

$$\frac{\partial p}{\partial t} = \frac{dx}{dt}a + \frac{dy}{dt}b + \frac{dz}{dt}c. \quad (1.11)$$



And, since  $x, y, z$  are functions of time, so also  $p$  is a function of time, but a geometrical one, whence follows the theorem:

If a moving point  $p$  is given as a geometrical function of time  $t$ , then the differential quotient of this function with the time is the velocity of this point in its magnitude and direction.

Up until now we have associated the concept of the geometric differential with three mutually orthogonal axes. We can easily disassociate the concept from this dependence.

In general, if

$$p = g + a_1\tau_1 + a_2\tau_2 + \dots,$$

where  $p$  and  $g$  are points,  $g$  a fixed point,  $a_1, a_2, \dots$  constant displacements,  $\tau_1, \tau_2, \dots$  arbitrary functions of time, then it is easy to show that

$$\frac{\partial p}{\partial t} = \frac{d\tau_1}{dt}a_1 + \frac{d\tau_2}{dt}a_2 + \dots$$

For suppose  $a_1, a_2, \dots$  are projected onto three mutually orthogonal axes  $a, b, c$ , and that  $a_1 = \alpha_1a + \beta_1b + \gamma_1c, \dots$ , where  $\alpha_1, \beta_1, \gamma_1, \dots$  are numerical magnitudes; then

$$p = g + a(\alpha_1\tau_1 + \alpha_2\tau_2 + \dots) + b(\beta_1\tau_1 + \beta_2\tau_2 + \dots) + c(\gamma_1\tau_1 + \gamma_2\tau_2 + \dots)$$

whence also from the definition

$$\begin{aligned} \frac{\partial p}{\partial t} &= a \left( \alpha_1 \frac{d\tau_1}{dt} + \alpha_2 \frac{d\tau_2}{dt} + \dots \right) + b \left( \beta_1 \frac{d\tau_1}{dt} + \beta_2 \frac{d\tau_2}{dt} + \dots \right) + c \left( \gamma_1 \frac{d\tau_1}{dt} + \gamma_2 \frac{d\tau_2}{dt} + \dots \right) \\ &= (\alpha_1a + \beta_1b + \gamma_1c) \frac{d\tau_1}{dt} + (\alpha_2a + \beta_2b + \gamma_2c) \frac{d\tau_2}{dt} + \dots \\ &= a_1 \frac{d\tau_1}{dt} + a_2 \frac{d\tau_2}{dt} + \dots \end{aligned}$$

That form, in which  $p$  is represented as a function of time, is the most general form in which  $p$  can be represented as a geometrical function of time, and therefore from this general form we immediately arrive at the differential quotients of that function with respect to time, that is at the velocity of the point  $p$

regarded as a function of time, if we treat the differentiation of spatial magnitudes the same as algebraic ones; and we would have arrived at this concept directly if we had proceeded purely geometrically with the development of the concept of the differential.

Now if the velocity is symbolized by  $v$ , then  $\frac{\partial p}{\partial t}$  is again a geometric function; then we call  $\frac{\partial v}{\partial t}$  (the rate at which the velocity of the point increases) the *acceleration* of the point  $p$ , which is therefore again interpreted as comprising not only its absolute magnitude, but also its direction. According as its direction makes an obtuse, right, or acute angle with the velocity, it will increase, leave unchanged, or decrease the absolute value of the velocity. In analogy with the terminology of the differential calculus we call the expression that is  $\frac{\partial}{\partial t} \left( \frac{\partial p}{\partial t} \right)$ , which we symbolize as  $\frac{\partial^2 p}{\partial t^2}$ , the second differential quotient of the spatial function  $p$  with respect to the time  $t$ , and this is therefore regarded as equal to the acceleration.

These are the essential characteristics of the pure theory of motion.

## § 1.6. The Differential Equations of Mechanics

In mechanics we consider motions as they are produced by their causes. We call the cause of the acceleration of a point the *accelerating force* acting on it, and if no other accelerating force acts at the same time, we regard it as equal to the acceleration. We say further that two accelerating forces simultaneously act on a point if the acceleration of the point is the sum of the accelerating forces, whereby it is established that the accelerating forces are always defined as displacements, and thus their sum is also interpreted in this sense. Thus if arbitrarily many accelerating forces  $P_1, P_2, \dots$  act on a point, then one has

$$\frac{\partial^2 p}{\partial t^2} = P_1 + P_2 + \dots \quad (1.12)$$

as the fundamental equation of mechanics.

Again, we can consider the accelerating forces themselves, either all or individually, as the actions of more general forces that relate to whole systems of points. Of this type are all the forces of nature. Thus for example gravitation tends to decrease the relative distance between two points, cohesion to maintain unchanged the relative distance between each pair of points of a system, elasticity to restore the volume lying between four neighboring points, and so forth. It is thus important to understand more precisely the concept of such tendencies.

The distance between two points, the spatial content between four points, and in general everything related to tendency of a force residing in a system of points can be interpreted as a function of these points; thus we establish the following definition of such forces:

We say that a force that is based on a system of equally heavy (equally movable) points  $p, q, r, \dots$ , tends to increase an algebraic function<sup>4</sup>  $F(p, q, r, \dots)$  of those points if the accelerating force produced by it acts on each of the points in that direction in which that function increases the most with equally large but infinitesimal motions of the points, and the absolute magnitudes of the accelerating forces to which each pair of points of the system are subject by virtue of that force stand in the proportion in which the two points taken individually induce the function to increase with equally large but infinitesimal motions in those directions.

An example will illustrate this.

Let the tendency of a force  $V$  be to separate two equally weighted points  $p$  and  $q$  from each other. Thus we can regard  $(p - q)^2$  as the algebraic function of the points  $p$  and  $q$ , which that force tends to increase. First, in order to find the direction of the accelerating force that  $V$  exerts on the point  $p$ , we have to ask what directions must be taken in equally large but infinitesimal paths to increase the value of  $(p - q)^2$  the most. This is obviously the direction of elongation of  $qp$  beyond  $p$ . Likewise the accelerating force to which  $q$  is subject acts in the prolon-

---

<sup>4</sup>An algebraic function will however be called such that is proportional to a variable numerical magnitude.

gation of  $pq$  beyond  $q$ . If we now ask for the absolute ratio of these two accelerating forces, it is equal to the ratio by in which  $(p - q)$  increases with equally large but infinitesimal paths in the direction of prolongation, while in each case the other point is thought of as stationary. This ratio is obviously that of equality; thus the accelerating forces are taken as absolutely equal.

Now we can find a very simple procedure according to which, whatever may be the algebraic function on which the tendency of the force is based, one can always find the directions and ratios of the accelerating forces that act thereby. Since I will not give an independent foundation for geometric differential calculus, I will again at first link the derivation of this procedure to orthogonal coordinates, although the procedure itself is completely independent of them.

Let the algebraic function  $F(p_1, p_2, \dots)$  of the points  $p_1, p_2, \dots$  be given. In order to transform it into an algebraic function of numerical magnitudes, we assume three mutually orthogonal axes through a point  $g$  and in them three equally long units  $a, b, c$ , by which we measure orthogonal projections of the displacements  $p_1 - g, p_2 - g, \dots$ , and therefore set

$$p_1 - g = x_1a + y_1b + z_1c; \quad p_2 - g = x_2a + y_2b + z_2c; \quad \dots$$

Now we can regard that function as an algebraic function of the points  $p_1, p_2, \dots$ , which is proportional to a function of the combined coordinates, each measured by the proper unit, that is, equal to such a function multiplied by a constant factor. Since only the ratios and directions of the accelerating forces can be derived from that function  $F(p_1, p_2, \dots)$ , and, as is clear from the definition, neither of these is changed if one multiplies the function by an arbitrary positive magnitude, one can set the algebraic function equal to a numerical function of the measures  $x_1, y_1, z_1; x_2, y_2, z_2; \dots$ , multiplied by the square of the unit. Thus we set

$$F(p_1, p_2, \dots) = a^2 f(x_1, y_1, z_1; x_2, y_2, z_2; \dots).$$

Now in order to find the directions and ratios of the accelerating forces we have to consider the functional increase that

results if the points  $p_1, \dots$  describe infinitesimally small paths. The infinitesimally small path of the point  $p_1$ , which we symbolize by  $\delta p_1$ , is

$$\delta p_1 = a \delta x_1 + b \delta y_1 + c \delta z_1,$$

where  $a \delta x_1$ ,  $b \delta y_1$ ,  $c \delta z_1$  are its orthogonal projections on the three axes. By this motion  $p_1$  is transformed into  $p_1 + \delta p_1$ , and thus  $g + ax_1 + by_1 + cz_1$  into

$$g + a(x_1 + \delta x_1) + b(y_1 + \delta y_1) + c(z_1 + \delta z_1);$$

that is,  $x_1$  thereby transforms into  $x_1 + \delta x_1$ ,  $y_1$  into  $y_1 + \delta y_1$  and  $z_1$  into  $z_1 + \delta z_1$ . Likewise, if  $\delta p_2$  is the motion of the point  $p_2$ , and

$$\delta p_2 = a \delta x_2 + b \delta y_2 + c \delta z_2,$$

then by this motion  $x_2$  transforms into  $x_2 + \delta x_2$ ,  $y_2$  into  $y_2 + \delta y_2$ ,  $z_2$  into  $z_2 + \delta z_2$ , and so forth. The increase of  $f$  flowing therefrom we thus find by differentiations of this function, regarding  $\delta x_1, \dots$  as differentials of the corresponding variables. If we also symbolize the resulting differential of the function by  $\delta$ , we obtain

$$\begin{aligned} \delta f(x_1, y_1, z_1; x_2, y_2, z_2; \dots) &= u_1 \delta x_1 + v_1 \delta y_1 + w_1 \delta z_1 \\ &+ u_2 \delta x_2 + v_2 \delta y_2 + w_2 \delta z_2 + \dots \quad (1.13) \end{aligned}$$

where  $u_1, v_1, w_1, \dots$  are the partial differential quotients with respect to the corresponding variables, that is  $x_1, y_1, z_1, \dots$ .

In order to find the directions and ratios of the accelerating forces, we again transform the differential equation by multiplication with  $a^2$  into geometrical form, and, just as previously  $F = a^2 f$ , we now set  $\delta F = a^2 \delta f$ ; thus

$$\delta F(p_1, p_2, \dots) = a^2 [u_1 \delta x_1 + v_1 \delta y_1 + w_1 \delta z_1 + u_2 \delta x_2 + v_2 \delta y_2 + w_2 \delta z_2 + \dots]$$

or, since  $a^2 = b^2 = c^2$ ,

$$\delta F(p_1, p_2, \dots) = u_1 a \times a \delta x_1 + v_1 b \times b \delta y_1 + w_1 c \times c \delta z_1 + u_2 a \times a \delta x_2 + v_2 b \times b \delta y_2 +$$

Thus, since

$$a \delta x_1 + b \delta y_1 + c \delta z_1 = \delta p_1, \quad a \delta x_2 + b \delta y_2 + c \delta z_2 = \delta p_2, \quad \dots$$

and  $a, b, c$  are mutually orthogonal, then

$$\delta F(p_1, p_2, \dots) = (au_1 + bv_1 + cw_1) \times \delta p_1 + (au_2 + bv_2 + cw_2) \times \delta p_2 + \dots$$

Finally, if we symbolize the magnitude  $au_1 + bv_1 + cw_1$  by  $q_1$ ,  $au_2 + bv_2 + cw_2$  by  $q_2$ , and so forth, and call these displacements  $q_1, q_2, \dots$  the *partial differential quotients* with respect to  $p_1, p_2, \dots$ , then we obtain the simple equation<sup>5</sup>

$$\delta F(p_1, p_2, \dots) = q_1 \times \delta p_1 + q_2 \times \delta p_2 + \dots \quad (1.14)$$

and the theorem:

The evolution ( $\delta F$ ) that an algebraic function ( $F$ ) of points ( $p_1, \dots$ ), which appears as the product of the square of a displacement with a variable numerical magnitude, undergoes when the points on which it depends describe infinitesimally small paths equals the sum that one obtains if one takes the partial derivatives with respect to the individual variable points, inner multiplies these partial differential quotients with the paths of the corresponding points and adds these inner products.

And

One finds the partial differential quotients of such a function ( $F$ ) with respect to one of the points ( $A$ ) on which it depends if one divides the function by the square of the unit, represents the quotient as a function of the coordinates of this point taken with respect to three mutually orthogonal and equally long units laid from an arbitrary point, differentiates with respect to the coordinates of that point  $p_1$ , and adds these differential quotients after one multiplies each of them by the corresponding measure.

The directions and ratios of the forces follow at once from equation (28).

If one considers the evolution of the function by different infinitesimally small but equally long paths  $\delta p_1$  of the point  $p_1$ ,

---

<sup>5</sup>The equation in the original uses a special notation for the differential.

then it is equal to  $q_1 \times \delta p_1$ , that is, with constant  $q_1$  proportional to the orthogonal projection of  $\delta p_1$  on  $q_1$ , thus with equally long  $\delta p_1$  the largest if  $\delta p_1$  has the same direction as  $q_1$ . Thus  $q_1, q_2, \dots$  express the directions of the accelerating forces. The evolutions of the function that result by two different points  $p_1$  and  $p_2$  acting in these directions  $q_1$  and  $q_2$  with equally long paths are in the same proportion as  $q_1 \times \delta p_1$  to  $q_2 \times \delta p_2$ , thus, since  $\delta p_1$  parallels  $q_1$  and  $\delta p_2$  parallels  $q_2$ , and  $\delta p_1$  and  $\delta p_2$  are equally long, as  $q_1$  to  $q_2$ ; that is:

The accelerating forces to which the points of a system are subject due to a force that tends to increase an algebraic function of those points, act in magnitude and direction as the partial differential quotients of this function with respect to these points.

Before I derive the formulas of mechanics that flow from this, I will show that the partial differential coefficients of a function of points, whose derivation was previously tied to certain orthogonal axes, are in fact independent of them.

Three mutually orthogonal units  $a, b, c$  and a point  $g$  were assumed, the displacements  $p_1 - g, p_2 - g, \dots$  were orthogonally projected onto their directions, each of these projections was measured by its similarly directed unit and the results of these measures symbolized by  $x_1, y_1, z_1, \dots$ , so that  $p_1 - g = x_1 a + y_1 b + z_1 c, \dots$ ; thus by substitution of these values for  $p_1, \dots$  into the algebraic function  $F$  of the points, and division by the square of the unit there resulted an algebraic function  $f$  of  $x_1, y_1, z_1, \dots$ . This was differentiated with respect to the variables  $x_1, y_1, z_1, \dots$ , these differential quotients were multiplied by their corresponding units  $a, b, c$ , and these products added.

Now if we let  $a, b, c$  transform into  $e, e', e''$ , where

$$\begin{aligned} e &= \alpha a + \beta b + \gamma c \\ e' &= \alpha' a + \beta' b + \gamma' c \\ e'' &= \alpha'' a + \beta'' b + \gamma'' c \end{aligned} \tag{1.15}$$

then if

$$p_1 = v_1 e + v_1' e' + v_1'' e'',$$

so that

$$v_1 e + v'_1 e' + v''_1 e'' = x_1 a + y_1 b + z_1 c,$$

then the values of  $x_1, y_1, z_1$  are as follows:

$$x_1 = v_1 \alpha + v'_1 \alpha' + v''_1 \alpha''$$

$$y_1 = v_1 \beta + v'_1 \beta' + v''_1 \beta''$$

$$z_1 = v_1 \gamma + v'_1 \gamma' + v''_1 \gamma'';$$

and one obtains corresponding values for the coordinates of the remaining points. If one wishes to find the differential quotients of a function  $F$  with respect to these new units, then after one has substituted the expressions just found into  $f(x_1, y_1, \dots)$  one has to differentiate with respect to  $v, v', v''$ , multiply the resulting differential quotients by  $e, e', e''$ , and add these products. Then it remains to show that this sum still equals  $q_1$ , that is

$$\frac{\partial f}{\partial v_1} e + \frac{\partial f}{\partial v'_1} e' + \frac{\partial f}{\partial v''_1} e'' = \frac{\partial f}{\partial x_1} a + \frac{\partial f}{\partial y_1} b + \frac{\partial f}{\partial z_1} c.$$

Now

$$\frac{\partial f}{\partial v_1} = \frac{\partial f}{\partial x_1} \frac{\partial x_1}{\partial v_1} + \frac{\partial f}{\partial y_1} \frac{\partial y_1}{\partial v_1} + \frac{\partial f}{\partial z_1} \frac{\partial z_1}{\partial v_1} = \frac{\partial f}{\partial x_1} \alpha + \frac{\partial f}{\partial y_1} \beta + \frac{\partial f}{\partial z_1} \gamma,$$

$$\frac{\partial f}{\partial v'_1} = \frac{\partial f}{\partial x_1} \alpha' + \frac{\partial f}{\partial y_1} \beta' + \frac{\partial f}{\partial z_1} \gamma',$$

$$\frac{\partial f}{\partial v''_1} = \frac{\partial f}{\partial x_1} \alpha'' + \frac{\partial f}{\partial y_1} \beta'' + \frac{\partial f}{\partial z_1} \gamma''$$

and thus

$$\frac{\partial f}{\partial v_1} e + \frac{\partial f}{\partial v'_1} e' + \frac{\partial f}{\partial v''_1} e'' = \frac{\partial f}{\partial x_1} (\alpha e + \alpha' e' + \alpha'' e'') + \frac{\partial f}{\partial y_1} (\beta e + \beta' e' + \beta'' e'') + \frac{\partial f}{\partial z_1} (\gamma e + \gamma' e' + \gamma'' e'')$$

Now if  $e, e', e''$  are equal in length to  $a, b, c$  and at right angles to each other, then it is easy to show that  $\alpha e + \alpha' e' + \alpha'' e''$  is equal to  $a$  and so forth. For, symbolizing by  $\cos(rs)$  the cosine of the angle between two equally long displacements  $r$  and  $s$ ,



$r \cos(rs)$  is the orthogonal projection of  $s$  on  $r$  in magnitude and direction. From equation (29) it then follows that

$$\alpha = \cos(ea), \quad \alpha' = \cos(e'a), \quad \alpha'' = \cos(e''a).$$

Now

$$a = e \cos(ea) + e' \cos(e'a) + e'' \cos(e''a),$$

that is,

$$a = \alpha e + \alpha' e' + \alpha'' e''.$$

Likewise it follows that

$$b = \beta e + \beta' e' + \beta'' e'', \quad c = \gamma e + \gamma' e' + \gamma'' e''.$$

Thus one obtains the equation to be proved,

$$\frac{\partial f}{\partial v_1} e + \frac{\partial f}{\partial v'_1} e' + \frac{\partial f}{\partial v''_1} e'' = \frac{\partial f}{\partial x_1} a + \frac{\partial f}{\partial y_1} b + \frac{\partial f}{\partial z_1} c.$$

If  $e, e', e''$  are not equally as long as  $a, b, c$ , but rather the length of the latter is the  $m$ -fold of that of the former, then according to the procedure given above in reference to the units  $e, e', e''$  one has to differentiate the function which we will symbolize by  $\phi$ . Thus, since  $F = a^2 f$ , and  $a^2 = m^2 e^2$ ,  $\phi = m^2 f$ . Then the projections of  $e, e', e''$  on  $a, b, c$  are equal to the cosines multiplied by  $m$ , and those of  $a, b, c$  on  $e, e', e''$  the cosines multiplied by  $me, me', me''$ , and

$$a = m^2(\alpha e + \alpha' e' + \alpha'' e'')$$

and so forth.<sup>6</sup>

If we collect the results obtained, there follows:

If  $F$  is the algebraic function of second dimensions which a force  $V$  that acts on a system of points  $p_1, p_2, \dots$  tends to increase, and  $q_1, q_2, \dots$  are the partial differential quotients of  $F$  with respect to the points  $p_1, p_2, \dots$  on which  $F$  depends, then

$$Aq_1, \quad Aq_2, \quad Aq_3, \quad \dots$$

---

<sup>6</sup>Since then  $a = m[e \cos(ea) + e' \cos(e'a) + e'' \cos(e''a)]$ , and  $\cos(ea) = m\alpha$ ,  $\cos(e'a) = m\alpha'$ ,  $\cos(e''a) = m\alpha''$ .

are the accelerating forces to which the points  $p_1, p_2, p_3, \dots$  are subject by virtue of that force  $V$ , where  $A$  symbolizes a common numerical factor, which depends on the positions of the points in a way defined by the nature of that force  $V$ .

Thus if the accelerating forces  $P_1, P_2, \dots$  as well as the accelerating forces  $Aq_1, Aq_2, \dots$  specified by the force  $V$  act on the points  $p_1, p_2, \dots$ , one has

$$\frac{\partial^2 p_1}{\partial t^2} = P_1 + Aq_1, \quad \frac{\partial^2 p_2}{\partial t^2} = P_2 + Aq_2, \quad \dots$$

and so forth.

If one inner multiplies the first of these equations by  $\delta p_1$ , the second by  $\delta p_2$ , and adds, then since  $q_1 \times \delta p_1 + q_2 \times \delta p_2 + \dots$  equals  $\delta F$ , one obtains the equation

$$\left( P_1 - \frac{\partial^2 p_1}{\partial t^2} \right) \times \delta p_1 + \left( P_2 - \frac{\partial^2 p_2}{\partial t^2} \right) \times \delta p_2 + \dots + A \delta F = 0 \quad (1.16)$$

which is valid for every value of  $\delta p_1, \delta p_2, \dots$ . Finally, if one assumes further, as may be assumed approximately for rigid bodies, that, with infinitesimally small distances from those positions for which  $F$  has a specific value, the force  $V$  already acts so intensely that the remaining forces cannot perceptibly separate the system from such positions, then we can regard  $F$  as approximately constant; we call such forces *constraint forces* if they can already overcome the remaining forces with infinitesimally small distances from those positions, so that only infinitesimally small deviations from those positions for which  $F$  has that specified value can therefore enter. Thus:

If arbitrary forces  $P_\alpha$  act on any equally heavy points  $p_1, p_2, \dots$  of a system, and these points are either all or in part subject to constant forces which with irresistible power attempt to maintain certain functions  $L, M, N, \dots$  of these points at zero, one has the equation

$$0 = \left( P_1 - \frac{\partial^2 p_1}{\partial t^2} \right) \times \delta p_1 + \left( P_2 - \frac{\partial^2 p_2}{\partial t^2} \right) \times \delta p_2 + \dots \\ + \lambda \delta L + \mu \delta M + \nu \delta N + \dots \quad (1.17)$$

which in combination with the equations

$$L = 0, \quad M = 0, \quad N = 0, \quad \dots$$

suffice for the complete specification of the acceleration of each of the points.

This is the reduced form, in our analysis, of the general equation of motion as it is placed at the foundation of all of mechanics by Lagrange, and thereby the whole of mechanics is subject to this new analysis.

## § 1.7. Application to a Problem in Statics

As an example of the application of this procedure I will arbitrarily choose a simple problem in statics, to wit the problem of finding the equilibrium state of a system of points  $p_1, p_2, \dots p_n$  that are so mutually related that the distance between each sequential pair of points remains constant, and which are in addition drawn by the accelerating forces  $P_1, P_2, \dots P_n$  respectively.

For substitution into (31) one has to set all the accelerations to zero, since equilibrium is maintained. We assume of the forces by which each sequential pair of points remains fixed that they seek to maintain constant the distances between each such pair of points, or rather, in order to get a function of second dimension, the square of the connecting displacements. Thus if, in short, we symbolize the points by  $p_\alpha$ , the forces by  $P_\alpha$ , where  $\alpha$  expresses the series of indices  $0, \dots, n$ , we obtain the equation

$$\sum P_\alpha \times \delta p_\alpha + \sum \lambda_\alpha \delta [(p_{\alpha+1} - p_\alpha)^2] = 0, \quad (1.18)$$

where the summation sign refers to the different values of  $\alpha$ , in the first term over the values  $0, \dots, n$ , and in the second over  $0, \dots, (n-1)$ ; that is, all values of  $\lambda$  in which  $\alpha$  does not have one of the values  $0, 1, 2, \dots (n-1)$  are set to zero. This equation in combination with the  $n$  equations

$$(p_{\alpha+1} - p_\alpha)^2 = a_\alpha^2, \quad (1.19)$$

which describe the constant distances whose squares are precisely the  $a_\alpha^2$ , completely specify the equilibrium state.

But

$$\begin{aligned} \frac{1}{2}\delta[(p_{\alpha+1} - p_\alpha)^2] &= (p_{\alpha+1} - p_\alpha) \times \delta(p_{\alpha+1} - p_\alpha) \\ &= (p_{\alpha+1} - p_\alpha) \times \delta p_{\alpha+1} - (p_{\alpha+1} - p_\alpha) \times \delta p_\alpha. \end{aligned}$$

Since (32a) is zero for each series of values of  $\delta p_\alpha$ , the terms of it multiplied by one and the same  $\delta p_\alpha$  must together give zero, whence one has the  $(n + 1)$  equations

$$P_\alpha + 2\lambda_{\alpha-1}(p_\alpha - p_{\alpha-1}) - 2\lambda_\alpha(p_{\alpha+1} - p_\alpha) = 0. \quad (1.20)$$

If one adds a sequential pair of these equations, for example that including  $P_\alpha$  and that including  $P_{\alpha+1}$ , one obtains

$$0 = P_\alpha + P_{\alpha+1} + 2\lambda_{\alpha-1}(p_\alpha - p_{\alpha-1}) - 2\lambda_{\alpha+1}(p_{\alpha+2} - p_{\alpha+1});$$

or more generally, if one adds all equations in (32c) between the indicator  $\alpha'$  and the assumed greater  $\alpha$ , then one has

$$0 = P_{\alpha'} + \dots + P_\alpha + 2\lambda_{\alpha'-1}(p_{\alpha'} - p_{\alpha'-1}) - 2\lambda_\alpha(p_{\alpha+1} - p_\alpha).$$

Thus if  $\alpha'$  is set to zero, so that  $\lambda_{\alpha'-1}$  is also zero according to the above,

$$P_0 + P_1 + \dots + P_\alpha - 2\lambda_\alpha(p_{\alpha+1} - p_\alpha) = 0. \quad (1.21)$$

Further, if  $\alpha$  is simultaneously set to  $n$ , so that  $\lambda_n$  is zero,

$$P_0 + P_1 + \dots + P_n = 0. \quad (1.22)$$

Now in order to eliminate  $\lambda_\alpha$  from equations (32d) in general one has only to outer multiply it by  $(p_{\alpha+1} - p_\alpha)$ , since the outer product (the parallelogram) of similarly directed displacements is zero. Thus one obtains

$$(P_0 + P_1 + \dots + P_\alpha) \cdot (p_{\alpha+1} - p_\alpha) = 0. \quad (1.23)$$

These  $n$  equations in (34), in combination with (33) and the  $n$  constraint equations in (32c), completely specify the equilibrium state, and, if the initial point or generally one of the  $n$

points is given, permits everything to be found at once by constructions from the formulas, which is the overall advantage of this new analysis.

Suppose for example that the distances  $a_\alpha$  between each sequential pair of points, the  $n$  forces  $P_0, P_1, \dots, P_{n-1}$  and the position of the point  $p_0$  are given, and the force  $P_n$  and the positions of the points  $p_1, \dots, p_n$  are to be found.

In fact equation (34) is just an expression of the fact that  $P_0 + \dots + P_\alpha$  is always parallel to  $p_{\alpha+1} - p_\alpha$ . From this results the following construction. One draws a  $(n+1)$ -gon  $A_0, A_1, \dots, A_n$  whose sides taken progressively from the vertex  $A_1$  are equal and equally directed to the forces  $P_0, P_1, \dots, P_{n-1}$ , that is

$$A_0 - A_1 = P_0, \quad A_1 - A_2 = P_1, \quad A_2 - A_3 = P_2, \quad \dots$$

whence

$$A_\alpha - A_1 = P_0 + P_1 + \dots + P_\alpha,$$

and equation (34) becomes

$$(A_\alpha - A_1) \cdot (p_{\alpha+1} - p_\alpha) = 0. \quad (1.24)$$

If the force sought,  $P_n$ , is set equal to  $X - A_{n-1}$ , where  $X$  is a point to be found, then equation (33) transforms into

$$A_{n-1} - A_n + X - A_{n-1} = 0,$$

that is:  $X = A_n$ , i.e. the side  $A_n - A_{n-1}$  of the auxiliary figure is the force  $P_n$  sought. Furthermore, to find the points  $p_1, \dots, p_n$ , if one scribes a circle (or in space a sphere) of radius  $a_0$  about  $p_0$ , draws a straight line parallel to  $P_0$  (or  $A_0 - A_1$ ) through  $P_0$ , then each of the two points where this straight line intersects the circle (or the sphere) replacing  $p_1$  satisfies the conditions of equilibrium. If one scribes a circle (a sphere) with  $a_1$  as radius about  $p_1$  and draws from  $p_1$  a straight line parallel to  $A_1 - A_2$ , then each of the points of intersection of this straight line and the circle (the sphere) replacing  $p_2$  satisfy the conditions of equilibrium.

Thus there result  $2^n$  equilibrium positions in all. But one easily convinces oneself that only one among them gives the stable

equilibrium position, which results if one replaces the straight lines with rays directed opposite to the sides  $A_\alpha - A_{\alpha+1}$ .

Of course the construction given could easily have been derived immediately from the idea. But if one pursues this conceptual derivation, one would see at once that it is identical to that given by our analysis, which is only so to speak its translation into conceptual language. And this is in general the essential advantage of the new analysis.

## § 1.8. Transition from Displacements to Points

I now continue the development of the new analysis which was interrupted earlier. I above derived the inner product of two displacements from the idea of congruence. Each displacement appears as the difference of two points, and thus the inner product of two displacements is in the form

$$(a - b) \times (c - d),$$

where  $a, b, c, d$  are points. In order to proceed from this by dropping the parentheses to the inner products of points, and of points with displacements, I will avail myself of a general derivation theorem which is generally of the greatest importance for the new analysis, thus:

**Theorem (12).** *If certain laws are generally valid for differences of each pair of magnitudes of a series of magnitudes  $(a, b, c, \dots)$  and one has any given equation  $(A)$  in which those magnitudes  $(a, b, c, \dots)$  are only found paired in differences, and one thus derives a new equation form  $(B)$  from this equation  $(A)$  if one also applies those laws to those magnitudes themselves instead of to their differences: then one always obtains from it a true equation if one replaces the series of magnitudes  $(a, b, c, \dots)$  by the series of differences  $(a - r, b - r, \dots)$  of the magnitudes of this series from any given magnitude  $(r)$ ; and if  $B$  again includes those magnitudes only paired in differences, then  $B$  is itself a true equation as well; and indeed all this even follows if one only assumes for such differences the law that  $(a - b) + (b - c) = a - c$ .*

Thus, since  $(a - b) + (b - r) = a - r$ , then also

$$a - b = (a - r) - (b - r); \quad (1.25)$$

consequently, by this law assumed for the differences, one can derive from equation  $A$  an equation  $A'$  in which, instead of the magnitudes  $a, b, c, \dots$  appearing in  $A$ , only the differences  $a - r, b - r, c - r, \dots$  occur. If I now obtain thereby an equation form  $B$  from  $A$  by applying the laws valid generally for differences of those magnitudes, then I proceed to a true equation  $B'$ , which differs from  $B$  only in that instead of the magnitudes  $a, b, \dots$  in  $B$  the differences  $a - r, b - r, \dots$  enter. Further, if in the equation form  $B$  the magnitudes  $a, b, \dots$  only appear paired in differences, then in  $B'$  the differences  $a - r, b - r, \dots$  would again only appear paired in differences. But according to the law represented by equation (35) I can replace each such difference of the differences  $a - r, b - r, \dots$  by the corresponding difference of the magnitudes  $a, b, \dots$  without the equation ceasing to be true. Thus however I obtain the form  $B$  from  $B'$ . Since therefore  $B'$  was a true equation, so also is  $B$  according to the last assumption made.

With this I now complete the definition which, by virtue of the theorem just proved, completely suffices for all analytic laws of point conjunctions if those of displacement conjunctions are known. That is

**Definition (8).** I regard an equation ( $B$ ) that includes points, not all of which are paired in differences, as true if and only if a true equation ( $A$ ) can be found in which the points only occur paired in differences, and which has the property that one can derive the given equation ( $B$ ) from it if one also applies to points the conjunctive laws previously proved for displacements in general.

From this it follows at once that, if one has a true point equation  $B$  in which at least not all the points are paired in differences, and by treating the points as displacements one obtains from it an equation  $C$  in which likewise not all the points are paired in differences, this is likewise a true equation. For

according to the definition  $B$  is only true if it is derivable from a true displacement equation  $A$  by application of the laws, valid for displacements to points as well. Now since in just such a way as  $B$  is derivable from  $A$ ,  $C$  is again derivable from  $B$ , so the last,  $C$ , is also derivable in such a way from  $A$ , whence  $C$  is also true. However, since the assumption of that theorem that  $(a - b) + (b - c) = a - c$  for the displacements  $a - b$ ,  $b - c$  is also generally valid if  $a$ ,  $b$ ,  $c$  are points we can, by virtue of the foregoing Theorem 12, go yet further and say in addition that if  $C$  is a displacement equation that results from  $B$  in the manner given, it is correct; for then  $C$  also follows from  $A$  in such a way, that is by application to points of the laws established for displacements; whence  $C$  is also true, according to the theorem adduced, if  $A$  is. Thus:

In a true equation containing points, if one applies to the points the laws generally valid for displacements, then the transformed equation is likewise true. I can also generalize this theorem as follows:

**Theorem (13).** *All generally valid laws of conjunction for displacements are also valid for points and generally for point magnitudes.*

That is, for point magnitudes since the same laws that are valid for displacements must also be valid for their multiples, since these multiples are likewise displacements, and also for multiples of points, that is for point magnitudes. Further, since each true point equation  $B$  is derivable from a true displacement equation  $A$  according to Definition 8, it follows by virtue of the same Theorem 12 that the condition formulated for the point equation  $B$  in that Theorem 12 is subject to the theorem:

**Theorem (14).** *A true point equation remains true if one replaces the collection of points  $(a, b, \dots)$  appearing therein with their differences  $(a - r, b - r, \dots)$  from one and the same point  $(r)$ .*

By application of these two Theorems 13 and 14 all laws of the conjunctions of points and point magnitudes follow most



easily, if one knows the laws of the conjunctions of displacements. I will therefore incidentally and for better comparison also derive the laws of addition and subtraction of point magnitudes from those of displacements.

## § 1.9. Sums of Point Magnitudes

If  $a, b, c, \dots$  are points, and  $\alpha, \beta, \gamma, \dots$  numerical magnitudes, and to begin with

$$\alpha + \beta + \gamma + \dots = 0$$

then if  $r$  is an arbitrary point one has

$$\begin{aligned} \alpha a + \beta b + \dots &= \alpha a + \beta b + \dots + (\alpha + \beta + \dots)r \\ &= \alpha(a - r) + \beta(b - r) + \dots, \end{aligned}$$

that is, calling  $\alpha, \beta, \dots$  the *weights* of the point magnitudes  $\alpha a, \beta b, \dots$ ,  $(\alpha + \beta + \dots)$  their collective weight, and  $\alpha(a - r) + \beta(b - r) + \dots$  their collective deviation from the point  $r$ , then:

**Theorem (15).** *A sum of point magnitudes whose collective weight is zero is a displacement; their collective deviation from a variable point is constant, equal to the displacement that is their sum.*

Further: If the collective weight is not zero, then let their sum be set equal to an as-yet unknown point magnitude  $\rho x$ , where  $\rho$  is a numerical magnitude,  $x$  a point; thus

$$\alpha a + \beta b + \dots = \rho x, \tag{1.26}$$

whence, according to Theorem 14,

$$\alpha(a - r) + \beta(b - r) + \dots = \rho(x - r). \tag{1.27}$$

If one subtracts this equation from (36), one has

$$(\alpha + \beta + \dots)r = \rho r;$$

thus, again according to Theorem 14, if  $r'$  is another point,

$$(\alpha + \beta + \dots)(r - r') = \rho(r - r'),$$

a displacement equation from which it follows that

$$\alpha + \beta + \dots = \rho. \quad (1.28)$$

If one substitutes this value in (36) one then has

$$\alpha(a - r) + \beta(b - r) + \dots = (\alpha + \beta + \dots)(x - r), \quad (1.29)$$

or, since  $(\alpha + \beta + \dots)$  is not zero,

$$x - r = \frac{\alpha(a - r) + \beta(b - r) + \dots}{\alpha + \beta + \dots}. \quad (1.30)$$

Now if  $x$  is defined by this equation, as is possible by a simple construction, then equation (38) is also true, for, since (36) immediately follows from (38) when the values found are substituted for  $\rho$  and  $x$ , (36) is also true according to Theorem 13, while the foregoing development shows that if (36) is to be true,  $\rho$  and  $x$  must necessarily be subject to the definitions given in (37) and (38).

Moreover, if we set  $r$  equal to  $x$  in (36a), then we have

$$\alpha(a - x) + \beta(b - x) + \dots = 0. \quad (1.31)$$

As is known, one calls this point  $x$  the *center of gravity* between the points  $a, b, \dots$  endowed with weights  $\alpha, \beta, \dots$  or, expressed mathematically, the *mean* between the point magnitudes  $\alpha a, \beta b, \dots$ . Thus:

**Theorem (16).** *A sum of point magnitudes whose collective weight is not zero is again a point magnitude whose weight is the collective weight of the added point magnitudes (37), and whose position is the mean between them (38); the collective deviation of the point magnitudes from an arbitrary point equals that of their sum from the same point (36a), so that the collective deviation from the mean is zero, and the mean is therefore*

*found (39) if one draws the displacements from a point to the positions of the point magnitudes, multiplies these displacements by the corresponding weights, adds the products, divides the sum by the collective weight, and draws a displacement equal to this quotient which has that point as initial point; the final point of this displacement is then the mean sought.*

If we say that the weight of a displacement is zero, and if we call displacements and point magnitudes collectively *magnitudes of first order*, then from Theorems 15 and 16 there follows at once the theorem: The weight of a sum of magnitudes of first order equals the sum of the weights of the summands, or more generally:

**Theorem (17).** *An equation between magnitudes of first order, that is, an equation in which the magnitudes of first order are only subject to addition, subtraction, multiplication, and division, remains true if one replaces the magnitudes of first order with their weights.*

## § 1.10. Inner Multiplication of Point Magnitudes. Inner Magnitudes

I now proceed to the inner product of two magnitudes of first order.

The inner product of two point magnitudes as well as that of a point magnitude with a displacement can be represented as the difference of two squares or as multiples of this difference.

Thus if in the following  $a, b, c$  are understood to be points,  $\alpha, \beta, \gamma$  numerical magnitudes, and  $p$  a displacement, then one has

$$a^2 - b^2 = (a + b) \times (a - b) = 2a \times (a - b).$$

This last expression is a product of a point and a displacement. Thus if one has any given product of a point magnitude and a displacement, say  $\gamma c \times p$ , which is equal to  $c \times \gamma p$ , then has only to set  $\alpha + \beta$  equal to  $c$  and  $2(a - b)$  equal to  $\gamma p$ , whence  $a$  and  $b$  are specified if  $\gamma, c, p$  are given, and one then obtains

$\gamma c \times p = a^2 - b^2$ ; thus the inner product of a point magnitude with a displacement is reduced to the difference of two inner point squares.

Likewise

$$\alpha(a^2 - p^2) = \alpha(a + p) \times (a - p),$$

where  $a + p$  and  $a - p$  are points. Thus if one had any given product of two point magnitudes  $\beta b \times \gamma c$ , which is equal to  $\beta\gamma b \times c$ , then one has only to set  $\beta\gamma$  equal to  $\alpha$ ,  $b$  equal to  $a + p$ , and  $c$  equal to  $a - p$ , whence  $\alpha$ ,  $a$ ,  $p$  are specified if  $\beta$ ,  $\gamma$ ,  $b$ ,  $c$  are given, and one obtains that  $\beta b \times \gamma c$  equals  $\alpha(a^2 - p^2)$ . Both results can be expressed in words thus:

The [inner] product of a point magnitude and a displacement is equal to the difference ( $a^2 - b^2$ ) of the squares of two points  $a$  and  $b$  which have the position of the point magnitude at the mean between them, and whose interval ( $a - b$ ) is equal to the displacement of that product multiplied by half the weight of the point magnitude. And the [inner] product of two point magnitudes is equal to the difference  $a^2 - p^2$  of the squares of a point  $a$  and a displacement  $p$ , multiplied by a coefficient  $\alpha$ , where the coefficient  $\alpha$  equals the product of the weights of the two point magnitudes, the point  $a$  the mean between their positions and the square of the displacement  $p$  equals the square of the interval of one of these positions from the mean.

Instead of  $\alpha(a^2 - p^2)$  we can also write  $\alpha a^2 - \alpha p^2$ ; thus we can bring an arbitrary sum of inner products to the form

$$\alpha_1 a_1^2 + \alpha_2 a_2^2 + \dots + A_1 + A_2 + \dots,$$

or, with the summation symbol, to the form

$$\sum \alpha a^2 + \sum A$$

where the  $\alpha$  symbolizes numerical magnitudes, the  $a$  points, and the  $A$  inner displacement products. Thus each given equation between inner products can be brought to the form

$$\sum \alpha a^2 + \sum A = 0. \quad (1.32)$$

Now if this equation is to be true, then according to Theorem 14 it must be true that

$$\sum \alpha(a-r)^2 + A = 0 \quad (1.33)$$

for each point  $r$ , that is

$$\sum \alpha a^2 + A - 2r \times \sum \alpha a + r^2 \sum \alpha = 0. \quad (1.34)$$

If this equation is subtracted from the given (41), one obtains

$$2r \times \sum \alpha a - r^2 \sum \alpha = 0. \quad (1.35)$$

Thus, again according to Theorem 14, for each point  $r'$

$$2(r-r') \times \sum \alpha(a-r') - (r-r')^2 \sum \alpha = 0. \quad (1.36)$$

Since one may choose  $r$  and  $r'$  arbitrarily, one may assume  $r$  different from  $r'$ , and choose  $r$  so that the displacement  $r-r'$  is perpendicular to the displacement  $\sum \alpha(a-r')$ ; then (according to Theorem 3) the first term in (45) will be zero, whence one obtains

$$(r-r')^2 \sum \alpha = 0;$$

and since  $r$  differs from  $r'$  here, whence  $(r-r')^2$  is not zero, then necessarily

$$\sum \alpha = 0. \quad (1.37)$$

If one substitutes this value into equation (45), which holds for all orientations of the points  $r$  and  $r'$ , one then always has

$$2(r-r') \times \sum \alpha(a-r') = 0,$$

for every orientation of the points  $r$  and  $r'$ . Now here one can also choose  $r$  so that it differs from  $r'$  but  $r-r'$  is not perpendicular to  $\sum \alpha(a-r')$ ; then  $\sum \alpha(a-r')$  must be zero (according to Theorem 3). Consequently, since according to (46)  $r' \sum \alpha$  is zero, it is necessary that

$$\sum \alpha a = 0. \quad (1.38)$$

This equation includes equation (46) (according to Theorem 17). Now if, besides this equation (47), equation (42) also holds for any given point  $r$ , then since from (42) there results (43), and this, by virtue of (47) and the included equation (46), is transformed into (41), the validity of this last follows.

Thus an equation of the form (41) is true if and only if equation (47) and, for every point  $r$ , equation (42) hold. This whole series of results contains an extraordinary wealth of relations, which I will now express in words along with the theorem itself.

I will call the magnitude  $\alpha$  the *mean magnitude* of the multiple point square  $\alpha a^2$  and say that the mean magnitude of an inner displacement product is zero; further, I will call the magnitude  $\alpha(a - r)^2$  the *deviation* of the multiple point square  $\alpha a^2$  from the point  $r$  and say that the deviation of an inner displacement product from an arbitrary point is equal to this inner displacement product itself. Given this, the results of the foregoing can be stated in the following theorem:

An equation whose terms are multiple point squares and inner displacement products is true if and only if the two equations, which result if one replaces the terms with their mean magnitudes or with their deviations from an arbitrary point, are true.

Or, since from equations (46) and (47) the equations (41) and (42) follow as well, if one defines by an inner magnitude a sum of multiple point squares and inner displacement products, and by the center of the inner magnitude the point which is the mean between the points that appear in it endowed with the corresponding weights, then:

**Theorem (18).** *An equation whose terms are inner magnitudes is true if and only if both the equation of their mean magnitudes and that of their deviations from an arbitrary point are true.*

Since the deviation from an arbitrary point is, as inner displacement product, itself an inner magnitude whose mean magnitude is zero, and the mean magnitude of a sum of inner magnitudes is a magnitude of first order, thus itself either a point magnitude or a displacement, there follows from the last theorem at once:

**Theorem (18a).** *An equation between inner magnitudes is true if and only if both the equation between their mean magnitudes and that between the deviations from an arbitrary point of those inner magnitudes are true.*

## § 1.11. First Type of Sum of Inner Magnitudes: Inner Displacement Product

From this theorem it follows immediately that, if one asks for the conditions under which a sum of inner magnitudes can equal zero, then three essentially different cases can enter here which furnish the three types of inner magnitude. Thus the sum of the mean magnitudes, or what is the same thing, the mean magnitude of the sum can be zero or a displacement or a point magnitude. We will separate the consideration of the special cases according to these three principal cases.

First

**Theorem (19).** *If the sum of the mean magnitudes belonging to the inner magnitudes, and thus also the mean magnitude belonging to the sum of those inner magnitudes itself, is zero, then this sum is an inner displacement product, namely that which is equal to the sum the deviations of those inner magnitudes from any given point.*

This follows not only from the above Theorem 18a, but also immediately, since then

$$\sum \alpha a^2 + A = \sum \alpha a^2 - 2r \times \sum \alpha a + r^2 \sum \alpha + A$$

as  $\sum \alpha a$  and thus also  $\sum \alpha$  are zero; thus that expression

$$= \sum \alpha (a - r)^2 + A$$

since

$$\sum \alpha a^2 - 2r \times \sum \alpha a + r^2 \sum \alpha = \sum \alpha (a - r)^2. \quad (1.39)$$

Simultaneously there is implicit in this the direct proof that in this case the sum of the deviations from any variable point is constant, that is  $\sum \alpha(a - r)^2$  is constant if  $r$  is variable and  $\sum \alpha a$  is zero.

### § 1.12. Second Type of Sum of Inner Magnitudes: Inner Planar Magnitudes

The two other cases now lead us to the concepts of new magnitudes. Thus, second, suppose the sum of the mean magnitudes is a displacement  $p$ , that is (according to Theorem 15), the sum of their weights is zero.

I proceed here from the simplest case, that is by considering the sum  $a^2 + (-1)b^2$ , or what is the same thing, the difference  $a^2 - b^2$ , that is the difference of two point squares. As we have showed above, this is equal to the inner product of a point and a displacement, thus

$$a^2 - (-1)b^2 = (a + b) \times (a - b),$$

and if we set  $a + b = 2c$  here, so that  $c$  is the mean between  $a$  and  $b$ , and we set the displacement  $a - b = p$ , then it follows that

$$a^2 - (-1)b^2 = 2c \times p = c \times 2p, \quad (1.40)$$

if

$$a + b = 2c \quad \text{and} \quad a - b = p. \quad (1.41)$$

By these equations the mean magnitude and the deviation of an inner product  $c \times 2p$  of a point and a displacement is defined, since they must equal the mean magnitude and deviation of the sum  $a^2 + (-1)b^2$  set equal to it. Now the mean magnitude of this sum is equal to  $\alpha + (-1)b = a - b = p$ , that is, the mean magnitude of an inner product of a point and a displacement is equal to half of this displacement. Furthermore, the deviation of the sum  $a^2 + (-1)b^2$  from  $r$  is  $(a - r)^2 + (-1)(b - r)^2$ . This is equal to  $a^2 - b^2 - 2(a - b) \times r$ , and thus to  $(a + b - 2r) \times (a - b)$ , whence upon substitution from (50) equal to  $(c - r) \times 2p$ . The



last expression found is thus also equal to the deviation of the inner product  $c \times 2p$  from the point  $r$ , that is, the deviation of an inner product  $c \times q$  of a point and a displacement from a point  $r$  is equal to an inner displacement product whose first factor is the displacement of that product and whose other factor is the deviation  $c - r$  of the point  $c$  of that product from the point  $r$ .

Now if I have an arbitrary sum of inner magnitudes  $\sum \alpha a^2 + \sum A$  such that the mean value of this sum is a displacement  $p$ , then the question is whether I can also equate this sum to an inner product  $c \times q$  of a point  $c$  with a displacement  $q$ .

This equality would obtain (according to Theorem 18) if and only if the mean magnitude as well as the deviation from any given point  $r$  for  $\sum \alpha a^2 + \sum A$  is just as great as for  $c \times q$ . The mean magnitude of  $c \times q$  will then equal  $p$ , that is necessarily  $q = 2p$ , and the question is now only whether  $c$  can be chosen so that the deviation of the product  $c \times 2p$  from a point  $r$  is equal to the deviation of that sum from the same point  $r$ . Let the latter deviation be  $A_r$ ; whence  $A_r$  represents an inner displacement product; it is then asked whether  $c$  can be chosen so that

$$(c - r) \times 2p = A_r. \quad (1.42)$$

If one first takes any given point  $c'$  with the property that  $(c' - r) \times 2p$  is not zero, that is  $c'$  is different from  $r$  and  $c' - r$  is not perpendicular to  $p$ , then since inner displacement products are always proportional to numerical magnitudes (Definitions 3 and 4), it is possible to set

$$(c' - r) \times 2p = mA_r,$$

where  $m$  denotes some nonzero numerical magnitude. If one then takes

$$(c - r) = \frac{1}{m}(c' - r),$$

that is if one takes a displacement whose initial point is  $r$  and which equals the  $m$ -th part of the displacement  $(c' - r)$ , and calls its final point  $c$ , then

$$(c - r) \times 2p = \frac{1}{m}(c' - r) \times 2p = \frac{1}{m} \cdot mA_r = A_r,$$

that is, the point  $c$  thus found satisfies equation (51). Thus in fact  $c \times 2p$  is now equal to that sum.

Before we express this result in the form of a theorem, we will look at the significance of an inner product of a point with a displacement; this significance depends on the answer to the question of when two such products  $a \times p$  and  $b \times q$  can be equated. Should they be equal, then first of all their mean magnitudes must be equal, that is it is necessary that  $p = q$ . Thus it is asked under what circumstances  $a \times p = b \times p$ . Obviously if and only if  $(a - b) \times p = 0$ , that is, either  $a = b$  or  $a - b$  is perpendicular to  $p$ , or in sum if  $a$  and  $b$  lie in a single plane orthogonal to  $p$ . Thus since two products of a point and a displacement are equal if and only if these displacements, as well as those planes orthogonal to the displacements laid through the points, coincide, that displacement and this plane define the concept of that inner product. We therefore call the inner product of a point with a displacement a *planar magnitude*, and indeed to distinguish it from the planar magnitude given by outer multiplication (Grassmann's *Ausdehnungslehre* § 114), an *inner planar magnitude*, and the surface in which the point factor of that product can freely be moved without the value of the product changing, the *factor surface* of the planar magnitude.

According to these definitions we can now summarize the results obtained in the following theorem:

**Theorem** (20). *The inner product of a point with a displacement is an inner planar magnitude whose mean magnitude is half of this displacement, and whose factor surface is the plane orthogonal to the displacement laid through the point. The deviation of the product from the point  $r$  is equal to an inner product whose one factor is equal to double its mean magnitude (or the displacement of that product), and whose other factor is the deviation of an arbitrary point of the factor surface from the point  $r$ . The sum of several inner magnitudes is an inner planar magnitude if and only if the mean magnitude of that sum is equal to a nonzero displacement.*

Herein is also implicit the special theorem that the sum of planar magnitudes again yields a planar magnitude if the mean

magnitude of the sum is not zero, while it was already proved in Theorem 19 that this sum is equal to an inner displacement product if its mean magnitude is zero.

In order to be able to compare the inner planar magnitude with the outer (Ausdehnungslehre § 114), let an arbitrary equation between inner planar magnitudes and inner displacement products

$$\sum a \times p + \sum q \cdot v = 0 \quad (1.43)$$

be given, in which the magnitudes  $a$  represent points,  $p, q, v$  displacements, and let  $P, V$  be the surface areas orthogonally proportional to the magnitudes  $p, v$  (cf. Definition 5). Then the outer products  $a \cdot P$  will be outer planar magnitudes, while the outer products  $q \cdot V$  represent spatial volumes. Now it can easily be shown that, if equation (52) is true, then the equation

$$\sum a \cdot P + \sum q \cdot V = 0 \quad (1.44)$$

must also be true, and conversely (52) follows again from (53). For according to Theorem 18 the former (52) is true if and only if  $\sum p = 0$  and

$$\sum (a - r) \times p + \sum q \times v = 0 \quad (1.45)$$

for any given point  $r$ , and likewise according to Grassmann's Ausdehnungslehre § 112, the latter (53) if and only if  $\sum P = 0$  and

$$\sum (a - r) \cdot P + \sum q \cdot V = 0 \quad (1.46)$$

for any given point  $r$ . Now we have shown above (Theorem 5) that, if those equations (54) are valid, then these (55) must also be valid and conversely, whence also (52) will be true if (53) is, and conversely<sup>7</sup>. Thus:

**Theorem (21).** *Inner products, of whose two factors at least one must be a displacement, the other an arbitrary magnitude of first order, stand in the same relation as the outer products that*

---

<sup>7</sup>The proof in the original refers to Theorem 5.

*result if one replaces those displacements with the orthogonally proportional surface areas, leaves the other factor in the product unchanged, and changes the symbol for inner multiplication into that for outer; that is, each true equation between those inner products remains true if one replaces them with these outer products and conversely.*

On account of this correspondence in meaning I have designated both types of magnitude with the same name of planar magnitude.

### § 1.13. Addition of Inner Planar Magnitudes

I will now examine the addition of two planar magnitudes and that of a planar magnitude and an inner displacement product in detail.

For the addition of two planar magnitudes we can distinguish two cases, namely that their planes intersect or not. In the first case let  $a$  be a common point of the two planes; then

$$a \times p + a \times q = a \times (p + q),$$

that is, the sum of two planar magnitudes that intersect is a planar magnitude whose plane has the same line of intersection with both of those of the planar magnitudes to be summed, and whose mean magnitude is the sum of those of the summands.

If the planes do not intersect, that is if they are parallel, then the mean magnitudes will be multiples of the same displacement  $p$ . Thus if  $a \times \alpha p$  and  $b \times \beta p$  are the two summands, then

$$a \times \alpha p + b \times \beta p = (\alpha a + \beta b) \times p = s \times (\alpha + \beta)p,$$

if  $(\alpha + \beta)s = \alpha a + \beta b$  and  $\alpha + \beta$  is not zero, that is,  $s$  is the mean between  $\alpha a$  and  $\beta b$ . Thus the sum of two parallel planar magnitudes, the sum of whose mean magnitudes is not zero, is a planar magnitude whose mean magnitude is the sum of those of the summands, and whose plane is parallel to those of the summands and lies so that if one draws a straight line

that intersects this plane at  $s$  and those of the planes of the summands at  $a$  and  $b$ ,  $s$  is the mean of the two point magnitudes that lie on  $a$  and  $b$  and whose weights are as those of the mean magnitudes belonging to the summands.

If the sum of the two mean magnitudes is zero, then since

$$a \times p - b \times p = (a - b) \times p$$

it follows that the sum is then an inner displacement, one factor of which is  $p$ , twice the mean magnitude of that of one summand, and whose other factor is an arbitrary displacement from a point on the plane of the other summand to a point in the plane of the first.

Finally, since

$$a \times p + q \times p = (a + q) \times p,$$

it follows that: The sum of an inner planar magnitude and an inner displacement product is again an inner planar magnitude whose mean value is equal to that of the former and whose plane therefore results when the plane of that former planar magnitude progresses by a displacement  $q$ , which with  $p$ , twice the mean magnitude, yields an inner product that is equal to the given inner displacement product.

### § 1.14. Third Type of Sum of Inner Magnitudes: Spherical Magnitude

I now proceed to the third case of addition of inner magnitudes, that is where the mean magnitude of the sum is a point magnitude. And it is this case that leads to completely new magnitudes, which cannot be regarded as proportional to the earlier species of magnitudes.

First of all, one can always set each such sum whose mean magnitude is a nonzero point magnitude  $\alpha a$  equal to  $\alpha a^2 + A$  only if  $A$  is the deviation of that sum from the point  $a$ ; for the deviation of the magnitude  $\alpha a^2 + A$  from the point  $a$  is equal to  $A$ , whence the mean magnitude and deviation from the point

$a$  is then the same on both sides so that the equation is true (according to Theorem 18). Since  $\alpha a$ , and hence also  $\alpha$ , is not zero here, one can instead of  $\alpha a^2 + A$  write  $\alpha(a^2 + A')$ , where  $A' = A/\alpha$ . For more precise treatment there appear here three essentially different cases, according as  $A'$  is negative, positive, or zero.

In the first case suppose  $A' = -p^2$ ; then  $\alpha(a^2 + A')$  will equal

$$\alpha(a^2 - p^2) = \alpha(a + p) \times (a - p),$$

that is, equal the multiple inner product of two points, whose mean gives  $a$  and whose distance from the mean squared gives  $p^2$ . The mean magnitude  $\alpha a$  of such a multiple point product is thus the mean of the two points multiplied by the coefficient  $\alpha$  (whence the name mean magnitude), and its deviation from the mean is the square of the distance of one of the two points from the mean multiplied by the coefficient  $-\alpha$ . It thus follows at once that two multiple inner point products are equal if the coefficient, the mean of the points and the square of the distance of these points from the mean of the one are equal to those of the other; or otherwise expressed, such a product retains its value if the coefficient is constant and the points remain the endpoints of the diameter of a fixed spherical surface. Since such a product is thus associated with the spherical surface we call the multiple inner product of two points a *spherical magnitude*, and that spherical surface on which the point factors can be moved without changing the value of the product, the *factor surface* of the spherical magnitude.

We now turn to the other case, where  $A'$  is positive and equal to  $p^2$ , whence

$$\alpha(a^2 + p^2) = \alpha(a^2 + p^2).$$

Thus  $a^2 + p^2$  cannot be divided into real point factors, whence that magnitude  $\alpha(a^2 + p^2)$  also has no real factor surface. On the other hand this magnitude can obviously be represented as the sum of point squares or of multiple point squares with positive coefficients, and most simply in the form  $\beta(b^2 + c^2)$ .

In fact one has

$$\alpha(a^2 + p^2) = \beta(b^2 + c^2)$$

if and only if, first, the mean magnitudes of the two sides are equal, thus

$$\alpha a = \beta(b + c),$$

that is

$$\frac{b + c}{2} = a \quad \text{and} \quad \beta = \frac{\alpha}{2};$$

thus  $a$  is the mean between  $b$  and  $c$ , and second, the deviations of the two sides from any given point, for example the mean  $a$ , are equally large. That of the left side is  $\alpha p$ , that of the right  $\beta((b - a)^2 + (c - a)^2)$ , or, since  $\beta = \alpha/2$  and  $(b - a)^2$  is equal to  $(c - a)^2$  because  $a$  is the mean between  $b$  and  $c$ , the deviation of the right side from  $a$  is equal to  $\alpha(b - a)^2$ , that is therefore, second, if  $(b - a)^2 = p^2$ . Thus that magnitude  $\alpha(a^2 + p^2)$  can in fact be represented as half the coefficient  $\alpha$  multiplying the sum of the squares of two points  $b$  and  $c$ , whose mean is  $a$  and whose squared distance from the mean is equal to  $p^2$ . It therefore follows that, without changing the value of the expression, one can replace the points  $b$  and  $c$  with any two others,  $b'$  and  $c'$ , which likewise have  $a$  as their mean and are the same distance from  $a$  as they are, or expressed otherwise, which are endpoints of a diameter of that spherical surface that has the line from  $b$  to  $c$  as a diameter.

This magnitude  $\alpha(a^2 + p^2)$  or  $\alpha(a^2 + A')$ , where  $A'$  is positive, is thus also associated with a spherical surface. We therefore call this magnitude a *spherical magnitude* as well, and will call the surface on which the points, as whose multiple sum of squares that magnitude can be represented, can be moved, the *mean surface* of the spherical magnitude.

In order to examine this idea of a mean surface (as an average), we will consider an arbitrary sum of  $n$  point squares, thus say

$$a_1^2 + a_2^2 + \dots + a_n^2$$

or written more briefly,  $\sum a^2$ , so that the mean magnitude of this sum is  $\sum a$  or  $ns$ , if  $s$  is the center of gravity between the points  $a_1, a_2, \dots$ ; and the deviation from this center of gravity  $s$  is  $\sum(a - s)^2$ . The arithmetic mean of the totality of the

deviations from  $s$  is

$$\frac{\sum(a-s)^2}{n}$$

which may be set  $= p^2$ . Then

$$\sum a^2 = n(s^2 + p^2),$$

since both sides have the same mean magnitude ( $\sum a = ns$ ) and the same deviation from  $s$ , ( $\sum(a-s)^2 = np^2$ ). Now however  $p$  is the radius of the spherical surface that we have called the mean surface, and its square is the arithmetic mean between the different deviations of the point squares  $a_1^2, a_2^2, \dots$ , or, if one imagines a spherical surface laid through each of the points  $a_1, a_2, \dots$  that has the mean  $s$  as center, then the mean surface is a spherical surface with the same center, whose content (that is, since it is a surface, its surface area) is the arithmetic mean between the contents of those spherical surfaces, whence the name mean surface.

We therefore call the magnitude  $\alpha a^2 + A$ , whether  $A$  is negative or positive, and indeed even when  $A$  is zero, a *spherical magnitude*; its mean magnitude is  $\alpha a$ , its center  $a$ , its weight  $\alpha$ , its deviation from the center is  $A$ . We call this deviation  $A$  the *content* or *capacity* of the spherical magnitude.<sup>8</sup>

Before I collect the results achieved in a theorem I will in this connection recall that we set  $A' = A/\alpha$ , and if  $A'$  was negative, the radius of the factor surface squared yielded  $-A'$  (that is a positive value), while if  $A'$  was positive, the radius of the mean surface squared yielded  $A'$  itself. Thus we obtain the theorem:

**Theorem (22).** *There are three types of inner magnitude: the inner displacement product, which is proportional to a numerical magnitude, the inner planar magnitude, which is proportional to*

---

<sup>8</sup>Hereby one is not to think of the cubic content of the sphere, but rather, since the magnitude appears as a quadratic, of the surface area, thus of the content of the spherical surface. In fact the content of the spherical magnitude is proportional to the surface area of the spherical surface multiplied by the coefficient only if one regards the spherical surface as positive if it is the mean surface, and negative if it is the factor surface.



*an outer planar magnitude, and the spherical magnitude. A sum of inner magnitudes yields an inner displacement product, an inner planar magnitude or a spherical magnitude, according as the mean magnitude is zero, a displacement or a point magnitude. In the latter case the content of the spherical magnitude is equal to the deviation of that sum from the center of this spherical magnitude. If this content is zero, then the spherical magnitude is transformed into a multiple or simple point square; if it is not zero, then the sum yields either a spherical magnitude with a real factor surface or a real mean surface, according as the content of the spherical surface divided by the weight has a negative or positive value. Both surfaces are spherical surfaces whose center is the center of the spherical magnitude; the square of the radius of the mean surface is equal to the content divided by the weight, the square of the radius of the factor surface is the opposite of this quotient.*

## § 1.15. Geometric Representation of the Sums of Inner Magnitudes

The problem is now only to fix the sum of two spherical magnitudes or a spherical magnitude with another inner magnitude by the most specific geometric interpretation possible.

First, suppose two inner magnitudes with real factor surfaces (and thus spherical magnitudes of this type, or planar magnitudes) are to be added, whose factor surfaces meet (intersect, are tangent, or coincide): then if  $a$  is a common point, the two of them can be represented in the forms  $a \times b$ ,  $a \times c$ , where  $b$  and  $c$  are arbitrary magnitudes of first order (displacements or point magnitudes, according as the inner magnitudes are to be planar or spherical magnitudes). Since their sum is  $a \times (b + c)$ , it follows that this sum is also an inner magnitude with a real factor surface, this factor surface having all the points in common with those of the summands that these have with each other. Thus:

**Theorem (23A).** *Two inner magnitudes with real factor surfaces that meet yield as sum again an inner magnitude with a*

*real factor surface, this factor surface having all the points in common with those of the summands that these have in common with each other, and whose mean magnitude is the sum of the mean magnitudes of the summands.*

Hereby there follows a most elementary construction of the factor surface of the sum under the conditions imposed. Thus according to Theorem 16, if, supposing the sum of the weights is not zero, one constructs the position of the mean magnitude, that is the mean between the centers of the two spherical magnitudes endowed with their corresponding weights, and about this point as center scribes a spherical surface that goes through one of the common points of the two factor surfaces, then this is the factor surface of their sum; and if, supposing the sum of the weights is zero, one lays a plane through the common points of the two factor surfaces, or if they are tangent, a plane that is likewise tangent at this point where the two factor surfaces, or if they are tangent, a plane that is likewise tangent at this point where the two factor surfaces are tangent, then this plane is the factor surface of the sum.

In order to reduce the summation of spherical magnitudes to geometric perceptions in more general cases as well, I will introduce yet further terminology by which I can always express the geometric relations between spherical magnitudes purely geometrically. Thus I will regard the spherical magnitude  $a^2 - p^2$ , which appears as the inner product of two points  $(a + p)(a - p)$ , as simply equal to a spherical surface whose center is  $a$  and whose radius is  $p$ . Then it is clear that  $a^2 + p^2$ , as being equal to  $a^2 - (\sqrt{-1}p)^2$ , appears as a spherical surface with a real center  $a$  and an imaginary radius  $\sqrt{-1}p$ ; I will call this an *ideal spherical surface*, and the spherical surface  $a^2 - p^2$  its *corresponding real spherical surface*. But I will note that, as a consequence of the definition given above, the capacity of a spherical surface  $a^2 - p^2$  is equal to the negative square of its radius, that is, equal to  $-p^2$ , so that that of the ideal spherical surface is positive.

According to this we can now also express Theorem 23A above thus:

**Theorem (23B).** *The multiple sum of two spherical surfaces*

*that meet is a multiple spherical surface or a planar magnitude, according as the sum of coefficients has a nonzero value or is zero; its surface has the same points in common with the given spherical surfaces as these have with each other, and its mean magnitude is the corresponding<sup>9</sup> multiple sum of the centers of the two spherical surfaces.*

In particular the surface of the difference of two intersecting spherical surfaces is the plane of their intersection, and the surface of the difference of two tangent spherical surfaces is the plane that is likewise tangent to them at their common point of tangency.

I now consider first the difference plane of two spherical surfaces in more general cases as well.

## § 1.16. Difference Plane of Two Spherical Surfaces

The difference of two spherical surfaces  $A$  and  $B$  is equal to a planar magnitude whose mean value is the difference of the two mean values of  $A$  and  $B$ , and whose deviation from an arbitrary given point is equal to the difference of the deviations of those spherical surfaces from the same point. Now since the deviation of the planar magnitude from any point of its plane, but from no others, is zero, this plane, that is the difference plane of the two spherical surfaces, is the locus of a point  $c$  with respect to which the difference of the deviations of the two spherical surfaces is zero, that is deviating equally far from the two spherical surfaces, i.e. a point which, if  $A = a^2 + A$ ,  $B = b^2 + B$ , satisfies the equation

$$(a - c)^2 + A = (b - c)^2 + B. \quad (1.47)$$

First, if both spherical surfaces are real, whence  $A$  is equal to  $-p^2$ ,  $B$  is equal to  $-q^2$ , where  $p$  and  $q$  are displacements that

---

<sup>9</sup>That by the corresponding multiple sum is understood that with the same coefficients indeed hardly needs mentioning.

are the radii of the spherical surfaces, then it follows that:

$$(a - c)^2 - p^2 = (b - c)^2 - q^2. \quad (1.48)$$

Now if  $c$  lies outside of the spherical surface  $A$ , then  $(a - c)^2 > p^2$ , whence it is also the case that  $(b - c)^2 > q^2$ . For it is clear (cf. fig. 11) that the left side expresses the tangent drawn from  $c$  to  $A$ , the right that drawn from  $c$  to  $B$ ; thus the two tangents are equally long. Thus if one scribes a spherical surface about  $c$  as center, whose radius is equal to these tangents, then this will intersect the given spherical surfaces  $A$  and  $B$  orthogonally. The part of the difference plane lying outside of the two spherical surfaces  $A$  and  $B$  is therefore the locus of the centers of all spherical surfaces orthogonally intersecting both of those spherical surfaces simultaneously.

If  $c$  lies inside of  $A$ , then  $(a - c)^2$  is smaller than  $p^2$ , whence  $(b - c)^2$  is also smaller than  $q^2$ . Then one can also write equation (57):

$$p^2 - (a - c)^2 = q^2 - (b - c)^2 = s^2.$$

Then  $s$  is (cf. fig. 12) half of that chord in each spherical surface that is bisected by  $c$ , and the equation states that these halves are equally long for the two spherical surfaces. Thus if I lay a spherical surface about  $c$  as center, whose radius is  $s$ , then it will be halved by each of the two spherical surfaces  $A$  and  $B$ , and the part of the difference plane lying inside the two spherical surfaces (if such exists) is thus the locus of the centers of all spherical surfaces simultaneously halved by  $A$  and  $B$ . Thus:

**Theorem (24).** *The difference plane of two [real] spherical surfaces is the locus of the centers of all spherical surfaces that are either orthogonally intersected or halved by those two spherical surfaces simultaneously. That is, it is that which one has called the plane of like power of the two spherical surfaces or their ideal intersection plane.*

If the spherical surfaces  $A$  and  $B$  do not intersect, the construction of this difference plane can easily be reduced to the case of two intersecting spherical surfaces by using auxiliary spherical

surfaces that simultaneously intersect the given two. The points of the straight line in which the planes of the two intersections of such a spherical surface with  $A$  and  $B$  intersect each other deviate<sup>10</sup> equally far from these two spherical surfaces; thus if with two such intersecting spherical surfaces one constructs two such straight lines, then the plane laid on those lines is the difference plane sought (fig. 13).

Second, if two spherical surfaces  $A$  and  $B$  are ideal, and their corresponding real spherical surfaces  $A'$  and  $B'$  are constructed, then if  $A = -p^2$  and  $B = -q^2$ , whence with  $A = a^2 + p^2$ ,  $B = b^2 + q^2$ ,  $p$  and  $q$  are therefore the radii of their corresponding real spherical surfaces, then one has

$$(a - c)^2 + p^2 = (b - c)^2 + q^2 = v^2; \quad (1.49)$$

then  $v$  is the radius of a spherical surface simultaneously halving  $A'$  and  $B'$ , whose center is  $c$  (cf. fig. 14).

Thus the difference plane of two ideal spherical surfaces is the locus of the centers of all spherical surfaces that halve their corresponding real spherical surfaces.

The construction of this difference plane follows easily, since if one draws from  $a$  and  $b$  the radii  $p$  and  $q$  perpendicular to the straight line  $ab$ , draws a straight line that perpendicularly bisects the displacement connecting the endpoints of these radii, and through the intersection  $c'$  of this straight line with  $ab$  one lays the plane orthogonal to  $ab$ , then this is the difference plane sought, since the spherical surface about  $c'$  as center through the endpoints of the radii halves the spherical surfaces  $A'$  and  $B'$ , and the difference plane is orthogonal to  $ab$ .

Finally, if one of the spherical surfaces, say  $A$ , is real, the other,  $B$ , ideal, and  $B'$  its corresponding real surface, then if  $A = a^2 - p^2$ ,  $B = b^2 + q^2$ , whence  $A = -p^2$  and  $B = q^2$ , then from (56) one has

$$(a - c)^2 - p^2 = (b - c)^2 + q^2 = v^2.$$

---

<sup>10</sup>By the deviation of a point from a spherical surface we understand the same as the deviation of the latter from the former.

Thus  $v$  is the radius of a spherical surface about  $c$  as center, which orthogonally intersects  $A$ , and halves  $B'$ . Consequently:

The locus of the centers of all spherical surfaces that orthogonally intersect a given spherical surface  $A$  and simultaneously halve another  $B'$  is a plane, the difference plane between the former  $A$  and the ideal spherical surface  $B$  corresponding to the latter.

One can easily reduce the construction of this plane to one of the above two, by letting the capacities  $-p^2$  and  $+q^2$  increase or decrease by equal amounts, and indeed by so much that both spherical surfaces become either real or ideal; the difference, and thus also the difference plane, will not be altered, and by this increase or decrease spherical surfaces result, concentric with those given, and whose square radii increase or decrease by exactly the same amount, which are thus always easy to construct.

Now the converse problem can even be formulated, that is if one,  $A$ , of two spherical surfaces  $A$  and  $B$  is given along with the center of the other and their difference plane (which, if the solution is to be possible, must have an orientation orthogonal to the line connecting the centers), to find the other  $B$  or, if it should be ideal, the real spherical surface  $B'$  corresponding to it, by a construction.

If the given spherical surface is real, then if it intersects or is tangent to the difference plane, the other spherical surface  $B$  is specified since the difference plane intersects it in the same circular line or is tangent to it at the same point. On the other hand, if the given difference plane does not intersect the real spherical surface  $A$  then one need only lay about the point  $d$ , where the difference plane intersects the line  $ab$  connecting the centers, a spherical surface that orthogonally intersects the given spherical surface  $A$ . Then if  $b$  lies outside of  $F_r$ , that spherical surface  $B$  laid about  $b$  that likewise orthogonally intersects is the one sought; but if  $b$  lies inside the spherical surface  $F_r$  then  $B$  is an ideal spherical surface, and the real one  $B'$  corresponding to it is the spherical surface, halved by  $F_r$  that has  $b$  as center.

If however  $A$  is an ideal spherical surface, and the real one  $A'$  corresponding to it is given by a construction, then one need

only draw the radius of the spherical surface  $A'$  perpendicular to  $ab$ , and through the endpoint of this radius lay a spherical surface that has the intersection  $d$  of the connecting line  $ab$  and the difference plane as center; and again either the spherical surface  $B$  sought is that which is orthogonally intersected by  $F_r$ , or the spherical surface  $B$  is ideal, and the corresponding real surface  $B'$  is that halved by  $F_r$ .

# **A New Branch of Mathematics**

The Ausdehnungslehre of 1844, and Other Works

by

Hermann Grassmann

Pages 401–450

Chunk 9 of 12





## Multiple Sums of Spherical Surfaces

Now that I have developed the characteristics of the difference plane, I proceed to the more general problem of constructing the surface  $F$  of a multiple sum  $\alpha A + \beta B$  of two spherical surfaces  $A$  and  $B$ , or, should it be ideal, its corresponding real surface  $F'$ .

If  $\alpha + \beta = 0$ , and thus  $\beta = -\alpha$ , then that multiple sum will equal  $\alpha(A + B)$  and yield a planar magnitude whose plane is identical to the difference plane  $A - B$  and thus is found according to what went before. We therefore assume that  $\alpha + \beta$  is not zero; then the sum is a spherical magnitude whose weight is  $(\alpha + \beta)$ , which can therefore be represented in the form  $(\alpha + \beta)F$ , where  $F$  is the spherical surface sought. One therefore has

$$\alpha A + \beta B = (\alpha + \beta)F. \quad (59)$$

If this equation is to be true, then (according to Theorem 18) first of all the mean magnitudes of the two sides must be equal, that is, the center of  $F$  must be the mean between the centers of  $A$  and  $B$  endowed with the coefficients  $\alpha$  and  $\beta$ , whereby the center of  $F$  is found. Second, according to the same theorem the deviations of the two sides from any given point  $r$  must be equal; if the deviation of a spherical surface from a point  $r$  is symbolized by  $A_r$  for short, then necessarily

$$\alpha A_r + \beta B_r = (\alpha + \beta)F_r; \quad (60)$$

and conversely, if the center of  $F$  is defined in the manner given, and the deviations in (59) from any given point are equal, then equation (59) is true, all this according to Theorem 18. Now if  $c$  is any given point of the difference plane between  $A$  and  $B$ , then according to the above one has  $A_c = B_c$ ; thus if this is substituted into (60), after one has replaced  $c$  by  $r$  one obtains

$$(\alpha + \beta)A_c = (\alpha + \beta)F_c,$$

that is, if a point deviates equally far from two or (since the conclusion can be extended to arbitrarily many terms) more spherical surfaces, it deviates just as much from the spherical surface of their multiple sum if the sum of coefficients is not zero; and for our case it follows that, if a spherical magnitude is the sum of two others, the difference planes between each pair of their corresponding spherical

surfaces coincide, and also conversely, if this is the case and simultaneously the mean magnitude of one spherical magnitude is the sum of those of the other two, then that spherical magnitude is also the sum of these two.

Thus the construction of the sum surface  $F$  follows easily. If one constructs the difference plane  $A - B$  and the mean between  $\alpha a$  and  $\beta b$  (where  $a$  and  $b$  are the centers of  $A$  and  $B$ ) and lays about this mean a spherical surface according to the method of construction given above, which likewise has the plane  $A - B$  as difference plane with  $A$ , then this is the spherical surface of the sum sought.

## Conclusion

With this I believe the most important laws on inner products of two magnitudes of first order have been given. To be sure there is still immediately implicit in what went before the seed of a fuller conception of the inner product, whose development in the same way also seems proper and thus appears necessary for the organic unity of the whole.

Thus we had already in Definitions 6 and 7 and the theorems following therefrom obtained the inner products of two extensions of higher order, each of which appears as the outer product of displacements, and furthermore, we had in the transference law (Theorem 12) and in the consequent Definition 8 presented the procedure according to which one can from such equations in which only displacements enter obtain the corresponding equations in which points replace the displacements. Thus instead of extensive magnitudes conjoined by inner products we obtain products of points, that is elementary magnitudes of higher order (cf. Grassmann's *Ausdehnungslehre*, part 2). And in Theorem 13 we have already shown that the set of laws generally valid for inner products of extensive magnitudes must also be valid for those elementary magnitudes, or generally for those magnitudes that result if one replaces the displacements, as whose products those extensive magnitudes appear, with arbitrary point magnitudes.

It is thus a question of presenting the intuitive concepts of these conjunctions whose laws are given by the theorems cited. Since thereby the nature of this presentation offers no more difficulties, as it proceeds in complete analogy with the procedure introduced for the inner product of two points, I believe that it may be omitted. On the other hand the conception of the quotient, which we

have thus far completely overlooked, and in particular the quotient of nonparallel displacements, leads to new conjunctions completely different from all previous ones, that is to the laws of powers, roots, and logarithms of spatial magnitudes in which the angle appears as the power of an exponent or as a logarithm. Yet since this, which one can already take from the completely new methods of conjunction that hereby apply to geometry, would have entailed a completely novel order of development, whose explication up to a point from which the unity of the whole could be apprehended, would involve a discussion of greater extent than the present one, and although this order of development can indeed also include the Leibnizian characteristic, yet it is not so closely linked to it as the analysis given here. Consequently I believe it should not further increase the extent of this essay. I also believe that the development given here will already suffice to show how correctly Leibniz had foreseen the characteristic benefit of a pure geometric analysis.

Leibniz emphasized that the solution of a geometric problem by means of this analysis simultaneously provides the solution, the construction, and the argument, and indeed in such a way as is given of necessity by the analysis itself.<sup>1</sup> Now since in the analysis presented here each equation is only the expression for a geometric relation clad in analytic form, and this relation is expressed purely and clearly in the equation, without being concealed by arbitrary magnitudes—as perhaps the coordinates of the standard analysis—, it can thus be read off without further ado; and since in addition each transformation of such an equation is only the expression of a construction accompanying it, it follows that in fact in the analysis given the analytic solution of a geometric problem proceeds simultaneously with the construction and the argument. Furthermore, since nothing arbitrary, having no necessary connection with the nature of the problem, as the coordinates of analytic geometry, was introduced, the type of solution must always both be conformable to the nature of the problem, and since it has the form of an analysis, also a necessity, whence there can be no question of searching about for methods of solution.

---

<sup>1</sup>“But this new characteristic ... cannot fail to give at the same time the solution, the construction, and the geometric demonstration, all in a natural way and by an analysis, that is to say, by specific methods.” And again he says: “Its principal utility consists in the consequences and reasonings that it makes possible by operations on the characters, which cannot be expressed in the figures ..., instead of which this method would lead surely and without pains.”

In order that this analysis satisfy all demands in the latter respect, the theory of equations, that is the way that unknown magnitudes can be eliminated from them, must admittedly still be fully developed. But one sees that this theory is at least possible according to the principles taken as fundamental.

In addition he emphasizes as an essential benefit of the geometric analysis that with it one must be able to treat mechanics almost as geometry, and in general only by means of it can one hope to penetrate deeper into the mathematical treatment of physics, and for example explore the inner constitution of natural bodies.<sup>2</sup> Now I believe I have shown in the application to mechanics presented above how in fact mechanics can be treated in a purely geometric way by this analysis, whence it already follows that this method of treatment, applied to physics generally, would simplify the mathematical treatment superbly, whereof, were I given the space, I could have given simple examples from optics, acoustics, electrodynamics, and other branches of physics. Finally, I also believe that one is thereby no longer far removed from discovering the inner constitution of natural bodies, that is the relative positions of their simple or compound atoms; in any case it is already clear from the applications that this analysis permits in crystal structure (cf. Grassmann's *Ausdehnungslehre* §171) that this new analysis is indispensable if one is not to annihilate the intuition by the introduction of coordinates and other apparatus disturbing the treatment, and entangling the method in useless details.

Finally, at the end of Leibniz's presentation is yet another remarkable point where he quite clearly expresses the applicability of this analysis to objects that are not of spatial nature, but adds that it is not possible to give a clear concept of this in a few words.<sup>3</sup>

Now in fact, as is demonstrated throughout Grassmann's *Ausdehnungslehre*, all concepts and laws of the new analysis can be developed completely independently of spatial intuitions, since they can

---

<sup>2</sup>He says: "I believe that by this means one can handle mechanics in nearly the same way as geometry" and "Finally, I do not expect that one can proceed very far in physics before having to find some such shortcut . . . There is however some hope of achieving it in part (that is, to investigate the inner constitution of composite matter), once this truly geometric analysis is established."

<sup>3</sup>"I have but one remark to add, which is that I see that it is possible to extend the characteristic to those cases that are not subject to the imagination; but that is too important and would take me too long to explain in a few words."

be tied to the abstract concept of a continuous transformation; and, once one has grasped this idea of the pure, conceptually interpreted continuous transformation, it is easy to see that the laws developed in this essay are also capable of this interpretation, stripped of spatial intuitions. Thus this thought of Leibniz is also realized, so that now, as it appears to me, nothing essential remains to be added to bring to light the correctness of all that he claimed about the geometric analysis, apart from isolated exaggerations, which however generally reside more in their expression than in the subject itself; and also in this respect to recognize the wonderful force of an intellect, which from the first beginnings of an unimaginably great series of developments was able to command a view of the full importance of this series and the essential and distinctive advantages that it must offer.

# THE GRASSMANNIAN THEORY OF POINT MAGNITUDES AND THE MAGNITUDE FORMS DEPENDING ON THEM

by A. F. Möbius

The study of the preceding article, and particularly of its last part (from § 14 on) may, despite the unmistakeable endeavor of its author for clarity, nevertheless be associated with some difficulties, which result because the author sought to base his new analysis on a method that lies rather far from the currently standard procedures of mathematical investigations, and because in analogy with arithmetic operations he treats objects as magnitudes that are not magnitudes as such, and of which one can in part form no representation. Since however this new analysis seems to merit every attention on account of the simplicity by which geometric investigations can be conducted with it, I have attempted in the following to base it more suitably, and thus, as I hope, in a more easily comprehensible way, on the spirit of geometry, and to show how those apparent magnitudes can be regarded as abbreviated expressions of real magnitudes. However I have not written my essay in the form of a commentary but rather as an independent article, so that it will also be understandable to anyone even prior to the reading of the foregoing. —The terminology introduced by Herr Grassmann I have retained unaltered, nor have I essentially deviated from his procedures in the merely mathematical developments. What beyond this has been introduced by me is of insufficient importance to mention particularly here.

### § 0.1.

By lines will always be understood only straight lines in the following.

Of two lines  $AB$  and  $CD$ , which are either parallel or parts of one and the same straight line, one says that they have the same (opposite) direction if, when one line, say  $CD$ , is produced parallel to itself from the coincidence of its initial point  $C$  with the initial point  $A$  of the other, the final points  $B$  and  $D$  of the two lines lie on the same side (opposite sides) of the now common initial point.

The placement of an equal sign between the expressions for two lines, e.g.

$$AB = CD,$$

will always here indicate that the lines  $AB$  and  $CD$  are not only of equal length, but that they also have the same direction. Likewise, by the formula

$$CD = \alpha AB$$

in which  $\alpha$  denotes a numerical coefficient, one understands that the absolute lengths of  $AB$  and  $CD$  are as 1 to  $\alpha$ , and that  $CD$  has the same or opposite direction as  $AB$ , according as  $\alpha$  is positive or negative. In regard to this,  $BA = -1AB$  is always replaced by the abbreviation  $-AB$ .

## Geometric Addition of Lines

### § 0.2.

If the lengths and directions of two or more lines  $AB$ ,  $CD$ ,  $EF$  are given and from an arbitrary point  $O$  one makes  $OP = AB$ ,  $PQ = CD$ ,  $QR = EF$ , then the straight line  $OR$ , whose initial and final points are the initial and final points of the broken line  $OPQR$  produced by the whole operation, is called the *geometric sum* of the lines  $AB$ ,  $CD$ ,  $EF$ , and the operation itself *geometric addition*. — In the same way  $SV$  is found as the sum of the lines furnished with coefficients  $\alpha AB$ ,  $\gamma CD$ ,  $\epsilon EF$  if, proceeding from an arbitrary point  $S$ , one makes  $ST = \alpha AB$ ,  $TU = \gamma CD$ ,  $UV = \epsilon EF$ . — If the final point of the broken line  $OPQR$  or  $STUV$  should coincide with its initial point, and thus forms a closed polygon, then the sum is  $= 0$ .

It is easy to show that the length and direction of the line that expresses the sum are independent both of the position of the initial point  $O$  or  $S$  and of the order in which the lines to be added are



adjoined, and thus depend only on the lengths and directions of these lines.

**Corollary.** If a line  $OR$  is the geometric sum of several other lines  $AB, CD, EF$ , and all these lines are projected by parallel planes on a straight line, then the projection of the first  $OR$  is the algebraic sum of the projections of the other lines. This becomes clear at once, since if, in order to find  $OR$ , one makes  $OP = AB, PQ = CD, QR = EF$  as before, and likewise projects  $OP, PQ, QR$  on two equal and equally directed lines, the projections are also equal and equally directed, and the projection of  $OR$  is equal to the algebraic sum of the projections of  $OP, PQ, QR$ .

### § 0.3.

That  $SV$  is the geometric sum of  $\alpha AB, \gamma CD, \epsilon EF$  will be expressed by the equation

$$SV = \alpha AB + \gamma CD + \epsilon EF.$$

Likewise, the equation

$$\alpha AB + \gamma CD - \epsilon EF = \lambda KL - \mu MN$$

signifies that the two lines represented by the sums of  $\alpha AB, \gamma CD, -\epsilon EF$ , and of  $\lambda KL, -\mu MN$  have equal lengths and the same direction.

From the nature of the geometric combination it follows that one can treat equations of this type as algebraic, to the extent that they remain in linear form with regard to the lines on which they occur; thus one can place terms on one side of the equal sign on the other with a sign change, multiply or divide all terms with the same number, and add two or more such equations together, or subtract one from the other, i.e. add it to the other with sign interchanged.

### § 0.4.

If, for lines to be added, the initial point of the second and each one following coincides with the final point of the one immediately preceding it, and thus they already form a broken line themselves, as e.g. the lines  $AB, BC, CD$ , then the whole operation given in § 2 is superfluous. One then has directly

$$AB + BC = AC, \quad AB + BC + CD = AD, \text{ and thus also}$$

$$AB + BC + CA = 0, \quad AC = AB - CB = BC - BA,$$

etc.—equations which are valid even if the points  $A, B, C, D$  should lie in space (i.e., even if they are not coplanar).—Cf. my *Elemente der Mechanik des Himmels*, §§1 and 2, and my article in *Crelles Journal für Mathematik*, vol. 28, p. 1.

## Inner Multiplication of Lines

### § 0.5.

By the product of two lines will here be understood, as usual, the number of surface units that result if, after fixing a certain line as unit length and the square constructed upon this line as the surface unit, one multiplies the two numbers according to which the two former lines are measured by the unit length.

The product of two lines  $AB$  and  $CD$ , multiplied by the cosine of the angle that is made by the directions of the two lines, is called the *inner product* of the two lines<sup>4</sup> and will be expressed by

$$AB \times CD,$$

or, what is the same thing: the inner product of two lines  $AB$  and  $CD$  is = the product of one of the lines with the orthogonal projection of the other line on it, and thus = the product of the lines  $AB$  and  $C'D'$ , if  $C'$  and  $D'$  are the orthogonal projections of the points  $C$  and  $D$ , or their prolongations, on the line  $AB$ ; and this product is to be taken as positive or negative, according as  $AB$  and  $C'D'$  have the same or opposite directions.

### § 0.6.

Since the inner product of two lines depends only on their lengths and the angle of their directions, it remains unchanged if instead of one of the lines, or instead of both, others are substituted that have the same lengths and the same directions. Thus if  $AB = EF$  and

---

<sup>4</sup>The product of a line  $a$  with another  $b$ , multiplied by the sine of the angle  $a \angle b$ , will be called the *outer product* of  $a$  with  $b$  (Grassmann's *Ausdehnungslehre* § 38). Since  $\sin b \angle a = -\sin a \angle b$ , the outer products of  $a$  with  $b$  and  $b$  with  $a$  are equal but opposite, while, since  $\cos b \angle a = \cos a \angle b$ , the value of an inner product, including its sign, remains unchanged upon interchange of its factors.

$CD = GH$ , then  $AB \times CD = EF \times GH$ . On the same grounds it becomes clear that

$$\alpha(AB \times CD) = \alpha AB \times CD = AB \times \alpha CD.$$

### § 0.7.

From the concept of the inner product it also follows that if we hold the lengths of two lines constant but regard their angle as variable, then their inner product will be zero if one of the lines makes a right angle with the other; that it attains its largest positive (negative) value if the lines have the same (opposite) directions, and that this value, apart from sign, is equal to the product of the lines themselves.—The inner product of two lines of equal length and the same direction is the square of one or the other of the lines, which as usual one designates by placing an exponent 2 as right superscript to the expression for the line.

### § 0.8.

Let  $a, b, f$  be three lines of arbitrary length and direction, and let  $a + b = c$ . According to § 2, the projection of  $c$  on any straight line is then equal to the algebraic sum of the projections of  $a$  and  $b$  on the same straight line.

Consequently, if  $a, b$  and  $c$  are orthogonally projected onto  $f$  one has the algebraic equation

$$c \cos \angle c = a \cos \angle a + b \cos \angle b, \text{ whence also}$$

$$f c \cos \angle c = f a \cos \angle a + f b \cos \angle b.$$

However, instead of  $f c \cos \angle c$  one can write  $f \times c = f \times (a + b)$ , instead of  $f a \cos \angle a$ ,  $f \times a$ , and so forth; whence

$$f \times (a + b) = f \times a + f \times b,$$

in which equation the plus sign on the left has the geometric significance given in § 3, while the plus sign on the right, as well as the equal sign, has the usual arithmetic significance.—Generally, as one may remark here, the significance of the equal sign in equations between inner products does not differ from conventional usage; for it

implies the equality of the number of surface units appearing on either side, while in equations whose terms are simply lines it implies equality of direction as well. Thus for example, if the two lines  $a$  and  $b$  are equally long, then  $a = b$  only if they also have the same direction; on the other hand it is always true that  $a^2 = b^2$ , whatever their directions may be.

If a fourth line  $g$  is added to the above three lines  $a$ ,  $b$ ,  $f$  then one has as well

$$(f+g) \times (a+b) = (f+g) \times a + (f+g) \times b = f \times a + g \times a + f \times b + g \times b,$$

and likewise the product of the sums of even more lines can be decomposed into partial products completely in accord with the rules of arithmetic; or, what is the same thing: calculations with arbitrarily directed lines, not only with regard to their geometric addition but also with regard to their inner multiplication, are handled completely as if all the lines were contained in one and the same straight line.

To adduce a very simple example, if one sets  $a + b = c$ , then  $a^2 + 2a \times b + b^2 = c^2$ . Thus if  $a$  and  $b$  are mutually perpendicular, then  $a \times b = 0$ , and thus  $a^2 + b^2 = c^2$ , whereby, as one immediately perceives, the Pythagorean theorem is proved.

## On Point Magnitudes and Magnitudes of First Order in General

### § 0.9.

If, in an equation between lines, two or more of them have a common initial point, and if the equation possesses the characteristic that it remains valid wherever the common initial point may be taken, while the final points of these lines and the initial and final points of the remaining lines are fixed points, then the former lines are called *point magnitudes*, and their fixed final points the *positions* of the point magnitudes.—The common, but indeterminate initial point of the point magnitudes will always be symbolized by  $X$  here.

Lines of fixed length and direction are called *displacements*.

Thus for example, wherever  $X$  may lie,  $XB - XA = AB$ , i.e. the difference of two point magnitudes whose positions are  $A$  and  $B$  is equal to and equally directed with a displacement of the line  $AB$ .

Further, if  $C$  is the midpoint of  $AB$ , then for each position of  $X$

one has  $XC - XA = AC = CB = XB - XC$ , and therefore

$$XA + XB = 2XC,$$

i.e. the sum of two point magnitudes is equal to twice a point magnitude whose position is the midpoint between the midpoints of the first two.

For calculation with point magnitudes one can for short drop the letter  $X$  designating the common initial point of these lines throughout, and thus for the two equations above also write

$$B - A = AB, \quad A + B = 2C.$$

Even so, I will make no use of this abbreviated notation in the following, where only the theory of point magnitudes, but no applications of this theory, will be presented.

## § 0.10.

In the equations that concern us at present every term is either a magnitude of *first order*, i.e. a line, or every term is a magnitude of *second order*, that is an inner product of two lines. Consequently, and since every line is either a displacement or a point magnitude, the general form of an equation of the first type will be

$$k + \Sigma \alpha XA + \alpha' XA' + \dots = 0. \quad (1)$$

Here  $k$  signifies the sum of all the terms that are displacements, a sum which, from what has been said about the addition of lines, is either likewise a displacement or is zero. The subsequent terms are point magnitudes associated with the numerical coefficients or weights  $\alpha$ ,  $\alpha'$ ,  $\dots$ , instead of which, by applying the summation symbol, the sum will be written  $\Sigma \alpha XA$  for short.

Now we seek first of all to ascertain the necessary and sufficient conditions for the existence of such an equation.—Wherever the point  $O$  may lie, one has

$$XA = OA - OX, \quad XA' = OA' - OX, \text{ etc.,}$$

and thus instead of (1) one can also write

$$k + \Sigma \alpha (OA - OX) = 0, \text{ i.e. } k + \Sigma \alpha OA - \Sigma \alpha OX = 0. \quad (2)$$

Now let  $O$  denote an arbitrarily specified point. Since the equation (2) identical to (1) must exist for each position of  $X$ , and thus also if  $X$  coincides with  $O$ , one then has as the first necessary condition the equation

$$k + \Sigma\alpha OA = 0.$$

Hereby, in combination with (2), if  $X$  does not coincide with  $O$ , it follows that  $\Sigma\alpha OX = 0$ , whence  $\Sigma\alpha = 0$ .

The necessary, and as one sees, simultaneously sufficient conditions for the validity of equation (1) are therefore: First, that it holds if a definite point  $O$  is substituted for  $X$ , and second, that the sum of the weights of the points is zero.

For a clearer understanding of the method followed in this investigation, one imagines (§8) all the displacements whose sum is  $k$ , and the points  $X, A, A', \dots$  as lying together in a straight line, and the auxiliary point  $O$  introduced as the initial point of the straight line. Then  $OX, OA, OA', \dots$  are the abscissas of the points  $X, A, A', \dots$ , which abscissas one sets  $= x, a, a', \dots$ . Thereby (2) is transformed into the algebraic equation  $k + \Sigma\alpha(a - x) = 0$ , i.e. into  $M - \mu x = 0$ , where  $M = k + \Sigma\alpha a$  and  $\mu = \Sigma\alpha$ ; and this equation must be true for every value of  $x$ , whence (2) or (1) must be true for every position of  $X$ , if—in accordance with the foregoing—each of the two magnitudes  $M$  and  $\mu$  is separately zero.

### § 0.11.

We observed previously that a sum of displacements is always reducible to a single displacement. Now we seek the simplest form to which a sum of point magnitudes  $\Sigma\alpha XA$  is reducible. If, as before, we introduce the arbitrarily specified point  $O$ , this sum will

$$= \Sigma\alpha(OA - OX) = \Sigma\alpha OA - \mu OX,$$

where  $\mu = \Sigma\alpha$ . Thus if, first of all, the sum of weights  $\Sigma\alpha = 0$ , then

$$\Sigma\alpha XA = \Sigma\alpha OA$$

= a sum of lines independent of  $X$  (and likewise of  $O$ ) = a displacement that may also  $= 0$ .

Second, if the sum of weights  $\Sigma\alpha$  is not  $= 0$ , then one places on  $O$  a line

$$OM = \frac{1}{\mu} \Sigma\alpha OA.$$

Thus  $\Sigma \alpha OA = \mu OM$ , and consequently

$$\Sigma \alpha XA = \mu OM - \mu OX = \mu XM$$

= a point magnitude whose weight  $\mu$  is the sum of the weights of the point magnitudes to be added, and whose position  $M$  is independent of both  $X$  and of  $O$  itself; it is the center of gravity of the weights  $\alpha, \alpha', \dots$  placed on the points  $A, A', \dots$ .

A sum of point magnitudes is thus either = 0, or = a displacement, or = a point magnitude; and the first or second case applies when the sum of weights of the point magnitudes = 0, the third when it is not = 0.

We can therefore conclude conversely that, according as a sum of point magnitudes is reducible to a displacement, which may also be zero, or to a point magnitude, its sum of weights is either zero, or not zero.

## § 0.12.

The sum of a displacement  $KL$  and a point magnitude  $\mu XM$  can always be reduced to a point magnitude whose weight  $\mu$  is equal to that of the former point magnitude. For if one places on  $M$  a line  $MN = \frac{1}{\mu} KL$ , then

$$KL + \mu XM = \mu XM + \mu MN = \mu XN.$$

By combining this theorem with the results obtained in the preceding § we arrive at the conclusion that, just as with a sum of point magnitudes alone, so also a sum of displacements and point magnitudes, and thus a sum of magnitudes of first order generally, according as the sum of weights of the point magnitudes is zero or not, is reducible to a displacement or to a point magnitude whose weight is equal to that sum of weights. In the same way it therefore follows at once that a displacement can be expressed as a sum of two point magnitudes whose sum of weights is zero, thus  $\alpha KL = \alpha XL - \alpha XK$ , and thus that one can represent a sum of displacements and point magnitudes as a sum of point magnitudes only, whose sum of weights is no different from that of the point magnitudes in the former sum.

## On Magnitudes of Second Order

### § 0.13.

A magnitude of second order, i.e. an inner product of two lines (§ 10), is either an inner product of two displacements, that is a fixed number of surface units, or an inner product of a displacement with a point magnitude, as  $AB \times XC$ , or an inner product of two point magnitudes, as  $XA \times XB$ .

However (§ 10),

$$AB \times XC = \frac{1}{4}(XC + AB)^2 - \frac{1}{4}(XC - AB)^2,$$

and if one places two lines  $CE = AB$  and  $CD = -AB = BA$  on  $C$ , so that  $C$  is the midpoint of a line  $DE$  in the same direction as  $AB$  and twice its length,

$$AB \times XC = \frac{1}{4}(XC + CE)^2 - \frac{1}{4}(XC + CD)^2 = \frac{1}{4}XE^2 - \frac{1}{4}XD^2.$$

Furthermore,

$$XA \times XB = \frac{1}{4}(XA + XB)^2 - \frac{1}{4}(XB - XA)^2 = XM^2 - \frac{1}{4}AB^2,$$

if  $M$  denotes the midpoint of  $AB$  (§ 9).

One thus infers that a sum of magnitudes is always representable as a sum of inner displacement products and squares of point magnitudes.

### § 0.14.

Accordingly,—and since a sum of inner displacement products, as a sum of numbers of surface units, can always be replaced by a single inner displacement product,—an equation in which every term is a magnitude of second order can always be brought to the form

$$A + \Sigma \alpha XA^2 = 0, \quad (1)$$

where  $A$  signifies an inner displacement product. To find the conditions for the validity of such an equation, we will, as in § 10 for equations of first order, introduce the point  $O$  to be arbitrarily chosen, and replace  $XA$  by  $OA - OX$ ; then the equation reads

$$A + \Sigma \alpha OA^2 - 2(\Sigma \alpha OA) \times OX + \Sigma \alpha OX^2 = 0.$$



For this to be valid for every position of  $X$ , it must first of all hold as well if  $X$  coincides with  $O$ ; thus it must follow that  $A + \Sigma\alpha OA^2 = 0$ , and consequently

$$2(\Sigma\alpha OA) \times OX - \Sigma\alpha OX^2 = 0. \quad (2)$$

Unlike algebra, however, we cannot conclude from this last equation that at least one of the two factors on the left,  $2(\Sigma\alpha OA) - \Sigma\alpha OX$  and  $OX$ , must be zero, since their inner product is also zero if the two lines that they express together make a right angle (§7). To avoid this ambiguity we will choose the position of  $X$  in such a way that the line  $OX$  is perpendicular to the displacement  $\Sigma\alpha OA$ , should this not be zero. The first term on the left in (2) will then  $= 0$ , and therefore, for this equation to remain true for such a position chosen for  $X$ , it is necessary that  $\Sigma\alpha OX^2 = 0$ , whence in every case  $\Sigma\alpha = 0$ . Thereby it is itself reduced to

$$2(\Sigma\alpha OA) \times OX = 0,$$

an equation that holds for every position of  $X$  if and only if  $\Sigma\alpha OA = 0$ .

The necessary and sufficient conditions for the validity of (1) are thus (a)  $A + \Sigma\alpha OA^2 = 0$ , (b)  $\Sigma\alpha = 0$ , (c)  $\Sigma\alpha OA = 0$ . One can however replace (b) and (c) by the single equation  $\Sigma\alpha XA = 0$ , as the necessary and sufficient condition for the validity of (b) and (c) (§10).

The conditions for the validity of equation (1) are thus first equation (a), or the condition that (1) still holds if a specific point is chosen; and second, that the sum  $\Sigma\alpha XA$ , which will be called the *mean magnitude* of  $A + \Sigma\alpha XA^2$ , is zero.

### § 0.15.

Before we proceed further, it will be a good idea to examine more closely the new concept of mean magnitude. According to the definition just established for the mean magnitude of  $A + \Sigma\alpha XA^2$ , the mean magnitude of  $A$ , i.e. of an inner displacement product, is zero. Since further (§13)

$$AB \times XC = \frac{1}{4}(XC + AB)^2 - \frac{1}{4}(XC - AB)^2, \text{ and}$$

$$XA \times XB = XM^2 - \frac{1}{4}AB^2, \text{ where } M \text{ is the midpoint of } AB,$$

the mean magnitude of  $AB \times XC$  is  $= \frac{1}{4}(XC + AB) - \frac{1}{4}(-XC - AB) = \frac{1}{2}AB$ , and the mean magnitude of  $XA \times XB$  is  $= XM$ .

Thus the mean magnitude of a sum of magnitudes of second order can always be given as the sum of the mean magnitudes of the individual terms, without having first to transform it into the form  $A + \Sigma \alpha XA^2$ .

*Note.* One can also define the mean magnitude of a given changing magnitude in the following way.—Assuming that  $X'$  is a point lying infinitely near  $X$ , then upon the transformation of  $X$  into  $X'$  the given magnitude changes its value by an infinitesimal amount. This change can be represented as an inner product, the first factor of which is the infinitesimally small line  $X'X$ . The other factor halved is called the mean magnitude of the one given.

Thus the change of  $AB \times XC$  is  $= AB \times X'C - AB \times XC = AB \times X'X$ , and thus  $AB$  is the mean magnitude of  $AB \times XC$ .

Further, the change of  $XA \times XB$  is  $= (X'X + XA) \times (X'X + XB) - XA \times XB = X'X \times (XA + XB) + X'X^2$ ; thus, upon dropping the second term as infinitesimally small compared to the first, the mean magnitude of  $XA \times XB$  is  $= \frac{1}{2}(XA + XB) = XM$ .

Likewise the mean magnitude of  $XA^2$  is found to be  $= XA$ ; that of  $XA$ ,  $= \frac{1}{2}$ , etc.

## § 0.16.

The theorem proved in § 14 can also be expressed as follows: two sums of magnitudes of second order are equal if and only if equality between them holds when a specific point is chosen for  $X$ , and if the mean magnitudes of the two sums are equal; —a theorem which, moreover, is also valid verbatim for two sums of magnitudes of first order, since according to the definition given in the preceding note the mean magnitude of such a sum is half the sum of the weights of the point magnitudes included in the former sum.

By means of these criteria for equality one can most easily accomplish the reduction to the simplest form of a given sum of magnitudes of second order, such as in the foregoing was expressed by  $A + \Sigma \alpha XA^2$ . We thus distinguish the only three possible cases (§ 11), according as the mean magnitude  $\Sigma \alpha XA$  of the given sum is either  $= 0$ , or  $=$  a displacement  $k$ , or equal to a point magnitude  $\mu XM$ , and for short we will designate the sum  $A + \Sigma \alpha XA^2$ , as a magnitude

depending on  $X$ , by  $[X]$ ; and thus  $A + \Sigma \alpha OA^2$  by  $[O]$ .

In the first case

$$[X] = [O], \quad (\text{a})$$

i.e. = an inner displacement product. For this equation remains true if the specific point  $O$  replaces  $X$ . Furthermore the mean magnitude of the left side of the equation is  $= 0$  by assumption; that of the right side is also, since the right is independent of  $X$ . Thus the mean magnitudes of the two sides are also equal; whence etc.

In the second case one can set

$$[X] = 2k \times XN. \quad (\text{b})$$

For the inner product  $k \times ON$ , in which  $k$  is a given displacement and  $O$  is a specific point, can assume all possible values by different choices of the point  $N$ . Thus  $N$  can always be specified so that  $[O] = 2k \times ON$ . Thus it happens that the first condition for the equality expressed by (b) is satisfied. But the second condition is also satisfied, since the mean magnitude of the left as well as of the right side of (b) is  $= k$ .

Third, if at last the mean magnitude of the given sum  $= \mu XM$ , then, in order that the mean magnitudes of the two sides be equal, one sets

$$[X] = \mu(F + XM^2), \quad (\text{c})$$

where  $F$  denotes an inner displacement product whose value is to be specified by the other conditions to be fulfilled, so that the equality still applies if  $X$  is replaced by a specific point  $O$ , whence  $\mu F = -[O] - \mu OM^2$  follows.—A sum of magnitudes of second order can therefore, according as its mean magnitude is zero, a displacement, or a point magnitude, be reduced to a inner product of a displacement with a point magnitude, or to the sum of a point square and an inner displacement product.

## § 0.17.

**Corollaries and examples.** 1. The condition  $\Sigma \alpha XA = 0$ , which obtains for the first of the three cases just considered, expresses that, for the points  $A, A', \dots$  with the weights  $\alpha, \alpha', \dots$  adjoined, each is the center of gravity of the others.<sup>5</sup> If this condition is satisfied,

---

<sup>5</sup>For since, if  $\Sigma \alpha XA = 0$ , then always  $\Sigma \alpha = 0$  as well, and thus  $\alpha' + \alpha'' + \dots = -\alpha$  (§ 11), whence after manipulation the former equation is identical to

then it follows from (a) that the sum  $[X]$ , and thus also  $\Sigma \alpha X A^2$ , is independent of  $X$ ; i.e. the sum of the squares of the distances of the points  $A, A', \dots$  from one and the same point  $X$ , multiplied by their respective weights, has the same magnitude for every position of  $X$ .— One can find several inferences derived by me from this previously known remarkable theorem in *Crelles Journal für Mathematik* 26, p. 26.

2. In the second case, where the mean magnitude of  $[X]$  is a displacement  $k$ , a point  $N$  was to be specified such that the equation  $[O] = 2k \times ON$  would be satisfied. This can however be done in innumerable ways. Thus for a point  $N_1$ , different from  $N$ , one also has  $[O] = 2k \times ON_1$ , if  $k \times ON_1 - k \times ON$ , i.e.  $k \times NN_1 = 0$ , that is if the line  $NN_1$  has an orientation perpendicular to the displacement  $k$  (§7), or expressed otherwise, if  $N_1$  and  $N$  lie in one and the same plane orthogonally intersecting the displacement  $k$ .

In the same way it becomes clear that the value of the product of a displacement with a point magnitude, as  $k \times XN$ , is not changed under a change of the position of  $N$  if and only if the new point  $N_1$  is contained in a plane laid through  $N$  that orthogonally intersects the displacement  $k$ . Thus for each point  $N_1$  of this plane, and for no others, is the projection of  $XN_1$  on  $k$  of the same magnitude (§5). One therefore calls the product of a displacement with a point magnitude a *planar magnitude*, and the plane specified in the manner imagined by the direction of the displacement and the position of the point magnitude the *factor surface* of the planar magnitude.

3. In the third case, where  $[X]$  is reducible to the  $\mu$ -fold sum of an inner displacement product and a point square  $XM^2$ , there are two sub-cases to distinguish, according as  $F$  is negative or positive.

If it is negative, then the sum  $XM^2 + F$  can be reduced to a single term. Thus one draws through  $M$  a line in an arbitrary direction, and on it specifies two points  $K$  and  $L$  equidistant in opposite directions from  $M$ , so that  $MK^2 = ML^2 = -F$ ; and thus

$$XM^2 + F = (XM + MK) \times (XM - MK) = (XM + MK) \times (XM + ML) = XK \times XL$$

and consequently  $[X] = \mu XK \times XL$  = the  $\mu$ -fold of the inner product of two displacements, whose positions  $K$  and  $L$  are so specified by  $M$  and  $F$  that  $KL^2 = -\frac{4F}{\mu}$  and  $KL$  is halved by  $M$ , or in other

---

$\alpha' XA' + \alpha'' XA'' + \dots = -\alpha XA = (\alpha' + \alpha'' + \dots)XA$ , whence  $A$  is the center of gravity of the weights  $\alpha', \alpha'', \dots$  situated on  $A', A'', \dots$

words that  $K$  and  $L$  are the two endpoints of any diameter of a sphere whose radius is  $= \sqrt{-F/\mu}$  and whose center is  $M$ . As one thus sees at once, the product of two point magnitudes  $XK \times XL$  has the remarkable property that it remains unchanged if  $K$  and  $L$  are replaced by the endpoints of any other diameter of the sphere described by  $KL$  as diameter. On these grounds the inner product of two point magnitudes is called a *spherical magnitude*, and the surface of the sphere the *factor surface* of the spherical magnitude. The mean magnitude of this last is  $XM$  or the point magnitude that has the center of the sphere as its position.

If  $F$  is positive, then the radius of the sphere is imaginary. We therefore investigate whether in this case the sum  $XM^2 + F$  can be associated with a sphere that also has  $M$  as center, and a radius whose square is  $= F$ . It is in fact found if  $K$  and  $L$  symbolize the endpoints of any given diameter of this sphere:

$$XM^2 + F = XM^2 + MK^2 = \frac{1}{2}(XM + MX)^2 + \frac{1}{2}(XM - MK)^2 = \frac{1}{2}(XM + MK)^2 +$$

The sum of the squares of two point magnitudes ( $XK^2 + XL^2$ ) therefore has the same property as that of an inner product of these magnitudes  $XK \times XL$ , and thus we also call it a *spherical magnitude*. As before the position of its mean magnitude is likewise the center of the circle. The sum  $[X]$  however is  $= \frac{\mu}{2}(XK^2 + XL^2)$ .

4. According to all this we can therefore express the result obtained at the close of the preceding § briefly as follows: According as the mean magnitude of a sum of magnitudes of second order is zero, a displacement, or a point magnitude, the sum itself is either an inner displacement product, a planar magnitude, or a spherical magnitude.

### § 0.18.

In the preceding § we have discovered conditions under which the functions of point magnitudes  $XK \times XL$  and  $XK^2 + XL^2$  remain unchanged when the positions of their points  $K$  and  $L$  change. It is easy to prove that those conditions are simultaneously the only ones under which the functions noted retain their values unchanged.

For if  $XK \times XL$  is not to change if the points  $F, G$  replace  $K, L$ , that is if the equation

$$XK \times XL = XF \times XG$$

is to obtain, then first of all (§ 16) the mean magnitudes of both sides must be equal, thus (§ 15)  $XM = XH$ , if  $M$  and  $H$  symbolize the midpoints of  $KL$  and  $FG$ . It thereby follows that  $XM - XH = HM = 0$ , i.e. the midpoints of  $KL$  and  $FG$  must coincide. The second condition to be fulfilled in order that the equation obtain is that it be true if a specific point is chosen for  $X$ . If for such a point we take the common midpoint  $M$  of the lines  $KL$  and  $FG$ , then since  $-MK = ML = \frac{1}{2}KL$  and  $-MF = MG = \frac{1}{2}FG$ , it is found that  $KL^2 = FG^2$ , i.e.  $KL$  and  $FG$  must have equal lengths. For the invariance of the equation it is therefore not only sufficient but also necessary that  $KL$  and  $FG$  are diameters of one and the same sphere.

In exactly the same way it is shown that the above conditions are necessary and sufficient for the validity of the equation

$$XK^2 + XL^2 = XF^2 + XG^2.$$

For there follows from this that the equation

$$XK + XL = XF + XG$$

between the mean magnitudes must be satisfied, i.e.  $XM = XH$ , where  $M$  and  $H$  have the same significance as before. As before,  $KL$  and  $FG$  must therefore have a common midpoint, and if this point is substituted for  $X$  in the first equation, then one has  $KL^2 = FG^2$ ; whence, etc.

As a supplement to these considerations we will seek to discover how the points  $A$  and  $B$  can be changed in general in the sum of two point squares with arbitrary coefficients, as for example  $\alpha XA^2 + \beta XB^2$ , without the sum itself being changed. —Thus if, when one transforms  $A$  and  $B$  into  $C$  and  $D$ , one has

$$\alpha XA^2 + \beta XB^2 = \alpha XC^2 + \beta XD^2, \quad (\text{a})$$

then already the equation

$$\alpha XA + \beta XB = \alpha XC + \beta XD$$

between the mean magnitudes of the two sums must obtain. Excluding the case that  $\alpha + \beta = 0$ , and where the sums in (a) are planar magnitudes (§§ 13 and 17, 2.), one can however set (§ 11)

$$\alpha XA + \beta XB = (\alpha + \beta)XM, \text{ whence also}$$

$$\alpha XC + \beta XD = (\alpha + \beta)XM.$$

Thus, if one lets  $X$  coincide with  $M$ , it follows that

$$\alpha MA + \beta MB = 0, \quad (b)$$

$$\alpha MC + \beta MD = 0. \quad (c)$$

Since according to (b) the lines  $MA$  and  $MB$  must have either the same or opposite directions (§ 1),  $M$  must be a point of the straight line  $AB$ , and likewise according to (c) a point of the straight line  $CD$ . The two straight lines must therefore intersect in a point  $M$ , such that

$$AM : MB : AB = CM : MD : CD = \beta : \alpha : \alpha + \beta; \quad (d)$$

or, what is the same thing: the center of gravity of the weights  $\alpha$  and  $\beta$ , whether they are placed on  $A$  and  $B$  or  $C$  and  $D$ , must be one and the same point  $M$ .

Second, the equation resulting from (a), when a specific point, say  $M$ , is substituted for  $X$ , must be true, that is the equation

$$\alpha MA^2 + \beta MB^2 = \alpha MC^2 + \beta MD^2.$$

If into this equation one substitutes the values of  $MB$  and  $MD$  those flowing from (b) and (c), then one has  $MA^2 = MC^2$ , whence also  $MB^2 = MD^2$  and  $AB^2 = CD^2$ , on account of proportion (d).

This leads us to the following result:

If one divides the line  $AB$  at  $M$  in the ratio  $\beta : \alpha$  and about  $M$  as center describes two spherical surfaces with  $MA$  and  $MB$  as radii, then the sum  $\alpha XA^2 + \beta XB^2$  will be unchanged under changes in the positions of  $A$  and  $B$  if and only if the new positions  $C$  and  $D$ , just as  $A$  and  $B$  themselves, are the intersections of an arbitrary straight line, laid through  $M$ , with the first and second spherical surfaces; but, according as the ratio  $\alpha : \beta$  is positive or negative, and thus  $A$  and  $B$  lie on the opposite or the same side of  $M$ ,  $C$  and  $D$  must have the same positions relative to  $M$ .

### § 0.19.

The theorem proved in § 16 concerning the reduction of a sum of magnitudes of second order can also be presented by a method similar

to that for the corresponding theorem for magnitudes of first order in §§ 11 and 12, and thus in a more direct way. However I hold it to be excessive to detain myself with the calculations necessary for this, and will only show how the reduction of a sum of second order can be carried out at once in the simpler cases.

1. The sum of a planar magnitude, say  $AB \times XC$ , and an inner displacement product is reducible to a planar magnitude.—For one can set the inner displacement product  $= AB \times CD$ , which product can have any value upon different assumptions for  $D$ . But then the sum becomes  $= AB \times XC + AB \times CD = AB \times XD$ .

2. The sum of two, and consequently also several, planar magnitudes is reducible to a planar magnitude.—For the sum

$$(S) \quad AB \times XF + DE \times XG$$

becomes, if one places a line  $BC = DE$  on  $B$ ,

$$= AB \times XF + BC \times (XF + FG) = AC \times XF + BC \times FG = \text{a planar magnitude, acc.}$$

The summation of two planar magnitudes can also be accomplished in the following somewhat simpler way. Since, if  $AB$  is the displacement factor and  $L$  any point of the factor surface of a planar magnitude, it appears itself as  $AB \times XL$ , one obtains as the sum of two planar magnitudes whose displacement factors are  $AB$  and  $DE$ , and  $L$  is a common point of both factor surfaces, and thus any point on the line of intersection of the two surfaces,

$$AB \times XL + DE \times XL = (AB + DE) \times XL = AC \times XL,$$

if, as before, one makes  $BC = DE$ .

This summation procedure fails to apply only in the case that the two factor surfaces are parallel. Then however the two displacement factors  $AB$  and  $DE$  orthogonal to them are parallel, and one can therefore (§ 1) set  $DE = \alpha AB$ . Consequently the sum (S)

$$= AB \times XF + \alpha AB \times XG = AB \times (XF + \alpha XG) = (1 + \alpha) AB \times XH,$$

and thus  $=$  a planar magnitude, if (§ 14) one sets  $XF + \alpha XG = (1 + \alpha)XH$ . If one lets  $X$  coincide with  $G$  here, then  $GF = (1 + \alpha)GH$ , and the point  $H$  on the straight line  $FG$  is to be defined so that one makes  $GH = \frac{1}{1 + \alpha} GF$ .



3. The sum of point squares  $\alpha XA^2, \alpha'XA'^2, \dots$ , whose coefficients have zero as sum, can be reduced to a planar magnitude. For if  $O$  is an arbitrary specific point, then  $\Sigma\alpha XO^2 = XO^2\Sigma\alpha = 0$ , since  $\Sigma\alpha = 0$ ; and therefore

$$\Sigma\alpha XA^2 = \Sigma\alpha(XA^2 - XO^2) = \Sigma\alpha(XA + XO) \times (XA - XO).$$

However, if  $M, M', \dots$  denote the midpoints of  $OA, OA', \dots$ , then  $XA + XO = 2XM, XA' + XO = 2XM', \text{ etc.};$  whence

$$\Sigma\alpha XA^2 = 2\Sigma\alpha OA \times XM,$$

i.e. = a sum of planar magnitudes, = a single planar magnitude according to 2.

4. Since  $XM = OM - OX$ , one can also write the last equation obtained as

$$\Sigma\alpha XA^2 = 2\Sigma\alpha OA \times OM - 2OX \times \Sigma\alpha OA.$$

Thus if both  $\Sigma\alpha$  and  $\Sigma\alpha OA$  are zero, then  $\Sigma\alpha XA^2 = 2\Sigma\alpha OA \times OM$ ; i.e. a sum of point squares  $\Sigma\alpha XA^2$  can be reduced to an inner displacement product if its mean magnitude  $\Sigma\alpha XA$  is zero (whence  $\Sigma\alpha OA = 0$  and  $\Sigma\alpha = 0$  follow).

5. If  $\Sigma\alpha$  is not  $= 0$ , then according to 3,

$$\Sigma\alpha XA^2 = \Sigma\alpha(XA^2 - XO^2) + \Sigma\alpha XO^2 = 2\Sigma\alpha OA \times XM + \Sigma\alpha XO^2 = \text{a planar m}$$

But this sum can be transformed into the sum of a point square and an inner displacement product in a way similar to that by which  $\alpha x + x^2$  is transformed into  $(x + \alpha)^2 - \alpha^2$ . Consequently a sum of point squares is always reducible to such a sum if the sum of coefficients is not zero. Cf. § 16.

# Editorial Notes



Grassmann's *Geometric Analysis* is his response to a prize problem posed by the Fürstlich Jablonowski'schen Gesellschaft (Princely Jablonowskian Academy) in Leipzig. The problem was posed in spring 1844 for the year 1845 and reads as follows:

“There still remain some fragments of a geometric characteristic invented by Leibniz (cf. *Christiaani Hugenii aliorumque seculi XVII. virorum celebrium exercitationes mathematicae et philosophicae*, ed. Uylenbroek, Hagae comitum 1833, fasc. II., p. 6) in which the relative orientations of places are designated directly by elementary symbols and determined by combinations thereof, without having to draw on the magnitudes of lines and angles, and which is therefore completely different from our algebraic and analytic geometry. It is asked whether this calculus can be recovered and further developed, or whether one similar to it can be suggested, as seems by no means impossible.”

Candidate papers were to have been accepted until the end of November, 1845. In Spring 1845 however the Academy passed another resolution:

“The two prize problems proposed for the year 1845,

1. On History ...
2. On Physics and Mathematics (see above) ...

are repeated by the Academy, and since they are proposed in relation to the bicentennial celebration of the birth of Leibniz, a born Leipziger, which falls in the last week of the month of June 1846, they are in consequence of this extended to the year 1846, and the deadlines for the prize essays indicated under I. and II. is postponed from the end of November, 1845, until the end of March, 1846, the prize being doubled to 45 gold ducats.”

Only one essay was received for the above prize problem, precisely Grassmann's *Geometric Analysis*, and this was awarded the prize. On 1 July 1846, in the first session of the newly established Königlich Sächsischen Gesellschaft der Wissenschaften (Royal Saxon

Academy of Science) Drobisch, in addition to his own speech, gave a speech as Secretary of the Jablonowski'schen Gesellschaft, announcing the result of the prize competition and reporting, probably on the basis of the evaluation written by Möbius, to the Jablonowski'schen Gesellschaft on Grassmann's work. Then in the year 1847 there appeared the *Geometric Analysis* as number 1 of the "Prize Essays awarded and published by the Fürstlich Jablonowski'schen Gesellschaft of Leipzig." It was accompanied by an expository article, "The Grassmannian theory of point magnitudes and the magnitude forms depending on them" presented by A. F. Möbius (cf. Möbius, *Werke*, Bd. I, pp. 613–33).

Leibniz's remarks on the advantages of such a geometric characteristic as he had in mind, and his suggestions about the realizability of this truly remarkable idea, are to be found in a supplement to a letter that Leibniz had written to Huygens from Hannover on 8 September 1679. In this letter he says (cf. the above work edited by Uylenbroek, fasc. I, p. 9):

"But after all the progress I had made in these matters (that is, in the theory of certain equations), I was no longer content with algebra, which provides neither the briefest methods nor the most beautiful constructions in geometry. This was because I believe we need another analysis properly geometrical or linear, by which we express place directly, just as algebra expresses magnitude. And I believe I have seen the means, and how to represent figures and even machines and movements in characters, just as algebra represents numbers or quantities; and I enclose an essay that seems to me worthy of consideration. There is no one with more power to assess it than you, sir, and your opinion I will regard more than that of many others." This essay is precisely the enclosure previously mentioned, and reads as follows:

I have found some elements of a new characteristic, entirely different from algebra, which will have great advantages for representing to the intellect everything that depends on the imagination precisely and naturally, but without figures.

Algebra is nothing but the characteristic of indeterminate numbers or quantities. But it does not directly express

situation, angles, and motion, whence it is often difficult to reduce what is in a figure to calculation, and even more difficult to find sufficiently convenient demonstrations and constructions, even after the algebraic calculation is completely done. But this new characteristic, following the visual figures, cannot fail to give at the same time the solution, the construction, and the geometric demonstration, all in a natural way and by an analysis, that is to say, by specific methods.

Algebra is obliged to assume the elements of geometry, whereas this characteristic carries analysis right through to the end. If it is achieved in the manner I conceive it, one would be able to cast into symbols, that would only be letters of the alphabet, the description of a mechanism, however it might be composed, which would give the intellect a method of knowing it distinctly and easily with all its parts, and even with their use and movements, without figures or models and without inconveniencing the imagination; and one would not need to have a figure of it present to the intellect if one desired an interpretation of the characters. By this means one would also be able to form exact descriptions of natural things, as for example plants and the structure of animals, and those who have not the ability to construct figures, provided they have this thing before them or in the intellect, would be able to explain perfectly and transmit their thoughts to posterity, whereas today they do not know how since the words of our language are not compelling enough or precise enough to explain adequately without figures.

But the least utility of this characteristic is with regard to description, for it would be better, when one is able and wishes to take the trouble, to construct the figure itself. Its principal utility consists in the consequences and reasonings that it makes possible by operations on the characters, which could not be expressed in figures, and still less in words, without circumlocutions, or at least without a tiresome obscurity, instead of which this method would lead surely and without pains. This utility would appear principally in the discovery of theorems, and of constructions that are more convenient and general than those known; and this would be so even in the ele-

mentary parts of geometry. It would also be very useful in that part of geometry called practical, but especially in mechanics, where it would greatly surpass all other methods.

I believe that by this means one can handle mechanics in nearly the same way as geometry, and I believe that one could construct machines that are described by characters just as one constructs figures in geometry.<sup>6</sup>

*The remaining editorial notes contain detailed historical and mathematical commentary on the Geometric Analysis, including notes on specific passages, terminology, and the relationship between Grassmann's work and that of Leibniz, Möbius, and others. The editors discuss Grassmann's concept of point magnitudes, the interpretation of his analytical methods, and the reception of his work in the mathematical community of the mid-nineteenth century.*

The fourth volume of Möbius's collected works contains, on pp. 663–97, an unpublished essay from 1862 entitled “On Geometric Addition and Multiplication.” In it Möbius gives a comparative account of his own and Grassmann's treatments of the topic. The essay was obviously written because Möbius was not satisfied with Grassmann's treatment (cf. *ibid.* p. 720 f.).

---

<sup>6</sup>Leibniz's essay continues for several more paragraphs.

# Selected Papers on Mathematics and Physics





# New Theory of Electrodynamics

[Poggendorfs *Annalen der Physik and Chemie* 64 (1), 1–18, 1845]

It is known that the motive forces that electric currents or magnets exert on each other, or the one on the other, can be obtained from a single assumption, insofar as our observations to date extend. Even so, the domain encompassed by those observations still permits a free scope for hypotheses on the mutual action of two current elements, as I will demonstrate here. As I subjected Ampère's assumption, according to which (as must be the case) the mutual action of two infinitely small current elements is taken as fundamental, to a more detailed examination, it seemed to me most implausible; and as I then sought to remove the arbitrariness in that assumption, another assumption suggested itself to me that appears to me the only possible one, and which in fact leads to a result completely different from that of Ampère.

1) I first lay down the following as a principle, that the force that attracts or repels a point must always be representable as the derivative of a potential, and therefore here the force that one current element exerts on another must be representable as the derivative of a potential with respect to distance.

2) If two infinitely small current elements  $a$  and  $b$  are given, then from the first it already follows that the mutual action of these elements is representable as the derivative of a function of the distance of the two elements with respect to the distance; and this function depends only on the directions of the two elements. Now if the distance  $r$  of the two elements is set as variable, but their directions as fixed, and therefore one of the elements is carried along the straight

line connecting it to the other, then the directions of the two elements relative to this connecting line must finally be the same, except that the two elements can lie either on one and the same side or on opposite sides of the straight line connecting them. In the first case, that of the two elements in the same direction, the point of the one element that is closest to the other is the initial point of the one if and only if it is the initial point of the other; in the second case, that of the two elements in opposite directions, the point of the one element that is closest to the other is the initial point of the one if and only if it is the final point of the other. The one case goes over to the other if one reverses the direction of one current. Since thereby, however, the action that the one current exerts on the other is also changed into its opposite, one can, upon reversing the direction of one current, also change the sign of the action. Therefore if one allows the sign of the one current to change at the same time as the direction of that current, the mutual action remains the same. Now if one changes the sign of the one current and its direction simultaneously, the first case again goes over to the second. The mutual action in both cases can therefore only differ by the sign of the one current, and since in both cases the sign of the current changed, the action must be the same in both cases. It therefore depends only on the angle that each element makes with the connecting line.

If these angles are called  $\alpha$  and  $\beta$ , then the mutual action of the two elements is thus

$$\frac{dP}{dr}, \quad (1)$$

where  $P$  is a function of  $r$ ,  $\alpha$ ,  $\beta$ , but in such a way that if one sets  $r' = \alpha r = r\alpha$  (i.e. if the two elements lie in the same direction as regards the connecting line, at distance  $r'$ ), then  $P$  changes into  $P'$  and

$$\frac{dP'}{dr'} = \frac{dP}{dr}. \quad (2)$$

Now one can however add to the angle  $\alpha$  of the one element an angle  $\alpha'$  so that the element lies in the direction of the connecting line. By this the action on the other element is increased by a certain amount  $Q$ . If one now likewise brings the other element into the direction of the connecting line by addition of  $\beta'$ , the action again increases by an amount  $Q'$ . But if both elements lie in the direction

of the connecting line, the action is equal to  $\frac{dP'}{dr'}$ ; thus

$$\frac{dP}{dr} + Q + Q' = \frac{dP'}{dr'}, \quad (3)$$

and therefore, since  $\frac{dP'}{dr'} = \frac{dP}{dr}$ ,  $Q + Q' = 0$ . Since however  $Q$  depends only on  $\alpha'$ , and  $Q'$  only on  $\beta'$ , while  $\alpha'$  and  $\beta'$  are independent of each other, it follows that  $Q$  and  $Q'$  must separately vanish. The action of two current elements is therefore independent of their directions relative to the connecting line, and depends only on the distance  $r$ , and indeed must simply equal  $\frac{dP}{dr}$ , where  $P$  is any function of  $r$ .

3) If one brings into play the principle that the action must be independent of the absolute directions in space, then one has the equation

$$\frac{\partial^2 P}{\partial \alpha \partial \beta} = 0, \quad (4)$$

from which follows  $P = f(\alpha) + \phi(\beta)$ , and thus the action  $= f'(\alpha) \frac{d\alpha}{dr} + \phi'(\beta) \frac{d\beta}{dr}$ . But since this must be  $= \frac{dP}{dr}$ , and thus independent of  $\alpha$  and  $\beta$ , it follows that  $f'(\alpha) \frac{d\alpha}{dr}$  must be independent of  $\alpha$ , and  $\phi'(\beta) \frac{d\beta}{dr}$  independent of  $\beta$ .

If one calls  $\lambda$  the angle of the two directions,  $\epsilon$  the angle that the plane laid through the two elements makes with the plane of the two connecting lines,  $\alpha$  the angle that the one element makes with the connecting line,  $\beta$  that which the other makes, and  $r$  the distance of the two elements, then by the spherical law of cosines

$$\cos \lambda = \cos \alpha \cos \beta + \sin \alpha \sin \beta \cos \epsilon,$$

and since  $\epsilon$  is independent of  $r$ , while  $\alpha$  and  $\beta$  are functions of  $r$ , it follows from (4), upon differentiation with respect to  $r$ , that

$$\frac{d^2 P}{d\alpha^2} \frac{d\alpha}{dr} + \frac{d^2 P}{d\beta^2} \frac{d\beta}{dr} = 0,$$

and since this equation must hold for all values of  $\frac{d\alpha}{dr}$  and  $\frac{d\beta}{dr}$ , it follows that  $\frac{d^2 P}{d\alpha^2} = 0$  and  $\frac{d^2 P}{d\beta^2} = 0$ , whence  $P = a\alpha + b\beta + c$ , where  $a$ ,  $b$ ,  $c$  depend only on  $r$ . But then the action  $= a \frac{d\alpha}{dr} + b \frac{d\beta}{dr} + \frac{dc}{dr}$ . Since however this must be  $= \frac{dP}{dr}$ , it follows that  $a = 0$ ,  $b = 0$ , and therefore  $P$  depends only on  $r$ , which leads back to the result obtained in No. 2.

4) The above development thus shows that if one assumes the action to be representable as the derivative of a potential, then it can depend only on the distance of the two elements, and not on their directions. This result seems so implausible that one must seek another way out. But the only way out is to abandon the assumption that the action is representable as the derivative of a potential. And this is in fact the path I have taken.

5) I therefore assume that the mutual action of two current elements is not representable as the derivative of a potential, but rather that the action on the element  $b$  consists of two parts, one of which acts along the element  $b$  itself, and the other perpendicular to it. If one calls the first part the longitudinal action, the second the transversal action, then each of these two parts can be represented as the derivative of a potential with respect to the corresponding direction. For the longitudinal action this direction is that of the element  $b$ , and thus the potential depends only on the position of the initial or final point of  $b$ . For the transversal action, the direction is perpendicular to  $b$  in the plane laid through the element  $b$  and the attracting element  $a$ .

I further assume that the action of the element  $a$  on  $b$  can be decomposed into the action of the initial point of  $a$  on  $b$  and the action of the final point of  $a$  on  $b$ , each of which acts as if it were a point current, i.e. as if it were the endpoint of an infinitely long current flowing through it. The action of the element  $a$  on  $b$  is then the difference of these two actions.

6) I now seek the action of a point current  $A$  on the element  $b$ . This action must be representable as the derivative of a potential with respect to the direction of  $b$ , and likewise as the derivative of a (different) potential with respect to the direction perpendicular to  $b$ . The first potential depends only on the distance  $r$  of  $A$  from the initial and final points of  $b$ , and indeed must be of the form  $\frac{f(r)}{r}$ , since the action must vanish at infinity and become infinite at  $A$ . The second potential must likewise have the form  $\frac{\phi(r)}{r}$ , and indeed, since the transversal action changes sign when one reverses the direction of  $b$ ,  $\phi$  must be an odd function of the angle that  $r$  makes with the direction of  $b$ .

By a suitable determination of the functions  $f$  and  $\phi$ , one obtains the following expression for the action of the element  $a = i ds$  on the

element  $b$ :

$$\frac{i ds}{r^2} \left( b_1 \sin \alpha - b_2 \frac{1}{2} \cos \alpha \right), \quad (1^*)$$

where  $b_1$  is the projection of  $b$  on a plane through  $A$  and the center of  $b$ ,  $b_2$  is the component of  $b_1$  perpendicular to  $r$ , and  $\alpha$  is the angle that  $r$  makes with the plane of these two directions.

7) This formula leads to the following law for the mutual action of two current elements  $a = i ds$  and  $b = i' ds'$ :

$$\frac{ii' ds ds'}{r^2} \left( \sin \alpha \sin \beta \cos \epsilon - \frac{1}{2} \cos \alpha \cos \beta \right), \quad (4)$$

where  $\alpha$  and  $\beta$  are the angles that  $a$  and  $b$  make with the connecting line  $r$ , and  $\epsilon$  is the angle between the planes laid through  $a$  and  $r$ , and through  $b$  and  $r$ .

This is the fundamental formula of my theory. It differs essentially from that of Ampère, which reads:

$$\frac{ii' ds ds'}{r^2} \left( \sin \alpha \sin \beta \cos \epsilon - \frac{1}{2} \cos \alpha \cos \beta \right), \quad (A)$$

and indeed the two formulas differ only in that Ampère has  $-\frac{1}{2} \cos \lambda$  instead of my  $-\frac{1}{2} \cos \alpha \cos \beta$ , where  $\lambda$  is the angle between the two elements. But this difference is essential, since Ampère's formula does not satisfy the principle that the action must be decomposable into a longitudinal and a transversal part.

8) I now show that my formula agrees with all experiments so far performed. These experiments consist either in that one observes the mutual interaction of two closed currents, or that one makes an element of one closed current moveable and observes, in part, the interaction it experiences from the whole current to which it belongs, and in part the change of this action that results if one introduces another closed current.

Now since, as one makes one element of a current moveable, the action that the whole current produces does not change, so the experiments previously performed extend only to the actions that closed currents exert, either on another current or on a current element.

On the other hand, no experiments have been performed to study the action of a current element, neither that which it experiences from another current element, nor that which it exerts on another.

9) I now show that for closed currents my formula agrees completely with that of Ampère. For if one integrates the action of a

closed current on an element, the difference between the two formulas vanishes, since the integral of  $\cos \lambda$  over a closed curve agrees with the integral of  $\cos \alpha \cos \beta$ .

10) The difference between the two formulas only becomes apparent when one considers the action of an open current or a current element. But since no experiments have yet been performed on such cases, the question as to which formula is correct remains open.

11) To decide this question, one could perform the following experiment. One constructs a closed current in which a small segment is replaced by a moveable conductor. One then observes the force that the remaining current exerts on this moveable conductor. According to Ampère's formula, this force depends only on the position of the endpoints of the moveable conductor, while according to my formula it also depends on the form of the remaining current. This difference should be observable.

12) I now show by an example how my formula leads to results different from those of Ampère. I consider the action of a rectilinear current  $AB$  on a small magnet or current element  $b$  that lies in a plane perpendicular to  $AB$  and parallel to a line drawn from a point  $C$  of  $AB$  to the center of  $b$ .

Let the distance of the center of  $b$  from  $AB$  be  $= r$ , the angle that the direction from  $AB$  to the center of  $b$  makes with the perpendicular from  $C$  to the center of  $b$  be  $= \alpha$ , and the length of the element  $b$  be infinitely small. Then according to my formula the action of the element  $i ds$  of the current  $AB$  on  $b$  is equal to

$$\frac{i ds b}{r^2} \sin \alpha,$$

while according to Ampère it is equal to

$$\frac{i ds b}{r^2} \left( 2 \sin \alpha - \frac{3}{2} \sin \alpha \cos^2 \alpha \right).$$

The two formulas therefore give different results. For  $\alpha = 0$ , i.e. when the element  $b$  lies in the direction perpendicular to  $AB$ , both give zero. But for  $\alpha = \frac{\pi}{2}$ , i.e. when  $b$  is parallel to  $AB$ , my formula gives  $\frac{i ds b}{r^2}$ , while Ampère's gives  $\frac{i ds b}{r^2} \cdot \frac{1}{2}$ .

13) If one now considers the action of the whole current  $AB$  on the magnet  $b$ , then one must integrate the above expressions from one end of  $AB$  to the other. If the current  $AB$  is infinitely long in both directions, then according to both formulas the action is the

same, namely a force that seeks to move the magnet in the direction perpendicular to the plane laid through  $AB$  and the magnet.

14) But if the current  $AB$  is bounded, then the two formulas give different results. According to Ampère, the action of a bounded current on a magnet is the same as if the current were closed by a straight line connecting its endpoints, while according to my formula this is not the case.

15) I now apply the two formulas to the case where an infinitely small current element  $i ds$  acts on a small magnet of length  $b$ , which lies in the direction of the line connecting the element and the magnet.

According to my formula, the action of the current element on the one pole of the magnet is  $= \frac{i ds b}{2r^2}$ , and on the other pole  $= -\frac{i ds b}{2r^2}$ , so that the two actions together yield a force  $= \frac{i ds b}{r^2}$  that moves the magnet in the direction from the current element toward the magnet.

According to Ampère, the action on the one pole is  $= \frac{i ds b}{r^2} \cdot \frac{1}{2}(2 - 3 \cos^2 \alpha)$ , where  $\alpha$  is the angle that the line connecting the element with the pole makes with the direction of the element. If the magnet lies in the direction of the line connecting it with the element, then  $\alpha = 0$  for the one pole and  $\alpha = \pi$  for the other, and the two actions together yield a force that is in general different from that given by my formula.

16) To compare the two formulas more closely, I consider the case where the magnet is parallel to the current element and perpendicular to the line connecting them. Let the distance of the center of the magnet from the current element be  $= r$ , the length of the magnet be  $= b$ , and the current element be infinitely small. Then according to my formula the action on the magnet is  $= \frac{i ds b}{2r^2}$ , attractive; and the action on the other pole is equally great, repulsive. Both actions together thus move the magnet in the direction perpendicular to the line connecting it with the element, with a force  $= \frac{i ds b}{r^2}$ .

According to Ampère, the action on the one pole is  $= \frac{i ds b}{r^2} \cdot \frac{1}{2} \sin^2 \alpha$ , where  $\alpha$  is the angle that the line connecting the element with the pole makes with the direction of the element. For the one pole  $\alpha$  is smaller, for the other larger than a right angle. The two actions are therefore both attractive, but of different magnitude, and together they move the magnet in a direction that in general does not coincide with the direction given by my formula.

In the case where the current element is perpendicular to the



plane laid through the magnet and the line connecting it with the center of the magnet, the two formulas give the same result.

17) Finally, I will point out two very improbable actions which a bounded current must exert on a magnet according to Ampère; thus first, with a bounded current a magnet would simultaneously assume a rotational motion about its magnetic axis, which in the case considered previously (no. 16) would attain its maximum; and second, a magnetic needle, freely moveable about its center will, in the vicinity of a bounded current, and assuming only this acts on it, in general not take up a position of stable equilibrium, but rather with its departure from the equilibrium position it will partly return back, partly flip into the opposite position, according as it departs to this or that side of that position.

18) Now if by the above development I may hope to have proved that the new theory is plausible in that respect, it is still desirable to effect a distinction beyond all doubts between this and the Ampèrian assumption in practice. May a more skilled physicist soon succeed in clearing away all the obstacles that seem to stand in the way of that crucial experiment mentioned above.

## Editorial Notes

1. Thus if one calls  $\phi$  the angle between the element  $b$  and its orthogonal projection on the plane laid on the element  $a$  and through the center of  $b$ , then the projection, which Grassmann denotes  $b_1$  hereafter, is given by  $b_1 = b \cos \phi$ . If in addition one calls  $\epsilon'$  the angle between  $a$  and  $b_1$ , and  $\beta'$  the angle between  $b$  and the center of the straight line connecting  $a$  and  $b$ , then  $\alpha = \beta' + \epsilon'$ , and, in the spherical triangle with sides  $\alpha$ ,  $\beta$ ,  $\epsilon$ ,  $\phi$  is the height on the side  $\alpha$  or its prolongation, whence  $\cos \alpha = \cos \alpha' \cdot \cos \phi$ ,  $\cos \epsilon = \cos \epsilon' \cdot \cos \phi$ .
2. One recalls that Ampère's formula (1) represents a force reckoned positive in the direction from the vertex of angle  $\beta$  toward the vertex of angle  $\alpha$ .
3. The expression "differentiate with respect to  $-ds$ " is somewhat unusual. Of the two currents that according to the first paragraph of no. 9 replace  $i ds$ , the first has the action

$$\frac{i b_1}{-2r} \cot \alpha = \frac{i b_1}{-2r} (1 + \cos \alpha),$$

and the second the action

$$\frac{i b_1}{-2r} (1 + \cos(\alpha + d\alpha)) = \frac{i b_1}{-2r} (1 + \cos \alpha) + \frac{i b_1}{-2r} \sin \alpha d\alpha,$$

and these expressions are to be added.

4. In Grassmann's unpublished papers is found a collection of paginated notes, provided with a table of contents, entitled "Elektrodynamik und Induktion." These notes apparently date from about the end of 1849, for in them Grassmann is concerned, among other things, with a work of Franz Neumann (*Allgemeine Gesetze der inducirten electrischen Ströme*, 1845), and with one by W. Weber (*Poggendorfs Annalen* 73, 1848). As no. 5 of this collection is included a copy of the present paper, and as no. 6 a "Note Thereon," which reads as follows:

"Formula 4 can easily be represented in the form

$$r_{\alpha\beta} = \left( \frac{r_{\alpha\cdot b}}{r^3} \right), \quad (1)$$

where  $r_{\alpha \cdot b}$  represents a displacement that lies on the same side with respect to  $b$  as  $a$  with respect to  $r$ . If  $\alpha$  and  $\beta$  are the centers of the elements and  $\alpha'$  and  $\beta'$  the feet of the common perpendicular to the lines  $A$  and  $B$  of the two elements, then

$$r = \beta - \alpha = \beta - \beta' + \beta' - \alpha' + \alpha' - \alpha,$$

and since the middle term on the right is perpendicular to both  $a$  and  $b$ , the corresponding part of the product (1) vanishes. Thus  $r_{\alpha \cdot b}$  can be replaced by the product of the two remaining terms, which yields a simpler representation of the fundamental formula."

whence

$$\frac{\mathbf{r} \cdot \mathbf{a} \cdot \mathbf{b}}{r^3} + \frac{(\beta - \beta')\mathbf{a} \cdot \mathbf{b}}{r^3} = \frac{\mathbf{p} \cdot \mathbf{a} \cdot \mathbf{b}}{r^3},$$

where  $\mathbf{p} = \beta - \alpha'$  is the common perpendicular from  $A$  to  $B$ . The last expression is  $-\alpha \cdot r \cdot p$ , the angle that the former expression represents is  $\frac{\mathbf{a} \cdot \mathbf{b}}{r^3}$ , whence we obtain for the action the expression

$$\frac{\mathbf{a} \cdot \mathbf{b}}{r^3} + \frac{\mathbf{a} \times \mathbf{b}}{r^3},$$

where  $\times$  represents the angle of the action.<sup>1</sup>

Now if by  $\alpha, \beta, \alpha', \beta'$  one means elementary points, then in fact

$$\mathbf{r} = \beta - \alpha = (\beta - \beta') + (\beta' - \alpha') - (\alpha - \alpha'),$$

and moreover, obviously

$$\beta - \beta' = \lambda \cdot \mathbf{b}, \quad \alpha - \alpha' = \mu \cdot \mathbf{a},$$

where  $\gamma$  and  $\mu$  denote numbers, whence the outer product  $[(\alpha - \alpha')\mathbf{a}] = 0$ ; if in addition one denotes the common perpendicular  $\alpha'\beta'$ , regarded as the displacement  $\beta' - \alpha'$ , as  $\mathbf{c}$  (in order to avoid  $\mathbf{p}$ ), then expression (1') can be written

$$\frac{1}{r^3}[\mathbf{r} \mid \mathbf{a} \mid \mathbf{b}] + \frac{1}{r^3}[\mathbf{c} \mid \mathbf{a} \mid \mathbf{b}].$$

However, according to A2, no. 180,

$$\begin{aligned} [\mathbf{b} \mid \mathbf{g} \mid \mathbf{b}] &= -[\mathbf{g} \mid \mathbf{b}]\mathbf{b}, \\ [\mathbf{c} \mid \mathbf{a} \mid \mathbf{b}] &= [\mathbf{c} \mid \mathbf{g}]\mathbf{b} - [\mathbf{a} \mid \mathbf{b}]\mathbf{c}, \end{aligned}$$

---

<sup>1</sup>Comprehension of this observation is made difficult because the meaning of the product sign employed is not evident; from the outset it is clear only that the letters  $\mathbf{r}, \mathbf{a}, \mathbf{b}$  appearing in the numerator and likewise the differences  $\beta - \beta'$  etc. (cf. A2, no. 222, Corollary) denote displacements. Fortunately one can infer the meaning of formula (1) above from one of Grassmann's later articles ("Zur Elektrodynamik"). There the attractive action of the element  $\mathbf{a}$  on the element  $\mathbf{b}$  is described in the form  $-\frac{1}{r^3}[\mathbf{r} \mid \mathbf{a} \mid \mathbf{b}]$ , where the underlined letters represent the displacements themselves, whose corresponding lengths are given by plain letters, and where  $\mid$  is the symbol for inner multiplication (cf. A2, chapter 4 and chapter 5, § 7), while  $\mathbf{r} \cdot \mathbf{a}$  denotes the outer (combinatorial) product of the two displacements  $\mathbf{r}$  and  $\mathbf{a}$  (A2, no. 254).

and since the displacement  $\mathbf{c}$  is perpendicular to both  $\mathbf{g}$  and  $\mathbf{b}$ ,  $[\mathbf{c} \mid \mathbf{g}] = [\mathbf{c} \mid \mathbf{b}] = 0$ : consequently (1') takes the form

$$\frac{1}{r^3}([\mathbf{b} \mid \mathbf{b}]\mathbf{a} - [\mathbf{a} \mid \mathbf{b}]\mathbf{b}) - \frac{1}{r^3}[\mathbf{a} \mid \mathbf{b}]\mathbf{c}.$$

Here the second term evidently represents a displacement parallel to the common perpendicular, and thus defines the shift that the element  $\mathbf{a}$  imparts to the element  $\mathbf{b}$ , parallel to the common perpendicular.

The first term represents a displacement orthogonal to the common perpendicular, thus defining the rotation about the common perpendicular that the element  $\mathbf{a}$  communicates to the element  $\mathbf{b}$ . If one divides the length of this displacement by the distance of the center  $\beta$  of the element  $\mathbf{b}$  from the rotational axis, then one obtains the angle of rotation. Now the distance of the point  $\beta$  from the rotational axis is equal to the length of the displacement  $\beta - \beta' = \lambda \cdot \mathbf{b}$ , and is thus equal to  $\lambda \cdot b$ , where no attention is paid to the sign of  $\lambda$ , since it ultimately drops out. Furthermore  $[\mathbf{b} \mid \mathbf{b}] = b^2$  and

$$a^2 \cdot b^2 - [\mathbf{a} \mid \mathbf{b}]^2 = [\mathbf{a} \cdot \mathbf{b}]^2$$

(A2, no. 151, 333, 177), whence the displacement  $[\mathbf{b} \mid \mathbf{b}]\mathbf{a} - [\mathbf{a} \mid \mathbf{b}]\mathbf{b}$ , which we denote by  $\mathbf{s}$  for short, has the length

$$s = \sqrt{b^4 \cdot a^2 - 2b^2[\mathbf{a} \mid \mathbf{b}]^2 + b^2[\mathbf{a} \cdot \mathbf{b}]^2} = b\sqrt{[\mathbf{c} \mid \mathbf{b}]^2},$$

and in addition

$$b \cdot \sqrt{[\mathbf{a}]} = a \cdot b \cdot 2 \sin(\angle \mathbf{ab})$$

(A2, no. 198), whence it follows that that angle equals

$$\frac{\lambda \cdot s}{\lambda \cdot b} = \frac{1}{r^3} ab \sin(\angle \mathbf{ab}).$$

Now if we recognize that  $ab \sin(\angle \mathbf{ab})$  is the content of the parallelogram defined by the two displacements  $\mathbf{a}$ ,  $\mathbf{b}$ , and that the magnitude, the position, and the sense is represented by the outer product (A2, no. 254), to which we add that the three displacements  $\mathbf{g}$ ,  $\mathbf{b}$ ,  $\mathbf{a}$  are all orthogonal to the common perpendicular  $\mathbf{c}$  and that  $[\mathbf{b} \mid \mathbf{b}] = b^2[\mathbf{b} \cdot \mathbf{a}]$ , then we perceive that  $\mathbf{a}$ , regarded from the common perpendicular, lies on the same side with respect to  $\mathbf{b}$  as does  $\mathbf{g}$ , and thus that the sense of rotation which endeavors

to produce corresponds to the sense of rotation of the parallelogram  $[\mathbf{b} \cdot \mathbf{a}] = -[\mathbf{a} \cdot \mathbf{b}]$ . Consequently, if we wish, we can represent the angle mentioned in magnitude and sense by the expression

$$-\frac{1}{r^3}[\mathbf{g} \cdot \mathbf{b}],$$

and thus we arrive at an expression

$$-\frac{1}{r^3}[\mathbf{g} \cdot \mathbf{b}] - \frac{1}{r^3}[\mathbf{g} | \mathbf{b}]\mathbf{c}$$

for the action of the element  $\mathbf{a}$  on the element  $\mathbf{b}$ , which agrees with that of Grassmann except for notation, and in the first term defines an angle whose plane is orthogonal to the common perpendicular, the second a displacement parallel to this perpendicular.

In regard to the difference in notation, it is to be noted that in the manuscript Grassmann uses the dot only as the symbol for inner multiplication of a plane area and a displacement, rather than as the symbol for the outer product of displacements; on the other hand the inner product of two displacements is indicated by  $\times$ , as in the Geometric Analysis.

If the elements  $\mathbf{a}$  and  $\mathbf{b}$  lie in the same plane, then  $\alpha'\beta' = \mathbf{c}$  is zero, whence expression (1'') reduces to its first term which, apart from sign, agrees with expression (6) above. That the minus sign is missing in formula (6) is explained by the fact that Grassmann had not thought to express in (6) the sense of rotation as well, particularly as he had previously defined this sense precisely enough.

It is now more comprehensible what Grassmann means toward the end of § 12. Thus the expression

$$\frac{\mathbf{a} \cdot \mathbf{b}}{r^3}$$

defines, depending on the meaning one gives the product  $\mathbf{a} \cdot \mathbf{b}$ , one or the other of the two motions into which the motion of the element  $\mathbf{b}$  can be decomposed: if one interprets  $\mathbf{a} \cdot \mathbf{b}$  as the outer product, then one gets the magnitude of the rotation about the common perpendicular; if one interprets it as the inner product, then one obtains a measure for the shift parallel to the common perpendicular. One obtains this shift itself if one multiplies  $[\mathbf{a} | \mathbf{b}] : r^3$  by  $-\mathbf{c} = \alpha' - \beta'$ . It is noteworthy that both the rotation and the shift only change sign if one interchanges  $\mathbf{a}$  with  $\mathbf{b}$ , since  $[\mathbf{b} \cdot \mathbf{a}] = -[\mathbf{a} \cdot \mathbf{b}]$  but  $[\mathbf{a} | \mathbf{b}] = [\mathbf{b} | \mathbf{a}]$ ,

and  $\beta' - \alpha' = -(\alpha' - \beta')$ , so that the analogy with gravitation is as complete as possible.<sup>2</sup>

---

<sup>2</sup>This point too is further developed in “Zur Elektrodynamik.”

## On the Theory of Compound Colors

[Poggendorfs *Annalen der Physik und Chemie* 89 (1), 69–84, 1853]

In the 87th volume of this journal, Helmholtz published<sup>3</sup> a series of observations, some of which were new and ingenious, and from which he comes to the conclusion, that the theory of compound colors, universally admitted since the time of Newton, is erroneous in its most essential points; and that there are properly only two prismatic colors, yellow and indigo, which when mixed give white. It may consequently not be regarded as superfluous to show that the Newtonian theory of compound colors is correct to a certain point, and especially that the opinion that every color has its complementary color, with which when mixed gives white, is founded upon mathematically incontestable facts, so that this opinion must be regarded as one of the most tenable in physics. I will then show that the positive observations of Helmholtz, instead of tending to the subversion of this theory, serve rather to prove and partly to complete it.

For this purpose it will be necessary to analyse the impression of color of which the eye is capable into its elements. The eye first of all distinguishes light as colored and colorless. In colorless light (white, gray) it only distinguishes a greater or less intensity, and this may be mathematically determined. In like manner we only distinguish the greater or less intensity of a homogeneous color. But even for the difference of the individual homogeneous colors we have a mathematically determinable standard, which is furnished most perfectly by the duration of vibration corresponding to each color; even in ordinary speech, this difference has been very suitably distinguished by the expression tint. Thus, in a homogeneous color we can distinguish both its intensity and its tint. If now a homogeneous color be mixed with colorless light, the impression of color becomes weakened by this intermixture. Ordinary language is rich in terms by which this difference of impression is denoted; on the other hand we want a short expression for it, and therefore we choose the phrase “intensity of the white.” In this manner we can distinguish in every mixed light three elements—the tint, the intensity of the color, and the intensity of the white.

---

<sup>3</sup>H. Helmholtz, “Ueber die Theorie der zusammengesetzten Farben,” *Pogg. Ann.* 87, 45–66 (1852), reprinted from Müllers *Archiv für Anatomie und Physiologie*, 1852, 461–487.



The various tints form a continuous series of such a kind, that when we start from one color of this series and proceed forward, we finally arrive at the original color. One circumstance, however, must not be left unnoticed here, namely, the difficulty of obtaining homogeneous red light, which forms the transition

between the violet and red of the ordinary spectrum, and which can only be produced by the prism under remarkably favorable circumstances (on a bright summer's noon).<sup>4</sup> This outermost color of the spectrum, which may equally well be regarded as extreme red or extreme violet, I will call purple.<sup>5</sup> If now we regard any compounded light, the eye can distinguish in it only the three elements above mentioned; that is to say, every impression of light may be imitated by mixing a homogeneous color of a certain intensity with colorless light of a certain intensity. Thus in every sensation of light we have to distinguish three things—the intensity of the color, the tint, and the intensity of the intermixed colorless light.

It would be easy to make an apparatus by means of which we should be enabled to determine every color according to these three elements. In order to give an idea of this, let us suppose two white screens of similar nature moveable upon a hinge, in such a manner that the white side of the screens shall be upon the outside of the angle formed at the hinge; let a divided circle be also provided to measure this angle. The colored light to be examined is now allowed to fall upon one of these screens in a plane perpendicular to the axis of rotation; on the other screen, and in the same plane, white light is suffered to fall in any direction; and in a direction perpendicular to the latter, homogeneous light; the light being so selected that it may have the

---

<sup>4</sup>In Pogg. Ann. 23, 435–443 (1834) there is an extract from a work of Brewster: "On a New Analysis of Sunlight into Three Fundamental Colors that form Coincident Spectra of Equal Length," (Edinburgh J. of Science, New Series, 5, 197). On p. 441f. he says: "In the summer of 1799, on a very hot day, about noon, Hassenfratz, with a sunbeam that he conducted through a 25 decimillimeter wide opening into a dark room and under the condition that upon falling upon a prism the incoming and outgoing angles were equal, obtained a spectrum 360 millimeters long, in which all the colors from purple to red were clearly visible. Upon the sun's setting, when it appeared green, the spectrum had a smaller length, and a more or less considerable fraction of the violet had vanished, as the purple, sometimes completely."

<sup>5</sup>One finds the rule as the solution to problem II, proposition VI in Newton's Optics: "Given the quantity and quality of each primary color in a mixture, to know the color of the compound."

same tint as the colored light. The angle of the screens is now changed until both appear of the same nature to the eye; and then it is clear that the color of the first screen is imitated by the mixture of the homogeneous and white light falling upon the second. The divided circle now gives the proportion of the intensities of these lights; and the tint is given by the duration of vibration of the homogeneous light, or by any other equivalent datum.

In the preceding arguments we have first of all assumed, that every impression of color of this kind may be analysed into three mathematically determinable elements—the tint, the intensity of the color, and the intensity of the intermixed white.

In the second place we assume, that if one of two mingling lights be continuously altered (whilst the other remains unchanged), the impression of the mixed light also is continuously changed.

This assumption may be thus enunciated: if a homogeneous color, whose intensity and tint continuously change, be mingled with another homogeneous color of unchanging intensity and tint, the intensity of the color of the mixture, its tint, and the intensity of the white which it contains, all change continuously. This is not the place to enter more fully into the discussion of these axioms, and therefore I will only add, that no transition from one color to another in a continuous series of colors can take place without all the three elements changing continuously, although very different relations may subsist between the changes of these elements, and although one or more of these elements may be zero before and after the transition.

If two lights of equal intensity and similar tint be mingled, the tint and the intensity of the color remain unchanged; only the intensity of the white is changed, and it is clear that it is doubled. If two lights of similar tint but unequal intensity be mingled, it is clear that the intensity of the color of the mixture is equal to the sum of the intensities of the colors of the two mingling lights; the tint remains unchanged; and the intensity of the white is equal to the sum of the intensities of the white in the mingling lights. If now two lights of equal intensity and different tints be mingled, it is easy to show, that the color of the mixture can never be colorless; for were the color of the mixture colorless, then according to the first assumption, the mixture must possess the same nature if one of the

mingling lights be continuously changed. But we may now suppose that one of the two mingling lights is that which remains unchanged, and the other that which is continuously altered; then the mixture is the same as the mixture of the unchanged light with a changing light, and is therefore by the second assumption continuously changed. Now let the changing light change continuously until it has the same tint as the unchanged light; then according to what has just been shown, the mixture can give no colorless light. Thus the lights of different tint can never give colorless light if they are of equal intensity.

This limitation may however be removed if the intensities are different. Let  $a$  be the tint of one mingling light,  $x$  the tint of the other, and let  $x$  be continuously changed in the sense of positive transition. Let the color  $a$  have the intensity  $\alpha$ , and the color  $x$  the intensity  $\xi$ . Then if  $a$  and  $x$  denote colors of such a nature that, when they are of equal intensity, they give colorless light, then if  $\alpha$  and  $\xi$  are not equal, their mixture will continuously change; and as, according to the assumption, it can never give colorless light, its tint will also change continuously; thus, as the mixture at first has the tint  $a$ , and finally the tint  $x$ , it has continuously passed from  $a$  to  $x$ . This transition may be either

in the positive or negative direction; in the latter case the mixture, starting from the tint  $a$ , must have again reached the tint  $a$  before arriving at the tint  $x$ ; but since the same reasoning applies whatever value between  $a$  and  $x$  we assign to  $x$ , the transition cannot be in the negative direction unless the tint of the mixture always remains  $a$ , however the color  $x$  changes, and this is impossible. The transition must therefore take place in the positive direction, and there will be in the series of tints between  $a$  and  $x$ , one, and only one, which the mixture can possess; that is to say, of two mingling colors of different intensity there can always be but one tint, which is determined by the ratio of the intensities; and since this must be the same for whatever degree of intensity, the color which is mixed with  $a$ , in order that, as the intensity changes, the mixture may give colorless light, is the same which, when mixed with  $a$ , in any degree of intensity, must give colorless light.

If now we test Helmholtz's experiments, we obtain from them, at least approximately, the color which is capable of furnishing colorless light with any other given color. For yellow, according to Helmholtz, this is indigo, a result which certainly does not require proof. Between orange and red, according to him, there is only one color which possesses this property, namely green; it is therefore their complementary color. Now as, by the observations of Helmholtz, orange and green give no white, it follows that the green which with red gives white cannot be the same as that which with orange gives white; but since the complementary color must be the same whatever the intensity, it follows, that between red and orange there are at least two different greens, each of which may serve as complementary color. The same applies to the other series of experiments. The table of complementary colors derived from them would be as follows:

|         |                  |         |               |         |        |
|---------|------------------|---------|---------------|---------|--------|
| Yellow, | Yellowish-green, | Green,  | Bluish-green, | Azure,  | Indigo |
| Indigo, | Violet,          | Purple, | Red,          | Orange, | Yellow |

in which the complementary colors stand one above another.

In this table I have not followed the positive direction of transition of tints followed by Helmholtz, but the negative, inasmuch as I have placed red after purple and orange after red. In the other series the positive direction is followed; for it must be observed that the colors mentioned in the first series do not possess perfectly uniform tints; for instance, the green which gives white with red inclines to yellow, and the green which with yellow gives white inclines to blue; the bluish green also, which with yellow gives white, inclines more to blue than the green which with orange gives white. With these restrictions we have the following theorem:

*If  $a$  and  $a'$  be two complementary colors, and  $b, b'$  the two colors complementary to two other colors which lie close to  $a$  and  $a'$  on the same side, then of the two colors  $b$  and  $b'$ ,  $b$  is that which lies upon the side of positive transition from  $a$ .* It is evident that if the color  $a$  be given,  $a'$ ,  $b$ , and  $b'$  may be found by observation. For instance, if  $a$  be yellow,  $a'$  is indigo; between  $a$  and  $a'$  on the side of positive transition lie the various tints of green and blue; if now we take two colors lying close to yellow, the one on the side of the green, the other on the side of the orange, then of the two complementary colors the one must fall on the side of violet, the other on the side of indigo; the first belongs to the greenish yellow, the second to the orange.

If the homogeneous colors of different tints and intensities be represented by radii vectores, drawn from a center to the circumference of a circle, the angle included between two such radii being proportional to the interval of the corresponding colors in the series of tints; and if at the extremity of each radius a mass be placed proportional to the intensity of the corresponding color, then we can determine the center of gravity of these masses. If now two colors  $\alpha a$  and  $\beta b$  be mingled, then (leaving the intermixture of white out of the question) the mixture of  $c$  and  $c'$  may be represented by the combination of the four colors  $\alpha a$ ,  $\beta b$ ,  $\alpha' a$ ,  $\beta' b$ . But  $\alpha a$  mixed with  $\alpha' a$  gives  $(\alpha + \alpha')a$ , and  $\beta b$  mixed with  $\beta' b$  gives  $(\beta + \beta')b$ . As, however, these latter have the original tints  $a$ ,  $b$ , or  $a'$ ,  $b'$ , their mixture is represented by the geometrical

sum of the radii vectores taken in their own direction; and therefore also the color of the mixture is represented by the radius vector of the center of gravity of the two masses  $(\alpha + \alpha')$  at the extremity of the radius vector of  $a$ , and  $(\beta + \beta')$  at the extremity of the radius vector of  $b$ .

The assumption that the colors  $a$  and  $a'$  are complementary is expressed by the equations

$$\alpha + \alpha' = w, \quad \beta + \beta' = w,$$

where  $w$  denotes the intensity of the white color produced; and the color of the mixture, its intensity, and the intensity of the white contained in it are given by the radius vector, the sum of the two masses, and twice the least of the two, respectively. If now a third color be added, the center of gravity of the first two is to be found with the third; and thus in general the mixture of any number of colors is represented by the center of gravity of the masses placed at the extremities of the corresponding radii vectores, the intensity of the color being equal to the sum of all the masses, and the intensity of the white equal to twice the smallest mass.

For let several colors be represented by  $a, b, c, \dots$ , their intensities by  $\alpha, \beta, \gamma, \dots$ , and let the masses placed at the extremities of the corresponding radii vectores be proportional to  $\alpha, \beta, \gamma, \dots$ . If we suppose the colors to be all different, the center of gravity is to be found of the whole of the masses; and if the colors be not all different, those which are the same must be collected together, and the radius vector corresponding to that color must have a mass equal to the sum of the masses of the same color. But the center of gravity of all the masses may be found by first seeking that of the masses belonging to the same color, and then that of all the others. Therefore the center of gravity will remain the same, whether we suppose the masses belonging to the same color to be collected at the extremity of the corresponding radius vector or not; the only difference is, that in the former case the intensity of the color is equal to the sum of all the masses, and in the latter case it is equal to the sum of the collected masses. The intensity of the white is in both cases equal to twice the least of the collected masses; for since the mixture of two colors  $a, a'$  gives white, the intensity of the white is equal to twice the least of the two intensities. The theorem is therefore proved.

Figure 1: [Figure 1: Circle of colors showing complementary pairs]

In order to render this theorem more evident, let us suppose that all the homogeneous colors are arranged in a circle in the order of their tints, and that at the extremity of each radius vector a mass equal to the intensity of the corresponding color is placed. Then the center of gravity of all the masses represents the mixture of all the colors, and the intensity of the color of the mixture is equal to the sum of all the masses, and the intensity of the white is equal to twice the least mass. If now the least mass be removed, the center of gravity of the remaining masses represents the mixture of the remaining colors, and the intensity of the white is equal to twice the least of the remaining masses. If therefore the least mass be removed, and the center of gravity of the remaining masses be the same as before, the mixture of the remaining colors will be the same as the mixture of all the colors, and the intensity of the white will be diminished by twice the least mass. If therefore the least mass be continually removed, the center of gravity will continually approach the center of the circle, and the intensity of the white will continually diminish, until at last the center of gravity coincides with the center, and the intensity of the white is zero.

This is the case when the sum of all the masses is zero; for then the center of gravity coincides with the center of the circle. In this case the mixture of all the colors gives a pure color, that is, a color without any admixture of white. If now the sum of all the masses be not zero, but the center of gravity coincide with the center, then the mixture of all the colors gives white. This is the case when the colors



are complementary in pairs, and the intensities of the complementary colors are equal.

Figure 2: [Figure 2: Center of gravity construction for color mixture]

In order to represent the mixture of two colors geometrically, Helmholtz employs a triangular figure, in which the fundamental colors are placed at the three angles, and the colors produced by mixing them are represented by points within the triangle. This method is quite correct, but it is not so general as the one I have given, for it assumes that there are three fundamental colors, whereas my method applies to any number of colors. Moreover, the triangular figure does not give the intensity of the white, which is an essential element of the mixture.

Figure 3: [Figure 3: Color triangle showing mixture proportions]

If the homogeneous colors of different tints and intensities be represented by radii vectores, drawn from a center to the circumference of a circle, the angle included between two such radii being proportional to the interval of the corresponding colors in the series of tints; and if at the extremity of each radius a mass be placed proportional to the intensity of the corresponding color, then we can determine the center of gravity of these masses. If now two colors  $\alpha a$  and  $\beta b$  be mingled, then the mixture is represented by the center of gravity of the masses  $\alpha$  at the extremity of  $a$  and  $\beta$  at the extremity of  $b$ , the intensity of the color being equal to the sum of the two masses, and the intensity of the white being equal to twice the smaller mass.

This rule, derived by a purely mathematical method from four sufficiently well founded assumptions, agrees in its essential features with Newton's empirical rule. Nevertheless, the manner in which Newton distributes the homogeneous colors on the circumference of his discs requires a thorough revision, towards which only the first steps are made by the researches of

## Editorial Notes

1. That is, in the article “Ueber die Theorie der zusammengesetzten Farben,” Pogg. Ann. 87, 45–66 (1852), reprinted from Muellers Archiv fuer Anatomie und Physiologie, 1852, 461–487.
2. In Pogg. Ann. 23, 435–443 (1834) there is an extract from a work of Brewster: “On a New Analysis of Sunlight into Three Fundamental Colors that form Coincident Spectra of Equal Length,” (Edinburgh J. of Science, New Series, 5, 197). On p. 441f. he says:  

“In the summer of 1799, on a very hot day, about noon, Hassenfratz, with a sunbeam that he conducted through a 25 decimilimeter wide opening into a dark room and under the condition that upon falling upon a prism the incoming and outgoing angles were equal, obtained a spectrum 360 millimeters long, in which all the colors from purple to red were clearly visible. Upon the sun’s setting, when it appeared green, the spectrum had a smaller length, and a more or less considerable fraction of the violet had vanished, as the purple, sometimes completely.”
3. One finds the rule as the solution to problem II, proposition VI in Newton’s Optics: “Given the quantity and quality of each primary color in a mixture, to know the color of the compound.”
4. In the article cited in note 1, on p. 58.

It is of historical interest that Grassmann had already communicated his theory of compound colors in October 1852 in a lecture to the Physikalischen Gesellschaft zu Stettin. In the “Mitteilungen Ueber die Taetigkeit der physikalischen Gesellschaft zu Stettin in den Jahren von 1835–1867,” one finds on p. 60f.:

“Prof. Grassmann on the latest discoveries in the field of compound colors. After the lecturer began with an historical survey of the various ideas about colors and their compounding, he showed that, in order to proceed to a decision among them one first of all had to separate the objective phenomena that can be illustrated by means of a prism from the subjective impressions of colors. In this regard he noted that the colors throughout the paintings of the most excellent artists, in particular Raphael. The lecturer then proceeded to the subjective color impressions, and showed that the eye is not

equipped to distinguish the manifold differences of colors united in the same space, but rather that these colors are blended into a total impression in which, besides the color tints between red and violet, only the admixture of white can be distinguished as well. In reference to this he noted the report in two articles by Helmholtz in the latest volume of *Pogg. Ann.* and noted that, according to Helmholtz's investigations, there are only two color pairs that yield white upon combination, namely first of all yellow and ultramarine, as Helmholtz demonstrated directly by observation, and second red and green. The latter, the lecturer contended, must follow from Helmholtz's investigations, since the investigation throughout appears to demonstrate this combination for red and green. Between these four colors, red, yellow, green, and blue, which are here interpreted as the characteristic fundamental colors, the lecturer arranges the other colors in such a way that each pair of these colors yields the mean color, but mixed with white."

Among Grassmann's unpublished papers there is a brief note: "On the theory of compound colors 1876," from which the following is reproduced:

"Helmholtz, in his *Treatise on Physiological Optics* (1867), pp. 277–308 has adopted the theory of compound colors from my article (1853), except that he replaces azure by the more suitable name cyan-blue (Berlin blue), and identifies purple as a color only realizable as a mixture of extreme red and extreme violet (perhaps according to table 327 as the octave of that red). Neglecting the deviations that depend partly on the sensitivity of the eye, partly on objective variations (diffraction, dispersion, fluorescence, phosphorescence), and which should be taken into account, he has even so introduced the following important

improvement: Instead of the four colors red, yellow, green, blue as fundamental colors, he takes only three, namely red, green, and blue, and obtains yellow as the mixture of red and green. In this respect experience would decide against my selection of four fundamental colors.

“But as for the rest, Helmholtz (pp. 295–299) declares my theory to be wrong, in that according to his experiments one and the same color when mixed with different other colors yields different complementary colors, and thus according to his experiments the existence of complementary colors is refuted. Against this I must advance the following considerations:

“1) The observations cited are irrelevant, since the colors to be mixed must be pure, that is completely homogeneous (of one wavelength), which was not the case in Helmholtz’s experiments.

“2) The observations of Helmholtz confirm my theory, since according to my theory two non-homogeneous colors of the same color yield upon mixing two colors that stand symmetrically with respect to the center of the color circle, and which therefore appear as different complementary colors.

“3) The refutation attempted by Helmholtz rests on a logical error, since he identifies the complementary color of a color with the color that when mixed with it yields white, whereas according to Newton’s definition the complementary color is that which when mixed with it in the proper ratio yields white.”

5. The main series of elementary observations needed to establish compound colors, was not carried out. On the other hand, an admittedly only fragmentary series of observations were performed that would contradict the laws of compound colors if they were not based on secondary processes or on incomplete or erroneous evaluations, or on the insensitivity of the optic nerve.

“Among these are, first, that the strength of the light sensation for different colors is not proportional to the living force of the ether

vibrations. This is based in part on the fact that the ether vibrations of the different colors are weakened in different degrees by the transparent middle of the eye, as many a time the infrared does not pass at all, partly due to the reverberation in the eye, partly due to the unequal sensitivity of the nerves for different colors in different illuminations; with weaker illumination the nerves are more sensitive to the more refrangible colors than for the less refrangible, and conversely for stronger illumination. The influence of brightness on the color of the transmitted light must also be noted here; as Helmholtz has correctly noted, pure spectral yellow mixed with white yields a greenish color, while at greater intensity it yields pure yellow."

6. Helmholtz mentions Grassmann's work on pages 283, 295, and 307 of his *Handbuch der physiologische Optik* (Leipzig: Voss, 1867). On page 307 he says: "The writer's improved method of investigating the mixtures of the colors of the spectrum removed the apparent contradiction of Newton's rule so far as it related to the center of gravity construction; but on the other hand, in spite of Grassmann's arguments, the circular form of the color field was left as a question that was still debatable. Finally the principles of Newton's law of mixture were experimentally confirmed by Koenig."
7. Unger compares the color scale from extreme red to extreme violet with the musical scale by putting extreme red into correspondence with an arbitrary fundamental note and then every color and note whose vibrational periods stand in the same ratio as those two. He then asserts that those colors that correspond to the notes of a chord produce the most pleasant sensations to the eye, namely the three colors red, green, and blue, which correspond to a major chord, and which he finds predominate

throughout the paintings of the most excellent artists, in particular Raphael. The lecturer then proceeded to the subjective color impressions, and showed that the eye is not equipped to distinguish the manifold differences of colors united in the same space, but rather that these colors are blended into a total impression in which, besides the color tints between red and violet, only the admixture of white can be distinguished as well. In reference to this he noted the report in two articles by Helmholtz in the latest volume of Pogg. Ann. and noted that, according to Helmholtz's investigations, there are only two color pairs that yield white upon combination, namely first of all yellow and ultramarine, as Helmholtz demonstrated directly by observation, and second red and green. The latter, the lecturer contended, must follow from Helmholtz's investigations, since the investigation throughout appears to demonstrate this combination for red and green. Between these four colors, red, yellow, green, and blue, which are here interpreted as the characteristic fundamental colors, the lecturer arranges the other colors in such a way that each pair of these colors yields the mean color, but mixed with white."



## On the Various Types of Multiplication

[Crelles Journal 49(2), pp. 123–141, 1855]

M. Cauchy has communicated to the Academie des Sciences on the 10, 17, and 24 of January of last year the principles of a calculus based on quantities that he calls *clefs algebriques* (cf. Comptes Rendus XXXVI (1853), 70, 129, 161; Oeuvres, I. Serie, XI, no. 514, XII, nos. 516, 517). He shows that with the help of this calculus one can solve problems in analysis and mechanics with the greatest of ease, problems that entail long and laborious calculations with the application of the usual methods. It is only some weeks since I was able to obtain copies of these communications, but at a glance I recognized that the principles established by him, and the results that he had derived, were absolutely the same as those that I had already published in 1844 (in a work entitled *Ausdehnungslehre oder Wissenschaft der extensiven Grosse*, Leipzig: 1844), where at the same time I had also given numerous applications to algebraic analysis, geometry, mechanics, and other branches of physics. After the publication of that work I was successful in simplifying and extending the principles of the calculus presented, but I decided to withhold their publication until such time as I would have the leisure to revise the whole work. However, as a consequence of the articles just mentioned, I find myself forced to publish here some of the results obtained, as I see that in his *clefs algebriques* M. Cauchy has in a way found the keys, not only to enter into the theory that I have published so far, but in addition to open the door to many results not yet published. What I consider more important still, the course he has followed is precisely that which I myself have followed, and which appears to me to be the only appropriate one for this subject.

Thus I am led to set out the principal types of multiplication to which this analysis leads; and for this purpose I limit myself at first to extensive quantities, and among these to the theory of the products of two factors.

### § 1. *Introduction*

I say that two or more quantities  $A, B, C, \dots$  have between them an algebraic relation when one or another of them is equal to a polynomial whose terms are proportional to the others, so that one has,

for example,

$$A = \beta B + \gamma C + \dots,$$

where  $\beta, \gamma, \dots$  denote simple numbers, rational or irrational, real or imaginary, zero or not. To suggest a concrete idea, imagine that the letters  $A, B$  denote any different objects, for example different species of plants. One can then say of the species denoted by  $A$  that it is composed of the species denoted by  $B$  taken  $\beta$  times and of the species  $C$  taken  $\gamma$  times, etc., by which however one must not understand that the species  $A$  is actually formed by combining specimens of the species  $B$  and  $C$ , but rather that the relation between  $A, B, C$  is such as if  $A$  were formed from  $B$  and  $C$  in the manner indicated.

In the same way, one can regard three lines, or any four points in space, as a system of relative units, from which one can compose all lines or all points of space provided the three original lines are not parallel to the same plane and the original points are not all in the same plane. In this connection we remark that one can regard three lines, or any four points in space, as a system of relative units, from which one can compose all lines or all points of space provided the three original lines are not parallel to the same plane and the original points are not all in the same plane.

To suggest a concrete idea, imagine that the letters  $A, B$  denote any different objects, for example different species of plants.

## § 2. *Multiplication of Extensive Quantities*

6

For example, the product of the extensive quantities  $\alpha a + \beta b$  and  $\gamma a + \delta b$ ,  $\alpha, \beta, \gamma, \delta$  denoting numbers,  $a$  and  $b$  relative units, is:

$$(\alpha a + \beta b)(\gamma a + \delta b) = \alpha\gamma \cdot aa + \alpha\delta \cdot ab + \beta\gamma \cdot ba + \beta\delta \cdot bb.$$

Thus it only remains to define the products of the relative units. One can distribute these unit products into classes according to the various systems of equations that may hold between them. If one imagines the unit products of one and the same class combined by addition, each class of unit products corresponds to one equation (or several) between these sums. But if there are  $n$  units, the number of their pairwise products is  $\frac{1}{2}n(n+1)$ . Thus, if one wishes that the class of unit products just mentioned contain all products of the units, it is necessary that there exist between these  $\frac{1}{2}n(n+1)$  products at least  $\frac{1}{2}n(n+1) - 1$  independent equations. But it is simpler and more fruitful to divide the products of the units into two classes, and to suppose that the unit products of each class are combined in one sum, whence each class corresponds to only a single equation. The products of one class then appear as having one factor in common, and those of the other class appear as having the other factor in common; and the quantities obtained by multiplication then depend on one or the other of the two factors. It is in this way that I have arrived at the two principal types of multiplication which I have called *outer* and *inner* multiplication (see *Ausdehnungslehre*, § 12), and which I have subsequently applied in Crelles Journal (vols. 31, 36, 42, 44), and which M. Cauchy and M. Saint-Venant have recently proposed, without adding anything new.

---

<sup>6</sup>The associative law of multiplication tacitly introduced here will be dropped later; most of Grassmann's species of multiplication are nonassociative.

However, having published several applications of that multiplication, not only in the work cited but also in Crelles Journal (vols. 31, 36, 42, 44), I will dispense with going into more detail on that subject.

With regard to algebraic multiplication, the rules of the operation are well known; but they are not the same when that multiplication is applied to quantities that are not at all algebraic; for example, if it is a question of algebraically multiplying, not the magnitudes of lines, but the lines themselves, given in magnitude and direction, or indeed of any points. To place the advantage of that multiplication into evidence, we remark for example that, if the letters  $a, b, c, d, e, f, g, h, i$  denote any nine points of a curve of third order, and  $x$  a tenth point of that curve, one obtains for the equation of that curve

$$[a \cdot b \cdot c \cdot x][d \cdot e \cdot f \cdot x][g \cdot h \cdot i \cdot x] = 0,$$

where, for example,  $[a \cdot b \cdot c \cdot x]$  denotes the parallelogram area of the four points  $a, b, c, x$  (the product of three of them being a point and this point combined with the fourth giving the parallelogram area). The simplicity of this equation, which is of the third degree in  $x$ , and which comprehends all possible cases of curves of the third order, appears to me to speak loudly for the introduction of this analysis.

### § 3. *Symmetric Multiplication*

**Definition.** I call a multiplication *symmetric* if the product is not changed on interchanging the factors.

**Theorem 1.** If the equation

$$\sum (a_{r,s} u_r u_s) + \sum (a_{r,r} u_r u_r) + u_1 u_1 + (a_{s,s} u_s u_s) = 0,$$

be any one of the constraint equations. Having cast it into the form

$$\sum (a_{r,s} u_r u_s) + \sum (a_{r,r} u_r u_r) + u_1 u_1 + (a_{s,s} u_s u_s) = 0,$$

where the indices  $r$  and  $s$  are different from 1, we substitute  $-u_1$  for  $+u_1$  and obtain:

$$\sum (a_{r,s} u_r u_s) - \sum (a_{r,1} u_r u_1) + (a_{1,s} u_1 u_s) + u_1 u_1 + \cdots = 0,$$

the latter of which is also true when one replaces the indices 1 and 2 with any two different ones from among the indices  $1, \dots, n$ .

Now, besides the above hypothesis, suppose in addition the other, that one may interchange any pair of the units without alteration of the conditional equations. Admitting this hypothesis, upon substituting  $u_1$  for  $u_2$  and vice versa, one obtains the following theorem:

**Theorem 0.0.1.** *If it is possible to change the sign of any one of the relative units, and to interchange any pair of them at will, without alteration of the constraint equations, then it is necessary that those equations, so long as one has them, comprise one or more of the following four systems:*

- (1)  $u_1u_2 = u_2u_1,$
- (2)  $u_1u_2 + u_2u_1 = 0,$
- (3)  $u_1u_1 = u_2u_2 = \cdots = u_nu_n,$
- (4)  $u_1u_2u_2 \cdots u_nu_n = 0,$

*in which one can replace the indices 1 and 2 by any two different ones from among the values  $1, \dots, n$ .*

The results at which we have arrived can be expressed by the following propositions.

**Theorem 0.0.2.** *Between two factors composed of the relative units  $u_1, u_2, \dots, u_n$ , there are sixteen species of symmetric multiplication.*

*Proof.* For if, as follows from the first theorem, each of the equations (1), (2), (3), (4) either holds or does not hold, then since there are four equations, the number of different cases is  $2^4 = 16$ .  $\square$

### § 4. *Circular Multiplication*

In the second case, following the proposed course, there is adjoined to the hypothesis established previously another, that for the units  $u_1$  and  $u_2$  there can be substituted the quantities  $x_1u_1 + x_2u_2$  and  $y_1u_1 + y_2u_2$ , composed by the coefficients  $x_1, x_2, y_1, y_2$  supposed different from zero, without the constraint equations being altered.

Upon writing, for brevity,

$$x_1u_1 + x_2u_2 = a, \quad y_1u_1 + y_2u_2 = b,$$

one has

$$\begin{aligned} a^2 &= x_1^2u_1u_1 + x_2^2u_2u_2 + x_1x_2(u_1u_2 + u_2u_1), \\ ab &= x_1y_1u_1u_1 + x_2y_2u_2u_2 + x_1y_2u_1u_2 + x_2y_1u_2u_1, \end{aligned}$$

from which one deduces the values of  $bb$  and  $ba$  by interchanging the letters  $x$  and  $y$ .

According to the preceding theorem one has for the constraint equations the four systems of equations

- (1)  $u_1u_2 = u_2u_1, \dots$
- (2)  $u_1u_2 = 0, \dots$
- (3)  $u_1u_1 = u_2u_2 = u_3u_3 = \dots$
- (4)  $u_1u_1 + u_2u_2 + \dots + u_nu_n = 0,$

which must hold either separately or combined in any given manner. It is in addition evident that the first of these systems is not altered by either of the substitutions.

To find the transformations of the other systems, suppose for simplicity that there are more than two units. Then upon substituting  $a$  and  $b$  in place of the units  $u_1$  and  $u_2$ , as is allowed by hypothesis, one has in place of equation (2) the following:

$$0 = ab + ba = 2x_1y_1u_1u_1 + 2x_2y_2u_2u_2 + (x_1y_2 + x_2y_1)(u_1u_2 + u_2u_1);$$

from which, since the last term vanishes as a consequence of equations (2), and since  $x_1, x_2, y_1, y_2$  are all different from zero, there results

$$u_1u_1 = 0, \quad u_2u_2 = 0;$$

from which it follows ( $x_1$  and  $y_1$  being different from zero) that

$$u_3u_3 = 0;$$

and since  $u_3u_3$  cannot be zero (that would imply the linear multiplication, which we exclude here), one has

$$u_3u_3 = u_1u_1 = u_2u_2.$$

Thus one obtains by interchanging indices, as permitted by hypothesis:

$$u_1u_1 = u_2u_2 = u_3u_3 = \cdots ;$$

that is to say, equations (2) imply equations (3).

Now suppose instead that equations (3) hold. Upon applying to the equation

$$u_1u_1 = u_3u_3$$

the indicated substitutions, one obtains

$$u_3u_3 = a^2 = x_1^2u_1u_1 + x_2^2u_2u_2 + x_1x_2(u_1u_2 + u_2u_1),$$

whence, upon replacing the squares  $u_2u_2$  and  $u_3u_3$  by the equivalent square  $u_1u_1$ , one finds

$$(x_1^2 + x_2^2 - 1)u_1u_1 + x_1x_2(u_1u_2 + u_2u_1) = 0.$$

Upon changing the sign of the unit  $u_1$ , as allowed by hypothesis, and subtracting the equation so obtained from the preceding equation, one has

$$2x_1x_2(u_1u_2 + u_2u_1) = 0;$$

from which, since  $x_1$  and  $x_2$  are different from zero, there results

$$u_1u_2 + u_2u_1 = 0,$$

that is to say: equations (3) imply equations (2).

It is therefore evident that, under the given hypothesis, equations (2) and (3) do not hold separately, and consequently instead of these two systems of equations one has only one, the result of their combination.

To find the conditions to which the coefficients  $x_1, y_1, x_2, y_2$  are subject, transform equation (4) as well by the transformations stated above.<sup>7</sup> One has

$$\begin{aligned} 0 &= aa + bb + u_3u_3 + \cdots + u_nu_n \\ &= (x_1^2 + y_1^2)u_1u_1 + (x_2^2 + y_2^2)u_2u_2 + (x_1x_2 + y_1y_2)(u_1u_2 + u_2u_1) \\ &\quad + u_3u_3 + \cdots + u_nu_n. \end{aligned}$$

Consequently, upon subtracting equation (4) one obtains

$$0 = (x_1^2 + y_1^2 - 1)u_1u_1 + (x_2^2 + y_2^2 - 1)u_2u_2 + (x_1x_2 + y_1y_2)(u_1u_2 + u_2u_1).$$

But as equations (2) and (3) hold simultaneously, one has

$$u_1u_2 + u_2u_1 = 0 \quad \text{and} \quad u_1u_1 = u_2u_2;$$

and as equation (3) cannot be satisfied unless  $u_1u_1$  is different from zero, it follows that

$$x_1^2 + x_2^2 = 1, \quad y_1^2 + y_2^2 = 1, \quad x_1x_2 + y_1y_2 = 0.$$

These are therefore the relations to which the coefficients  $x, y$  are subject. But these relations express that the substitutions

$$u'_1 = x_1u_1 + x_2u_2, \quad u'_2 = y_1u_1 + y_2u_2$$

are circular transformations. The result at which we have arrived can therefore be expressed by the following propositions.

**Theorem 0.0.3.** *If, first of all, one may change the sign of any one of the relative units, and interchange any pair at will, and if as well one may have two quantities, composed of two units by means of coefficients different from zero, which, being substituted for those units, also satisfy the constraint equations, it is necessary that those*

---

<sup>7</sup>This is not clear. Perhaps what is meant is: We will not examine the (otherwise evident) relations between the magnitudes  $x, y$ , since in the next case (4) the same relations enter.



*equations, so long as one has them, comprise one or more of the following three systems:*

$$[1] \quad u_r u_s = u_s u_r,$$

$$[2] \quad u_r u_s + u_s u_r = 0, \quad u_1 u_1 = u_2 u_2 = \cdots = u_n u_n,$$

$$[3] \quad u_1 u_1 + u_2 u_2 + \cdots + u_n u_n = 0,$$

*where the symbols  $u_1, u_2, \dots, u_n$  represent the relative units and where  $r$  and  $s$  denote any two different indices.*

**Theorem 0.0.4.** *Each of the systems established by the preceding theorem may hold separately, or be combined with any of the others, and in all there exist eight species of multiplication that satisfy the given hypothesis.*

**Theorem 0.0.5.** *There exist eight species of multiplication for which the equations obtained remain unaltered by any circular transformation of the units.*

## § 5. *Linear Multiplication*

We now suppose that the hypothesis established previously remains in force, and adjoin to it yet another, namely that one may replace each of the relative units by any quantity composed of them, without the constraint equations ceasing to hold. Then if the equations are to remain satisfied when one replaces  $u_1$  by  $x_1u_1 + x_2u_2 + \dots$ , it is necessary that certain relations obtain between the coefficients. In order to discover these relations, recall the two systems of equations relative to a linear multiplication, to wit

$$\begin{aligned} [1] \quad & u_r u_s = u_s u_r, \\ [2^*] \quad & u_1 u_1 = u_2 u_2 = \dots = 0, \quad u_1 u_2 = u_2 u_1, \dots, \end{aligned}$$

or indeed the two of them combined.

**Theorem 0.0.6.** *If one may replace each unit contained in the constraint equations by a quantity arbitrarily composed of the units, without those equations ceasing to hold, it is necessary that the constraint equations, so long as one has them, comprise one of the two systems*

$$\begin{aligned} [1] \quad & u_r u_s = u_s u_r, \\ [2^*] \quad & u_1 u_1 = u_2 u_2 = \dots = 0, \quad u_1 u_2 = u_2 u_1, \dots, \end{aligned}$$

or indeed the two of them combined.

We designate the multiplications established by this theorem as *linear multiplications*, since, when applied to geometry, they are effected by linear constructions, without the necessity of recourse to the circle. It is evident that there exist, to be precise, four species of linear multiplication, and that

by means of linear multiplication one is in a position to effect with the greatest of ease the elimination of unknowns between equations, whether linear or nonlinear, and to solve at a stroke a great many problems in geometry as well as mechanics, otherwise difficult to solve; as I have already shown in 1844, and as M. Cauchy and M. Saint-Venant have recently proposed, without adding anything new.

## § 6. *Two Important Cases of Circular Multiplication*

Among the circular multiplications, two deserve particular attention, as they lead to types of multiplication of which one has already made various applications to analysis and geometry. These two are the ones established by the following equations:

$$\begin{aligned} \text{(inner multiplication)} \quad & u_r u_s = u_s u_r, \quad u_1 u_1 = u_2 u_2 = \cdots, \\ \text{(outer multiplication)} \quad & u_r u_s = -u_s u_r. \end{aligned}$$

I have called this inner multiplication (see *Ausdehnungslehre*, p. XI, and especially *Geometrische Analyse*, pp. 17–57, where I have given applications to geometry and mechanics), intending thus to express the particular relation that exists between it and outer multiplication. To find this relation, let us take as units three perpendicular lines, all of equal magnitude, and project any line, denoted by the letter  $a$ , on two other lines  $b$  and  $c$ , supposed perpendicular to each other; let  $b_1$  and  $c_1$  be these projections: then it is easy to convince oneself that the inner product of the two lines  $a$  and  $b$  is equal to the product  $ab_1$ , while the outer product of the same lines is equal to the product  $ac_1$ ; then since the lines  $a$  and  $b_1$  are situated one inside the other, and the lines are situated one outside the other, it is proper to designate these products by the names I have given them.

The other circular multiplication is specified by the equations

$$u_r u_s = u_s u_r$$

and

$$u_1 u_1 + u_2 u_2 + \cdots + u_n u_n = 0.$$

This remarkable multiplication is represented, if one does not suppose there are two relative units, by the multiplication of two complex quantities such as  $a + b\sqrt{-1}$ , or, if one takes four units, by the remarkable multiplication of quaternions, which in its applications to geometry and mechanics, as I have shown in *Crelles Journal* (vol. 49), is equivalent to my outer and inner multiplication.

Again it remains for us to discuss the eight symmetric multiplications that are not also circular. But we observe that these multiplications are of no use, either in analysis, or geometry, and we have not mentioned them at all, as it never seemed appropriate, in establishing the various types of circular multiplication.

## Editorial Notes

1. The associative law of multiplication tacitly introduced here will be dropped later; most of Grassmann's species of multiplication are nonassociative.

The limitation of the products to two factors is a real restriction of the problem. Thus in particular for products of three or more factors yet further equations enter, e.g. the equations of the associative law or those formed by analogy with Jacobi's identity,

$$uv \cdot w + vw \cdot u + wu \cdot v = 0,$$

which define a linear species of multiplication in Grassmann's sense.

In general the requirements that Grassmann imposes on these units seem somewhat arbitrary. Therein lies an essential limitation, that only the unit products of equal order may be combined by linear equations, thus excluding precisely such cases for which the subject is the most interesting and important, and presents the greater difficulty. One sees this carried out in *Theorie der Transformationsgruppen* by S. Lie (ed. F. Engel, Leipzig: 1893), vol. III., p. 748ff.

Whatever the relation between Grassmann's species of multiplication and specific groups, the following is evident without further ado: If, by a system of constraint equations between the unit products,

one defines the multiplication of two extensive quantities  $a = \sum \alpha_i e_i$  and  $b = \sum \beta_i e_i$  by the formula

$$a \cdot b = \sum \alpha_i \beta_j e_i e_j,$$

then the totality of the quantities  $\sum \alpha_i e_i$  forms a group with respect to this multiplication, and indeed a group in the sense of Lie and Engel if one introduces the commutator  $ab - ba$  and the associator  $(ab)c - a(bc)$  as the two characteristic operations.

2. The first requirement, that one may change the signs of the units independently of one another, leads to the symmetry or skew-symmetry of the constraint equations. The second requirement, that the equations remain valid upon the substitution of linear combinations of the units, leads to the invariance of the constraint equations under the general linear group. The third requirement, that the equations remain valid upon circular transformations, leads to the invariance under the orthogonal group. The fourth requirement, that the equations remain valid upon arbitrary linear substitutions, leads to the invariance under the general linear group.

3. I.e. the equation

$$\sum a_{r,s} u_r u_s = 0 \quad (r < s).$$

4. The theorem is to be understood to mean that one may interchange the indices  $1, \dots, n$  in both the units and in the coefficients, and both arbitrarily.
5. This is not clear. Perhaps what is meant is: We will not examine the (otherwise evident) relations between the magnitudes  $x, y$ , since in the next case (4) the same relations enter.
6. The assertion claimed for the case of two units is not to be considered a proof, and accordingly theorems 4 and 5 have an exception in the case  $n = 2$ .

For two units there are twelve systems of constraint equations that satisfy Grassmann's first and second requirements.

If we set aside the trivial linear species of multiplication

- (0) (no constraint equations),
- (1)  $u_1u_2 = u_2u_1$ ,
- (2)  $u_1u_2 + u_2u_1 = 0$ ,
- (3)  $u_1u_1 = u_2u_2$ ,
- (4)  $u_1u_1 + u_2u_2 = 0$ ,

all with the transformation group

$$u'_1 = x_1u_1 + x_2u_2, \quad u'_2 = \pm(x_2u_1 + x_1u_2),$$

then one obtains the twelve cases by combining these equations. For example, the combination (2, 3) gives

$$u_1u_2 + u_2u_1 = 0, \quad u_1u_1 = u_2u_2,$$

with the transformation group

$$u'_1 = x_1u_1 + x_2u_2, \quad u'_2 = \pm(-x_2u_1 + x_1u_2).$$

From these four cases, (3), (4), (2, 3), (2, 4), there arise yet another four if one imposes condition (1):  $u_1u_2 = u_2u_1$ . The group of allowed transformations will obviously not change thereby.

Since the last mentioned group does not include the circular evolutions, theorem 6 still holds for the case of two units.

## On Electrodynamics

[Crelles Journal 83 (1), 57–64, 1877]

The law for the reciprocal action of two current elements, which I advanced in 1845 in Poggendorfs Annalen 64 (1), in opposition to Ampere's law as the more probable, has through the latest pioneering works of Herr Clausius, particularly in his articles in this journal, vol. 82, p. 1ff, found not just new support but, one may say, a secure foundation. In fact the force law for current elements which Herr Clausius derived from his general theory on p. 130 of the article cited agrees precisely with the law that I presented in the article mentioned above.

Since Herr Clausius, who clearly missed my article, does not mention this agreement in his works (cf. Poggendorfs Annalen 156; 157; and the Transactions of the Natural History Association of Prussian Rhineland and Westphalia),<sup>8</sup> I will discuss it briefly here and add a few, as I believe, not unimportant consequences.

It is very easy to demonstrate the agreement between the two laws with the analysis treated in my *Ausdehnungslehre* (of 1844 and 1862); however, since I cannot assume that the reader is familiar with that analysis, I will at first rely on the theorems of conventional analysis, in particular on the theorem that, if  $a_1, a_2, a_3$  are the orthogonal coordinates and  $a$  the length of one displacement, and  $b_1, b_2, b_3$  the corresponding coordinates and  $b$  the length of a second displacement, then the cosine of the angle between the directions of the two displacements, which I will denote by  $\cos(ab)$ , is given by

$$\cos(ab) = \frac{a_1b_1 + a_2b_2 + a_3b_3}{ab}.$$

Now let  $x, y, z$  be the coordinates of the initial point of a current element  $ds$ , and  $x', y', z'$  those of the initial point of a second current element  $ds'$ ; let  $dx, dy, dz$  be the coordinates of  $ds$  (that is, the projections of  $ds$  on the coordinate axes),  $dx', dy', dz'$  those of  $ds'$ ; let  $r$  be the distance between the two initial points, so that

$$r^2 = (x - x')^2 + (y - y')^2 + (z - z')^2;$$

---

<sup>8</sup>H. Grassmann, "Zur Elektrodynamik," Pogg. Ann. 64 (1845), 1–18; R. Clausius, "Ueber die Ableitung eines neuen elektrodynamischen Grundgesetzes," J. Reine Angew. Math. 82 (1877), 1–20.



if in addition  $i$  and  $i'$  are the two current intensities,  $\epsilon$  the angle between the two current elements, and  $k$  a constant numerical factor, then according to Clausius (op. cit. p. 130)

$$X = \frac{ki i'}{r^3} \left( -r \frac{dr}{ds} \frac{dx'}{ds'} + \frac{dx}{ds} \cdot r \frac{dr}{ds'} \right).$$

Now

$$\frac{1}{r} \frac{dr}{ds} = \frac{d(\log r)}{ds} = \frac{1}{2r^2} \frac{d(r^2)}{ds} = \frac{1}{2r^2} \frac{d[(x-x')^2 + (y-y')^2 + (z-z')^2]}{ds}.$$

Consequently

$$\frac{1}{r} \frac{dr}{ds} = -[(x-x')dx + (y-y')dy + (z-z')dz] : r^3 \cdot \frac{1}{ds}.$$

Furthermore

$$\cos \epsilon = (dx \cdot dx' + dy \cdot dy' + dz \cdot dz') : ds \cdot ds'.$$

Inserting these values into (1), one obtains

$$X = \frac{ki i'}{r^3} [(x-x') \cdot dx \cdot dx' - dx' \cdot ((x-x')dx + (y-y')dy + (z-z')dz)]$$

with corresponding expressions for  $Y$  and  $Z$ .

If one takes the  $z$ -axis orthogonal to the plane in which  $r$  and  $ds'$  lie, then  $z-z'=0$  and  $dz'=0$ , whence also  $Z=0$ . We can therefore call this plane the plane of action; the formula for  $X$  then becomes

$$X = \frac{ki i'}{r^3} \{ (x-x')(dx \, dx' + dy \, dy') - [(x-x')dx + (y-y')dy] dx' \}$$

and correspondingly for  $Y$ .

Now, as in my article in Poggendorfs Annalen 64, p. 9, let the product  $i' ds'$  be denoted by  $a$ , the product  $i ds$  by  $b$ , and the (orthogonal) projection of  $b$  on the plane of action by  $b_1$ , and further let the angles

$$\angle ra = \alpha, \quad \angle rb_1 = \beta, \quad \angle b_1 a = \gamma$$

be defined, so that

$$\alpha = \angle rb_1 + \angle b_1 a = \beta + \gamma.$$

One takes the  $y$ -axis in the direction  $b_1$ , and the  $x$ -axis in the perpendicular direction  $c$  lying in the plane of action, so that  $\angle bc = 90^\circ$ . Then  $i' dx'$  and  $i' dy'$  are the coordinates of  $a$ ,  $i dx$  and  $i dy$  those of  $b$ , whence

$$\cos \gamma = \cos(b_1 a) = \frac{ii'(dx dx' + dy dy')}{ab},$$

and

$$\cos \beta = \cos(rb_1) = \frac{i(x - x')dx + i(y - y')dy}{rb_1}.$$

If we insert these values, then

$$X = \frac{k}{r^3}[(x - x')abi \cos \gamma - i' dx' rb_1 \cos \beta],$$

$$Y = \frac{k}{r^3}[(y - y')abi \cos \gamma - i' dy' rb_1 \cos \beta].$$

Now  $y - y'$  is the projection of  $r$  on  $b_1$ , and is therefore equal to  $r \cos(rb_1) = r \cos \beta$ , and  $i' dy'$  is the projection of  $a$  on  $b_1$ , and therefore equal to  $a \cos(b_1 a) = a \cos \gamma$ , whence

$$Y = \frac{kab}{r^3}(\cos \beta \cos \gamma - \cos \gamma \cos \beta) = 0.$$

Further,  $x - x'$  is the projection of  $r$  on  $c$ , and therefore equal to

$$r \cos(rc) = r \cos(rb_1 + b_1 c) = r \cos(90^\circ + \beta) = -r \sin \beta,$$

and  $i' dx'$  is the projection of  $a$  on  $c$ , and therefore equal to

$$a \cos(ac) = a \cos(ab_1 + b_1 c) = a \cos(b_1 c - b_1 a) \\ = a \cos(90^\circ - \gamma) = a \sin \gamma,$$

whence

$$X = \frac{ka}{r^3}(\sin \beta \cos \gamma + \cos \beta \sin \gamma) = \frac{kab}{r^3} \sin(\beta + \gamma),$$

or

$$X = \frac{kab}{r^2} \sin \alpha. \quad (2)$$

Since  $Y$  and  $Z$  are zero this expression represents the entire force. It is identical to the expression that I communicated in Poggendorfs Annalen 64, p. 9, formula 4, except that  $k$ , whose value depends on the choice of units, was there itself set to unity.

But formula (2) achieves a completely new significance from the principle found by Herr Clausius. It no longer represents an hypothesis which could perhaps stand with equal rights among other hypotheses; rather it follows necessarily from Clausius' presentation. In order to demonstrate this and to add yet other consequences I will briefly review the course of development Clausius follows.

Herr Clausius proves that the principle that the founder of a unified theory of electrodynamics, Herr W. Weber, has advanced, is only in harmony with experiment provided one assumes that in every galvanic current the positive and negative electricity is moved in opposite directions with equal velocities. In particular he shows that if, in a galvanic current, the opposed electricities are moved with unequal velocities (if for example the negative is at rest), then upon taking Weber's law as a foundation the constant current must act dispersively on resting electricity, which contradicts experiment.

I remark that one can handle this proof in a most elementary way by means of a linear current, which consists of two concentric circular arcs and two straight displacements, provided the common center of the two circular arcs lies on the point at which the electricity rests, and the prolongations of the two displacements go through the same point. In this case a simple inspection of Weber's formula,

$$\frac{1}{c^2} \left( \frac{dr}{dt} \right)^2 + \frac{2r}{c^2} \frac{d^2r}{dt^2}$$

shows that the positive current (excluding the static action which is always cancelled by the negative) exerts on the resting positive electricity an attractive action going through the center of the current, which is proportional to the square of the velocity while the negative electricity is just as strongly repelled, and that on the other hand the negative current exerts actions in the opposite sense, but so that these actions are proportional to the square of the velocity with which the negative electricity flows. The net action is thus only zero if both velocities are equally great; in all other cases a dispersion of resting electricity must occur, contrary to experiment.

Herr Clausius proves the corresponding result for Riemann's principle. They would both agree with the facts of the case if it can be

assumed that in every galvanic current positive and negative electricity flows with equal velocities (in opposite directions). This assumption is however not acceptable, since for example in electrolytes the electricities

are moved by the ions, and these in general have unequal velocities.

Now Herr Clausius proceeds from the fundamental assumption of both Weber's and Riemann's laws, that the force depends only on the distance between the two particles and on the directions and magnitudes of their velocities and accelerations. He thereupon combines only the results of more reliable observations with the principle of the conservation of energy, and arrives at his fundamental formula (66), in which an unknown function of  $r$  still appears. This unknown function drops out, however, when one specifies the force with which a current element  $ds'$  acts on another  $ds$ , and so one arrives at equation (1) and its equivalent (2), which must therefore be regarded as perfectly well grounded, so long as one does not perhaps resort to higher than the second time derivatives of the moving electricities.

Here it is to be noted that formulas (1) and (2) also remain valid if the two opposite electricities flowing in opposite directions do not have the same velocity, since by the intensity of the current is meant the sum of the positive and, flowing in the opposite direction, negative electricity, which in unit time flows through a cross section of the lead.

In this regard I include a very simple derivation of the action of a constant closed current on a current element, which at the same time leads to results which, I believe, were previously unknown.

If one replaces  $a$  by its value  $i'ds'$  in (2), then upon integration one immediately finds the force  $v$  that a flowing displacement  $BC$  of intensity  $i'$  exerts on a current element whose initial point lies on  $A$  and whose projection on the plane  $ABC$  equals  $b_1$ . Thus if  $AD = h$  is the altitude of the triangle  $ABC$ , and the parts of the triangle are labelled as usual, then

$$v = \frac{ki'cb_1}{r^2}(\cos \beta + \cos \gamma).$$

In particular, if  $b_1$  is directed similarly to  $BC$ , then  $v$  has the direction  $AD$ ; and if  $b_1$  is rotated by an arbitrary angle in the plane of action, the direction of  $v$  is rotated by the same angle, while the value of  $v$  remains the same. If  $a$  is the third angle of the triangle and  $m$  is the line that bisects this angle and extends to the opposite side, then it is known that

$$\cos \beta + \cos \gamma = 2 \sin \frac{a}{2},$$

and the above formula becomes

$$v = \frac{2ki'b_1 \sin \frac{1}{2}a}{m}. \quad (3)$$

But in order to be able to employ this formula directly, it is necessary to represent in it the direction of the force  $v$  as well. To this end I must resort to a few concepts of the geometric analysis that I have presented in my *Ausdehnungslehre* of 1844 and 1862, that is to the concept of displacements, of plane areas, to the addition of displacements and plane areas and to the inner product of a plane area with a displacement.

Thus I set two bounded straight lines equal as displacements only if they have equal lengths and directions, and I set two plane elements equal as plane areas only if the two planes are parallel and the plane elements have equal and equally signed surface areas. Two displacements may be added by laying them contiguously together, whereupon the displacement from the initial point of the first to the final point of the second is the sum of the two displacements. Two plane areas may be added if one lays them contiguously as parallelograms, that is, lays them so that the base of the second coincides with the deck (the side opposite the base) of the first, whence the parallelogram whose base is the base of the first and whose deck is the deck of the second is the sum of the two plane areas. Finally, by the inner product of a plane area  $F$  of unit content with a displacement  $b$ , written  $[F|b]$ , I mean a displacement  $g$  as long as the projection  $b_1$  of  $b$  on  $F$  that stands perpendicular to  $b_1$  (and hence  $b$ ) in the plane  $F$  and which, laid contiguously with  $b_1$ , inflects to the same side toward which the circumference of  $F$  is traversed. If one replaces  $F$  by  $\lambda F$ , where  $\lambda$  expresses the plane area, then  $[\lambda F|b] = \lambda g$ .

If we apply this to the above case, and fix  $F$  as of the same sign as the plane area of the triangle  $ABC$ , then it is clear according to the above that the direction of the force is of opposite sign to  $[F|b]$  and thus, if the direction of the force is simultaneously to be expressed, then in (3)  $b_1$  is to be replaced by  $-[F|b]$ , that is

$$v = \frac{2ki' \sin \frac{1}{2}a}{m} [F|b]. \quad (4)$$

Now let the galvanic current flow through an arbitrary spatial polygon, one side of which is  $BC$ , while  $A$  remains the initial point of the current element  $b$ . For the triangle adjacent to  $ABC$  let  $a_1$  replace  $a$ ,  $m_1$  replace  $m$ ,  $F_1$  replace  $F$ , etc.; then the force  $V$  with which the whole polygon acts on the current element  $b$  is

$$V = -2ki'[Q|b], \quad \text{where} \quad Q = \sum \frac{\sin \frac{1}{2}a}{m} F + \dots \quad (5)$$

This equation includes the following important theorem:

“Given an arbitrary closed current in space, for every point  $A$  there is a certain plane through  $A$ , called the *plane of action* of the current with respect to the point  $A$ , which has the characteristics: first, that every element ( $b$ ) emanating from  $A$  will experience no action from the current if it is orthogonal to the plane; second, if the element stands at a slant to the plane it will experience the same action as its (orthogonal) projection ( $b_1$ ) on this plane would experience; third, that the force it experiences lies in this plane and is orthogonal to the projection ( $b_1$ ) of the current element and thus also to the current itself; and fourth, if  $g$  is the force which that current element ( $b$ ) emanating from  $A$  experiences at any given position and the projection ( $b_1$ ) of the current element on the plane of action is rotated through any given angle in this plane, then the force  $g$  is rotated through the same angle without changing its value.”

This plane of action is extremely easy to construct if the current is a polygonal current in space. For it is parallel to  $Q$ , and according to formula (5)  $Q$  is immediately found by the addition of the plane areas.

Much more convenient than the procedure followed here is the method in which one introduces the geometric analysis at once and from the outset. Then however the outer product of two displacements must be introduced. Thus by the outer product  $[ab]$  of two displacements  $a$  and  $b$  I mean the plane area of the parallelogram that has  $a$  as base and  $b$  as the side connected to it. Then from formula (1) there immediately follows the formula

$$P = \frac{k}{r^3} [r \cdot a \cdot b] \quad (2^*)$$

(which I will not prove here), where  $r$ ,  $a$ ,  $b$  represent the displacements themselves, whose lengths we denote by  $r$ ,  $a$ ,  $b$ , and where  $P$  is the force in its magnitude and direction.<sup>9</sup>

---

<sup>9</sup>The author indicates by  $[r \cdot a \cdot b]$  the outer product of the three displacements  $r$ ,  $a$ ,  $b$ , which is equal to the volume of the parallelepiped constructed from them.



If we set  $g = i'ds'$ , here, where again  $ds'$  simultaneously represents the direction of the current element  $ds'$  and we take  $ds'$  as an element of an arbitrary closed current, then upon extending the integration to the whole closed current we obtain

$$V = ki' \left[ \frac{Q}{r^3} \middle| b \right], \quad \text{where} \quad Q = \int \frac{r \cdot ds'}{r^3}. \quad (5^*)$$

Here as everywhere there is not the least difficulty in transforming the formulas of geometric analysis directly into the ordinarily quite complicated formulas of conventional analysis. To this end one has only to assume for the three mutually orthogonal coordinate axes three displacements whose directions are those of the positive axes and whose lengths are unity, these being  $e_1, e_2, e_3$ ; then if  $a_1, a_2, a_3$  are the coordinates of a displacement  $a$  one has only to replace  $a$  by  $a_1e_1 + a_2e_2 + a_3e_3$ . If one applies this procedure to each displacement and then carries out the additions and multiplications according to the standard laws of algebra, except that one may not combine or interchange the factors of a product without further consideration, then there remain in the formulas no displacements other than  $e_1, e_2, e_3$ , whose multiplication, whether it be outer or inner, is performed according to the definitions of these products.

In our formula (5\*) one then finally obtains  $V$  in the form

$$V = V_1e_1 + V_2e_2 + V_3e_3,$$

where  $V_1, V_2, V_3$  are the algebraic expressions sought.

...determined, and the general integration is thus completed. Only if  $K$  is equal to one of the  $2n$  values  $A_1, \dots, A_{2n}$  does the case mentioned above enter, that the  $y_1, \dots, y_n$  of the mean integration become infinite or undefined; in that case one can first take  $K$  to be infinitesimally different from that value  $A$  and define the mean integration with respect to this  $K$ .

The mean integration always remains independent of the solution of that equation of  $2n$ -th degree.

Now in order to be able to carry this over to the equations of motion, we must give equations (10) yet another form. For since the terms  $ge^{Kt}$ , which are to represent the forces, become infinite with  $t$  for real  $K$ , they do not correspond to any natural case under this assumption. One therefore replaces  $ge^{Kt}$  by the two terms

$$c \cos Kt + c' \sin Kt,$$

that is

$$\frac{c - c'i}{2}e^{Kti} + \frac{c + c'i}{2}e^{-Kti}.$$

These two terms are only distinguished by the algebraic sign of  $i = \sqrt{-1}$ .

Now if in (10) one first replaces  $g_ie^{\kappa_it}$  by

$$\frac{c_i - c'_i i}{2}e^{\kappa_i ti}$$

and so forth, then in (11) one must replace  $g_i$  by

$$\frac{c_i - c'_i i}{2}$$

and so forth, and in addition  $K$  by  $iK$  and  $K^2$  by  $-K^2$ ; they defined by (11) will then be imaginary, say  $v + wi$ , so that

$$u_1 = (v_1 + w_1 i)e^{Kt}, \dots, u_n = (v_n + w_n i)e^{Kt}.$$

Second, if one replaces  $g$  in (10) by

$$\frac{c + c'i}{2},$$

then there appear values that are distinguished from the above for  $u_1, \dots, u_n$  only by the algebraic sign of  $i$ ; if these are symbolized by  $u'_1, \dots, u'_n$ , then

$$u'_1 = (v_1 - w_1 i)e^{-Kt}, \dots, u'_n = (v_n - w_n i)e^{-Kt};$$

consequently

$$u_i + u'_i = 2v_i \cos Kt - 2w_i \sin Kt = a_i \cos Kt + b_i \sin Kt \quad (1)$$

if

$$2v_i = a_i \quad \text{and} \quad -2w_i = b_i.$$

## 502 SELECTED PAPERS

It must now be shown that in the motion defined by linear equations, of the form (10), the mean integration simultaneously yields the mean motion as defined above.

In the motion of an assembly of  $m$  equally heavy points in space the  $n$  in equations (10) and (11) equals  $3m$ , the equation in  $A$  is thus of  $6m$ -th degree. We assume orthogonal coordinate axes. Then the total living force is

$$T = \frac{1}{2} \sum \delta u_i^2,$$

whence

$$\int T dt = \frac{1}{2} \int \sum \delta u_i^2 dt,$$

where the sum is on  $u_1, \dots, u_{3m}$ .

Now in the general integration  $u_i$  consists in part of terms of the mean integration, which are of the form

$$a_i \cos Kt + b_i \sin Kt,$$

and in part of  $6m$  terms of the form

$$z_i e^{A_i t};$$

and  $\delta u_i$  thus includes terms of the form

$$Kb_i \cos Kt - Ka_i \sin Kt$$

and of the form

$$A_i z_i e^{A_i t},$$

and  $\delta u_i^2$  therefore includes the squares of these terms and the double product of every pair. One sees immediately that the terms of the form  $A_i^2 z_i^2 e^{2A_i t}$  for real  $A_i$  will themselves be infinite with infinite  $t$ , whence  $\int T dt$  is certainly less if these terms are missing than when they are present. Thus for the demonstration of mean motion we can drop these terms; and likewise if

$$A_1 = \alpha + \beta i,$$

and  $\alpha$  is not zero. Consequently there are only terms to consider where

$$A_1 = \beta i;$$

then another value of  $A$  is say

$$A_2 = -\beta i$$

and the real terms in  $u_i$  resulting thereby are of the form

$$p \cos \beta t + q \sin \beta t,$$

thus in  $\delta u_i$  of the form

$$\beta(q \cos \beta t - p \sin \beta t),$$

thus of a form corresponding to the terms of the mean integration.

If we consider first the squares, for example

$$(Kb_i \cos Kt - Ka_i \sin Kt)^2,$$

whence in  $T dt$  the term

$$\frac{1}{2}K^2(b_i \cos Kt - a_i \sin Kt)^2 dt,$$

then this yields

$$\frac{1}{4}K^2\{b_i^2(1 + \cos 2Kt) dt + a_i^2(1 - \cos 2Kt) dt - 2a_i b_i \sin 2Kt dt\}.$$

Upon integration this gives

$$\frac{1}{4}K^2(a_i^2 + b_i^2)t + P_i,$$

where  $P_i$  represents a purely finite periodic function.

If we now consider the double product of two such terms, for example

$$K(b_i \cos Kt - a_i \sin Kt)$$

and

$$\beta(q \cos \beta t - p \sin \beta t),$$

then in  $T dt$  this gives the term

$$K\beta dt(b_i q \cos Kt \cos \beta t + a_i p \sin Kt \sin \beta t - b_i p \cos Kt \sin \beta t - a_i q \sin Kt \cos \beta t)$$

$$= K\beta dt \left\{ \frac{b_i q + a_i p}{2} \cos(K - \beta)t + \frac{b_i q - a_i p}{2} \cos(K + \beta)t - \frac{b_i p + a_i q}{2} \sin(K + \beta)t \right\}$$

thus if  $K$  is not equal to  $\beta$  this yields purely finite periodic terms. Now we can take  $t$  to be so large that compared to terms of the form

$$\frac{1}{4}K^2(a_i^2 + b_i^2)t, \dots$$

the periodic terms vanish; then

$$\int T dt = \frac{1}{4} \sum K^2(a^2 + b^2)t + \frac{1}{4} \sum \beta^2(p^2 + q^2)t,$$

where the first sum relates to all the terms of the mean integrations and the second to the others. Here the  $a$  and  $b$  are invariable values, whereas  $p$  and  $q$  may be zero, so that for sufficiently great  $t$  the integral

$$\int T dt$$

will be the least if the  $p, q$  are all zero, that is the integral is the mean integral. It is thus proven that for linear differential equations the motion of the mean integration is at the same time the mean motion.

#### 504 SELECTED PAPERS

I call a term of the form

$$a \cos Kt + b \sin Kt,$$

where  $K$  is positive, whether  $a$  and  $b$  are numbers or displacements, an *elliptic term*, and  $K$  its *indicator*. Thus if in this case  $a$  and  $b$  represent displacements in different directions, and

$$a \cos Kt + b \sin Kt$$

is a displacement  $r$  that extends from a fixed point, then in the time interval  $2\pi : K$  the endpoint describes an ellipse, and in fact so that the displacement  $r$  itself describes equal areas in equal times, that is in the time element  $dt$  the area

$$\frac{1}{2}[ab]K dt;$$

the displacements  $a$  and  $b$  are conjugate radii of the ellipse. Thus if one sets

$$\cos Kt = u, \quad \sin Kt = v,$$

then that radius

$$r = ua + vb$$

and

$$u^2 + v^2 = 1,$$

which is the equation of an ellipse with conjugate radii  $a$  and  $b$ .

Furthermore, in the time element  $dt$  the endpoint of  $r$  describes the displacement

$$\delta r \, dt,$$

that is

$$(b \cos Kt - a \sin Kt) K \, dt,$$

and  $r$  itself the surface area

$$\frac{1}{2} [\delta r \cdot r] \, dt,$$

that is

$$\frac{1}{2} K (a \cos Kt + b \sin Kt) \cdot (b \cos Kt - a \sin Kt) \, dt.$$

Since

$$[aa] = [bb] = 0, \quad [ab] = -[ba]$$

and

$$\cos^2 Kt + \sin^2 Kt = 1,$$

the enclosed outer product yields the value  $[ab]$ , whence the surface area described in the time element  $dt$  is

$$= \frac{1}{2} [ab] K \, dt.$$

We can now express the law of mean motion for our case as follows: If the motion of an assembly of points is represented by linear differential equations, then there correspond to the elliptic terms that appear in the expression of the force elliptic terms with the same indicator in all displacements drawn from a fixed point to the moving points, and indeed the coefficients of these terms are completely defined by the given equations, and besides these terms no others enter in the mean motion.

I remark in addition that the stability or instability of the mean motion can be obtained most easily from the principles developed above.

## § 6. Application to the Theory of Tides

Here we first suppose that the system subject to tides is an assembly of  $m$  points of unit mass. Then, for relative motion with respect to the center of gravity, equation (3) of §3 holds, that is

$$\delta^2 y_1 = q_1 + \frac{1}{m}p - \frac{1}{m}p_1, \dots, \delta^2 y_m = q_m - \frac{1}{m}p,$$

in which I have separated the inner forces  $q_1$ , etc. from the outer forces  $p_1$ , etc., and set

$$p_1 + \dots + p_m = p.$$

Now let the system be subjected to a uniform rotation about a fixed axis through the center of gravity and assume, as is permitted in the theory of tides in first approximation (which is all that is considered here), that the points are displaced only infinitesimally from the configuration that they would assume in this uniform rotation. Further, let  $n$  be the angular velocity of this rotation, whence  $nt$  is the rotation during the time  $t$ . If  $a$  is a displacement laid in the plane of rotation (and thus orthogonal to the axis of rotation), then it is transformed by the rotation through the angle  $nt$  into

$$a \cos nt + a' \sin nt,$$

where  $a'$  is orthogonal to  $a$ , lying to its positive side in the plane of rotation, and is equal to  $a$  in length. Following the usual analogy, we denote this displacement by  $ae^{int}$ , where  $i$  is the planimetric representation of  $\sqrt{-1}$ . Then  $a$  is thereby transformed into

$$a(\cos nt + i \sin nt) = ae^{int},$$

and thus<sup>1</sup>

$$\delta(ae^{int}) = aiae^{int}.$$

## 506 SELECTED PAPERS

**§ 7.** In the equations developed above for the rotation of the system there appears the displacement  $r$ , whose direction is that of the attracting center (the sun or moon), and in whose length  $\rho$  is expressed the distance of this center from the center of gravity of the system. Now this distance changes as well, though in a periodic way. Thus one can also represent  $r$  as a sum of elliptic terms. Since  $r$  is

---

<sup>1</sup>See editorial note 4.



a displacement of constant direction, which rotates uniformly about the axis of the system, then one can, in consequence of the formula developed above, also set

$$r = ae^{\alpha t} + \dots,$$

where the omitted terms are of higher order. Further, let the direction of the axis of the system with respect to the sun be denoted by  $x$  and with respect to the moon by  $x'$ .

Now  $r$  is always a principal radius of the ellipsoid that is described by the point of the fluid mass on which the attraction of the central body acts most strongly. If we decompose this ellipsoid into two, of which one has radius  $r$  and the second has principal radii that are orthogonal to  $r$  and differ from  $r$  by only infinitesimal amounts, and if we denote the increments of the latter radii by  $dr$  and  $dr'$ , then these are proportional to the attractions in these directions, and are therefore, if  $L$  is the attraction at distance 1 and  $\rho$  the distance,

$$\frac{L}{\rho^3} \cdot dr \quad \text{and} \quad \frac{L}{\rho^3} \cdot dr'.$$

Now if  $\mu$  is the mass of the central body, then

$$L = \mu, \quad \text{and} \quad \frac{L}{\rho^3} = \frac{\mu}{\rho^3}.$$

If  $\nu$  is the mean motion of the central body, then

$$\rho^2 \nu = \text{const.}, \quad \text{whence} \quad \rho = \frac{\text{const.}}{\sqrt{\nu}}.$$

Thus if one substitutes for  $\mu$  its value in terms of  $\rho$  and  $\nu$ , one obtains

$$\frac{L}{\rho^3} = \nu^2.$$

For the sun,  $\nu = n$ , and for the moon we denote it by  $n'$ . Then for the sun

$$\frac{L}{\rho^3} = n^2,$$

and for the moon

$$\frac{L}{\rho^3} = n'^2.$$

Now the increments  $dr$  and  $dr'$  are in the direction of the attraction that produces them, thus in the directions of the principal

radii orthogonal to  $r$ . The direction of one of these principal radii makes an angle of  $45^\circ$  with the plane of the orbit; if we denote the displacement in this direction by  $s$  and the other by  $s'$ , then these two displacements together are equivalent to the rotation of  $r$  by the angle  $n'$ , but have opposite senses. Then the attractive forces corresponding to them are, since the one has the sense of  $r$  and the other the opposite sense,

$$2\frac{L}{\rho^3} \cdot r.$$

## 508 SELECTED PAPERS

If we substitute these values in equations (14), then we obtain for the sun the equation

$$\delta^2 u_1 + 2\delta u_1 \cdot \alpha + (x_1 + u_1)\alpha^2 = q'_1 + \frac{1}{m}(p_1 - \mu p)e^{-\alpha t} + \frac{3n^2}{m} \{ [\cos \eta \cos \vartheta + \sin \eta \sin \vartheta \cos(\omega_1 + \gamma t)] r e^{-\gamma t} - x_1 \} \quad (2)$$

and for the moon the corresponding equation with  $n'$  instead of  $n$  and  $\gamma'$  instead of  $\gamma$ . But  $q'_1$  depends on the relative separations of the points, and thus here on

$$x_1 - x_r + (u_1 - u_r),$$

that is, on

$$x_1 - x_r + (u_1 - u_r),$$

where  $u_1 - u_r$  is infinitely small compared to  $x_1 - x_r$ . Thus  $q'_1$  separates into two terms, the first of which does not include  $u$  while the second is a linear function of  $u$ . Let the latter be denoted by  $q''_1$ , the former by  $\phi_1$ ; then we can separate the above equation into two equations, that is

$$x_1 \alpha^2 = q''_1, \dots, x_m \alpha^2 = q''_m, \quad (3)$$

which define the state of equilibrium, and

$$\delta^2 u_1 + 2\delta u_1 \cdot \alpha + u_1 \alpha^2 - \phi_1 = \frac{1}{m}(p_1 - \mu p)e^{-\alpha t}, \quad (4)$$

$$\delta^2 u_m + 2\delta u_m \cdot \alpha + u_m \alpha^2 - \phi_m = \frac{1}{m}(p_m - \mu p)e^{-\alpha t}, \quad (5)$$

which have exactly the form of the equations obtained in §5, and their mean integration thus yields the motion of the tides.

The problem is thus to develop the right sides of these equations (13) in elliptic terms. We first assume only a single star, and specifically let it be nearly spherical and the distance of its center from the center of gravity of the system be infinitely great compared to the dimensions of the system. The attraction that a sphere exerts on an exterior point by its gravitation is the same as if all its mass were concentrated at its center. If  $L$  is this attraction at a distance 1, then at a distance  $\rho$  it equals

$$\frac{L}{\rho^2}.$$

Now if  $r$  is the displacement from the center of gravity of the system to the center of the sphere at time  $t = 0$  and if  $\rho$  is the length of  $r$ , then at this time  $p_1$ , omitting terms of higher degree of smallness, is, including magnitude and direction,

$$\frac{L(r - x_1)}{\rho^3}$$

or

$$\frac{L(r - x_1)}{(r - [r|x_1])^3}.$$

Upon development this yields

$$\frac{L}{\rho^3} \left( r - 3 \frac{[r|x_1]}{\rho^2} r - x_1 \right).$$

Then, since  $x$  is assumed to extend from the center of gravity of the system, and thus

$$[r|x] = 0,$$

one obtains

$$\frac{L}{\rho^3} \left( 3 \frac{[r|x_1]}{\rho^2} r - x_1 \right).$$

Consequently, at time  $t = 0$  the right side of equation (13) equals

$$\frac{L}{\rho^3} \left( \frac{3[r|x_1]}{\rho^2} r - x_1 \right).$$

Now if, for the duration of a day, the distance of the star and its declination are assumed constant, while in time  $t$  its right ascension changes by  $\beta t$ ,<sup>2</sup> then in time  $t$ ,  $r$  transforms into

$$re^{\beta t},$$

---

<sup>2</sup>See editorial note 6.

and  $x_1$  into

$$x_1 e^{\beta t},$$

and the right side of equation (13) becomes

$$\frac{L}{\rho^3} \left( \frac{3[r e^{\beta t} | x_1 e^{\beta t}]}{\rho^2} r e^{\beta t} - x_1 e^{\beta t} \right) e^{-\alpha t}.$$

Now according to the concept of the inner product, its value does not change if both its factors rotate about the same axis and by the same angle, for example  $-\beta t$ ;<sup>3</sup> and if  $\alpha - \beta = \gamma$ , one finds that the right side of equation (13) equals

$$\frac{L}{\rho^3} \left( \frac{3[r | x_1]}{\rho^2} r e^{-\gamma t} - x_1 \right).$$

## 510 SELECTED PAPERS

If the length of  $x_1$  equals  $\mu\rho$ , then

$$[r | x_1 | r] = \mu\rho^2 \cos \Phi,$$

where  $\Phi$  is the angle between  $r$  and  $x_1 e^{int}$ . If  $\eta$  is the angle that the axis  $\alpha$  forms with  $r$ , and  $\vartheta$  the angle that it makes with  $x_1$ , and if  $\omega_1$  is the angle of the plane  $\alpha r$  to the plane  $\alpha x_1$ , so that  $\omega_1 + \gamma t$  is the angle that the plane  $\alpha r$  forms with the plane  $\alpha x_1 e^{int}$ ;<sup>4</sup> then

$$\cos \Phi = \cos \eta \cos \vartheta + \sin \eta \sin \vartheta \cos(\omega_1 + \gamma t),$$

whence we find that the above expression

$$= \frac{L}{\rho^3} \{ 3\mu [\cos \eta \cos \vartheta + \sin \eta \sin \vartheta \cos(\omega_1 + \gamma t)] r e^{-\gamma t} - x_1 \},$$

where one can replace  $r$  by  $r_1 + r_2$ , where  $r_1$  lies on the axis,  $r_2$  on the equator, and thus one can replace  $r e^{-\gamma t}$  by

$$r_1 + r_2 e^{-\gamma t}.$$

If one then replaces

$$\cos(\omega_1 + \gamma t)$$

---

<sup>3</sup>See editorial note 7.

<sup>4</sup>See editorial note 8.

by its equivalent value<sup>5</sup>

$$\frac{e^{i(\omega_1+\gamma t)} + e^{-i(\omega_1+\gamma t)}}{2}$$

then at this point one perceives that the whole expression consists of three elliptic terms with indicators 0,  $\gamma$ , and  $2\gamma$ ,<sup>6</sup> where  $\gamma$  is the apparent angular velocity of the star, and thus  $2\pi : \gamma$  is its apparent period.

## 512 SELECTED PAPERS

Now upon introducing a second star that, as is the case for tides, has an influence on the motion, and whose apparent period is  $2\pi : \gamma'$ , there enter two more elliptic terms with indicators  $\gamma'$  and  $2\gamma'$ . If we denote these five elliptic terms for the first point by

$$P_{1,0}, P_{1,\gamma}, P_{1,2\gamma}, P_{1,\gamma'}, P_{1,2\gamma'},$$

then the first of equations (13) becomes

$$\delta^2 u_1 + 2\delta u_1 \cdot \alpha + u_1 \alpha^2 - \phi_1 = P_{1,0} + P_{1,\gamma} + P_{1,2\gamma} + P_{1,\gamma'} + P_{1,2\gamma'}. \quad (6)$$

It therefore follows that since with tides one only sees the mean motion, for  $u_1$  there is likewise a sum of five elliptic terms, thus

$$u_1 = U_1 + U_{1,\gamma} + U_{1,2\gamma} + U_{1,\gamma'} + U_{1,2\gamma'}, \quad (7)$$

where the elliptic terms are at the same time functions of  $x$ , for example

$$U_\gamma = a_\gamma \cos \gamma t + b_\gamma \sin \gamma t,$$

where  $a_\gamma$  and  $b_\gamma$  are functions of  $x$ . If one wants the surface of the sea (taken as a point in our system) to remain constant, then one must have

$$[x_1 u_1] = 0,$$

and this must also be so if one substitutes for  $u_1$  the value from (15). Since this must be satisfied for all  $t$ , the coefficients of  $\cos \gamma t$ ,  $\sin \gamma t$ , etc., must each individually vanish, and this yields the conditions for the form of the functions  $a_\gamma$ ,  $b_\gamma$ , etc.

## EDITORIAL NOTES

<sup>5</sup>See editorial note 9.

<sup>6</sup>See editorial note 10.

1. Grassmann's recollection of the date appears to have been faulty, since an examination certificate including an evaluation of his dissertation is dated 1 May 1840, and that year Whitsunday fell on 7 June. [Justus Grassmann]
2. Grassmann did not live to write this article.
3. The coefficients of  $e^{A_1 t}$  all include one and the same arbitrary constant, as do those of  $e^{A_2 t}$  etc.
4. Cf. "Rotation About a Point" [this volume].
5. The following may serve to explain. For any given displacement  $p$  denote by  $Pp$  the projection on the axis of rotation,  $Sp$  that on the plane of rotation, so that  $p = Pp + Sp$ .

Then under the rotation the displacement  $a$  transforms into the displacement  $Pa + Sa \cdot e^{\alpha t}$ , as was previously shown in the text. If one sets the displacement  $= ae^{\alpha t}$ , then

$$\delta ae^{\alpha t} = Sa \cdot e^{\alpha t} \alpha, \quad \delta^2 ae^{\alpha t} = -n^2 Sa \cdot e^{\alpha t}.$$

Putting  $\delta a$  for the displacement  $Sa \cdot \alpha$  and  $\delta^2 a$  for the displacement  $-n^2 Sp \cdot a$  gives

$$Sa \cdot e^{\alpha t} = ae^{\alpha t} \alpha, \quad \delta^2 ae^{\alpha t} = ae^{\alpha t} \alpha^2.$$

If one rotates the displacement  $ae^{\alpha t}$  yet again by the angle  $\beta t$ , then one has rotated the displacement  $a$  by the angle  $(\alpha + \beta)t$  in all. Thus

$$ae^{\alpha t} \cdot e^{\beta t} = ae^{(\alpha + \beta)t},$$

a formula which is used subsequently (for  $\beta = -\alpha$ ).

6. More precisely: If the right ascension changes by  $n'$  in unit time, then  $\beta = in'$ .
7. Better: Rotates by  $-n't$ , whereby the displacements are multiplied by  $e^{int}$ .
8. More accurately: Since  $n - n' = \gamma$ , this angle is  $\omega_1 + \gamma t$ , and in the following equation replaces  $\gamma$  after the "cos" symbol.
9. Here  $\omega_1$  also replaces  $\omega$ .
10. Better said: The indicators are 0,  $\gamma$ , and  $2\gamma$ .

## FROM GRASSMANN'S UNPUBLISHED PAPERS

**Mechanics According to the Principles of Extension  
Theory\***

*Second Article*

§ 1. In this article I limit myself to the application of the concepts of point magnitudes of first and second order to mechanics.

Instead of the displacement  $x$ , whose initial point is fixed, it is natural to designate its final point, since its initial point introduces something rather arbitrary into the treatment. However, this point must then be regarded as a magnitude with two aspects; first, the position of the point, and second, a numerical magnitude (according to the *Ausdehnungslehre* of 1844 the weight). The time differential  $\delta$  assumes however the invariance of the weight. The formulas of the first article are not essentially changed by this new method of treatment; but completely new relations appear in the point magnitudes of second order, relations that were presented in part by Poinsot, Möbius, Plücker, v. Staudt, Felix Klein (*Math. Ann.* 4, 403ff), but were not reduced to their essentials.

Specifically, according to the *Ausdehnungslehre* of 1844 ( $A_1$  § 106ff,  $A_2$  nos. 247, 249) as point of second order there appears the product of two points, that is the line segment lying between the elementary points that comprise its factors, so that two products  $[ab]$  and  $[cd]$ , in which  $a, b, c, d$  are elementary points, may be set equal as line segments if and only if  $[ab]$  and  $[cd]$  lie on the same straight line and have equal directions and lengths.

**Part Three    515**

In a finite domain the line segments  $A, B, C, \dots$  yield zero if their sum forms a closed polygon; that is, then

$$2[AB] + 2[AC] + 2[BC] + \dots = 0$$

(cf. A<sub>1</sub> §124; no. 286). The infinitely distant line segment is then Poincot's force couple (cf. also *Math. Ann.* 4, 404, footnote \*\*\*).

The representation of the connection between linear geometry and the mechanics of rigid bodies of F. Klein (*Math. Ann.* 4, 403–415) (as well as the theory of null systems of v. Staudt) lacks the unifying thread presented by extension theory. This consists in the sum  $S$  of line segments (not reducible to a straight line) that encompasses both v. Staudt's null system and Plücker's linear complex. The plane to which a point  $a$  corresponds is  $[aS]$ , the point to which a plane  $\alpha$  corresponds is  $[\alpha S]$ , the null line  $P$  is that corresponding to the equation  $[PS] = 0$  (and forms the complex defined by  $S$ ).

The involutorial position of two complexes, more properly two magnitudes of second order,  $S$  and  $S_1$ , is defined by the equation  $[SS_1] = 0$ . If  $S$  is represented (with the help of the four points  $a, b, c, d$  in the form)

$$S = x[ab] + x'[cd] + y[ac] + y'[db] + z[da] + z'[bc],$$

\* If  $p, a, b, c, d$  are simple points, then it follows that

$$\begin{aligned} [p \cdot ab \cdot ab] &= 0, \\ [p \cdot ab \cdot ac] + [p \cdot ac \cdot ab] &= 0 = p[ab \cdot ac] + p[ac \cdot ab], \\ [p \cdot ab \cdot cd] + [p \cdot cd \cdot ab] &= p[abcd]. \end{aligned}$$

It thereby follows that, if  $P$  and  $Q$  are any two line segments, it is always the case that

$$[P|Q] + [Q|P] = p[PQ]$$

and further, that for any given sum  $S$  of line segments

$$[pSS] = p[SS].$$

With the help of the supplements there follows for a planar element  $\pi$  the equation

$$[\pi SS] = \pi[SS].$$

## 516 SELECTED PAPERS

[where the coefficients  $x, x', y, y', z, z'$  are numbers] and  $S_1$  correspondingly with the index 1, then the equation is

$$xx'_1 + x_1x' + yy'_1 + y_1y' + zz'_1 + z_1z' = 0.$$



Thus incidence [of the two magnitudes  $S$  and  $S_1$ ].\*

§ 2. Each irreducible, that is not reducible to a single, force is reducible to two forces, and indeed there are thereby four arbitrary numerical conditions, namely:

- a) a point of one force and the plane of the other, provided only both point and plane are distinct [see  $A_2$  no. 285 or]
- b) the situation of one force, provided it does not have the situation of a null line.

Thus if  $[ab]$  is the [line of a] force and

$$S = x[ab] + [cd],$$

where  $x$  is a number, then it follows that  $S - x[ab]$  is reduced to a line segment, whence necessarily

$$[S - x[ab]]^2 = 0,$$

from which

$$S^2 - 2x[Sab] = 0,$$

and

$$x = \frac{S^2}{2[Sab]}$$

follows, provided it is not true that  $[Sab] = 0$ , that is that  $[ab]$  is a null line.

Further one has

$$[aS] = [acd], \quad [bS] = [bcd],$$

$$[aS \cdot bS] = [acd \cdot bcd] = [cd] \cdot [abcd]$$

whereby  $[cd]$  is determined. This assumes  $[abcd] \neq 0$ . If  $[abcd] = 0$ , then  $[abS] = 0$ , whence  $[ab]$  would be a null line. If this is excluded, then it follows that

$$S^2 = 2[abcd], \quad [cd] = \frac{2[aS \cdot bS]}{S^2}.$$

\* If  $S_1$  is a linear magnitude, then

$$x_1x'_1 + y_1y'_1 + z_1z'_1 = 0,$$

and the condition that  $S_1$  is a null line is then given by the equation in the text, which shows that the null lines form a complex. (Ed.)

**Part Three 517**

Conversely, if this equation is satisfied, then  $S = p[ab] + [cd]$ , where  $p$  is a numerical factor.

§ 3. Now if, as is easily shown, an infinitely small rotational motion about an axis can also be completely represented by a line segment, since the magnitude of the infinitely small rotation can be represented by the length of this line segment and furthermore that the result of two infinitely small rotations taken about different axes is always independent of their order, so that the fundamental theorem of addition is valid, it is then self-evident that all laws for the action of forces on a fixed body are identical to the laws of these rotations, assuming that those forces and these rotations are interpreted as point magnitudes of second order (cf. also no. 347, fn.).

§ 4. The centrifugal moment of a force  $p$ —regarded as a displacement—with respect to a point distant from the point of application by [the displacement]  $r$ , is\*

$$[p|r]]$$

and thus the centrifugal moment for a series of forces

$$= \sum [p|r].$$

If the reference point is changed, say to  $q$ , then that sum

$$= \sum [p|(r - q)] + \sum [p|q]$$

or, if  $\sum p = s$ ,

$$= \sum [p|(r - q)] + [s|q].$$

When is  $\sum [p|(r - q)] = 0$ , that is  $\sum [p|r] = [s|q]$ ? If  $[p|s] = \alpha s^2$ , then necessarily  $\sum [s|(\alpha s - q)] = 0$ , that is  $\alpha s - q$  must be orthogonal to  $s$ , whereby the location of  $q$  is determined;  $q$  lies on the plane orthogonal to  $s$  through the point  $\alpha s$ . It is self-evident that the line on which the final point of  $q$  lies is orthogonal to  $s$ . The locus of the points  $q$  is the principal line of the system of forces.

**518 SELECTED PAPERS**

§ 5. *Rotation of a displacement about an axis.*

$x^\alpha$  denotes the displacement that forms the same angle with  $a$  as the displacement  $x$  and results by a rotation about  $a$  by the rotation angle  $\alpha$ ; the sign is chosen so that  $[ax]$  is rotated in the positive sense through  $\alpha$  if  $[ax] \neq 0$ .

If now  $[ax] = 0$ , then  $x^\alpha$  is the displacement that has the same direction and length as  $x$ , but starting from the final point of the displacement  $ae^{\alpha x}$ , and hence

$$x^\alpha = xe^{\alpha x}.$$

§ 6. If  $x_1^\alpha, x_2^\alpha$  are two displacements that form equal angles with  $a$ , then

$$x_1^\alpha + x_2^\alpha = (x_1 + x_2)^\alpha.$$

§ 7. If  $b$  is orthogonal to  $a$ , then  $b^\alpha$  can be represented as a shift, whence

$$b^\alpha = b \cos \alpha + [ab] \cdot \frac{\sin \alpha}{a^2}.$$

### Part Three 519

§ 8. If  $x = [a|x] \cdot a + (x - [a|x] \cdot a)$ , where the second term is orthogonal to  $a$  (since its inner product with  $a$  is zero), then by § 7

$$\begin{aligned} (x - [a|x] \cdot a)^\alpha &= (x - [a|x] \cdot a) \cos \alpha + [a(x - [a|x] \cdot a)] \frac{\sin \alpha}{a^2} \\ &= (x - [a|x] \cdot a) \cos \alpha + [ax] \frac{\sin \alpha}{a^2}, \end{aligned}$$

and thus, if one denotes the second term by  $x_a$ ,

$$x^\alpha = [a|x] \cdot ae^{\alpha x} + x_a \cos \alpha + [ax] \frac{\sin \alpha}{a^2}. \quad (8)$$

§ 9. *Corollary.* If  $c$  is orthogonal to  $a$  and  $b$  and  $\alpha$  the angle from  $a$  to  $b$ , then

$$a^\alpha b^\alpha = c^2 \cdot a.$$

§ 10. *Fundamental theorem for the combination of rotations.*

### 520 SELECTED PAPERS

*Proof.* It is always true that

$$x = [a|x]a + (x - [a|x]a).$$

Here the second term is orthogonal to  $a$  since its inner product with  $a$  is zero, whence one obtains (according to § 3)

$$x_a = [a|x] \cdot a - x,$$

and therefore

$$[a|x]^\alpha = [a|x]ae^{\alpha x},$$

since  $[a|x]$  is a number. In the same way one finds

$$x_b = [b|x] \cdot b - x,$$

$$x_b^\beta = [b|x]be^{\beta x},$$

and the theorem thus reads:

$$(x_a^\alpha + [a|x]ae^{\alpha x})_b^\beta + [b|x]be^{\beta x} = (x_b^\beta + [b|x]be^{\beta x})_a^\alpha + [a|x]ae^{\alpha x}.$$

Since  $x_a^\alpha$  forms the same angle with  $a$  as  $x_a$ , the angle is also formed with  $b$ , whence

$$(x_a^\alpha)_b^\beta = x_a^{\alpha+\beta}$$

and since in this formula  $\alpha$  and  $\beta$  are arbitrary, one can also write

$$(x_b^\beta)_a^\alpha = x_b^{\beta+\alpha}.$$

### Part Three 521

Thus

$$[a|a_2^\alpha]a_2^\alpha = [a|a_2 \cdot qa_2] = [a|a_2(q_1[a_1a_2] + q_3[a_3a_2])].$$

Since however

$$[a|a_2|a_3a_2] = 0, \quad [a|a_2|a_1a_2] = 1,$$

thus

$$\frac{1}{a^2} - q = a_2 \sin \alpha + a_3(\cos \alpha - 1) = a_3 - a_3^{\alpha a} = [[a_1a_2]] = ([[a_1a_2]])^{\alpha a},$$

$$q = [[a_1]a_2] - ([[a_1]a_2])^{\alpha a} = [[a|a_2]] - ([[a|a_2]])^{\alpha a}.$$

### 522 SELECTED PAPERS

§ 16. In the same way one obtains, if  $b^\beta$  and  $c^\gamma$  are rotations,

$$(b^\beta \cdot c^\gamma)_a^\alpha - b^\beta \cdot (c^\gamma)_a^\alpha = \frac{1}{a^2}([b|c]] - [[b|c]]^{\alpha a}) = \frac{q}{a^2}, \quad (9)$$

where  $q = [a|a_2^\alpha]$  is the same value as above. If  $[bc] = 0$ , then  $q = 0$ .

§ 17. If  $A$  is a line segment of unit length,  $x$  a point,  $\alpha$  an angle, then

$$xA^\alpha$$

denotes the rotation about the axis  $A$  by the angle  $\alpha$ , so that, if  $xA^\alpha = x_1$ ,

$$[x_1A] = [xA] \quad \text{and} \quad \frac{[x_1xA]}{[xA]^2} = \tan \alpha.$$

§ 18. If  $A$  and  $B$  are of unit length, then

$$(xA^\alpha)B^\beta = x \cdot A^\alpha B^\beta$$

and

$$A^\alpha B^\beta = (A^\alpha B^\beta)_A^\alpha \cdot A^\alpha,$$

where according to § 12

$$(A^\alpha B^\beta)_A^\alpha = B^{\alpha+\beta}.$$

### Part Three 523

Thus if  $q = [A|A_2^\alpha]$ ,

$$A^\alpha B^\beta = B^{\alpha+\beta} \cdot A^\alpha \cdot q.$$

§ 19. Every rotation can be represented as a product of rotations about three mutually orthogonal axes, and can also be represented as a shift, whence combined from infinitely small rotations about two axes.

§ 20. Every line that yields zero upon outer multiplication by the shifting system is an axis of the system.

§ 21. Every shifting system can be reduced to two lines. If  $A$  and  $B$  are the reduced lines and  $a, b$  their lengths, then all axes of the system are cut by the line

$$C = \frac{A}{a} \pm \frac{B}{b}.$$

The system shifts all points of this line in their own direction.

### 524 SELECTED PAPERS

§ 22. The product  $A^\alpha B^\beta$ , if  $A$  and  $B$  are of equal length and at right angles, represents a quarter revolution about the axis  $[AB]$  if  $\alpha = \beta = \frac{1}{2}\pi$ .

---

# THE POSITION OF THE HAMILTONIAN QUATERNIONS IN EXTENSION THEORY

[*Math. Comm.* 12, 375–386, 1877]

Since extension theory makes only one arbitrary assumption, that is that there exist magnitudes that can be numerically derived from more than one unit, and proceeds from this in a completely objective way, all expressions that are numerically derivable from a number of independent units, and in particular the Hamiltonian quaternions, have their definition in extension theory and only find their scientific foundation in it. This was not previously recognized, and even Dillner, in his instructive article on quaternions (*Math. Ann.* 11, 168ff.) does not mention extension theory even once, although he derives a whole series of theorems from the theory of quaternions which have already found their basis much more simply and emerge more naturally from the nature of the subject in my *Ausdehnungslehre* of 1844, and likewise in the later revision of 1862. Also it is reprehensible, and little serviceable to the theory of quaternions, that according to Hamilton's procedure one designates simple and long familiar concepts by new, often unsuitable terms, as "vector" for "displacement," "tensor" for "length" or "numerical value" (A<sub>2</sub> no. 414), etc.

The Hamiltonian quaternions arise from one of the multiplications that I presented in my article "Sur les différents genres de multiplication" in Crelle's Journal 49, 136ff., and have associated with the three groups of equations

$$e_r e_s = e_s e_r, \quad (10)$$

$$e_r e_s = -e_s e_r \quad (r \neq s), \quad (11)$$

$$e_r^2 = \lambda \quad (12)$$

where in addition the law that  $[abc] = 0$  if two of the factors are equal, and conceptually that  $[abc]$  is equal to the parallelepiped in which three coterminal edges are equal to  $a, b, c$ .

## 526 SELECTED PAPERS

brackets) the central product. It then follows at once that the central product  $ab$  of two displacements can be represented in the form

$$ab = \lambda[a|b] + \mu[ab] \quad (13)$$

where  $\lambda$  and  $\mu$  are arbitrary, non-zero constants.

From the constraint equations (2) it follows that for the central product of two displacements there are  $\frac{1}{2}n(n-1)+1$  mutually independent unit products, one of which (say  $e_1^2$ ) is the basis of the inner product  $[a|b]$ , the others ( $e_1e_2, e_1e_3, e_2e_3$ , etc.) of the outer product. In space, where the number of mutually independent displacements equals three, that is  $n=3$ , the number of mutually independent products to which central multiplication reduces equals four. The constraint equations of central multiplication thus become

$$\begin{aligned} e_3e_2 &= -e_2e_3, & e_1e_3 &= -e_3e_1, & e_2e_1 &= -e_1e_2, \\ e_1^2 &= e_2^2 = e_3^2. \end{aligned} \quad (14)$$

But the essential characteristic of central multiplication in space as a domain of third order is that the number of mutually independent unit products in (a) equals the number of units, and one can therefore reduce them to these. Thus the units of the products are maintained the same as originally, even if one includes the numerical unit fundamental to (b). This simple relation vanishes in domains of higher order, so that in extension theory, which treats domains of arbitrary order, central multiplication retains no simple meaning. I therefore limit myself to space and assume that the three basis units  $e_1, e_2, e_3$  are three equally long and mutually perpendicular displacements, whose common length I take as the unit of length.

From the supplements (A<sub>2</sub> nos. 94ff, or “Note on the addition of a note to the article “On the different kinds of multiplication”)[in this volume]

### Part Three 527

are interpreted as the supplements of  $e_2e_3, e_3e_1, e_1e_2$  and are thus defined as

$$e'_1 = [e_2e_3], \quad e'_2 = [e_3e_1], \quad e'_3 = [e_1e_2], \quad (15)$$

$$[e_2e_3] = \lambda e_1, \quad [e_3e_1] = \lambda e_2, \quad [e_1e_2] = \lambda e_3,$$

where the stroke  $|$  is the symbol for the supplements, and it is assumed that  $e_1, e_2, e_3$  form a normal set. Further, it is established there that for an arbitrary displacement  $a$ , which is derivable from the original units by the numbers  $a_1, a_2, a_3$ , its complement is derivable from the supplements of those units by the same numbers, that

is

$$\lambda(a_1e_1 + a_2e_2 + a_3e_3) = a_1[e_2e_3] + a_2[e_3e_1] + a_3[e_1e_2], \quad (16)$$

$$\lambda(a_1[e_2e_3] + a_2[e_3e_1] + a_3[e_1e_2]) = a_1e_1 + a_2e_2 + a_3e_3, \quad (17)$$

and it is shown (A<sub>2</sub>, no. 37ff.) that the same relation obtains if one replaces the original units with the units of any other normal set. With the help of this concept we can now write the fundamental equation (4) in the form

$$ab = \lambda[a|b] + \mu[ab], \quad (18)$$

where  $\lambda$  and  $\mu$  are constant numbers. If  $\lambda$  and  $\mu$  are changed in the same ratio, for example by the factor  $\nu$ , then the product is only changed by the same numerical factor; its form thus remains unchanged. We can therefore set one of these numbers to unity without any essential change. We set  $\mu = 1$ . We then define  $\lambda$  such that each central product of three factors is subject to the law of combinability (the associative principle), that is  $abc = a(bc)$ . This will be satisfied if it is valid for the unit products of central multiplication. These unit products can be reduced to those of inner and outer multiplication.

## 528 SELECTED PAPERS

There then follows for the central multiplication of three units, retaining the cyclic assumption on  $r, s, t$ ,

$$(e_re_s)e_t = e_ie_t = \lambda = e_re_r = e_r(e_se_t)$$

and likewise

$$(e_ie_s)e_r = -e_re_t = -\lambda = -e_ie_s = e_i(e_se_r);$$

that is, combinability is valid for three different factors, just as for three similar factors. Thus also for similar ones separated by a dissimilar. For

$$e_r(e_se_r) = -(e_se_r)e_r = (e_re_s)e_r.$$

On the other hand  $(e_re_r)e_s = \lambda e_s$  and  $e_r(e_re_s) = e_re_t = -e_s$ . Thus if combinability is to be valid for this case as well, then necessarily  $\lambda = -1$ . Conversely, if  $\lambda = -1$ , then combinability of the factors also follows for the remaining products of the units. For then

$$(e_se_s)e_r = \lambda e_r = -e_r = -e_se_t = e_s(e_re_s);$$



further,

$$(e_r e_s) e_s = e_r e_i = -e_r = \lambda e_r = e_r (e_s e_s)$$

and

$$(e_s e_r) e_r = -e_i e_r = -e_s = \lambda e_s = e_s (e_r e_r).$$

Thus there follows combinability for each set of three unit factors, whence for every three factors and consequently for arbitrarily many ( $A_1$ , § 3). For central multiplication we therefore set  $\lambda = -1$ , while  $\mu = 1$  was chosen previously, whence

$$ab = -[a|b] + [ab]. \quad (19)$$

From this fundamental equation follow all laws of quaternions, and indeed most of them with the greatest of ease. In addition the natural nomenclature now follows naturally of itself. We can call  $-[a|b]$  the *inner*,  $[ab]$  the *outer* part of the quaternion. If  $a$  and  $b$  are parallel, the outer part will be zero, and the factors will be interchangeable as with every inner product. If  $a$  and  $b$  are mutually perpendicular, then the inner part vanishes.

### Part Three 529

If  $a + \alpha$  is a quaternion, then it is known that one designates the quaternion  $a - \alpha$  as that conjugate to it. Of fundamental significance is the law

$$\begin{aligned} \text{If } (a + \alpha)(\beta + b) &= \gamma + c, \\ \text{then also } (\beta - b)(a - \alpha) &= \gamma - c. \end{aligned}$$

Thus the inner part of the first product is  $a\beta - [\alpha|b]$ , whence this  $= \gamma$ , but  $-[\alpha|b]$  is also the inner part of the second product. On the other hand the outer part of the first product is  $\alpha b + \beta a + [ab] = c$ , the outer part of the second is  $-\alpha b - \beta a + [b\alpha]$ , or since  $[b\alpha] = -[\alpha b]$  equals  $-c$ , this second product  $= \gamma - c$ . It is clear at once that this can be extended to arbitrarily many factors. Thus

**(II)** *The product of arbitrarily many quaternions is conjugate to the oppositely-ordered product of the conjugate quaternions.*

This theorem represents formula (14) of Dillner, from which his formula (13) follows if one sets the inner parts  $(\alpha, \beta, \dots)$  to zero.

If the displacement is  $a = a_1 e_1 + a_2 e_2 + a_3 e_3$ , then  $[\alpha|a]$ , which I have symbolized briefly as  $a^2$ , equals  $a_1^2 + a_2^2 + a_3^2$  and represents the square of the length of that displacement. In accordance with this analogy, if  $q = \alpha + a_1 e_1 + a_2 e_2 + a_3 e_3 = \alpha + a$ , I call

$$\sqrt{\alpha^2 + a_1^2 + a_2^2 + a_3^2}, \quad \text{that is } \sqrt{\alpha^2 + a^2} = \sqrt{q \cdot q},$$

the length of the quaternion  $q$  (according to Hamilton the *tensor*).— Now if one multiplies the first formula in II by the second, one obtains

$$(\alpha + a)(\beta + b)(\beta - b)(\alpha - a) = (\gamma + c)(\gamma - c),$$

that is

$$(\alpha^2 + a^2)(\beta^2 - b^2) = \gamma^2 - c^2.$$

Since  $\beta^2 - b^2 = \beta^2 + b^2$  is a number, and thus its position is a matter of

### 530 SELECTED PAPERS

indifference, one can write  $(\beta^2 + b^2)(\alpha^2 + a^2)$  instead; but then the right side is also of this form, and one thus has

(III) *The length of the product of arbitrarily many quaternions equals the product of their lengths.*

Now if  $\rho$  is the length of a quaternion  $q = \alpha + \beta a$ , where  $a$  is a displacement of unit length, then  $\rho^2 = \alpha^2 + \beta^2$ . If  $\alpha = \rho \cos \gamma$  then  $\beta = \rho \sin \gamma$ , whence

$$q = \rho(\cos \gamma + a \sin \gamma).$$

Here  $a$  is called the *measure* and  $\gamma$  the *angle* of the quaternion, while  $\cos \gamma + a \sin \gamma$  is the *quaternary unit*. The product of two quaternary units of equal measure leads to very simple results. Thus if  $a$  is the measure of two quaternary units and  $\alpha$  and  $\beta$  their angles, then one finds

$$\begin{aligned} & (\cos \alpha + a \sin \alpha)(\cos \beta + a \sin \beta) \\ \text{(IV)} \quad & = \cos \alpha \cos \beta - \sin \alpha \sin \beta + a(\cos \alpha \sin \beta + \sin \alpha \cos \beta) \\ & = \cos(\alpha + \beta) + a \sin(\alpha + \beta), \end{aligned}$$

since  $a^2 = -a^2 = -1$ ; that is

*One multiplies quaternary units of equal measure by adding their angles.*

Included implicitly in this is that one exponentiates a quaternary unit by a whole number upon multiplying its angle by this number. We can also retain the same determination for exponentiation by a rational and a negative number, but with the condition that the angle of the quaternion to be exponentiated remains within the limits of a single revolution, for example lying between  $\pi$  and  $-\pi$  (cf. my *Arithmetik*, Stettin: 1860, nos. 426–433).

A definition for these conjunctions is necessary, and likewise the condition given above, since otherwise one offends against the rule of logic that one may not define the same thing in two different ways, particularly if the two definitions are contradictory. The latter would however be the case for exponentiation by fractional numbers, if one did not restrict the angle.

### Part Three 531

If the measure of a quaternion equals a unit  $e_1$ , that is the displacement of length 1, then  $q$  can be represented in the form

$$q = a_0 + a_1 e_1;$$

if one lets

$$\frac{1}{q} = \beta_0 + \beta_1 e_1 + \beta_2 e_2 + \beta_3 e_3,$$

then there must obtain the equation

$$(\alpha_0 + \alpha_1 e_1)(\beta_0 + \beta_1 e_1 + \beta_2 e_2 + \beta_3 e_3) = 1,$$

or if one multiplies out according to the fundamental formula  $ab = -[a|b] + [ab]$ :

$$(\alpha_0 \beta_0 - \alpha_1 \beta_1) + (\alpha_0 \beta_1 + \alpha_1 \beta_0) e_1 + (\alpha_0 \beta_2 - \alpha_1 \beta_3) e_2 + (\alpha_0 \beta_3 + \alpha_1 \beta_2) e_3 = 1,$$

whence

$$\alpha_0 \beta_0 - \alpha_1 \beta_1 = 1, \quad \alpha_0 \beta_1 + \alpha_1 \beta_0 = 0, \quad \alpha_0 \beta_2 - \alpha_1 \beta_3 = 0, \quad \alpha_0 \beta_3 + \alpha_1 \beta_2 = 0.$$

From the last two equations it follows that

$$(\alpha_0^2 + \alpha_1^2) \beta_2 = 0, \quad (\alpha_0^2 + \alpha_1^2) \beta_3 = 0;$$

if  $\alpha_0^2 + \alpha_1^2 = 0$ , that is  $\alpha_0 = \alpha_1 = 0$ , then  $q = 0$ , and division is impossible. If this is excluded, then  $\beta_2 = \beta_3 = 0$  and

$$\frac{1}{q} = \beta_0 + \beta_1 e_1 = \frac{\alpha_0 - \alpha_1 e_1}{\alpha_0^2 + \alpha_1^2};$$

that is

$$\frac{1}{\alpha_0 + \alpha_1 e_1} = \frac{\alpha_0 - \alpha_1 e_1}{(\alpha_0 + \alpha_1 e_1)(\alpha_0 - \alpha_1 e_1)} = \frac{\alpha_0 - \alpha_1 e_1}{\alpha_0^2 + \alpha_1^2}, \quad (20)$$

and this agrees with formula (IV) if one substitutes  $-\gamma$  for  $\gamma$  and divides by  $\rho^2$ .

§ 2. I now turn to the applications. The applications of quaternions presented by Hamilton, and more explicitly by his successors, fall in part under those I have given for extension theory. Here we select only those that appear characteristic of quaternions, that is those that depend on the form of the quaternion product. Thus in particular the applications in which a quaternion is regarded as a quotient and thereby the rotation in space about a point is made to depend on the quaternion, is characteristic. I first present the application to the generation of ellipses and spheres. The theorem that a quaternion can be represented as a quotient of two displacements appears to me to be more simply proved in the following way.

If  $\alpha$  is a displacement of unit length and  $\beta$  a number, then I seek a displacement  $x$  satisfying the equation

$$x(\alpha \tan \beta) = x\alpha^{2\beta}.$$

Since  $\alpha^2 = -1$ , one can also write this equation as

$$x\alpha^{2\beta} \cdot \alpha = x \cdot \alpha\alpha^{2\beta},$$

and from this it follows that  $x$  must lie in the same plane as  $\alpha$  and  $\alpha^{2\beta}$ . If one therefore sets

$$x = \alpha \cdot a + \alpha^{2\beta} \cdot b,$$

## 532 SELECTED PAPERS

then

$$x \cdot \alpha \tan \beta = -a + b(\alpha \tan \beta) = -a - b\alpha \tan \beta$$

and

$$x\alpha^{2\beta} = a(\alpha \cdot \alpha^{2\beta}) + b = -a + b.$$

If both are to be equal, then  $a$  must equal  $b \cos^2 \beta$ , whence

$$x = \alpha \cdot b \cos^2 \beta + \alpha^{2\beta} \cdot b = ab \cos \beta \cdot \alpha^\beta,$$

and if one sets  $b \cos \beta = c$ ,

$$x = ac \cdot \alpha^\beta = c \cdot \alpha^{\frac{1}{2}\pi + \beta}$$

and therefore

$$\alpha^{\frac{1}{2}\pi+\beta} : \alpha = \alpha \tan \beta, \quad (21)$$

whence the theorem is proven.

According to this theorem, if  $a$  and  $b$  are the legs and  $p$  the altitude of a right triangle, then

$$a : b = \left( \frac{a}{p} \right) \tan \alpha,$$

if  $\alpha$  is the angle opposite  $a$ ; and if  $c$  is the hypotenuse, then

$$c : a = \left( \frac{a}{c} \right) \tan \beta : \left( \frac{a}{p} \right) \tan \alpha.$$

Here the quotients are quaternions, and in particular those that serve to represent rotations about a point.

On the same grounds calculation with quaternions is to be rejected for the rotation of a spatial structure about an axis (Dillner, nos. 39–42), and for the translation (shift) of a spatial structure, since for the rotation about a point the product of the quotients must again be a quotient of the same kind, which is not the case for these other applications.

### Part Three 533

§3. *To generate the straight line, circle, ellipse, parabola, hyperbola.*

If the radius drawn from a fixed point to a point moving uniformly in a circle is multiplied by a constant quaternion, then the point describes an ellipse. For if  $r$  is the radius,  $q$  a constant quaternion, then  $rq = r_1$  and  $r_1$  is the new radius. If  $q = \alpha + a$ , then the inner part of  $rq$ , that is  $\alpha r$ , is that radius of the ellipse which is the projection of  $r_1$  on  $r$ ; and the outer part  $[ra]$  is the area described by the radius  $r_1$  in unit time, which is therefore constant.

If  $q = \alpha$ , then  $r_1 = \alpha r$ , and the ellipse reduces to a circle; if  $q = a$ , then  $r_1 = [ra]$ , and since  $r_1$  is now always orthogonal to  $r$ , the point  $r_1$  describes a circle with radius  $r$  that is displaced from the original circle by a quarter turn. If  $q = 1 + a$ , where  $[ra]$  is the area described by the radius in unit time, then the ellipse is described by the endpoint of the radius vector.

To generate the parabola, let  $s$  be the constant displacement from the center to the focus. Then

$$r_1 = q(r - s) + s = qr + (1 - q)s$$

is the equation of the parabola.

The hyperbola can also be generated by means of quaternions, but the formulas become more complicated.

§ 4. *To generate the sphere.*

The locus of all points  $x$  satisfying the equation

$$[x|x] + [a|x] + \alpha = 0$$

is a sphere. For if  $c$  is the center and  $\rho$  the radius, then

$$[x - c|x - c] = \rho^2,$$

that is

$$[x|x] - 2[c|x] + [c|c] - \rho^2 = 0.$$

Conversely, from the first equation it follows that  $x$  always lies on the sphere whose center is  $-\frac{1}{2}a$  and whose radius is determined by  $\rho^2 = [\frac{1}{2}a|\frac{1}{2}a] - \alpha$ .

If  $x$  satisfies the equation

$$q \cdot x \cdot q' = x_1,$$

where  $q$  and  $q'$  are constant quaternions, then the locus of all points  $x_1$  corresponding to the points of a sphere is again a sphere.

§ 5. *Applications to the geometry of position.*

If  $a, b, c, \dots$  are the displacements from a fixed point to the points  $A, B, C, \dots$ , then the quaternion

$$Q = \frac{(b-a)(c-b)(d-c) \cdots}{(c-a)(d-b)(e-c) \cdots}$$

remains unchanged if the points  $A, B, C, \dots$  are transformed by the same quaternion. This follows immediately from the combinability of the quaternion product.

### 534 SELECTED PAPERS

In particular, if  $A, B, C$  are three points on a straight line and  $A_1, B_1, C_1$  are the corresponding points, then

$$\frac{(b-a)(c-b)}{(c-a)}$$

is a quaternion that depends only on the ratios of the segments, and this quaternion is the same for  $A_1, B_1, C_1$ .

§ 6. *Applications to spherical trigonometry.*

The formulas of spherical trigonometry can be expressed very simply by means of quaternions. If  $a, b, c$  are the unit displacements from the center of the sphere to the vertices of a spherical triangle, then the sides  $\alpha, \beta, \gamma$  and the angles  $A, B, C$  of the triangle satisfy the equations

$$[ab] = \sin \gamma \cdot e_c, \quad [bc] = \sin \alpha \cdot e_a, \quad [ca] = \sin \beta \cdot e_b,$$

where  $e_a, e_b, e_c$  are the unit vectors perpendicular to the planes  $OBC$ ,  $OCA$ ,  $OAB$  respectively.

From these equations it follows that

$$abc = a \cdot [bc] = a \cdot (\sin \alpha \cdot e_a) = \sin \alpha \cdot [a|e_a] = \sin \alpha \cdot \sin \beta \sin C.$$

Likewise one finds

$$abc = \sin \beta \sin \gamma \sin A = \sin \gamma \sin \alpha \sin B.$$

Thus

$$\frac{\sin \alpha}{\sin A} = \frac{\sin \beta}{\sin B} = \frac{\sin \gamma}{\sin C}. \quad (22)$$

Furthermore, from

$$(a \cdot b)c = (-\cos \gamma + [ab])c = -\cos \gamma \cdot c + \sin \gamma \cdot [e_c c]$$

and

$$a(bc) = a(-\cos \alpha + [bc]) = -\cos \alpha \cdot a + \sin \alpha \cdot [ae_a]$$

it follows, since  $a(bc) = (ab)c$  by combinability, that

$$-\cos \gamma \cdot c + \sin \gamma \cdot [e_c c] = -\cos \alpha \cdot a + \sin \alpha \cdot [ae_a].$$

### Part Three 535

Now since  $[abc] = [bca] = [cab]$ , one can also replace  $\sin \alpha' : \sin \alpha$  with  $\sin \beta' : \sin \beta$  and  $\sin \gamma' : \sin \gamma$ . If  $\sin \alpha' : \sin \alpha = K$ , then

$$[abc] = \sin \alpha \sin \beta \sin \gamma \cdot K,$$

and likewise

$$\sin \beta' = K \sin \beta, \quad \sin \gamma' = K \sin \gamma.$$

If one takes the inner product of both sides of the above equation with  $a$ , then, since  $[a|a] = 1$ ,  $[a|[e_c c]] = [ae_c c] = [cae_c] = [ca] \sin \gamma \cdot e_c] = \sin \gamma \cdot [ca|e_c]$ , and since  $[a|c] = -\cos \beta$ ,  $[a|[ae_a]] = [aae_a] = 0$ :

$$-\cos \gamma \cos \beta + \sin^2 \gamma \cdot [e_c|a] = -\cos \alpha.$$

But  $[e_c|a] = \frac{[cab]}{\sin \gamma} = \frac{[abc]}{\sin \gamma}$ , whence

$$\cos \alpha = \cos \beta \cos \gamma - \sin \beta \sin \gamma \cos A.$$

Likewise one finds

$$\cos \beta = \cos \gamma \cos \alpha - \sin \gamma \sin \alpha \cos B, \quad (23)$$

$$\cos \gamma = \cos \alpha \cos \beta - \sin \alpha \sin \beta \cos C. \quad (24)$$

### § 7. *On the transformation of rectangular coordinates.*

If  $e_1, e_2, e_3$  are three mutually orthogonal unit displacements, and  $e'_1, e'_2, e'_3$  are three other mutually orthogonal unit displacements, then the equations of transformation are

$$e'_1 = \cos \alpha_1 e_1 + \cos \beta_1 e_2 + \cos \gamma_1 e_3,$$

$$e'_2 = \cos \alpha_2 e_1 + \cos \beta_2 e_2 + \cos \gamma_2 e_3,$$

$$e'_3 = \cos \alpha_3 e_1 + \cos \beta_3 e_2 + \cos \gamma_3 e_3,$$

where  $\alpha_i, \beta_i, \gamma_i$  are the angles that  $e'_i$  makes with  $e_1, e_2, e_3$ .

### 536 SELECTED PAPERS

If  $x = x_1 e_1 + x_2 e_2 + x_3 e_3 = x'_1 e'_1 + x'_2 e'_2 + x'_3 e'_3$ , then

$$x'_1 = [x|e'_1] = x_1 \cos \alpha_1 + x_2 \cos \beta_1 + x_3 \cos \gamma_1,$$

etc.

These formulas of transformation can also be expressed by means of quaternions. If  $q$  is a quaternion of length 1, and  $a$  is an arbitrary displacement, then  $q \cdot a \cdot q^{-1}$  is a displacement of the same length as  $a$ , and the transformation  $a' = qa q^{-1}$  represents a rotation about an axis. If one chooses  $q$  appropriately, one can obtain any rotation, and thus also the transformation from one system of orthogonal axes to another.

### § 8. *Applications to the theory of curves and surfaces.*

The osculating plane of a curve at a point  $P$  is the plane through three consecutive points of the curve. If  $r$  is the displacement from a fixed point to  $P$ , and  $dr$  is the displacement from  $P$  to a consecutive point, then the osculating plane is the plane of the displacements  $dr$  and  $d^2r$ .

The curvature of the curve at  $P$  is given by

$$\frac{1}{\rho} = \frac{|[dr \cdot d^2r]|}{|dr|^3}.$$



For surfaces, the normal at a point  $P$  is the displacement perpendicular to all tangents at  $P$ . If the surface is given by the equation  $f(r) = 0$ , then the normal is proportional to the gradient of  $f$ .

#### § 9. *Conclusion.*

From the preceding developments it is clear that the Hamiltonian quaternions form a special case of the multiplication presented in extension theory, and that all laws of quaternions follow from the fundamental equation  $ab = -[a|b] + [ab]$ . The methods of extension theory are however more general, since they are not limited to space of three dimensions, and they also lead more directly to the goal, without requiring the introduction of new concepts and terms for old and familiar things.

I hope that these brief indications will suffice to show the position of the Hamiltonian quaternions in extension theory, and to demonstrate that the latter provides the natural foundation for the former.

## EDITORIAL NOTES

From the composition of this article it appears that Grassmann did not have access to Hamilton's papers, since he cites only secondary sources (Dillner and Hankel). His opinion of the work of the Irish mathematician might otherwise have been rather more favorable.

Although we recognize that we will probably be unable to do either party justice, we will nevertheless not shun the problem of taking a position on the questions touched upon by Grassmann.

We must agree with Grassmann that, compared to quaternions, extension theory appears as the more comprehensive and wide-ranging discipline, and we also find that quaternionic methods have been applied by Hamilton himself, and even more by his followers, to a set of subjects for whose treatment they are not at all suitable. On the other hand it is only fair to acknowledge that the theory of quaternions is also an original creation, and in its own characteristic domain (which includes kinematics and a part of mathematical physics) it realizes the notion of a geometric method of calculation in a way more complete than does Grassmann's. The two methods of calculation are complementary.

The circumstance that the two authors worked independently of

each other meant of course that no one was prevented, as with Grassmann, from pointing out the ultimate “place in extension theory” of quaternions as a development of this discipline in a particularly important direction.

In this respect we must not overlook

#### 540 SELECTED PAPERS

that extension theory necessarily leads to quaternions—which is completely correct as such. However one involuntarily asks whether the whole argument would have been possible if quaternions had not already been proposed. In particular, the completely inadequate motivation for replacing the product  $[ab]$  in equation (4) by its complement (in (4b)) seems almost like a conjurer’s trick. It would have been simpler to move the unavoidable intellectual leap to the beginning and therefore to have begun the development with equation (I).

In our opinion another difficulty is even more weighty, as it must always entail a serious impediment to the beginner. In basing the theory of quaternions on calculations with displacements there are at hand only three coordinated units, the three independent displacements in space. The fourth unit, which is identified with the unit of ordinary calculations, appears as a magnitude of completely different characteristics. Now that this can nevertheless be combined with displacements, that is with directed magnitudes, in the form of a sum, wreaks decided violence to the law urged universally, but particularly by Grassmann, that only similar magnitudes are to be united as a “sum.” It is this step that ought to have been more carefully motivated, and treated completely differently than by the otherwise frequently used analogy with the geometric interpretation of  $x + iy$ .

To us it seems questionable whether this defect can be fully remedied

#### Part Three 541

1. Here Grassmann makes use of a particular orientation of the coordinate system. More simply and pertinently, it is known that the uniqueness of division follows from equation (III).
2. The title of Frege’s *Habilitationsschrift* is “Rechnungsmethoden, die sich auf eine Erweiterung des Grössenbegriffes gründen.”

3. In order to state clearly the relation of Grassmann's symbolism for rotations to Euler's formulas, we decompose the quaternion  $q = \alpha + a$  into a product of two vectors (displacements according to Grassmann)  $b$  and  $c$ :

$$q = bc. \quad (25)$$

These will then enclose a fixed angle that is the angle of the quaternion, and one of them can still be chosen arbitrarily in the plane of the quaternion. If therefore  $q'$  is another quaternion, then one can simultaneously set

$$q' = cd, \quad (26)$$

and therefore

$$qq' = c^2 \cdot bd, \quad (27)$$

whence the product  $qq'$  again appears in the same form as  $q$  and  $q'$ .

Now let  $e_0$  be the elementary point where the three mutually orthogonal displacements  $e_1, e_2, e_3$  meet, and simultaneously the fourth quaternion unit (not allowed to be so designated by Grassmann), and further let

$$x = x_0e_0 + x_1e_1 + x_2e_2 + x_3e_3$$

be an arbitrary ( $x_0$ -fold) point of space; then the expression for it being acted upon by the quaternion  $q$ , that is rotated about its displacement  $a$  by twice the angle of the quaternion, is

$$z = Kq \cdot x \cdot q = (\alpha - a)x(\alpha + a). \quad (28)$$

But according to (A) one can also write this formula thus:

$$z = c \cdot bxb \cdot c; \quad (29)$$

and therefore one can set

$$z = cyc, \quad y = bxb. \quad (30)$$

Each of the

## 542 SELECTED PAPERS

In Grassmann's method of notation, the following replace formulas (D) and (E):

$$(D') \quad xe^{2\angle bc} = z,$$

$$(E') \quad ye^{i\pi} = z, \quad x^{br} = y,$$

while the following replaces formula (B):

$$(B') \quad e^{2\angle bc} \cdot e^{2\angle cd} = e^{2\angle bd} \quad (\text{cf. (20)}). \quad (31)$$

We are not able to perceive wherein the advantages claimed of the Grassmannian symbolism lie, especially since no new results are obtained thereby. In any case, formula (C), or the equivalent Eulerian formula, is not made superfluous by formula (D'), since this expresses a completely different idea.

To be sure, Grassmann has obtained a classification in his system, subordinating rotations to the concept of the quotient, that is to linear transformations. However this was clear from the beginning; it requires no formulas at all. In addition one does not see why only the rotation and not also quaternion multiplication itself is subordinate to the concept of the quotient.

Besides, the Grassmannian notation is most unclear. The symbol  $e^{\angle}$  can represent a quaternion (formula V) as well as a quotient of the type belonging to the rotation (formula 16); juxtaposition ( $x \cdot e^{\angle a}$ ) thereby represents two completely different types of multiplication.

## BIBLIOGRAPHY OF GRASSMANN'S PRINCIPAL WORKS

A number of political articles from the years 1848–49, short reviews, and reports from the Pommerschen Hauptverein für Evangelisierung Chinas (mostly from 1858–61) have been omitted. Entries marked with an asterisk are translated in this volume.

*Die Lehre vom Satze.* Privately printed, 1831.

“Ableitung der Krystallgestalten aus dem allgemeinen Gesetze der Krystallbildung.” *Programme der Stettiner Ottoschule* (1839).

*Theorie der Ebbe und Flut.* Dissertation (1840).

“Theorie der Centralen.” *Crelles J.* 24, 262–282, 372–380 (1842); 25, 57–73 (1843).

*Grundriss der deutschen Sprachlehre.* H. G. Effenbarts Erbin, Stettin, 1842.

*Leitfaden für den ersten Unterricht in der Lateinischen Sprache.* Privately printed, 1843. (2nd ed., Effenbarts Erbin, Stettin, 1846).

*Leitfaden für den ersten Unterricht in der Deutschen Sprache* (with Robert Grassmann). Privately printed, 1843 (2nd ed., 1848; 3rd ed., 1852).

\**Die Lineale Ausdehnungslehre, ein neuer Zweig der Mathematik, dargestellt und durch Anwendungen auf die übrigen Zweige der Mathematik, wie auch auf die Statik, Mechanik, die Lehre von Magnetismus und die Krystallonomie erläutert.* Otto Wigand, Leipzig, 1844 (2nd ed., 1878).

\*“Neue Theorie der Elektrodynamik.” *Poggendorfs Ann.* 64, 1–18 (1845).

\*“Kurze Übersicht über das Wesen der Ausdehnungslehre.” *Grunerts Archiv* 6, 337–350 (1845).

*Deutsches Lesebuch für Schüler von 8–12 Jahren* (with W. Langbein). L. Oehmigke, Berlin, 1846 (6 subsequent editions).

\*“Geometrische Analyse, geknüpft an die von Leibniz erfundene geometrische Charakteristik.” *Leibnizens gesammelte Werke*, herausg. von Pertz, Bd. Math. Schriften, Erste Abteilung, Bd. 1. Asher, Halle, 1846 (2nd ed., 1890).

\*“Neue Theorie der Elektrodynamik.” *Poggendorfs Ann.* 64, 1–18 (1845).

#### 544 SELECTED PAPERS

“Erzeugung der Kurven vierter Ordnung durch Bewegung gerader Linien.” *Crelles J.* 44, 1–25 (1852).

\*“Zur Theorie der Farbenmischung.” *Poggendorfs Ann.* 89, 69–84 (1853).

“Übersicht der Akustik und der niedern Optik.” *Programme des Königlichen und Stadtgymnasiums zu Stettin* (1854).

“Allgemeiner Satz über die lineale Erzeugung aller algebraischen Oberflächen.” *Crelles J.* 49, 1–9 (1855).

“Grundsätze der stereometrischen Multiplikation.” *Crelles J.* 49, 10–20 (1855).

“Über die verschiedenen Arten der linealen Erzeugung algebraischer Oberflächen.” *Crelles J.* 49, 21–36 (1855).

“Die stereometrische Gleichung zweiten Grades und die dadurch dargestellten Oberflächen.” *Crelles J.* 49, 37–46 (1855).

“Die stereometrischen Gleichungen dritten Grades und die dadurch erzeugten Oberflächen.” *Crelles J.* 49, 47–65 (1855).

\*“Sur les différents genres de multiplication.” *Crelles J.* 49, 123–141 (1855).

“Die lineale Erzeugung von Kurven dritter Ordnung.” *Crelles J.* 52, 254–275 (1856).

*Lehrbuch der Arithmetik für höhere Lehranstalten.* Enslin, Berlin, 1861. Also under the title *Lehrbuch der Mathematik für höhere Lehranstalten. Erster Teil: Arithmetik.*

“Über die Verbindung der stummen Konsonanten mit folgendem v und die davon abhängigen Erscheinungen.” *Zeitschrift für vergleichende Sprachforschung* 9, 1–35 (1860).

*Die Ausdehnungslehre. Vollständig und in strenger Form bearbeitet.* Enslin, Berlin, 1862.

“Über die Verbindung der Konsonanten mit folgenden j und die davon abhängigen Erscheinungen.” *Zeitschrift für vergleichende Sprachforschung* 11, 13–18 (1862).

\*“Zur Theorie der Farbenmischung.” *Poggendorfs Ann.* 89, 69–84 (1853).

### Part Three 545

“Die <sup>abndc</sup> und die Verwandtschaft der europäischen mit semitischen, speziell mit hebräischen, Anlauten.” *Zeitschrift für vergleichende Sprachforschung* 12, 81–102, 174–187 (1863).

“Die Verwandtschaft der indogermanischen, semitischen und der abylonischen Sprachen.” *Zeitschrift für vergleichende Sprachforschung* 13, 62–87 (1864).

\*“On the Addition and Multiplication of Series of Numbers, and on the Convergence of the Binomial Series.” *Quarterly Journal of Pure and Applied Mathematics* (Cambridge) 7, 114–132 (1866).

\*“Zur Theorie der Farbenempfindungen.” *Poggendorfs Ann.* 130, 84–88 (1867); *Ann. d. Chemie u. Pharm.* Suppl. 5, 70–72 (1867).

\*“Versuch einer allgemeinen Multiplikationstabelle.” *Göttinger Nachrichten* (1867), 65–80.

\*“Zur Electrodynamik.” *Crelles J.* 69, 1–3 (1868).

\*“Die fundamentale Uranschauung in der Mannigfaltigkeitslehre.” *Göttinger Nachrichten* (1868), 497–507.

\*“Die lineale Erzeugung von Kurven dritter Ordnung.” *Programme des Königlichen und Stadtgymnasiums zu Stettin* (1869), 1–

28.

“Leitfaden der deutschen Sprache.” Robert Grassmann, Stettin, 1870 (5th ed., 1882).

“Die Feihoss, Totxocr, dehas.” *Zeitschrift für vergleichende Sprachforschung* 19, 309–310 (1870).

“Zur Theorie der Kurven dritter Ordnung.” *Göttinger Nachrichten* (1872), 505–509.

## INDEX

**accelerating forces:** their measure, 350; addition, 350

**acceleration** [*Beschleunigung*], 168; of shadow as second differential quotient, 490

**addition:** and subtraction, general concept, 37; of areas, 108; of volumes, 109; of relative magnitudes, 185; of point magnitudes and of linear magnitudes, 326, 362. See also point, displacement, extension

**additive magnitude** [*Summengrösse*]. See magnitude.

**affine** assembly of magnitudes, 248; its structure, 248; relation between products with, 250; distinguished position of outer and regressive multiplication, 251; direct and reciprocal, 251; when one of two is the shadow of the other, 253; in the plane, 254; if two are affine, then their systems are collinearly or reciprocally related, 254–258

**algebraic:** equation, 333; relation between several quantities, 452. See also function

**angle:** as logarithm, 14, 382; of a quaternion, 530

**Ampere**, A. M., 417, 418, 430, 473

**Ampère’s law of electrodynamics:** its hypothetical basis and failure, 417–419; removal of the arbitrariness from hypothesis, 419–422

**analytic:** conjunction, 36; form, 38

## 548 INDEX

**collective motion** [*Gesammbewegung*] or collective force, 71

**collective weight** of a series of point magnitudes, 362

**collinearly related systems**, 255. See also affine and metric relations

**collineation:** its difference from congruence, 320–324; equations, 323; C. functions 329, 330

**color:** three elements of, 435; tint, 435; complementary, 435, 437–438

**combinability** [*Vereinbarkeit*], law of, 527. See also conjunction

**combination:** concept of, 26; of two systems, 241; of point magnitudes, 324; when zero, 324; equations, 325; as multiplication, 324, 329; measure value of, 327

**combinatory product**, 286; and outer product, 288; and regressive product, 294

**combined** [*Zusammengesetztes*] (inner) system, 143, 239

**common** [*Gemeinschaftliches*] system of two systems, 200, 201

**congruence**, 320: difference from collineation, 322–324

**conic section** [*Kugelschnitt*]: geometric equation of, 236, 285; through five given points, 236

**conjunction** [*Verknüpfung*]: concept, 34; combinability of factors, 34; interchangeability of elementary, 35; synthetic and analytic, 36; of magnitudes, 37; addition as, 37; algebraic forms of, 38

**cross ratio** [*Doppelquotient*], 259

**crystal structures** [*Krystallgestalten*]: representation of, 269, 270; harmonic relations, 270

**current**, 479–480

**curves in the plane**, 235

**D'Alembert's principle**, 498

**d'Arrest**, 538

**degree** [*Grad*]: of dependence, 200; of multiplication, 202. See also product

**dependence**, linear, 326

**dependent** [*Abhängig*]. See extension, elementary magnitude

**derived** [*Ableiten*] by linear constructions, 323

**de Tilly**, 409

**deviation** [*Abweichung*]: of a point, 67; of an element, 156; of a displacement, 161; of an elementary magnitude, 180; of an extension, 181; of a rigid elementary magnitude, 179; of a sum, 366; of an inner product, 368, 371; of a spherical magnitude, 373; of a spherical surface, 376; of a point from a spherical surface, 378

## Index 549

**elementary assembly measure**, 259; between elementary magnitudes of higher order, 179; between extensive magnitudes, 233;



independent of measure, 259; between inner products, 340, 347; between inner magnitudes, 366; between inner products of magnitudes of first order, 369. See also curves, harmonic, magnitude

**equilibrium** [*Gleichgewicht*]: condition for, 92; of a body system, 171; equality in, 179; in space, 182

**elevation** [*Höhenseite*] of a parallelogram, 85

**elimination** of an unknown from equations of higher degree, 152

**elliptic terms** in the expression for the force and mean motion, 504

**Engel, F.**, 409, 469, 539

**equal measure, quaternary units of**, 530

**equality** [*Gleichheit*] and difference, concepts, 26, 33, 320

**division**: general law, 41; arithmetic, 38; outer, 113; regressive, 224. See also quotient

**domain** [*Gebiet*], 29, 47, 285

**double system**, 130

**Drobisch, M. W.**, 403

**element**: generating, 28; concept of, 45, 284; independent, 170

**elementary** [*Einfache*]: extensive form, 29; conjunction, 35; extensive structure, 47; factor, 82; extensive domain, 279; domain, 279

**elementary assembly** [*Elementarverein*], 156

**elementary magnitude**: concept of first order, 157; deviation and weight of first order, 157; addition, 157; representation of, 158; multiplication of, 159; represented as multiple element or displacement, 160; final symbol for, 160; addition with a multiple element, 161; compared with the outer product of two elements, 169, 174, 182, 327; multiples and fractions of, 485

**divergence** [*Ausweichung*]: of a rigid elementary magnitude, 174; of an elementary magnitude of first order, 178; of a sum of elementary magnitudes, 179; of an extensive magnitude, 180

## 550 INDEX

**exponentiation of a quaternion**, 530

**extension** [*Ausdehnung*]: extensive form, 27; elementary, 29; extensive magnitude of first order (displacement), 47; of higher order as product, 77–78; of higher order and their addition, 97; elementary representation of, 106; independent, 107, 200; of supplementary order, 140; of a rigid elementary magnitude, 179

**extension theory** [*Ausdehnungslehre*], concept of, 25

**extensive domain** [*Ausdehnungsgebiet*], elementary, 279

**extensive:** magnitude, 47, 103; structure of first order 47; system, 260; quantity, 452

**factor, combinatory:** of first order, 286; of  $(n - 1)$ -th order, 288, 294. See also outer factor, product

**factors** [*Glieder*] of a conjunction, 34

**final element** [*Endelement*], 47

**finite** [*Endlich*] magnitude, 225

**force** [*Kraft*], 69; summation of, 69; as linear magnitude, 190; equivalent, 192; in a plane, 192–194; in space, 194; reduction of a sum, 195; analogy with gravitation, 425; and Ampère's law, 426–427; assembly of and moment, 195; accelerating, 350; constant, 351; elementary, 491; elliptic terms in expression for, 504

**Fraunhofer**, J., 439, 447

**Frege**, G. W., 533, 541

**function:** algebraic and homogeneous, 333; geometric, 349; algebraic of points, 352; partial differential quotients with respect to points, 353, 354; accelerating forces defined by, 352–356

**fundamental conjunctions** [*Grundverknüpfungen*], 144

**fundamental evolution** [*Grundänderung*], 47, 452; independent, 37

**fundamental laws of mechanics**, 490–492

**fundamental measures** [*Grundmasse*]: for extensive magnitudes, 147; for elementary magnitudes, 185

**Gauss**, K. F., 166, 540

**geometric:** exponential magnitude, 14; sum, 108; function, 349

**geometry:** not a branch of pure mathematics, 24; present foundation of, 60–62; new formulation, 62–63, 279

**Gordan**, P., 312

**Grassmann**, J. G., 9, 27, 298

**Grassmann**, R., 24

**Grassmann's law of electrodynamics**, 422, 473, 475, 479; special case, 423; general case, 425, 479

THE PLACE OF HAMILTON'S QUATERNIONS  
IN THE AUSDEHNUNGSLEHRE<sup>†</sup>

<sup>†</sup>*Math. Ann.* **12** (1877), 375–386. Translated by Lloyd C. Kannenberg.

... displacement of length 1, then  $q$  can be represented in the form  $q = a_0 + a_1e_1$ ; if one lets

$$\frac{1}{q} = \beta_0 + \beta_1e_1 + \beta_2e_2 + \beta_3e_3,$$

where  $e_1, e_2, e_3$  form a normal set, then for the specification of  $\beta_0, \dots, \beta_3$  one obtains the equation

$$(\beta_0 + \beta_1e_1 + \beta_2e_2 + \beta_3e_3)(a_0 + a_1e_1) = 1,$$

which comprises the four equations

$$\begin{aligned} \beta_0a_0 - \beta_1a_1 &= 1 & \beta_2a_0 - \beta_3a_1 &= 0 \\ \beta_1a_0 + \beta_2a_1 &= 0 & \beta_2a_0 + \beta_3a_1 &= 0. \end{aligned}$$

From the first two follow

$$\beta_0 = \frac{a_0}{a_0^2 + a_1^2}, \quad \beta_1 = \frac{-a_1}{a_0^2 + a_1^2},$$

where  $a_0^2 + a_1^2 = p^2$ ,  $p$  being the length of  $q$ ; from the latter two there follows  $\beta_2 = 0$ ,  $\beta_3 = 0$ , whence

$$\frac{1}{a_0 + a_1e_1} = \frac{a_0 - a_1e_1}{p^2}.$$

In particular if  $q = a_0 + a_1e_1$  is a quaternary unit, whence  $a_0^2 + a_1^2 = 1$ , then

$$1 : (a_0 + a_1e_1) = a_0 - a_1e_1.^1$$

From this there easily follows  $1 : q^\mu = q^{-\mu}$ , in conformity with algebra.

The section in Dillner treating calculations in double axis systems, that is on the distributive law (nos. 14 to 23), is superfluous as a consequence of my fundamental definition I.

Of the usual applications of quaternions, to be rejected are those to the composition of forces (Dillner, no. 4) and to the rotational moment of mechanical work (Dillner, no. 24), since the

---

<sup>1</sup>That is, in (4b) the left side is homogeneous of second degree in  $e_1, e_2, e_3$ , the right homogeneous of first degree in  $e_0, e_1, e_2, e_3$ , where  $e_0$  symbolizes the newly introduced unit. Since in conformity with what has already been developed  $\mu$  cannot be zero, one can at least set  $\mu$  equal to unity without loss of generality.

conjunction of displacements yields these concepts most simply, whereas quaternions mix in irrelevancies. In fact  $a + b$  is the force composed from  $a$  and  $b$ ,  $[ab]$  the moment of the force  $b$  on the lever arm  $a$ ,  $[abc]$  the moment of the force  $c$  on the lever arm  $b$  attached to the fixed axis  $a$ ,  $[a|b]$  the work of the force  $b$  with respect to the path  $a$ .

On the same grounds calculation with quaternions is to be rejected for the rotation of a spatial structure about an axis (Dillner, no. 39–42), and for the transformation of rectilinear coordinates (Dillner, no. 44–48, Hankel<sup>2</sup> § 58). The latter problem is solved most easily and directly for orthogonal coordinate systems by inner multiplications; in this connection I refer to my article “Die Mechanik nach den Principien der Ausdehnungslehre” in these *Annalen*, XII, p. 222.

The former problem is most easily solved by the complete quotient of displacements (A2, nos. 377–390). By such a quotient I understand an expression that transforms each displacement, under multiplication (by this expression), into a specific displacement. To this end it suffices to establish which three displacements three noncoplanar displacements in space are transformed into by that multiplication. Thus for example should the displacements  $e_1, e_2, e_3$  (that stand in no numerical relation, that is are noncoplanar) be transformed by such a quotient  $Q$  into the displacements  $a_1, a_2, a_3$  under multiplication by  $Q$ , that is  $e_1Q = a_1, e_2Q = a_2, e_3Q = a_3$ , then according to the general law of multiplication

$$(a_1e_1 + a_2e_2 + a_3e_3)Q = a_1a_1 + a_2a_2 + a_3a_3, \quad (12.1)$$

and the product of any displacement in space with  $Q$  is then completely specified. I then write

$$Q = \frac{(a_1 \ a_2 \ a_3)}{(e_1 \ e_2 \ e_3)} \quad (12.2)$$

and call  $(e_1 \ e_2 \ e_3)$  the *denominator*,  $(a_1 \ a_2 \ a_3)$  the corresponding *numerator*.

The problem of finding the displacement  $x$  (that is, its direction) which is thus transformed into a multiple of itself, so that  $xQ = px$ , is of fundamental significance. This problem (A2, no. 388ff.) is most easily solved by means of an equation of third degree through outer multiplication. Thus if  $Q$  has the above value and  $x = x_1e_1 + x_2e_2 + x_3e_3$ , then the equation

$$xQ - px = 0$$

is transformed into the equation

$$x_1(a_1 - pe_1) + x_2(a_2 - pe_2) + x_3(a_3 - pe_3) = 0. \quad (12.3)$$

Here  $x_1, x_2, x_3$  cannot all simultaneously be zero, since otherwise  $x$  would be zero, which of course is excluded. Now if for example  $x_1$  is different from zero, then one outer multiplies the equation by  $a_2 - pe_2$  and  $a_3 - pe_3$ ; then upon division by  $x_1$  one obtains the equation

$$[(a_1 - pe_1)(a_2 - pe_2)(a_3 - pe_3)] = 0, \quad (12.4)$$

since the outer product of three displacements is always zero if two of the displacements are equal (see below).

---

<sup>2</sup>Hermann Hankel, *Vorlesung über die komplexen Zahlen*, Leipzig: Voss, 1867.

This is a cubic equation with respect to  $p$ . I assume that the three roots  $p_1, p_2, p_3$  of this equation are different from each other, since the case of equal roots can be derived as a limiting case from the more general case of unequal roots. The ratios  $x_1 : x_2 : x_3$  are then completely defined by these values  $p_1, p_2, p_3$  by virtue of the former equation (12.3), and can immediately be found upon multiplication by each of the magnitudes  $a_1 - p_1 e_1$ ,  $a_2 - p_2 e_2$ ,  $a_3 - p_3 e_3$ . One thus obtains three axes  $c_1, c_2, c_3$ , defined by their directions, that belong to the three roots  $p_1, p_2, p_3$  and, as one sees immediately, stand in no numerical relation to one another.

Thus  $Q$  is represented in the *normal form*

$$Q = \frac{(p_1 c_1 \ p_2 c_2 \ p_3 c_3)}{(e_1 \ e_2 \ e_3)}. \quad (12.5)$$

I call  $c_1, c_2, c_3$  the *axes of the quotient* and  $p_1, p_2, p_3$  the corresponding *principal numbers*. This representation is of fundamental importance for the theory of linear relationships and for a set of algebraic problems, for example for those that Dr. Gottlob Frege has treated in his dissertation upon the attainment of the *venia docendi*, Jena 1874, pp. 20–23.<sup>3</sup>

Should the quotient  $Q$ , upon which the problem mentioned depends, produce only a rotation, then two of the three axes will be imaginary, one real, and its corresponding principal number will = 1. If  $a$  is this real axis, then the two imaginary axes are of the form  $b + ci$  and  $b - ci$ , where  $a, b, c$  are mutually perpendicular. The two principal numbers belonging to these imaginary axes have the numerical value one, and are thus of the form  $\cos \alpha - i \sin \alpha$  and  $\cos \alpha + i \sin \alpha$ ; consequently

$$\begin{aligned} a &= Qa \\ (b + ci)Q &= (b + ci)(\cos \alpha + i \sin \alpha) = b \cos \alpha + c \sin \alpha \\ &\quad + i(c \cos \alpha - b \sin \alpha) \\ (b - ci)Q &= (b - ci)(\cos \alpha + i \sin \alpha) = b \cos \alpha + c \sin \alpha \\ &\quad - i(c \cos \alpha - b \sin \alpha). \end{aligned} \quad (12.6)$$

These two equations (12.6) added and divided by 2 yield

$$bQ = b \cos \alpha + c \sin \alpha$$

and subtracting the second from the first and dividing by  $2i$  yields

$$cQ = c \cos \alpha - b \sin \alpha, \quad (12.7)$$

that is, upon multiplication by  $Q$ ,  $b$  is rotated through angle  $\alpha$  toward  $c$  in the  $bc$  plane, and  $c$  is rotated by the same angle. Every multiple sum of  $b$  and  $c$ , that is every displacement in the  $bc$  plane, is thus obviously rotated by the same angle.

It is very convenient to establish the following two symbolic expressions for this quotient, between which one may choose as necessary,

$$Q = a^\alpha = e^{Lbb'} \quad (12.8)$$

---

<sup>3</sup>The title of Frege's Habilitationsschrift is "Rechnungsmethoden, die sich auf eine Erweiterung des Grössenbegriffes gründen."

where  $a$  is of unit length along the axis of rotation,  $\alpha$  is the angle of rotation, and  $b'$  is the displacement into which  $b$ , which is perpendicular to the axis of rotation, is transformed by the rotation. Here  $\alpha$  and  $Lbb'$  are only distinguished in that the former represents the angle as a number, the latter the same angle regarded as part of the plane of rotation. Then  $xe^\alpha$  signifies the displacement  $x'$  that forms the same angle with  $a$  as  $x$ , but to the opposite side, so that  $Lxx' = 2Lax'$ ; likewise  $x' \cdot b^\alpha$  represents the displacement laid similarly in such a way that  $Lx'x'' = 2Lx'b$ ; consequently

$$xa^{Laa} \cdot b^{Lbb} = xe^{2Lax'+2Lx'b} = xe^{2Lab}. \quad (12.9)$$

Thus one obtains

$$a^{Laa'} \cdot b^{Lbb'} = e^{2Lab}. \quad (12.10)$$

This theorem is important for the composition of rotations. Thus it yields

$$e^{2Lab} \cdot e^{2Lbc} = a^{Laa'} \cdot b^{Lbb'} \cdot b^{Lbb'} \cdot c^{Lcc'} = a^{Lcc'} = e^{2Lac}, \quad (12.11)$$

whence

$$e^{2Lab} \cdot e^{2Lbc} = e^{2Lac}, \quad (12.12)$$

a formula that must replace Dillner's formula (81), so complicated and mixed with alien ingredients.<sup>4</sup>

The most beautiful application of quaternions is that to spherical trigonometry. Yet I believe that even here the conjunction of displacements is superior to calculation with quaternions. In this connection the concept of the product  $[abc]$  of three displacements  $a, b, c$  is already included implicitly in the above. In fact if  $[bc]$  is the complement of the displacement  $a_1$ , thus  $[bc] = |a_1$ , then  $[abc] = [a_1a_1]$ , that is, it is equal to the inner product of the displacements  $a$  and  $a_1$ . I call  $[abc]$  the *outer product* of the three displacements  $a, b, c$  (A1 § 31ff., A2 no. 262). From the above there easily follow the formulas

$$[abc] = [bca] = [cab] = -[acb] = -[bac] = -[cba],$$

in addition the law that  $[abc] = 0$  if two of the factors are equal, and conceptually that  $[abc]$  is equal to the parallelepiped in which three coterminal edges are equal to  $a, b, c$ .

Now I replace a spherical polygon  $ABCD \dots$  with the series of displacements  $a, b, c, d, \dots$  drawn from the center of the sphere to the vertices  $A, B, C, D, \dots$  and set the length of the radius of the sphere equal to one. If one then replaces  $ab, bc, cd, \dots$  by radii orthogonal to  $ab, bc, cd, \dots$  lying on the same side (for example on the left), one obtains the corresponding polar angle. Thus let  $a, b, c$  be the radii drawn to the vertices of a spherical triangle, and  $c'$  be the radius orthogonal to  $a, b$  such that  $[abc']$  is positive, and likewise  $b'$  be orthogonal to  $c, a$ ,  $a'$  to  $b, c$ , with  $[cab']$ ,  $[bca']$  positive. Then  $a'b'c'$  is the polar angle; but  $a$  is also orthogonal to  $b', c'$ ;  $b$  to  $c', a'$ ; and  $c$  to  $a', b'$ , all on the same side, whence also  $abc$  is the polar angle of  $a'b'c'$ . Now let the angles  $bc, ca, ab, b'c', c'a', a'b'$  be symbolized by  $\alpha, \beta, \gamma, \alpha', \beta', \gamma'$ , so that these six angles can be considered as positive numbers. In order to obtain

---

<sup>4</sup>Naturally one of the limits is also excluded. Our modern interpretation is instead expressed by saying that with a fractional  $\mu$ ,  $(\cos \alpha + \mu \sin \alpha)$  is to be regarded as a multivalued function.

the relations between these magnitudes in the easiest way I will make use of the product of two surface areas in space, whereby I set (according to A1 § 132, no. 103)

$$[ab \cdot bc] = [abc] \cdot b$$

(A2)

It then follows (A2 no. 97) that the product of the supplements of two displacements  $a, b$  in space is the supplement of the product of these displacements, that is:

$$|[ab]| = |[a|b]|.$$

For the above six radii  $a, b, c, a', b', c'$  there follows from this first of all  $a' \sin \alpha = |[bc]|$ . For if  $Lbc$ , in the plane  $bc$  equals  $90^\circ$  and  $c_1$  is likewise a radius, then  $a', b, c_1$  form a normal set and therefore

$$a' = |[bc_1]| = [bc] : \sin \alpha.$$

In the same way,

$$\begin{aligned} b' \sin \beta &= |[ca]|, & c' \sin \gamma &= |[ab]|; \\ a \sin \alpha' &= |[b'c']|, & b \sin \beta' &= |[c'a']|, \\ c \sin \gamma' &= |[a'b']|. \end{aligned}$$

Thus

$$a = \frac{|[bc]|}{\sin \alpha'} = \frac{[ca \cdot ab]}{\sin \alpha'} = \frac{[abc] \cdot a}{\sin \alpha'} = \frac{\sin \alpha \cdot a}{\sin \beta \sin \gamma} = \frac{\sin \alpha \cdot a}{\sin \alpha' \sin \beta \sin \gamma}, \quad (12.13)$$

whence

$$[abc] = \sin \alpha' \sin \beta \sin \gamma = \sin \alpha \sin \beta' \sin \gamma' = \sin \alpha \sin \beta \sin \gamma' \cdot K,$$

where  $\sin \alpha' : \sin \alpha = K$ .

Now since  $[abc] = [bca] = [cab]$ , one can also replace  $\sin \alpha' : \sin \alpha$  with  $\sin \beta' : \sin \beta$  and  $\sin \gamma' : \sin \gamma$ . If  $\sin \alpha' : \sin \alpha = K$ , then

$$[abc] = \sin \alpha \sin \beta \sin \gamma \cdot K, \text{ and likewise}$$

$$\begin{aligned} [a'b'c'] &= \sin \alpha' \sin \beta' \sin \gamma' \cdot K, \\ K &= \frac{\sin \alpha' \sin \beta' \sin \gamma'}{\sin \alpha \sin \beta \sin \gamma}. \end{aligned} \quad (12.14)$$

Further, since  $a \sin \alpha = [bc]$ , one has  $[a'bc] = a' \cdot a \sin \alpha = \sin \alpha$ , and thus

$$\begin{aligned} [a'bc] &= \sin \alpha, & [b'ca] &= \sin \beta, & [c'ab] &= \sin \gamma, \\ [ab'c'] &= \sin \alpha', & [bc'a'] &= \sin \beta', & [ca'b'] &= \sin \gamma'. \end{aligned} \quad (12.15)$$

Moreover  $[b' \cdot bc] = [b' \cdot a]q \sin \alpha = \sin \alpha \cos \gamma'$ , and thus generally

$$\begin{aligned} [b'bc] &= \sin \alpha \cos \gamma', & [c'bc] &= \sin \alpha \cos \beta', \text{ etc.}, \\ [bb'c'] &= \sin \alpha' \cos \gamma, & [cb'c'] &= \sin \alpha' \cos \beta, \text{ etc.} \end{aligned} \quad (12.16)$$

Now if  $a, b, c$  are three arbitrary noncoplanar displacements, then any other displacement  $d$  can be derived from them numerically. Thus if

$$d = xa + yb + zc,$$

then since  $[bbc]$  and  $[cbc]$  are zero, one obtains  $[dbc] = x[abc]$  upon outer multiplication by  $[bc]$ , whence  $x = [dbc] : [abc]$  and correspondingly for the others, whence

$$d[abc] = a[dbc] + b[adc] + c[abd]$$

or symmetrically

$$a[bcd] - b[cda] + c[dab] - d[abc] = 0 \quad (12.17)$$

for four arbitrary displacements  $a, b, c, d$ .

We can utilize this equation immediately to solve the important problem "Establish the equation between every set of four of the radii  $a, b, c, a', b', c'$ ."

For  $a, b, c, a'$  one first finds the equation

$$a[bca'] - b[ca'a] + c[a'ab] - a'[abc] = 0,$$

that is

$$(VI) \quad a \sin \alpha + b \sin \beta \cos \gamma' + c \sin \gamma \cos \beta' = a'[abc],$$

or, if one inner multiplies by an arbitrary radius  $r$  and replaces  $[abc]$  by its value,

$$\cos ra \cdot \sin \alpha + \cos rb \cdot \sin \beta \cos \gamma' + \cos rc \cdot \sin \gamma \cos \beta' = \cos ra' \cdot \sin \alpha \sin \beta \sin \gamma \cdot K. \quad (12.18)$$

One obtains six such formulas. These are the formulas implied by Dillner under his formula (17).

Further one finds for  $a, b, a', b'$  the equation

$$a[ba'b'] - b[a'b'a] + a'[b'ab] - b'[aba'] = 0,$$

that is

$$(VII) \quad \sin \gamma'(a \cos \alpha - b \cos \beta) + \sin \gamma(a' \cos \alpha' - b' \cos \beta') = 0$$

or

$$\sin \gamma'(\cos ra \cos \alpha - \cos rb \cos \beta) + \sin \gamma(\cos ra' \cos \alpha' - \cos rb' \cos \beta') = 0. \quad (12.19)$$

In particular, if one sets  $r = a$ , then since

$$\cos \alpha = [a|a'] = \sin \beta \sin \gamma', \frac{[abc]}{\sin \alpha},$$

after division by  $\sin \gamma'$  one obtains the familiar formula

$$\cos \alpha - \cos \beta \cos \gamma + \sin \beta \sin \gamma \cos \alpha' = 0. \quad (12.20)$$



One obtains three formulas such as (VII).

Finally, for  $a, b, c', a'$  one finds

$$a[bc'a'] - b[c'a'a] + c'[a'ab] - a'[abc'] = 0,$$

that is

$$(VIII) \quad \sin \beta'(a - b \cos \gamma) = \sin \gamma(a' - c' \cos \beta')$$

or

$$\sin \beta'(\cos ra - \cos rb \cos \gamma) = \sin \gamma(\cos ra' - \cos rc' \cos \beta'). \quad (12.21)$$

There are six such formulas.

One thus recognizes the great wealth of relations that result from these methods. Each geometric equation can be transformed into a theorem in spherics in this way.

As an example I mention the Gaussian theorem for the spherical quadrangle cited by Hankel on p. 193, to wit

$$\sin AB \sin CD \cos(AB, CD) = \cos AC \cos BD - \cos AD \cos BC,$$

which is a transformation of the formula (A2 no. 176), that is the formula

$$[ab|cd] = [a|c][b|d] - [a|d][b|c].$$

As a second example I mention the transformation of the equation  $a_1 + b_1 + c_1 + \dots = s_1$ , where  $a_1, b_1, c_1, \dots, s_1$  are displacements. Now if  $a_1 = aa$ ,  $b_1 = bb$ ,  $c_1 = cc$ ,  $s_1 = ss$ , where  $a, b, c, \dots, s$  are the lengths and thus  $a, b, c, \dots, s$  the radii of a sphere, all of unit length, then upon inner multiplication by an arbitrary radius  $r$  one has

$$a \cos ra + b \cos rb + \dots = s \cos rs.$$

If one sets  $r = s$  here, then one has

$$a \cos sa + b \cos sb + \dots = s,$$

whence

$$\frac{a \cos ra + b \cos rb + \dots}{a \cos sa + b \cos sb + \dots} = \cos rs. \quad (12.22)$$

A special case of this formula was developed by d'Arrest in the *Berichten der sächsischen Gesellschaft der Wissenschaften* of 1862 (p. 37ff.).

## Editorial Notes

From the composition of this article it appears that Grassmann did not have access to Hamilton's papers, since he cites only secondary sources (Dillner and Hankel). His opinion of the work of the Irish mathematician might otherwise have been rather more favorable.

Although we recognize that we will probably be unable to do either party justice, we will nevertheless not shun the problem of taking a position on the questions touched upon by Grassmann.

We must agree with Grassmann that, compared to quaternions, extension theory appears as the more comprehensive and wide-ranging discipline, and we also find that quaternionic methods have been applied by Hamilton himself, and even more by his followers, to a set of subjects for whose treatment they are not at all suitable. On the other hand it is only fair to acknowledge that the theory of quaternions is also an original creation, and in its own characteristic domain (which includes kinematics and a part of mathematical physics) it realizes the notion of a geometric method of calculation in a way more complete than does Grassmann's. The two methods of calculation are complementary.

The circumstance that the two authors worked independently of each other meant of course that no one was prevented, as with Grassmann, from pointing out the ultimate "place in extension theory" of quaternions as a development of this discipline in a particularly important direction. In this respect we must not overlook the fact that in order to carry through this development an innovation is required which, however self-evident it may seem to us today, was completely alien to Grassmann's original point of view. We mean the reduction of the product of second order to the original units under the assumption of the associative law of multiplication (Lie and Engel, *Theorie der Transformationsgruppen* III, Leipzig: 1893, p. 748). This portentous step, abandoning the domain of the familiar complex numbers, Hamilton is known to have taken; with Grassmann the so-called unit products always appear as magnitudes of a new type.

Turning now to Grassmann's presentation of the theory of quaternions, it cannot be denied that he does not do it justice. By this we mean not only that his polemic position seems to us to be excessive, but even more that the ideas adopted from Hamilton are not sufficiently identified as such. It seems as if Grassmann is utterly unconscious of the fact that the crucial innovation is not to be found in the article "Sur les différents genres de multiplication."

The presentation sought appears in detail in the first part of Grassmann's article. It was to be shown that the fundamental point of view of extension theory necessarily leads to quaternions—which is completely correct as such. However one involuntarily asks whether the whole argument would have been possible if quaternions had not already been proposed. In particular, the completely inadequate motivation for replacing the product  $[ab]$  in equation (4) by its complement (in (4b)) seems almost like a conjurer's trick. It would have been simpler to move the unavoidable intellectual leap to the beginning and therefore to have begun the development with equation (I).

In our opinion another difficulty is even more weighty, as it must always entail a serious impediment to the beginner. In basing the theory of quaternions on calculations with displacements there are at hand only three coordinated units, the three independent displace-

ments in space. The fourth unit, which is identified with the unit of ordinary calculations, appears as a magnitude of completely different characteristics. Now that this can nevertheless be combined with displacements, that is with directed magnitudes, in the form of a sum, wreaks decided violence to the law urged universally, but particularly by Grassmann, that only similar magnitudes are to be united as a “sum.” It is this step that ought to have been more carefully motivated, and treated completely differently than by the otherwise frequently used analogy with the geometric interpretation of  $x + iy$ .

To us it seems questionable whether this defect can be fully remedied in the way Grassmann has approached it. In any case it would be remedied if one based quaternions on rotations (as seems natural to us), and established the multiplication theorem as the parameter group of the group of rotations.

The double conception of quaternions, and thus the different interpretations of scalar and vector magnitudes, then only appears as a later stage of development, as consequences of the circumstance that the group of rotations is the adjoint of its own parameter group.

This interpretation of quaternions based on the abstractions of group theory has of course only appeared distinctly in modern times, but one already finds its origins with Gauss and Cayley.

1. That is, in (4b) the left side is homogeneous of second degree in  $e_1, e_2, e_3$ , the right homogeneous of first degree in  $e_0, e_1, e_2, e_3$ , where  $e_0$  symbolizes the newly introduced unit. Since in conformity with what has already been developed  $\mu$  cannot be zero, one can at least set  $\mu$  equal to unity without loss of generality.

2. Naturally one of the limits is also excluded. Our modern interpretation is instead expressed by saying that with a fractional  $\mu$ ,  $(\cos \alpha + \mu \sin \alpha)$  is to be regarded as a multivalued function.

3. Here Grassmann makes use of a particular orientation of the coordinate system. More simply and pertinently, it is known that the uniqueness of division follows from equation (III).

4. The title of Frege's Habilitationsschrift is “Rechnungsmethoden, die sich auf eine Erweiterung des Grössenbegriffes gründen.”

5. In order to state clearly the relation of Grassmann's symbolism for rotations to Euler's formulas, we decompose the quaternion  $q = a + \alpha$  into a product of two vectors (displacements according to Grassmann)  $b$  and  $c$ :

$$(A) \quad q = bc.$$

These will then enclose a fixed angle that is the angle of the quaternion, and one of them can still be chosen arbitrarily in the plane of the quaternion. If therefore  $q'$  is another quaternion, then one can simultaneously set

$$(A') \quad q' = cd,$$

and therefore

$$(B) \quad qq' = c^2 \cdot bd,$$

whence the product  $qq'$  again appears in the same form as  $q$  and  $q'$ .

Now let  $e_0$  be the elementary point where the three mutually orthogonal displacements  $e_1, e_2, e_3$  meet, and simultaneously the fourth quaternion unit (not allowed to be so designated by Grassmann), and further let

$$x = x_0e_0 + x_1e_1 + x_2e_2 + x_3e_3$$

be an arbitrary ( $x_0$ -fold) point of space; then the expression for it being acted upon by the quaternion  $q$ , that is rotated about its displacement  $\alpha$  by twice the angle of the quaternion, is

$$(C) \quad z = Kq \cdot x \cdot q = (\alpha - a)x(\alpha + a).$$

But according to (A) one can also write this formula thus:

$$(D) \quad z = c \cdot b \cdot x \cdot b \cdot c;$$

and therefore one can set

$$(E) \quad z = c y c, \quad y = b x b.$$

Each of the last two formulas represents a simple inversion (rotation through  $180^\circ$ ) about one of the axes  $b, c$ ; (E) and (D) together contain the Hamiltonian theorem that every rotation can be represented in  $\infty^1$  ways as the result of two inversions. (A), (A') and (B) thus contain the resulting law for the geometric composition of rotations.

In Grassmann's method of notation, the following replace formulas (D) and (E):

$$(D') \quad x e^{2Lbc} = z, \quad (E') \quad y e^{Lb} = z, \quad x e^{Lb} = y,$$

while the following replaces formula (B):

$$(B') \quad e^{2Lbc} \cdot e^{2Lcd} = e^{2Lbd} \quad (\text{cf. (20)}).$$

We are not able to perceive wherein the advantages claimed of the Grassmannian symbolism lie, especially since no new results are obtained thereby. In any case, formula (C), or the equivalent Eulerian formula, is not made superfluous by formula (D'), since this expresses a completely different idea.

To be sure, Grassmann has obtained a classification in his system, subordinating rotations to the concept of the quotient, that is to linear transformations. However this was clear from the beginning; it requires no formulas at all. In addition one does not see why only the rotation and not also quaternion multiplication itself is subordinate to the concept of the quotient.

Besides, the Grassmannian notation is most unclear. The symbol  $a^\alpha$  can represent a quaternion (formula V) as well as a quotient of the type belonging to the rotation (formula 16); juxtaposition ( $xa^\alpha$ ) thereby represents two completely different types of multiplication.

## Bibliography of Grassmann's Principal Works

*A number of political articles from the years 1848–49, short reviews, and reports from the Pommer-schen Hauptverein für Evangelisierung Chinas (mostly from 1858–61) have been omitted. Entries marked with an asterisk are translated in this volume.*

*Die Lehre vom Satze.* Privately printed, 1831.

“Ableitung der Krystallgestalten aus dem allgemeinen Gesetze der Krystallbildung.” *Programme der Stettiner Ottoschule* (1839).

*Theorie der Ebbe und Flut.* Dissertation (1840).

“Theorie der Centralen.” *Crelles J.* 24, 262–282, 372–380 (1842); 25, 57–73 (1843).

*Grundriss der deutschen Sprachlehre.* H. G. Effenbarts Erbin, Stettin, 1842.

*Leitfaden für den ersten Unterricht in der Lateinischen Sprache.* Privately printed, 1843. (2nd ed., Effenbarts Erbin, Stettin, 1846).

*Leitfaden für den ersten Unterricht in der Deutschen Sprache* (with Robert Grassmann). Privately printed, 1843 (2nd ed., 1848; 3rd ed., 1852).

\**Die Lineale Ausdehnungslehre, ein neuer Zweig der Mathematik, dargestellt und durch Anwendungen auf die übrigen Zweige der Mathematik, wie auch auf die Statik, Mechanik, die Lehre von Magnetismus und die Krystallonomie erläutert.* Otto Wigand, Leipzig, 1844 (2nd ed., 1878).

\*“Neue Theorie der Elektrodynamik.” *Poggendorfs Ann.* 64, 1–18 (1845).

\*“Kurze Übersicht über das Wesen der Ausdehnungslehre.” *Grunerts Archiv* 6, 337–350 (1845).

*Deutsches Lesebuch für Schüler von 8–12 Jahren* (with W. Langbein). L. Oehmigke, Berlin, 1846 (6 subsequent editions).

“Grundzüge zu einer rein geometrischen Theorie der Kurven mit Anwendung einer rein geometrischen Analyse.” *Crelles J.* 31, 111–132 (1846).

\**Geometrische Analyse, geknüpft an die von Leibniz erfundene geometrische Charakteristik. Gekrönte Preisschrift.* Mit einer erläuternden Abhandlung von A. F. Möbius. Weidmann, Leipzig, 1847.

“Über die Erzeugung der Kurven dritter Ordnung durch gerade Linien und über geometrische Definitionen dieser Kurven.” *Crelles J.* 36, 177–184 (1848).

“Der allgemeine Satz über die Erzeugung aller algebraischen Kurven durch Bewegung gerader Linien.” *Crelles J.* 42, 187–192 (1851).

“Die höhere Projektivität und Perspektivität in der Ebene; dargestellt durch geometrische Analyse.” *Crelles J.* 42, 193–203 (1851).

“Die höhere Projektivität in der Ebene; dargestellt durch Funktionsverknüpfungen.” *Crelles J.* 42, 204–212 (1851).

“Erzeugung der Kurven vierter Ordnung durch Bewegung gerader Linien.” *Crelles J.* 44, 1–25 (1852).

\*“Zur Theorie der Farbenmischung.” *Poggendorfs Ann.* 89, 69–84 (1853).

“Übersicht der Akustik und der niedern Optik.” *Programme des Königlichen und Stadtgymnasiums zu Stettin* (1854).

“Allgemeiner Satz über die lineale Erzeugung aller algebraischen Oberflächen.” *Crelles J.* 49, 1–9 (1855).

“Grundsätze der stereometrischen Multiplikation.” *Crelles J.* 49, 10–20 (1855).

“Über die verschiedenen Arten der linealen Erzeugung algebraischer Oberflächen.” *Crelles J.* 49, 21–36 (1855).

“Die stereometrische Gleichung zweiten Grades und die dadurch dargestellten Oberflächen.” *Crelles J.* 49, 37–46 (1855).

“Die stereometrischen Gleichungen dritten Grades und die dadurch erzeugten Oberflächen.” *Crelles J.* 49, 47–65 (1855).

\*“Sur les différents genres de multiplication.” *Crelles J.* 49, 123–141 (1855).

“Die lineale Erzeugung von Kurven dritter Ordnung.” *Crelles J.* 52, 254–275 (1856).

*Lehrbuch der Arithmetik für höhere Lehranstalten.* Enslin, Berlin, 1861. Also under the title *Lehrbuch der Mathematik für höhere Lehranstalten. Erster Teil: Arithmetik.*

“Über die Verbindung der stummen Konsonanten mit folgendem  $v$  und die davon abhängigen Erscheinungen.” *Zeitschrift für vergleichende Sprachforschung* 9, 1–35 (1860).

*Die Ausdehnungslehre. Vollständig und in strenger Form bearbeitet.* Enslin, Berlin, 1862.

“Über die Verbindung der Konsonanten mit folgenden  $j$  und die davon abhängigen Erscheinungen.” *Zeitschrift für vergleichende Sprachforschung* 11, 1–52, 81–103 (1862).

“Über die Aspiraten und ihr gleichzeitiges Vorhandensein im An- und Auslaute der Wurzeln.” *Zeitschrift für vergleichende Sprachforschung* 12, 81–138, 160 (1863).

“Über die Casusbildung im Indogermanischen.” *Zeitschrift für vergleichende Sprachforschung* 12, 241–266 (1863).

*Lehrbuch der Trigonometrie für höhere Lehranstalten.* Enslin, Berlin, 1865. Also under the title *Lehrbuch der Mathematik für höhere Lehranstalten. Zweiter Teil: Trigonometrie.*

“Die Italischen Götternamen.” *Zeitschrift für vergleichende Sprachforschung* 16, 101–119, 161–196 (1867).

“Grundriss der Mechanik (für den Unterricht in Prima).” *Programme des Königlichen und Stadtgymnasiums zu Stettin* (1867).

“Verschiedene mathematische Bemerkungen. I. Bildung rationaler Dreiecke. II. Angenäherte Konstruktion von  $\pi$ . III. Lösung der Gleichung  $x^3 + y^3 + z^3 + u^3 = 0$  in ganzen Zahlen.” *Grunerts Archiv* 49, 1–3; 49–50 (1868).

“Elementare Auflösung der allgemeinen Gleichung vierten Grades.” *Grunerts Archiv* 51, 93–96 (1870).

*Deutsche Pflanzennamen.* R. Grassmann, Stettin, 1870.

“Feihoss, Totxoös, dehās.” *Zeitschrift für vergleichende Sprachforschung* 19, 309–310 (1870).

“Zur Theorie der Kurven dritter Ordnung.” *Göttinger Nachrichten* (1872), 505–509.

“Über zusammengehörige Pole und ihre Darstellung durch Produkte.” *Göttinger Nachrichten* (1872), 567–576.

*Wörterbuch zum Rig-Veda*. F. A. Brockhaus, Leipzig, 1872–75.

“Die neuere Algebra und die Ausdehnungslehre.” *Math. Ann.* 7, 538–548 (1874).

*Rig-Veda, Übersetzt und mit kritischen und erläuternden Anmerkungen*. F. A. Brockhaus, Leipzig, 1876–77.

“Vier pommersche Volkslieder.” *Deutsche Lieder. Festgruss an Ludwig Erk zum fünfzigjährigen Dienstjubiläum*. A. Birlinger and W. Creelius, eds. Henninger, Heilbronn, 1876.

\*“Zur Elektrodynamik.” *Crelles J.* 83, 57–64 (1877).

“Ursprung der präpositionen im Indogermanischen.” *Zeitschrift für vergleichende Sprachforschung* 23, 559–579 (1877).

\*“Die Mechanik nach den Prinzipien der Ausdehnungslehre.” *Math. Ann.* 12, 222–240 (1877).

\*“Der Ort der Hamilton’schen Quaternionen in der Ausdehnungslehre.” *Math. Ann.* 12, 375–386 (1877).

“Über die physikalische Natur der Sprachlaute.” *Wiedemanns Ann.* 1, 606–629 (1877).

“Bemerkungen zur Theorie der Farbenempfindungen.” Appendix to *Elementen der reinen Empfindungslehre*, by W. Preyer. Dufft, Jena, 1877.

“Verwendung der Ausdehnungslehre für die allgemeine Theorie der Polaren und den Zusammenhang algebraischer Gebilde.” *Crelles J.* 84, 273–283 (1878).

*Ober den Abfall vom Glauben. Mahnungen an die wissenschaftlich Gebildeten der Neuzeit*. O. Brandner, Stettin, 1878.

## Index

- accelerating forces: their measure, 350; addition, 350
- acceleration [*Beschleunigung*], 168; of a point as second differential quotient, 490
- addition: and subtraction, general concept, 37; of areas, 108; of volumes, 109; of relative magnitudes, 185; of point magnitudes and of linear magnitudes, 326, 362. *See also* point, displacement, extension
- additive magnitude [*Summengrösse*]. *See* magnitude.
- affine assembly of magnitudes, 248; its structure, 248; relation between products with, 250; distinguished position of outer and regressive multiplication, 251; direct and reciprocal, 251; when one of two is the shadow of the other, 253; in the plane, 254; if two are affine, then their systems are collinearly or reciprocally related, 254–258
- algebraic: equation, 333; relation between several quantities, 452. *See also* function
- Ampère, A. M., 417, 418, 430, 473
- Ampère's law of electrodynamics: its hypothetical basis and failure, 417–419; removal of the arbitrariness from hypothesis, 419–422
- analytic: conjunction, 36; form, 38
- angle: as logarithm, 14, 382; of a quaternion, 530
- angular current, 418; action on an element derived from Ampère's law, 418–422
- associative principle, 527
- axis of equilibrium, 164
- barycentric reference system, 186
- base system [*Grundsystem*]. *See* shadow
- Bolyai, J., 409
- boundary elements [*Granzelemente*] of a vertex structure, 174
- Brewster, D., 446
- capacity [*Gehalt*] of a spherical magnitude, 374. *See also* content
- Cauchy, A. L., 451, 454, 466
- Cayley, A., 540
- center of mass or gravity [*Schwerpunkt*], 71, 164; motion of, 71, 363; of spherical magnitude, 374; equation of motion of, 493. *See also* mass
- characteristic [*Eigentümlich*] value or factor. *See* product
- characteristic [*Charakteristik*], Leibnizian, 316–318, 382–384; application to the analytic representation of spheres, planes, and straight lines, 318–320; defect, 320
- Clausius, R., 473–477, 481
- Clausius's law identical to Grassmann's, 473–476
- clefs algébriques of Cauchy, 451, 454
- collective deviation [*Gesammtabweichung*]: from a point, 67; theorems on, 67; of an elementary assembly, 156; of a series of point magnitudes, 362



collective moment: of several forces, 91; of motion, 91; general concept of, 109; if no forces act, 110; dependence of on reference point, 110; of parallel forces, 164

collective motion [*Gesammbewegung*] or collective force, 71

collective weight of a series of point magnitudes, 362

collinearly related systems, 255. *See also* affine and metric relations

collineation: its difference from congruence, 320–324; equations, 323; C. functions 329, 330

- color: three elements of, 435; tint, 435; complementary, 435, 437–438
- combinability [*Vereinbarkeit*], law of, 527. *See also* conjunction
- combination: concept of, 26; of two systems, 241; of point magnitudes, 324; when zero, 324; equations, 325; as multiplication, 324, 329; measure value of, 327
- combinatory product, 286; and outer product, 288; and regressive product, 294
- combined [*Zusammengesetztes*] (inner) system, 143, 239
- common [*Gemeinschaftliches*] system of two systems, 200, 201
- congruence, 320; difference from collineation, 322–324
- conic section [*Kugelschnitt*]: geometric equation of, 236, 285; through five given points, 236
- conjunction [*Verknüpfung*]: concept, 34; combinability of factors, 34; interchangeability of elementary, 35; synthetic and analytic, 36; of different orders, 39
- constraint forces, 357; equation for an assembly, 498
- constructions, 63; linear, 235, 255, 323; corresponding, 323
- content [*Inhalt*] of a spherical magnitude, 374. *See also* capacity
- coordinates: transformation for parallel, 112; general, 187, 269. *See also* reference system
- cross ratio [*Doppelquotient*], 259
- crystal structures [*Krystallgestalten*]: representation of, 269, 270; harmonic relations, 270
- current, 479–480
- curves in the plane, 235
- D'Alembert's principle, 498
- d'Arrest, 538
- degree [*Grad*]: of dependence, 200; of multiplication, 202. *See also* product
- dependence, linear, 326
- dependent [*Abhängig*]. *See* extension, elementary magnitude
- derived [*Ableiten*] by linear constructions, 323
- de Tilly, 409
- deviation [*Abweichung*]: of a point, 67; of an element, 156; of a displacement, 161; of an elementary magnitude, 180; of an extension, 181; of a multiple point square, 366; of an inner displacement product, 366; of a sum, 366; of an inner product, 368, 371; of a spherical magnitude, 373; of a spherical surface, 376; of a point from a spherical surface, 378
- difference [*Verschiedenheit*]. *See* equality
- difference plane [*Differenzebene*]: of two spherical surfaces, 376–380; for multiple sums of spherical surfaces, 381
- differential calculus: application of extension theory to, 167; geometric, 351. *See also* function
- differential quotients, 488–490
- Dillner, G., 525, 529–534, 539

displacement [*Strecke*], 47, 288; tentative symbol for, 48; addition and subtraction of, 48–50, 51–58; construction of, 54–57; and evolution, 58; representation of in a system of  $m$ -th order, 60; problems on, 65; motion of, 73–76; final symbol for, 160; addition with a multiple element, 161; compared with the outer product of two elements, 169, 174, 182, 327; multiples and fractions of, 485

divergence [*Ausweichung*]: of a rigid elementary magnitude, 174; of an elementary magnitude of first order, 178; of a sum of elementary magnitudes, 179; of an extensive magnitude, 180

division: general law, 41; arithmetic, 38; outer, 113; regressive, 224. *See also* quotient

domain [*Gebiet*], 29, 47, 285

double system, 130

Drobisch, M. W., 403

element: generating, 28; concept of, 45, 284; independent, 170

elementary [*Einfache*]: extensive form, 29; conjunction, 35; extensive structure, 47; factor, 82; extensive domain, 279; domain, 279

elementary assembly [*Elementarverein*], 156

elementary magnitude: concept of first order, 157; deviation and weight of first order, 157; addition, 157; representation of, 158; multiplication of, 159; represented as multiple element or displacement, 159, 161; dependent and independent, 170; product of  $n$  of first order, 171; of  $n$ -th order, 172, 288; rigid, 173; condition for equality, 180; and extensive magnitude, 180, 182; reference measures for, 185. *See also* magnitude

elementary representation of an extension, 106

elementary system of  $n$ -th order, 170; its relation to an extensive magnitude, 179; equality in, 179; in space, 182

elevation [*Höhenseite*] of a parallelogram, 85

elimination of an unknown from equations of higher degree, 152

elliptic terms in the expression for the force and mean motion, 504

Engel, F., 409, 469, 539

equal measure, quaternary units of, 530

equality [*Gleichheit*] and difference, concepts, 26, 33, 320

equally directed [*Gleichläufig*], 48

equally signed [*Gleichbezeichnet*]: magnitudes, 288; parts of lines, planes, etc., 289

equations: of first degree, 93; of  $m$ -th order, 139; when a factor can be dropped, 140, 243; shadow of, 140, 142, 243; derivation of a replacement assembly of, 145; between reference terms or indicators, 148; between elementary magnitudes of higher order, 179; between extensive magnitudes, 233; independent of measure, 259; between inner products, 340, 347; between inner magnitudes, 366; between inner products of magnitudes of first order, 369. *See also* curves, harmonic, magnitude

equilibrium [*Gleichgewicht*]: condition for, 92; of a body immersed in a fluid, 164; of forces in space, 194; of a system of mutually fixed points acted on by accelerating forces, 357–360

Euclid, 32, 61, 62

Euler, L., 541

evolution [*Änderung*]: continuous, 28, 47; of an element, 47; opposed, 48; addition of, 51–58; positive or negative circular, 461

exponential magnitude, geometric, 14, 15

exponentiation of a quaternion, 530

- extension [*Ausdehnung*]: extensive form, 27; elementary form, 29; extensive magnitude of first order (displacement), 47; of higher order as product, 77–78; of higher order and their addition, 97; elementary representation of, 106; independent, 107, 200; of supplementary order, 140; of a rigid elementary magnitude, 179
- extension theory [*Ausdehnungslehre*], concept of, 25
- extensive domain [*Ausdehnungsgebiet*], elementary, 279
- extensive: magnitude, 47, 103; structure of first order, 47; system, 260; quantity, 452
- facteurs symboliques of Cauchy, 454
- factor, combinatory: of first order, 286; of  $(n - 1)$ -th order, 288, 294. *See also* outer factor, product
- factors [*Glieder*] of a conjunction, 34
- final element [*Endelement*], 47
- finite [*Endlich*] magnitude, 225
- force [*Kraft*], 69; summation of, 69; as linear magnitude, 190; equivalent, 192; in a plane, 192–194; in space, 194; reduction of a sum, 195; analogy of two assemblies of, 145; sum, 195; assembly of and moment, 195; accelerating, 350; constant, 351; elementary, 491; inner and outer, 492; as partial differentials of a potential, 495; constraint equations for, 497. *See also* function, moment
- force couple [*Kraftpaare*], 193
- form, thought, 24; different types of, 24–28
- formal: product. *See* product. sum. *See* sum
- forms, general theory of, 33–43
- Fraunhofer, J., 439, 447
- Frege, G. W., 533, 541
- function: algebraic and homogeneous, 333; geometric, 349; algebraic of points, 352; partial differential quotients with respect to points, 353, 354; accelerating forces defined by, 352–356
- fundamental conjunctions [*Grundverknüpfungen*], 144
- fundamental evolution [*Grundänderung*], 47, 452; independent, 37
- fundamental laws of mechanics, 490–492
- fundamental measures [*Grundmasse*]: for extensive magnitudes, 147; for elementary magnitudes, 185

- Gauss, K. F., 166, 540
- geometric: exponential magnitude, 14; sum, 108; function, 349
- geometry: not a branch of pure mathematics, 24; present foundation of, 60–62; new formulation, 62–63, 279
- Gordan, P., 312
- Grassmann, J. G., 9, 27, 298
- Grassmann, R., 24
- Grassmann's law of electrodynamics, 422, 473, 475, 479; special case, 423; general case, 425, 479; analogy with gravitation, 425; and Ampère's law, 426–427
- guide system [*Leitsystem*]. *See* shadow
- Hamilton, W. R., 311, 525, 529, 539
- Hankel, H., 19, 301, 532, 539
- harmonic: equations, coefficients (weights), systems, 261; mean, 262, 267; sum, 262; equation and pole system, 265–267
- Hassenfratz, 446
- Hegel, G. W. F., 16
- Helmholtz, H., 279, 280, 310, 409, 435, 439–441, 445, 447–449
- Henrici, O., 311
- Hering, E., 449
- Huygens, C., 404, 408
- ideal spherical surfaces, 375
- incident [*Eingeordnet*] magnitudes, 215
- independent [*unabhängig*]. dependent in zeroth degree, 200; magnitudes of first order, 226; magnitudes of  $(n - 1)$ -th order, 227. *See also* evolution, extension, element, elementary magnitude
- indicators [*Zeiger*] of an extensive magnitude, 148; of an elementary magnitude, 185; equations between, 188, 189; of an elliptic term, 504
- indifferent form, 38
- inferior system [*Untersystem*], 230
- infinite [*Unendliche*] magnitude, 225
- initial element [*Anfangselement*], 47
- inner element [*Zwischenelement*]: concept, 175; analytic representation, 175–176
- inner square: of a displacement, 334; of a point, 364
- intensity of color, 435
- intensive magnitude, 27
- interchange law [*Vertauschungsgesetz*]: in outer multiplication, 81, 107; geometric basis for, 85; in statics, 93; in regressive multiplication, 209
- interior elements of a vertex structure, 175. *See also* product, magnitude
- internal forces, 92; restriction on, 109

- Jacobi, K., 303
- Kirchhoff, G. R., 485
- Klein, F., 300, 514, 515
- Kölpe, O., 449
- Lagrange, J., 10, 72, 88, 110, 191, 303
- Laplace, P., 10, 110, 511
- Leibniz, G. W., 316–317, 330, 383–384, 403–404, 408–409
- length (tensor) of a quaternion, 529
- Lie, S., 300–301, 409, 469, 539
- Liebisch, 307
- linear [*Linear, Lineal*], relations, 252; related systems, 254. *See also* constructions, dependence
- linear magnitudes [*Liniengrößen*]: parts of definite lines, 182, 288; sum of two, 183, 289; and displacement, 329, 330. *See also* addition, combination
- living force and work, 494
- Lobachevsky, N. I., 409
- magnetic axis, 165
- magnitude [*Grösse*]: spatial and elementary of first order, 288; of  $n$ -th order, 288; seven types of, 289; product with a number, 332; algebraic homogeneous equation, 333; corresponding algebraic conjunction, 333; points and displacements, 363; inner, 366, 374. *See also* sum
- mass [*Masse*], 70, 492
- mathematics, concept and classification of, 23–28
- Maxwell, J. C., 449
- mean [*Mitte*]: of a series of points, 68, 162; of an assembly of points with weights, 163; as axis, 163
- mean: integration of a linear differential equation, 500; motion of an assembly, 499–506
- mean magnitude: of an inner point square and an inner displacement product, 366; of an inner product of a point and a displacement, 369; basis for nomenclature, 372; surface of two point magnitudes, 372, 373
- measure [*Masswerth*], 233; value, 324; of a quaternion, 530
- mechanics, principles of, 69–71; differential equations of, 356
- metric: relations of collinear point structures, 259; value. *See* measure
- mixed [*Gemischtes*] product. *See* product
- mixture of colors, 437–443; and Newton's law, 439–440
- Möbius, A. F., 11, 166, 186, 255, 302, 307, 323, 347, 385f, 403, 411–414, 442–443, 453, 514–515

moment: of a force, 90; of motion, 91; dependence between, 112; of parallel forces, 164; as deviation, 190–191; as force magnitude, 194; general theorem on, 194

motion [*Bewegung*], 91, pure theory of, 347

multiple: of a magnitude, 148; sum of displacements, 155, 375

multiple equation [*Vielfachengleichung*], 112

multiplication: general concept, 39; general laws of, 40; real concept, 42; arithmetic, 38; of extensive quantities, 453; symmetric, 455–459; circular, 459–463; linear, 463–466; algebraic, 465; outer, 465; application of, 466; inner and circular, 467; of complex quantities, 467; outer, inner, and central, 525, 526. *See also* product

multiplicative relation to addition, 82

multiplying a magnitude, 212; and sum, 293, 294; inner, 369

nearest covering [*Nachstumfassendes*] system, 201, 210

Neumann, F., 307, 430

Newton, I., 69–70, 316, 435, 439, 445, 446, 449

nonzero or nontrivial [*Geltender*] value, magnitudes of, 78, 369

normal set, 526; of three pairwise perpendicular displacements of unit length, 487; transformation of, 487

Noth, H., 20

number, concept of, 26

numerical magnitudes [*Zahlengrößen*], 126; conjunction of, 126–132; as extensions, 133

numerical relation, 196

numerically derivable, 279

Ohm, G., 426

open: squares, 277. *See also* product

oppositely directed [*Gegenläufig*], 48

order [*Stufenzahl*]. *See* extension, system, product

orthogonally proportional surface areas and displacements, 339–341

outer [*Äusseres*]: factor, 201. *See also* product, division

outer [*Ausseres*]: product of two point magnitudes and displacements, 171; application to space, 182, 288; magnitude, 254, 324; as geometric function of time, 349; conjunction and displacements, 360; magnitude, 324; square, inner, 364. *See also* element, elementary magnitude



Poinsot, L., 514, 515

point: addition of, 291–292, 293, 326–329. *See also* element, elementary magnitude

point assembly [*Punktvereine*], 163

point representative [*Punktträger*], 276

pole system of a harmonic equation, 261

Poncelet, V., 267, 268

position of a point magnitude, 364

potential, 495

Preyer, W., 20, 449

principal measure [*Hauptmass*] of a system of  $n$ -th order, 147, 210

principal system: of an equation between extensive magnitudes, 140; of a regressive product, 203

product [*Ergebniss*] of a conjunction, 34

product [*Produkt*]: outer, 77, 81–84, 103–106, 171–174, 486–488; regressive, 199–217, 282, 294; outer and regressive, 216–222, 226–237, 294–295; with respect to a double system, 230; open, 271–277; inner, 334–344, 360–365, 487. *See also* combinatory

progressive [*fortschreitend*] multiplication, 127

progressively, sides of a triangle taken, 345

projection: of a system, 241; different types in a plane, 241; relation to shadow, 260. *See also* shadow

proportion: as equality of two quotients, 125; pure geometric representation in geometry, 134–138; between parallel displacements, 134, 136; between numerical magnitudes and spatial magnitudes, 332

pure factors or magnitudes, 213; value, magnitudes of, 78, 369

quaternary units, 530

quaternion, 528, 529. *See also* length, quaternary unit, measure, angle, exponentiation, quotient

quotient: of extensions, 114; geometric example of, 116; unique, 116; of similar magnitudes, 121; of relative magnitudes, 224; general uniqueness, 225; of differently directed displacements, 314; of a measure, 334, 339; of nonparallel displacements, 382; of quaternions, 530

Raphael, 447

ray, 419–422

reciprocal, related systems, 255, 257. *See also* affine

reference axes [*Richtaxen*], 147

reference domains [*Richtgebiete*], 147

reference elements [*Richtelemente*], 185

reference measures [*Richtmasse*]: for extensive magnitudes, 147; for elementary magnitudes, 185; in space, 185; and products, 250

reference systems: for extensive magnitudes, 147; for elementary magnitudes, 185–187

reference terms [*Richtstücke*], 147

regressive [*Eingewandt*] factors, 208. *See also* product

relation [*Verwandtschaft*]. *See* affine

relative [*Bezüglich*]. *See* product, relative

relative magnitudes [*Beziehungsgrößen*], 217; generalization of, 224; with respect to a double system, 230. *See also* addition, degree

relative number [*Beziehungszahl*] of a regressive product, 202

relative system [*Beziehungssystem*], 203

replaceable equations, 147

Reye, K., 21

Riemann, B., 279, 310, 476–477

rigid [*Starr*]. *See* elementary magnitude

rotation [*Schwenkung*], 13; about a fixed point, 517–524; formal, represented by a quotient of displacements, 533

Saint-Venant, B., 466

Schlegel, V., 20, 299

Schweins, 517

sense, 39, 48

sequence number [*Gefolgzahl*], 178

Servois, F. J., 301

shadow [*Abschattung*]: outer, 142–145, 149–150; general concept of, 239; analytic representation, 240; of a relative magnitude, 240; of a sum, 242; of a product, 244; sense of, 243, 247; as a special case of the affinity, 252

similar [*Ahnlich*], 137

similar [*Gleichartige*]: magnitudes (forms), 39; displacements, 48; extensions, 75; elementary magnitudes, 159; relative magnitudes, 224; spatial magnitudes, 332

spatial function, 350. *See also* magnitude

specific. *See* product

spherical magnitude: concept, 371–374; as multiple inner point product, 371; as sum of inner point squares, 373. *See also* sum

spherical space of Helmholtz, 280

square, inner, 487; of a displacement, 334. *See also* open

statics, new foundation for, 192–198; a problem from, 353–360

Straight line, 63, 188

Study, E., 307, 413

Sturm, R., 20

subordinate [*Untergeordnet*], 115, 245

subordinate form [*Form der Unterordnung*]. *See* product (regressive)

subtraction. *See* addition

sum: of extensions of equal order, 97, 102–105; harmonic, 262; of inner magnitudes, 366–367; and spherical magnitude, 371–374; of inner planar magnitudes, 369–370; of inner point squares, 372; and factor

surfaces, 375

superior system [*Obersystem*], 230

superordinate [*Übergeordnet*], 230

supplement of a displacement, 527

supplementary number [*Erganzzahl*], 210

supplementary: order, 140; reference measure, 147

surface motion, 494

Sylvester, J., 303

symmetric, 233

synthetic conjunction, 36

system or domain [*Gebiet*], 29; of first order, 28; of second and higher order, 29; generation of, 50, 58; of an assembly of magnitudes of first order, 252

thought form [*Denkform*]. *See* form

tides, 506–512

total intensity of a mixture of colors, 443

Unger, F. W., 446

uniqueness [*Eindeutigkeit*] of analysis, 37

unit, absolute and relative, 452

Uylenbroeck, 316, 403, 404, 408

Varignon, P., 88, 302

velocity [*Geschwindigkeit*]: of a point, 168, 348; of the projection, 348; as differential quotient of a geometric function, 349

vertex structure [*Eckgebilde*], 174; comparison with a product of displacements, 176–179; equality of, 178; calculation of, 178

volume, 289

von Staudt, K., 305, 514–515

Weber, W., 430, 476

weight: of an elementary assembly, 156; of a point magnitude, 362; of a displacement, 362; of a sum of factors of first order, 362; of a spherical magnitude, 374

work and living force, 494

zero, 38, 78

## Hermann Günther Grassmann

(1809–1877)

Hermann Günther Grassmann was born and died in Stettin, Germany. With Hamilton and Boole he ranks as one of the pioneers of modern algebra. As well as his mathematical contributions, he was an accomplished linguist, folklorist, musician, botanist, and physicist. He translated the Hindu Rig Veda into German verse, harmonized folksongs, invented a heliostat, and wrote important works on color, electricity, and acoustics. His Sanskrit dictionary on the Rig Veda is still widely used.

Grassmann's most famous mathematical work, *Die lineale Ausdehnungslehre*, was largely ignored at the time of its publication, but praised by Gauss and Möbius, who were however baffled by its philosophical orientation and unusual terminology. Grassmann greatly influenced the philosophers Paul Carus (whom he taught) and Alfred North Whitehead.

This volume includes the full text of Grassmann's famous masterwork of 1844, *Linear Extension Theory: A New Branch of Mathematics*, which here appears in English for the first time, along with his *Geometric Analysis*, and important papers on mathematics and physics.

“Grassmann is a pivotal figure in the historical evolution of mathematical language to characterize human understanding of the physical world... What sets Grassmann ahead of other creative mathematicians is his systematic vision of a universal geometric calculus. This vision marks him as one of the great conceptual synthesizers of all time. Accordingly, the chief presentation of that vision, *Die lineale Ausdehnungslehre*, deserves a place alongside Euclid's *Elements* and Descartes's *Géométrie* in the library of immortal mathematical masterworks.”

— DAVID HESTENES  
Professor of Physics and Astronomy  
Arizona State University

“Mathematicians as well as historians and philosophers of mathematics have long recognized the need for an English translation of Hermann Grassmann's masterpiece, *Die lineale Ausdehnungslehre* of 1844. Lloyd Kannenberg has now not only very effectively fulfilled this need but in addition has translated other key documents both by and about Grassmann. The publication of this volume is a major contribution to scholarship.”

— MICHAEL J. CROWE  
Professor of the History and Philosophy of Science  
University of Notre Dame